



DeepLearning.AI

Probability and Statistics for Machine Learning and Data Science

-

Week 3: Sampling and Point Estimates

W3 Lesson 1



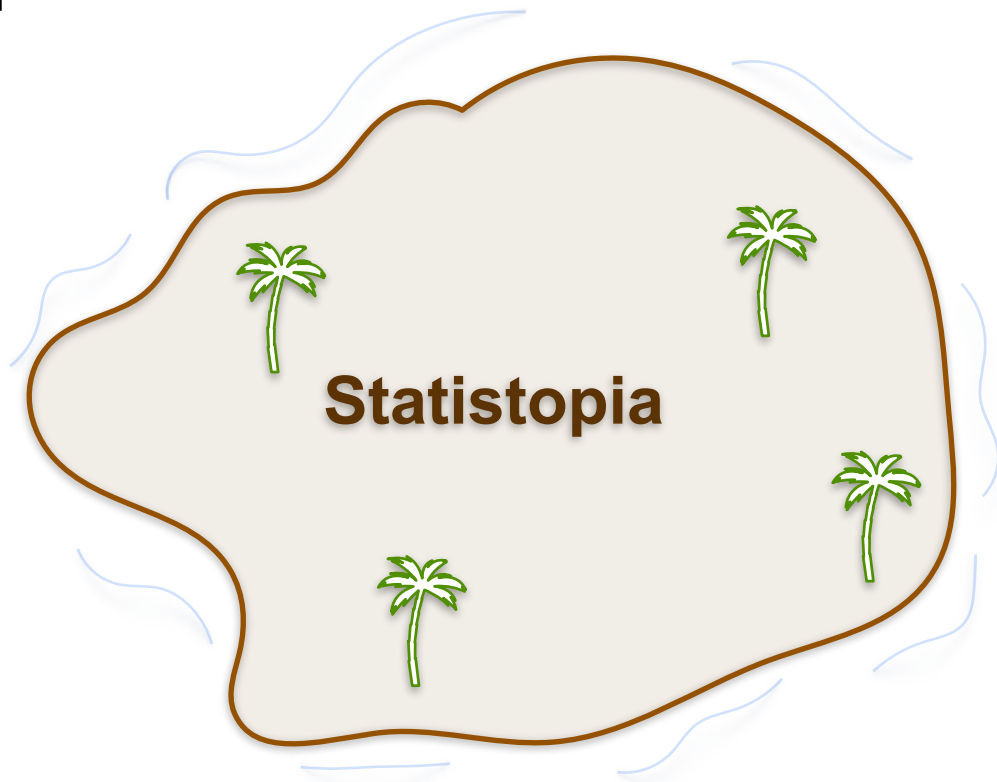
DeepLearning.AI

Sample and Population

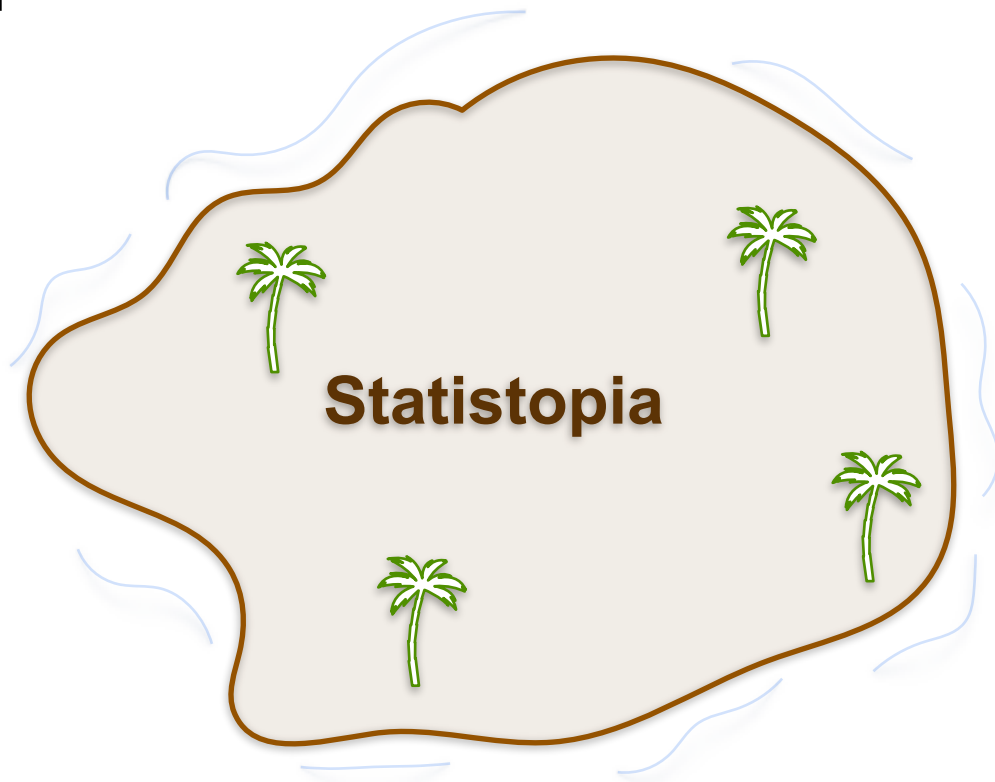
Population and Sample

Population and Sample

Population and Sample



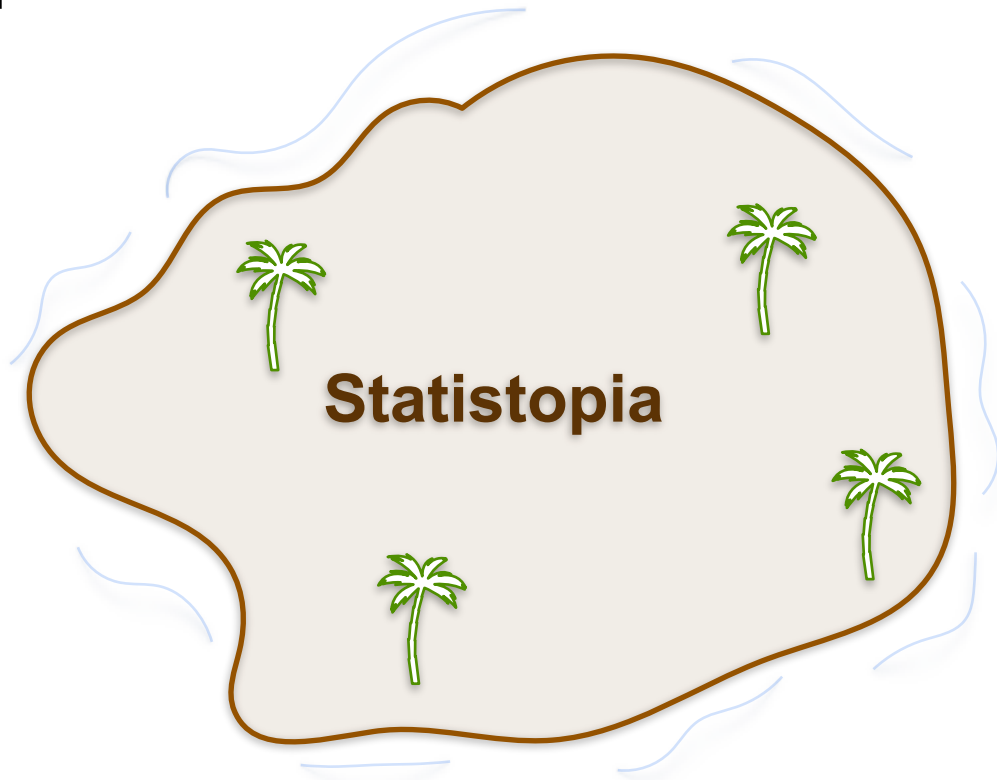
Population and Sample



Population and Sample



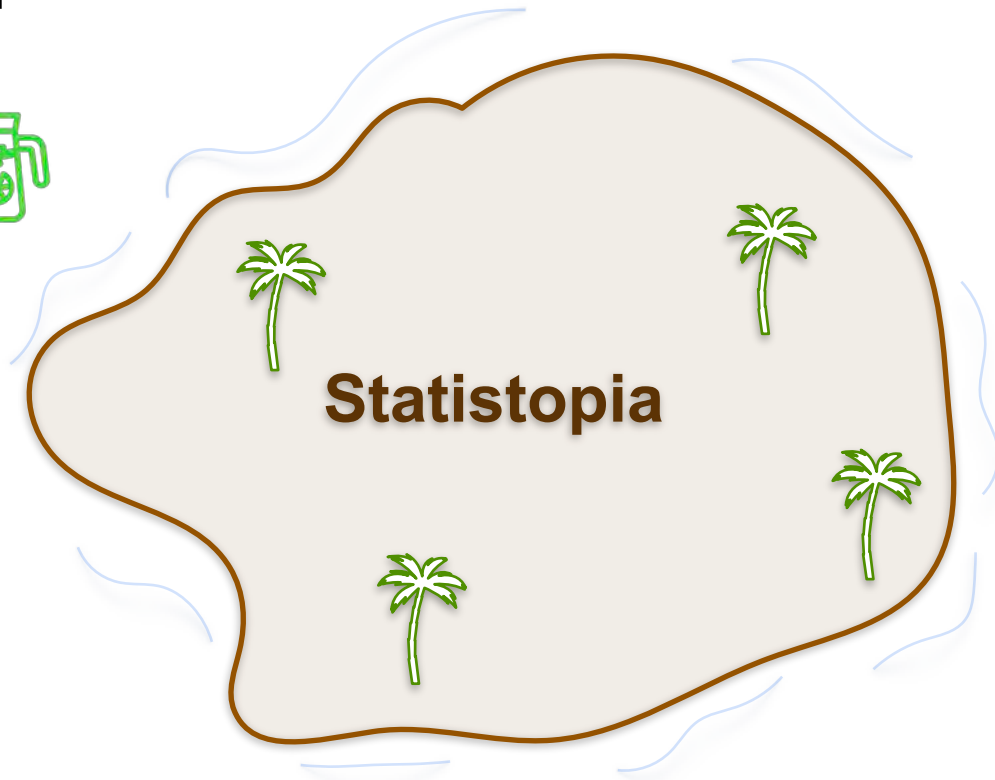
Find the **average height** of
the people living on
Statistopia



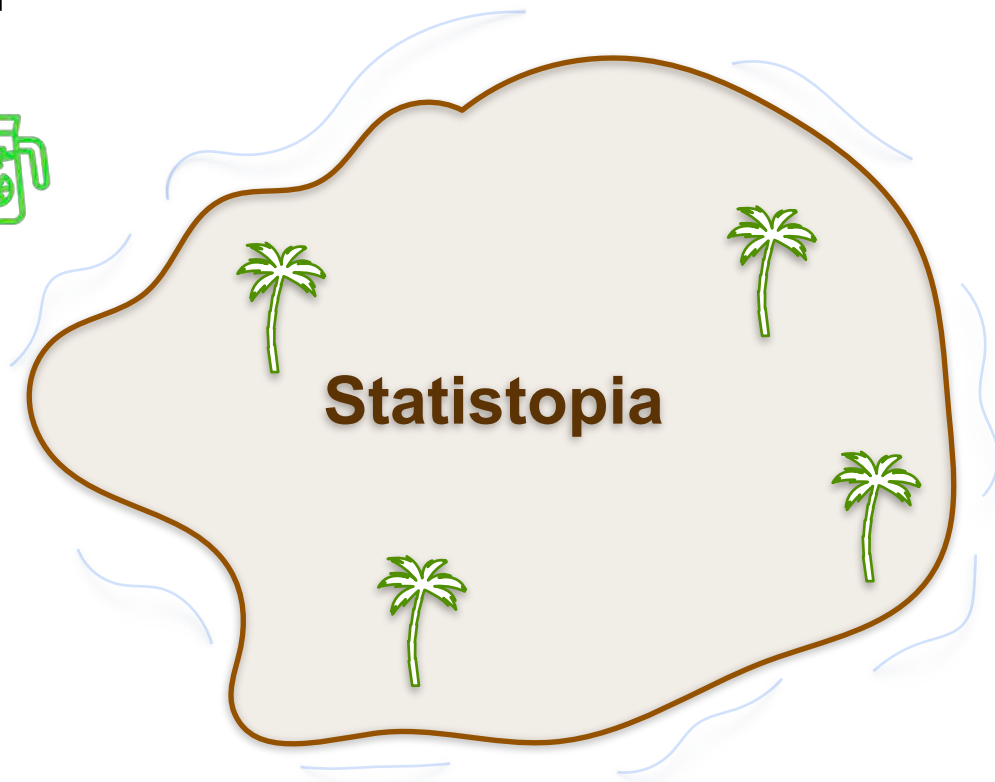
Population and Sample



Find the **average height** of
the people living on
Statistopia



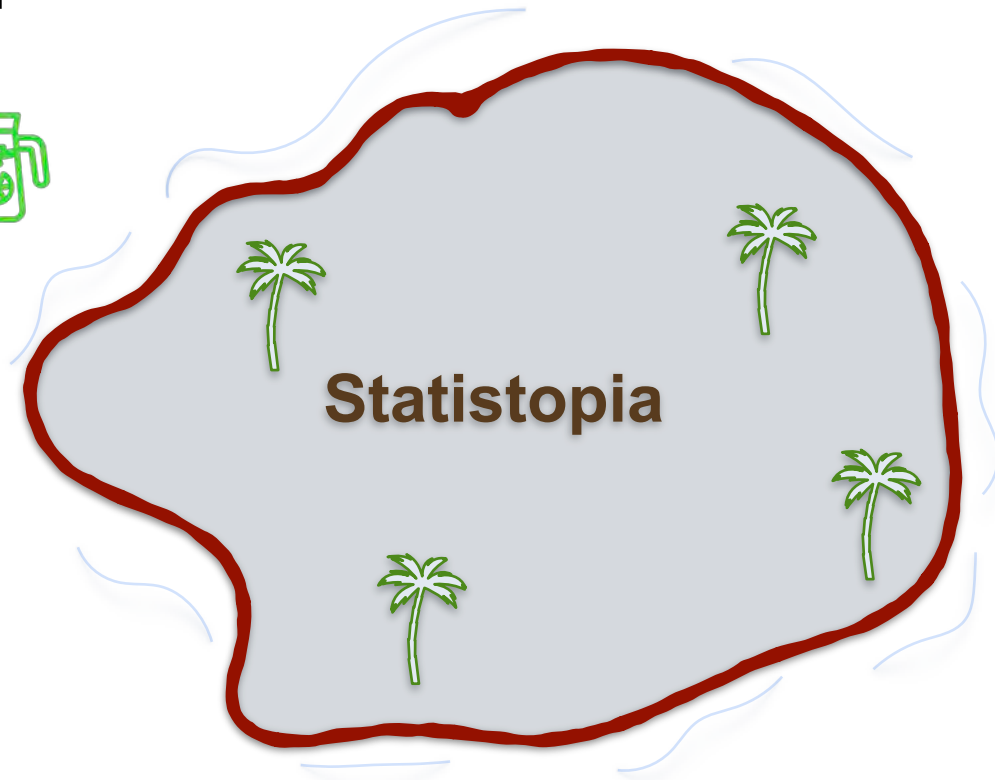
Population and Sample



Population and Sample



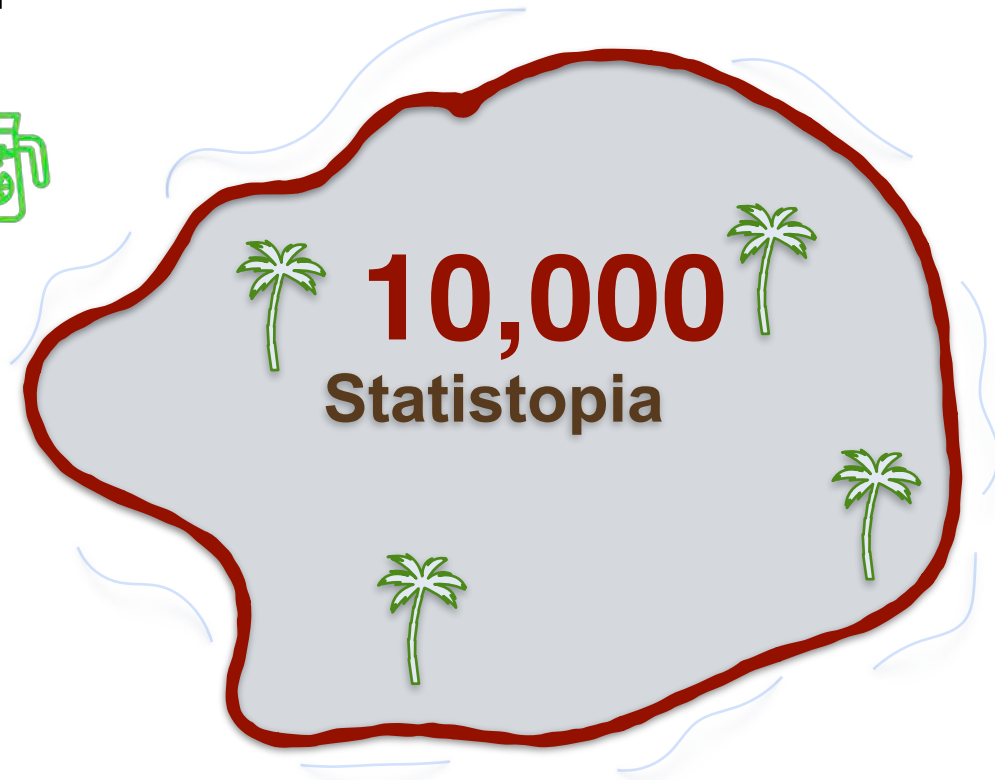
- Ask everyone on the island for their height.
- Divide by the total number



Population and Sample



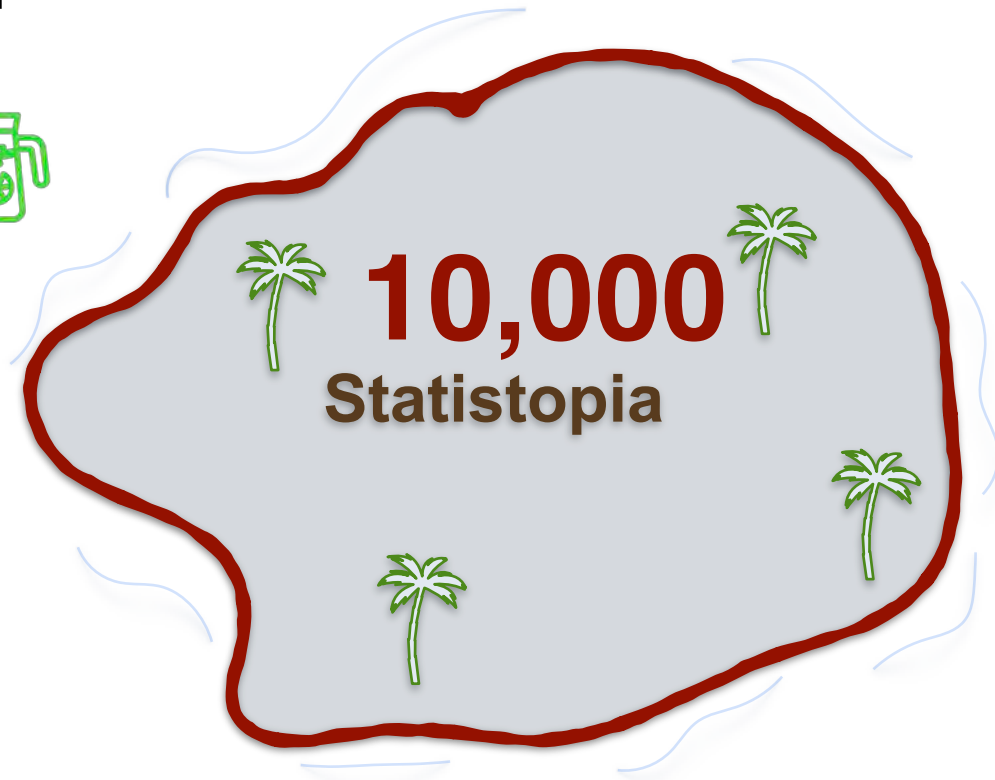
- Ask everyone on the island for their height.
- Divide by the total number



Population and Sample



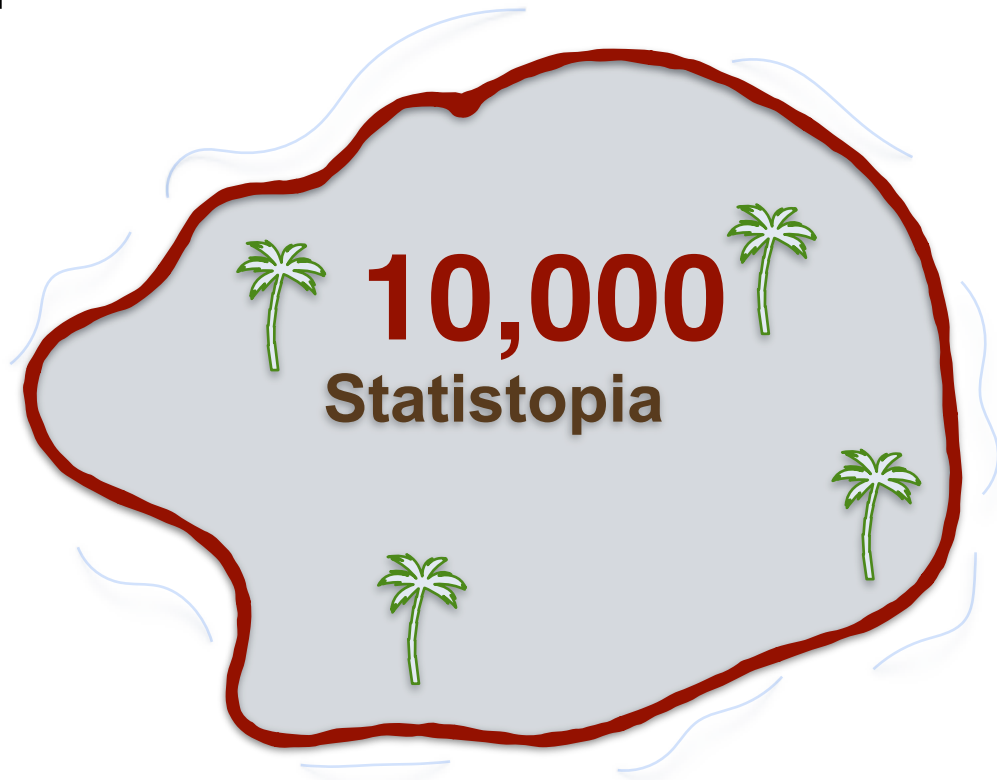
- Ask everyone on the island for their height.
- Divide by the total number



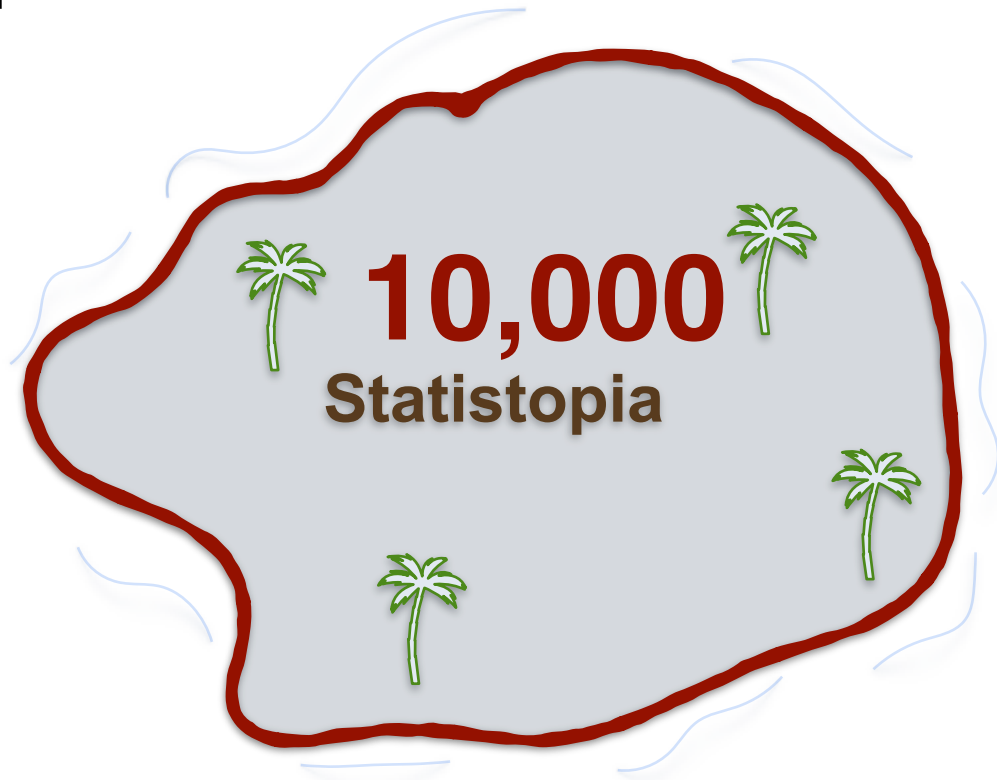
Population and Sample



- Ask everyone on the island for their height.
- Divide by the total number



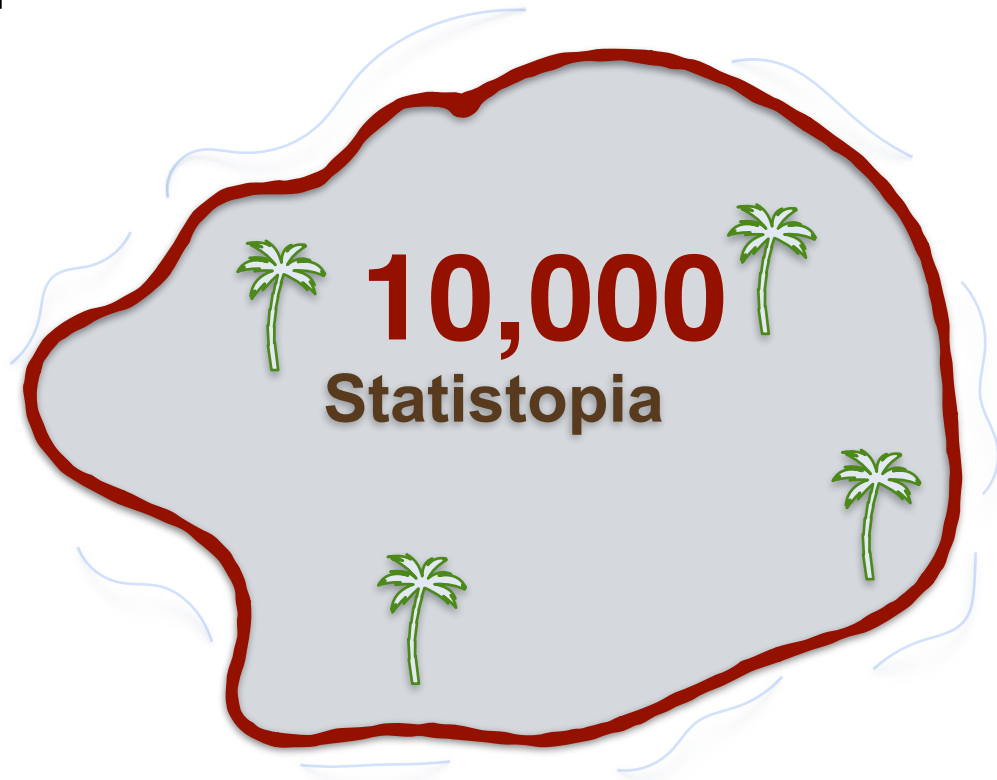
Population and Sample



Population and Sample



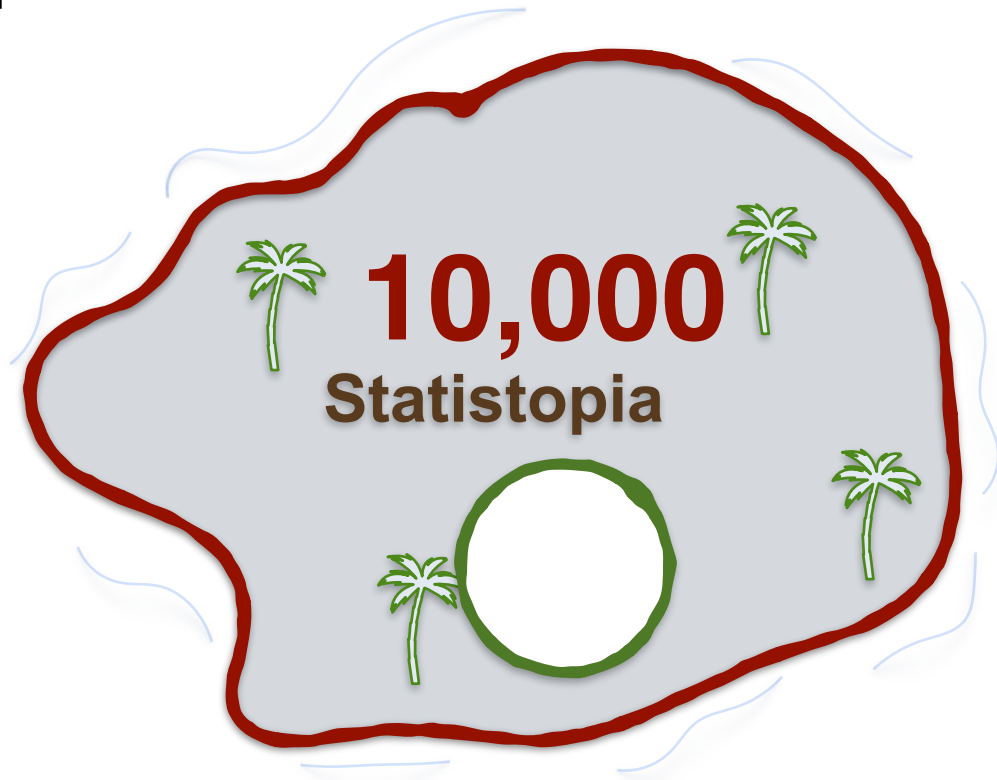
- Only ask a subset of the group to estimate the average height



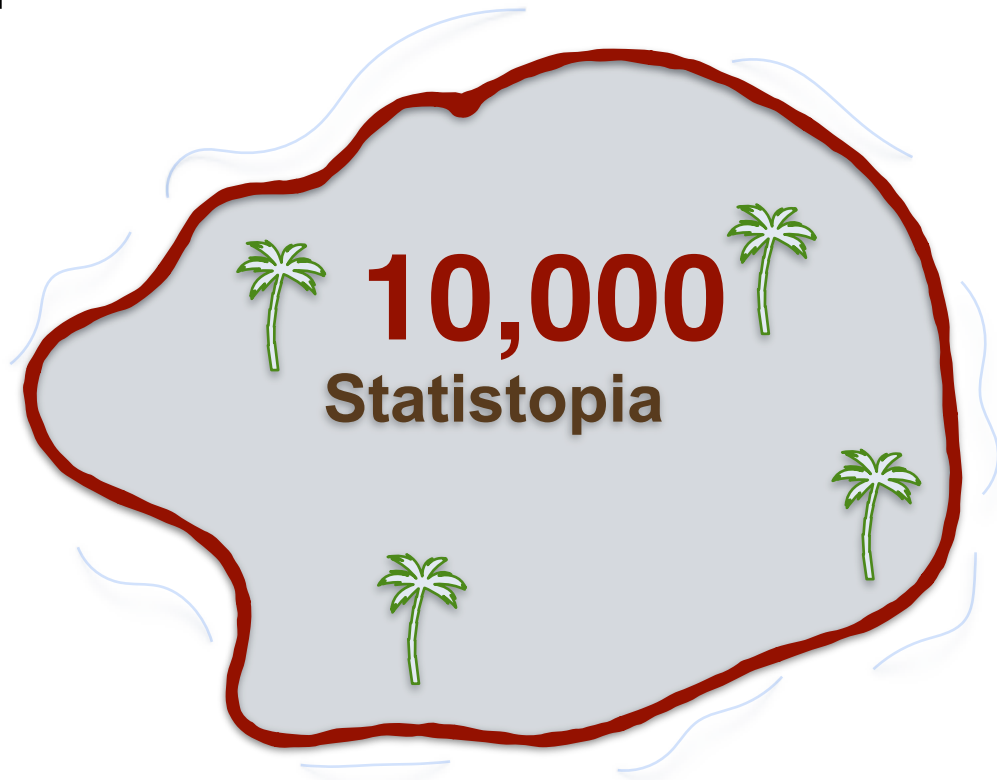
Population and Sample



- Only ask a subset of the group to estimate the average height



Population and Sample

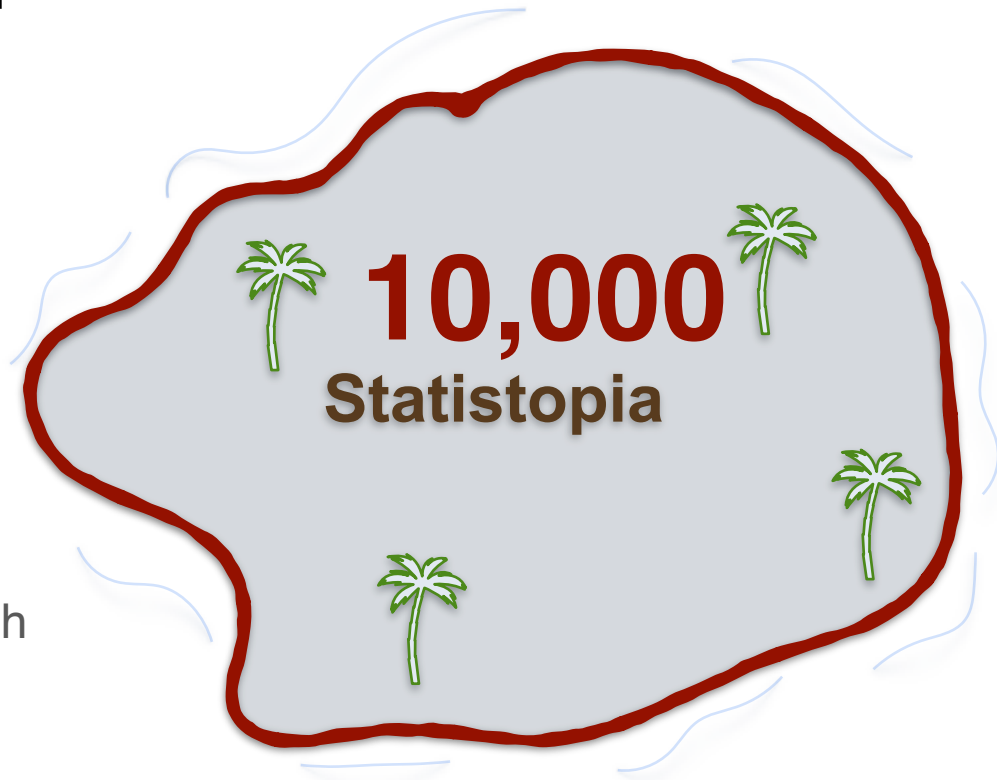


Population and Sample



Population:

the entire group of individuals or elements you want to study which share a common behaviour



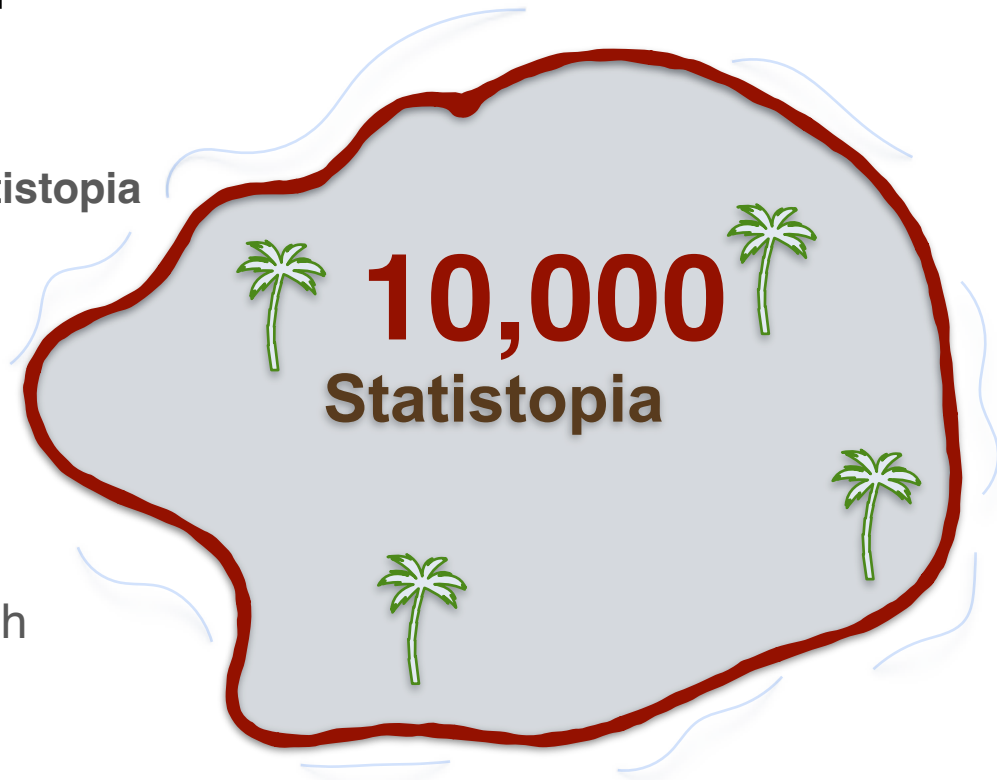
Population and Sample



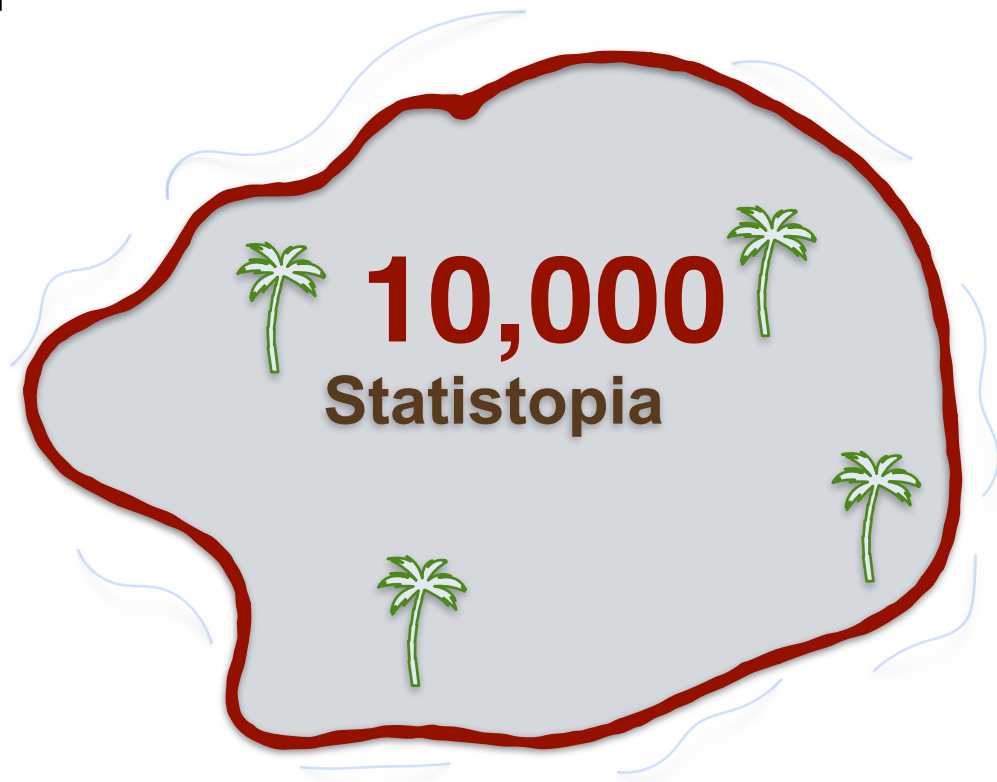
The people of statistopia

Population:

the entire group of individuals or elements you want to study which share a common behaviour



Population and Sample

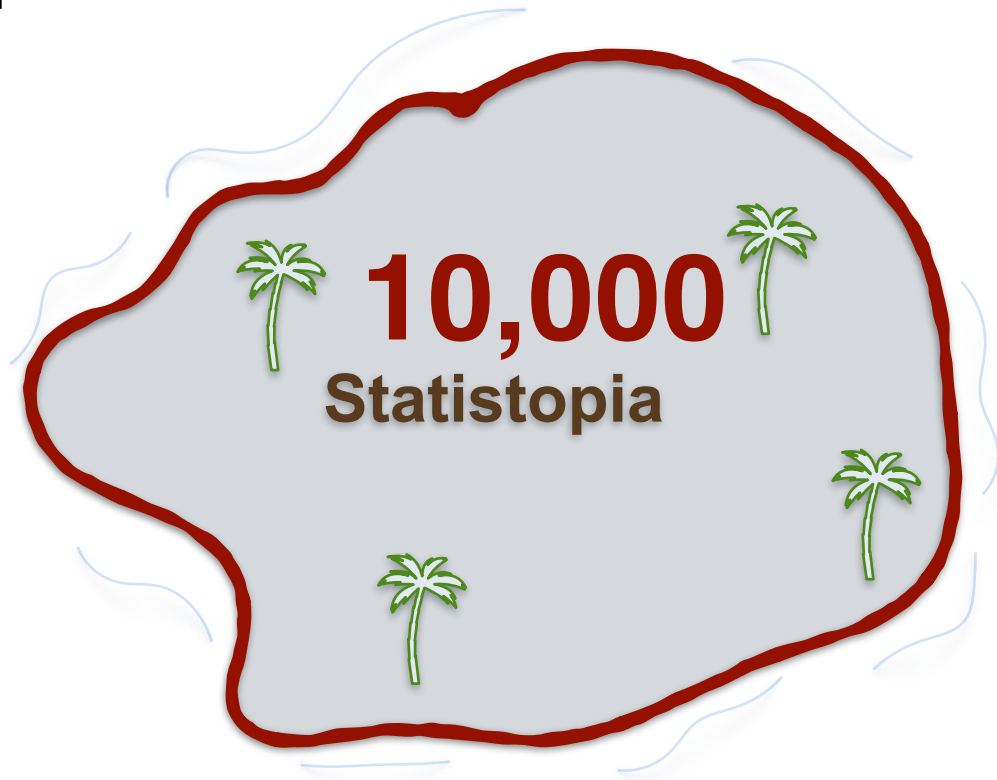


Population and Sample



Sample:

subset of the population you use to draw conclusions about the population as a whole



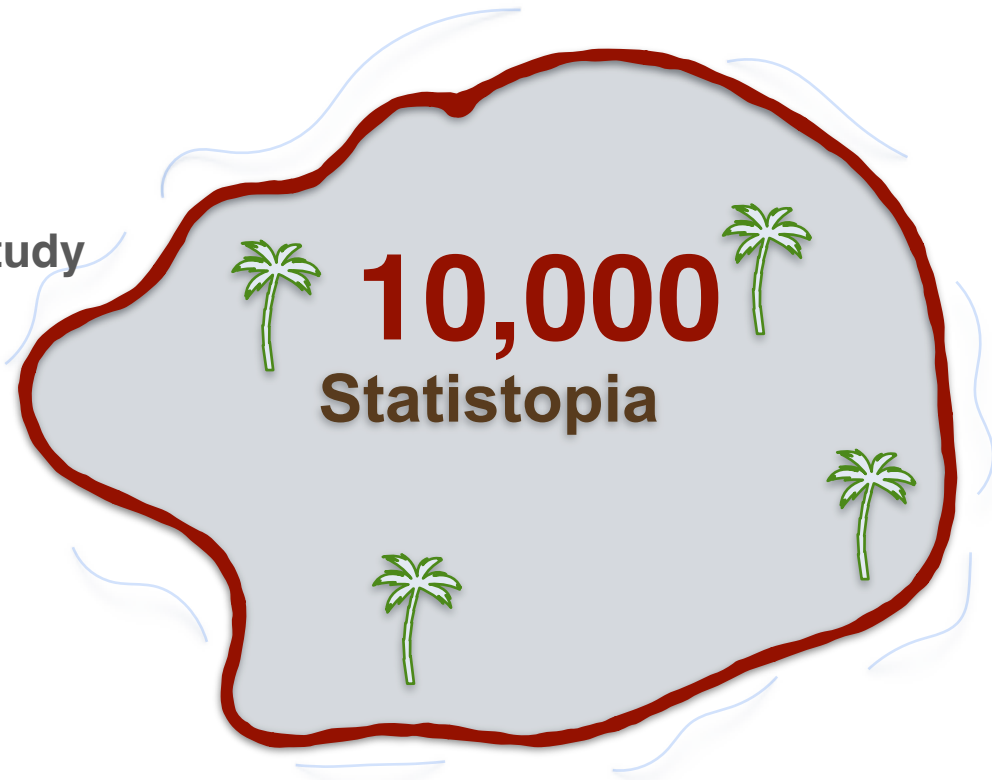
Population and Sample



The people you
select for your study

Sample:

subset of the population you use
to draw conclusions about the
population as a whole



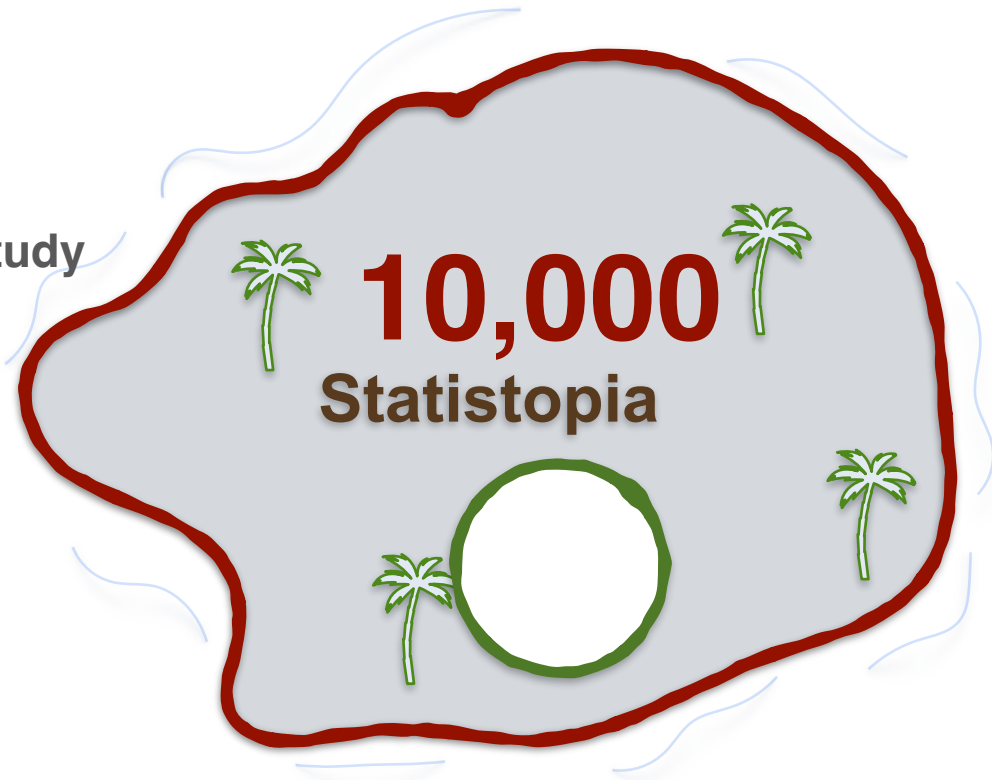
Population and Sample



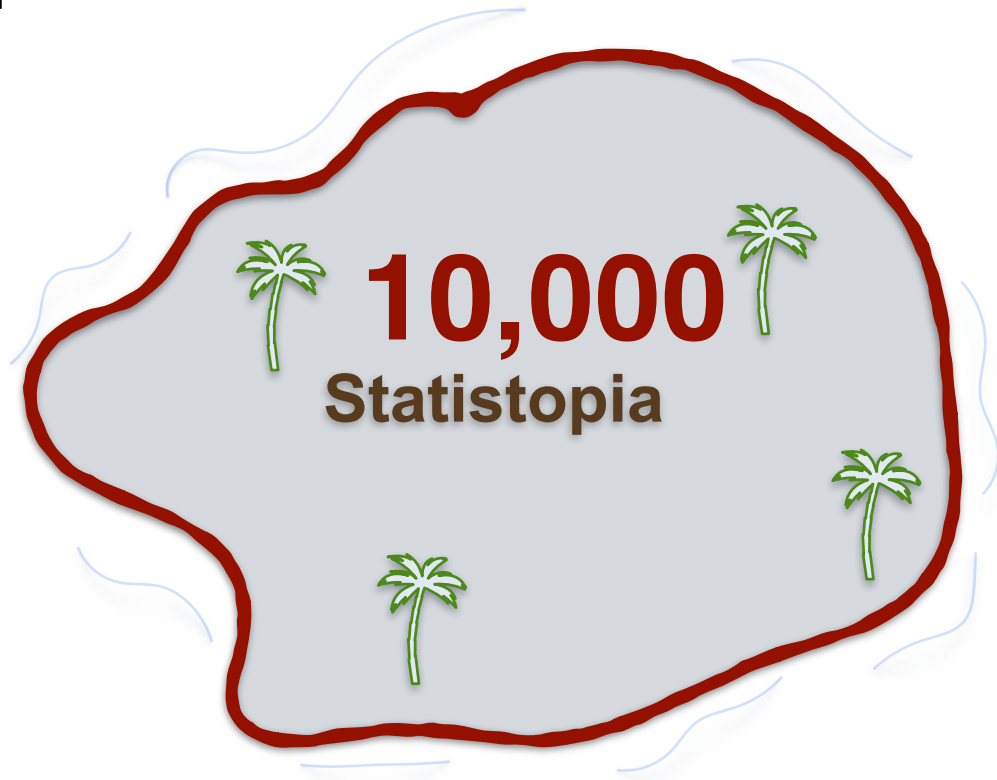
The people you
select for your study

Sample:

subset of the population you use
to draw conclusions about the
population as a whole



Population and Sample

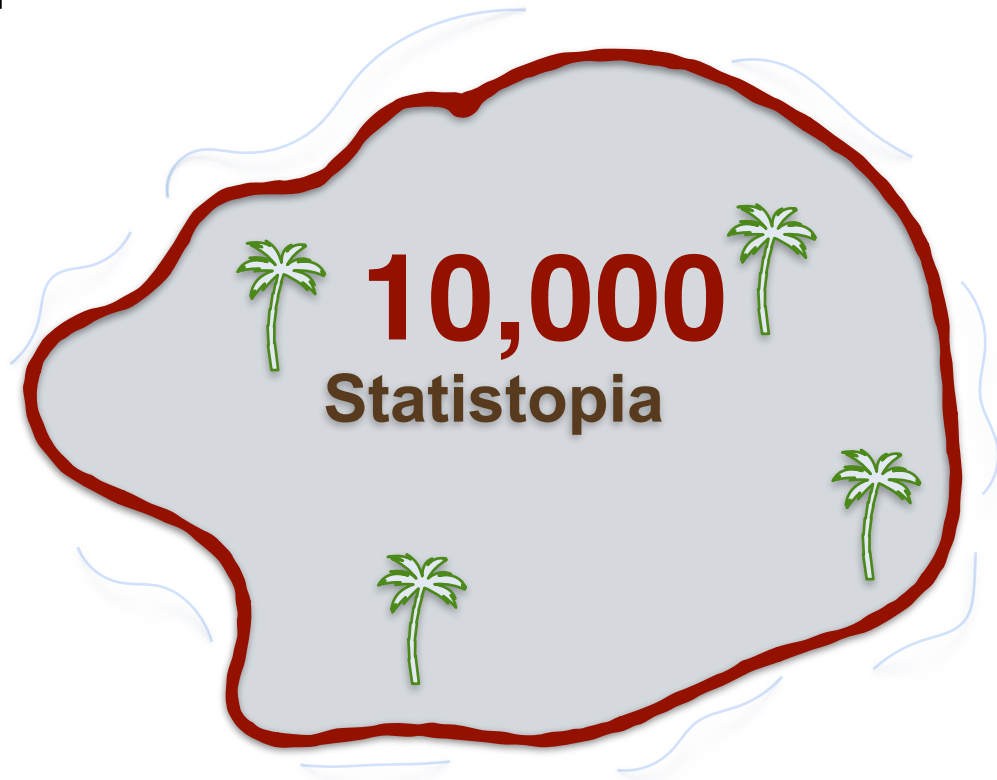


Population and Sample



Population Size (N)

10,000

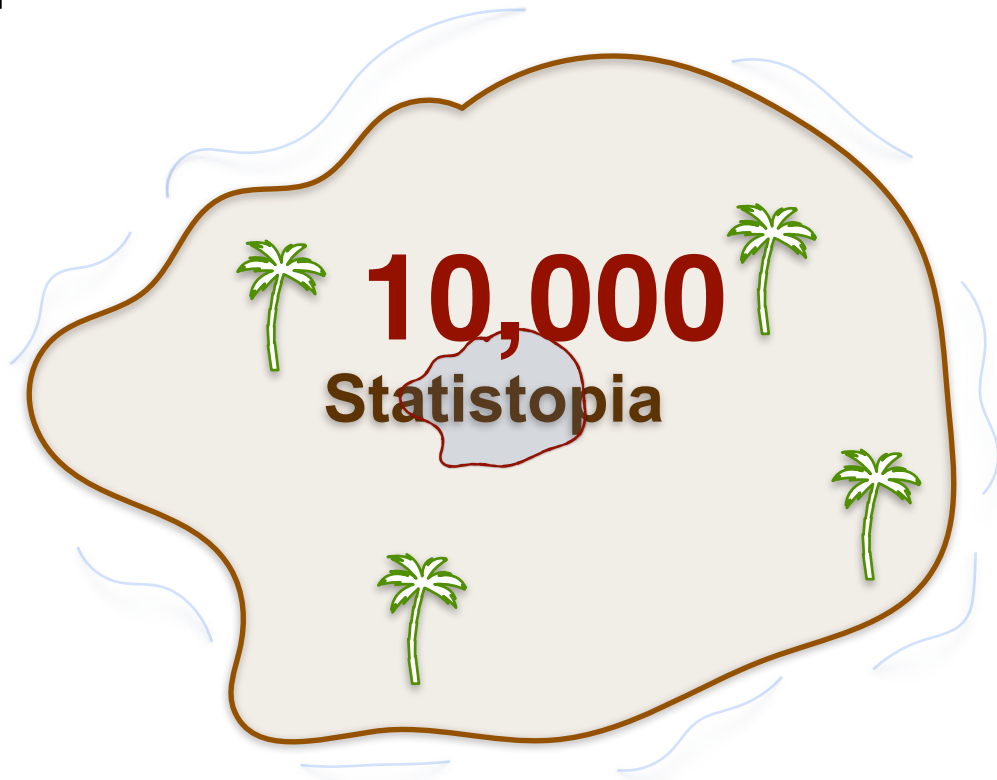


Population and Sample



Population Size (N)

10,000



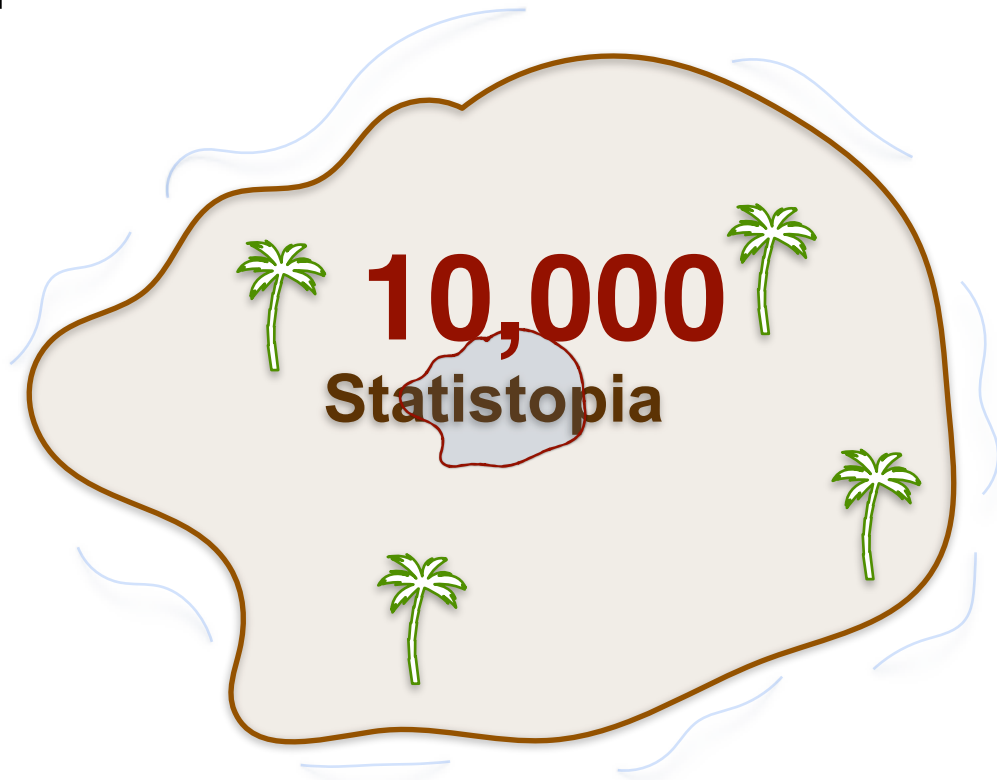
Population and Sample



Population Size (N)

10,000

Sample Size (n)



Population and Sample

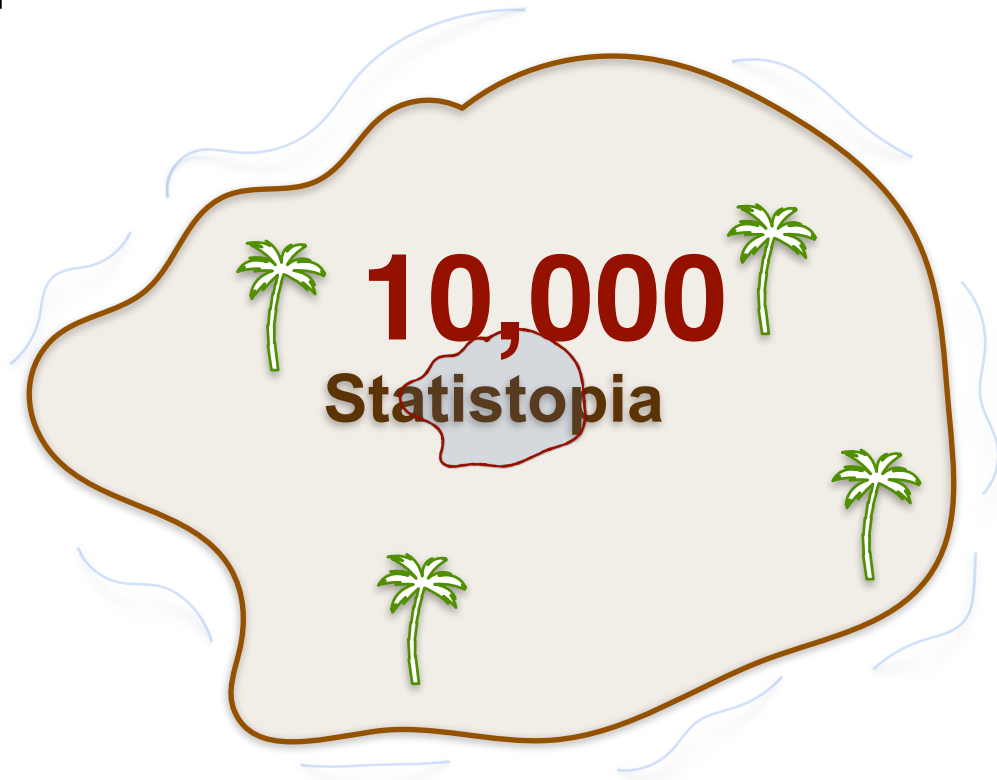


Population Size (N)

10,000

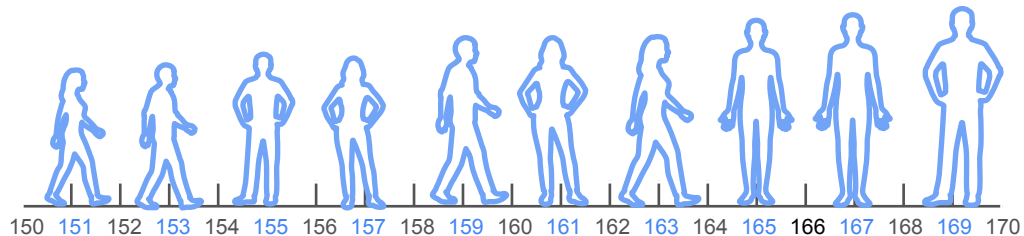
Sample Size (n)

1 - 9,999



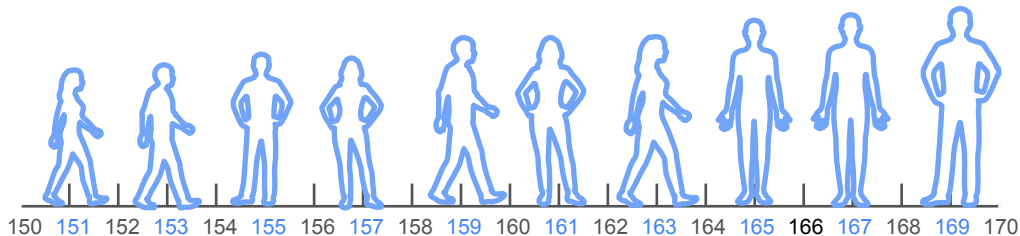
Population and Sample

Population and Sample

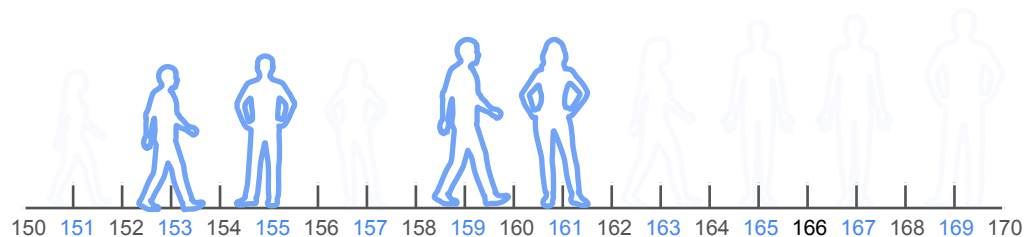


Population and Sample

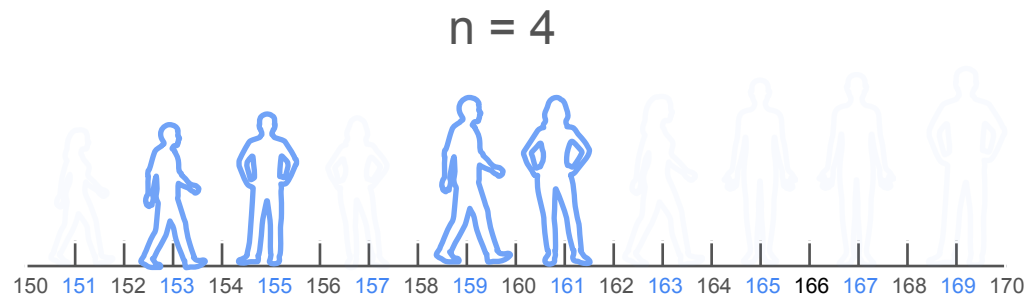
$N = 10$



Random Sampling

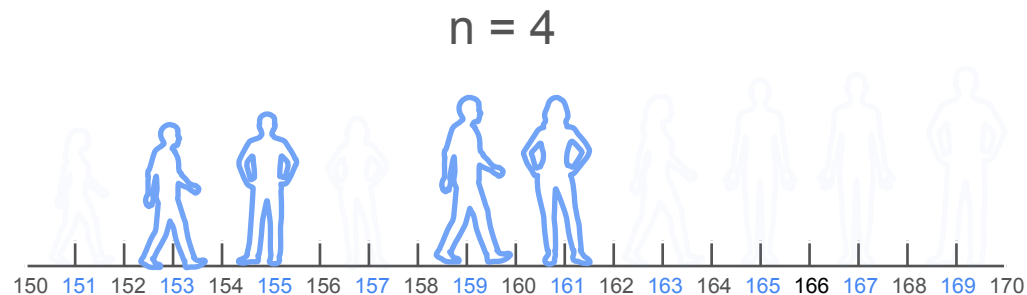


Random Sampling



Random Sampling

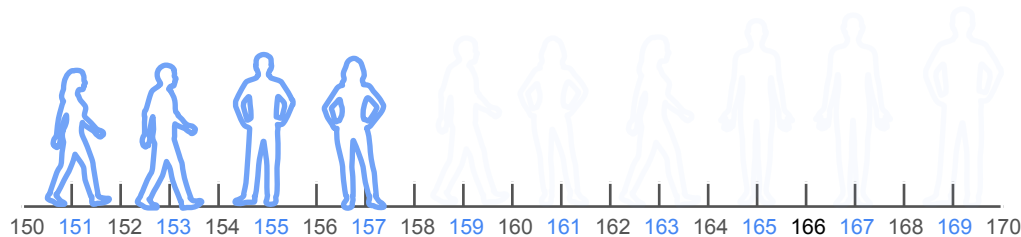
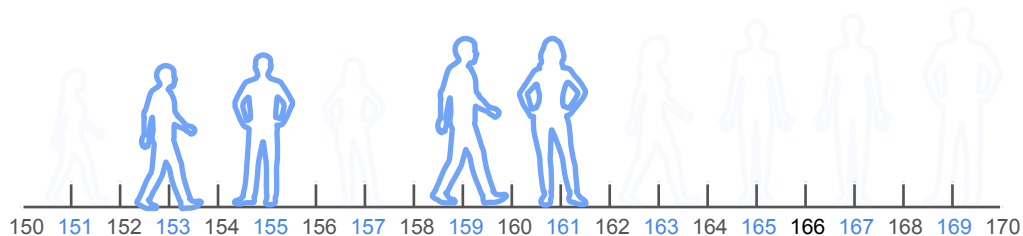
A



Random Sampling

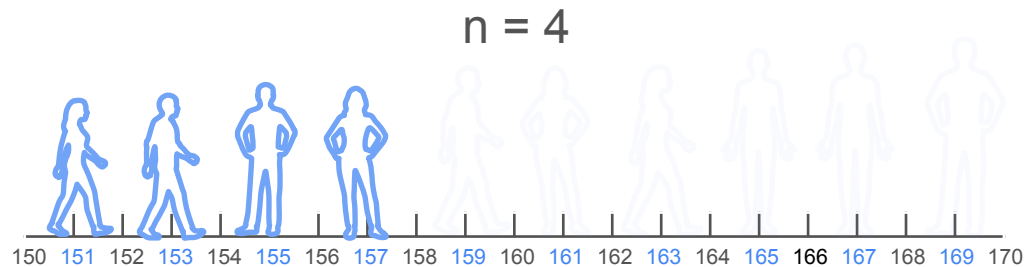
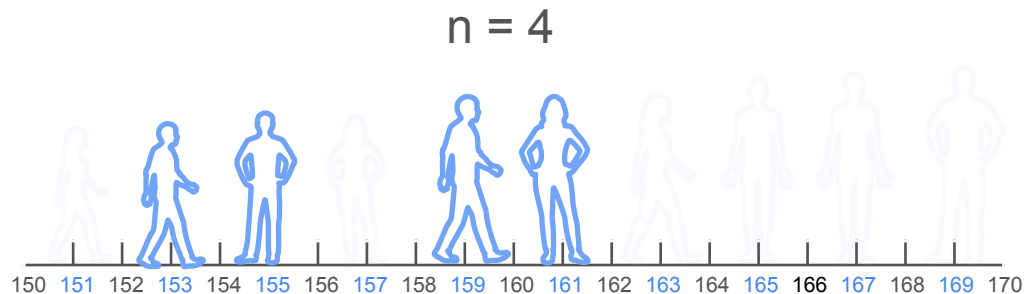
A

$n = 4$



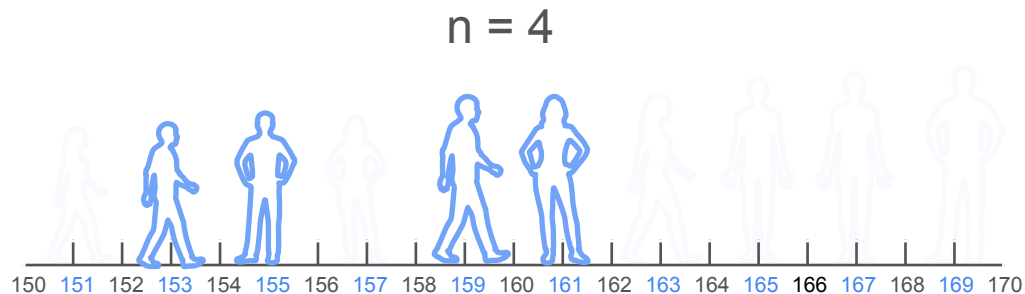
Random Sampling

A

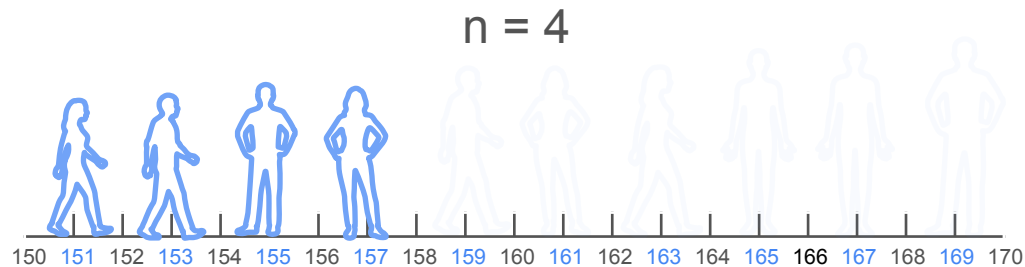


Random Sampling

A

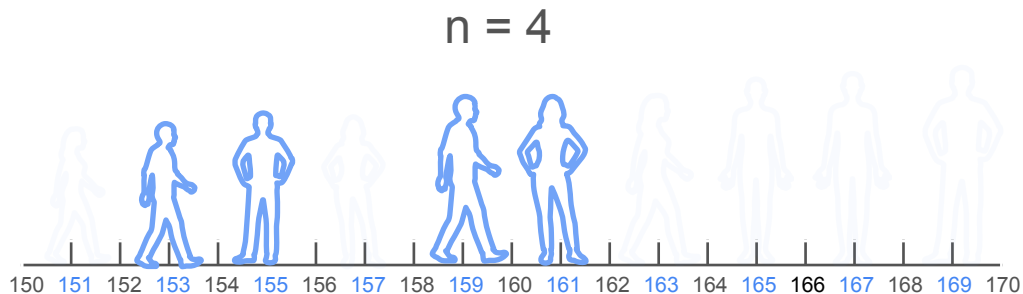


B



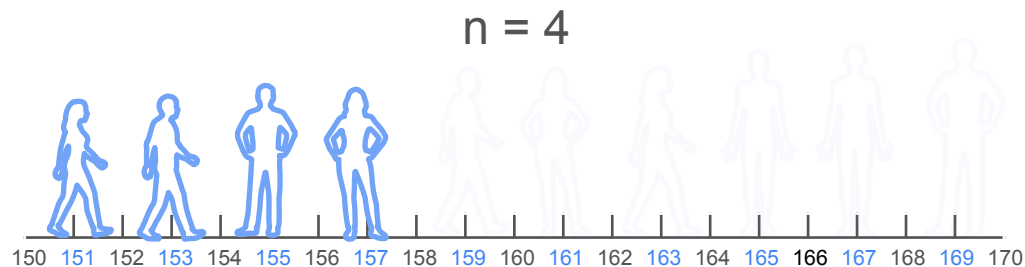
Random Sampling

A



Which is the better sample
to estimate the population
mean height?

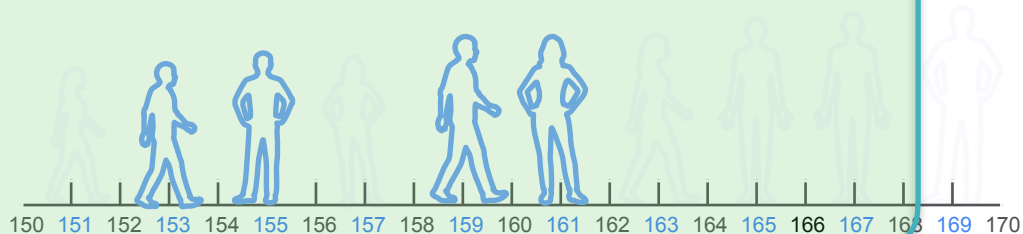
B



Random Sampling

A

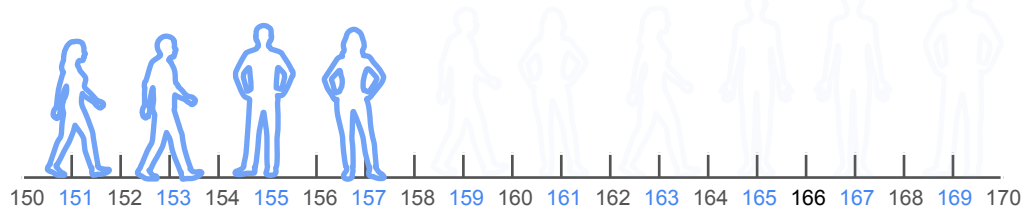
$n = 4$



Which is the better sample to estimate the population mean height?

B

$n = 4$



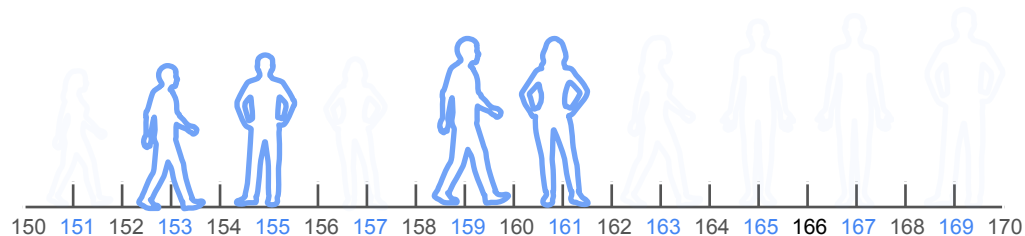
Independent Sample

Independent Sample

Example 1

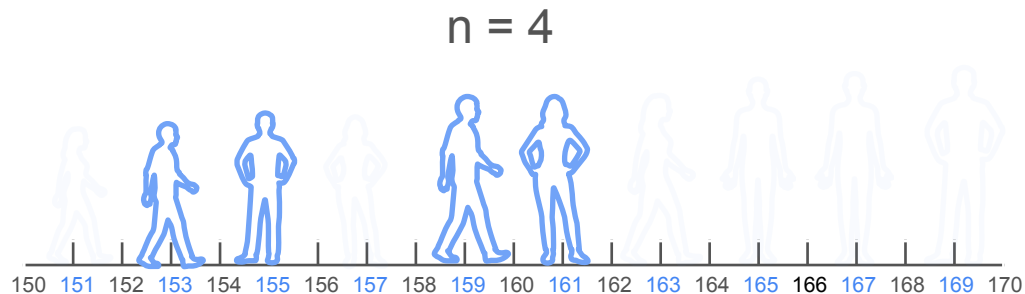
Independent Sample

Example 1



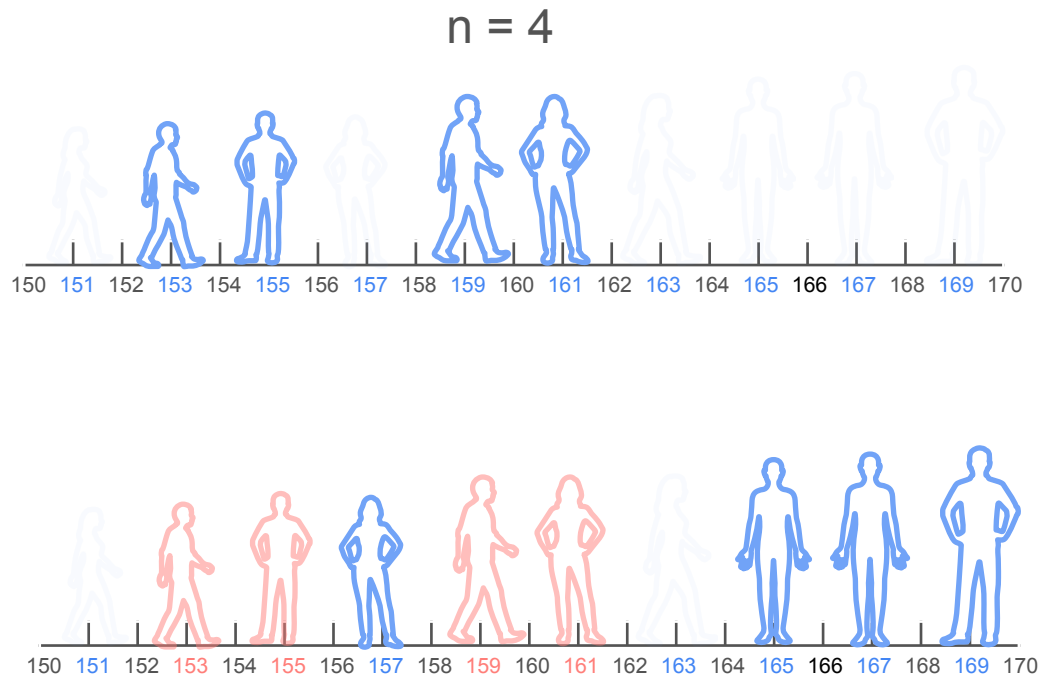
Independent Sample

Example 1



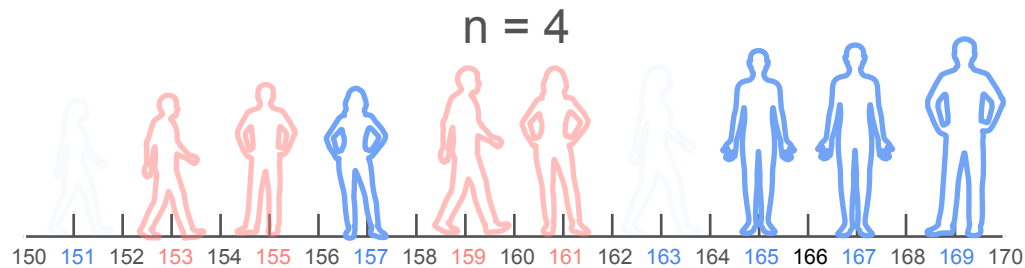
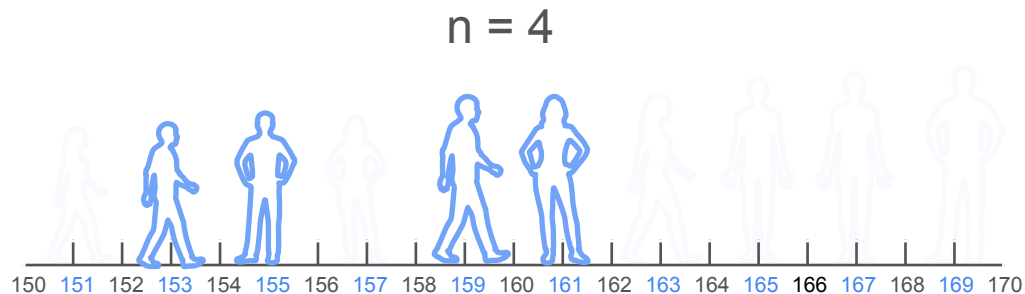
Independent Sample

Example 1



Independent Sample

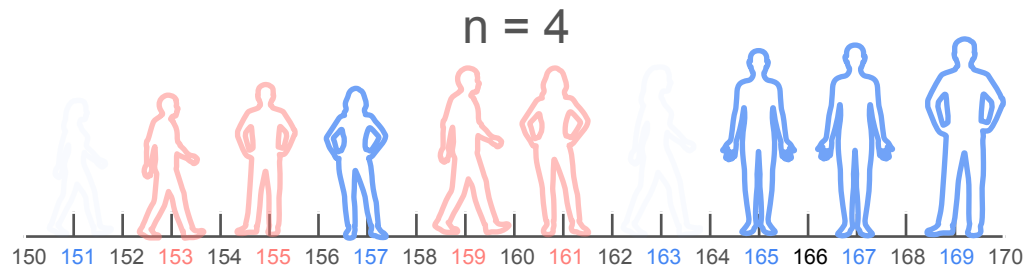
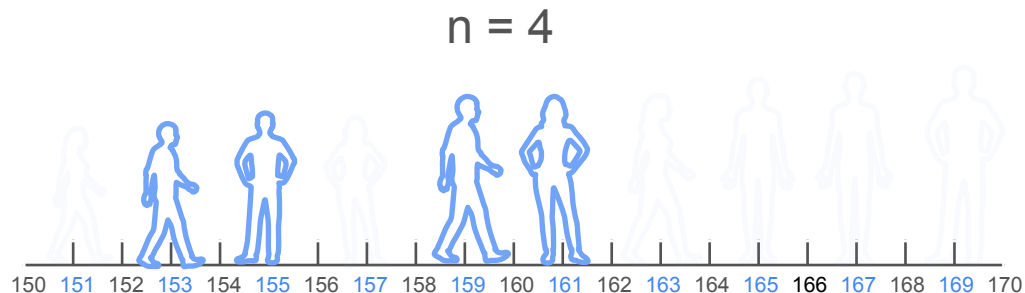
Example 1



Independent Sample

Example 1

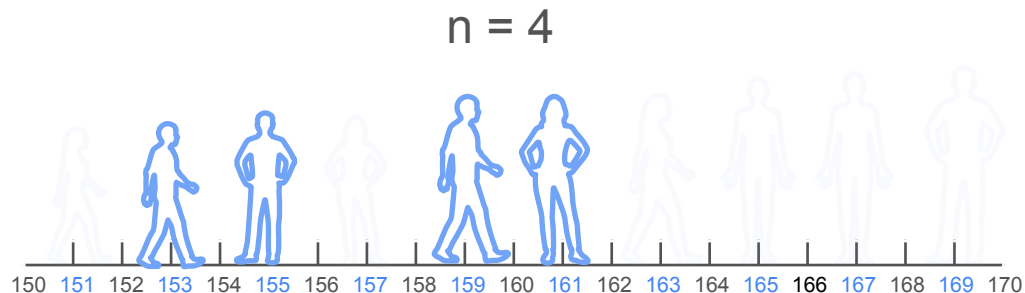
1st sample set



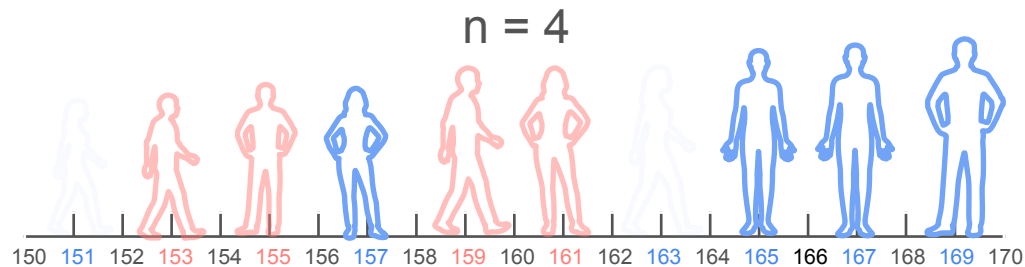
Independent Sample

Example 1

1st sample set



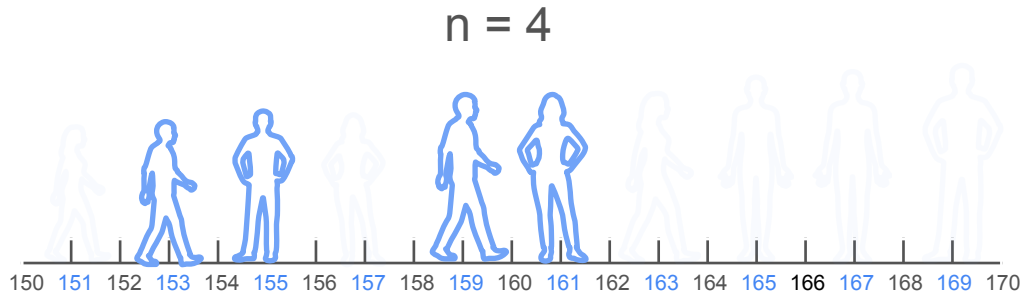
2nd sample set



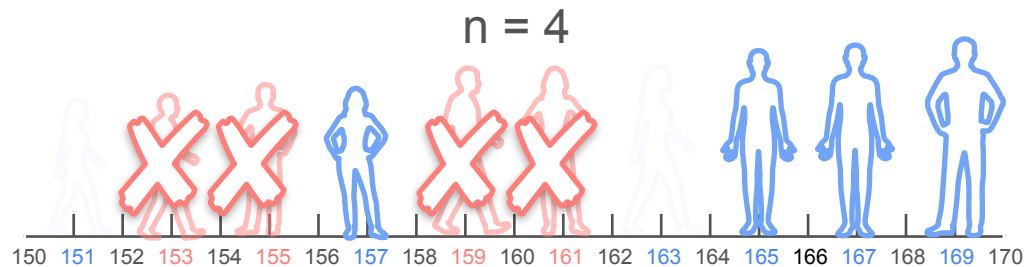
Independent Sample

Example 1

1st sample set



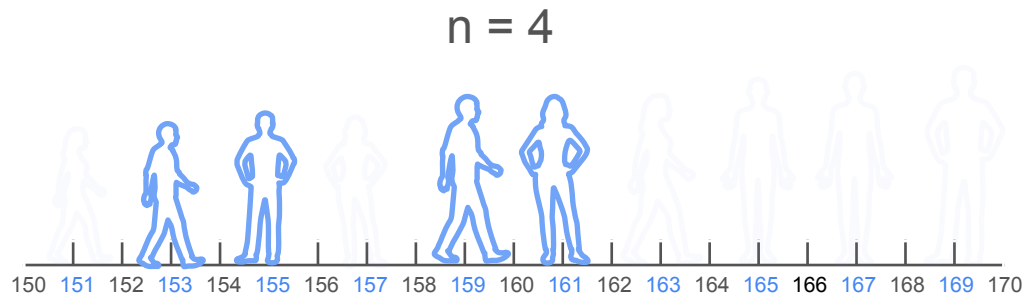
2nd sample set



Independent Sample

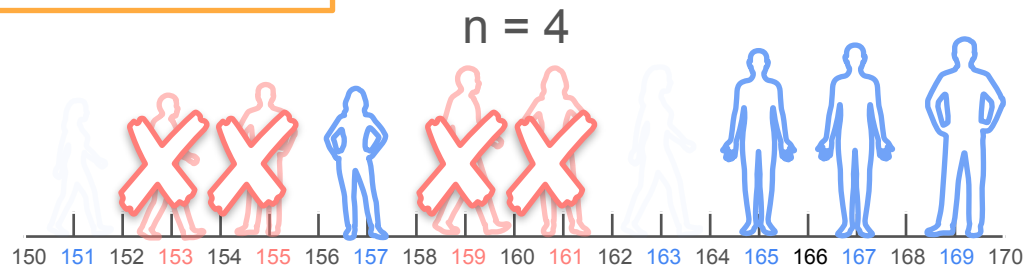
Example 1

1st sample set



Why is sample set two not a good sample?

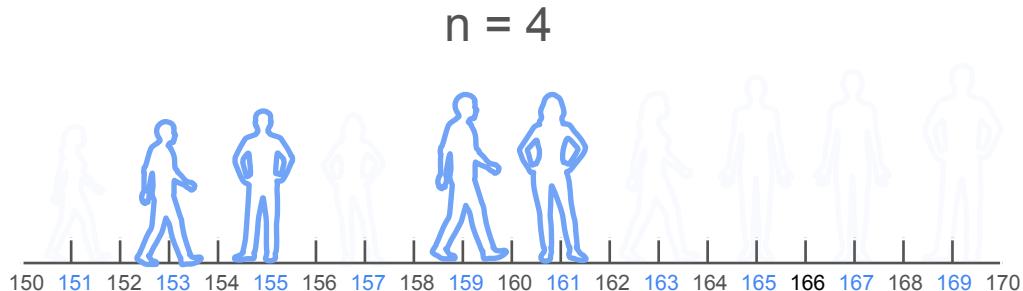
2nd sample set



Independent Sample

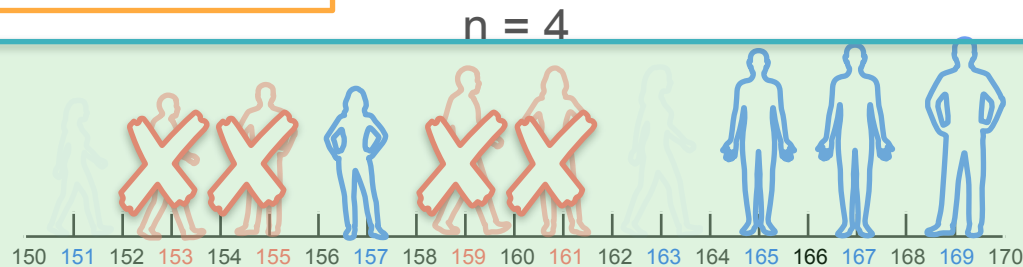
Example 1

1st sample set



Why is sample set two not a good sample?

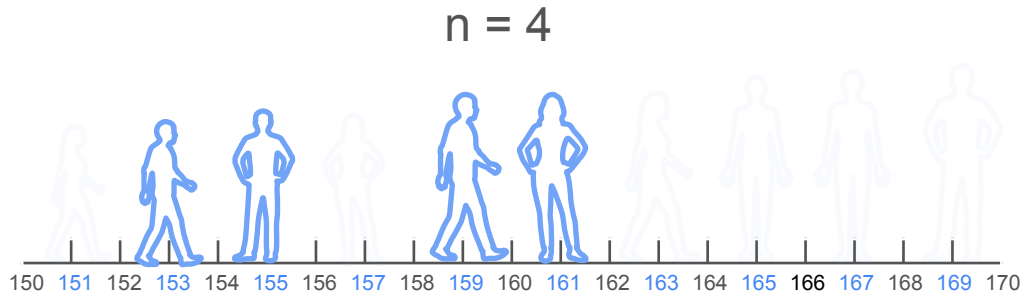
2nd sample set



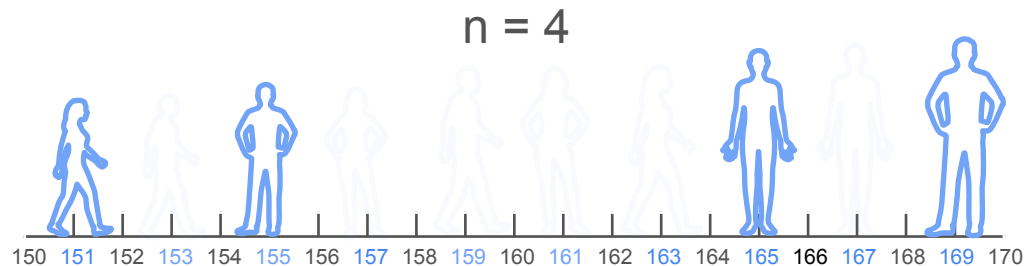
Independent Sample

Example 2

1st sample set



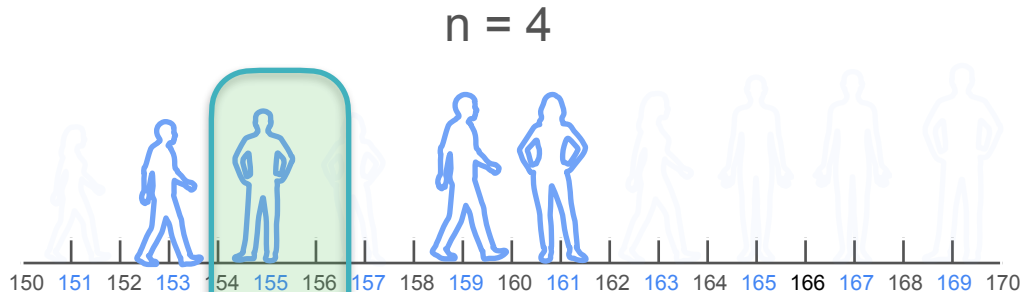
2nd sample set



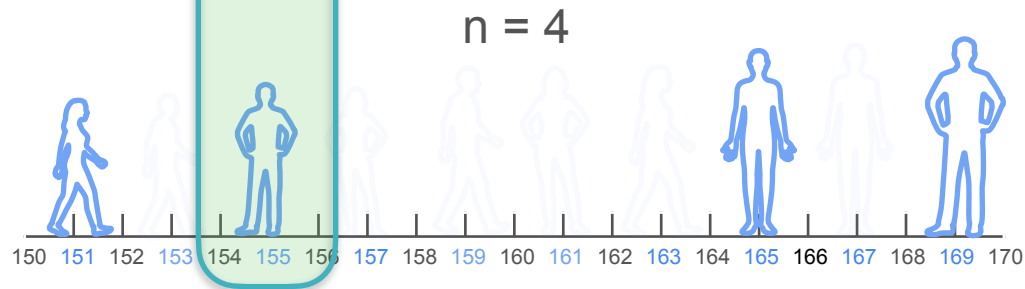
Independent Sample

Example 2

1st sample set



2nd sample set



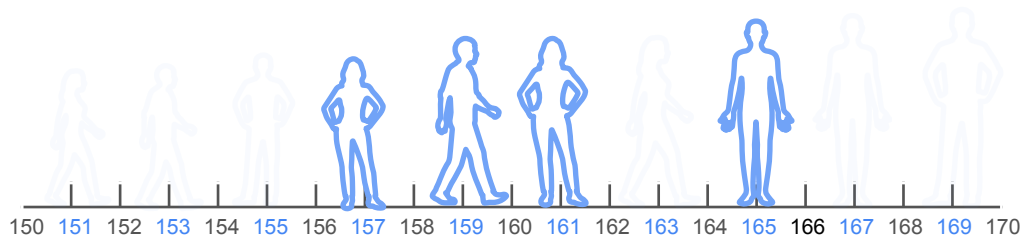
Identically Distributed Samples

Identically Distributed Samples

Example 1

Identically Distributed Samples

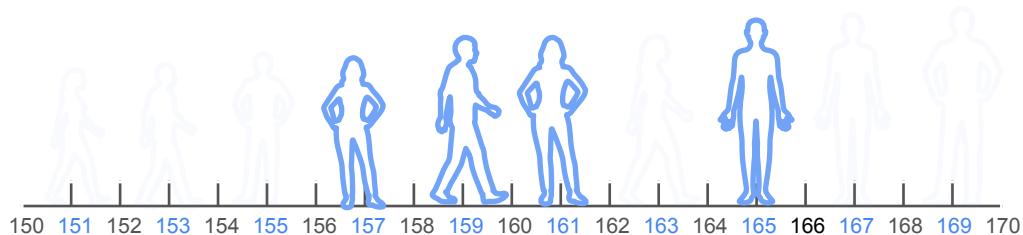
Example 1



Identically Distributed Samples

Example 1

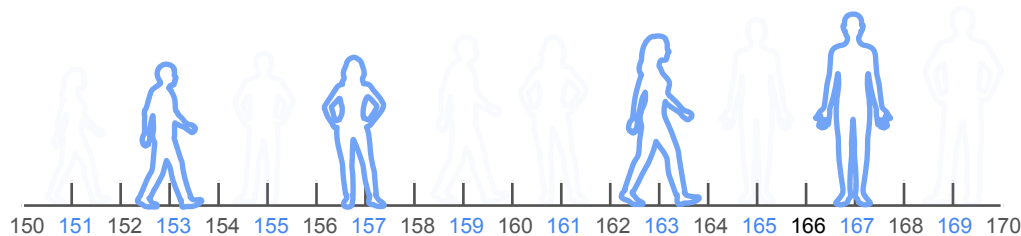
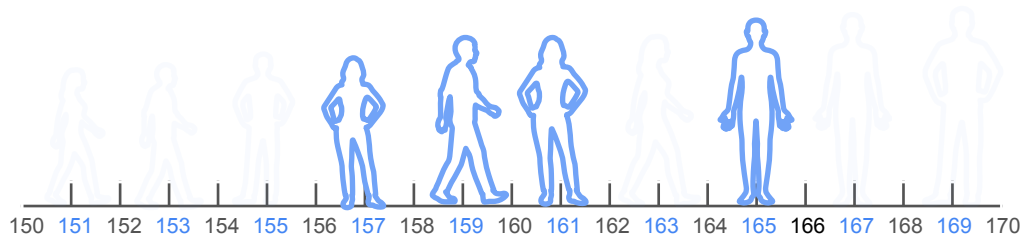
$n = 4$



Identically Distributed Samples

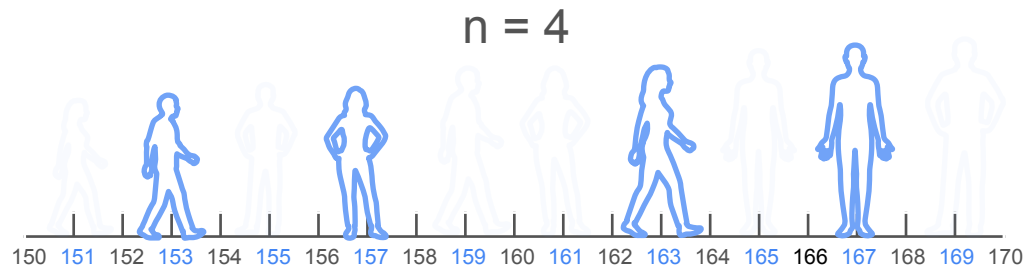
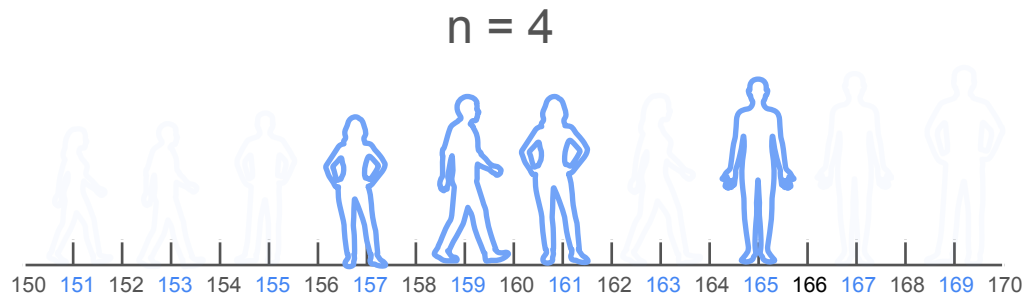
Example 1

$n = 4$



Identically Distributed Samples

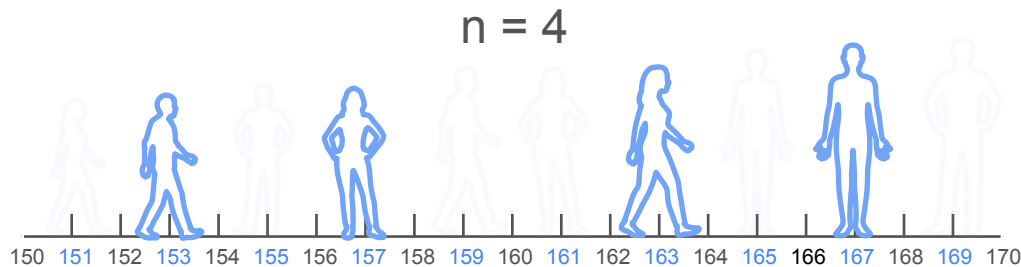
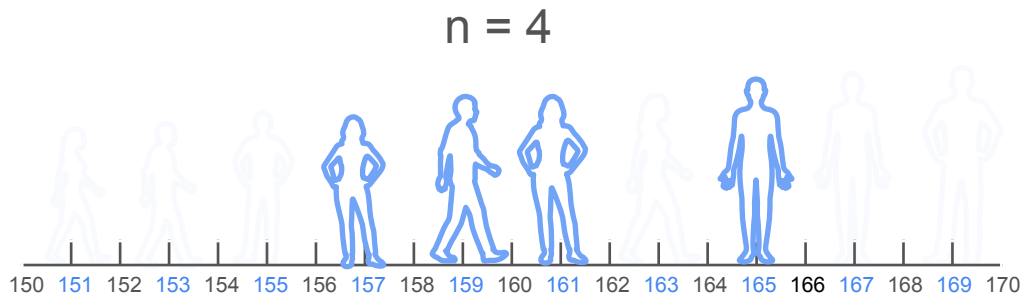
Example 1



Identically Distributed Samples

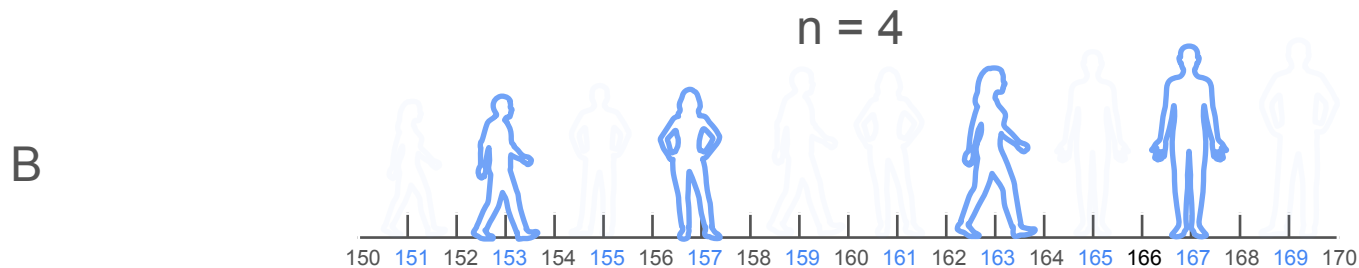
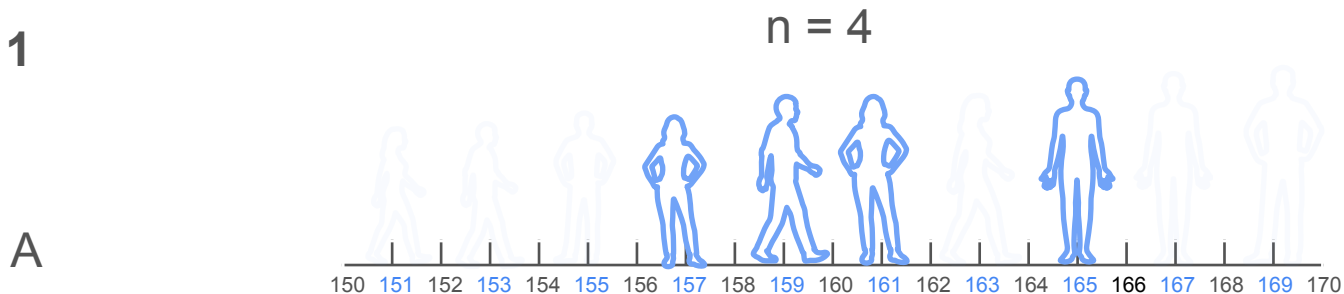
Example 1

A



Identically Distributed Samples

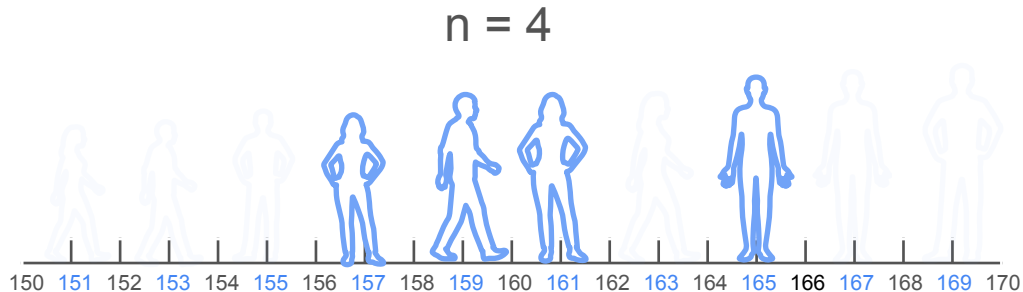
Example 1



Identically Distributed Samples

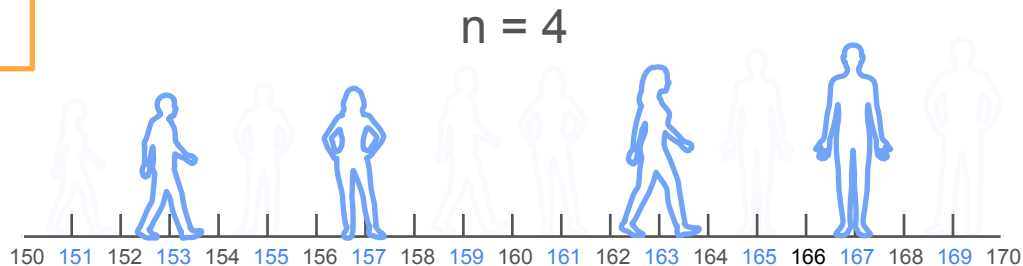
Example 1

A



Which of the following samples
are identically distributed?

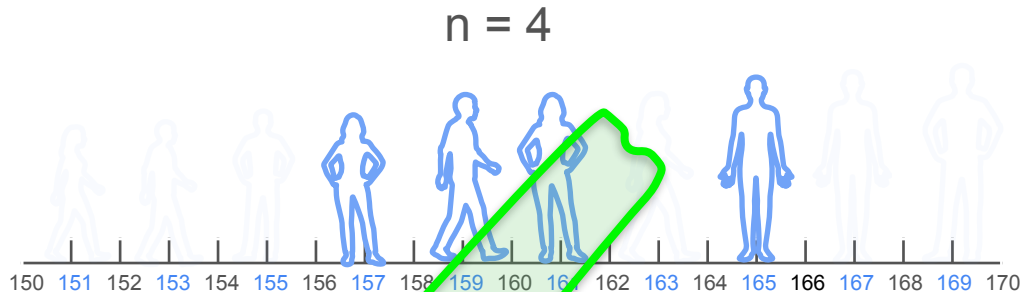
B



Identically Distributed Samples

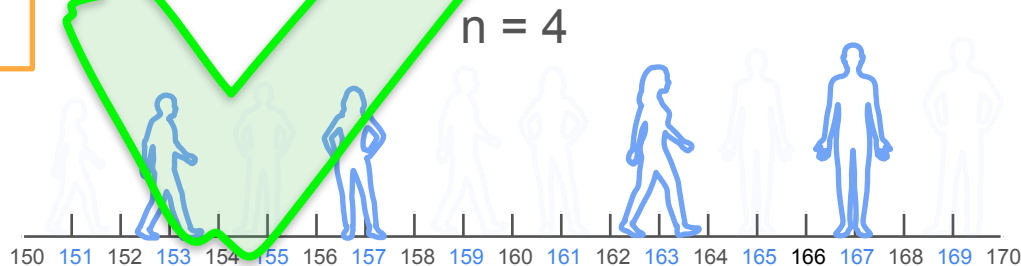
Example 1

A



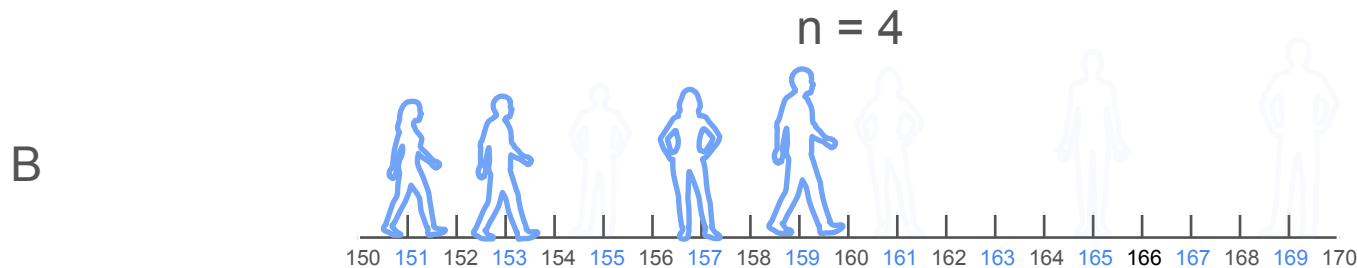
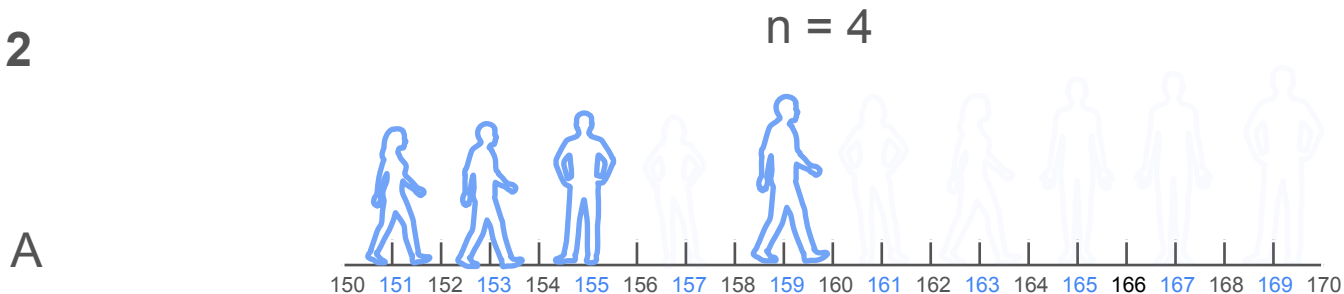
Which of the following samples
are identically distributed?

B



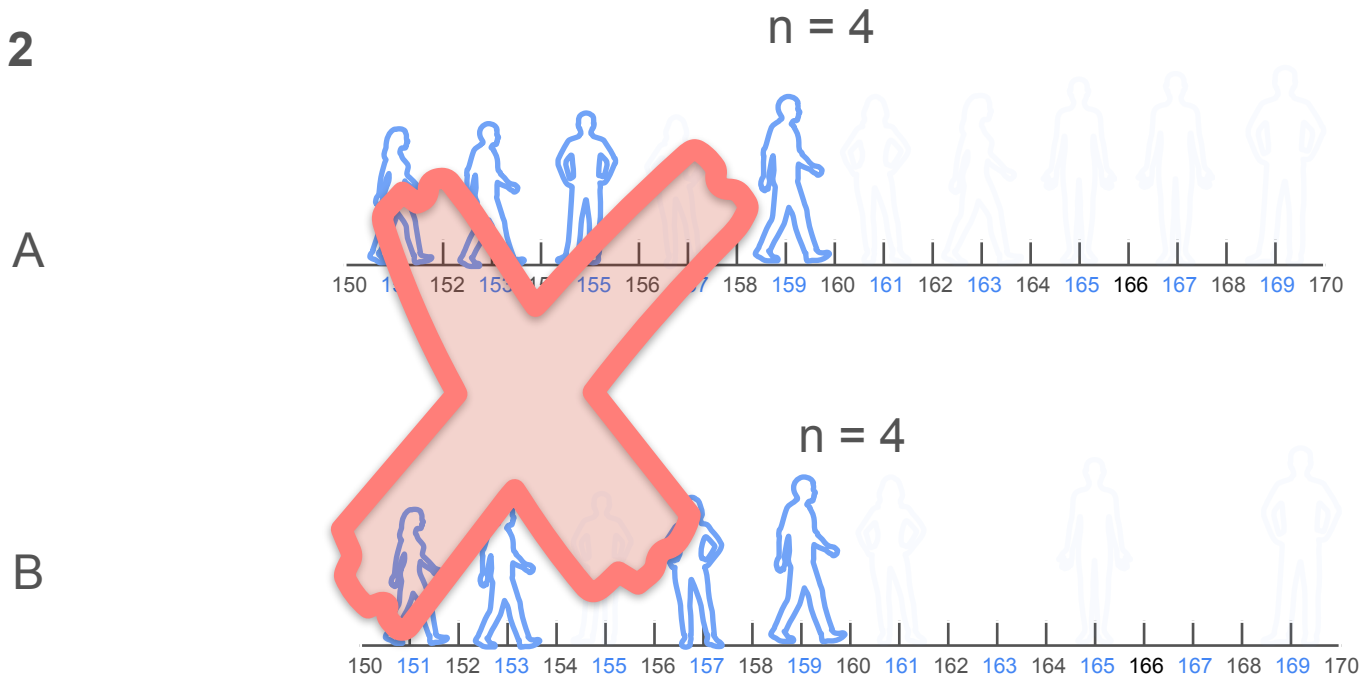
Identically Distributed Samples

Example 2



Identically Distributed Samples

Example 2



The Avocado Toast Trend

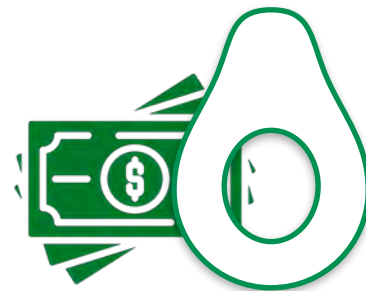
The Avocado Toast Trend



The Avocado Toast Trend



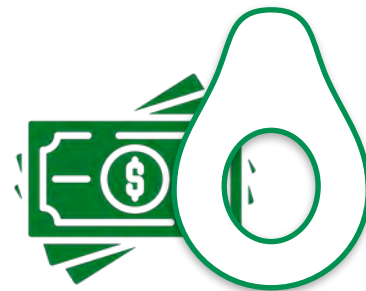
Study the price of avocados
in the United States



The Avocado Toast Trend



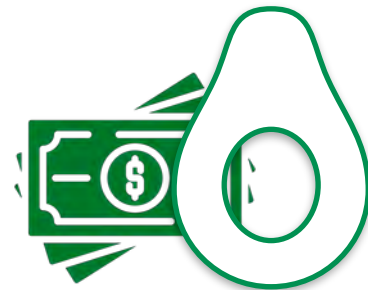
Study the price of avocados
in the United States



The Avocado Toast Trend

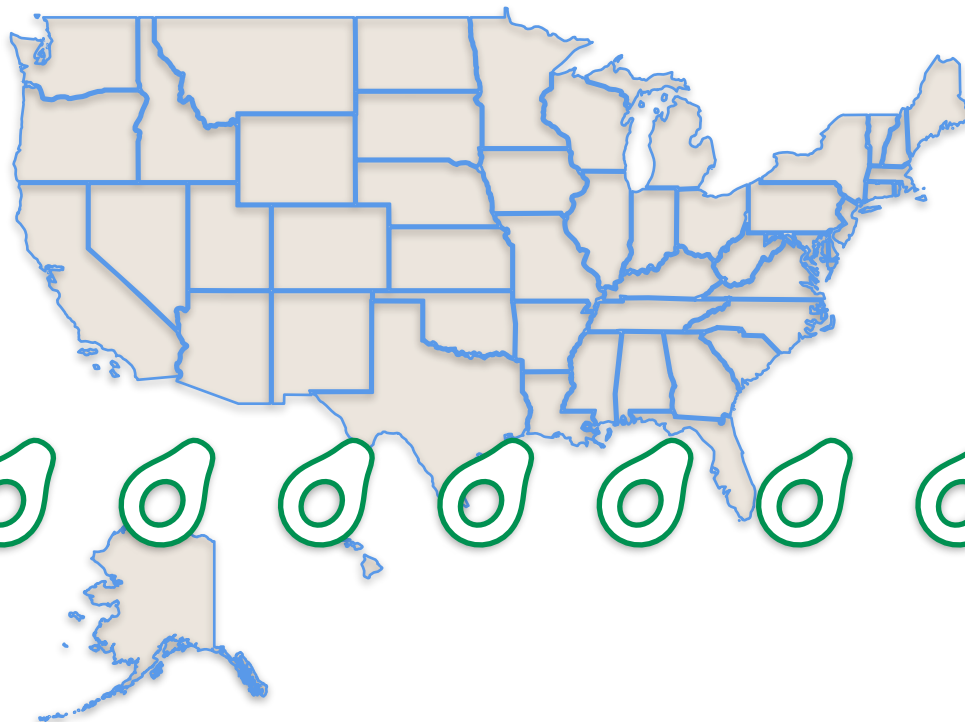


Study the price of avocados
in the United States



What is the population of your study?

The Avocado Toast Trend

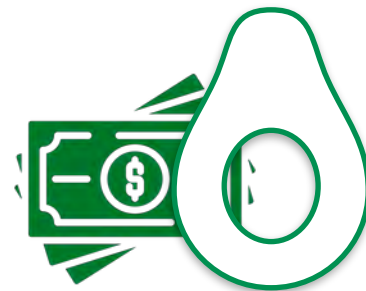


**All avocados
sold in the US**

The Avocado Toast Trend



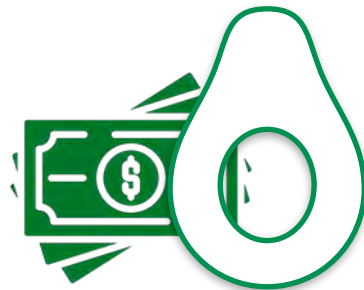
Study the price of avocados
in the United States



The Avocado Toast Trend



Study the price of avocados
in the United States

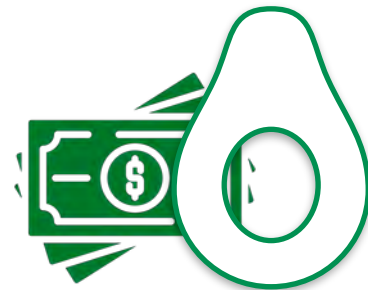


What is the sample of your study?

The Avocado Toast Trend



Study the price of avocados
in the United States

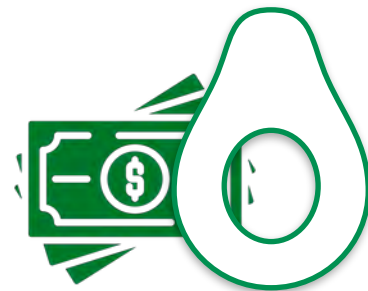


What is the sample of your study?

The Avocado Toast Trend



Study the price of avocados
in the United States



What is the sample of your study?

**Avocados sold
in the 4 stores
you selected**

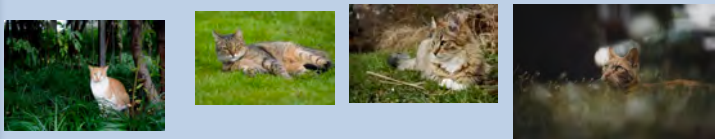
Population and Sample in Machine Learning

Population and Sample in Machine Learning

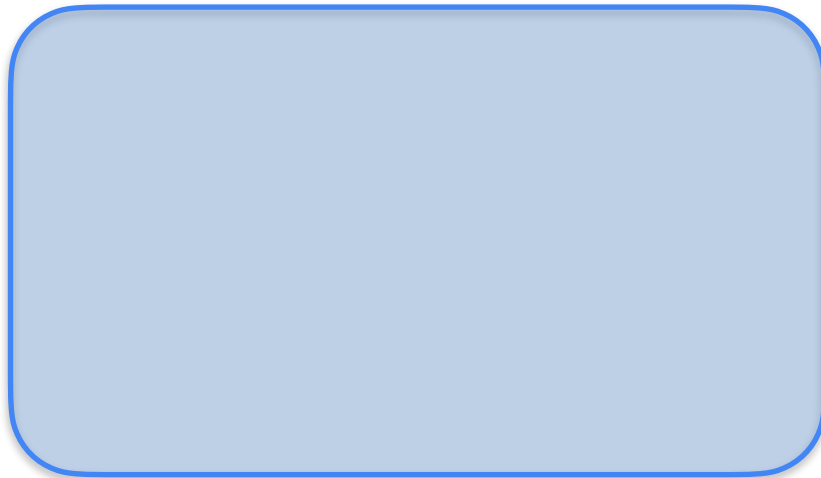
Every dataset you work with in machine learning is a sample

NOT the population

Cats



Not cats

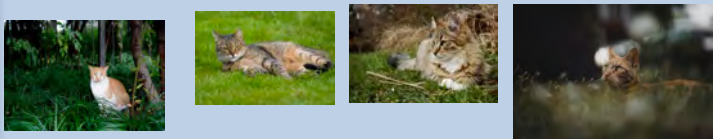


Population and Sample in Machine Learning

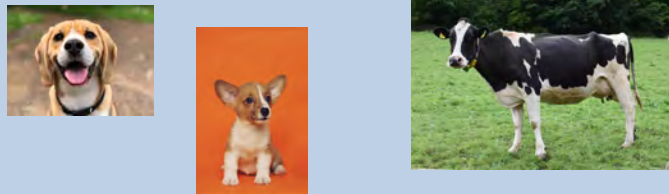
Every dataset you work with in machine learning is a sample

NOT the population

Cats



Not cats



Population and Sample in Machine Learning

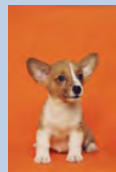
Every dataset you work with in machine learning is a sample

NOT the population

Cats



Not cats



Population and Sample in Machine Learning

Every dataset you work with in machine learning is a sample

NOT the population

Cats



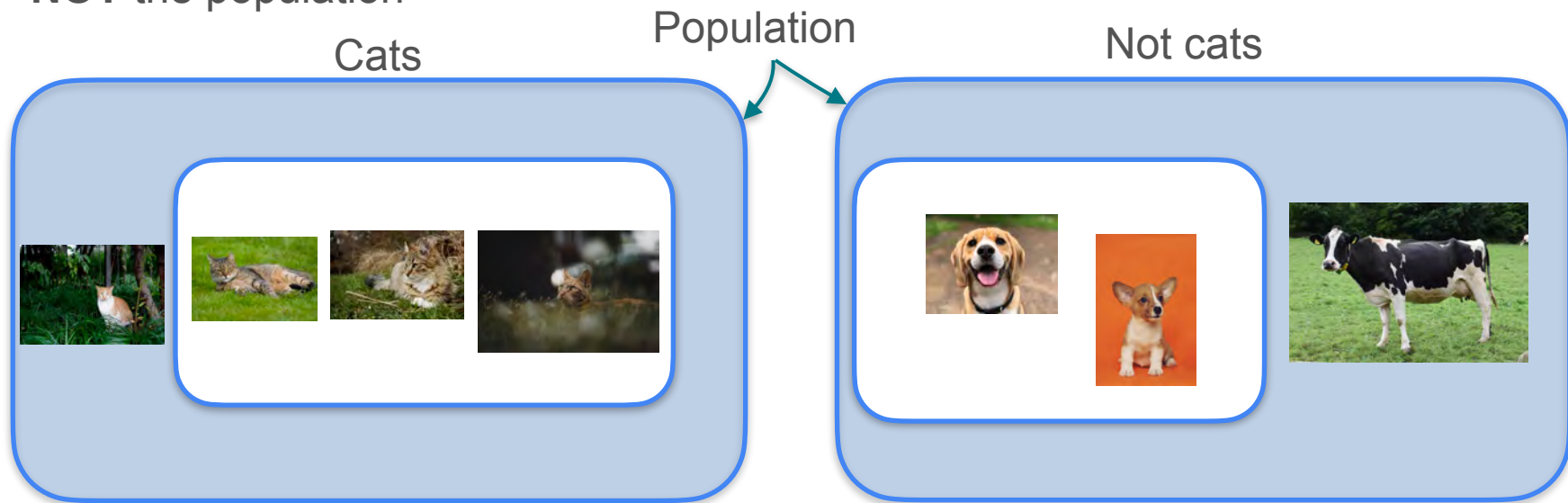
Not cats



Population and Sample in Machine Learning

Every dataset you work with in machine learning is a sample

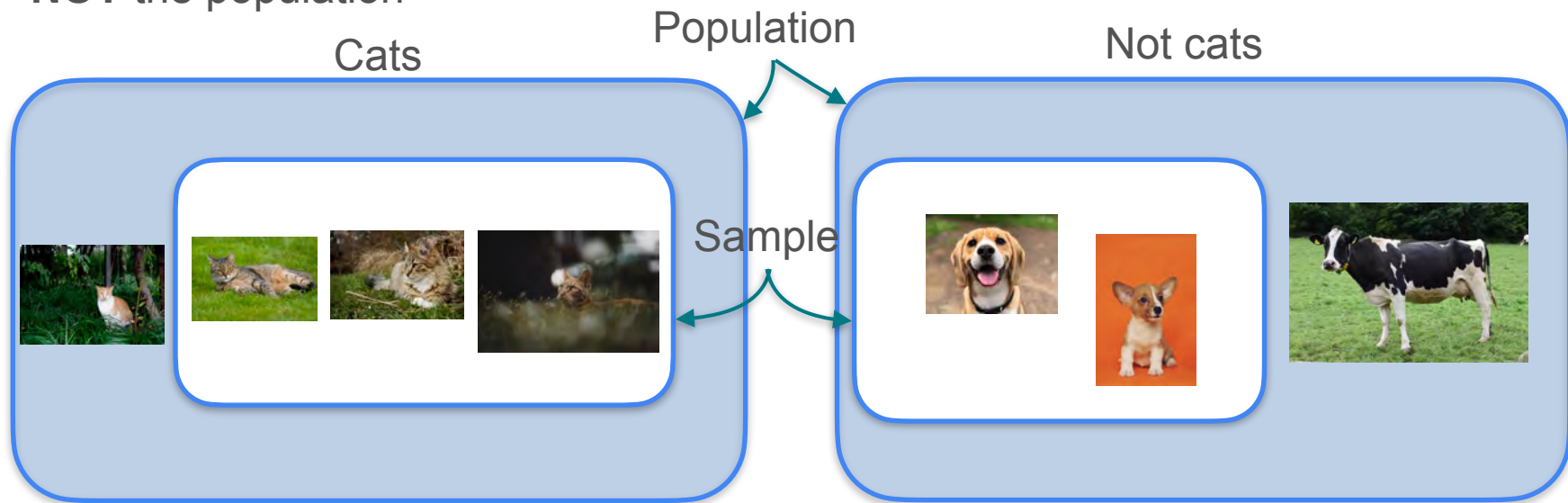
NOT the population



Population and Sample in Machine Learning

Every dataset you work with in machine learning is a sample

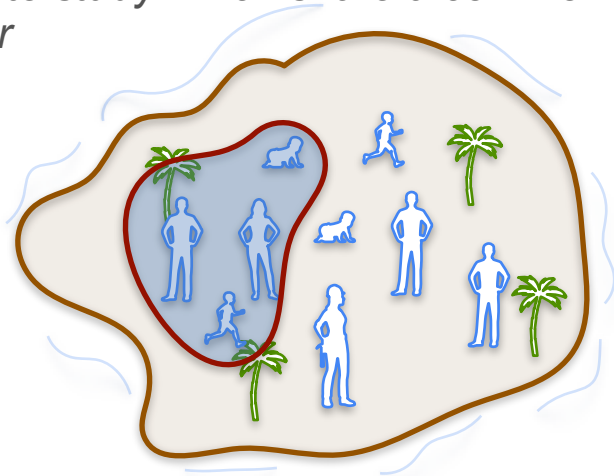
NOT the population



Recap

Population

the entire group of individuals or elements you want to study which share a common behaviour



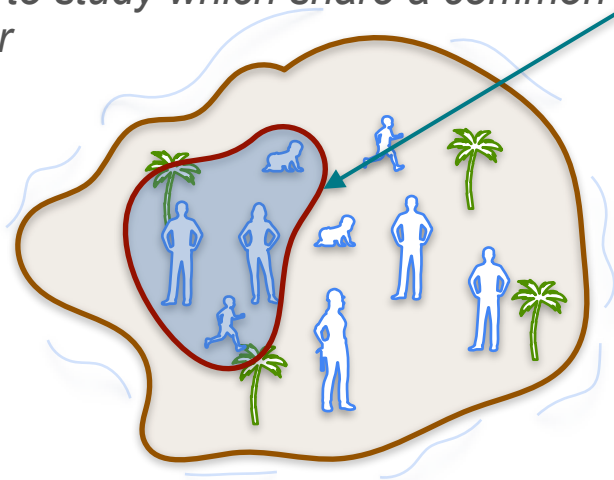
Sample

subset of the population you use to draw conclusions about the population as a whole

Recap

Population

the entire group of individuals or elements you want to study which share a common behaviour



Sample

subset of the population you use to draw conclusions about the population as a whole

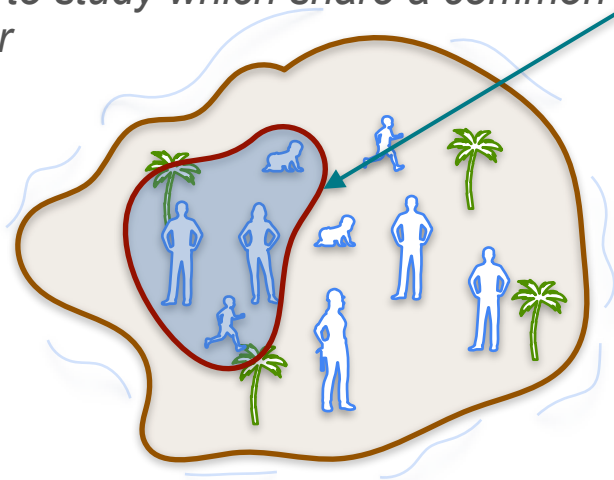
Population Size: N

Sample Size:

Recap

Population

the entire group of individuals or elements you want to study which share a common behaviour



Sample

subset of the population you use to draw conclusions about the population as a whole

Population Size: N

Sample Size: n



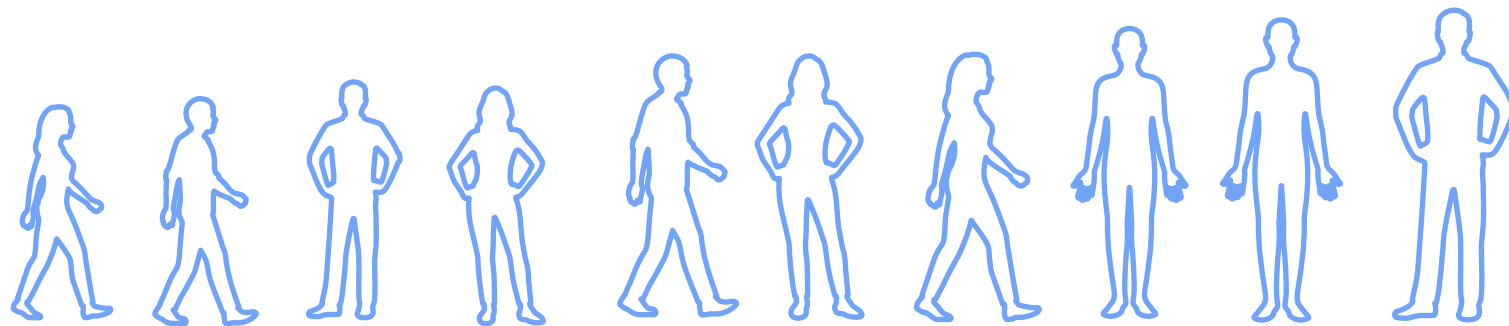
DeepLearning.AI

Sample and Population

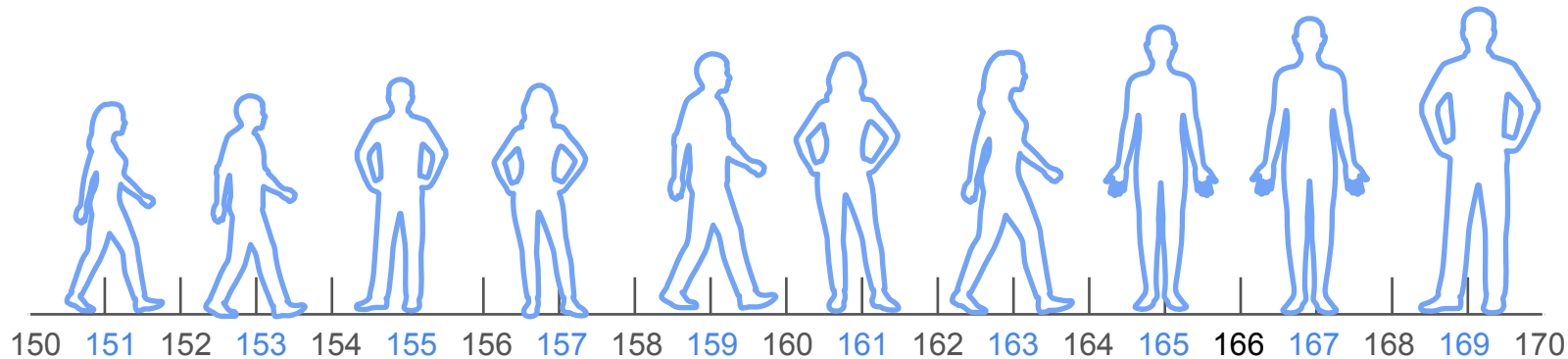
Sample Mean, Proportion, and Variance

Population and Sample Mean

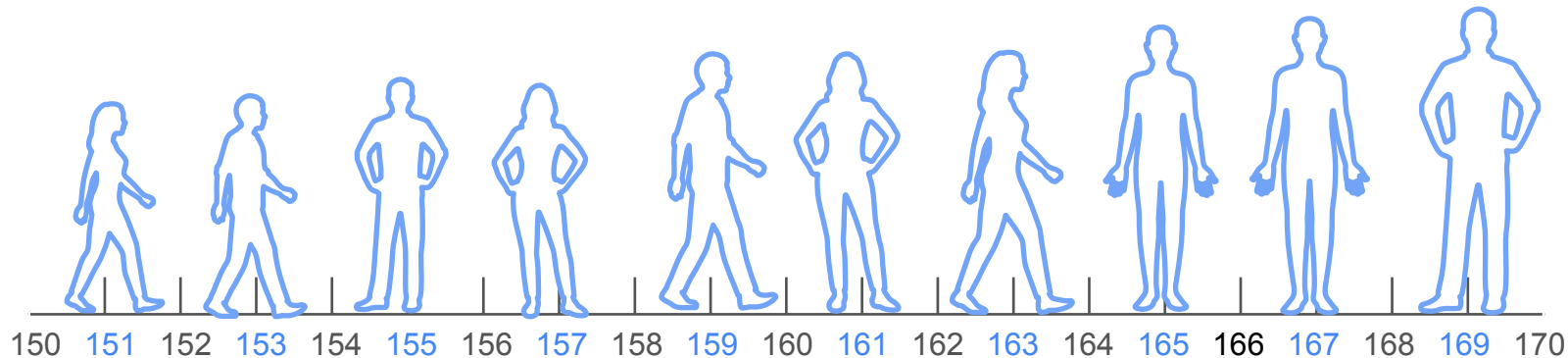
Population and Sample Mean



Population and Sample Mean

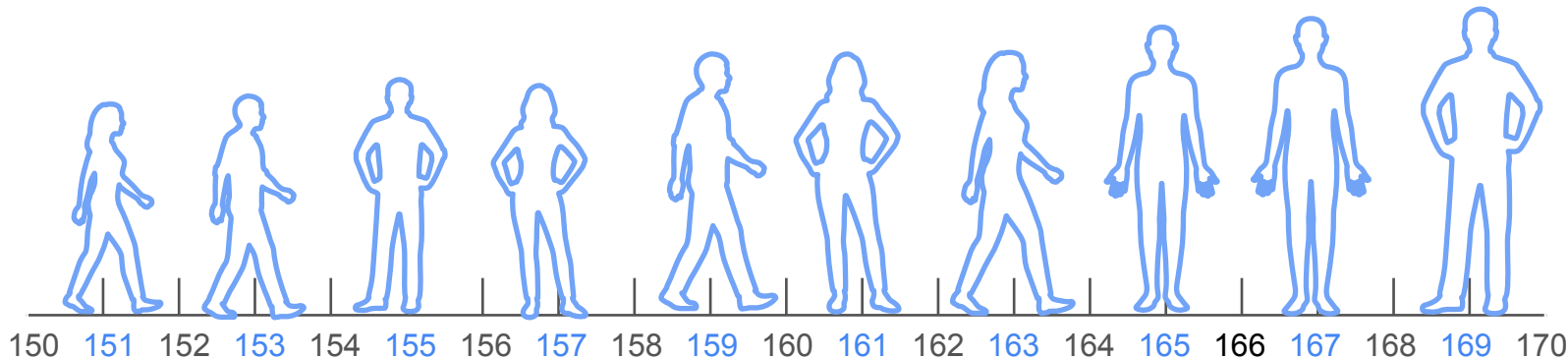


Population and Sample Mean



What is the average height in statistopia?

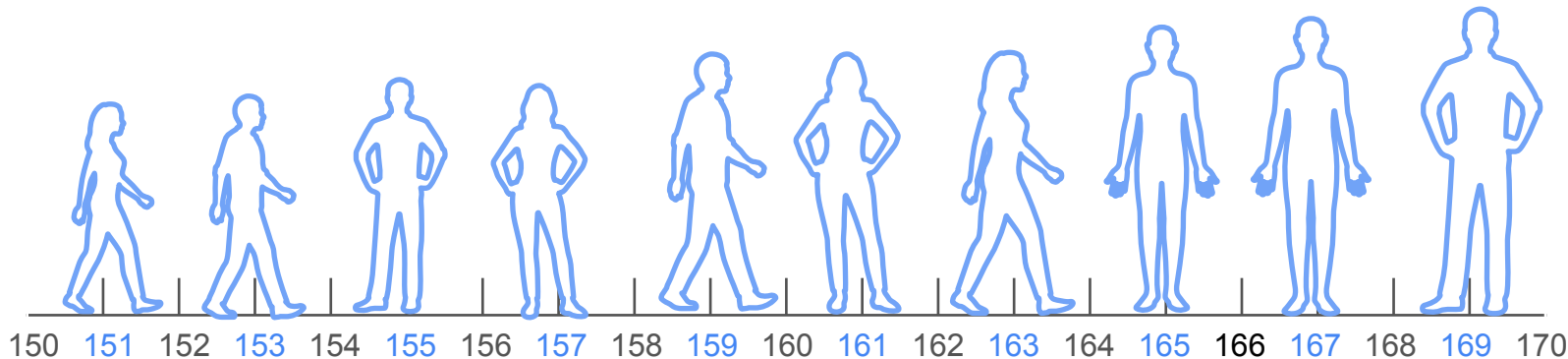
Population and Sample Mean



What is the average height in statistopia?

$$\frac{151 + 153 + 155 + 157 + 159 + 161 + 163 + 165 + 167 + 169}{10}$$

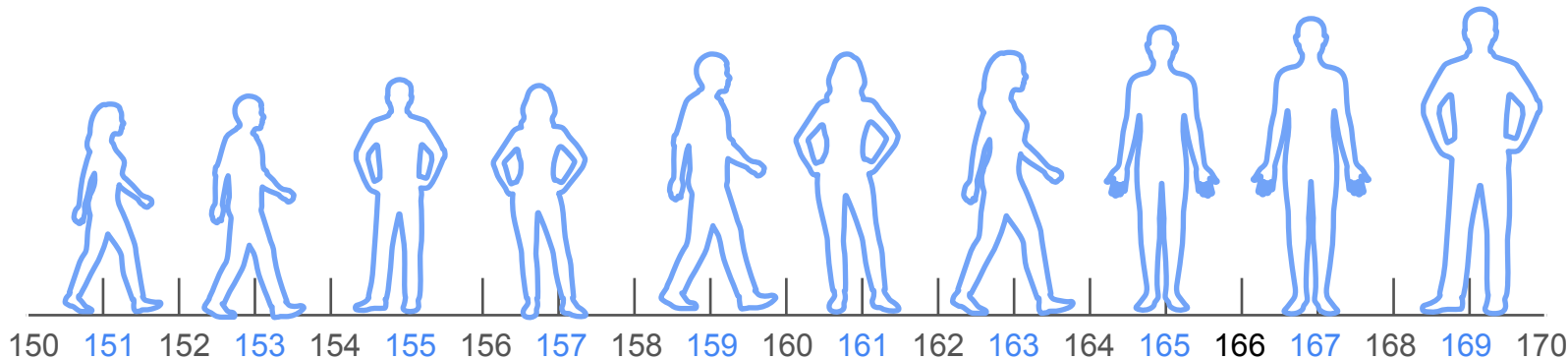
Population and Sample Mean



What is the average height in statistopia?

$$\frac{151 + 153 + 155 + 157 + 159 + 161 + 163 + 165 + 167 + 169}{10}$$
$$= \frac{1600}{10} = 160cm$$

Population and Sample Mean



What is the average height in statistopia?

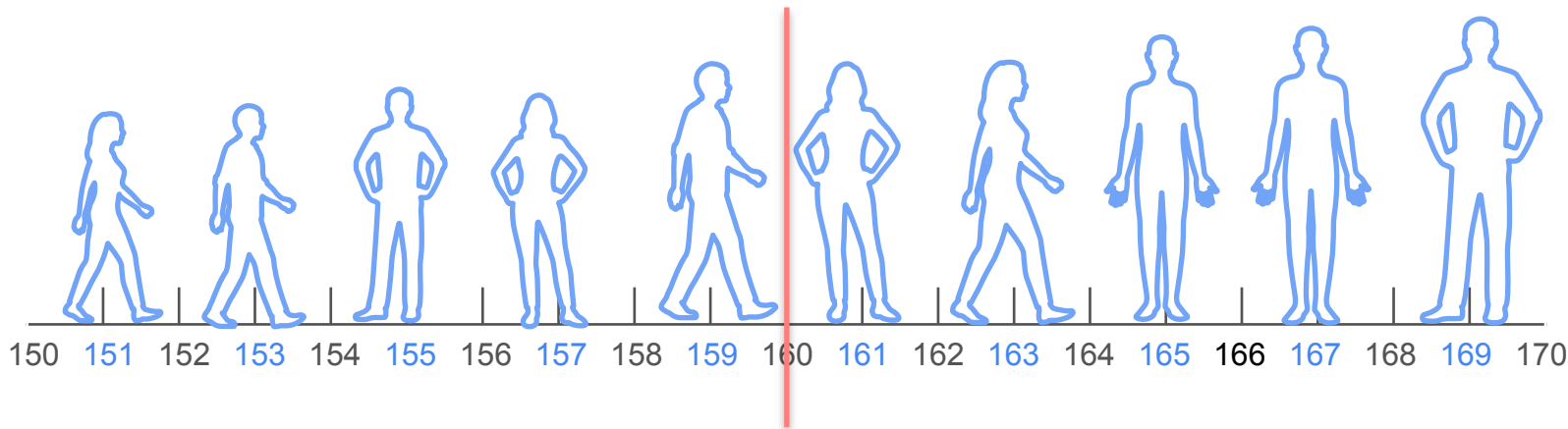
$$\frac{151 + 153 + 155 + 157 + 159 + 161 + 163 + 165 + 167 + 169}{10}$$

$$= \frac{1600}{10} = 160cm$$

Population mean

μ

Population and Sample Mean



What is the average height in statistopia?

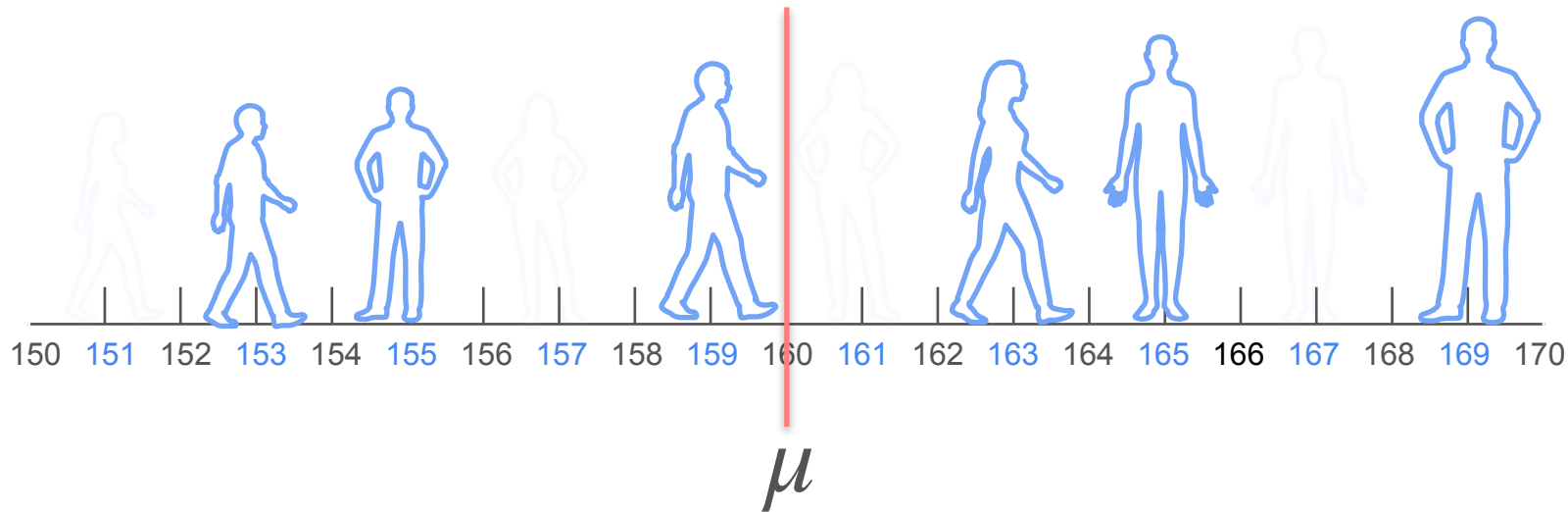
$$\frac{151 + 153 + 155 + 157 + 159 + 161 + 163 + 165 + 167 + 169}{10}$$

$$= \frac{1600}{10} = 160cm$$

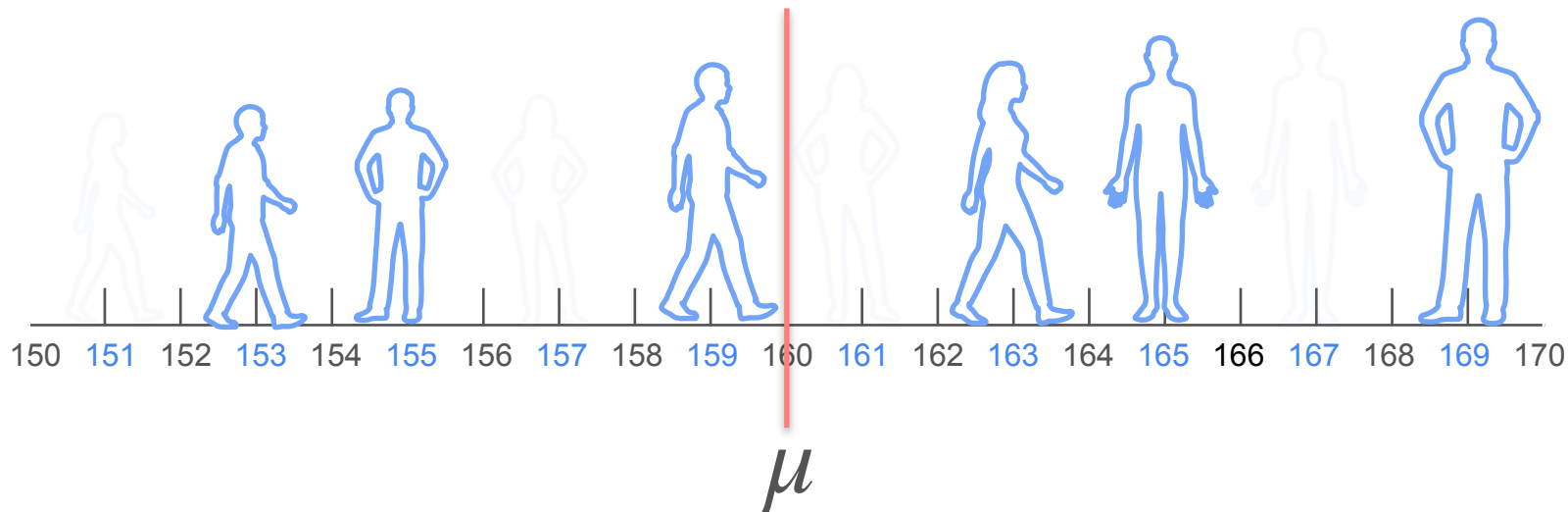
Population mean

μ

Population and Sample Mean



Population and Sample Mean



What is the average height in statistopia?

Population and Sample Mean



$$n = 6$$

$$\mu$$

What is the average height in statistopia?

Population and Sample Mean



$$n = 6$$

$$\mu$$

What is the average height in statistopia?

$$\frac{153 + 155 + 159 + 163 + 165 + 169}{6}$$

Population and Sample Mean



$$n = 6$$

$$\mu$$

What is the average height in statistopia?

$$\frac{153 + 155 + 159 + 163 + 165 + 169}{6} = \frac{964}{6} = 160.97$$

Population and Sample Mean



$$n = 6$$

$$\mu \quad \bar{x}_1$$

What is the average height in statistopia?

$$\frac{153 + 155 + 159 + 163 + 165 + 169}{6} = \frac{964}{6} = 160.97 \quad \bar{x}$$

Population and Sample Mean



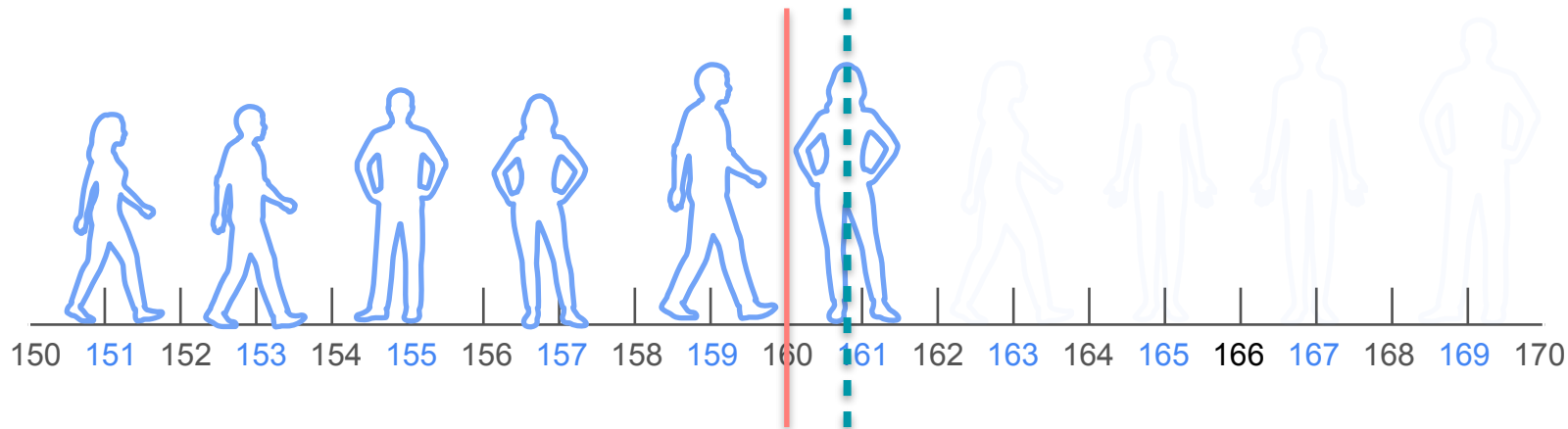
$$n = 6$$

$$\mu \quad \bar{x}_1$$

What is the average height in statistopia?

$$\frac{153 + 155 + 159 + 163 + 165 + 169}{6} = \frac{964}{6} = 160.97 \quad \bar{x}$$

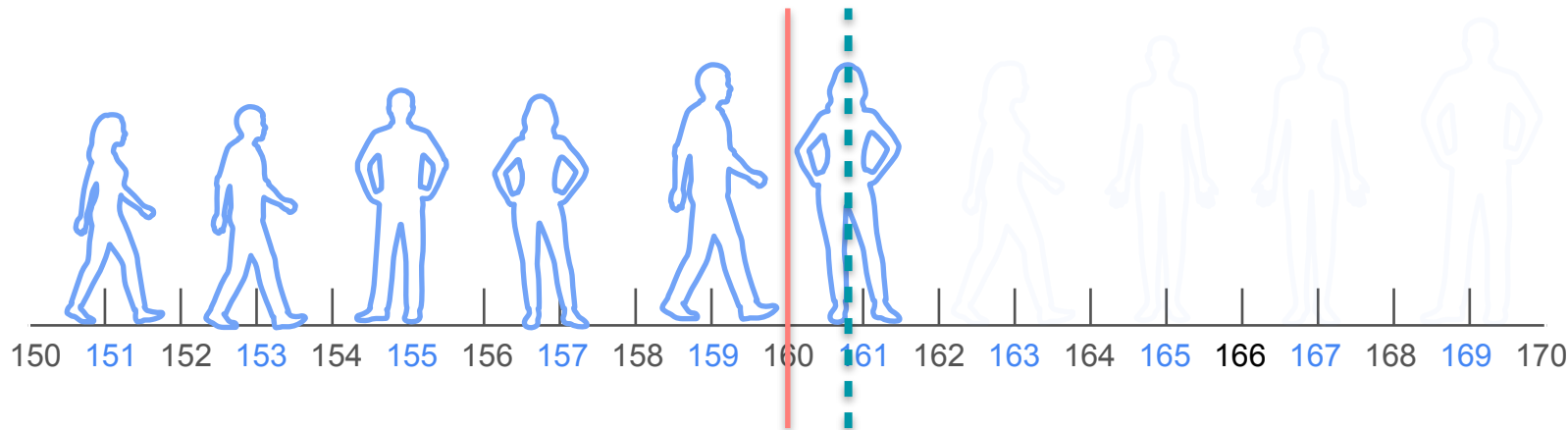
Population and Sample Mean



$$n = 6$$

$$\mu \quad \bar{x}_1$$

Population and Sample Mean

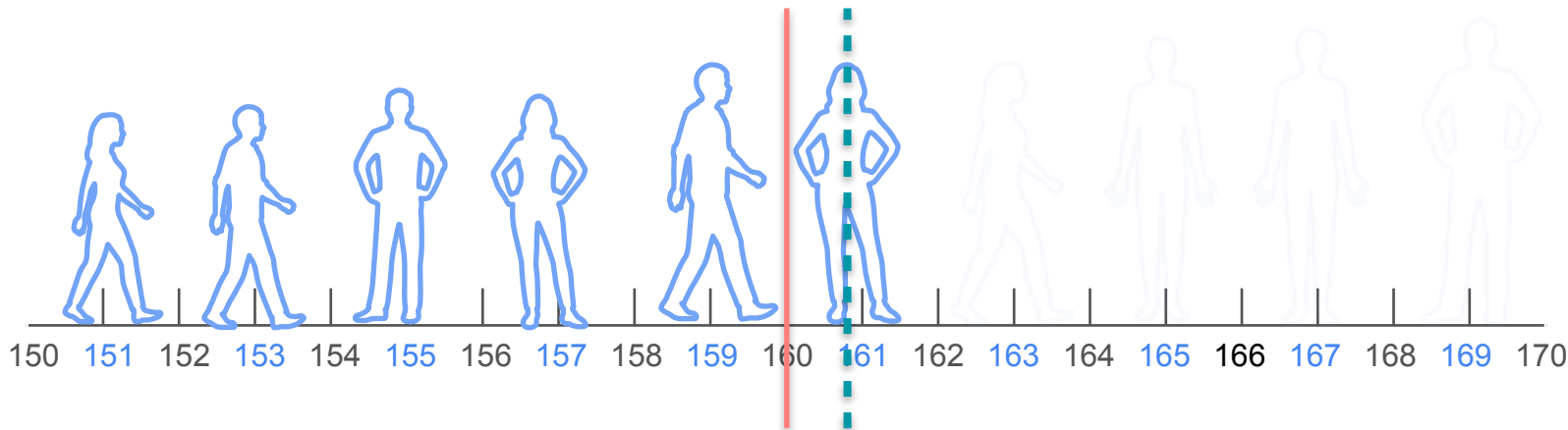


$$n = 6$$

$$\mu \quad \bar{x}_1$$

What is the average height in statistopia?

Population and Sample Mean



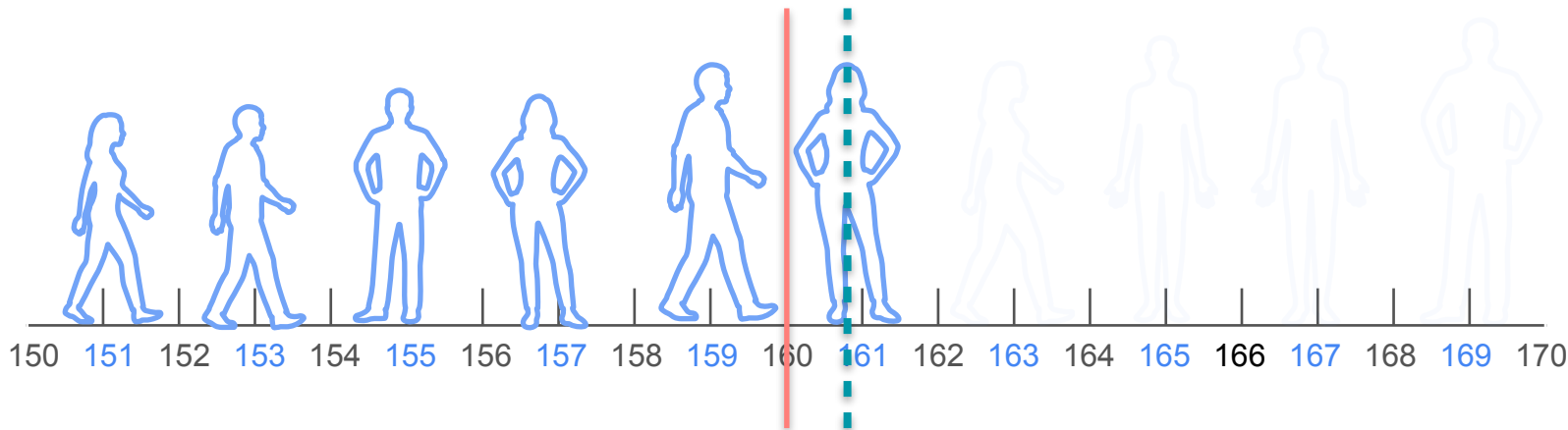
$$n = 6$$

$$\mu \quad \bar{x}_1$$

What is the average height in statistopia?

$$\frac{151 + 153 + 155 + 157 + 159 + 161}{6}$$

Population and Sample Mean



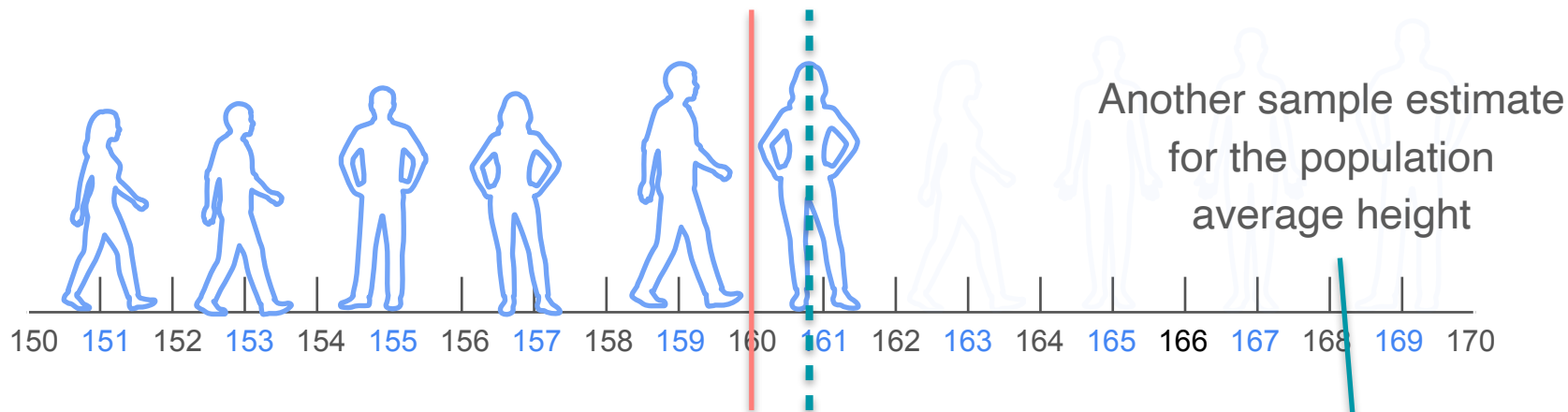
$$n = 6$$

$$\mu \quad \bar{x}_1$$

What is the average height in statistopia?

$$\frac{151 + 153 + 155 + 157 + 159 + 161}{6} = \frac{936}{6} = 156cm$$

Population and Sample Mean

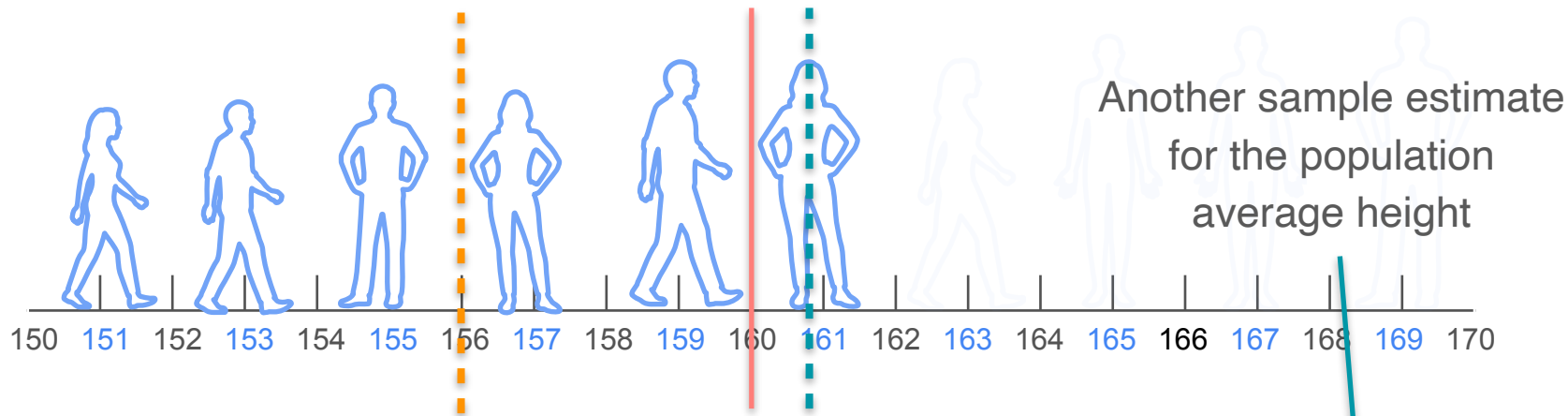


$$n = 6$$

What is the average height in statistopia?

$$\frac{151 + 153 + 155 + 157 + 159 + 161}{6} = \frac{936}{6} = 156cm$$

Population and Sample Mean



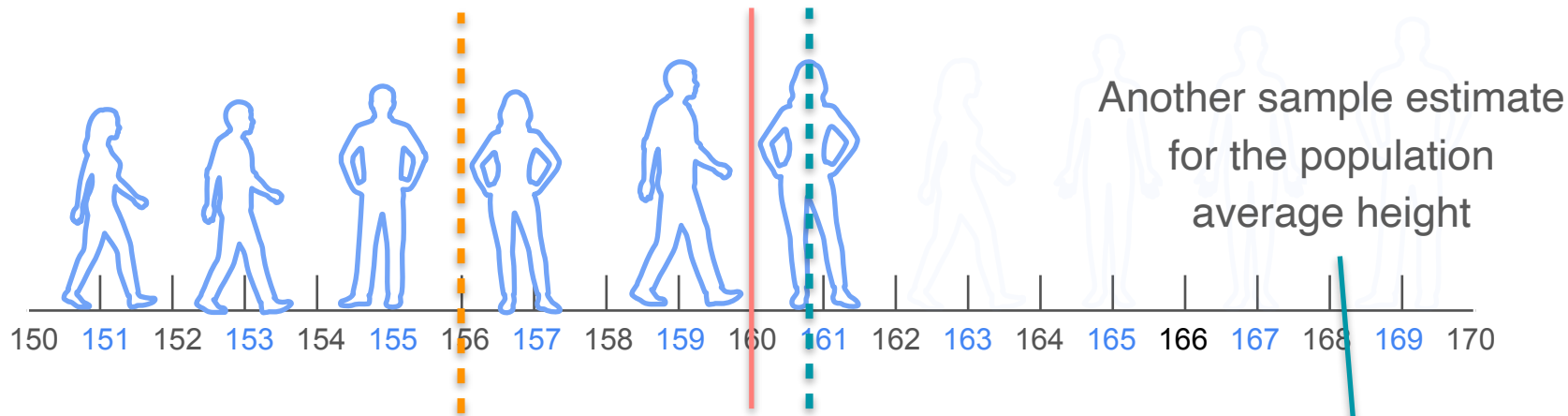
$$n = 6$$

$$\mu \quad \bar{x}_1$$

What is the average height in statistopia?

$$\frac{151 + 153 + 155 + 157 + 159 + 161}{6} = \frac{936}{6} = 156cm$$

Population and Sample Mean



$$n = 6$$

$$\bar{x}_2$$

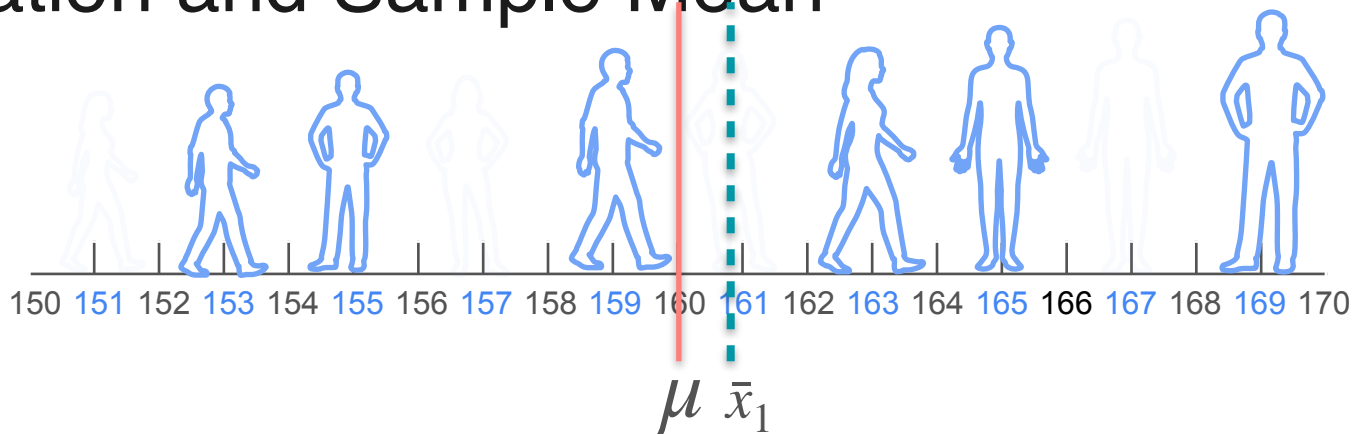
$$\mu \quad \bar{x}_1$$

What is the average height in statistopia?

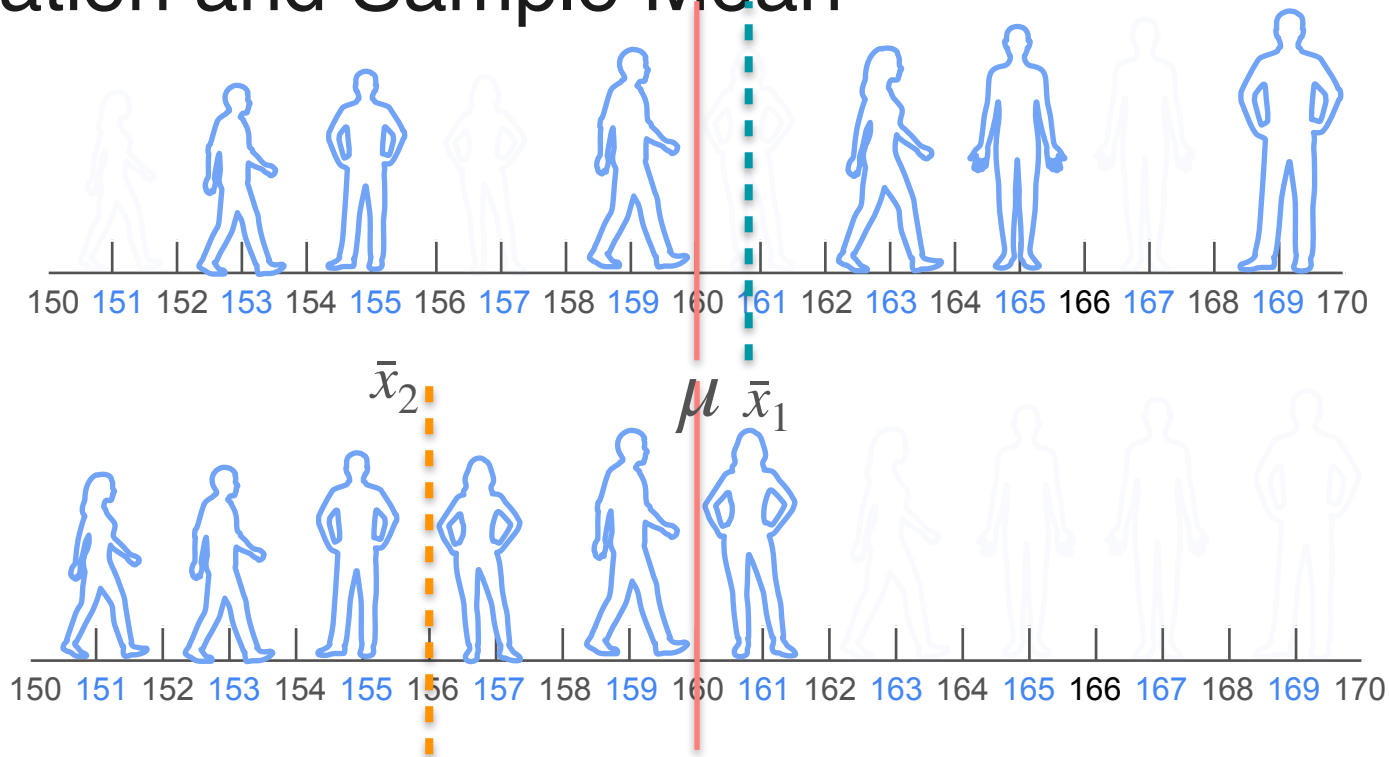
$$\frac{151 + 153 + 155 + 157 + 159 + 161}{6} = \frac{936}{6} = 156cm$$

Population and Sample Mean

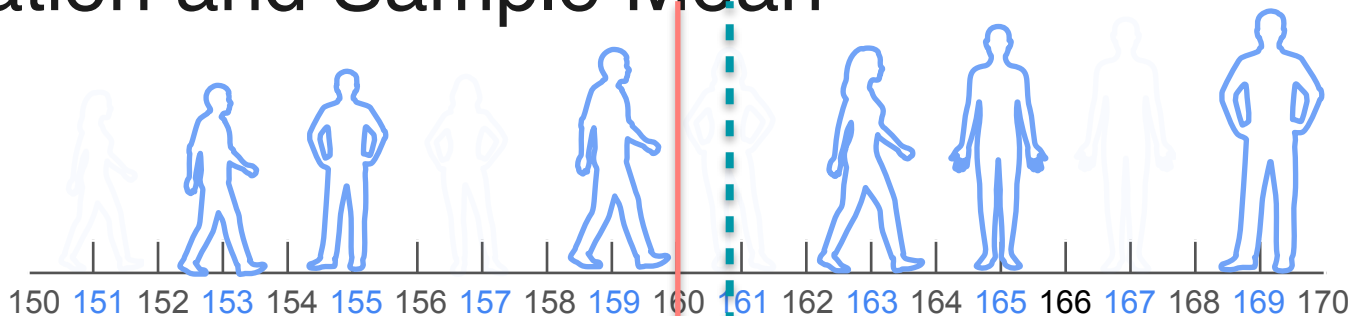
Population and Sample Mean



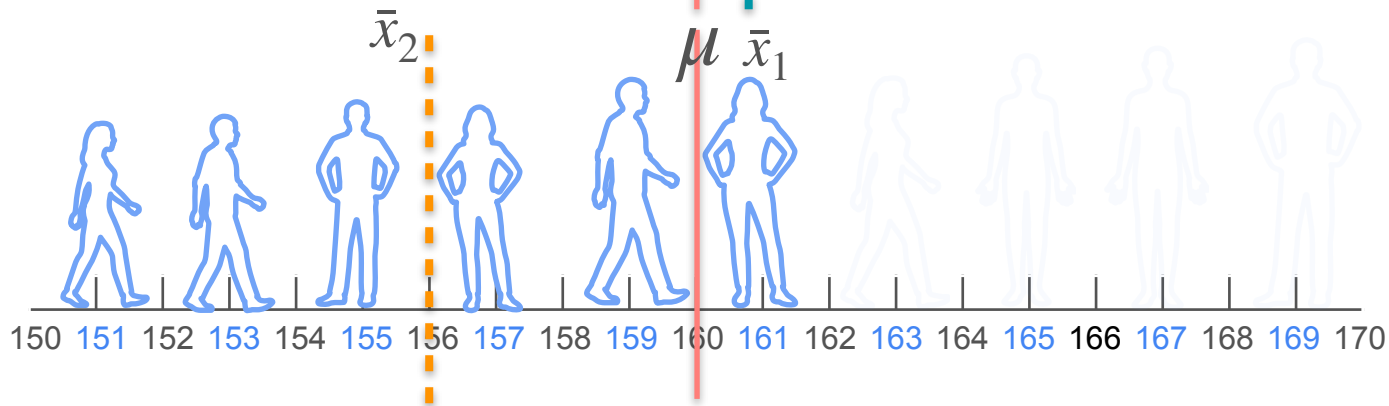
Population and Sample Mean



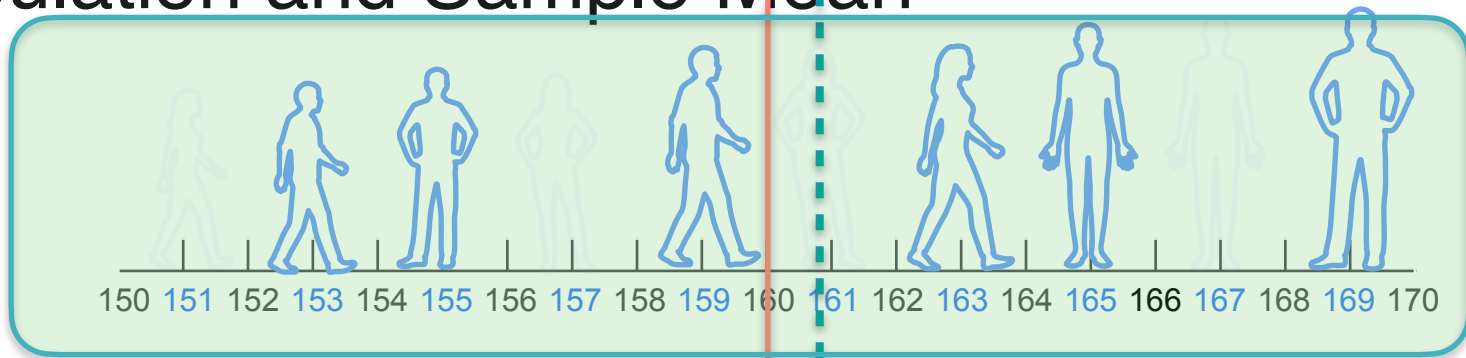
Population and Sample Mean



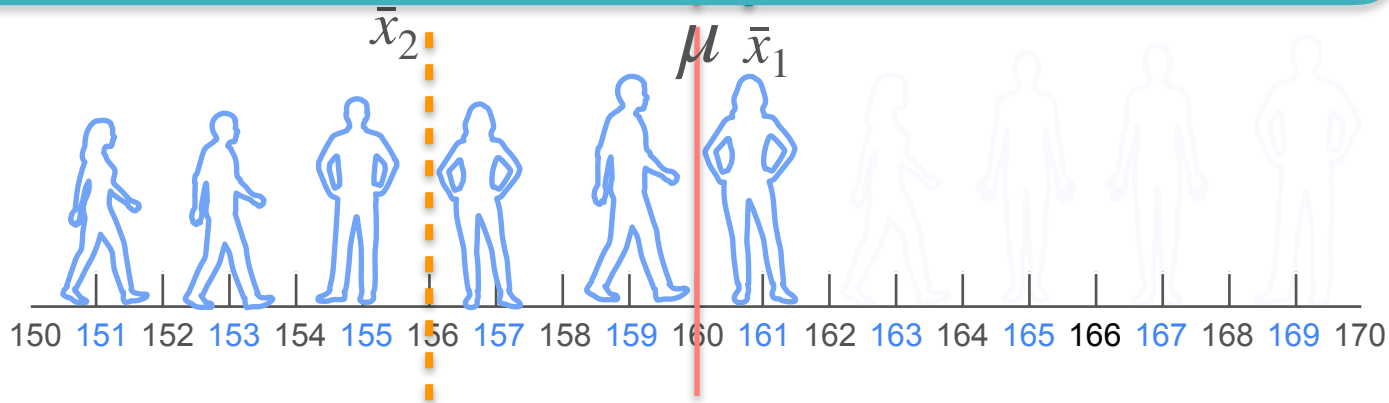
$n = 6$



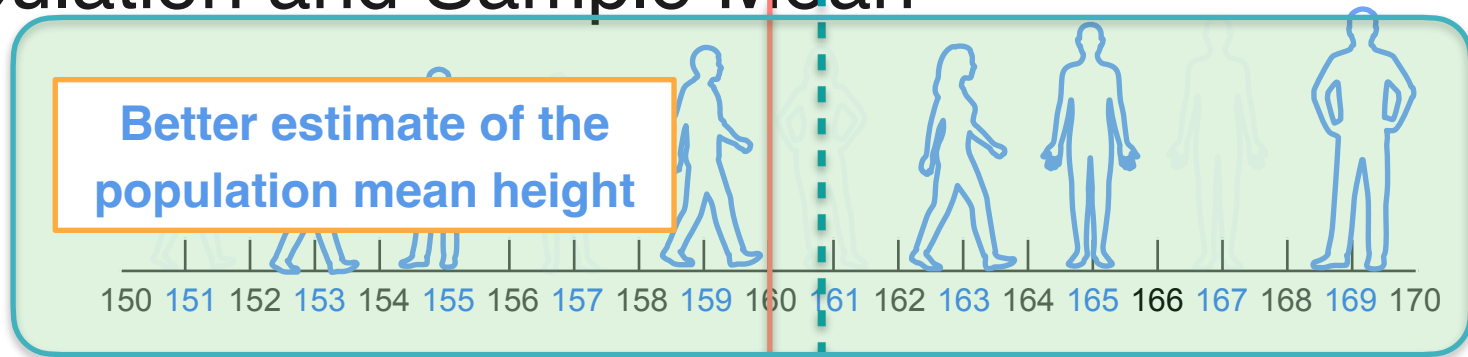
Population and Sample Mean



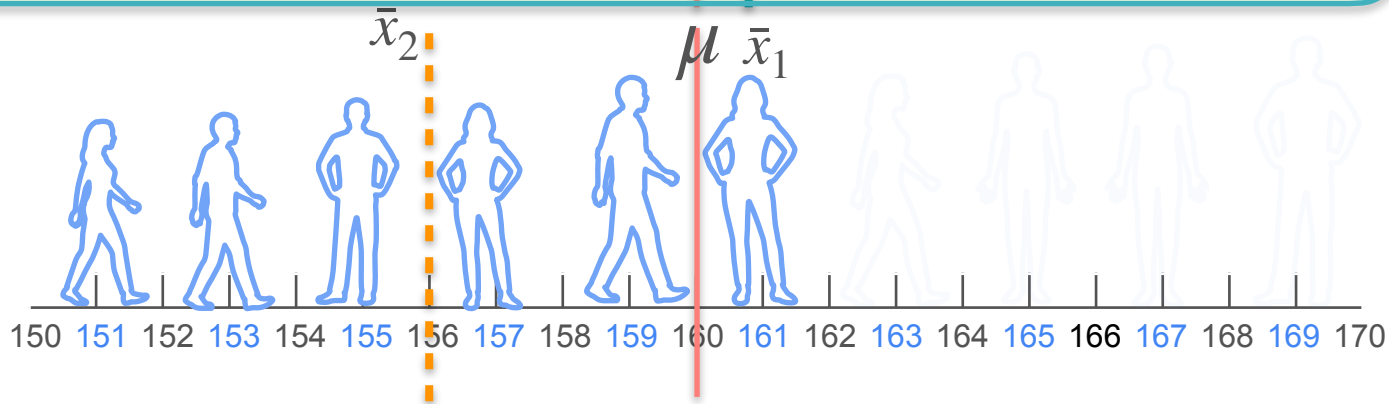
$n = 6$



Population and Sample Mean



$n = 6$



Population and Sample Mean



$$n = 6$$

$$\mu \quad \bar{x}_1$$

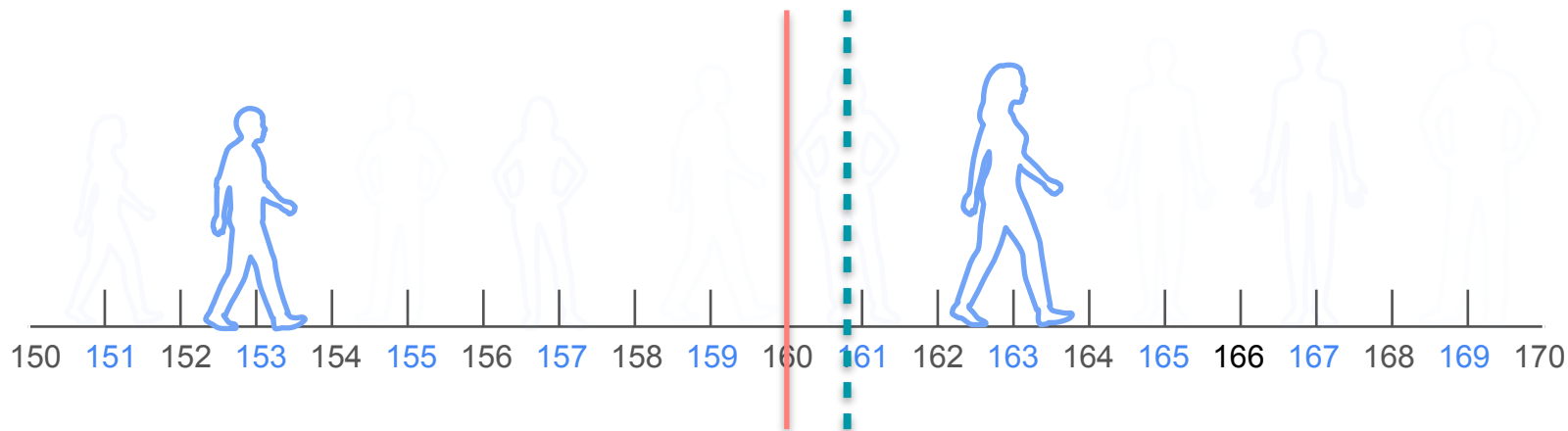
Population and Sample Mean



$$n = 6$$

$$n = 2$$

Population and Sample Mean



$$n = 6$$

$$n = 2$$

$$\mu \quad \bar{x}_1$$

Population and Sample Mean



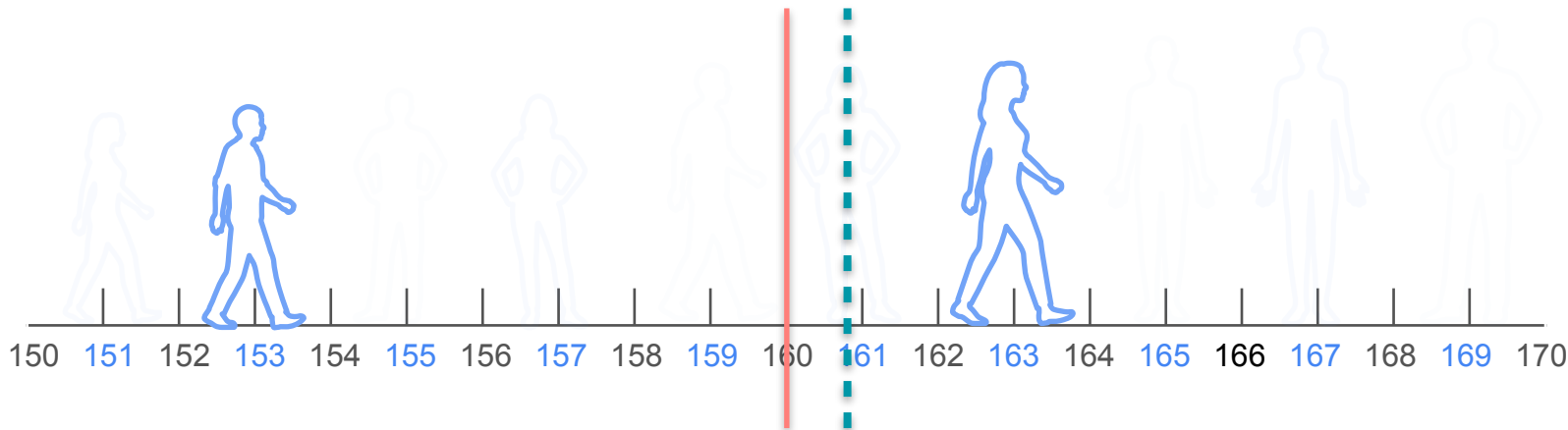
$$n = 6$$

$$n = 2$$

$$\mu \quad \bar{x}_1$$

What is the average
height in statistopia?

Population and Sample Mean



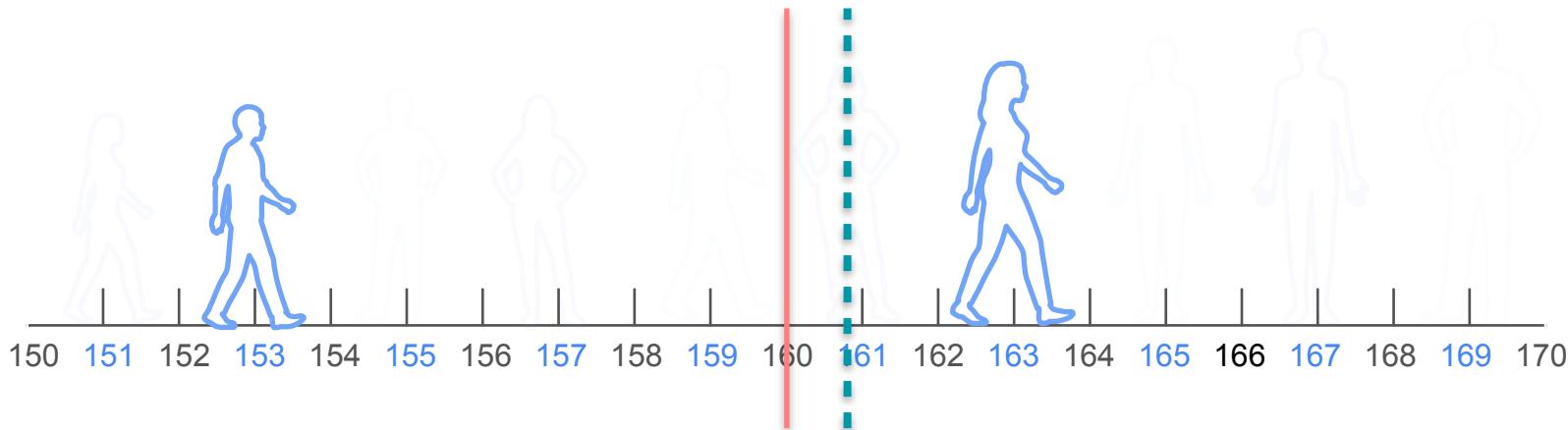
$$n = 6$$

$$n = 2$$

What is the average height in statistopia?

$$\frac{153 + 163}{2}$$

Population and Sample Mean



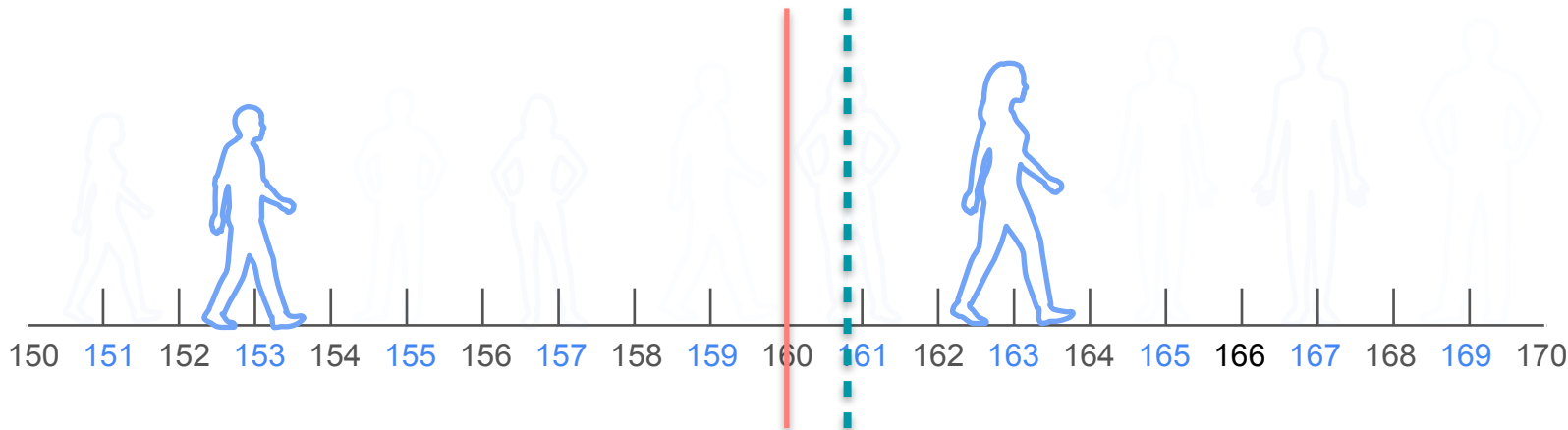
$$n = 6$$

$$n = 2$$

What is the average height in statistopia?

$$\frac{153 + 163}{2} = \frac{316}{2} = 158cm$$

Population and Sample Mean



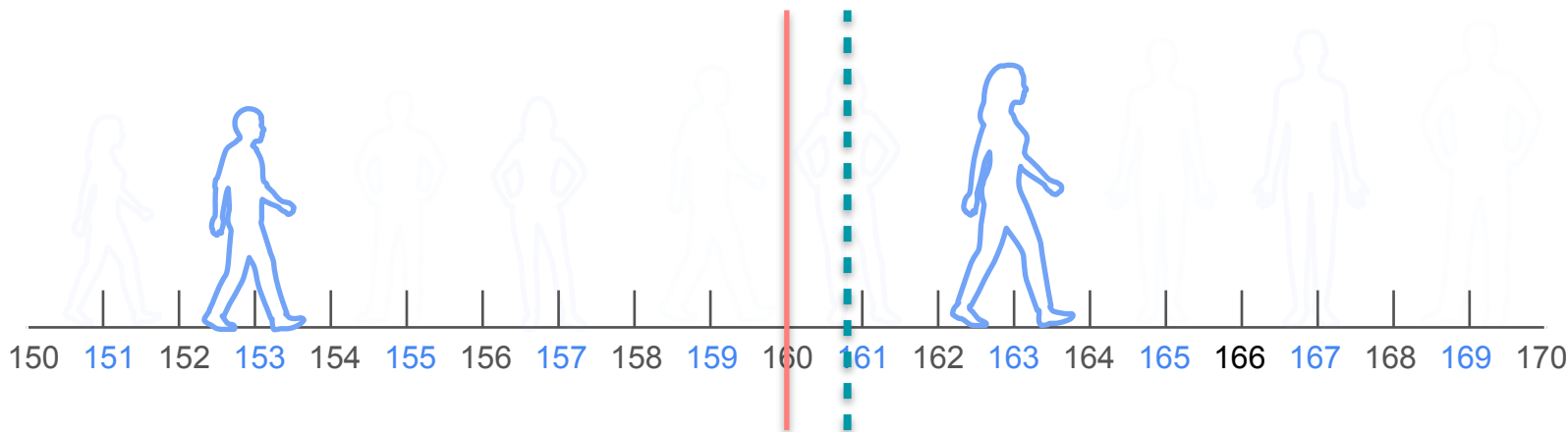
$$n = 6$$

$$n = 2$$

What is the average height in statistopia?

$$\frac{153 + 163}{2} = \frac{316}{2} = 158cm$$

Population and Sample Mean



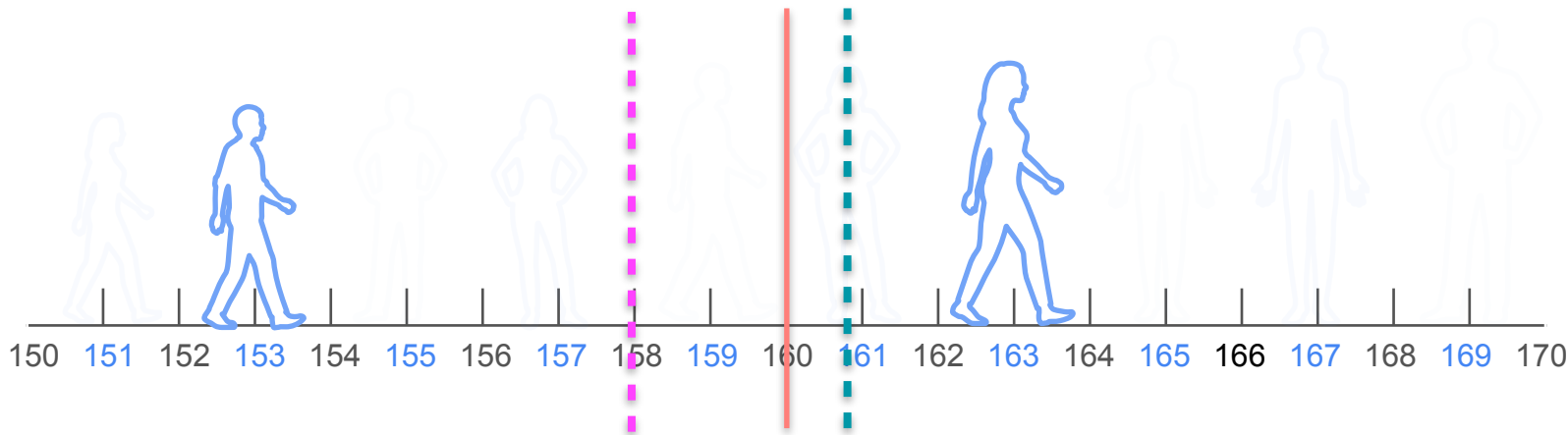
$$n = 6$$

$$n = 2$$

What is the average height in statistopia?

$$\frac{153 + 163}{2} = \frac{316}{2} = 158cm$$

Population and Sample Mean



$$n = 6$$

$$n = 2$$

What is the average height in statistopia?

$$\frac{153 + 163}{2} = \frac{316}{2} = 158cm$$

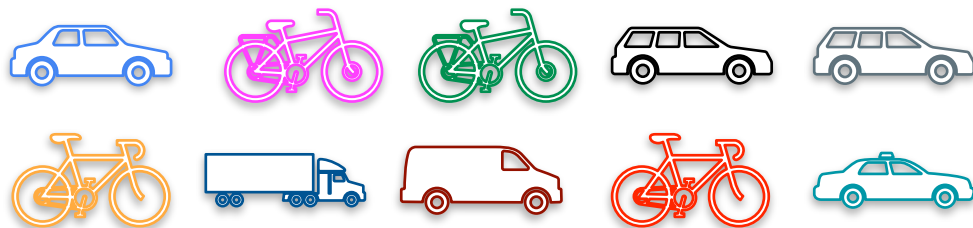
Proportion

Proportion

Population size: 10

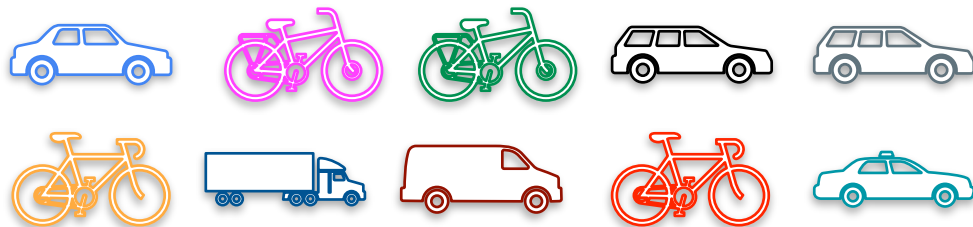
Proportion

Population size: 10



Proportion

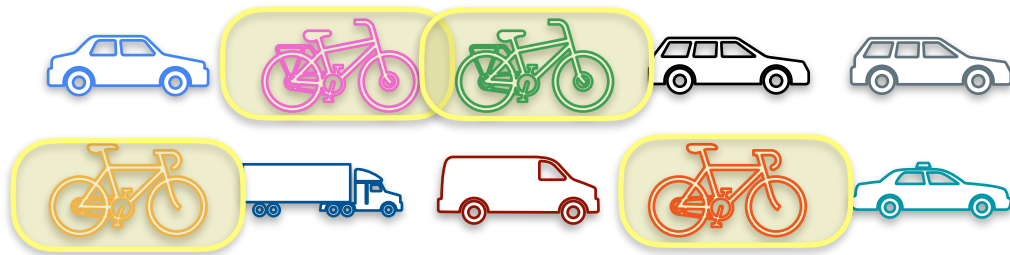
Population size: 10



What proportion of people own a bicycle?

Proportion

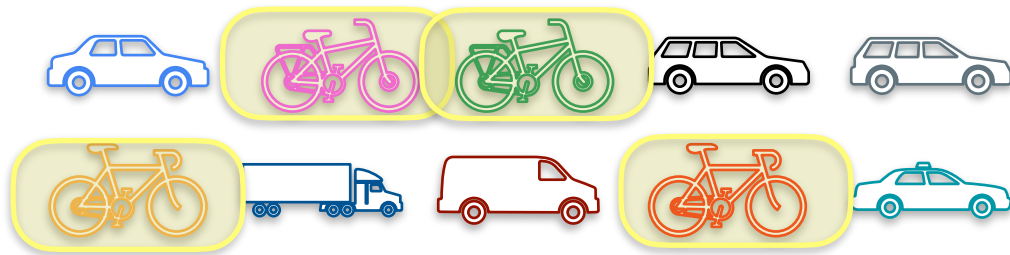
Population size: 10



What proportion of people own a bicycle?

Proportion

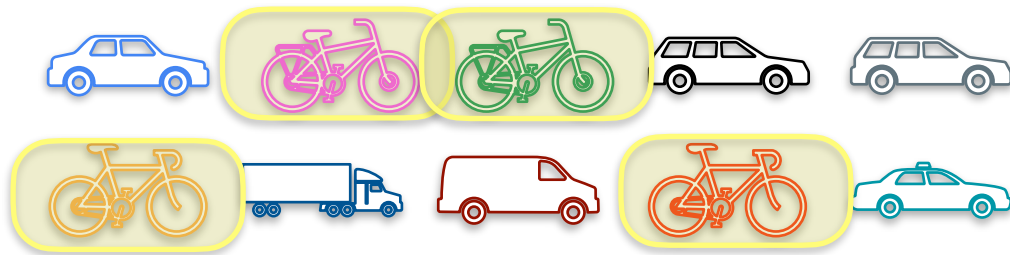
Population size: 10



What proportion of people own a bicycle?

Proportion

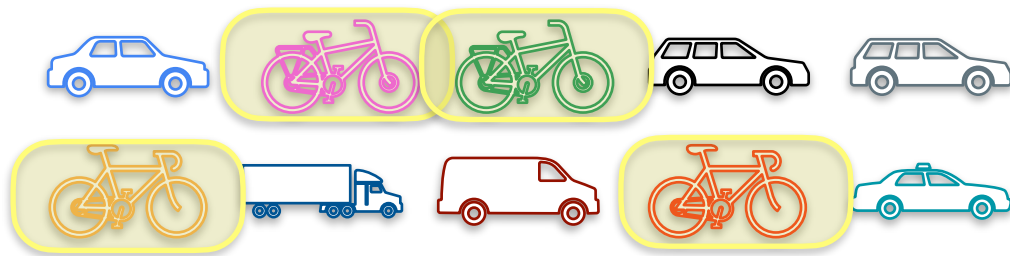
Population size: 10



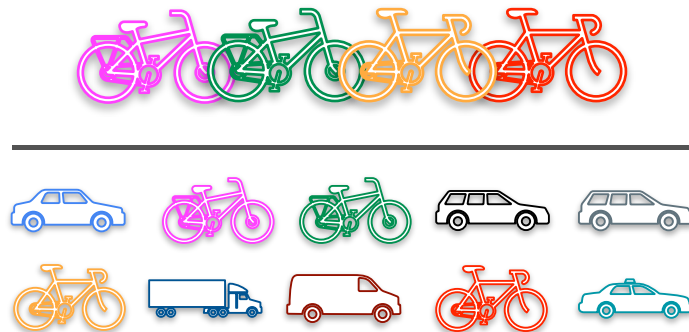
What proportion of people own a bicycle?

Proportion

Population size: 10

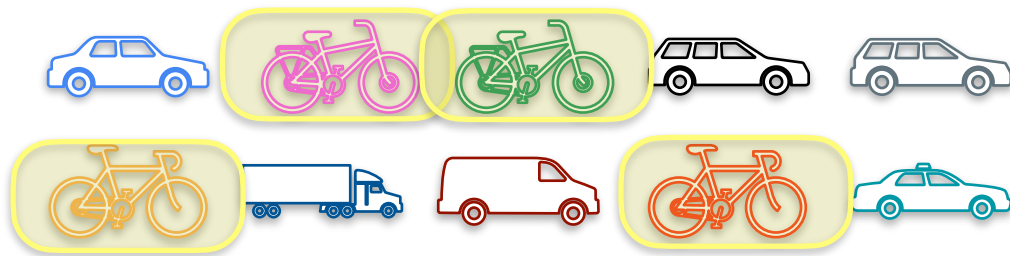


What proportion of people own a bicycle?

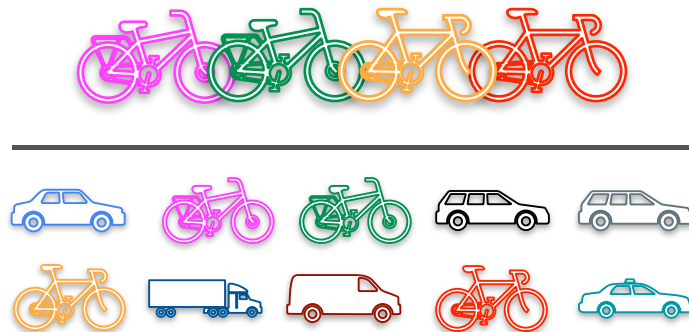


Proportion

Population size: 10



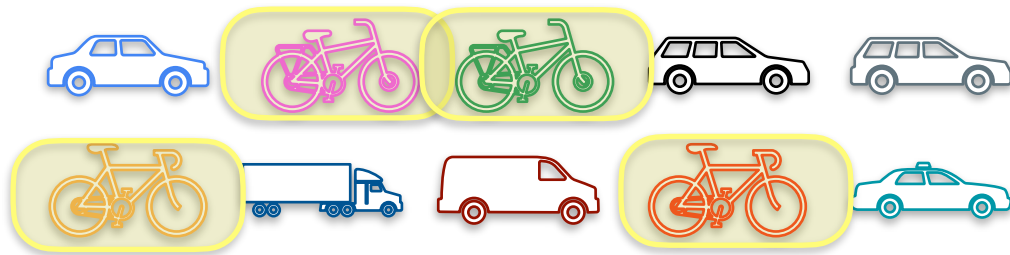
What proportion of people own a bicycle?



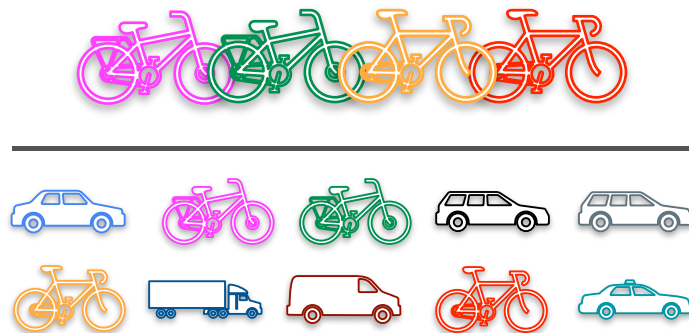
$$= \frac{4}{10} = 0.4 = 40 \%$$

Proportion

Population size: 10



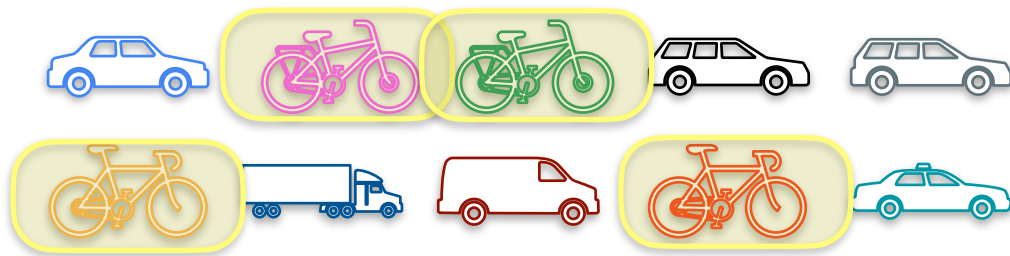
What proportion of people own a bicycle?



$$\text{population proportion} = \frac{4}{10} = 0.4 = 40 \%$$

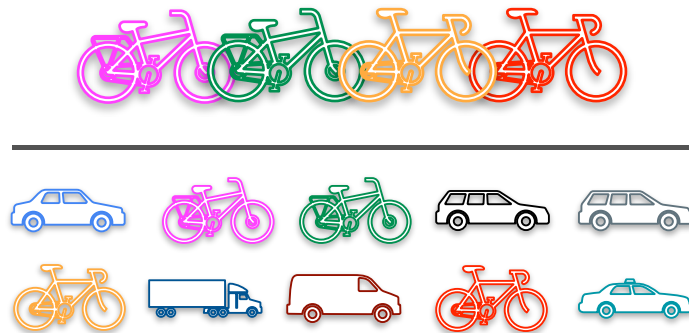
Proportion

Population size: 10



What proportion of people own a bicycle?

$$p \text{ population proportion} = \frac{4}{10} = 0.4 = 40 \%$$



Proportion

$$p = \frac{\text{number of items with a given characteristic } (x)}{\text{population } (n)}$$

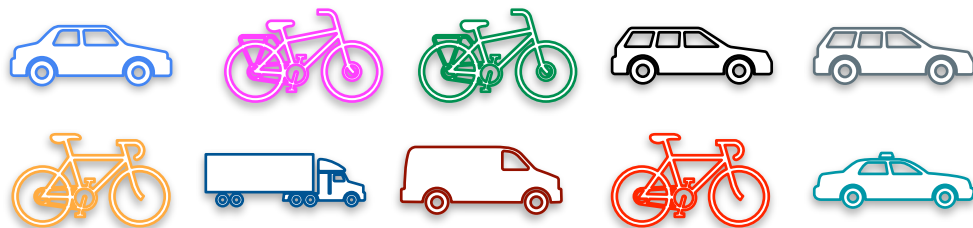
Proportion

population proportion

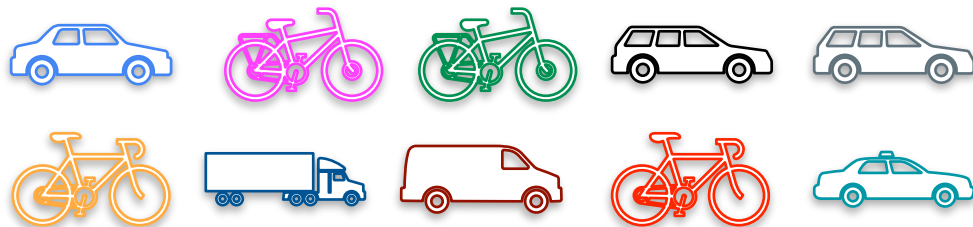
$$p = \frac{\text{number of items with a given characteristic } (x)}{\text{population } (n)}$$

Proportion

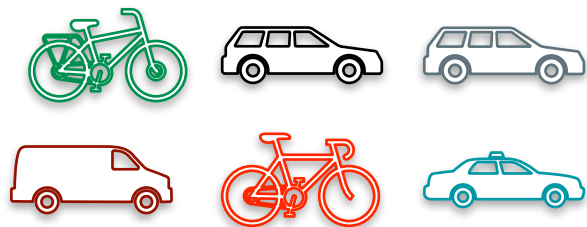
Population size: 10



Proportion

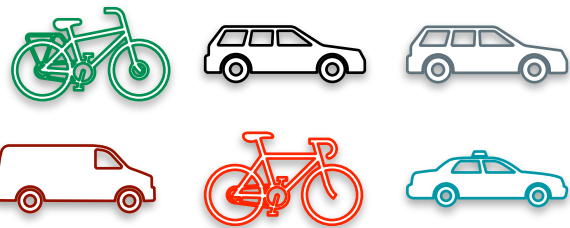


Sample Proportion



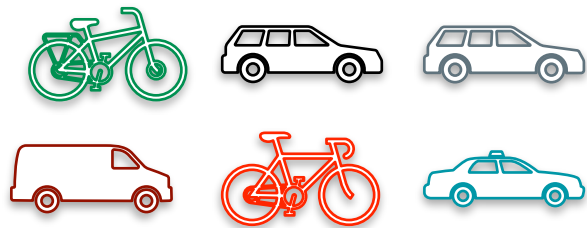
Sample Proportion

Sample size: 6



Sample Proportion

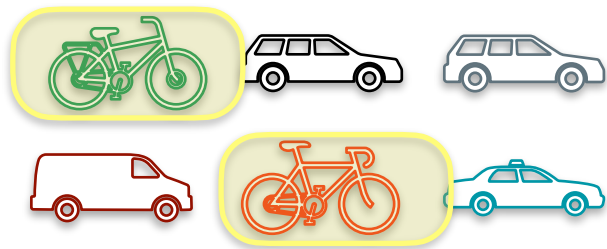
Sample size: 6



What proportion of people own a bicycle?

Sample Proportion

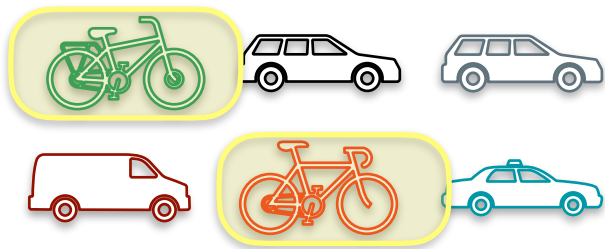
Sample size: 6



What proportion of people own a bicycle?

Sample Proportion

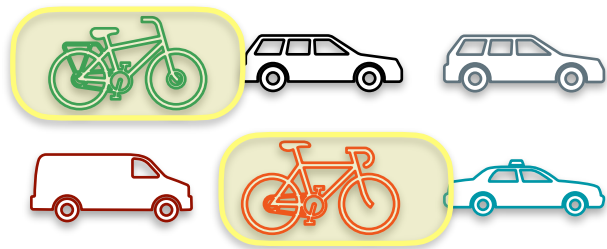
Sample size: 6



What proportion of people own a bicycle?

Sample Proportion

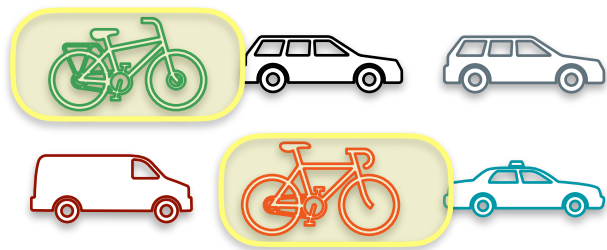
Sample size: 6



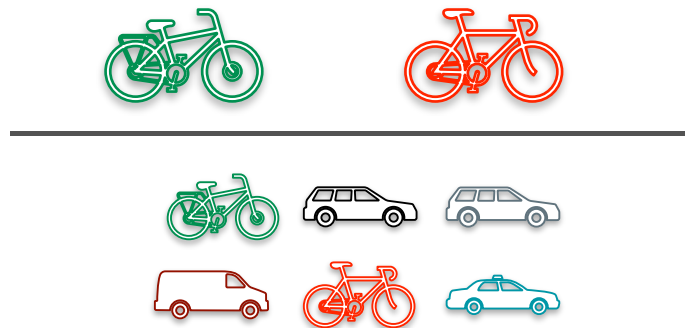
What proportion of people own a bicycle?

Sample Proportion

Sample size: 6

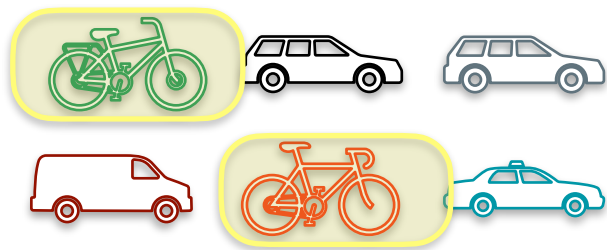


What proportion of people own a bicycle?

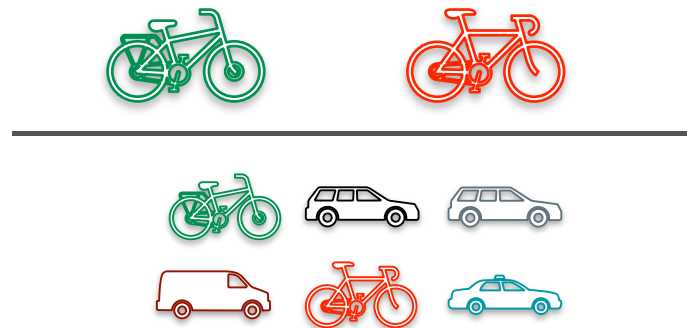


Sample Proportion

Sample size: 6



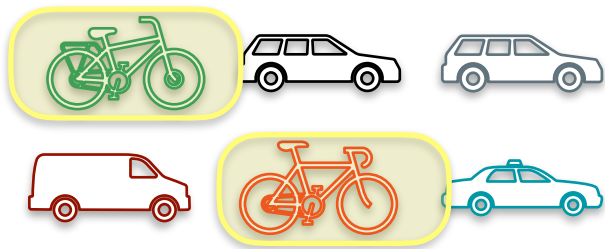
What proportion of people own a bicycle?



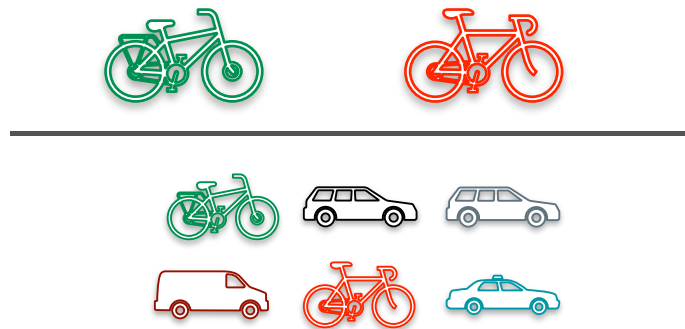
$$= \frac{2}{6} = 0.333 = 33.3 \%$$

Sample Proportion

Sample size: 6



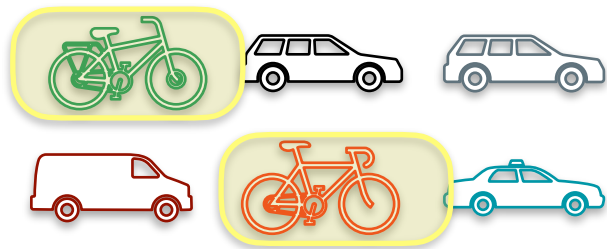
What proportion of people own a bicycle?



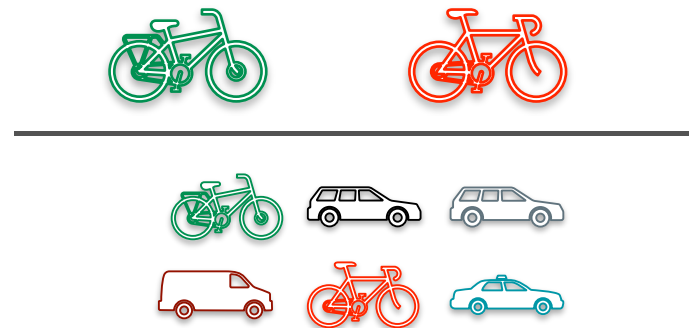
$$\text{sample proportion} = \frac{2}{6} = 0.333 = 33.3 \%$$

Sample Proportion

Sample size: 6



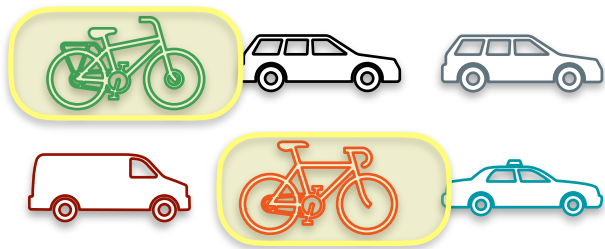
What proportion of people own a bicycle?



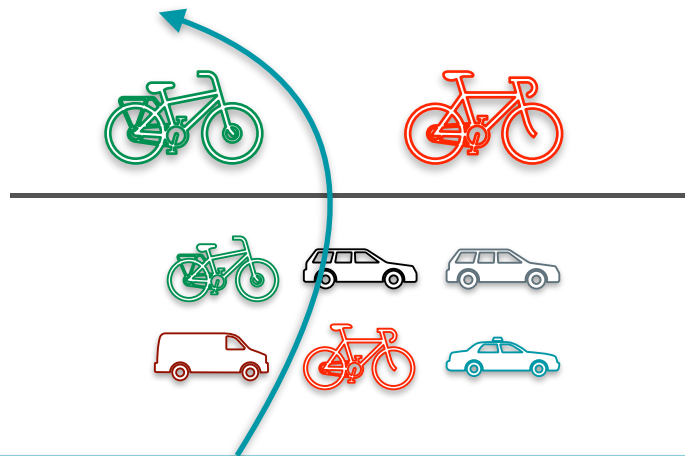
$$\hat{p} \text{ sample proportion} = \frac{2}{6} = 0.333 = 33.3 \%$$

Sample Proportion

Sample size: 6



What proportion of people own a bicycle?

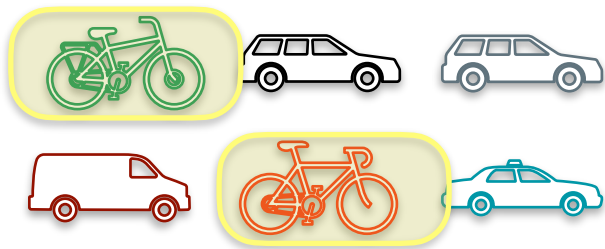


$$\hat{p} \text{ sample proportion} = \frac{2}{6} = 0.333 = 33.3 \%$$

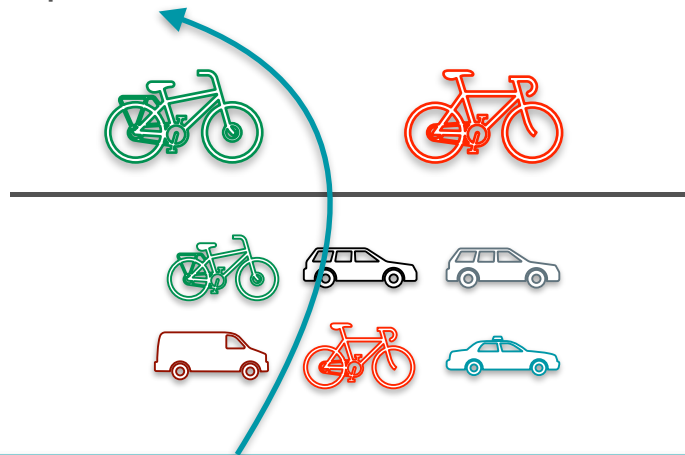
Sample Proportion

Sample size: 6

estimate of the population proportion



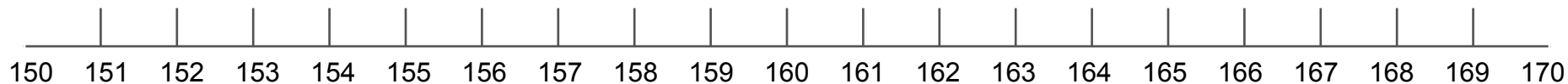
What proportion of people own a bicycle?



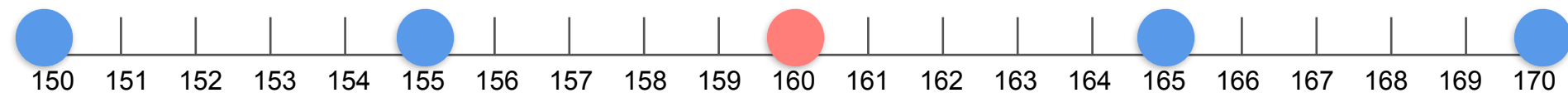
$$\hat{p} \text{ sample proportion} = \frac{2}{6} = 0.333 = 33.3 \%$$

Sample Variance

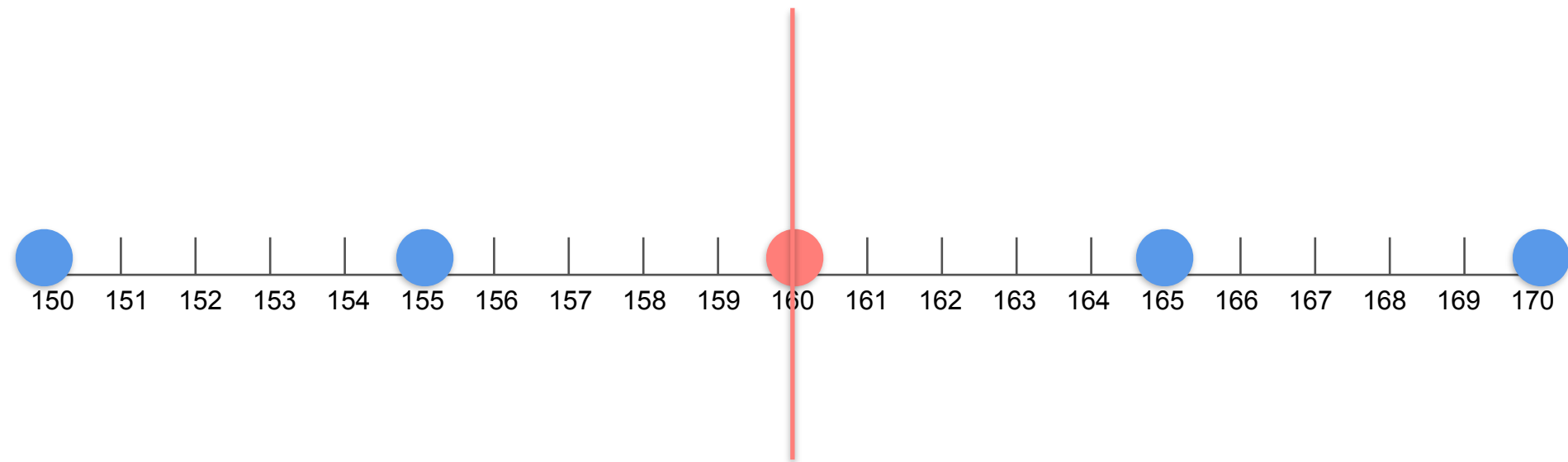
Sample Variance



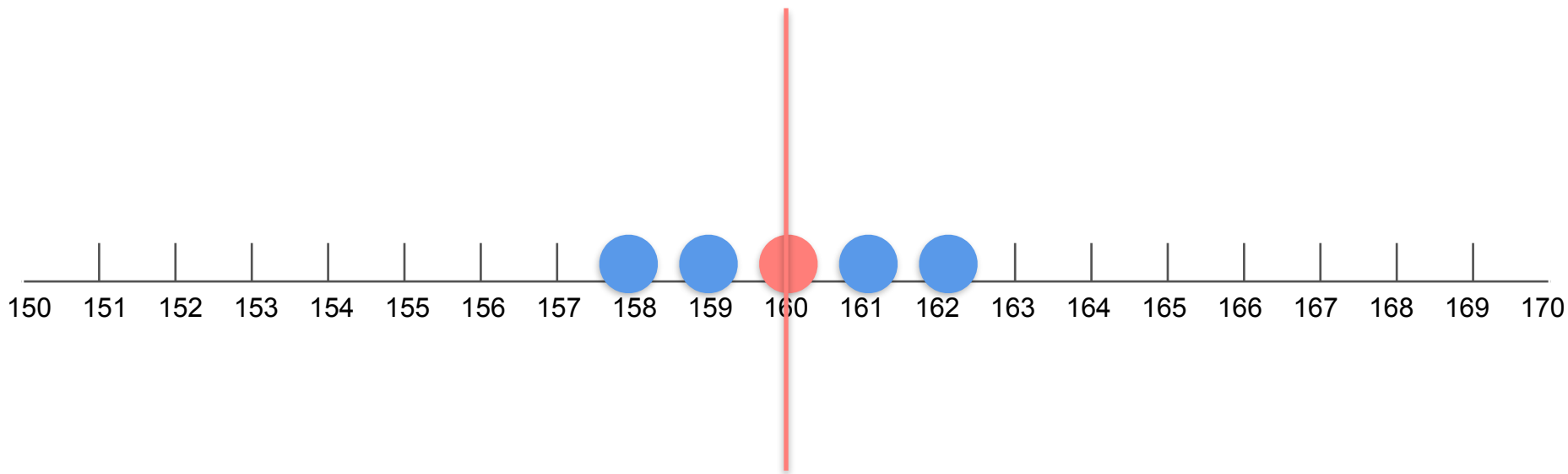
Sample Variance



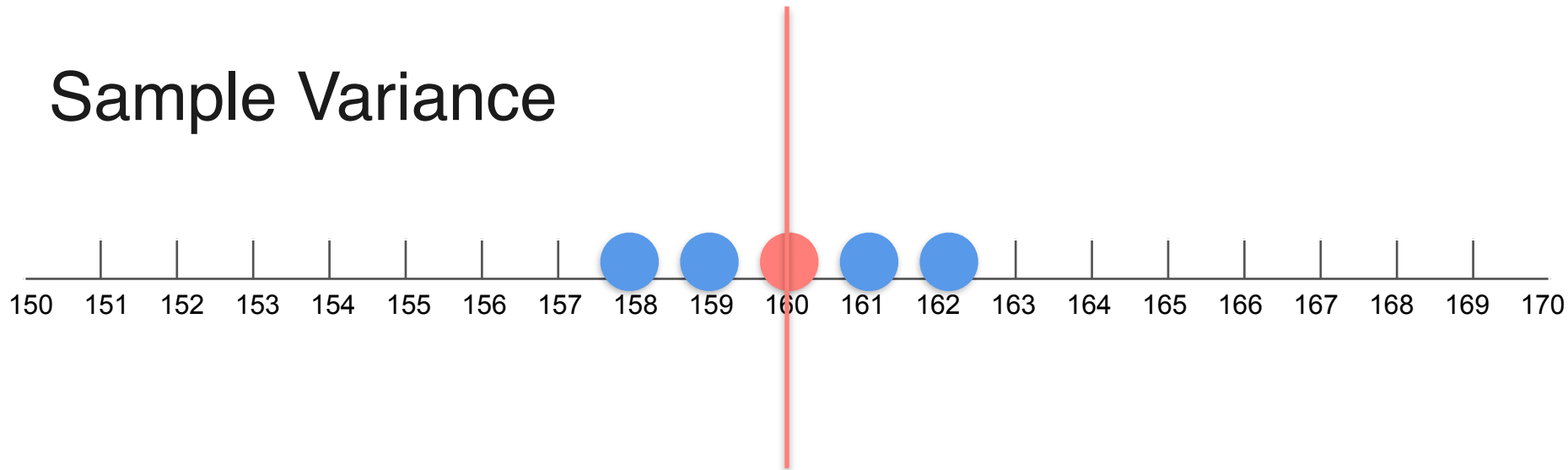
Sample Variance



Sample Variance



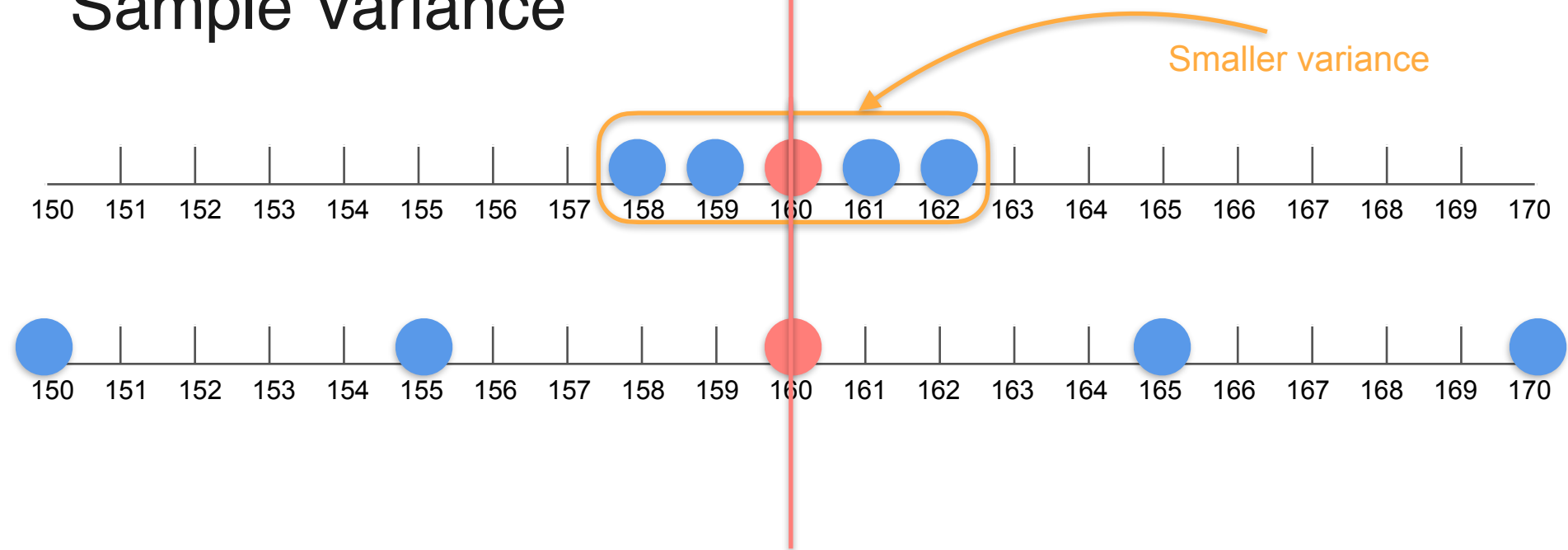
Sample Variance



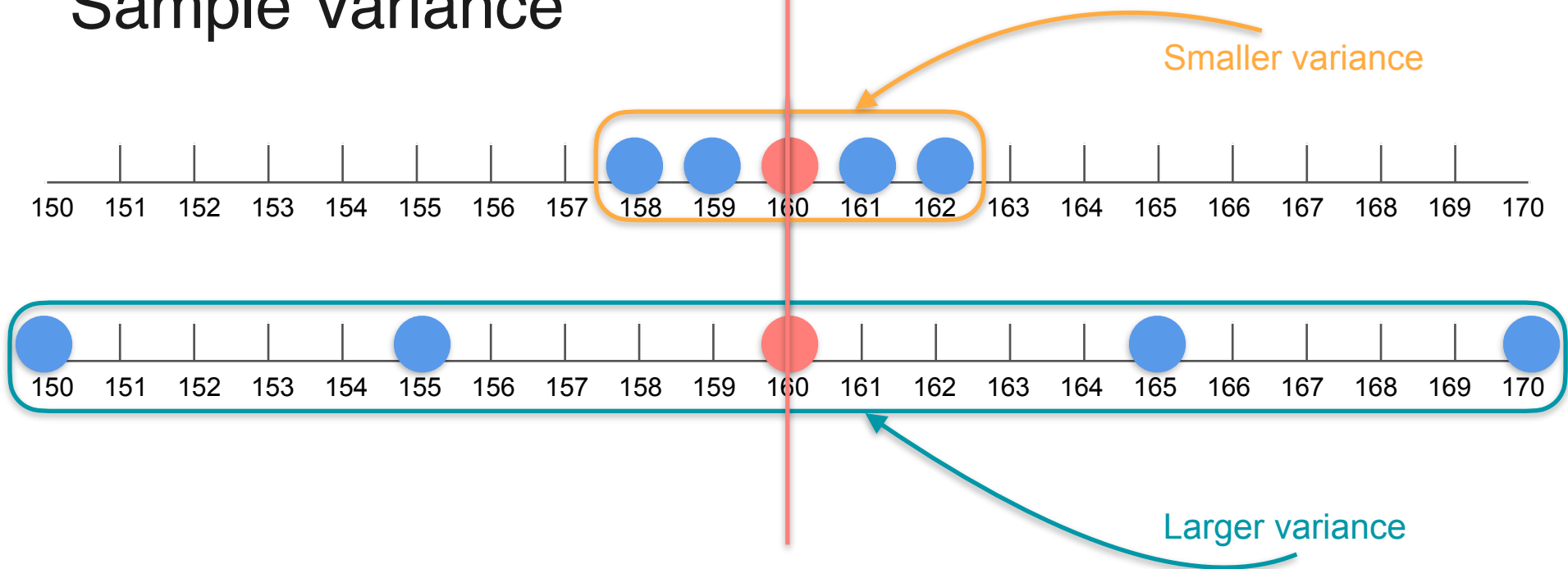
Sample Variance



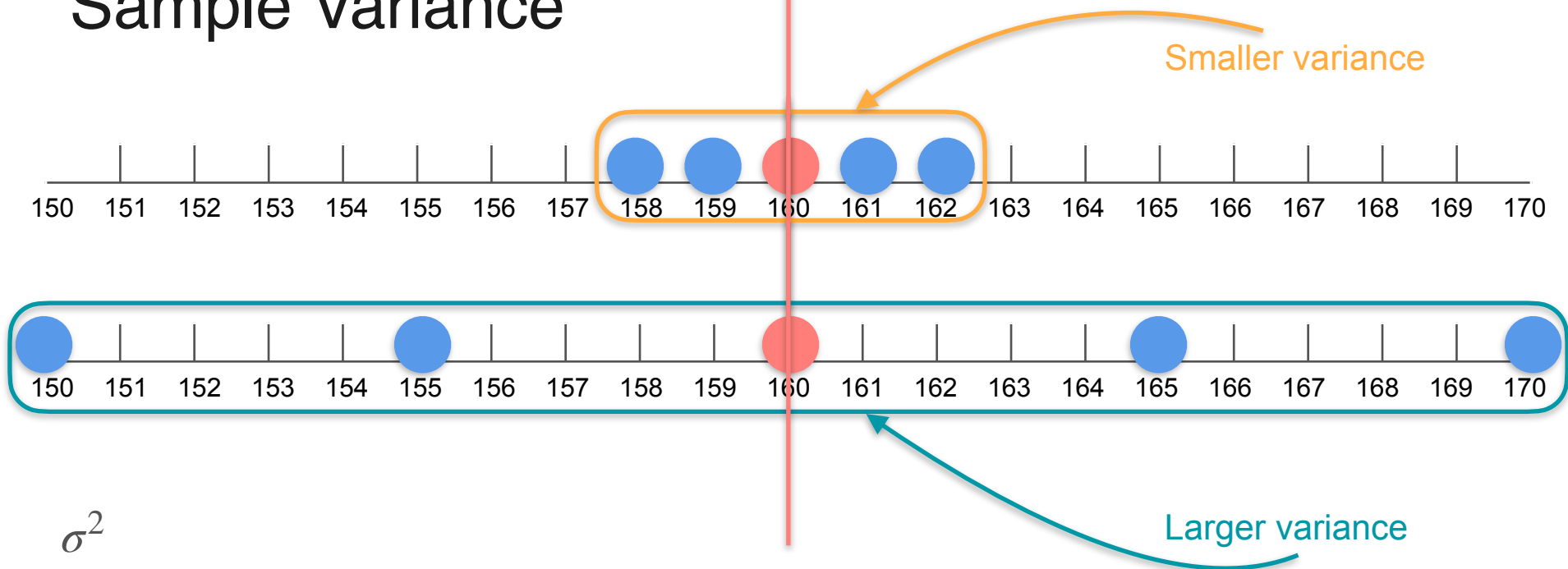
Sample Variance



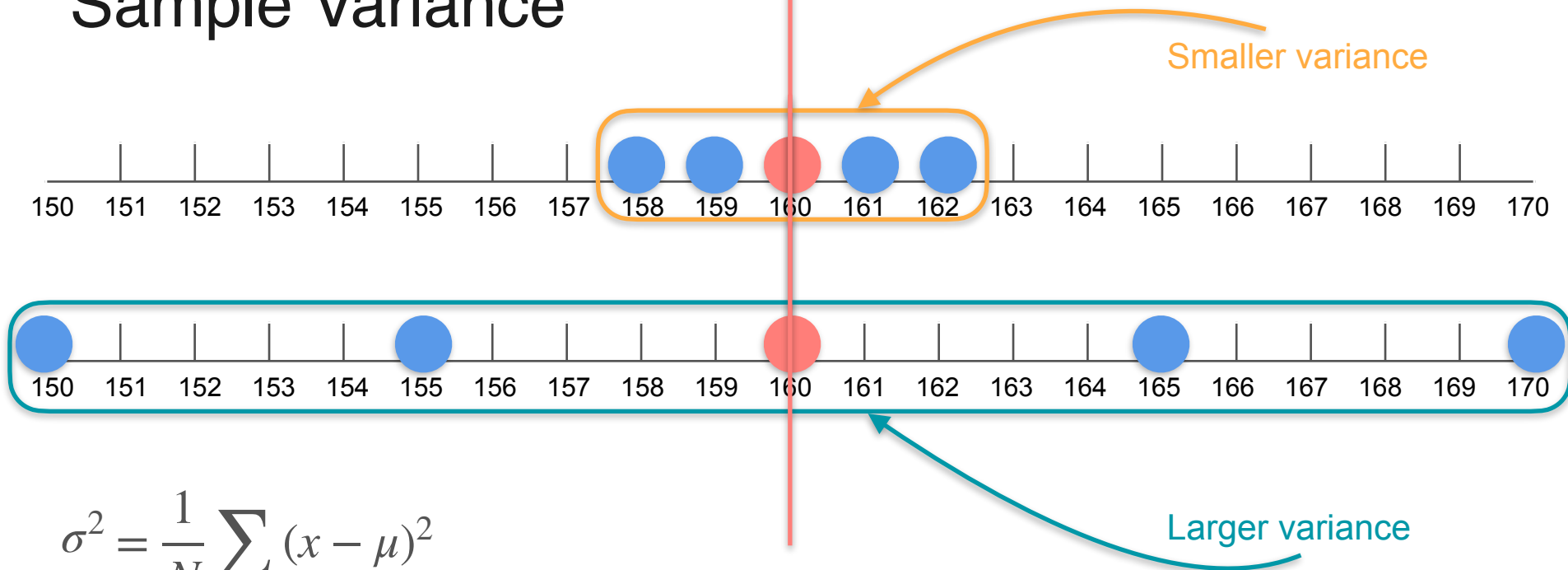
Sample Variance



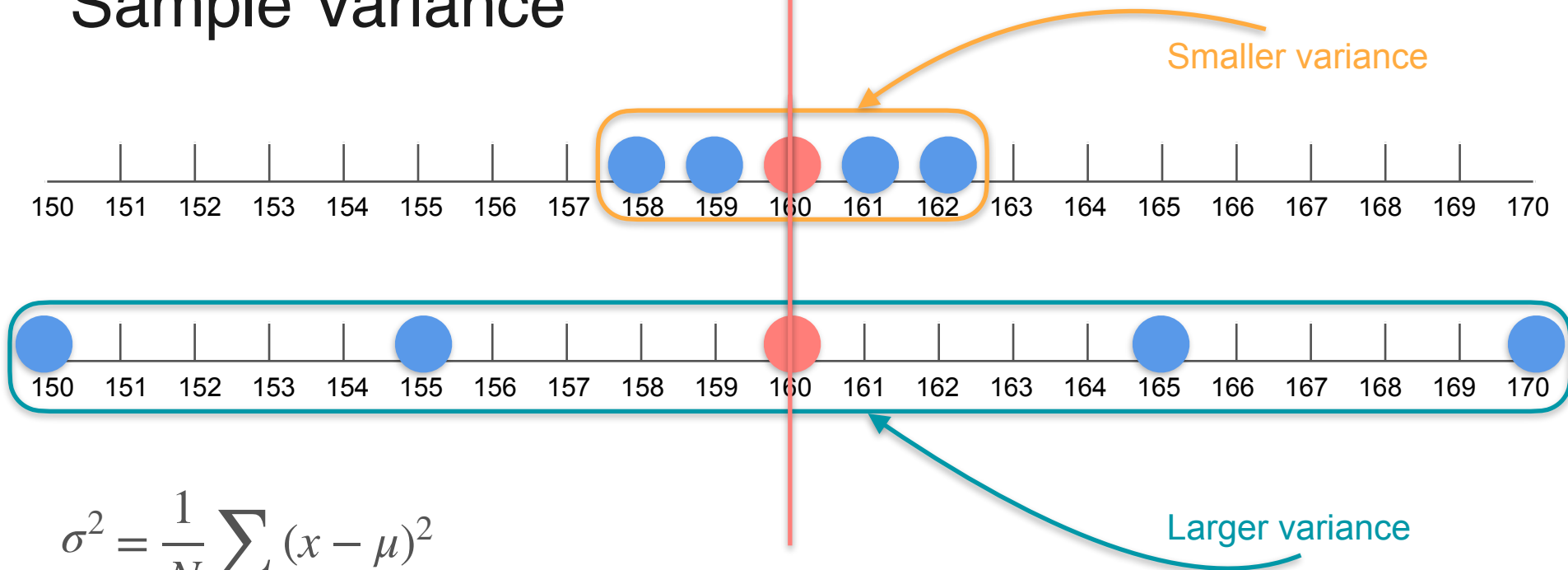
Sample Variance



Sample Variance



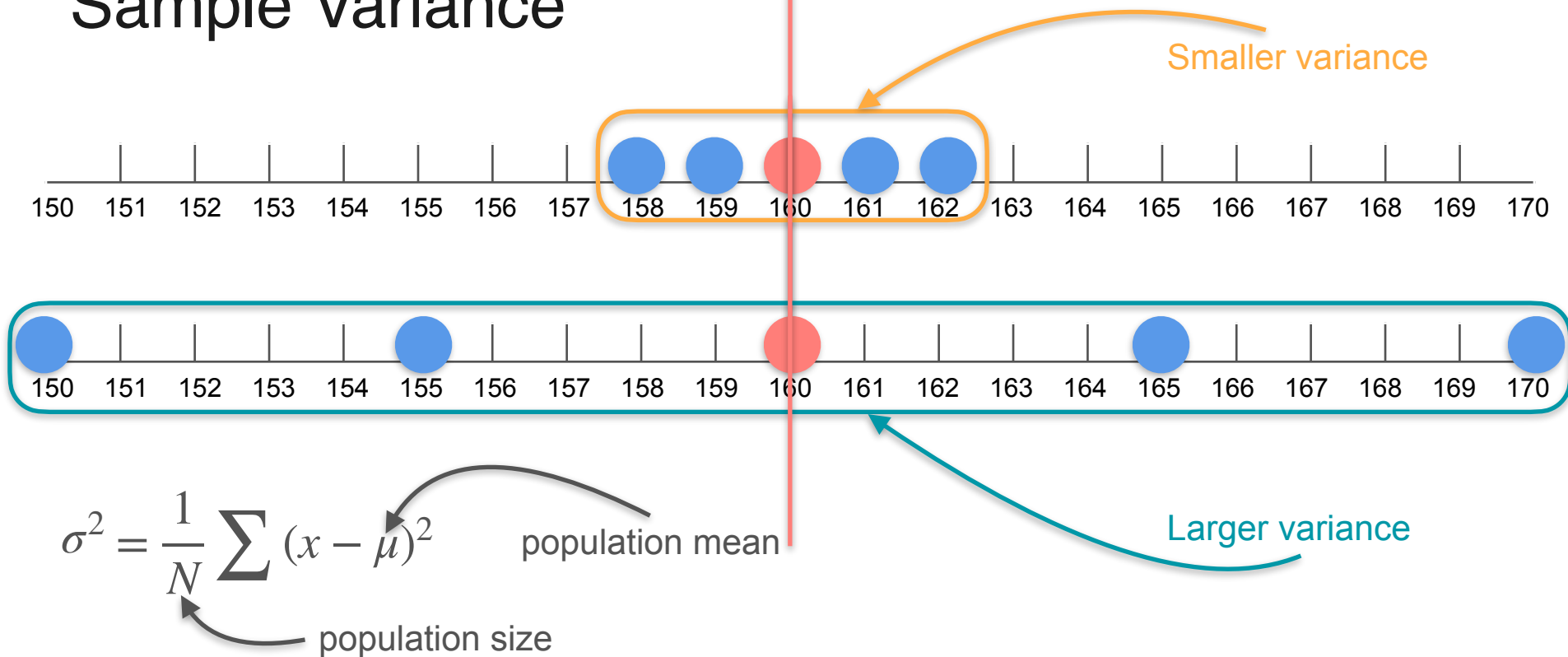
Sample Variance



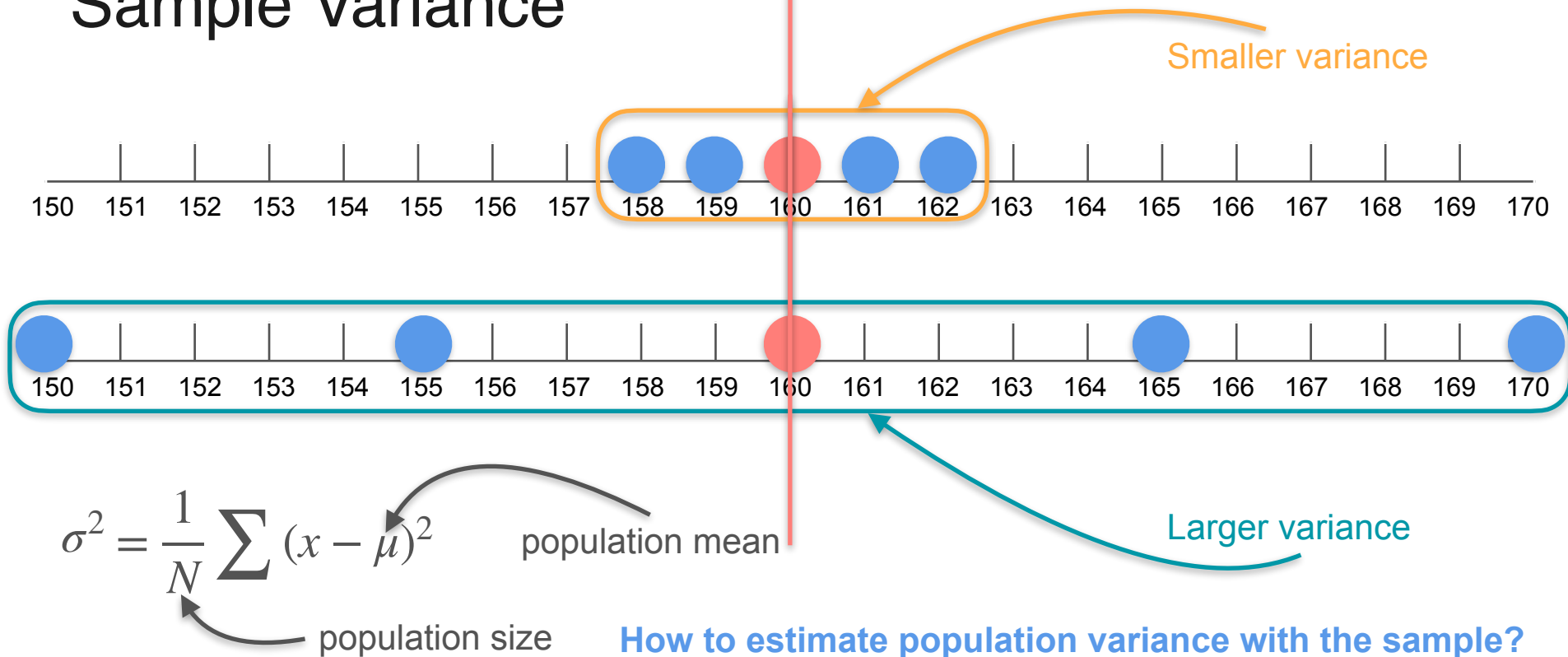
$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

population size

Sample Variance



Sample Variance



Variance Estimation

Variance Estimation



Variance Estimation



μ

Variance Estimation



$$\mu = \frac{1 + 2 + 3}{3}$$

Variance Estimation



$$\mu = \frac{1 + 2 + 3}{3} = \frac{6}{3}$$

Variance Estimation



$$\mu = \frac{1 + 2 + 3}{3} = \frac{6}{3} = 2$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

x	$x - \mu$	$(x - \mu)^2$
-----	-----------	---------------

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

x
1
2
3

$$x - \mu \quad (x - \mu)^2$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

x
1
2
3

$$x - 2$$

$$x - \mu \quad (x - \mu)^2$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$x - 2$		$(x - \mu)^2$
x	$x - \mu$	
1	-1	1
2	0	
3	1	

Variance Estimation

1 2 3

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$x - 2$$

x	$x - \mu$	$(x - \mu)^2$
1	-1	1
2	0	0
3	1	1

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$x - 2$

x	$x - \mu$	$(x - \mu)^2$
1	-1	1
2	0	0
3	1	1

$$\sum (x - \mu)^2$$

Variance Estimation

1 2 3

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$x - 2$

x	$x - \mu$	$(x - \mu)^2$
1	-1	1
2	0	0
3	1	1

$$\sum (x - \mu)^2 = 2$$

Variance Estimation

1 2 3

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$x - 2$$

x	$x - \mu$	$(x - \mu)^2$
1	-1	1
2	0	0
3	1	1

$$\frac{\sum (x - \mu)^2}{N} = 2$$

Variance Estimation

1 2 3

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$x - 2$$

x	$x - \mu$	$(x - \mu)^2$
1	-1	1
2	0	0
3	1	1

$$\frac{\sum (x - \mu)^2}{N} = \frac{2}{3}$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$x - 2$		
x	$x - \mu$	$(x - \mu)^2$
1	-1	1
2	0	0
3	1	1

$$\frac{\sum (x - \mu)^2}{N}$$

$$= \frac{2}{3}$$

σ^2
Population variance

Variance Estimation

1 2 3

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

Variance Estimation

1 2 3

$n = 2$
Samples

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

0
0,25
1
0,25
0
0,25
1
0,25
0

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

0
0,25
1
0,25
0
0,25
1
0,25
0

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

0
0,25
1
0,25
0
0,25
1
0,25
0

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

0
0,25
1
0,25
0
0,25
1
0,25
0

estimated
variance

$$= 0.333$$

$$= \frac{1}{3}$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

0
0,25
1
0,25
0
0,25
1
0,25
0

estimated
variance

$$= 0.333$$

$$= \frac{1}{3}$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

0
0,25
1
0,25
0
0,25
1
0,25
0

estimated
variance

$$= 0.333$$

$$= \frac{1}{3}$$

Variance Estimation

1 2 3

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1

1,5

2

1,5

2

2,5

2

2,5

3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

0

0,25

1

0,25

0

0,25

1

0,25

0

estimated
variance

$$= 0.333$$

$$= \frac{1}{3}$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1

1,5

2

1,5

2

2,5

2

2,5

3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

0

0,25

1

0,25

0

0,25

1

0,25

0

estimated
variance

$$= 0.333$$

$$= \frac{1}{3}$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1

1,5

2

1,5

2

2,5

2

2,5

3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

0

0,25

1

0,25

0

0,25

1

0,25

0

estimated
variance

$$= 0.333$$

$$= \frac{1}{3}$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n - 1}$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n - 1}$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n - 1}$$

0
0,5
2
0,5
0
0,5
2
0,5
0

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n - 1}$$

0
0,5
2
0,5
0
0,5
2
0,5
0

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n - 1}$$

0
0,5
2
0,5
0
0,5
2
0,5
0

estimated
variance

$$= 0.667$$

$$= \frac{2}{3}$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n - 1}$$

0
0,5
2
0,5
0
0,5
2
0,5
0

estimated
variance

$$= 0.667$$

$$= \frac{2}{3}$$

Variance Estimation

1	2	3
---	---	---

$$\mu = 2$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{2}{3}$$

$n = 2$
Samples

1	1
1	2
1	3
2	1
2	2
2	3
3	1
3	2
3	3

$\bar{x}$

1
1,5
2
1,5
2
2,5
2
2,5
3

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n - 1}$$

0
0,5
2
0,5
0
0,5
2
0,5
0

estimated
variance

$$= 0.667$$

$$= \frac{2}{3}$$

Variance Estimation

Population Variance Formula

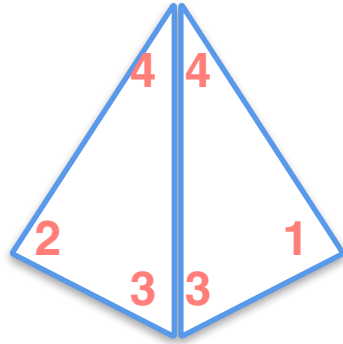
$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

Sample Variance Formula

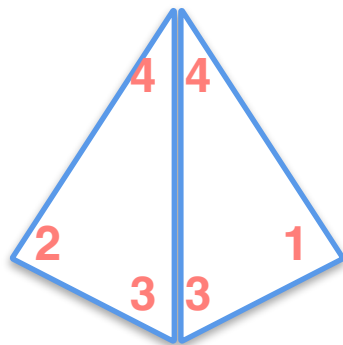
$$Var(x) = \frac{1}{n - 1} \sum (x - \bar{x})^2$$

Variance Estimation

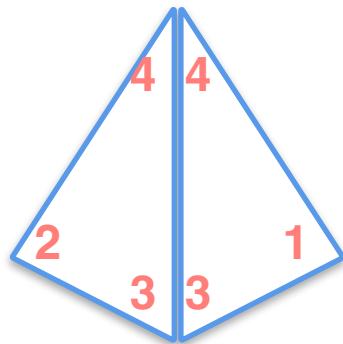
Variance Estimation



Variance Estimation

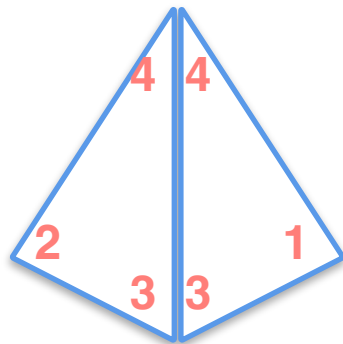


Variance Estimation



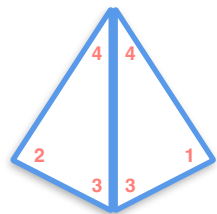
$$\mu = \frac{1 + 2 + 3 + 4}{4}$$

Variance Estimation



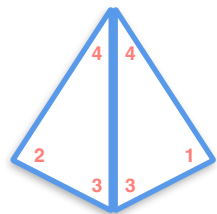
$$\mu = \frac{1 + 2 + 3 + 4}{4} = 2.5$$

Variance Estimation



$$\mu = 2.5$$

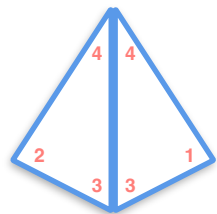
Variance Estimation



$$\mu = 2.5$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

Variance Estimation



$$\mu = 2.5$$

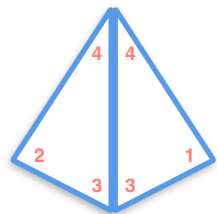
x

$x - \mu$

$(x - \mu)^2$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

Variance Estimation



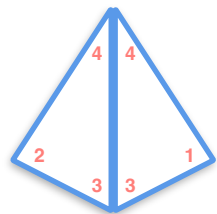
$$\mu = 2.5$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

x
1
2
3
4

$$x - \mu \quad (x - \mu)^2$$

Variance Estimation



$$\mu = 2.5$$

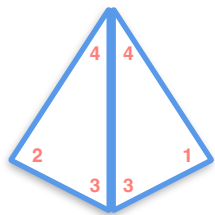
$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

x
1
2
3
4

$$x - 2.5$$

$$x - \mu \quad (x - \mu)^2$$

Variance Estimation



1 2 3 4

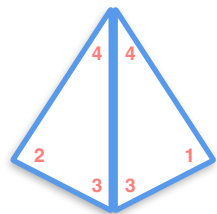
$$\mu = 2.5$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$x - 2.5$$

x	$x - \mu$	$(x - \mu)^2$
1	-1,5	
2	-0,5	
3	0,5	
4	1,5	

Variance Estimation



1 2 3 4

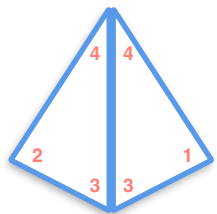
$$\mu = 2.5$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$x - 2.5$$

x	$x - \mu$	$(x - \mu)^2$
1	-1,5	2,25
2	-0,5	0,25
3	0,5	0,25
4	1,5	2,25

Variance Estimation



$$\mu = 2.5$$

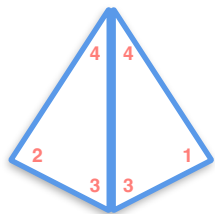
$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$x - 2.5$$

x	$x - \mu$	$(x - \mu)^2$
1	-1,5	2,25
2	-0,5	0,25
3	0,5	0,25
4	1,5	2,25

$$\sum (x - \mu)^2$$

Variance Estimation



1 2 3 4

$$\mu = 2.5$$

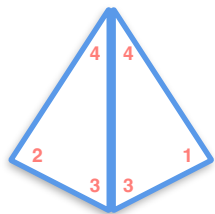
$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$x - 2.5$$

x	$x - \mu$	$(x - \mu)^2$
1	-1,5	2,25
2	-0,5	0,25
3	0,5	0,25
4	1,5	2,25

$$\sum (x - \mu)^2 = 5$$

Variance Estimation



1 2 3 4

$$\mu = 2.5$$

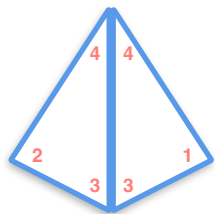
$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$x - 2.5$$

x	$x - \mu$	$(x - \mu)^2$
1	-1,5	2,25
2	-0,5	0,25
3	0,5	0,25
4	1,5	2,25

$$\frac{\sum (x - \mu)^2}{N} = 5$$

Variance Estimation



1 2 3 4

$$\mu = 2.5$$

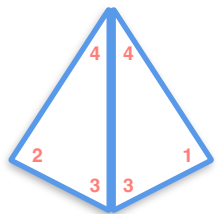
$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$x - 2.5$$

x	$x - \mu$	$(x - \mu)^2$
1	-1,5	2,25
2	-0,5	0,25
3	0,5	0,25
4	1,5	2,25

$$\frac{\sum (x - \mu)^2}{N} = \frac{5}{4}$$

Variance Estimation



$$\mu = 2.5$$

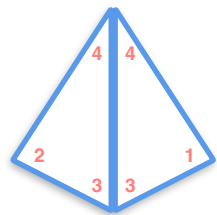
$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$x - 2.5$		
x	$x - \mu$	$(x - \mu)^2$
1	-1,5	2,25
2	-0,5	0,25
3	0,5	0,25
4	1,5	2,25

$$\frac{\sum (x - \mu)^2}{N} = \frac{5}{4}$$

Population variance σ^2

Variance Estimation

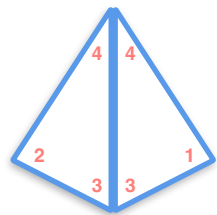


$$\mu = 2.5$$

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{5}{4}$$

Variance Estimation



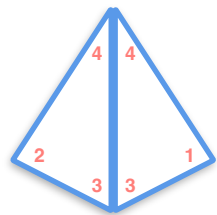
$$\mu = 2.5$$

$n = 2$
Samples

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{5}{4}$$

Variance Estimation



$$\mu = 2.5$$

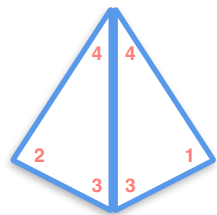
$n = 2$
Samples

x

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{5}{4}$$

Variance Estimation



1 2 3 4

$$\mu = 2.5$$

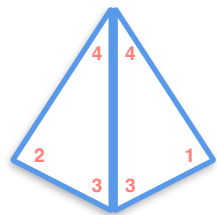
$n = 2$
Samples

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{5}{4}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

Variance Estimation



1 2 3 4

$$\mu = 2.5$$

$n = 2$
Samples

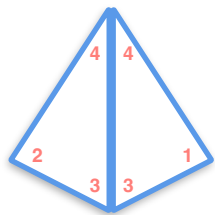
$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{5}{4}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

$\bar{x}$

Variance Estimation



$$\mu = 2.5$$

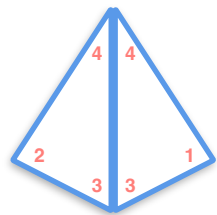
$n = 2$
Samples

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{5}{4}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
$\bar{x}$	1	1,5	2	2,5	1,5	2	2,5	3	2	2,5	3	3,5	2,5	3	3,5	4

Variance Estimation



1 2 3 4

$$\mu = 2.5$$

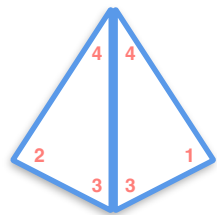
$n = 2$
Samples

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{5}{4}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
$\bar{x}$	1	1,5	2	2,5	1,5	2	2,5	3	2	2,5	3	3,5	2,5	3	3,5	4

Variance Estimation



$$\mu = 2.5$$

$n = 2$
Samples

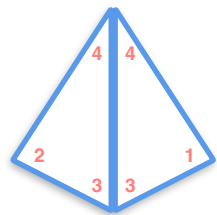
$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\sigma^2 = \frac{5}{4}$$

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
$\bar{x}$	1	1,5	2	2,5	1,5	2	2,5	3	2	2,5	3	3,5	2,5	3	3,5	4

Variance Estimation



$$\mu = 2.5$$

$n = 2$
Samples

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

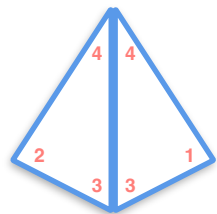
$$\sigma^2 = \frac{5}{4}$$

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
$\bar{x}$	1	1,5	2	2,5	1,5	2	2,5	3	2	2,5	3	3,5	2,5	3	3,5	4

$var(x)$

Variance Estimation



$$\mu = 2.5$$

$n = 2$
Samples

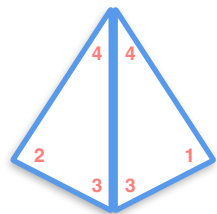
$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

$$\sigma^2 = \frac{5}{4}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
$\bar{x}$	1	1,5	2	2,5	1,5	2	2,5	3	2	2,5	3	3,5	2,5	3	3,5	4
$var(x)$	0	0,25	1	2,25	0,25	0	0,25	1	1	0,25	0	0,25	2,25	1	0,25	0

Variance Estimation



$$\mu = 2.5$$

$n = 2$
Samples

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

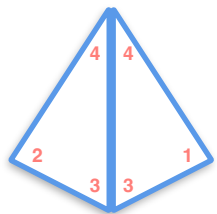
$$\sigma^2 = \frac{5}{4}$$

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

$$var(x) = \frac{5}{8}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
$\bar{x}$	1	1,5	2	2,5	1,5	2	2,5	3	2	2,5	3	3,5	2,5	3	3,5	4
$var(x)$	0	0,25	1	2,25	0,25	0	0,25	1	1	0,25	0	0,25	2,25	1	0,25	0

Variance Estimation



1 2 3 4

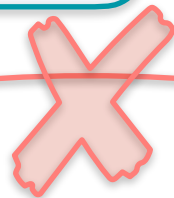
$$\mu = 2.5$$

$n = 2$
Samples

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$\text{Var}(x) = \frac{\sum (x - \bar{x})^2}{n}$$

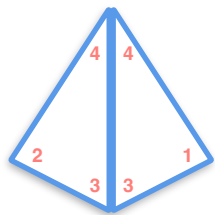
$$\sigma^2 = \frac{5}{4}$$



$$\text{var}(x) = \frac{5}{8}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
$\bar{x}$	1	1,5	2	2,5	1,5	2	2,5	3	2	2,5	3	3,5	2,5	3	3,5	4
$\text{var}(x)$	0	0,25	1	2,25	0,25	0	0,25	1	1	0,25	0	0,25	2,25	1	0,25	0

Variance Estimation



$$\mu = 2.5$$

$n = 2$
Samples

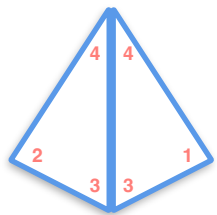
$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

$$\sigma^2 = \frac{5}{4}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
$\bar{x}$	1	1,5	2	2,5	1,5	2	2,5	3	2	2,5	3	3,5	2,5	3	3,5	4

Variance Estimation



$$\mu = 2.5$$

$n = 2$
Samples

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

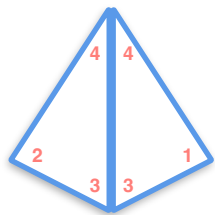
$$\sigma^2 = \frac{5}{4}$$

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n}$$

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n - 1}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
$\bar{x}$	1	1,5	2	2,5	1,5	2	2,5	3	2	2,5	3	3,5	2,5	3	3,5	4

Variance Estimation



$$\mu = 2.5$$

$n = 2$
Samples

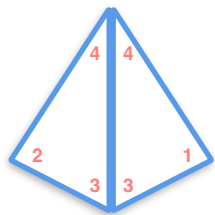
$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$\sigma^2 = \frac{5}{4}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
$\bar{x}$	1	1,5	2	2,5	1,5	2	2,5	3	2	2,5	3	3,5	2,5	3	3,5	4

Variance Estimation



$$\mu = 2.5$$

$n = 2$
Samples

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

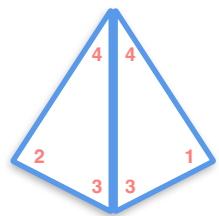
$$Var(x) = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$\sigma^2 = \frac{5}{4}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
$\bar{x}$	1	1,5	2	2,5	1,5	2	2,5	3	2	2,5	3	3,5	2,5	3	3,5	4

$var(x)$

Variance Estimation



$$\mu = 2.5$$

$n = 2$
Samples

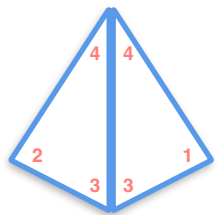
$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$\sigma^2 = \frac{5}{4}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
$\bar{x}$	1	1,5	2	2,5	1,5	2	2,5	3	2	2,5	3	3,5	2,5	3	3,5	4
$var(x)$	0	0,5	2	4,5	0,5	0	0,5	2	2	0,5	0	0,5	4,5	2	0,5	0

Variance Estimation



1 2 3 4

$$\mu = 2.5$$

$n = 2$
Samples

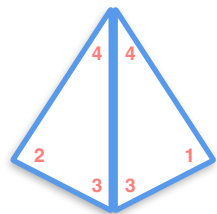
$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$\sigma^2 = \frac{5}{4}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
$\bar{x}$	1	1,5	2	2,5	1,5	2	2,5	3	2	2,5	3	3,5	2,5	3	3,5	4
$var(x)$	0	0,5	2	4,5	0,5	0	0,5	2	2	0,5	0	0,5	4,5	2	0,5	0

Variance Estimation



$$\mu = 2.5$$

$n = 2$
Samples

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

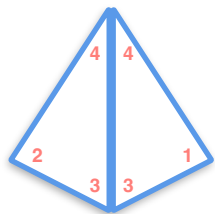
$$Var(x) = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$\sigma^2 = \frac{5}{4}$$

$$Var(x) = \frac{5}{4}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
$\bar{x}$	1	1,5	2	2,5	1,5	2	2,5	3	2	2,5	3	3,5	2,5	3	3,5	4
$var(x)$	0	0,5	2	4,5	0,5	0	0,5	2	2	0,5	0	0,5	4,5	2	0,5	0

Variance Estimation



1 2 3 4

$$\mu = 2.5$$

$n = 2$
Samples

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2 \quad \text{Var}(x) = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$\sigma^2 = \frac{5}{4}$$

$$\text{Var}(x) = \frac{5}{4}$$

x	1,1	1,2	1,3	1,4	2,1	2,2	2,3	2,4	3,1	3,2	3,3	3,4	4,1	4,2	4,3	4,4
$\bar{x}$	1	1,5	2	2,5	1,5	2	2,5	3	2	2,5	3	3,5	2,5	3	3,5	4
$\text{var}(x)$	0	0,5	2	4,5	0,5	0	0,5	2	2	0,5	0	0,5	4,5	2	0,5	0

Variance Estimation

Population Variance Formula

$$\sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

Sample Variance Formula

$$Var(x) = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$



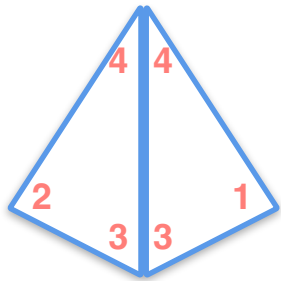
DeepLearning.AI

Sample and Population

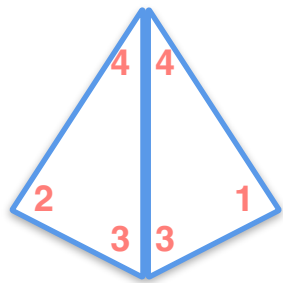
Law of Large Numbers

Law of Large Numbers

Law of Large Numbers

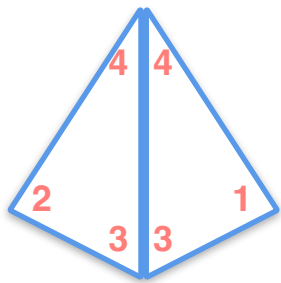


Law of Large Numbers



1 2 3 4

Law of Large Numbers



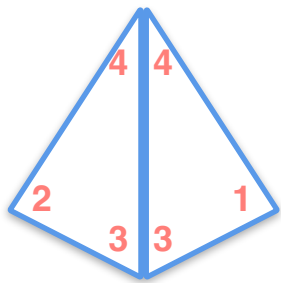
1 2 3 4

$$\mu = 2.5$$

Experiment:

Toss the 4-sided dice twice and record the average of your outcomes

Law of Large Numbers



1 2 3 4

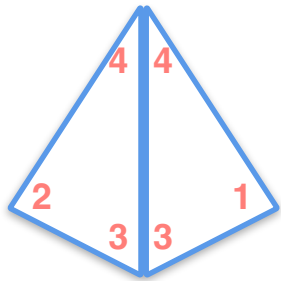
$$\mu = 2.5$$

Experiment:

Toss the 4-sided dice twice and record the average of your outcomes

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

Law of Large Numbers



1 2 3 4

$$\mu = 2.5$$

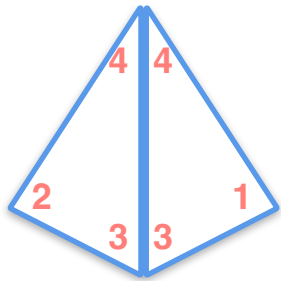
Experiment:

Toss the 4-sided dice twice and record the average of your outcomes

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

	1	2	3	4
1	1	1.5	2	2.5
2	1.5	2	2.5	3
3	2	2.5	3	3.5
4	2.5	3	3.5	4

Law of Large Numbers



1 2 3 4

$$\mu = 2.5$$

Experiment:

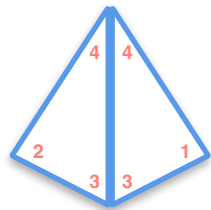
Toss the 4-sided dice twice and record the average of your outcomes

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

	1	2	3	4
1	1	1.5	2	2.5
2	1.5	2	2.5	3
3	2	2.5	3	3.5
4	2.5	3	3.5	4

	1	2	3	4
1	μ 2.5			
2				
3				
4				

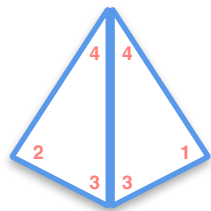
Law of Large Numbers



$$\begin{matrix} 1 & 2 & 3 & 4 \\ \mu = 2.5 \end{matrix}$$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

Law of Large Numbers

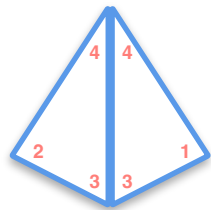


1 2 3 4
 $\mu = 2.5$

1 trial

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

Law of Large Numbers



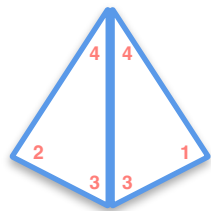
1 2 3 4
 $\mu = 2.5$

1 trial

4,3

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

Law of Large Numbers

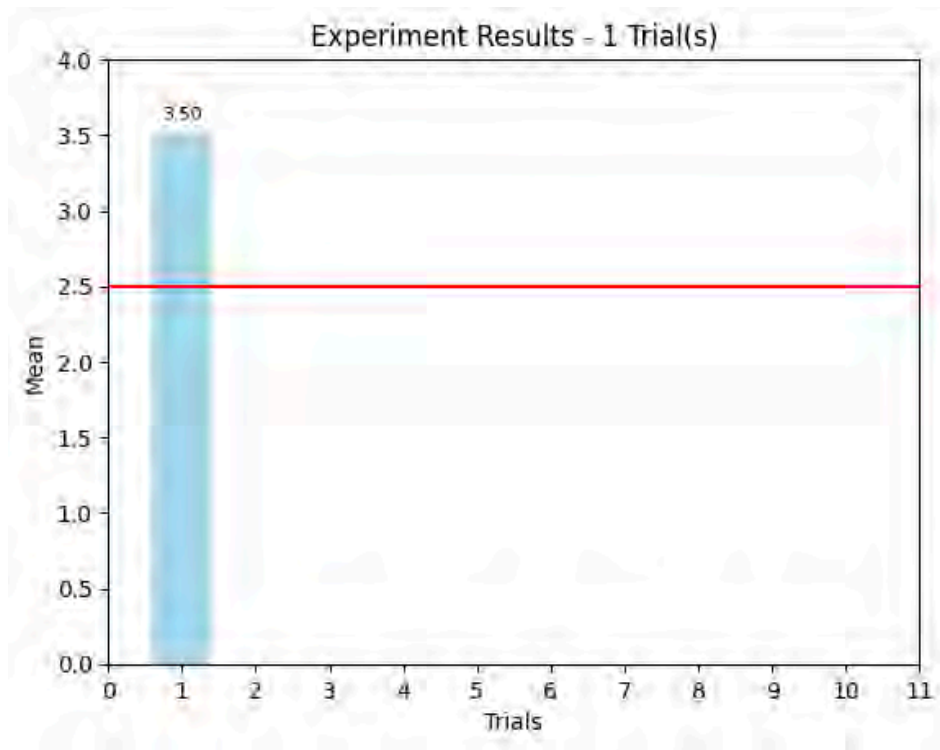


1 2 3 4
 $\mu = 2.5$

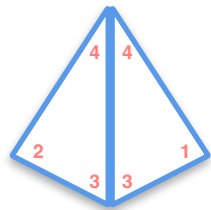
1 trial

4,3

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4



Law of Large Numbers

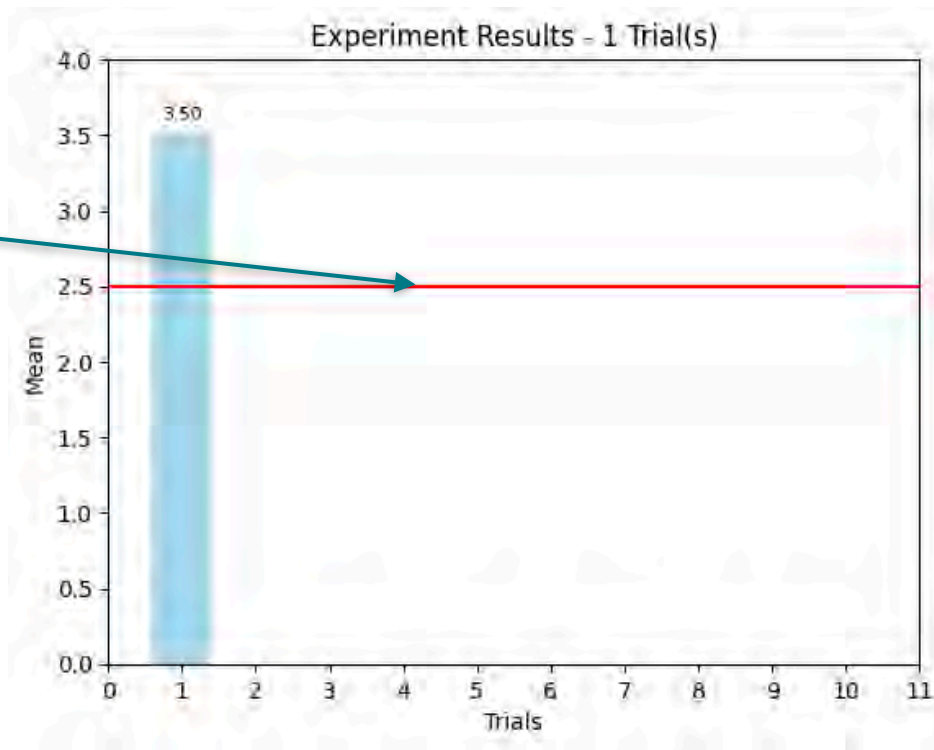


1 2 3 4
 $\mu = 2.5$

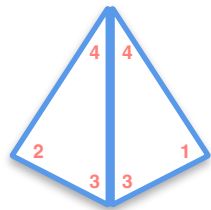
1 trial

4,3

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4



Law of Large Numbers



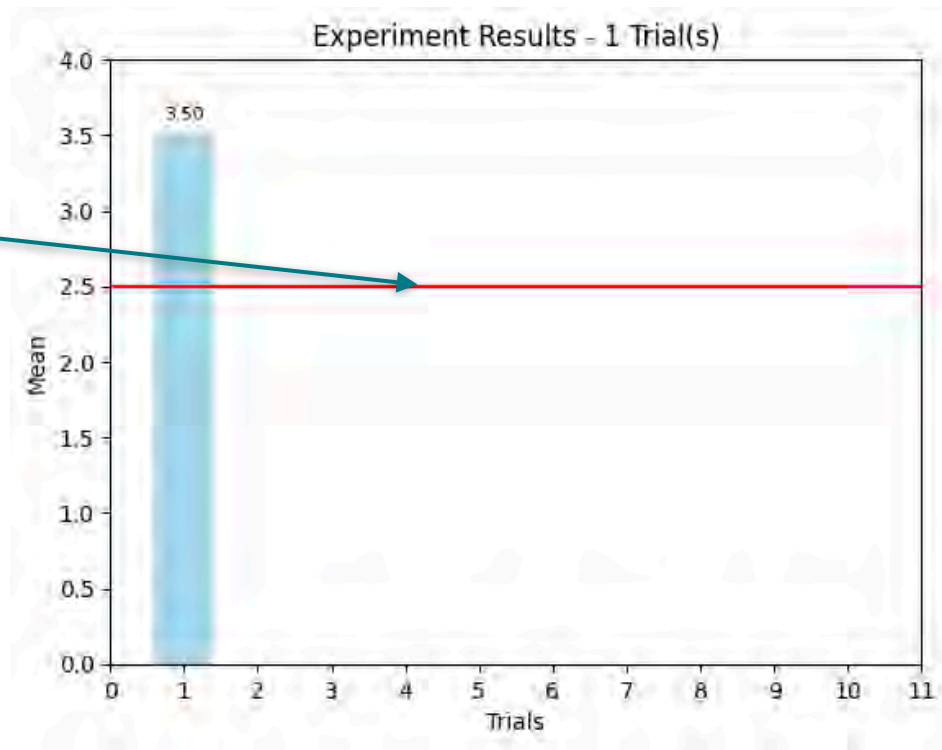
1 2 3 4
 $\mu = 2.5$

1 trial

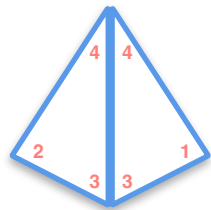
4,3

$\bar{x}_1$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4



Law of Large Numbers



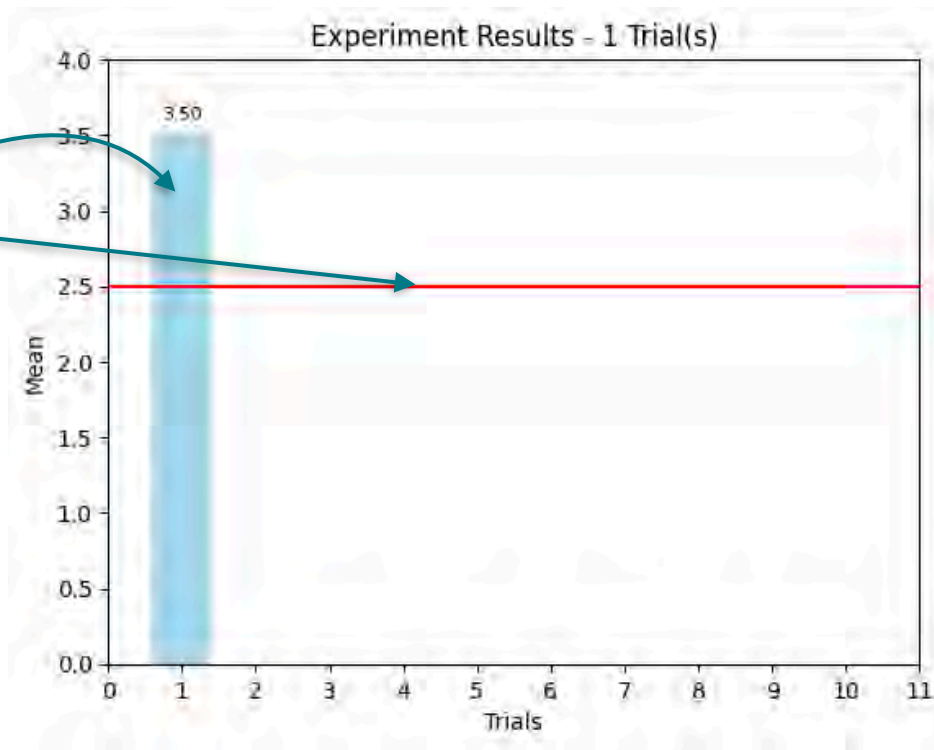
1 2 3 4
 $\mu = 2.5$

1 trial

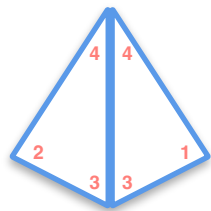
4,3

$\bar{x}_1$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4



Law of Large Numbers



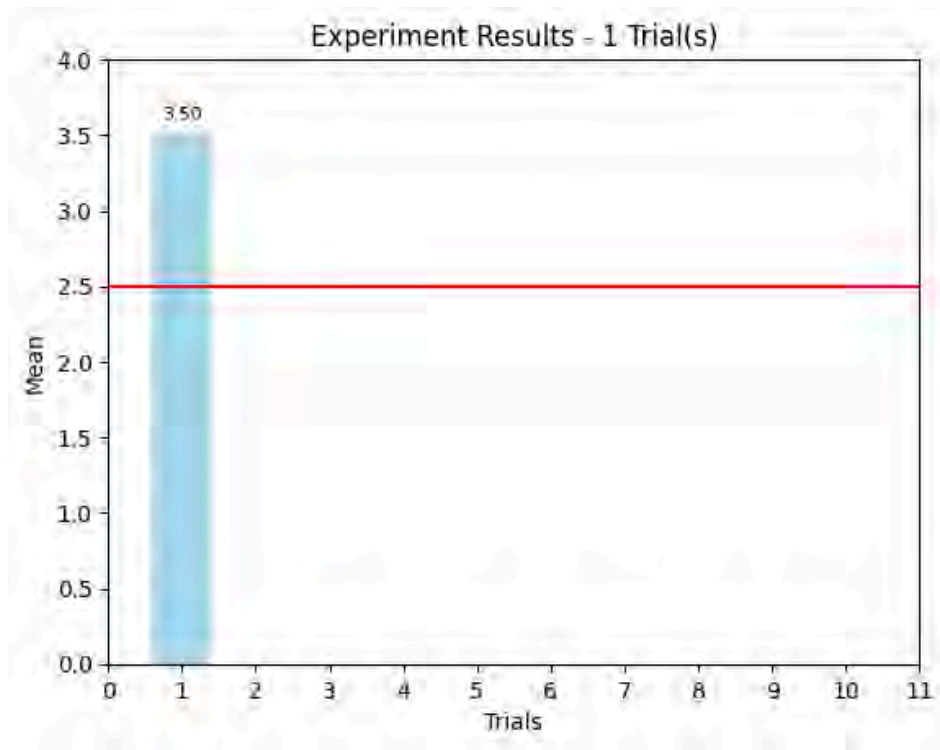
1 2 3 4
 $\mu = 2.5$

1 trial

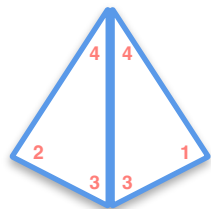
4,3

2 trials

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4



Law of Large Numbers



1 2 3 4
 $\mu = 2.5$

1 trial

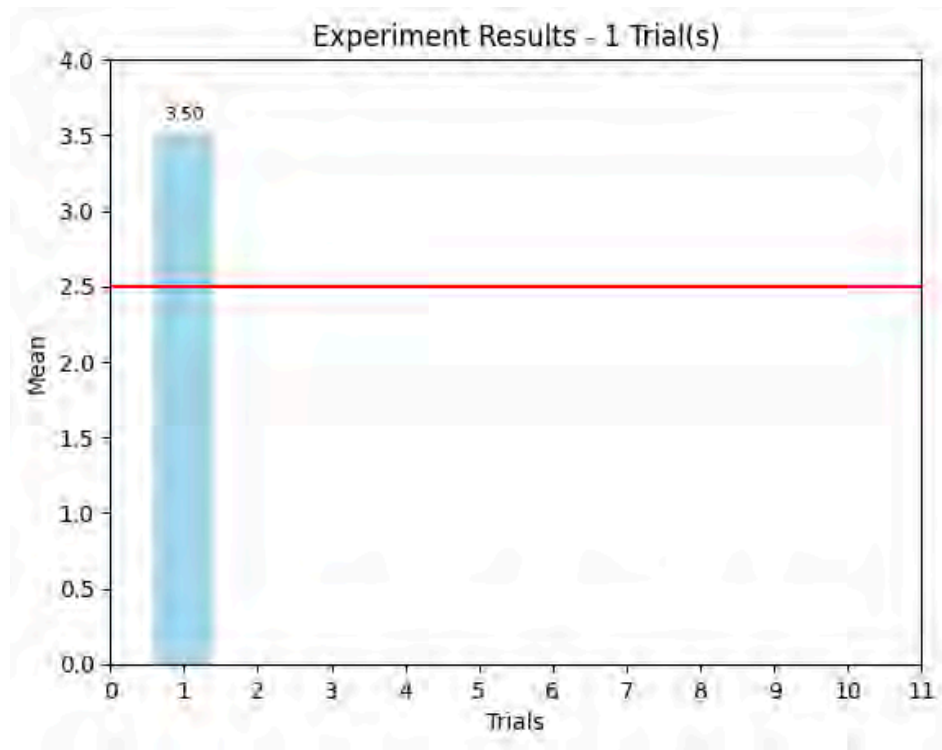
4,3

2 trials

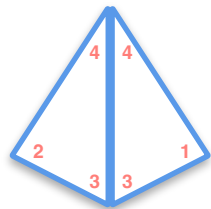
3,4

1,3

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4



Law of Large Numbers



1 2 3 4
 $\mu = 2.5$

1 trial

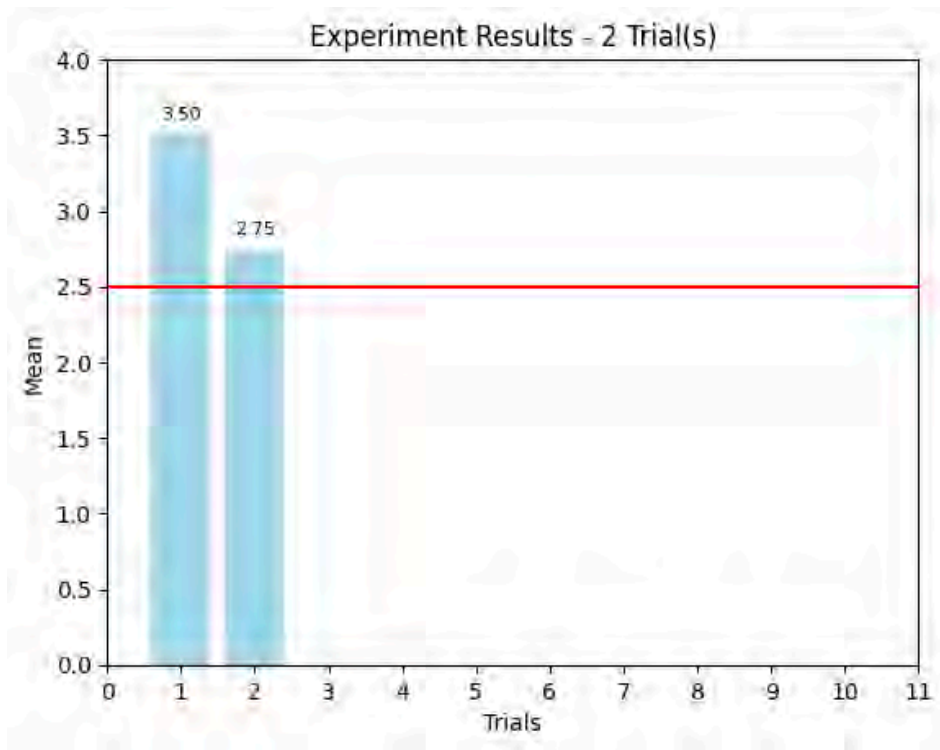
4,3

2 trials

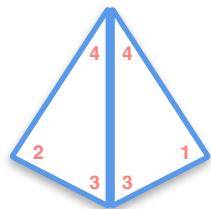
3,4

1,3

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4



Law of Large Numbers



1 2 3 4
 $\mu = 2.5$

1 trial

4,3

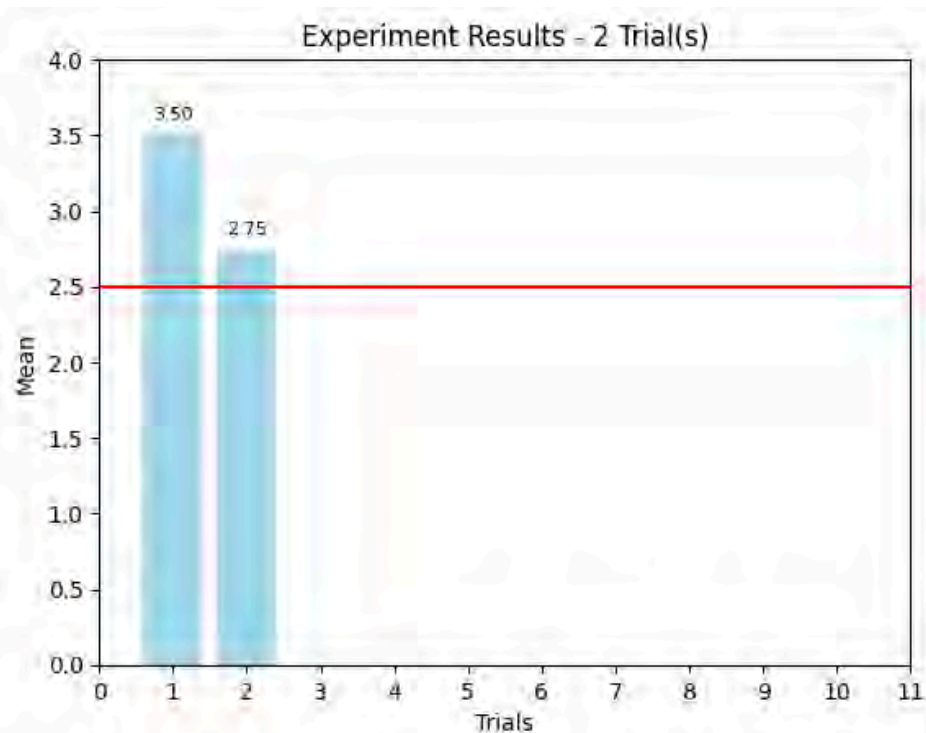
2 trials

3,4

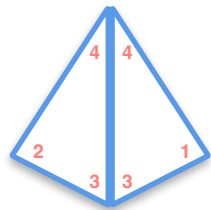
1,3

3 trials

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4



Law of Large Numbers



1 2 3 4
 $\mu = 2.5$

1 trial

4,3

2 trials

3,4

1,3

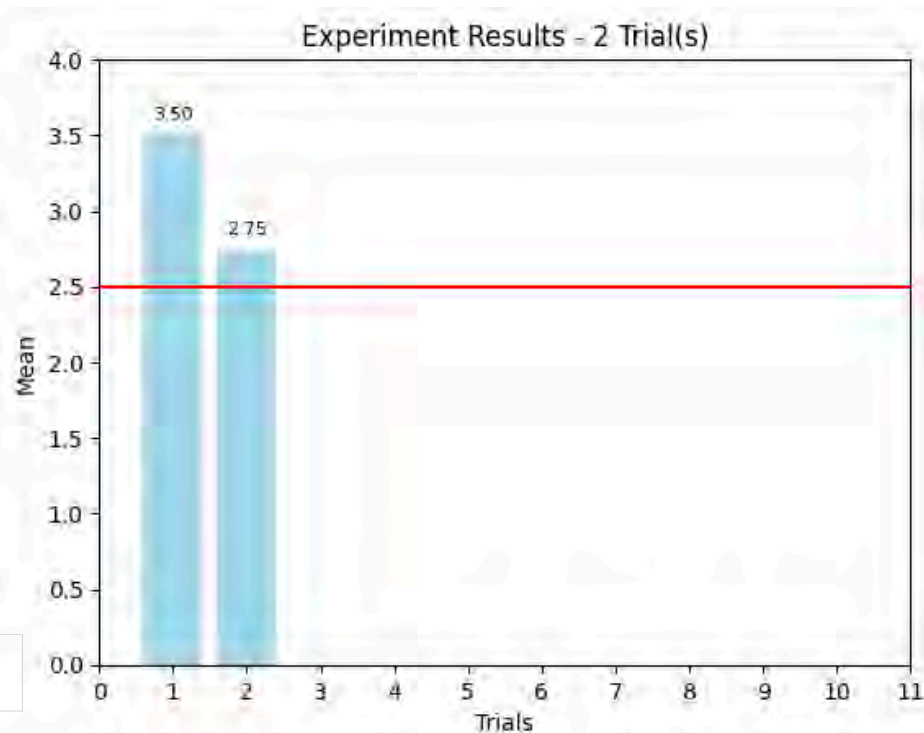
3 trials

3,1

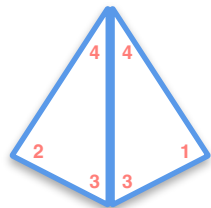
1,4

1,1

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4



Law of Large Numbers



1 2 3 4
 $\mu = 2.5$

1 trial

4,3

2 trials

3,4

1,3

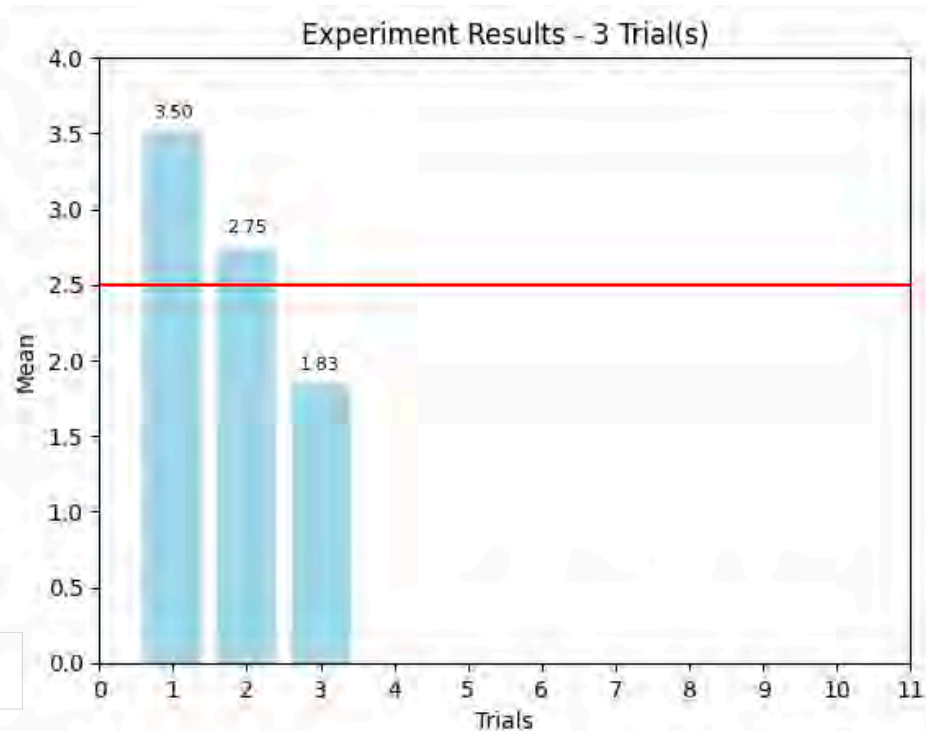
3 trials

3,1

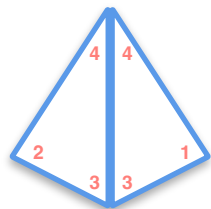
1,4

1,1

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4



Law of Large Numbers



1 2 3 4
 $\mu = 2.5$

2 trials

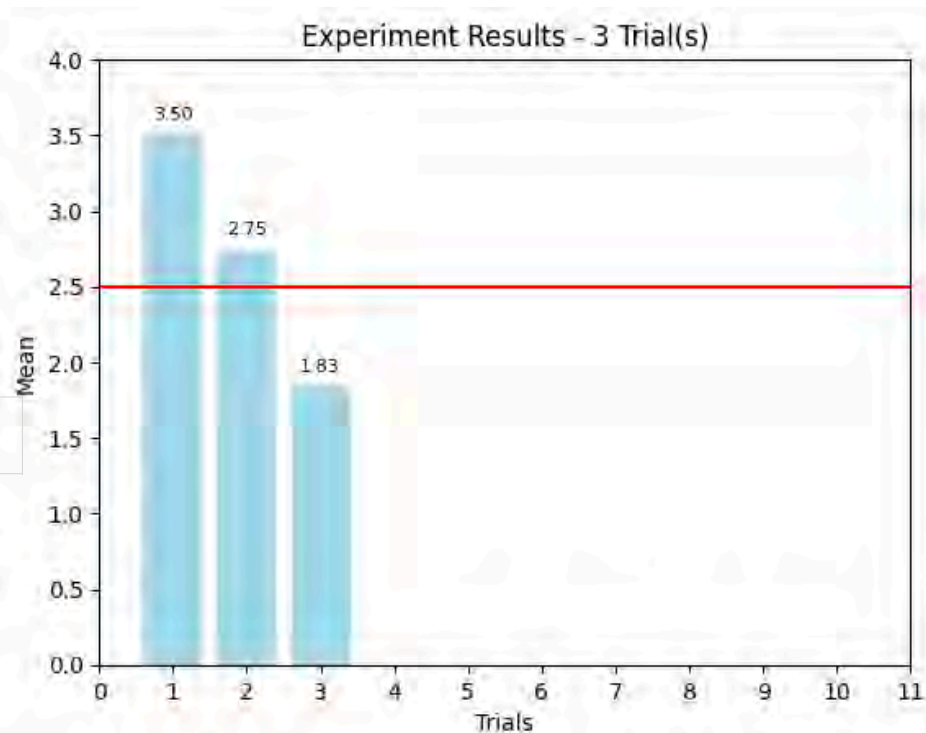
3,4
1,3

3 trials

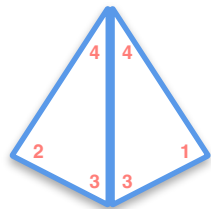
3,1	1,4	1,1
-----	-----	-----

4 trials

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4



Law of Large Numbers



1 2 3 4
 $\mu = 2.5$

2 trials

3,4
1,3

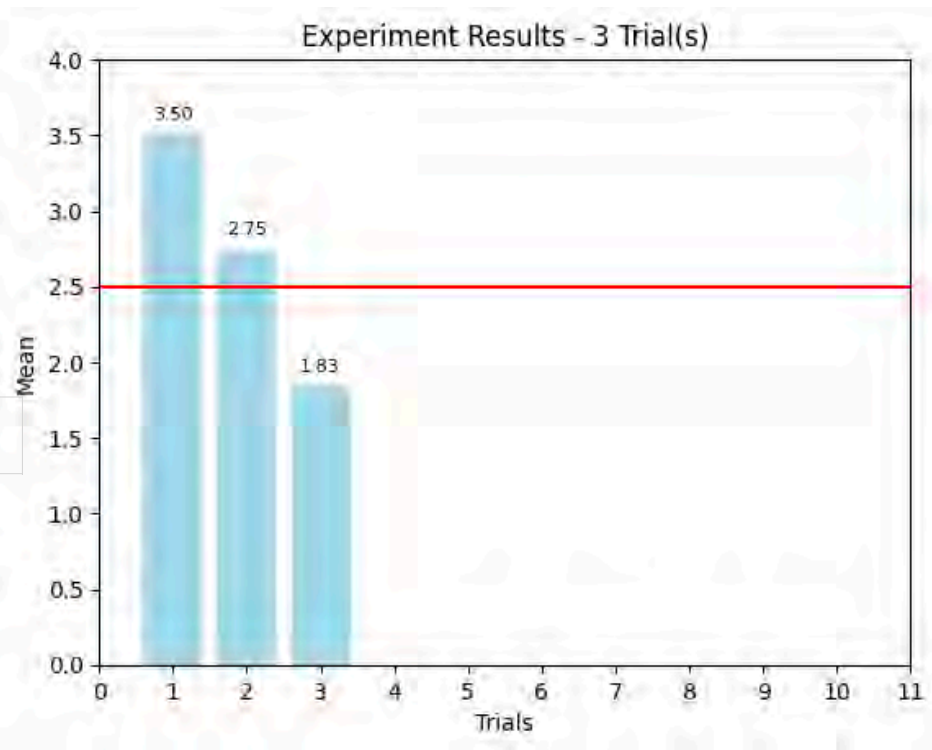
3 trials

3,1	1,4	1,1
-----	-----	-----

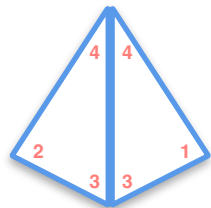
4 trials

3,1	3,1
1,2	3,2

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4



Law of Large Numbers



$$\mu = 2.5$$

2 trials

3,4
1,3

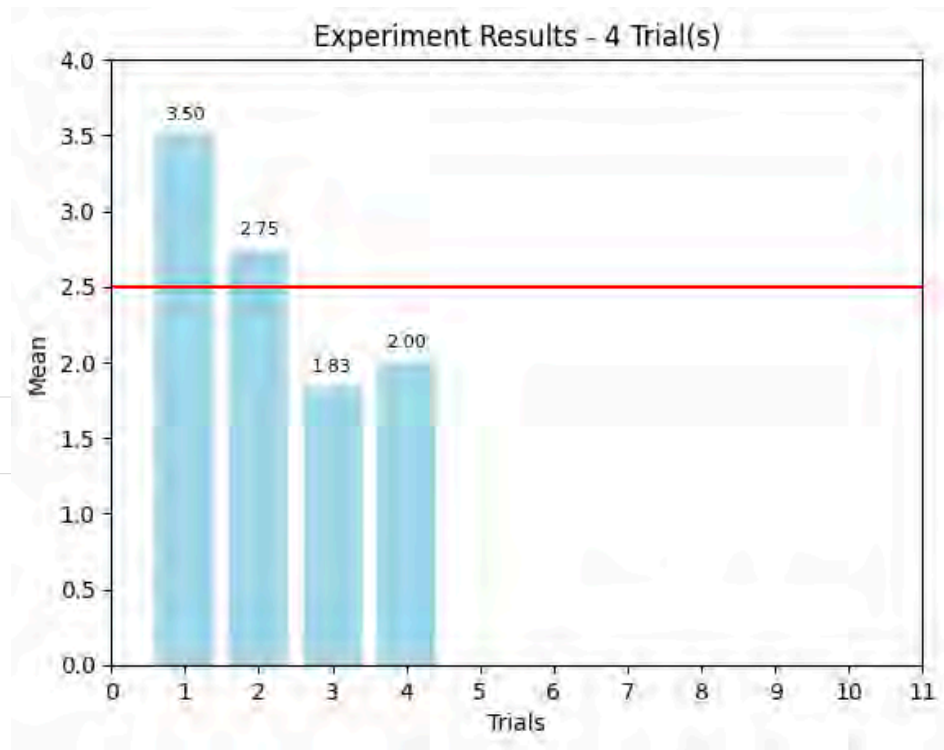
3 trials

3,1	1,4	1,1
-----	-----	-----

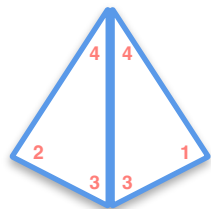
4 trials

3,1	3,1
1,2	3,2

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

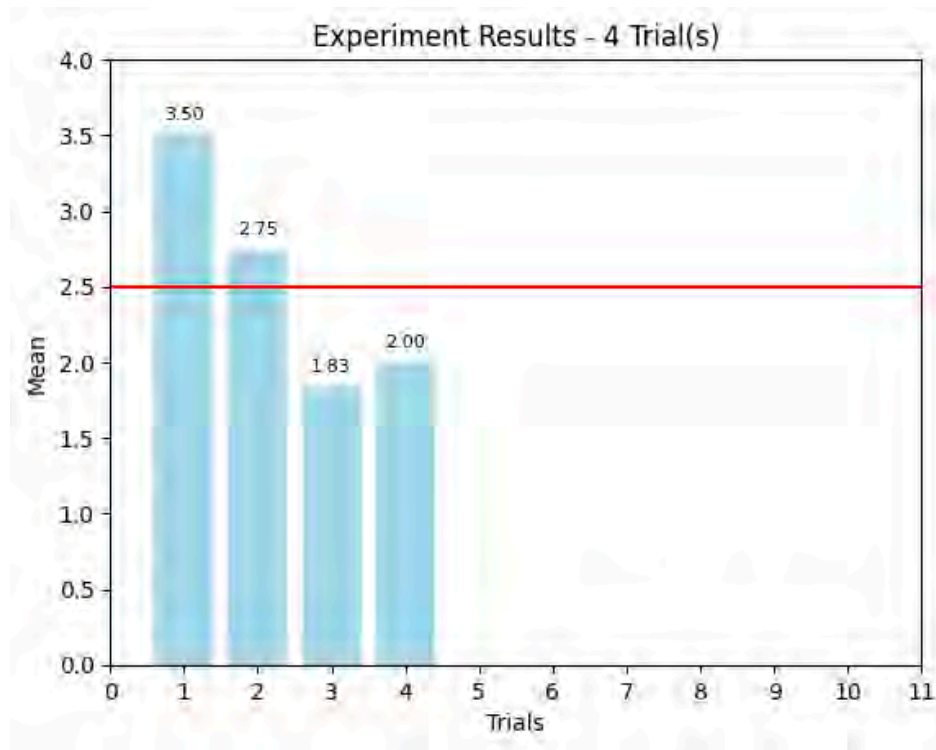


Law of Large Numbers

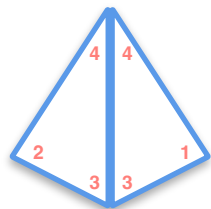


$$\begin{matrix} 1 & 2 & 3 & 4 \\ \mu = 2.5 \end{matrix}$$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

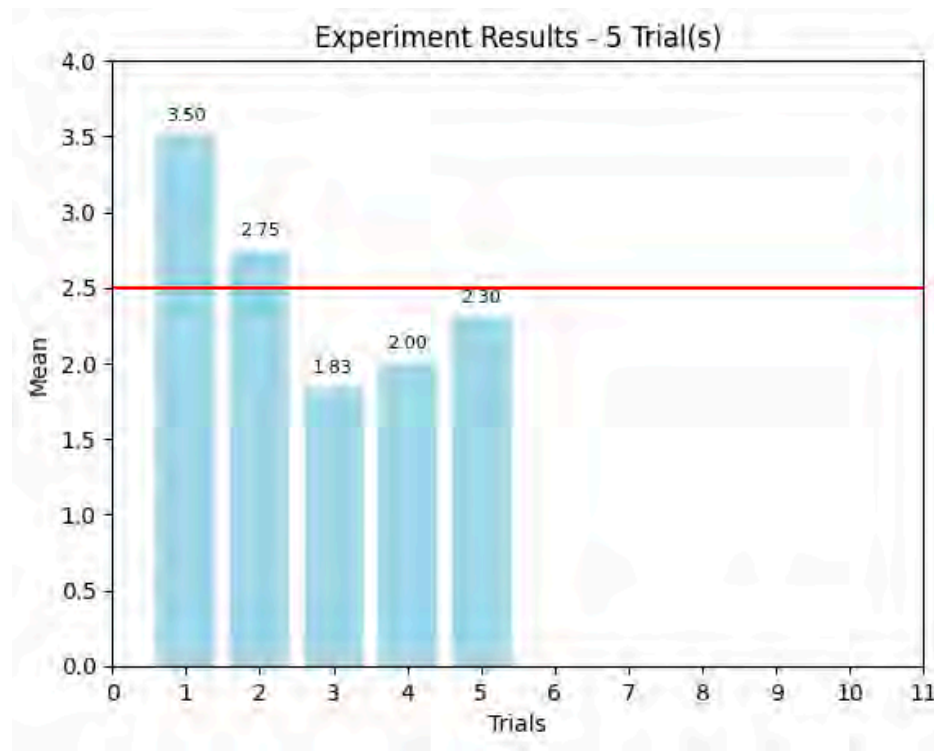


Law of Large Numbers

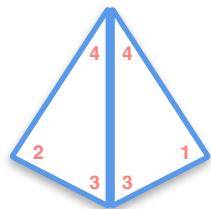


$$\begin{matrix} 1 & 2 & 3 & 4 \\ \mu = 2.5 \end{matrix}$$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

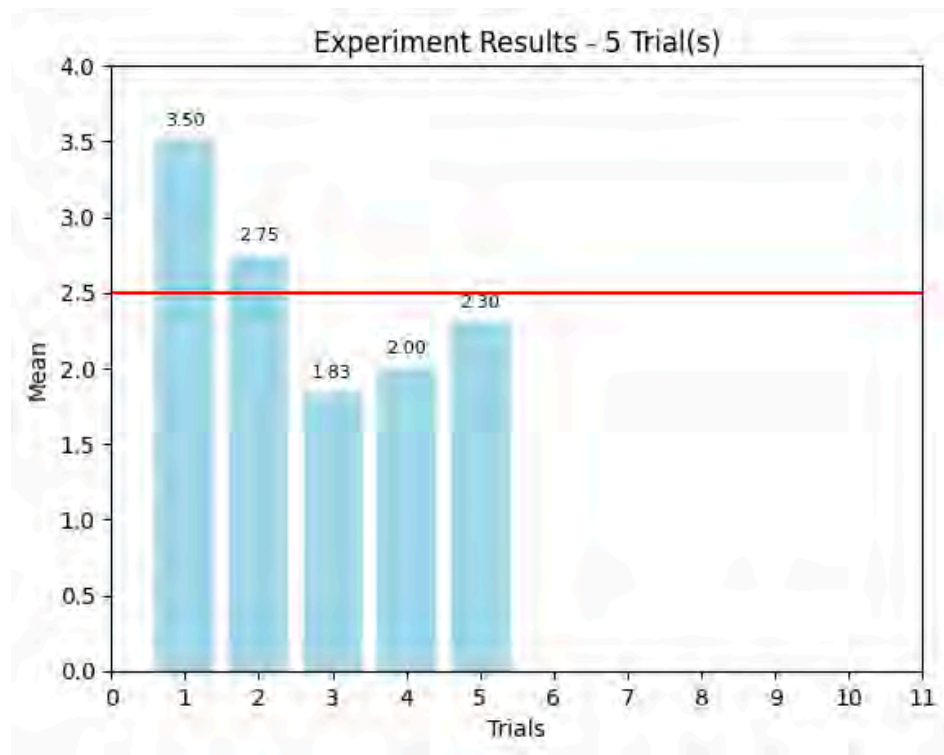


Law of Large Numbers

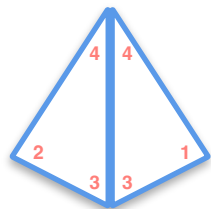


$$\mu = 2.5$$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

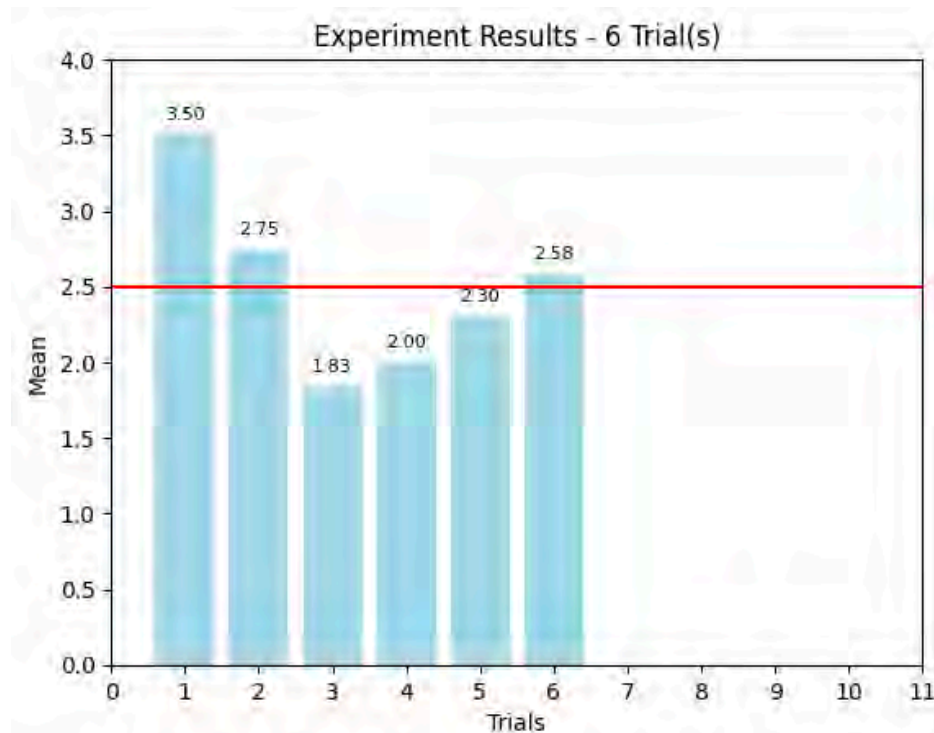


Law of Large Numbers

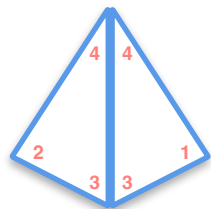


$$\begin{matrix} 1 & 2 & 3 & 4 \\ \mu = 2.5 \end{matrix}$$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

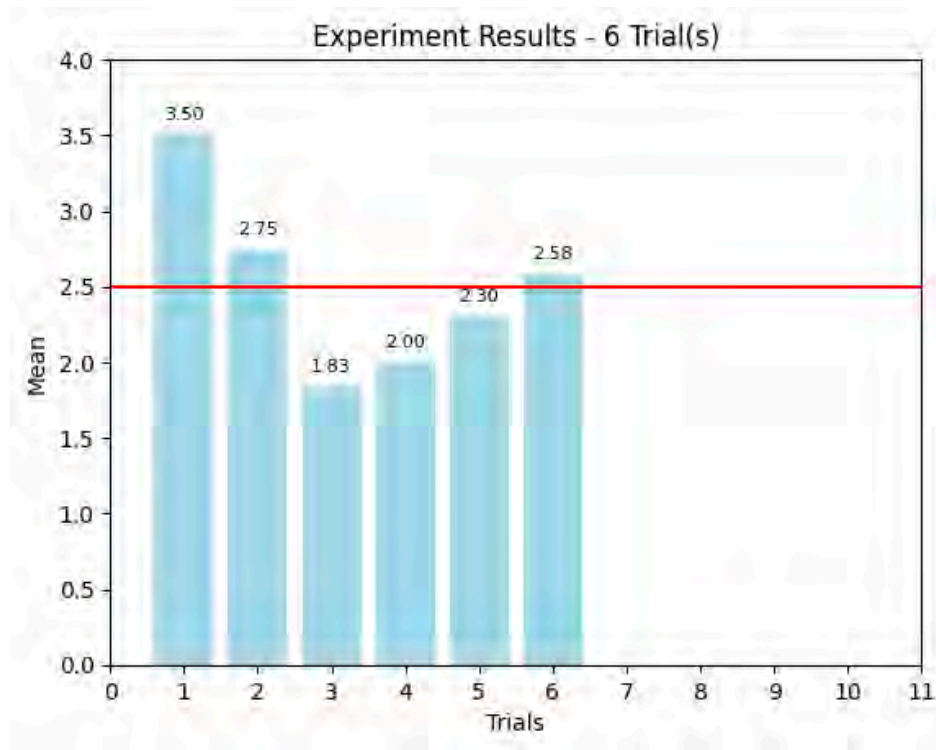


Law of Large Numbers

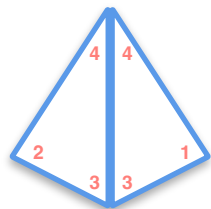


$$\begin{matrix} 1 & 2 & 3 & 4 \\ \mu = 2.5 \end{matrix}$$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

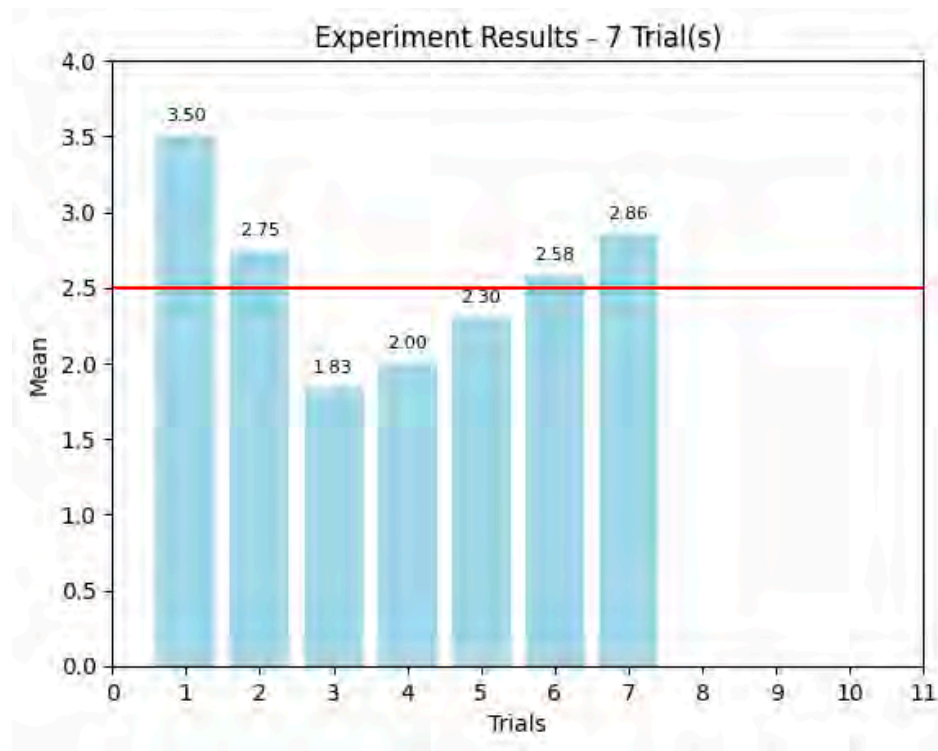


Law of Large Numbers

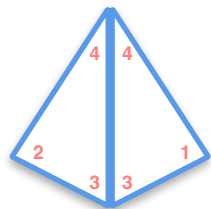


$$\mu = 2.5$$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

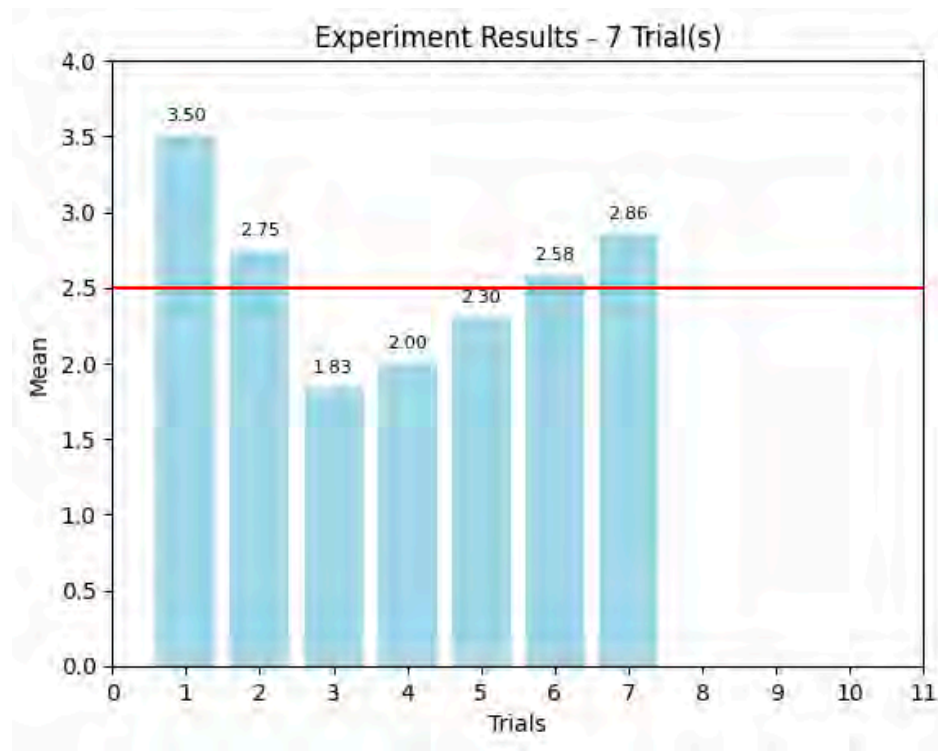


Law of Large Numbers

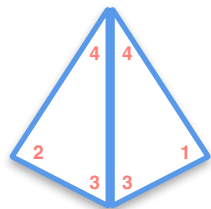


$$\mu = 2.5$$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

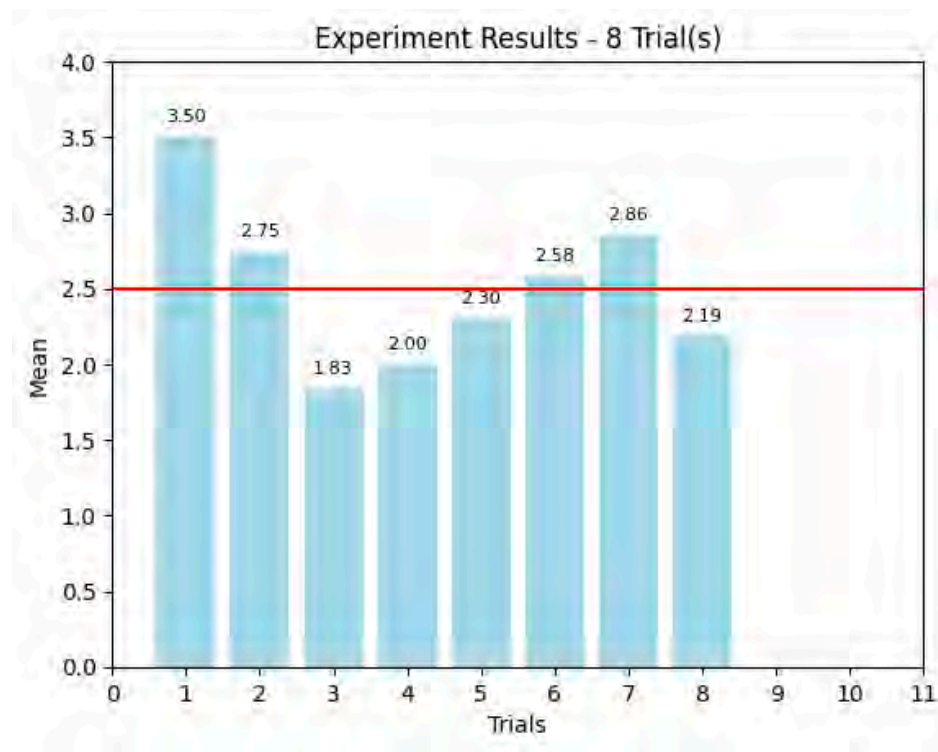


Law of Large Numbers

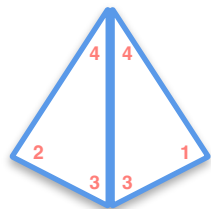


$$\begin{matrix} 1 & 2 & 3 & 4 \\ \mu = 2.5 \end{matrix}$$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

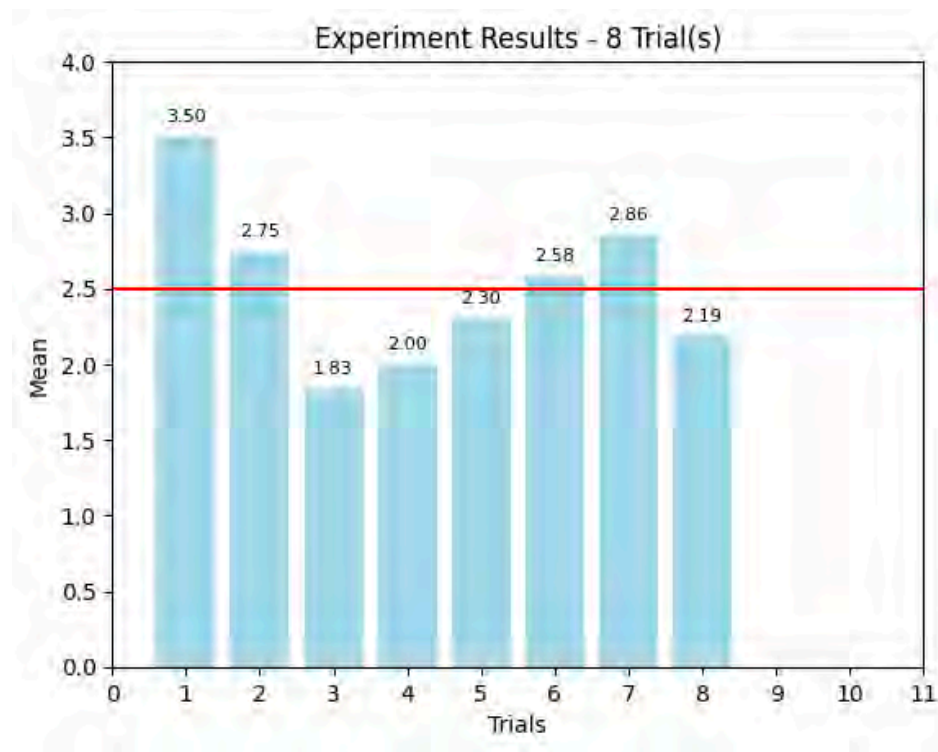


Law of Large Numbers

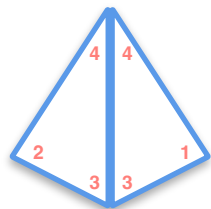


$$\begin{matrix} 1 & 2 & 3 & 4 \\ \mu = 2.5 \end{matrix}$$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

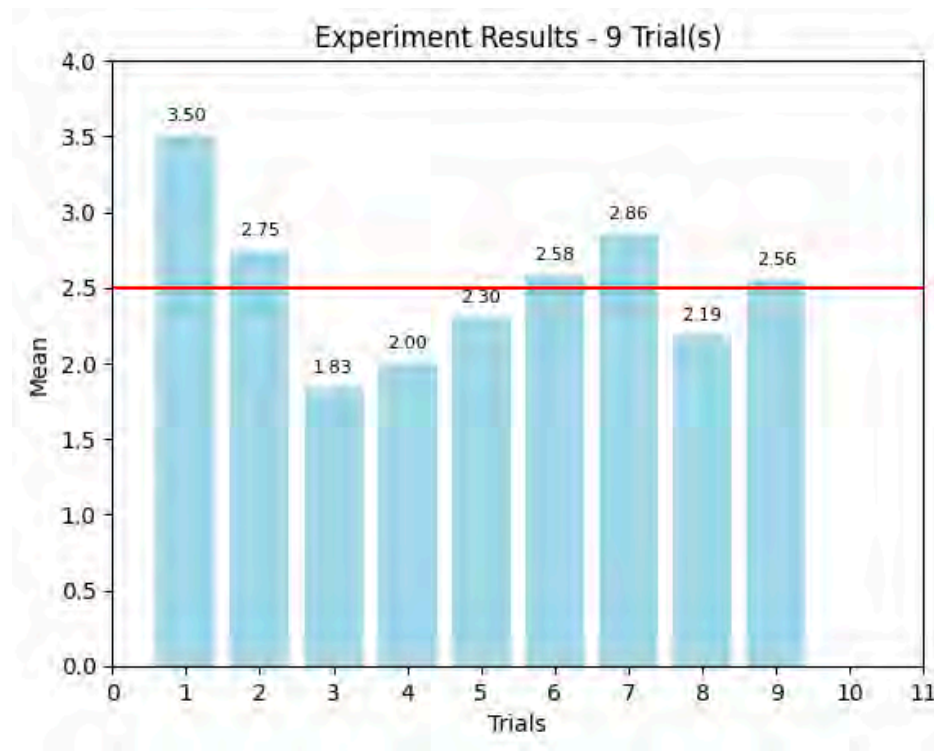


Law of Large Numbers

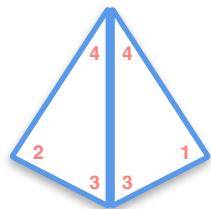


$$\begin{matrix} 1 & 2 & 3 & 4 \\ \mu = 2.5 \end{matrix}$$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

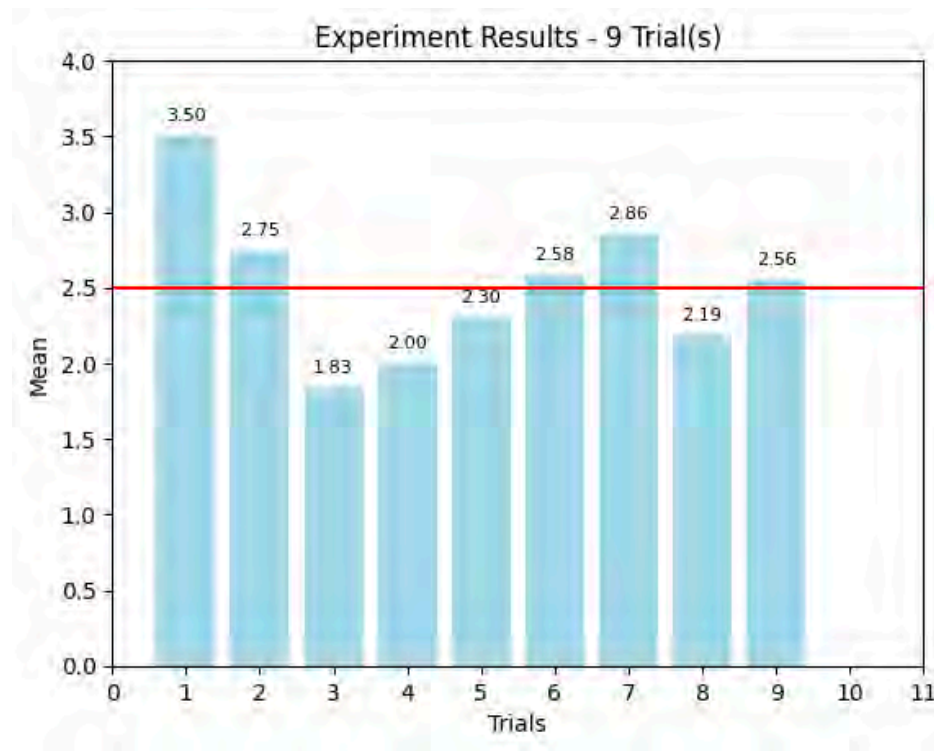


Law of Large Numbers

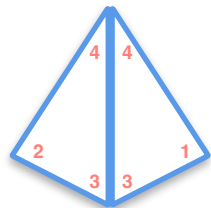


$$\begin{matrix} 1 & 2 & 3 & 4 \\ \mu = 2.5 \end{matrix}$$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

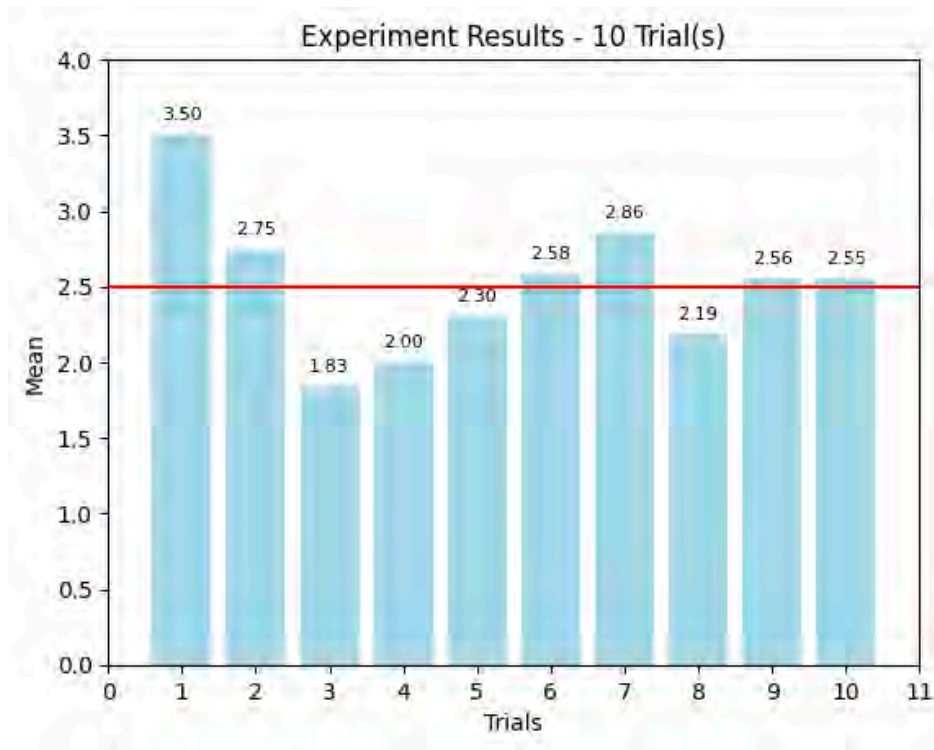


Law of Large Numbers

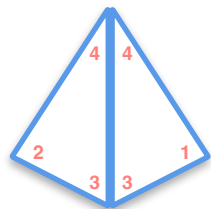


$$\begin{matrix} 1 & 2 & 3 & 4 \\ \mu = 2.5 \end{matrix}$$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4

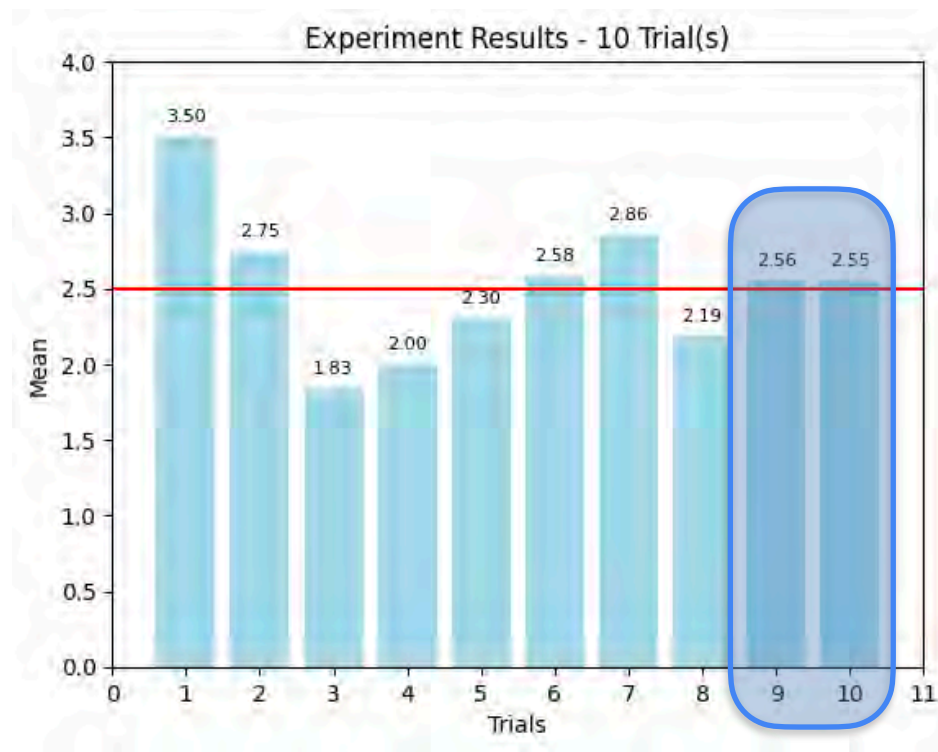


Law of Large Numbers

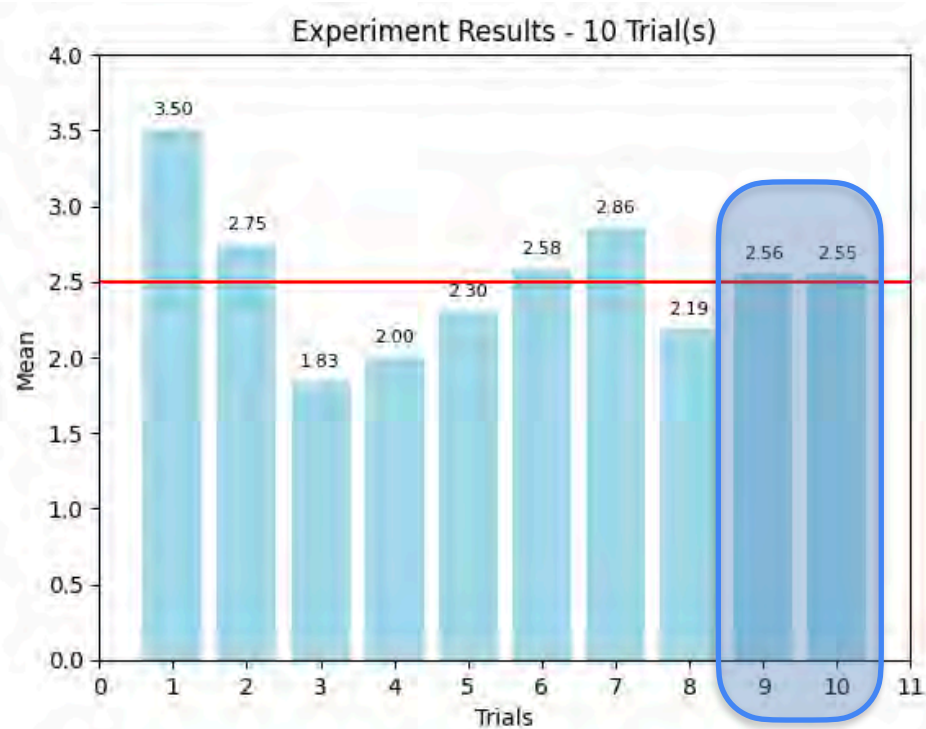


$$\mu = 2.5$$

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,4
3	3,1	3,2	3,3	3,4
4	4,1	4,2	4,3	4,4



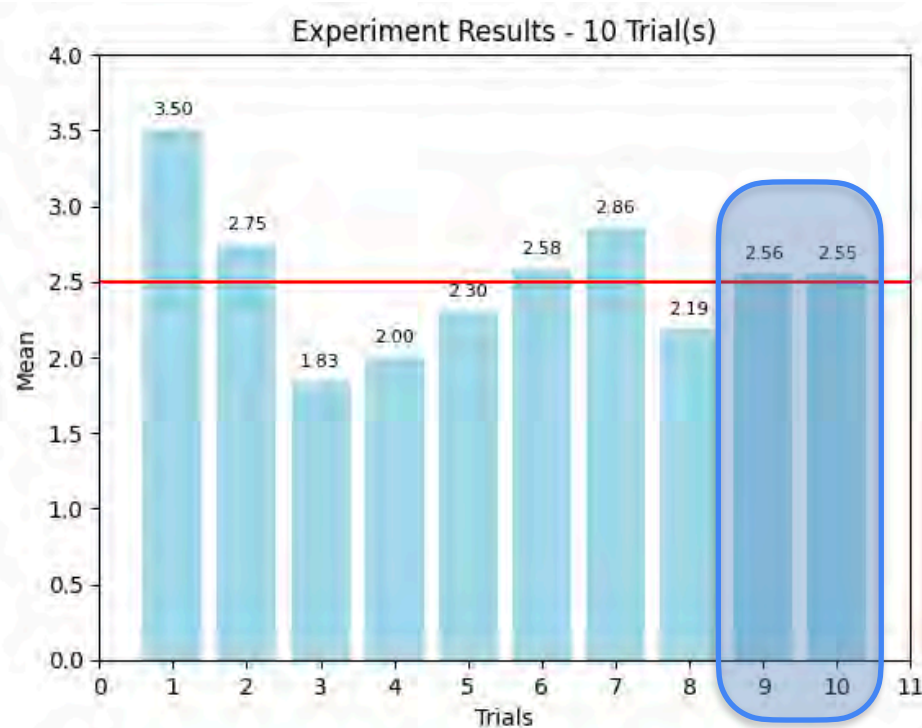
Law of Large Numbers



Law of Large Numbers

As the sample size increases, the average of the sample will tend to get closer to the average of the entire population.

Law of Large Numbers



Law of Large Numbers

Law of Large Numbers

Law of Large Numbers

Law of Large Numbers

n : number of samples

Law of Large Numbers

Law of Large Numbers

n : number of samples

X_i : some estimate X for a sample size i

Law of Large Numbers

Law of Large Numbers

n : number of samples

X_i : some estimate X for a sample size i

as $n \rightarrow \infty$

Law of Large Numbers

Law of Large Numbers

n : number of samples

X_i : some estimate X for a sample size i

as $n \rightarrow \infty$

$$\frac{1}{n} \sum_{i=1}^n X_i \rightarrow \mathbb{E}[X] = \mu_X$$

Law of Large Numbers

as $n \rightarrow \infty$

Law of Large Numbers

n : number of samples

X_i : some estimate X for a sample size i

as $n \rightarrow \infty$

$$\frac{1}{n} \sum_{i=1}^n X_i \rightarrow \mathbb{E}[X] = \mu_X$$

Law of Large Numbers

Law of Large Numbers

n : number of samples

X_i : some estimate X for a sample size i

$$\text{as } n \rightarrow \infty$$
$$\frac{X_1 + X_2 + X_3 + \dots + X_n}{n} \longrightarrow \mathbb{E}[X] = \mu_X$$

UNDER CERTAIN CONDITIONS

$$\text{as } n \rightarrow \infty \quad \frac{1}{n} \sum_{i=1}^n X_i \rightarrow \mathbb{E}[X] = \mu_X$$

Law of Large Numbers

UNDER CERTAIN CONDITIONS

Law of Large Numbers

UNDER CERTAIN CONDITIONS

- Sample is randomly drawn.

Law of Large Numbers

UNDER CERTAIN CONDITIONS

- Sample is randomly drawn.

Law of Large Numbers

UNDER CERTAIN CONDITIONS

- Sample is randomly drawn.
- Sample size must be sufficiently large.

Law of Large Numbers

UNDER CERTAIN CONDITIONS

- Sample is randomly drawn.
- Sample size must be sufficiently large.

Law of Large Numbers

UNDER CERTAIN CONDITIONS

- Sample is randomly drawn.
- Sample size must be sufficiently large.
- Independent observations.



DeepLearning.AI

Sample and Population

Central Limit Theorem

Central Limit Theorem (CLT) - Example 1

If $n = 1$

Central Limit Theorem (CLT) - Example 1



If $n = 1$

Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5$$



$$P(T) = 0.5$$

If $n = 1$

Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5$$



$$P(T) = 0.5$$

Random
variable $\rightarrow X$

If $n = 1$

Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5$$

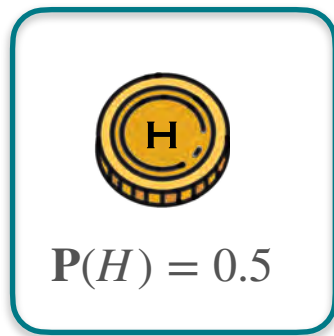


$$P(T) = 0.5$$

Random variable $\rightarrow X$ number of heads when a coin is flipped n times

If $n = 1$

Central Limit Theorem (CLT) - Example 1



$$P(T) = 0.5$$

Random variable $\rightarrow X$ number of heads when a coin is flipped n times

$$\text{If } n = 1 \quad X = 1$$

Central Limit Theorem (CLT) - Example 1



Random variable $\rightarrow X$ number of heads when a coin is flipped n times

If $n = 1$

$X = 1$

$X = 0$

Central Limit Theorem (CLT) - Example 1



Random variable $\rightarrow X$ number of heads when a coin is flipped n times

If $n = 1$ $X = 1$ $X = 0$

Discrete Random Variable

Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5$$

$$X = 1$$



$$P(T) = 0.5$$

$$X = 0$$

Central Limit Theorem (CLT) - Example 1



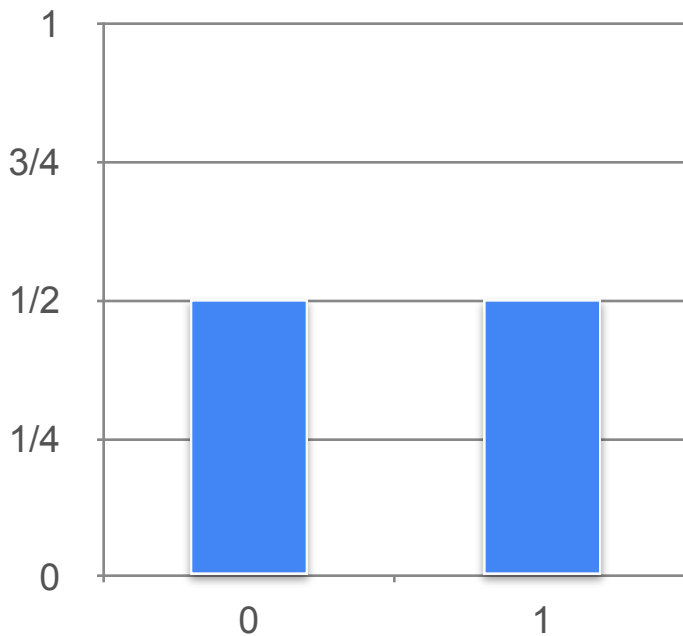
$$P(H) = 0.5$$

$$X = 1$$



$$P(T) = 0.5$$

$$X = 0$$



Central Limit Theorem (CLT) - Example 1



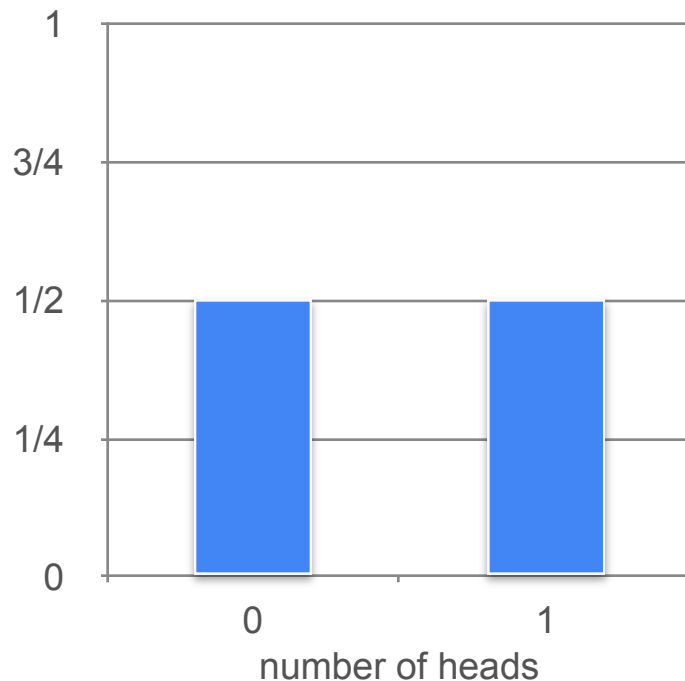
$$P(H) = 0.5$$

$$X = 1$$



$$P(T) = 0.5$$

$$X = 0$$



Central Limit Theorem (CLT) - Example 1



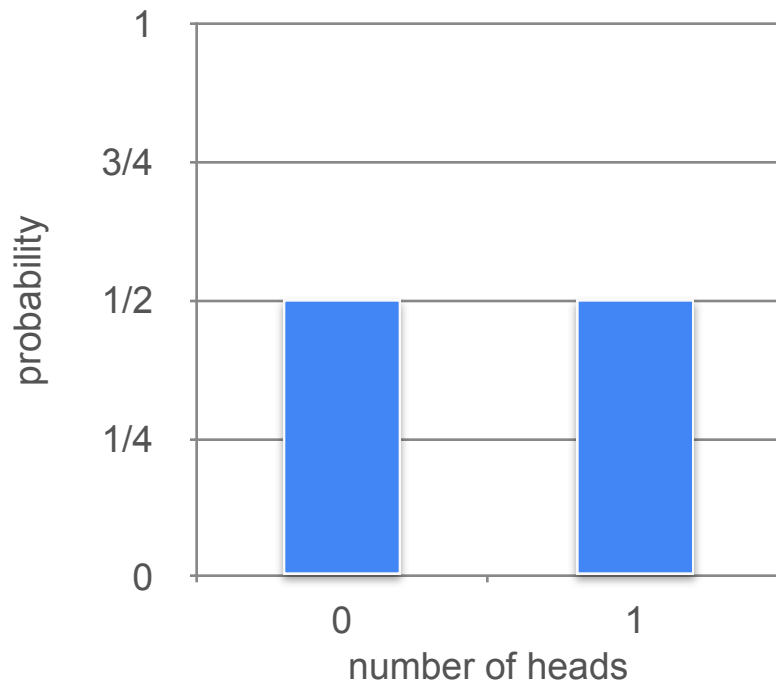
$$P(H) = 0.5$$

$$X = 1$$



$$P(T) = 0.5$$

$$X = 0$$



Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5$$

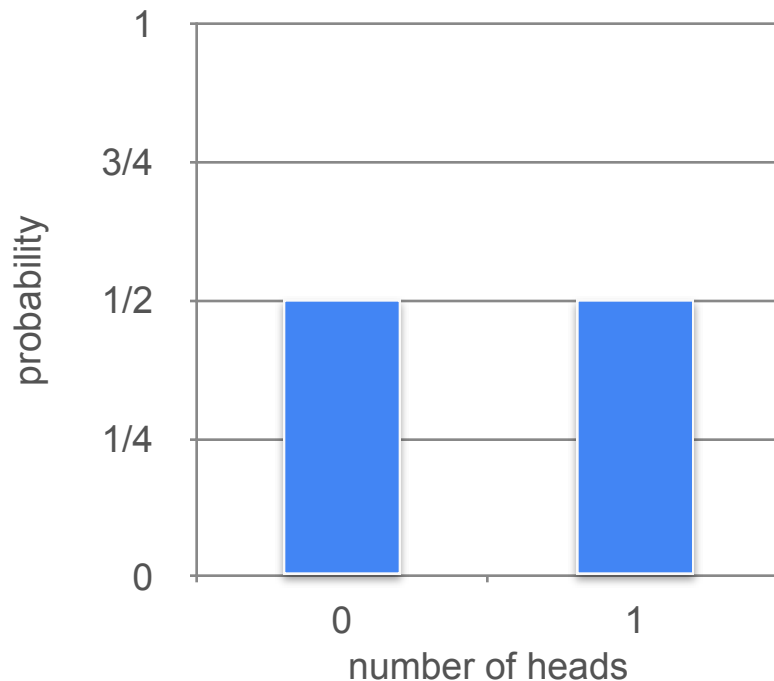
$$X = 1$$



$$P(T) = 0.5$$

$$X = 0$$

What can we say about the probability distribution when the number of coin flips increases?



Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5$$

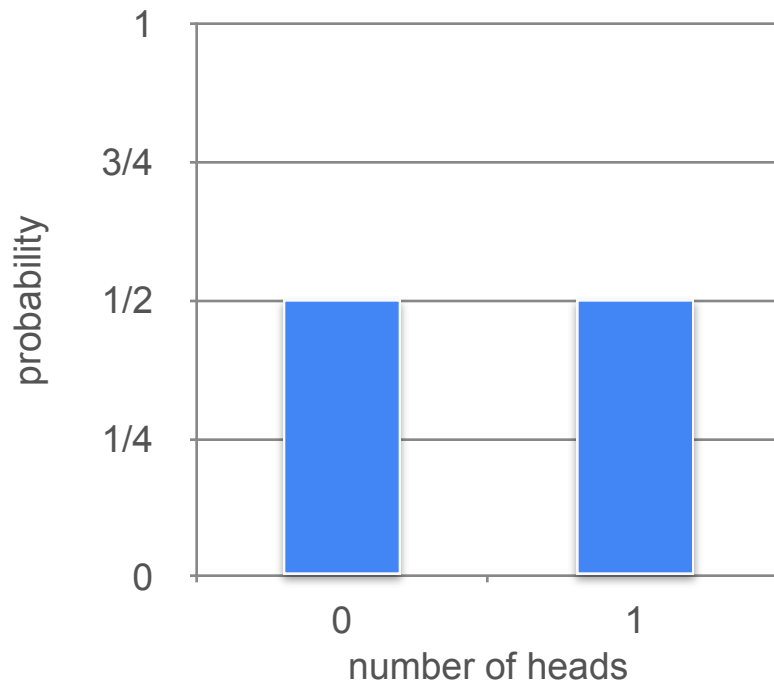
$$X = 1$$



$$P(T) = 0.5$$

$$X = 0$$

What can we say about the probability distribution when the number of coin flips increases?



Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5$$

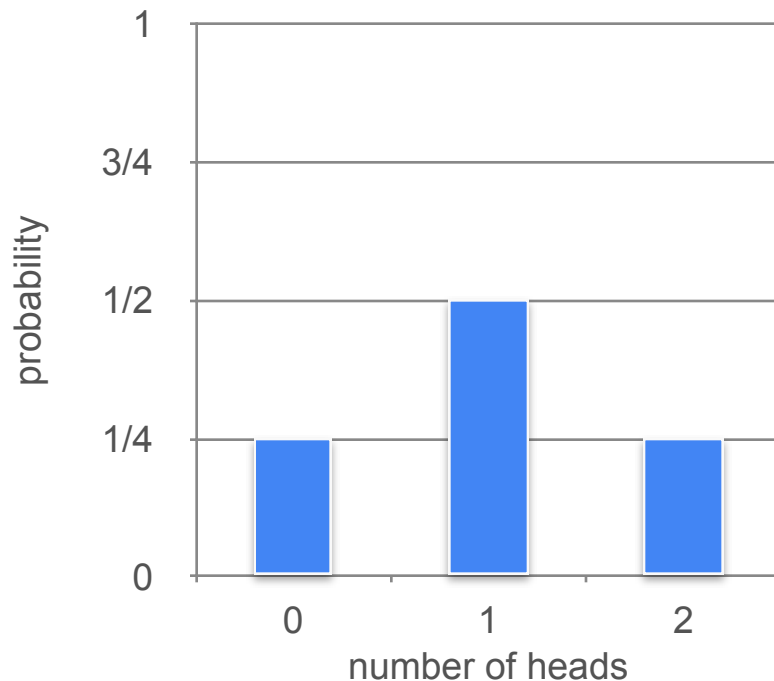
$$X = 1$$



$$P(T) = 0.5$$

$$X = 0$$

What can we say about the probability distribution when the number of coin flips increases?



Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5$$

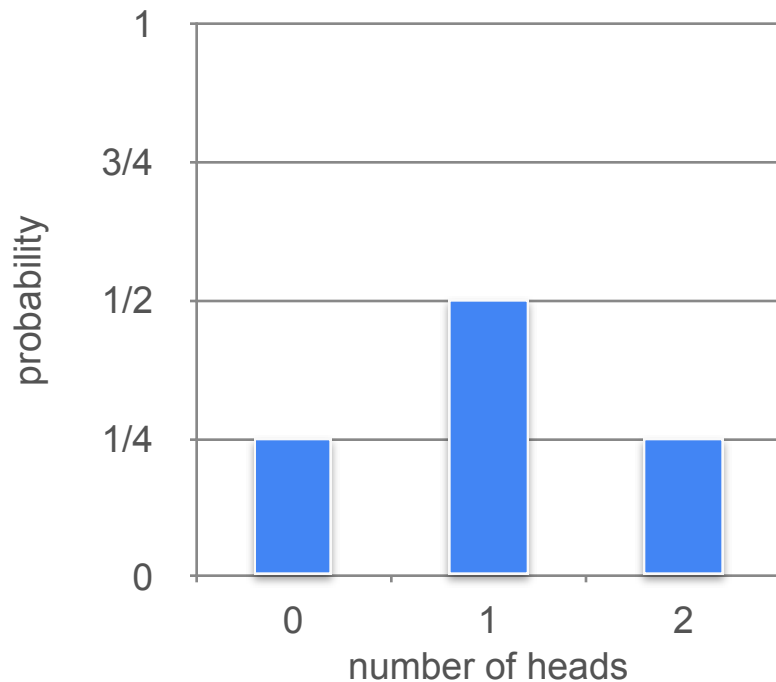
$$X = 1$$



$$P(T) = 0.5$$

$$X = 0$$

What can we say about the probability distribution when the number of coin flips increases?



Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5$$

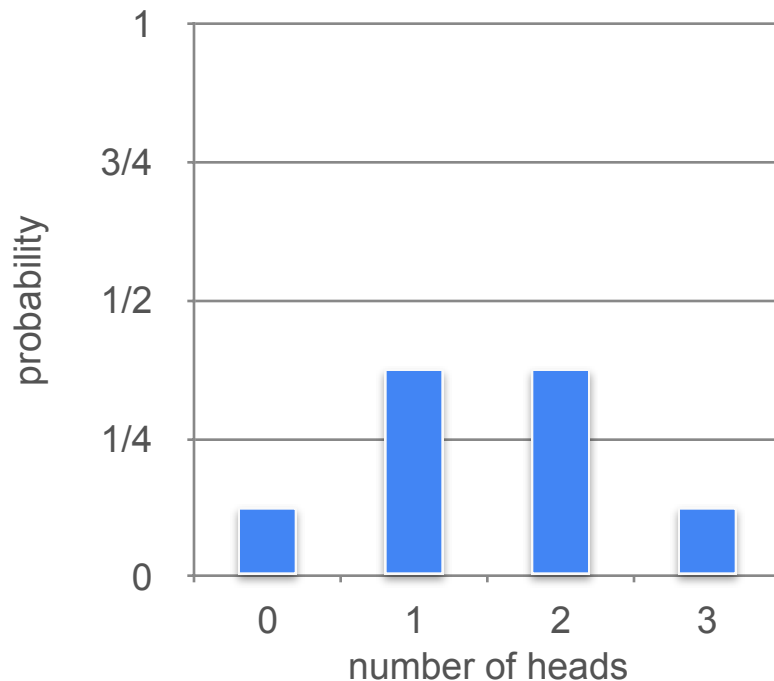
$$X = 1$$



$$P(T) = 0.5$$

$$X = 0$$

What can we say about the probability distribution when the number of coin flips increases?



Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5$$

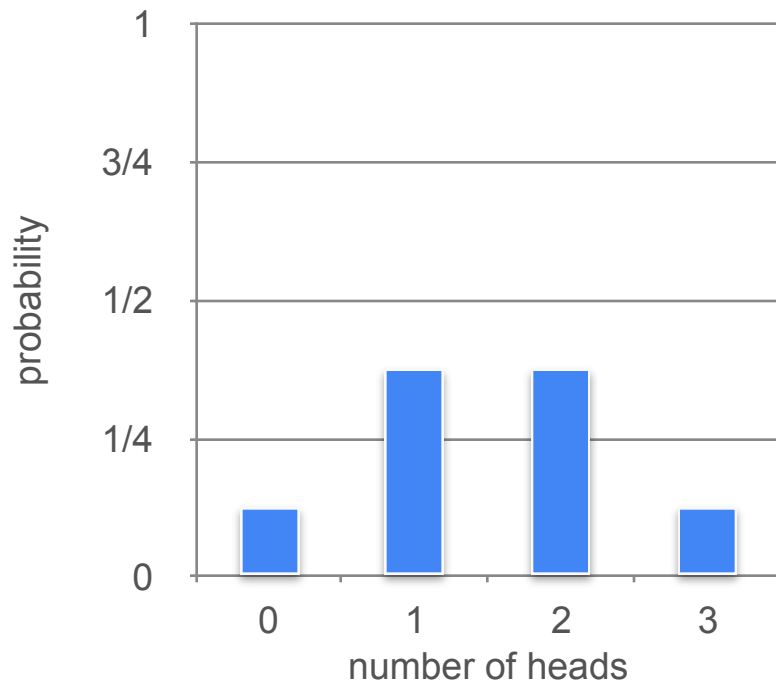
$$X = 1$$



$$P(T) = 0.5$$

$$X = 0$$

What can we say about the probability distribution when the number of coin flips increases?



Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5$$

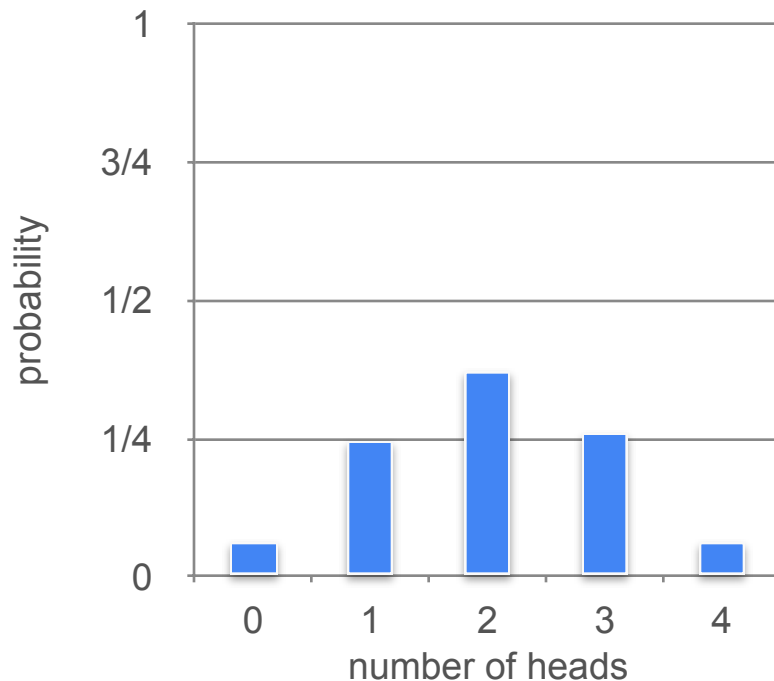
$$X = 1$$



$$P(T) = 0.5$$

$$X = 0$$

What can we say about the probability distribution when the number of coin flips increases?



Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5$$

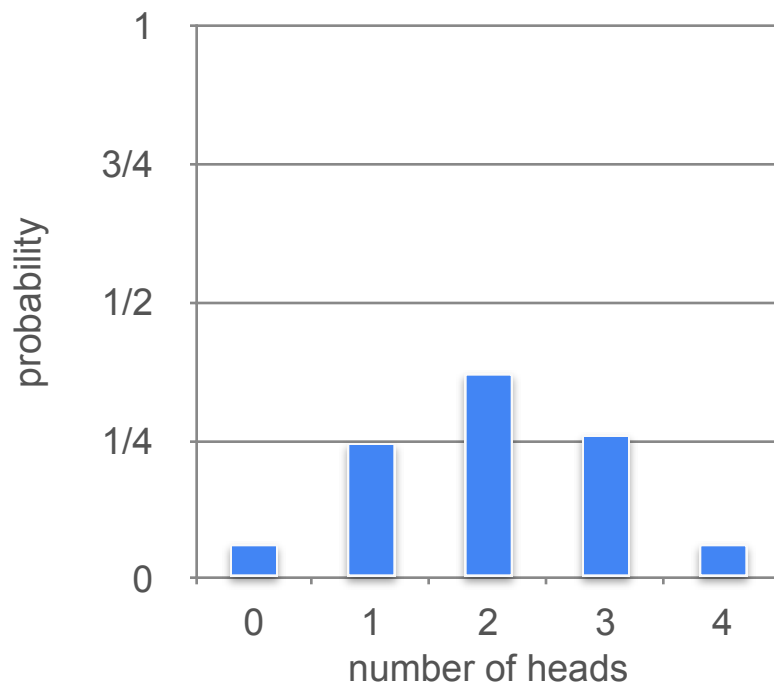
$$X = 1$$



$$P(T) = 0.5$$

$$X = 0$$

What can we say about the distribution when the number of coins we flip increases?



Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5$$

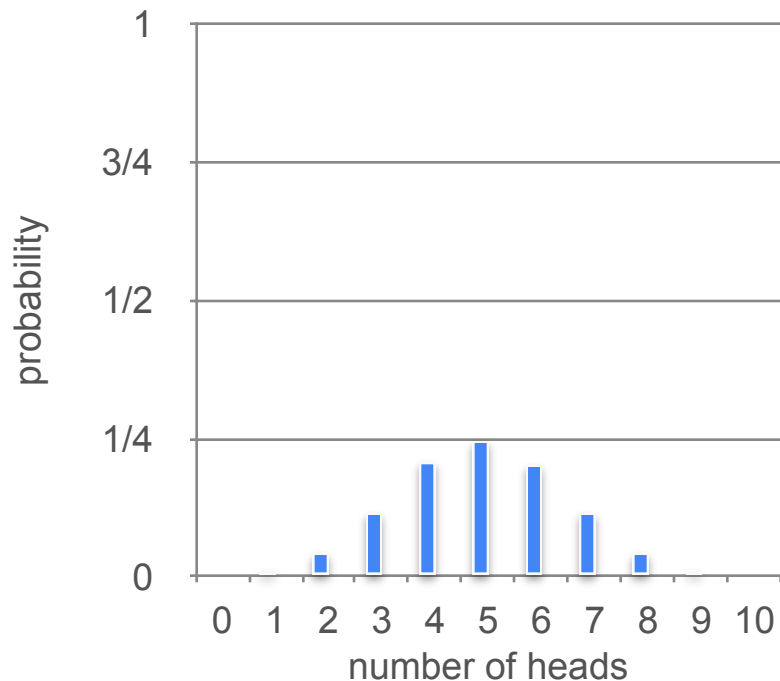
$$X = 1$$



$$P(T) = 0.5$$

$$X = 0$$

What can we say about the distribution when the number of coins we flip increases?



Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5$$

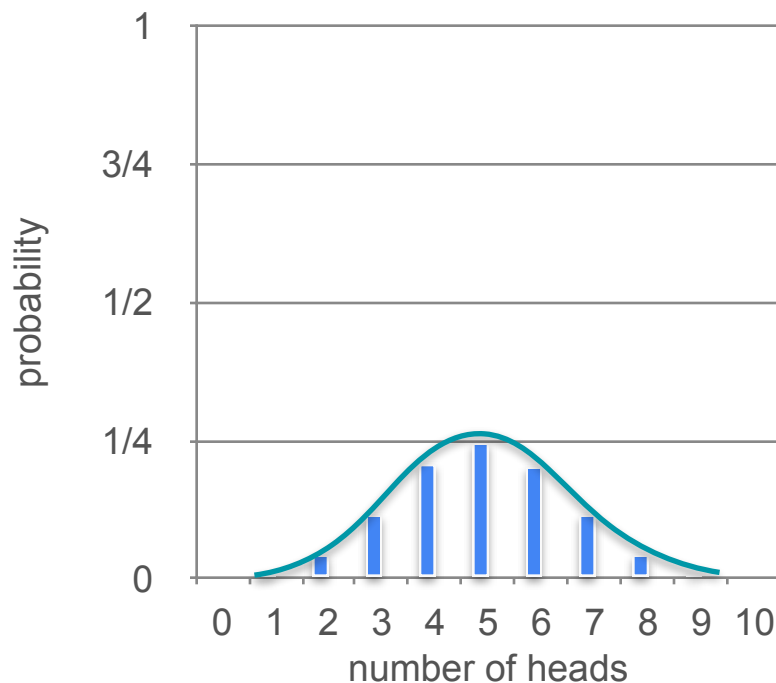
$$X = 1$$



$$P(T) = 0.5$$

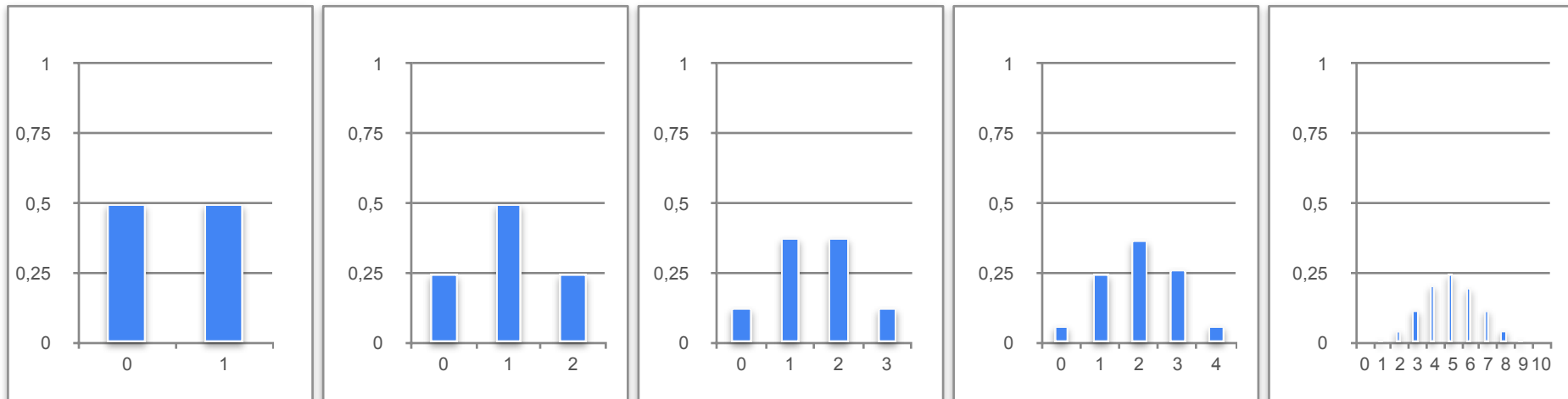
$$X = 0$$

What can we say about the distribution when the number of coins we flip increases?

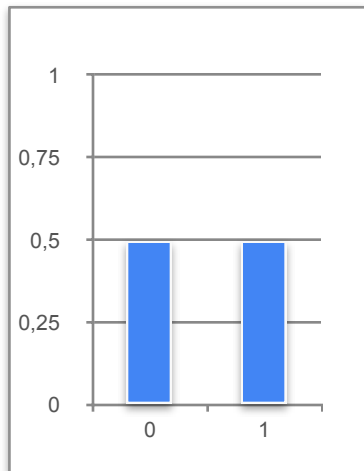


Central Limit Theorem (CLT) - Example 1

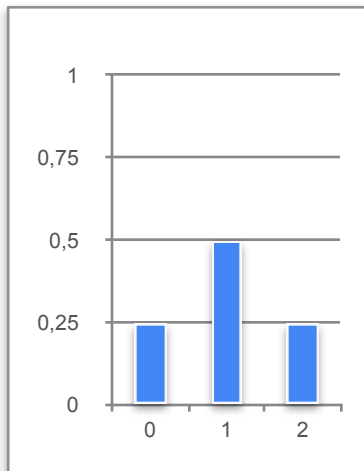
Central Limit Theorem (CLT) - Example 1



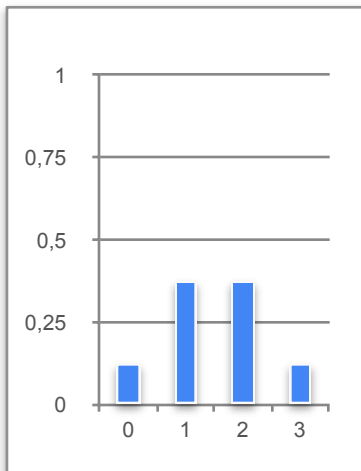
Central Limit Theorem (CLT) - Example 1



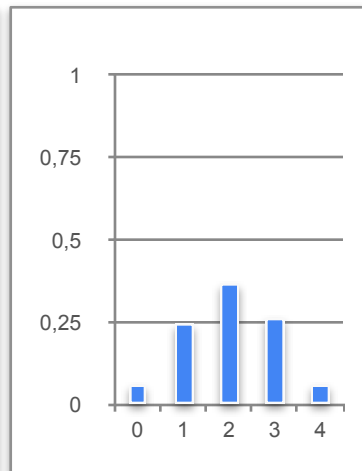
$n = 1$



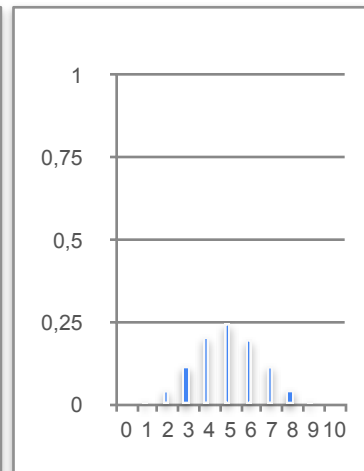
$n = 2$



$n = 3$

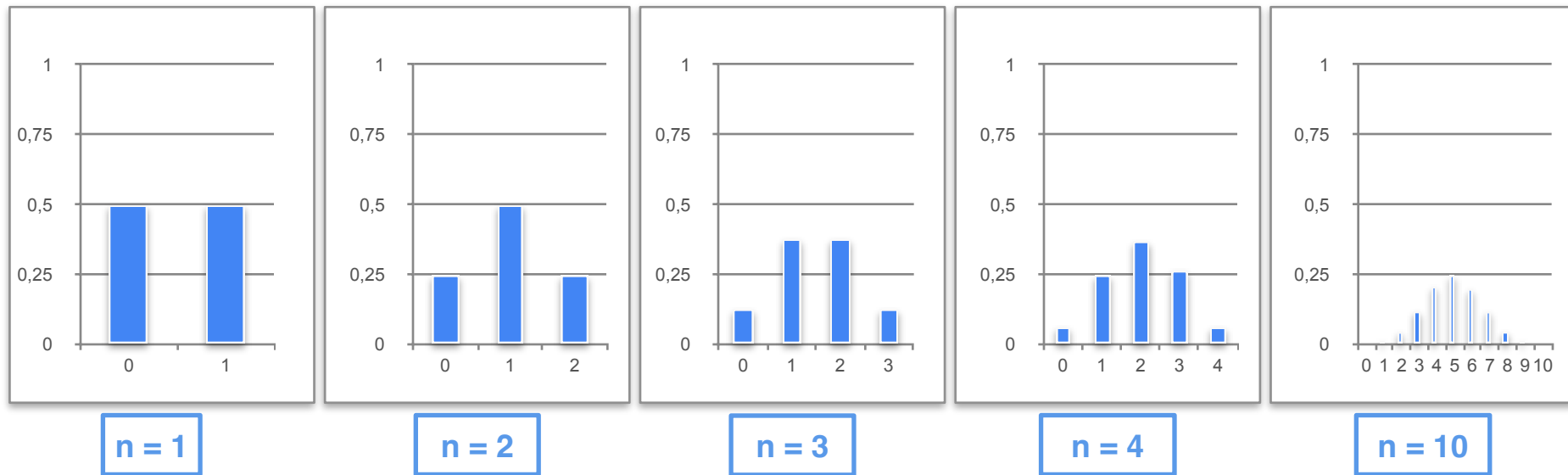


$n = 4$



$n = 10$

Central Limit Theorem (CLT) - Example 1



As n increases, the probability distribution becomes closer to a Gaussian distribution

Central Limit Theorem (CLT) - Example 1



Random variable $\rightarrow X$ number of heads when a coin is flipped n times

Central Limit Theorem (CLT) - Example 1



Random variable $\rightarrow X$ number of heads when a coin is flipped n times

$$\mu = np$$

Central Limit Theorem (CLT) - Example 1



Random variable $\rightarrow X$ number of heads when a coin is flipped n times

$$\mu = np = nP(H)$$


Central Limit Theorem (CLT) - Example 1



Random variable $\rightarrow X$ number of heads when a coin is flipped n times

$$\mu = np = nP(H)$$



$$\sigma^2 = np(1 - p)$$

Central Limit Theorem (CLT) - Example 1



Random variable $\rightarrow X$ number of heads when a coin is flipped n times

$$\mu = np = nP(H)$$



$$\sigma^2 = np(1 - p) = nP(H)P(T)$$



Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5 \quad P(T) = 0.5$$

$$\mu = np = nP(H)$$



$$\sigma^2 = np(1 - p) = nP(H)P(T)$$



Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5 \quad P(T) = 0.5$$

$$\mu = np = nP(H)$$



$$\sigma^2 = np(1 - p) = nP(H)P(T)$$



Central Limit Theorem (CLT) - Example 1



$n = 1$

$$\mathbf{P}(H) = 0.5 \quad \mathbf{P}(T) = 0.5$$

$$\mu = np = n\mathbf{P}(H)$$



$$\sigma^2 = np(1 - p) = n\mathbf{P}(H)\mathbf{P}(T)$$



Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5 \quad P(T) = 0.5$$

$$\mu = np = nP(H)$$



$$\sigma^2 = np(1 - p) = nP(H)P(T)$$



$$n = 1$$

$$\mu = np$$



Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5 \quad P(T) = 0.5$$

$$\mu = np = nP(H)$$



$$\sigma^2 = np(1 - p) = nP(H)P(T)$$



$$n = 1$$

$$\mu = np$$



$$\mu = 1 \times 0.5$$

Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5 \quad P(T) = 0.5$$

$$\mu = np = nP(H)$$


$$\sigma^2 = np(1 - p) = nP(H)P(T)$$




$$n = 1$$

$$\mu = np$$

$$\mu = 1 \times 0.5 = 0.5$$


Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5 \quad P(T) = 0.5$$

$$\mu = np = nP(H)$$



$$\sigma^2 = np(1 - p) = nP(H)P(T)$$



$$n = 1$$

$$\mu = np$$



$$\mu = 1 \times 0.5 = 0.5$$

$$\sigma^2 = np(1 - p)$$



Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5 \quad P(T) = 0.5$$

$$\mu = np = nP(H)$$



$$\sigma^2 = np(1 - p) = nP(H)P(T)$$



$$n = 1$$

$$\mu = np$$



$$\mu = 1 \times 0.5 = 0.5$$

$$\sigma^2 = np(1 - p)$$



$$\sigma^2 = (1 \times 0.5)(0.5)$$

Central Limit Theorem (CLT) - Example 1



$$P(H) = 0.5 \quad P(T) = 0.5$$

$$\mu = np = nP(H)$$



$$\sigma^2 = np(1 - p) = nP(H)P(T)$$



$$n = 1$$

$$\mu = np$$



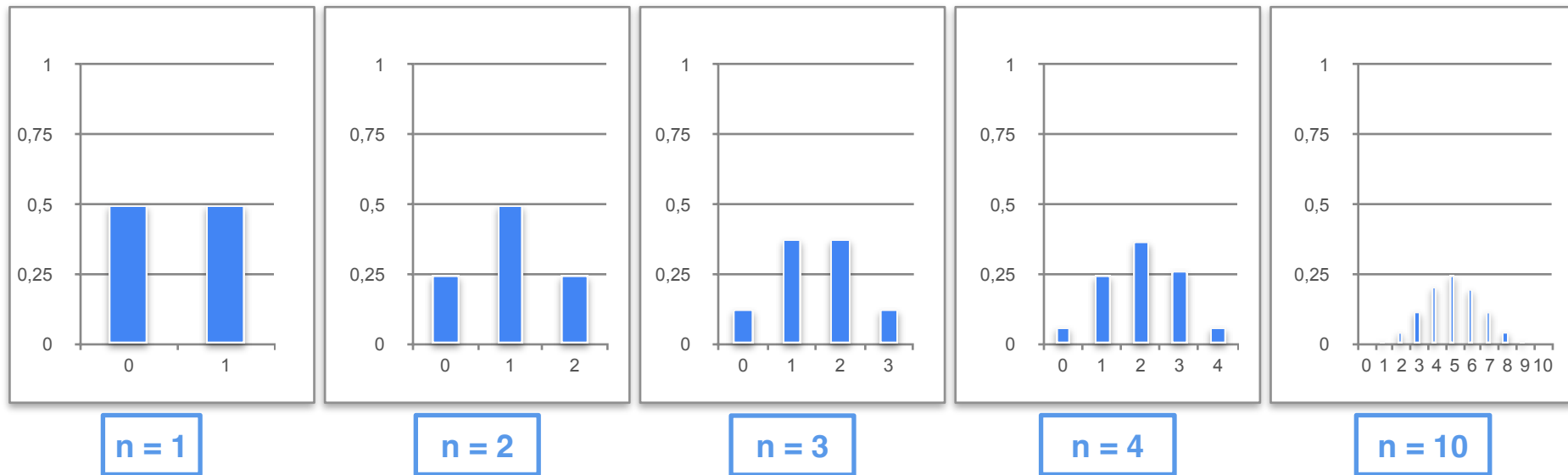
$$\mu = 1 \times 0.5 = 0.5$$

$$\sigma^2 = np(1 - p)$$



$$\sigma^2 = (1 \times 0.5)(0.5) = 0.25$$

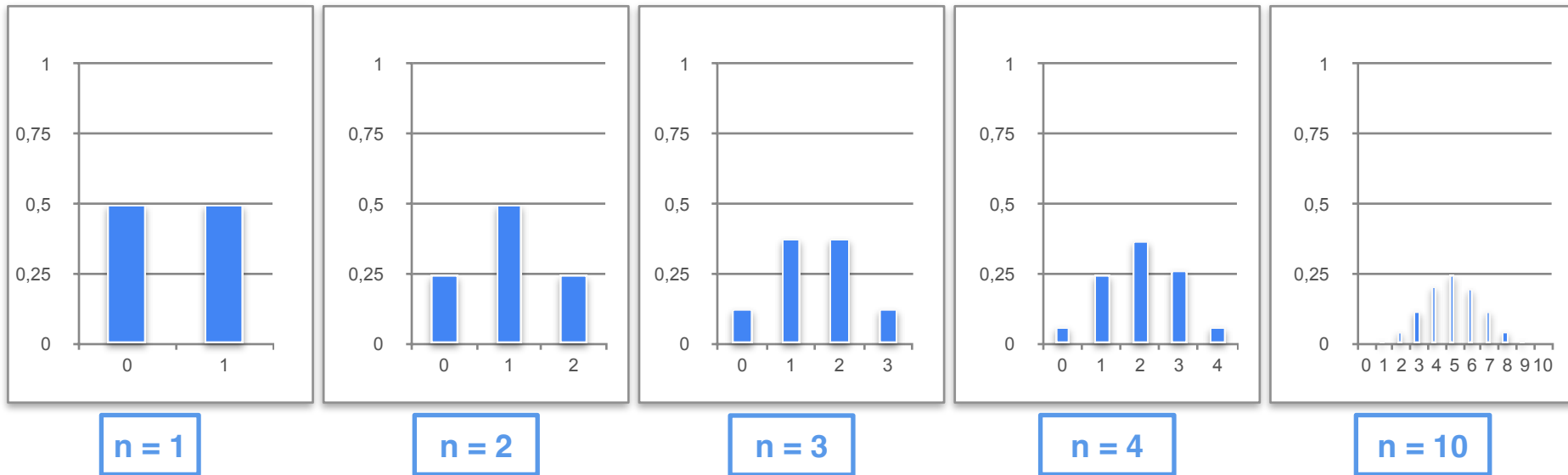
Central Limit Theorem (CLT) - Example 1



As n increases, the probability distribution becomes closer to a gaussian distribution

Central Limit Theorem (CLT) - Example 1

$$\mu = np$$

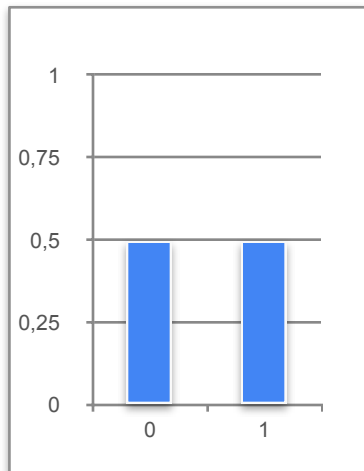


As n increases, the probability distribution becomes closer to a gaussian distribution

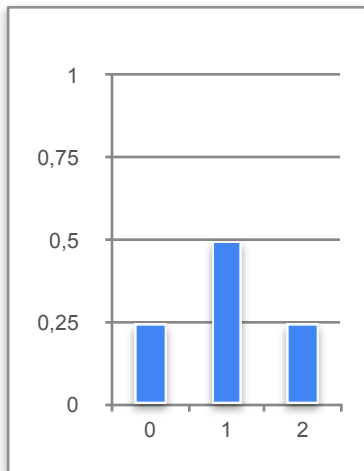
Central Limit Theorem (CLT) - Example 1

$$\mu = np$$

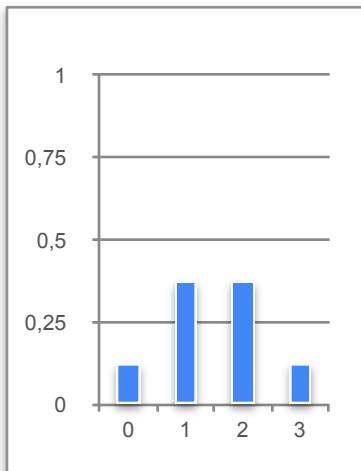
$$\sigma^2 = np(1 - p)$$



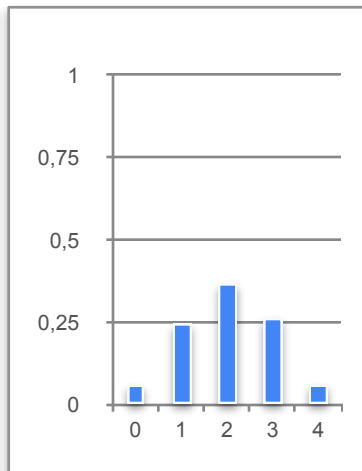
$n = 1$



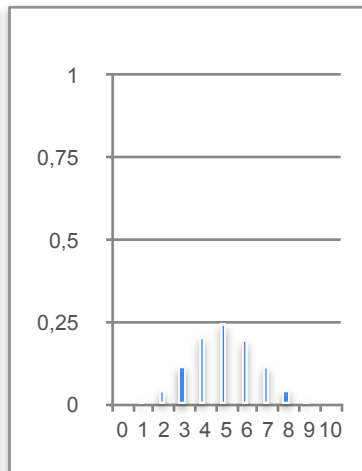
$n = 2$



$n = 3$



$n = 4$



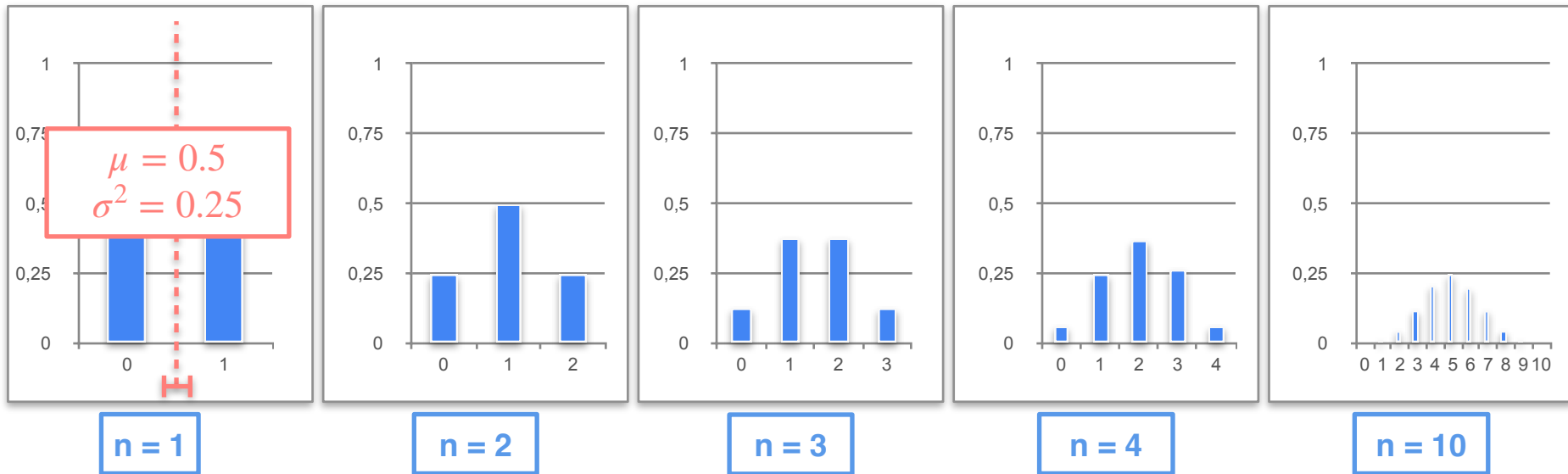
$n = 10$

As n increases, the probability distribution becomes closer to a gaussian distribution

Central Limit Theorem (CLT) - Example 1

$$\mu = np$$

$$\sigma^2 = np(1 - p)$$

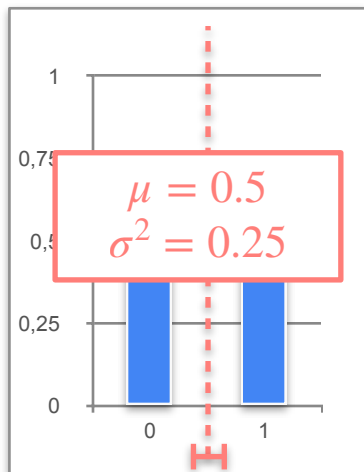


As n increases, the probability distribution becomes closer to a gaussian distribution

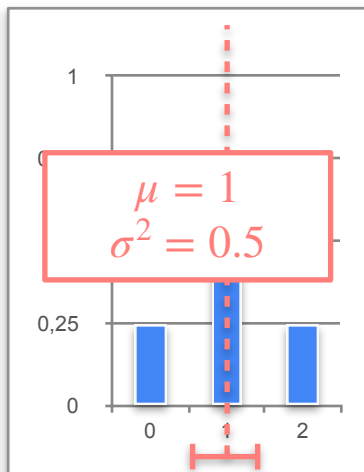
Central Limit Theorem (CLT) - Example 1

$$\mu = np$$

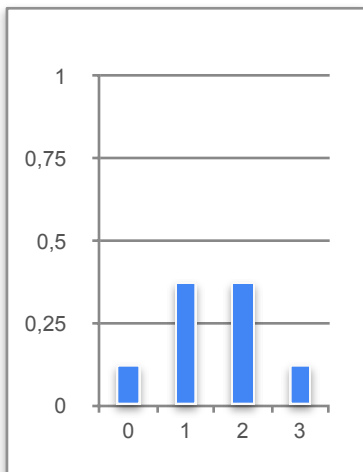
$$\sigma^2 = np(1 - p)$$



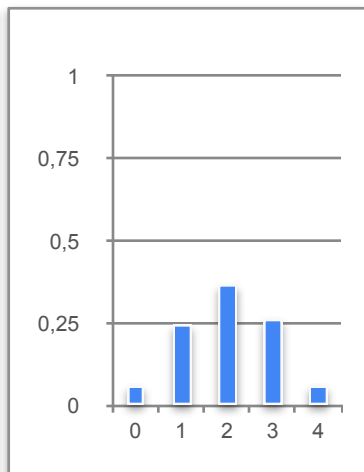
$n = 1$



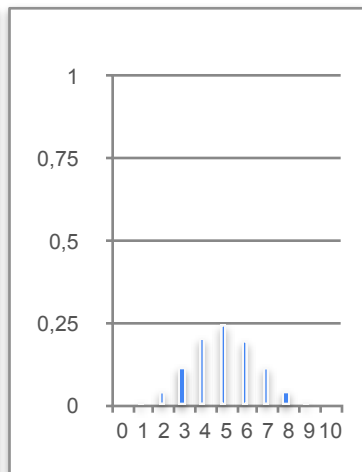
$n = 2$



$n = 3$



$n = 4$



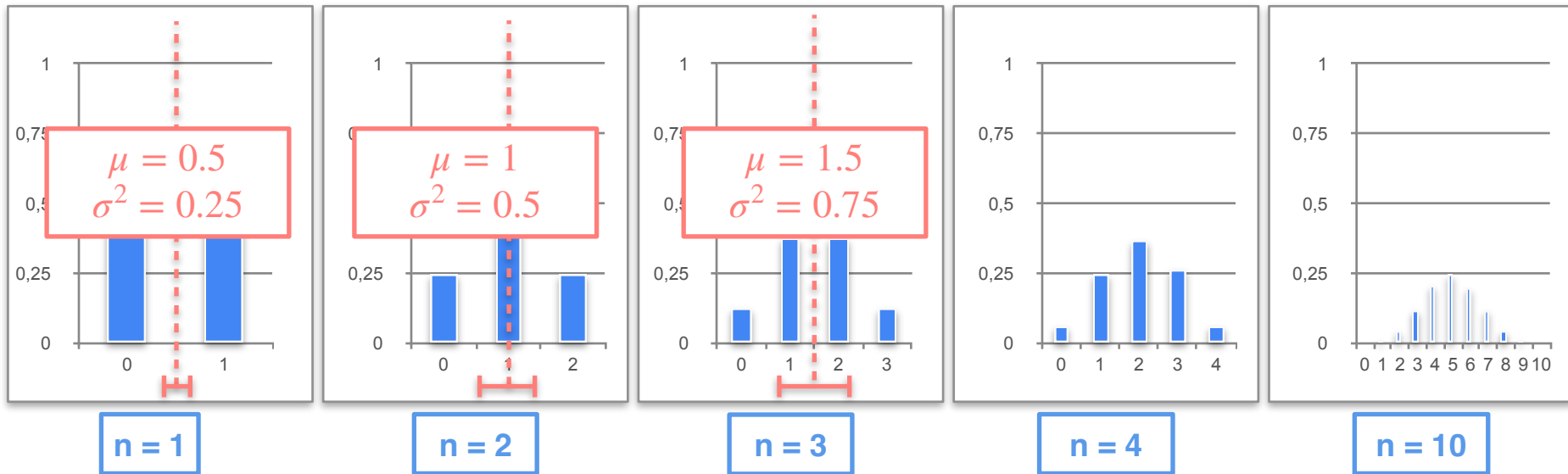
$n = 10$

As n increases, the probability distribution becomes closer to a gaussian distribution

Central Limit Theorem (CLT) - Example 1

$$\mu = np$$

$$\sigma^2 = np(1 - p)$$

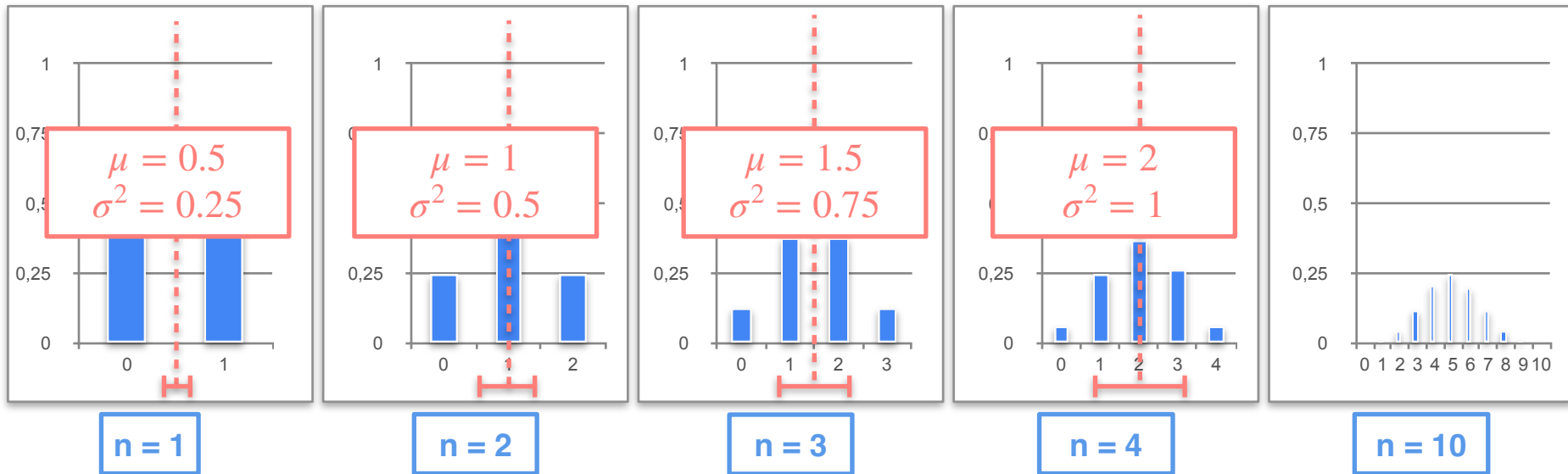


As n increases, the probability distribution becomes closer to a gaussian distribution

Central Limit Theorem (CLT) - Example 1

$$\mu = np$$

$$\sigma^2 = np(1 - p)$$

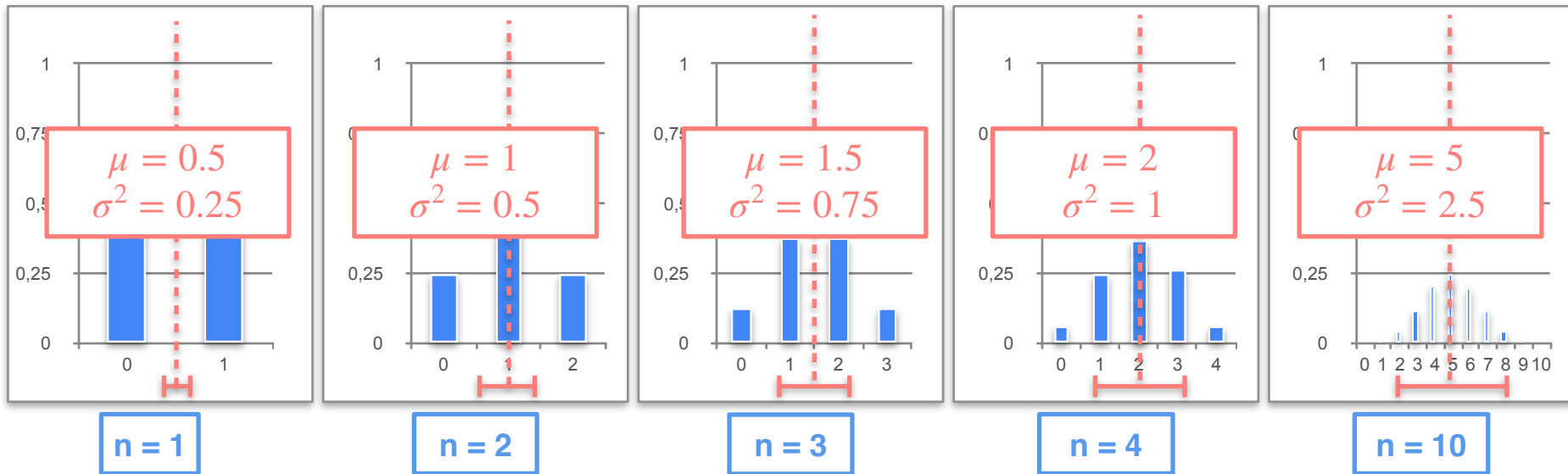


As n increases, the probability distribution becomes closer to a gaussian distribution

Central Limit Theorem (CLT) - Example 1

$$\mu = np$$

$$\sigma^2 = np(1 - p)$$



As n increases, the probability distribution becomes closer to a gaussian distribution

Central Limit Theorem (CLT) - Example 1

$$\mu = 0.5$$
$$\sigma^2 = 0.25$$

$$\mu = 1$$
$$\sigma^2 = 0.5$$

$$\mu = 1.5$$
$$\sigma^2 = 0.75$$

$$\mu = 2$$
$$\sigma^2 = 1$$

$$\mu = 5$$
$$\sigma^2 = 2.5$$

Central Limit Theorem (CLT) - Example 1

$$\mu = 0.5$$
$$\sigma^2 = 0.25$$

$$\mu = 1$$
$$\sigma^2 = 0.5$$

$$\mu = 1.5$$
$$\sigma^2 = 0.75$$

$$\mu = 2$$
$$\sigma^2 = 1$$

$$\mu = 5$$
$$\sigma^2 = 2.5$$

as n become sufficiently large we will get a normal distribution with parameters

Central Limit Theorem (CLT) - Example 1

$$\mu = 0.5$$
$$\sigma^2 = 0.25$$

$$\mu = 1$$
$$\sigma^2 = 0.5$$

$$\mu = 1.5$$
$$\sigma^2 = 0.75$$

$$\mu = 2$$
$$\sigma^2 = 1$$

$$\mu = 5$$
$$\sigma^2 = 2.5$$

as n become sufficiently large we will get a normal distribution with parameters

$$\mu = np$$

Central Limit Theorem (CLT) - Example 1

$$\mu = 0.5$$
$$\sigma^2 = 0.25$$

$$\mu = 1$$
$$\sigma^2 = 0.5$$

$$\mu = 1.5$$
$$\sigma^2 = 0.75$$

$$\mu = 2$$
$$\sigma^2 = 1$$

$$\mu = 5$$
$$\sigma^2 = 2.5$$

as n become sufficiently large we will get a normal distribution with parameters

$$\mu = np$$

$$\sigma^2 = np(1 - p)$$

Central Limit Theorem (CLT) - Example 1

$$\mu = np$$

$$\mu = 0.5$$
$$\sigma^2 = 0.25$$

$$\mu = 1$$
$$\sigma^2 = 0.5$$

$$\mu = 1.5$$
$$\sigma^2 = 0.75$$

$$\mu = 2$$
$$\sigma^2 = 1$$

$$\mu = 5$$
$$\sigma^2 = 2.5$$

as n become sufficiently large we will get a normal distribution with parameters

$$\mu = np$$

$$\sigma^2 = np(1 - p)$$

Central Limit Theorem (CLT) - Example 1

$$\mu = np$$

$$\sigma^2 = np(1 - p)$$

$$\mu = 0.5$$
$$\sigma^2 = 0.25$$

$$\mu = 1$$
$$\sigma^2 = 0.5$$

$$\mu = 1.5$$
$$\sigma^2 = 0.75$$

$$\mu = 2$$
$$\sigma^2 = 1$$

$$\mu = 5$$
$$\sigma^2 = 2.5$$

as n become sufficiently large we will get a normal distribution with parameters

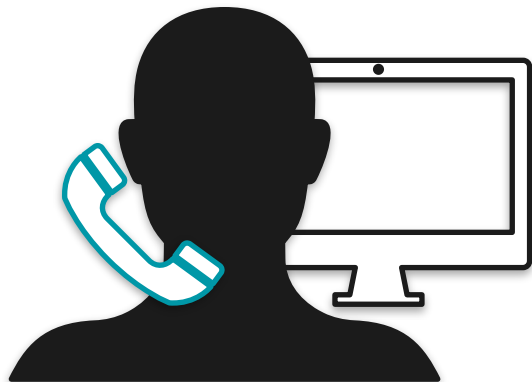
$$\mu = np$$

$$\sigma^2 = np(1 - p)$$

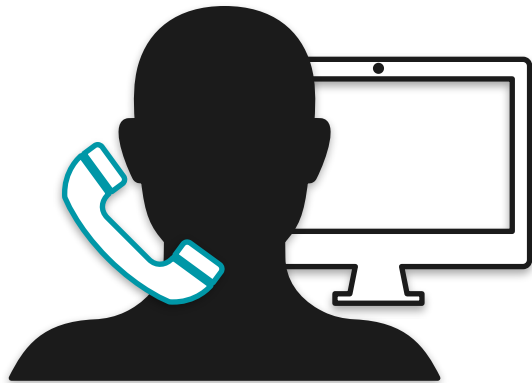
Video 4b: Central Limit Theorem

- Example 2: Continuous Random Variable (Sample mean of a sampling distribution)

Uniform Distribution: Motivation

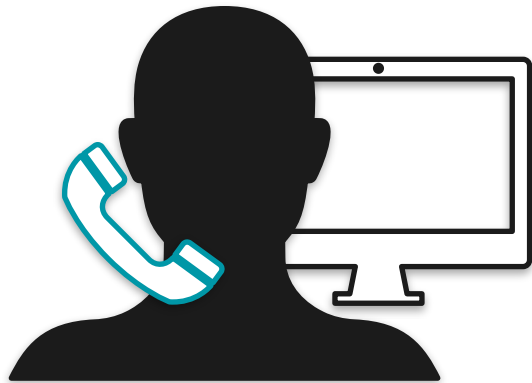


Uniform Distribution: Motivation



You're calling a tech support line.
They can answer any time between zero and 15 minutes and if they don't answer in this time, the line is disconnected.

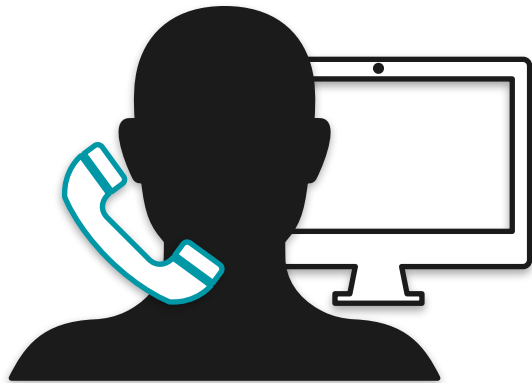
Uniform Distribution: Motivation



You're calling a tech support line.
They can answer any time between zero and 15 minutes and if they don't answer in this time, the line is disconnected.

X = "Wait time for a called to be answered "

Uniform Distribution: Motivation



You're calling a tech support line.
They can answer any time between zero and 15 minutes and if they don't answer in this time, the line is disconnected.

X = "Wait time for a called to be answered "

$$X \sim \mathcal{U}(0,15)$$

Central Limit Theorem (CLT) - Example 2

$$n = 1 \qquad Y_1 = \frac{X_1}{1}$$

Central Limit Theorem (CLT) - Example 2

$$n = 1 \quad Y_1 = \frac{X_1}{1}$$

$$n = 2 \quad Y_2 = \frac{X_1 + X_2}{2}$$

Central Limit Theorem (CLT) - Example 2

$$n = 1 \quad Y_1 = \frac{X_1}{1}$$

$$n = 2 \quad Y_2 = \frac{X_1 + X_2}{2}$$

$$n = 3 \quad Y_3 = \frac{X_1 + X_2 + X_3}{3}$$

$$\vdots \quad \vdots$$

Central Limit Theorem (CLT) - Example 2

$$n = 1 \quad Y_1 = \frac{X_1}{1}$$

Record the average of all n experiments

$$n = 2 \quad Y_2 = \frac{X_1 + X_2}{2}$$

$$n = 3 \quad Y_3 = \frac{X_1 + X_2 + X_3}{3}$$

$$\vdots \quad \vdots$$

Central Limit Theorem (CLT) - Example 2

$$n = 1 \quad Y_1 = \frac{X_1}{1}$$

$$n = 2 \quad Y_2 = \frac{X_1 + X_2}{2}$$

$$n = 3 \quad Y_3 = \frac{X_1 + X_2 + X_3}{3}$$

$$\vdots \quad \vdots$$

Record the average of all n experiments

$$Y_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Central Limit Theorem (CLT) - Example 2

$$n = 1 \quad Y_1 = \frac{X_1}{1}$$

$$n = 2 \quad Y_2 = \frac{X_1 + X_2}{2}$$

$$n = 3 \quad Y_3 = \frac{X_1 + X_2 + X_3}{3}$$

$$\vdots \quad \vdots$$

Record the average of all n experiments

$$Y_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Can we say anything about the distribution of this average?

Central Limit Theorem (CLT) - Example 2

$$n = 1 \quad Y_1 = \frac{X_1}{1}$$

Central Limit Theorem (CLT) - Example 2

$$n = 1 \quad Y_1 = \frac{X_1}{1}$$

What happens to the distribution of these averages as n increases?

Central Limit Theorem (CLT) - Example 2

$$n = 1 \quad Y_1 = \frac{X_1}{1}$$

Create many samples of Y_1 so
you can get a pretty histogram

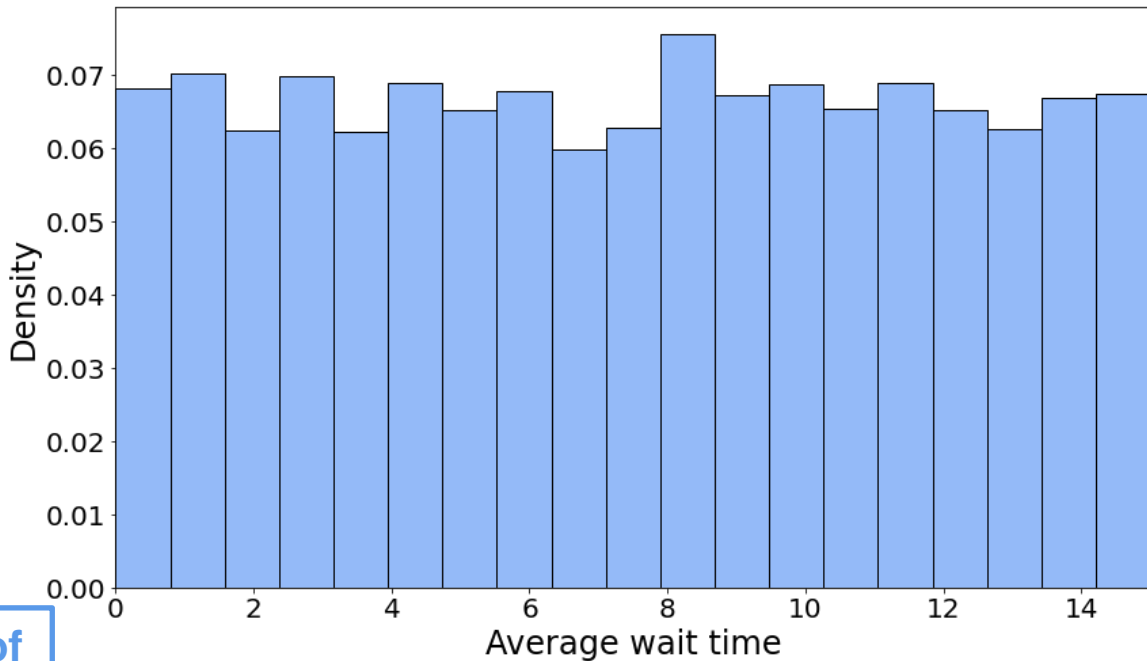
**What happens to the distribution of
these averages as n increases?**

Central Limit Theorem (CLT) - Example 2

$$n = 1 \quad Y_1 = \frac{X_1}{1}$$

Create many samples of Y_1 so you can get a pretty histogram

What happens to the distribution of these averages as n increases?

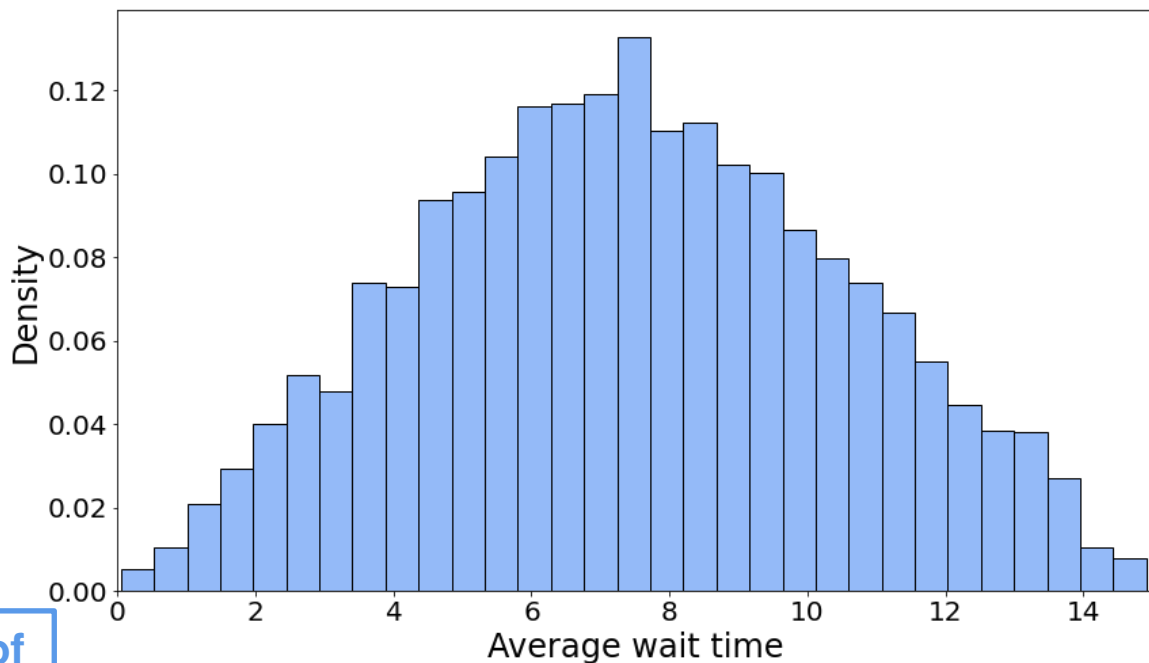


Central Limit Theorem (CLT) - Example 2

$$n = 2 \quad Y_2 = \frac{X_1 + X_2}{2}$$

Create many samples of Y_2 so you can get a pretty histogram

What happens to the distribution of these averages as n increases

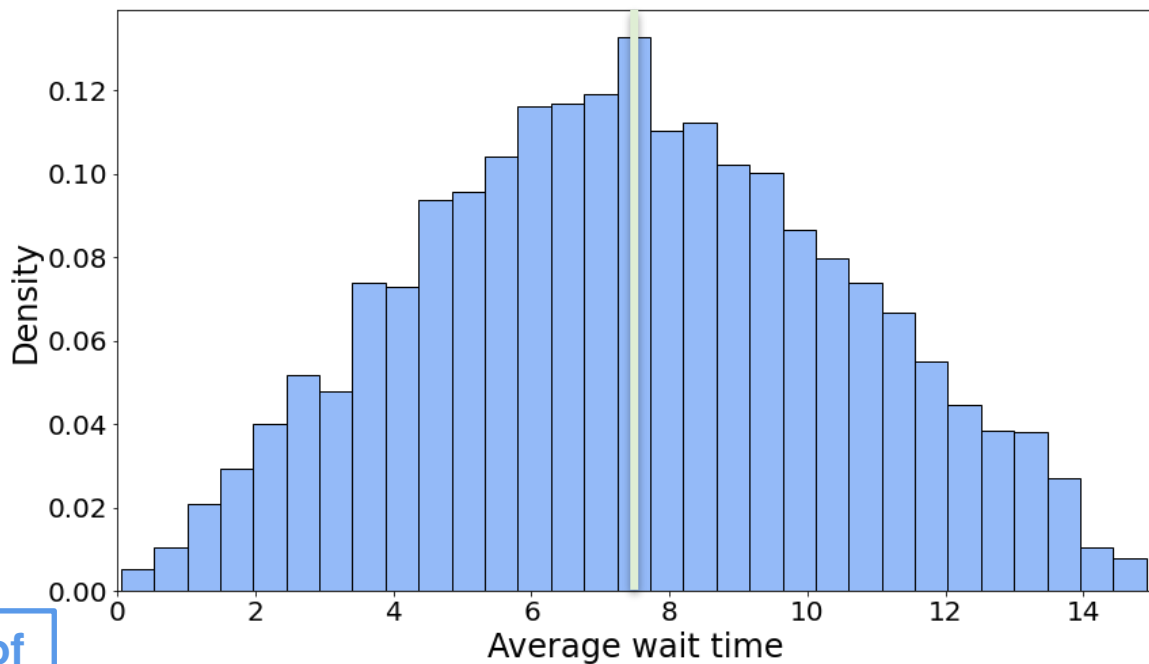


Central Limit Theorem (CLT) - Example 2

$$n = 2 \quad Y_2 = \frac{X_1 + X_2}{2}$$

Create many samples of Y_2 so you can get a pretty histogram

What happens to the distribution of these averages as n increases

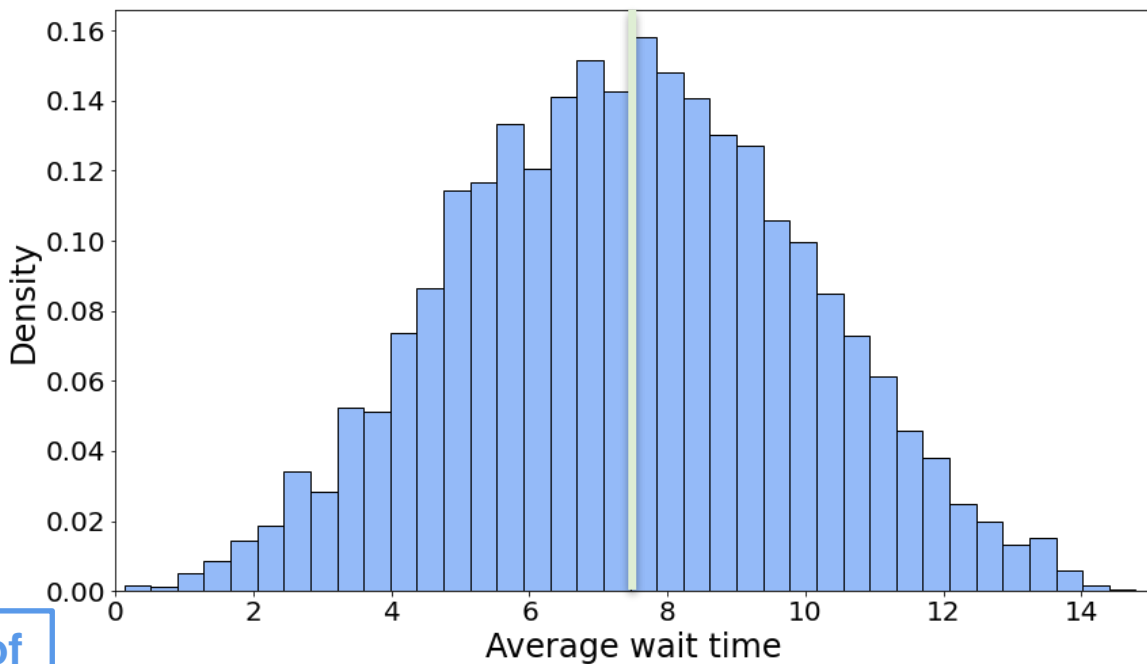


Central Limit Theorem (CLT) - Example 2

$$n = 3 \quad Y_3 = \frac{X_1 + X_2 + X_3}{3}$$

Create many samples of Y_3 so you can get a pretty histogram

What happens to the distribution of these averages as n increases

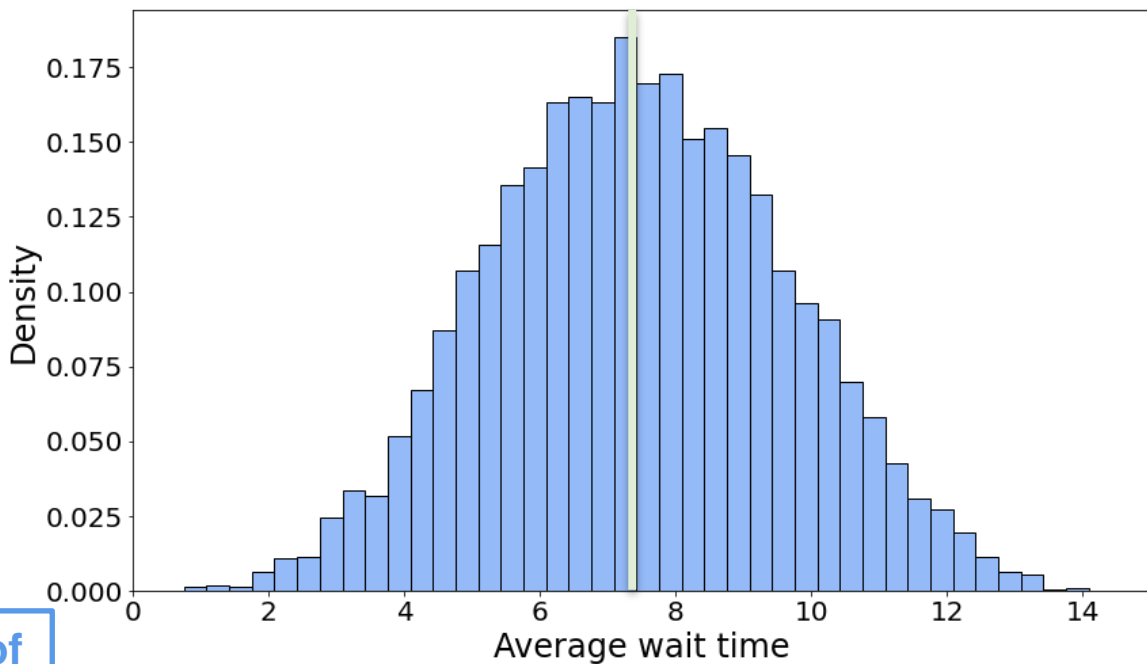


Central Limit Theorem (CLT) - Example 2

$$n = 4 \quad Y_4 = \frac{X_1 + X_2 + X_3 + X_4}{4}$$

Create many samples of Y_4 so you can get a pretty histogram

What happens to the distribution of these averages as n increases

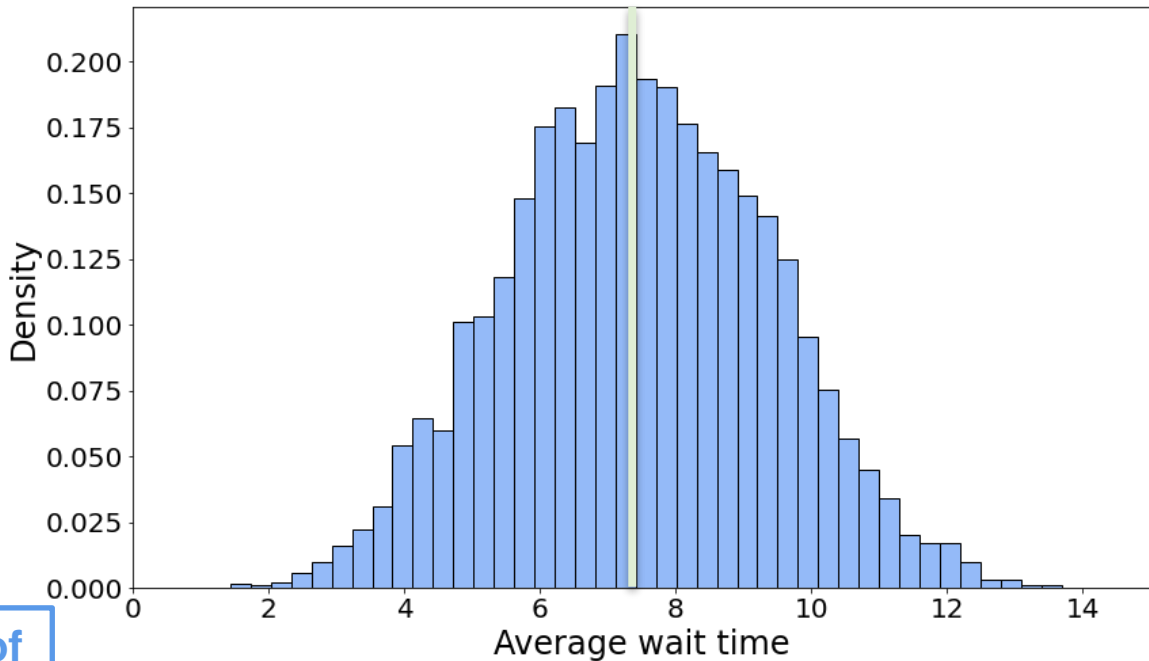


Central Limit Theorem (CLT) - Example 2

$$n = 5 \quad Y_5 = \frac{X_1 + X_2 + X_3 + X_4 + X_5}{5}$$

Create many samples of Y_5 so you can get a pretty histogram

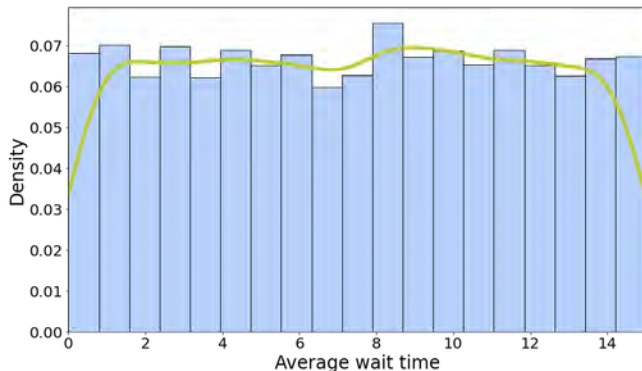
What happens to the distribution of these averages as n increases



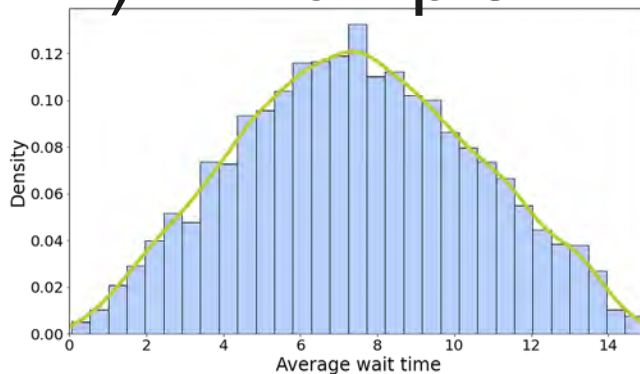
Central Limit Theorem (CLT) - Example 2

Central Limit Theorem (CLT) - Example 2

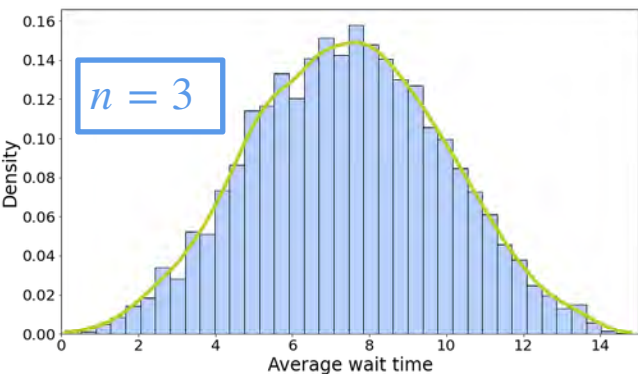
$n = 1$



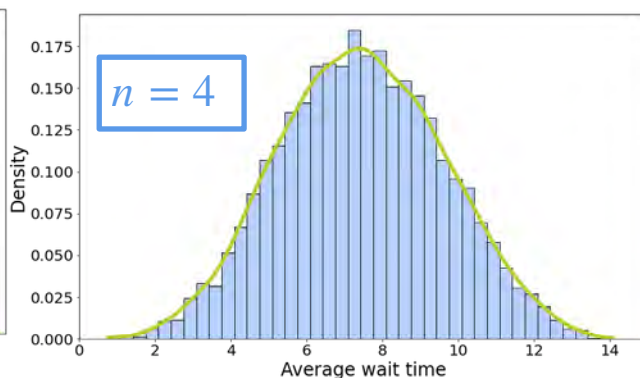
$n = 2$



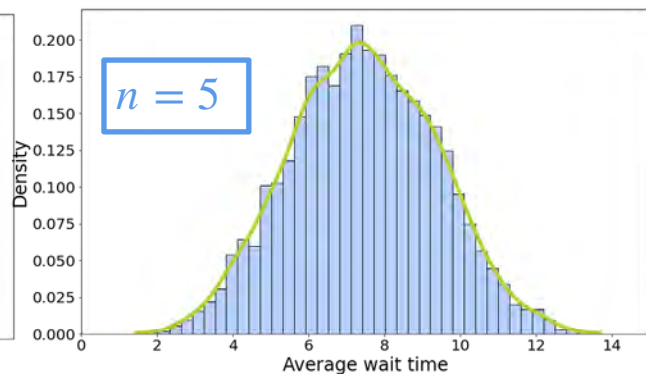
$n = 3$



$n = 4$



$n = 5$



Central Limit Theorem (CLT) - Example 2

$$= \frac{1}{n^2}$$

Central Limit Theorem (CLT) - Example 2

$$\mathbb{E}[Y_n] = \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n X_i \right]$$

$$= \frac{1}{n^2}$$

Central Limit Theorem (CLT) - Example 2

$$\mathbb{E}[Y_n] = \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n X_i \right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i]$$

$$= \frac{1}{n^2}$$

Central Limit Theorem (CLT) - Example 2

$$\mathbb{E}[Y_n] = \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n X_i \right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} n \mathbb{E}[X] = \mathbb{E}[X]$$

$$= \frac{1}{n^2}$$

Central Limit Theorem (CLT) - Example 2

$$\mathbb{E}[Y_n] = \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n X_i \right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} n \mathbb{E}[X] = \mathbb{E}[X] = 7.5$$

$$= \frac{1}{n^2}$$

Central Limit Theorem (CLT) - Example 2

$$\mathbb{E}[Y_n] = \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n X_i \right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} n \mathbb{E}[X] = \mathbb{E}[X] = 7.5$$

$$\text{Var}[Y_n] = \text{Var} \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{1}{n^2}$$

Central Limit Theorem (CLT) - Example 2

$$\mathbb{E}[Y_n] = \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n X_i \right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} n \mathbb{E}[X] = \mathbb{E}[X] = 7.5$$

$$\text{Var}[Y_n] = \text{Var} \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i)$$

Central Limit Theorem (CLT) - Example 2

$$\mathbb{E}[Y_n] = \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n X_i \right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} n \mathbb{E}[X] = \mathbb{E}[X] = 7.5$$

$$\begin{aligned} \text{Var}[Y_n] &= \text{Var} \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) \\ &= \frac{1}{n^2} n \text{Var}(X) = \frac{\text{Var}(X)}{n} \end{aligned}$$

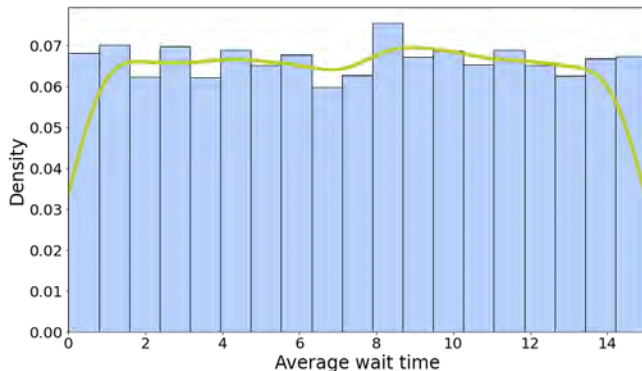
Central Limit Theorem (CLT) - Example 2

$$\mathbb{E}[Y_n] = \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n X_i \right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} n \mathbb{E}[X] = \mathbb{E}[X] = 7.5$$

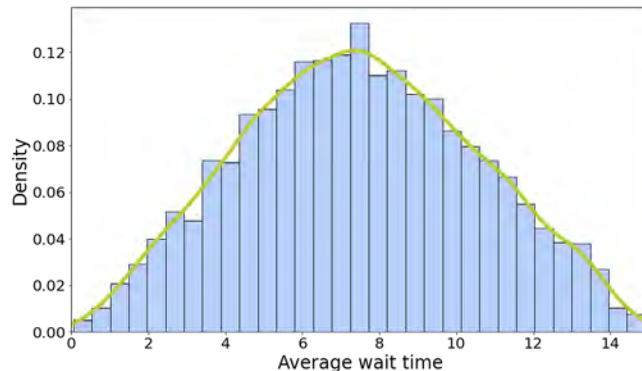
$$\begin{aligned} \text{Var}[Y_n] &= \text{Var} \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) \\ &= \frac{1}{n^2} n \text{Var}(X) = \frac{\text{Var}(X)}{n} = \frac{18.75}{n} \end{aligned}$$

Central Limit Theorem (CLT) - Example 2

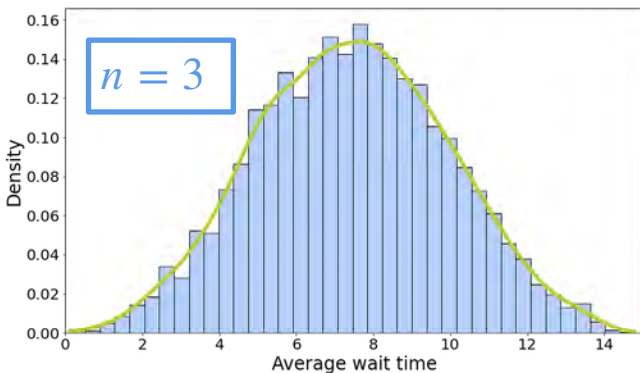
$n = 1$



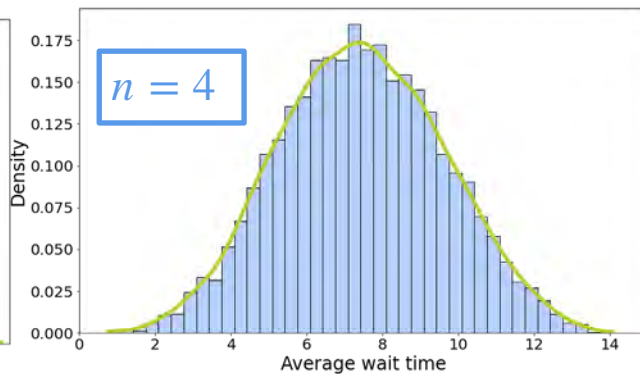
$n = 2$



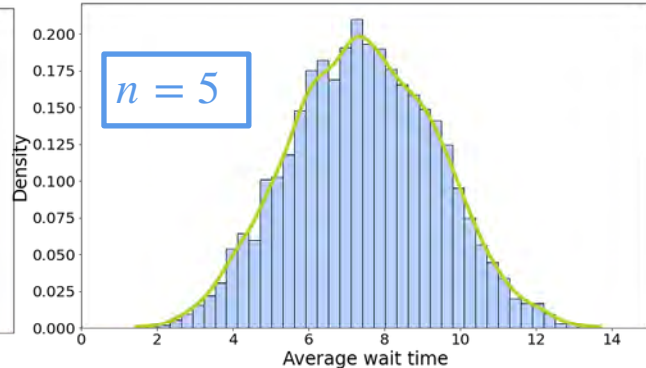
$n = 3$



$n = 4$

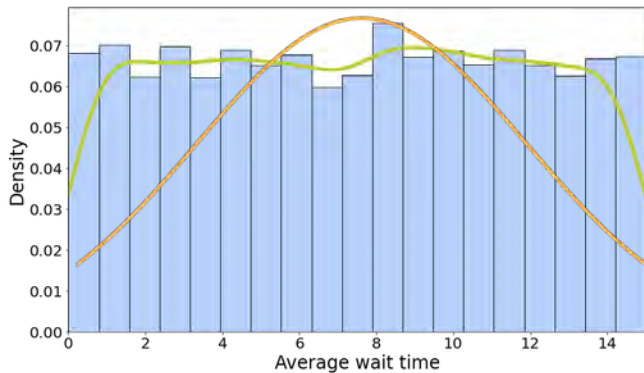


$n = 5$

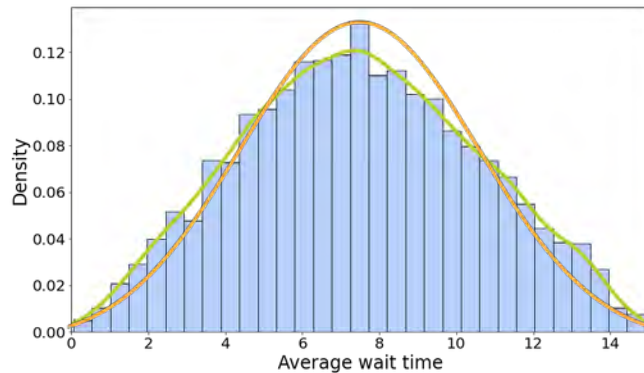


Central Limit Theorem (CLT) - Example 2

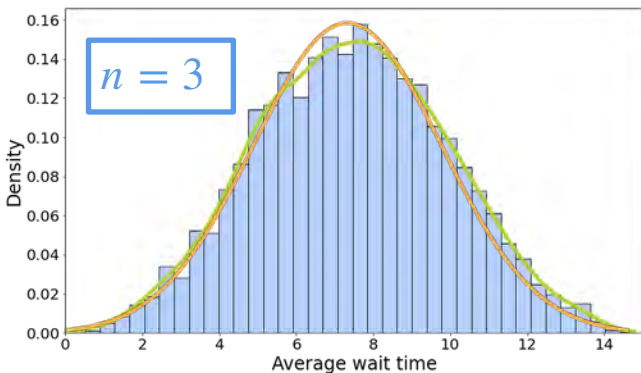
$n = 1$



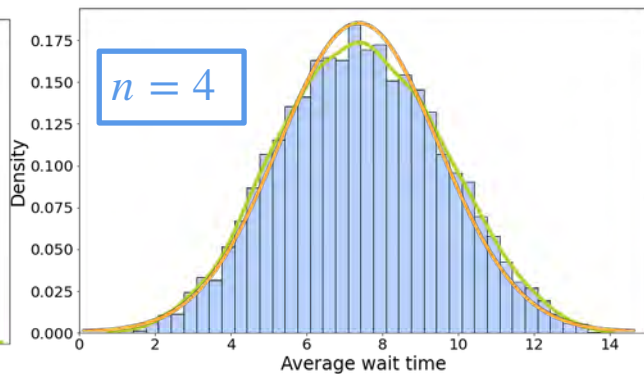
$n = 2$



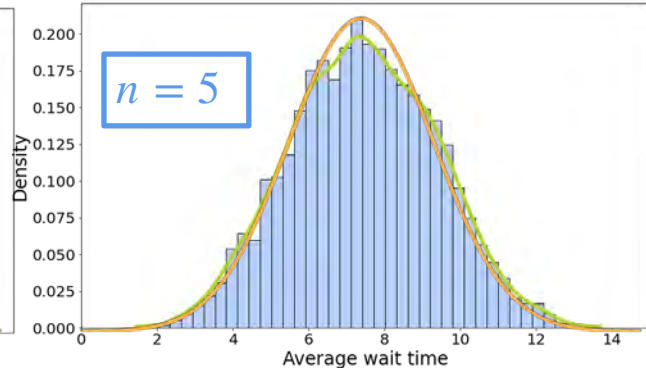
$n = 3$



$n = 4$

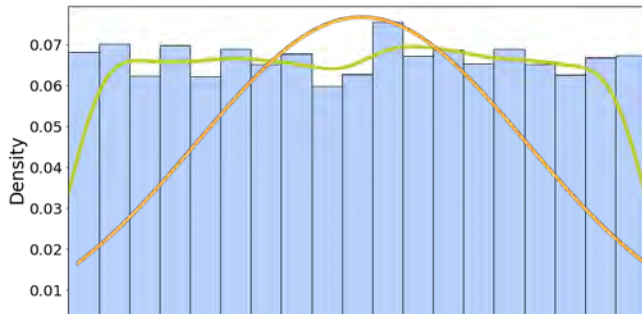


$n = 5$

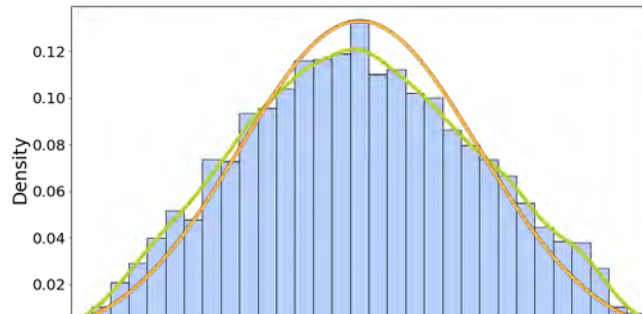


Central Limit Theorem (CLT) - Example 2

$n = 1$

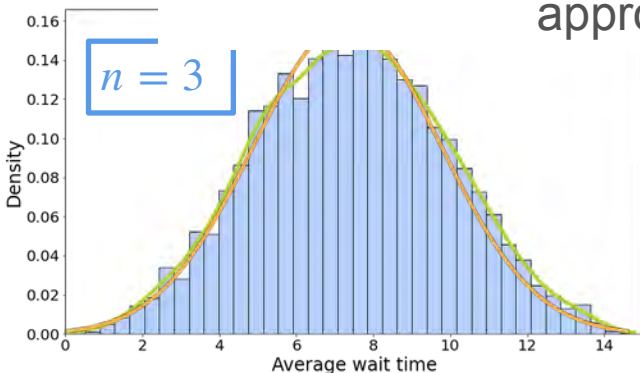


$n = 2$

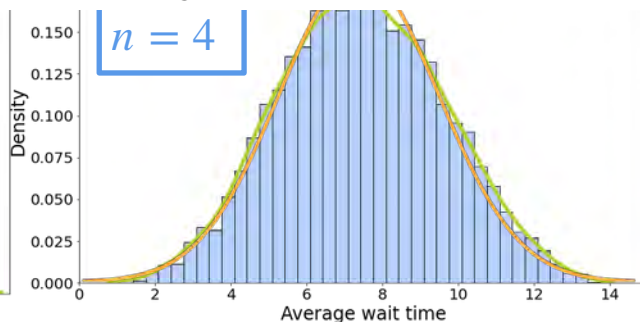


When you average a large enough number of variables, the distribution will approximately follow a normal distribution

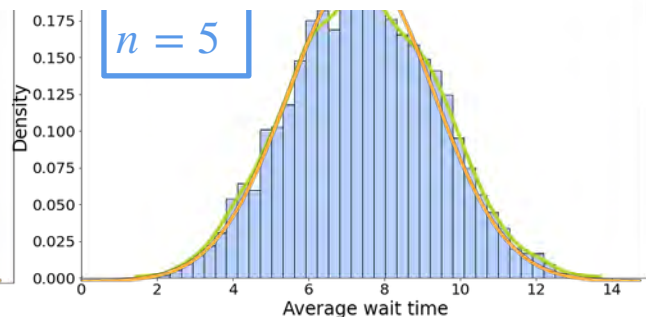
$n = 3$



$n = 4$

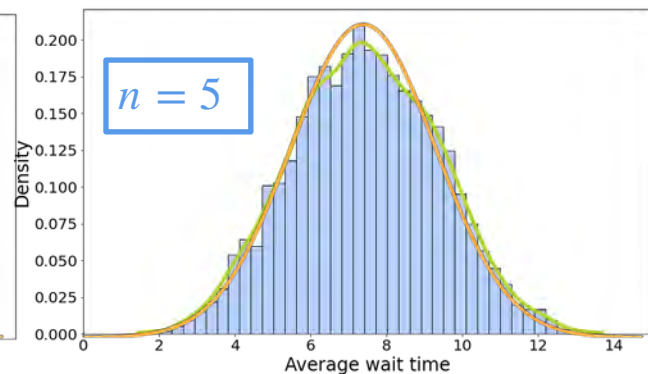
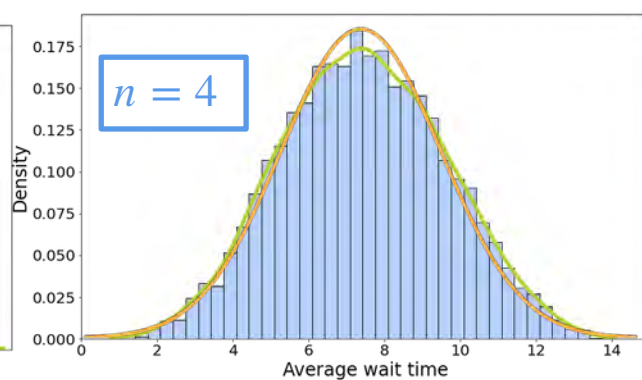
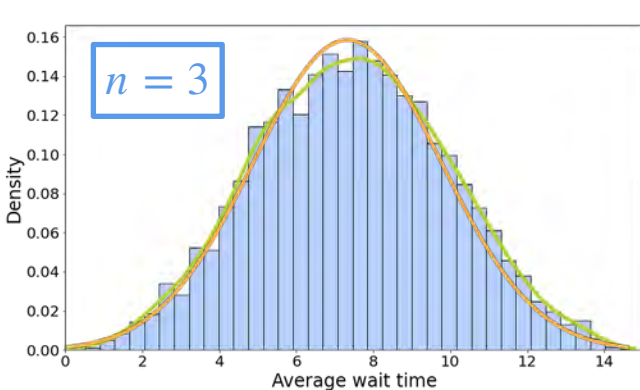


$n = 5$



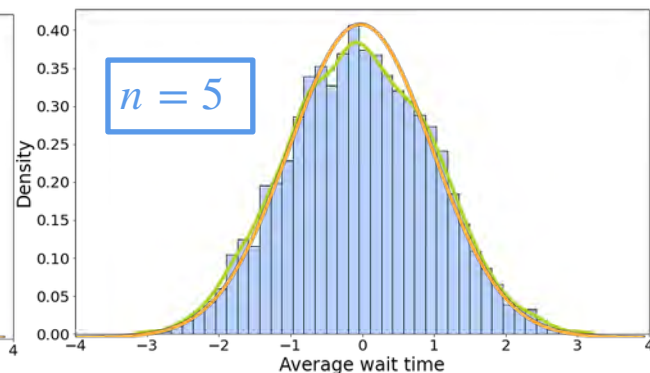
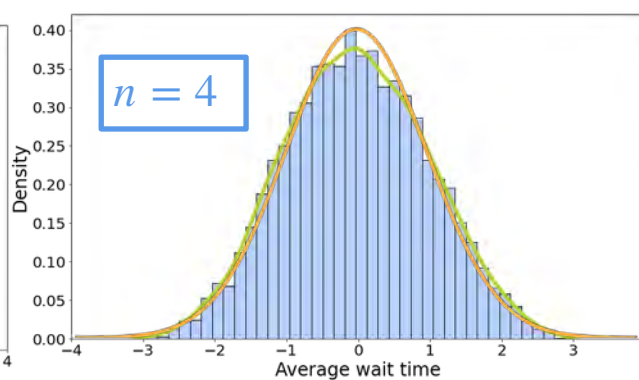
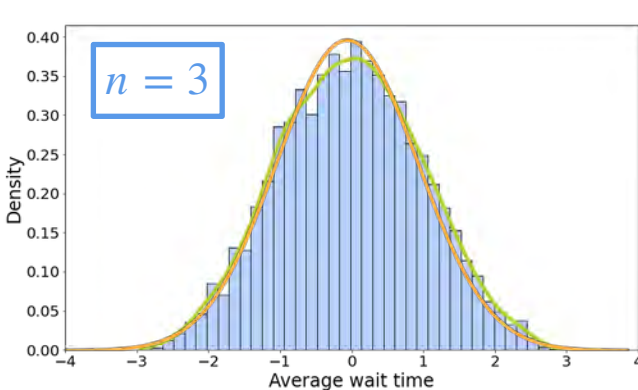
Central Limit Theorem (CLT) - Example 2

$$\frac{Y_n - 7.5}{\sqrt{18.75/n}}$$



Central Limit Theorem (CLT) - Example 2

$$\frac{Y_n - 7.5}{\sqrt{18.75/n}} \xrightarrow{n \uparrow} \mathcal{N}(0,1)$$



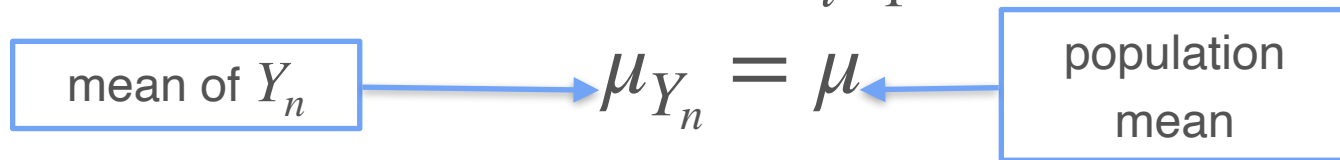
Central Limit Theorem (CLT) - Example 2

Central Limit Theorem (CLT) - Example 2

$$Y_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Central Limit Theorem (CLT) - Example 2

$$Y_n = \frac{1}{n} \sum_{i=1}^n X_i$$



Central Limit Theorem (CLT) - Example 2

$$Y_n = \frac{1}{n} \sum_{i=1}^n X_i$$

mean of Y_n $\rightarrow \mu_{Y_n} = \mu \leftarrow$ population mean

variance of Y_n $\rightarrow \sigma_{Y_n}^2 = \frac{\sigma^2}{n} \leftarrow$ population variance

Central Limit Theorem (CLT) - Formal Definition

Central Limit Theorem (CLT) - Formal Definition

$$\text{As } n \rightarrow \infty \quad \frac{\frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X]}{\sigma_X} \sqrt{n} \sim \mathcal{N}(0, 1^2)$$

Central Limit Theorem (CLT) - Formal Definition

$$\text{As } n \rightarrow \infty \quad \frac{\frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X]}{\sigma_X} \sqrt{n} \sim \mathcal{N}(0, 1^2)$$

$$\text{As } n \rightarrow \infty \quad \frac{1}{n} \left(\frac{\sum_{i=1}^n X_i - \frac{1}{n} n \mathbb{E}[X]}{\sigma_X} \right) \sqrt{n} \sim \mathcal{N}(0, 1^2)$$

Central Limit Theorem (CLT) - Formal Definition

$$\text{As } n \rightarrow \infty \quad \frac{\frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X]}{\sigma_X} \sqrt{n} \sim \mathcal{N}(0, 1^2)$$

$$\text{As } n \rightarrow \infty \quad \frac{1}{n} \left(\frac{\sum_{i=1}^n X_i - \cancel{\frac{1}{n}} n \mathbb{E}[X]}{\sigma_X} \right) \sqrt{n} \sim \mathcal{N}(0, 1^2)$$

Central Limit Theorem (CLT) - Formal Definition

$$\text{As } n \rightarrow \infty \quad \frac{\frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X]}{\sigma_X} \sqrt{n} \sim \mathcal{N}(0, 1^2)$$

$$\text{As } n \rightarrow \infty \quad \frac{1}{\cancel{\sqrt{n}}} \left(\frac{\sum_{i=1}^n X_i - \cancel{\frac{1}{n}} \mathbb{E}[X]}{\sigma_X} \right) \cancel{\sqrt{n}} \sim \mathcal{N}(0, 1^2)$$

Central Limit Theorem (CLT) - Formal Definition

$$\text{As } n \rightarrow \infty \quad \frac{\frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X]}{\sigma_X} \sqrt{n} \sim \mathcal{N}(0, 1^2)$$

As $n \rightarrow \infty$

Central Limit Theorem (CLT) - Formal Definition

$$\text{As } n \rightarrow \infty \quad \frac{\frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X]}{\sigma_X} \sqrt{n} \sim \mathcal{N}(0, 1^2)$$

$$\text{As } n \rightarrow \infty \quad \frac{\sum_{i=1}^n X_i - n\mathbb{E}[X]}{\sqrt{n}\sigma_X} \sim \mathcal{N}(0, 1^2)$$

W3 Lesson 2



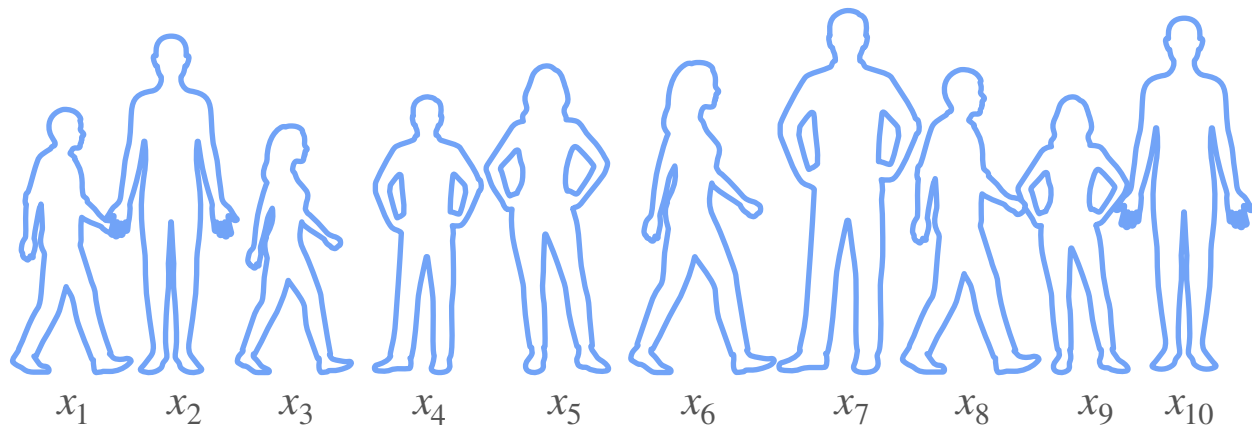
DeepLearning.AI

Point Estimation

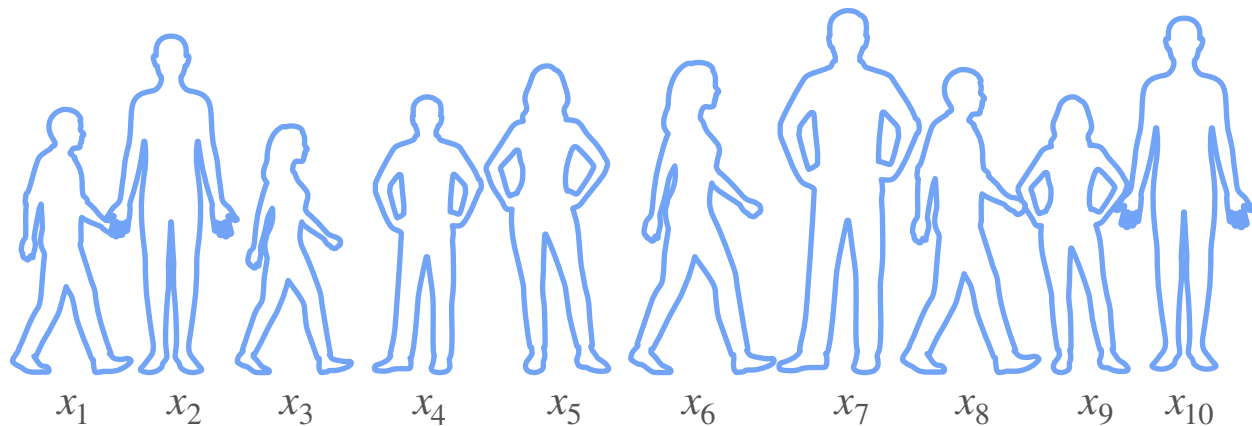
What is Point Estimation?

Point Estimates

Point Estimates



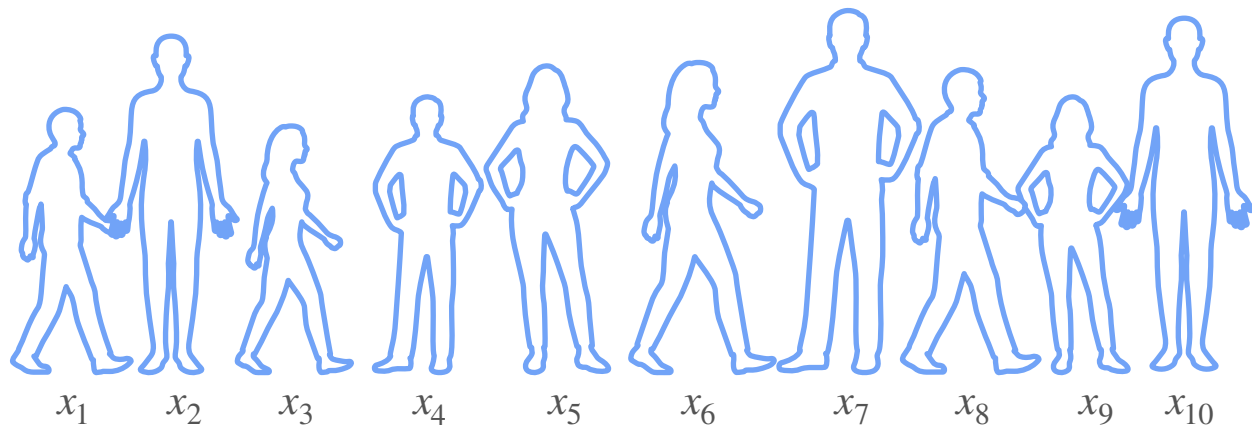
Point Estimates



Mean of the
population?

$$\mu \approx \frac{1}{10} \sum_{i=1}^{10} x_i = \bar{x}$$

Point Estimates



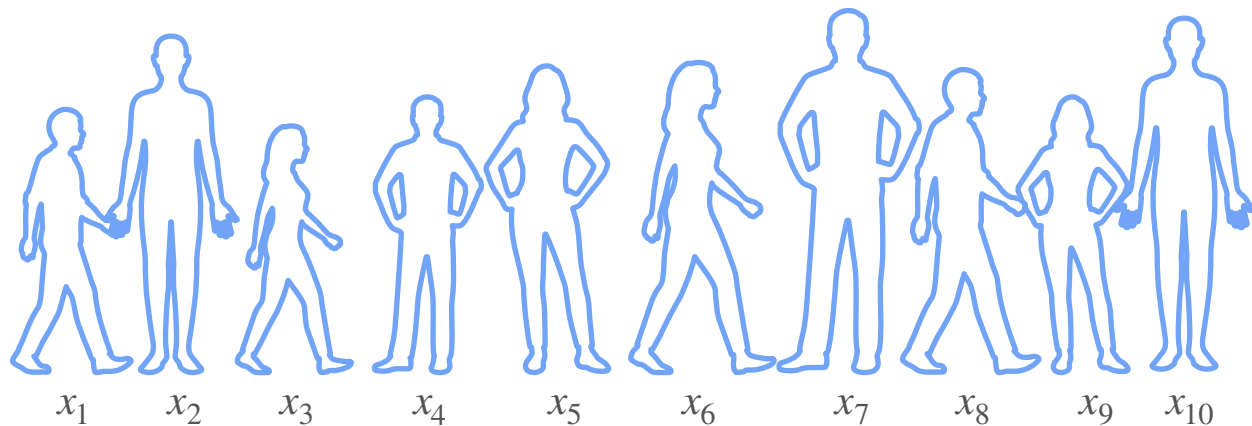
Mean of the
population?

$$\mu \approx \frac{1}{10} \sum_{i=1}^{10} x_i = \bar{x}$$

Variance of the
population?

$$\sigma^2 \approx \frac{1}{10 - 1} \sum_{i=1}^{10} (x_i - \bar{x})^2 = s^2$$

Point Estimates



Mean of the
population?

$$\mu \approx \frac{1}{10} \sum_{i=1}^{10} x_i = \bar{x}$$

Variance of the
population?

$$\sigma^2 \approx \frac{1}{10 - 1} \sum_{i=1}^{10} (x_i - \bar{x})^2 = s^2$$

$\bar{x}$ and s^2 are point estimates

Point Estimates

Point Estimates

A **point estimate** is a **single numerical value** based on **sample data** that is used to **approximate an unknown parameter** of a population or model parameter.

Point Estimates

Point Estimates



$P(H)?$

Point Estimates



$P(H)?$



Point Estimates



$P(H)?$

10 throws
7 heads
3 tails



Point Estimates



$P(H)?$

10 throws
7 heads
3 tails

$$P(H) = \frac{7}{10}$$



Point Estimates

	x	y
0		
1		
2		
$\vdots$		
50		

Point Estimates

	x	y
0		
1		
2		
$\vdots$		
50		



Point Estimates

	x	y
0		
1		
2		
$\vdots$		
50		

$$y = \beta_0 + \beta_1 x$$

$$\beta_0, \beta_1 = ??$$

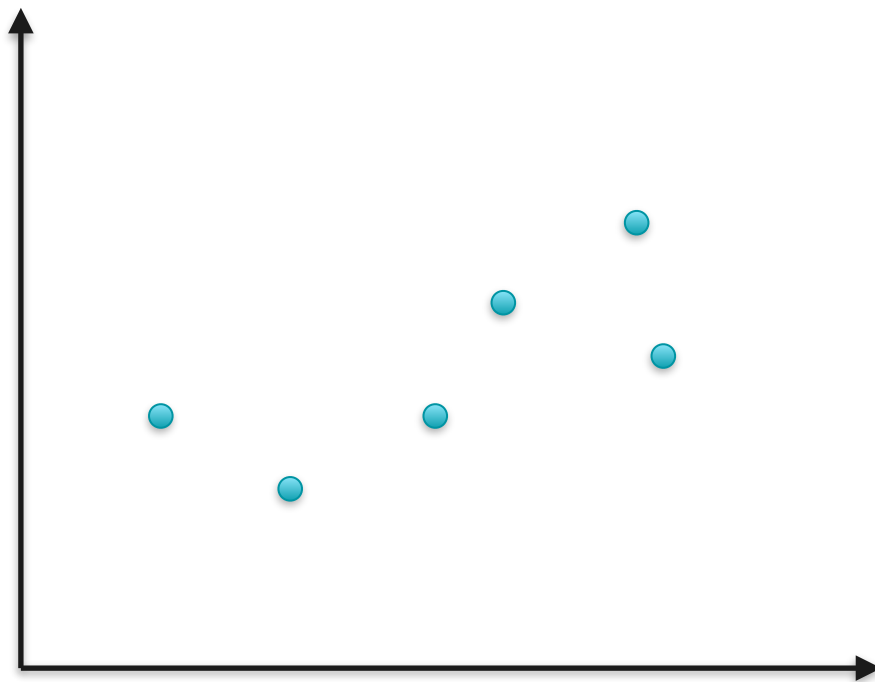


Point Estimates

	x	y
0		
1		
2		
$\vdots$		
50		

$$y = \beta_0 + \beta_1 x$$

$$\beta_0, \beta_1 = ??$$

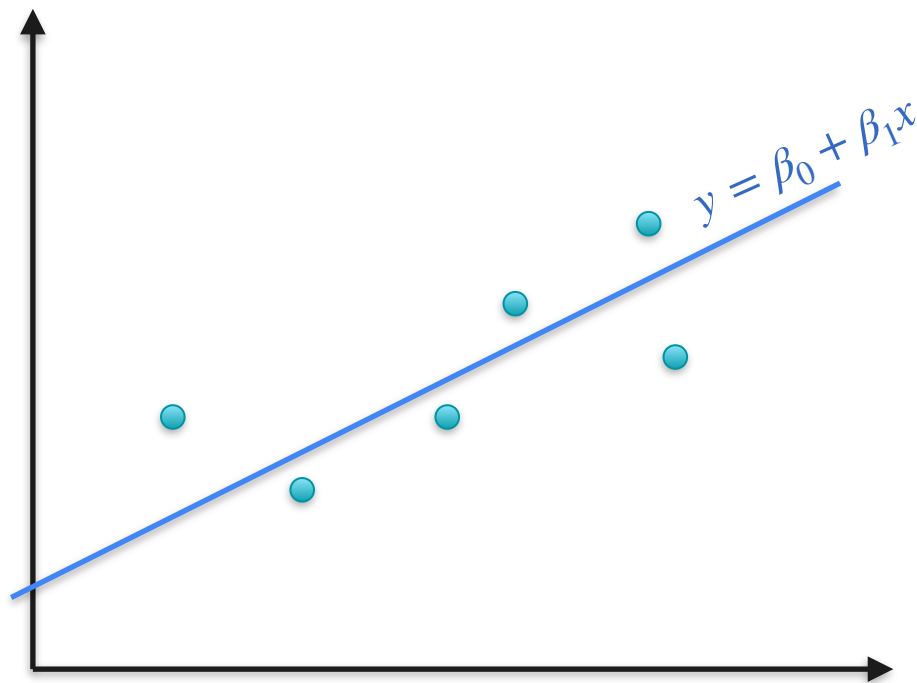


Point Estimates

	x	y
0		
1		
2		
$\vdots$		
50		

$$y = \beta_0 + \beta_1 x$$

$$\beta_0, \beta_1 = ??$$





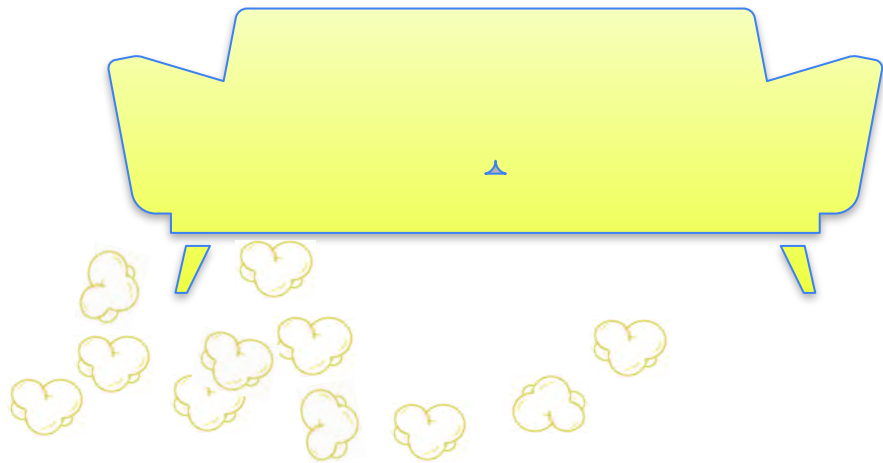
DeepLearning.AI

Point Estimation

Maximum Likelihood Estimation: Motivation

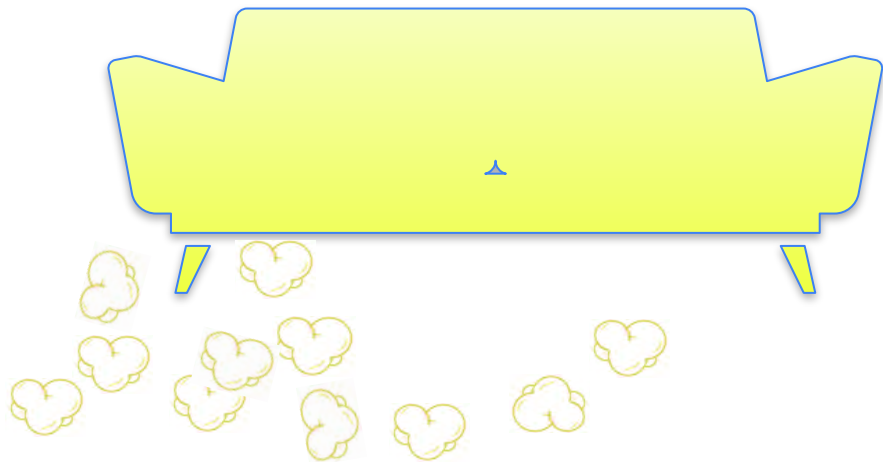
There's Popcorn on the Floor. What Happened?

There's Popcorn on the Floor. What Happened?

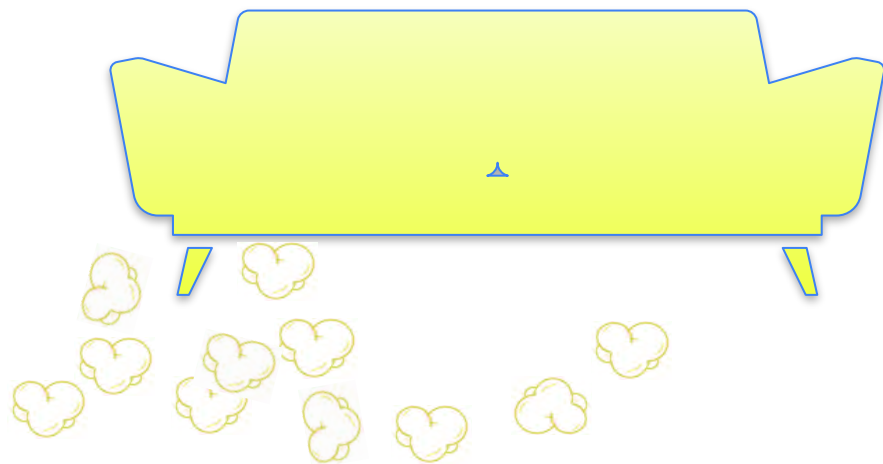


There's Popcorn on the Floor. What Happened?

Movies



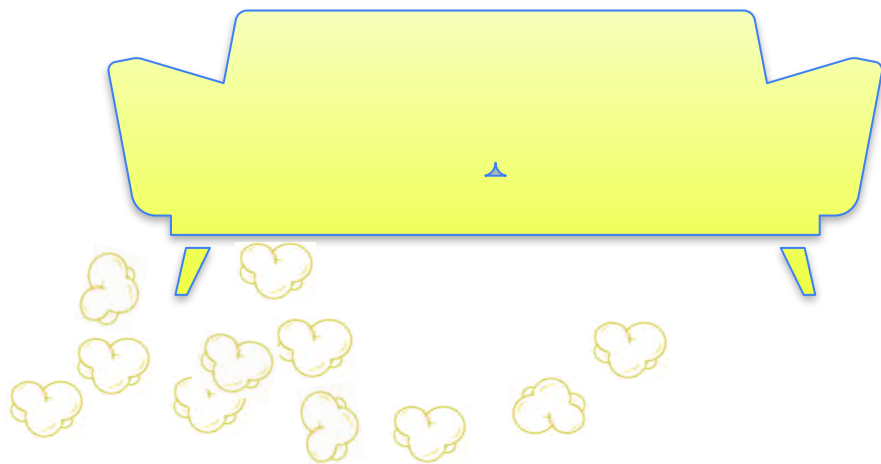
There's Popcorn on the Floor. What Happened?



Movies 

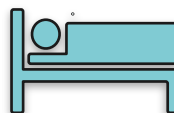
Board Games 

There's Popcorn on the Floor. What Happened?



Movies 

Board Games 

Nap 

Quiz

What do you think happened?

- A. People were watching a movie
- B. People were playing boardgames
- C. People were taking a nap

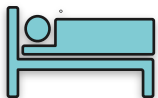
There's Popcorn on the Floor. What Happened?



Movies



Board Games



Nap



There's Popcorn on the Floor. What Happened?

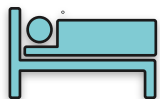


Movies

High



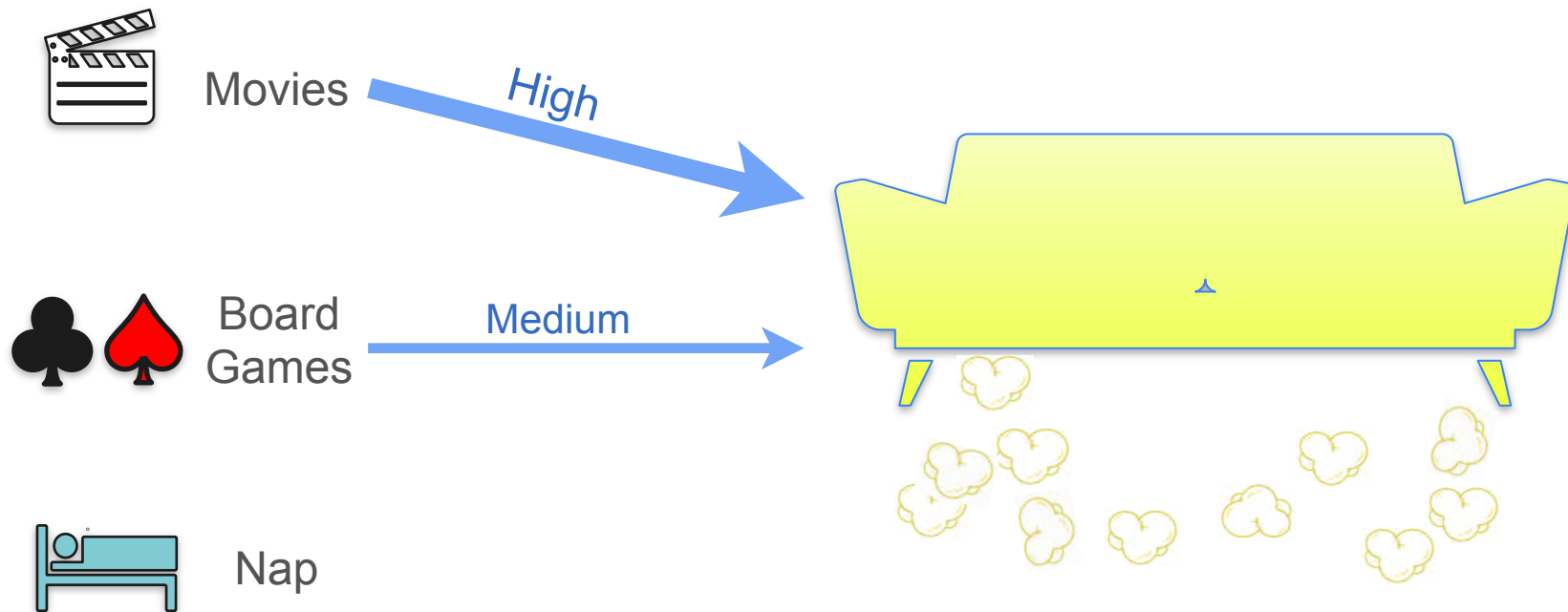
Board
Games



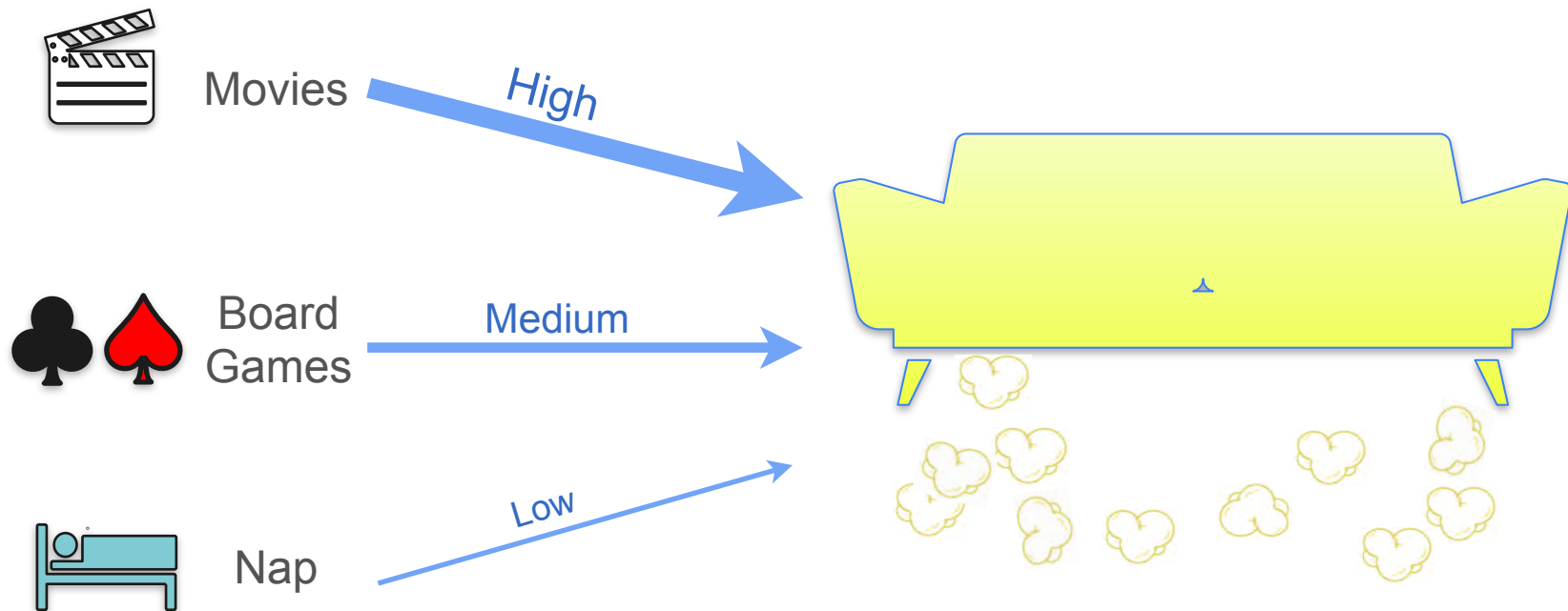
Nap



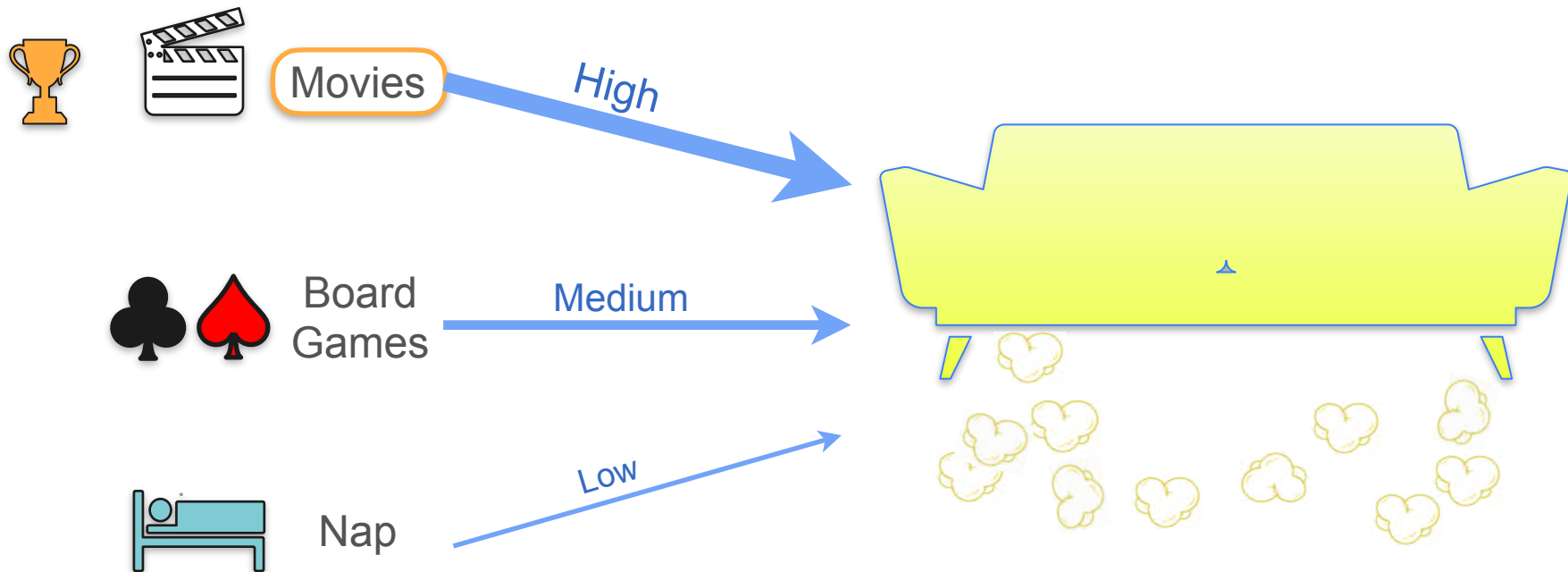
There's Popcorn on the Floor. What Happened?



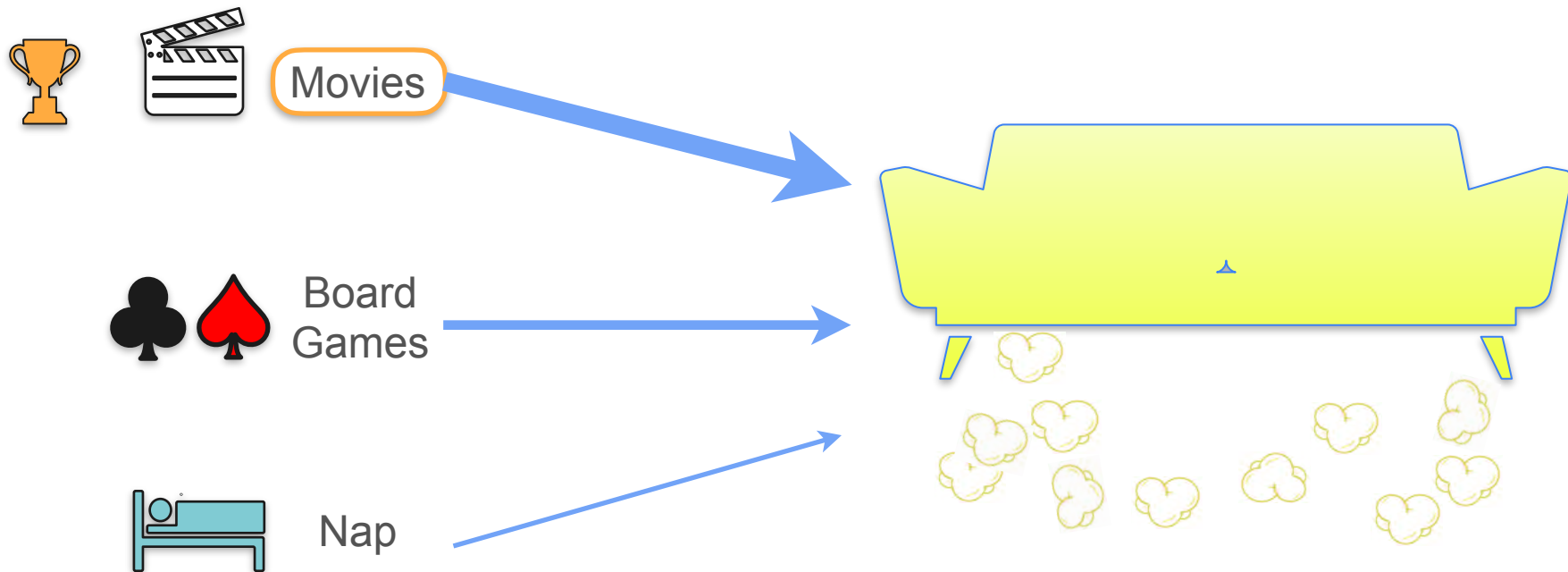
There's Popcorn on the Floor. What Happened?



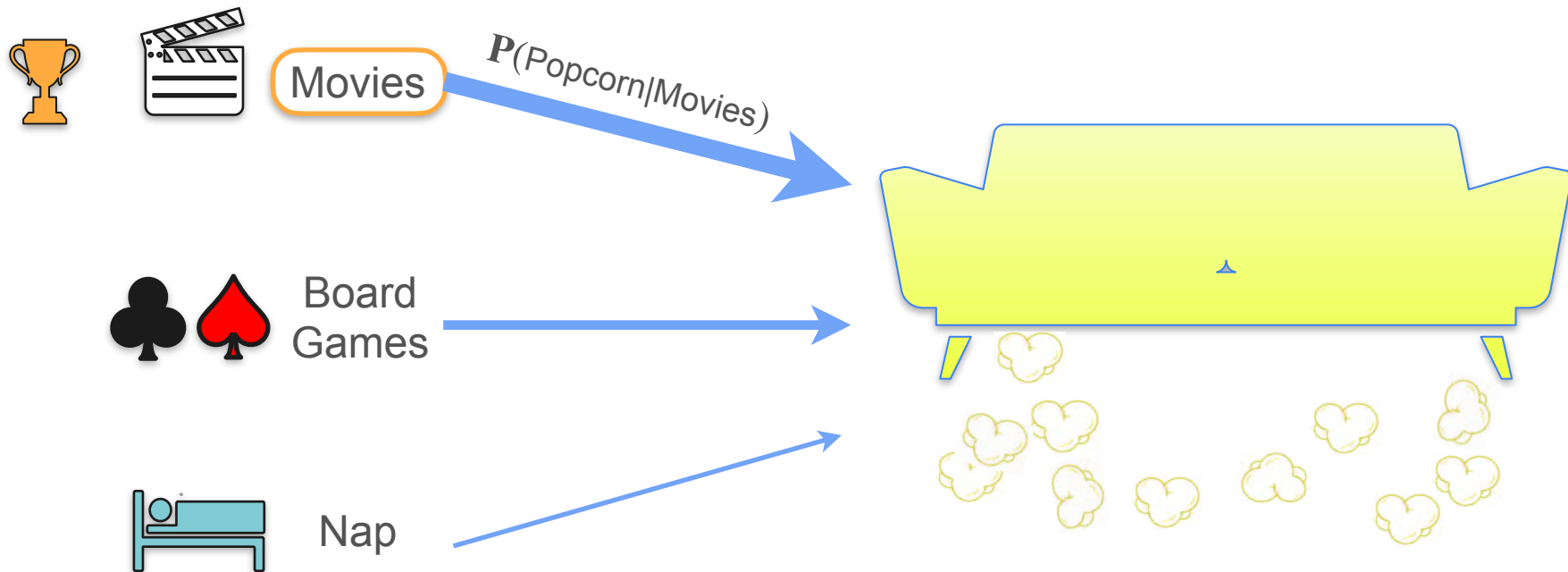
There's Popcorn on the Floor. What Happened?



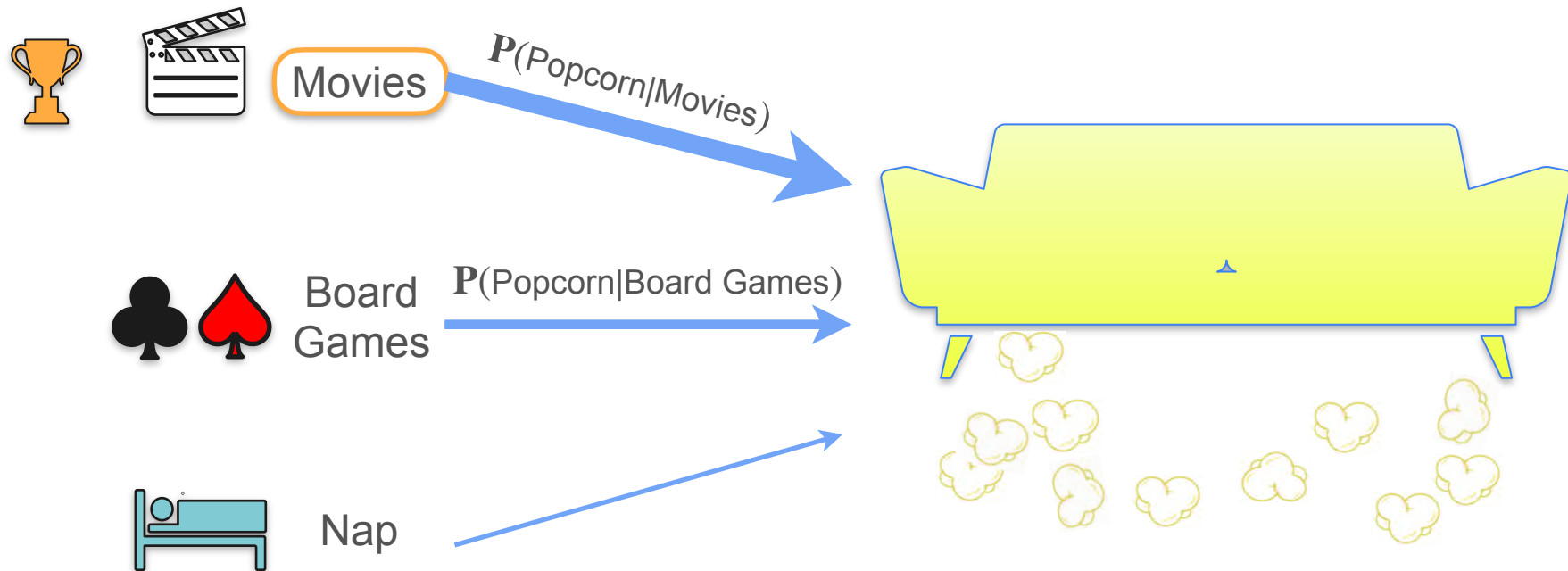
There's Popcorn on the Floor. What Happened?



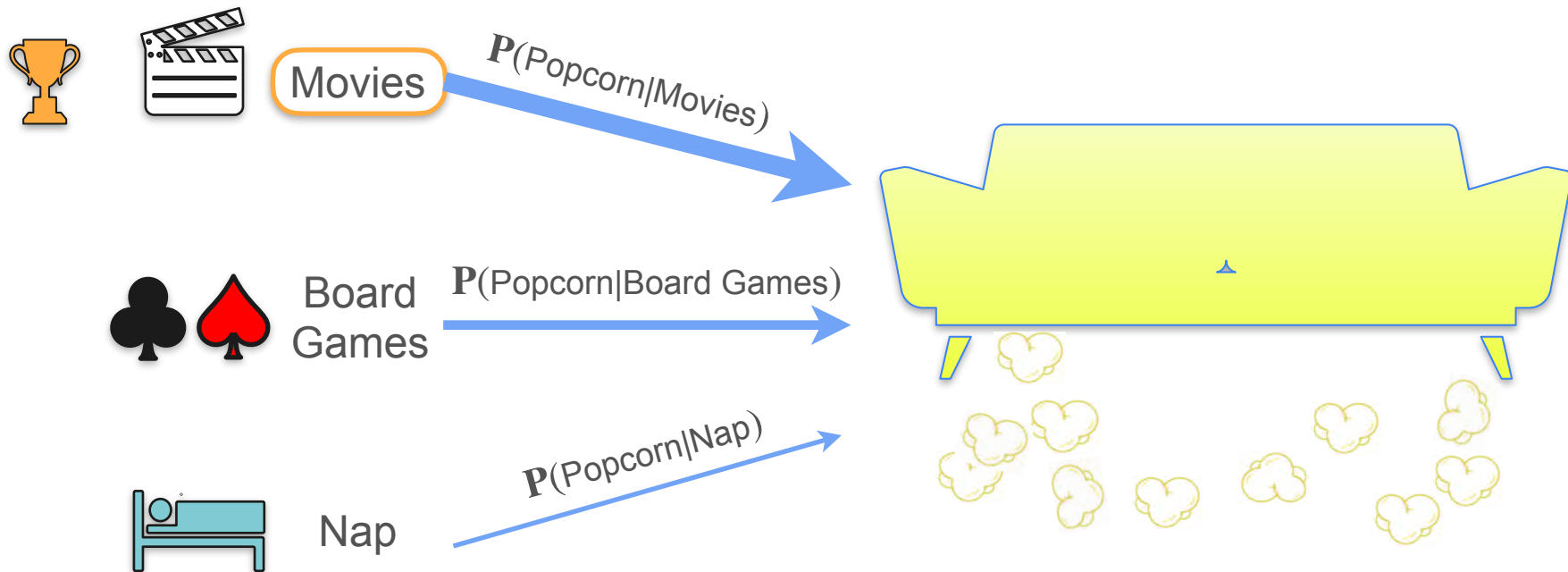
There's Popcorn on the Floor. What Happened?



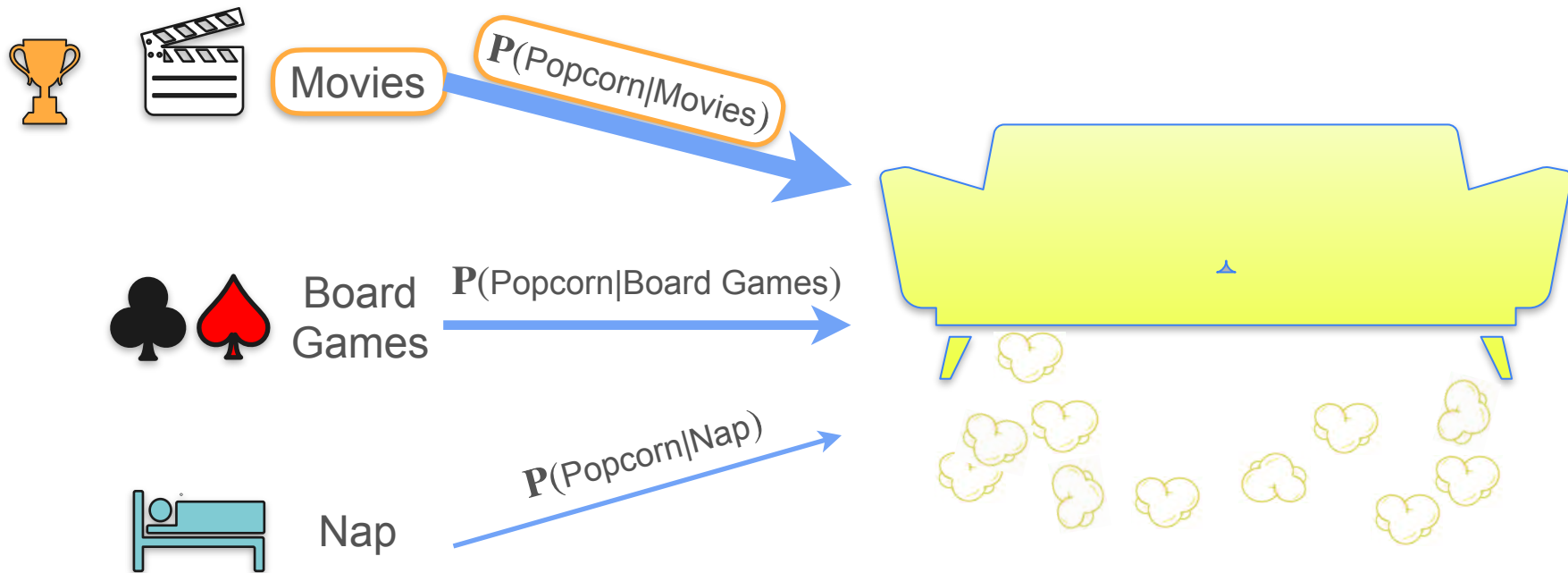
There's Popcorn on the Floor. What Happened?



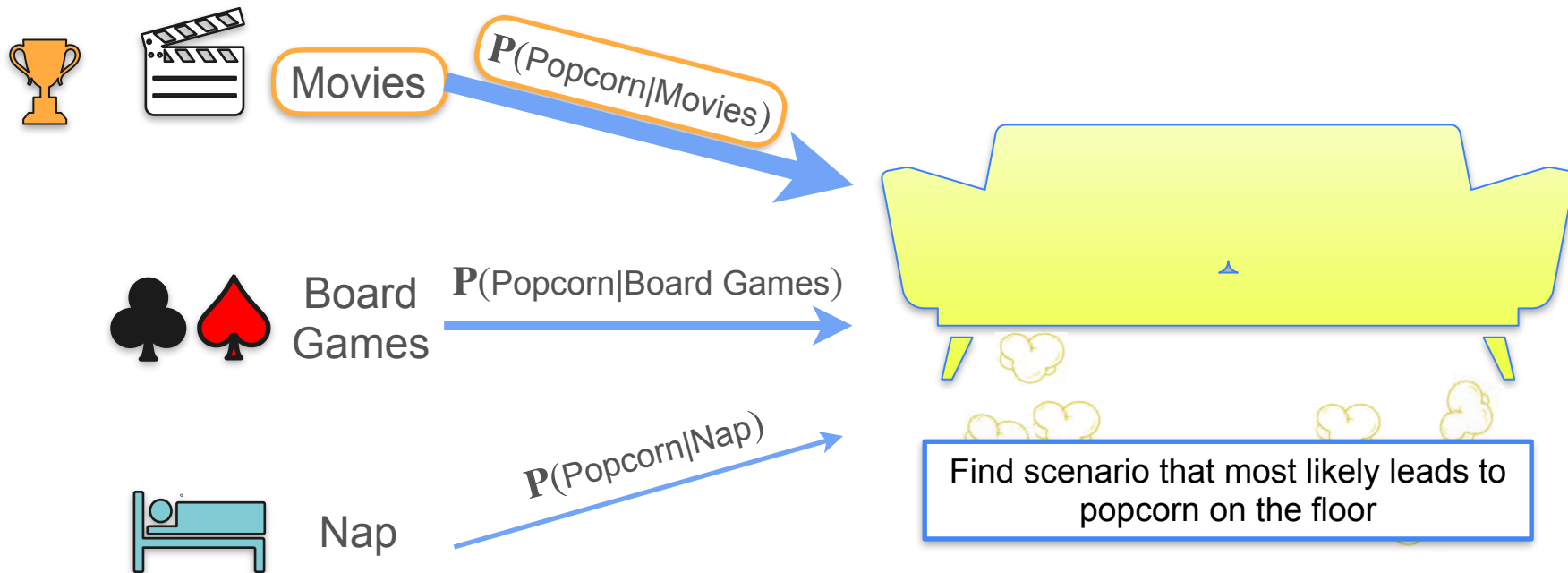
There's Popcorn on the Floor. What Happened?



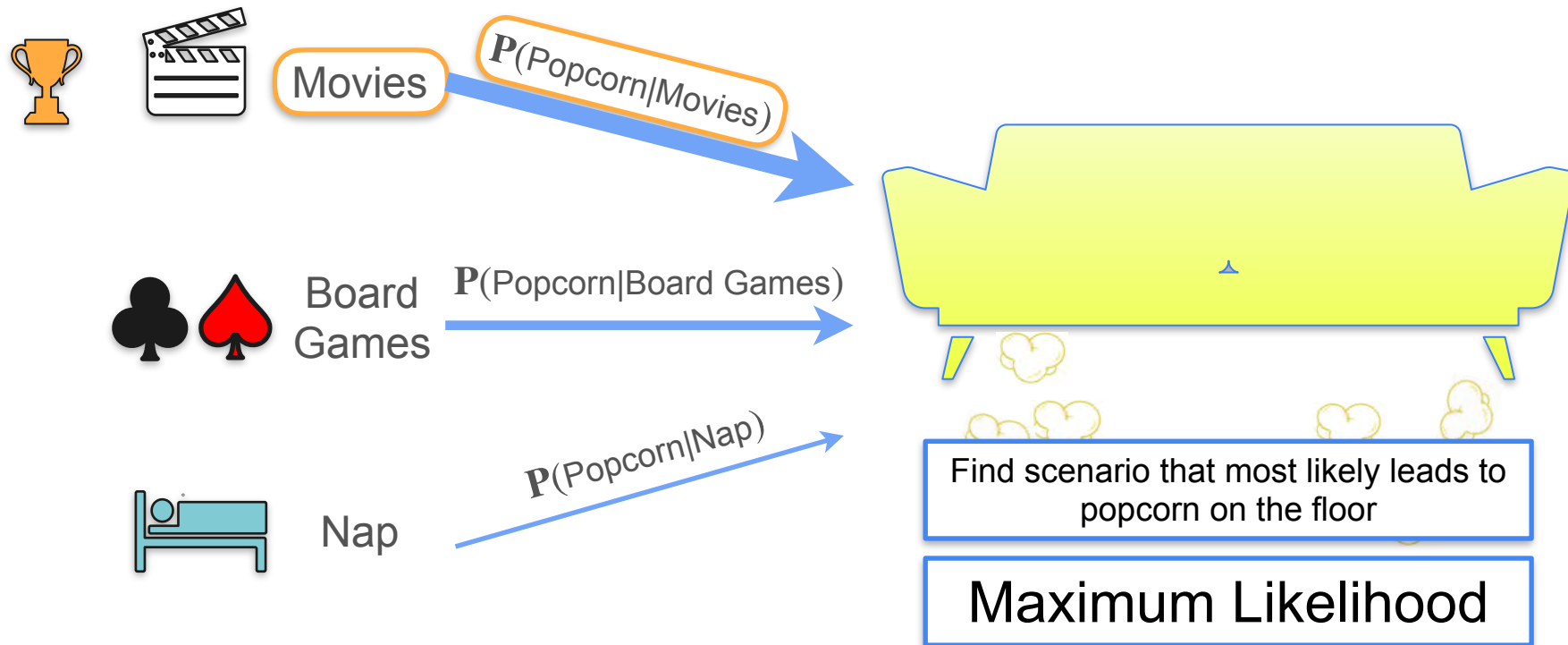
There's Popcorn on the Floor. What Happened?



There's Popcorn on the Floor. What Happened?



There's Popcorn on the Floor. What Happened?



Maximum Likelihood

Maximum Likelihood



Data

Maximum Likelihood



Model 1



Data

Maximum Likelihood



Model 1



Model 2



Data

Maximum Likelihood



Model 1



Model 2

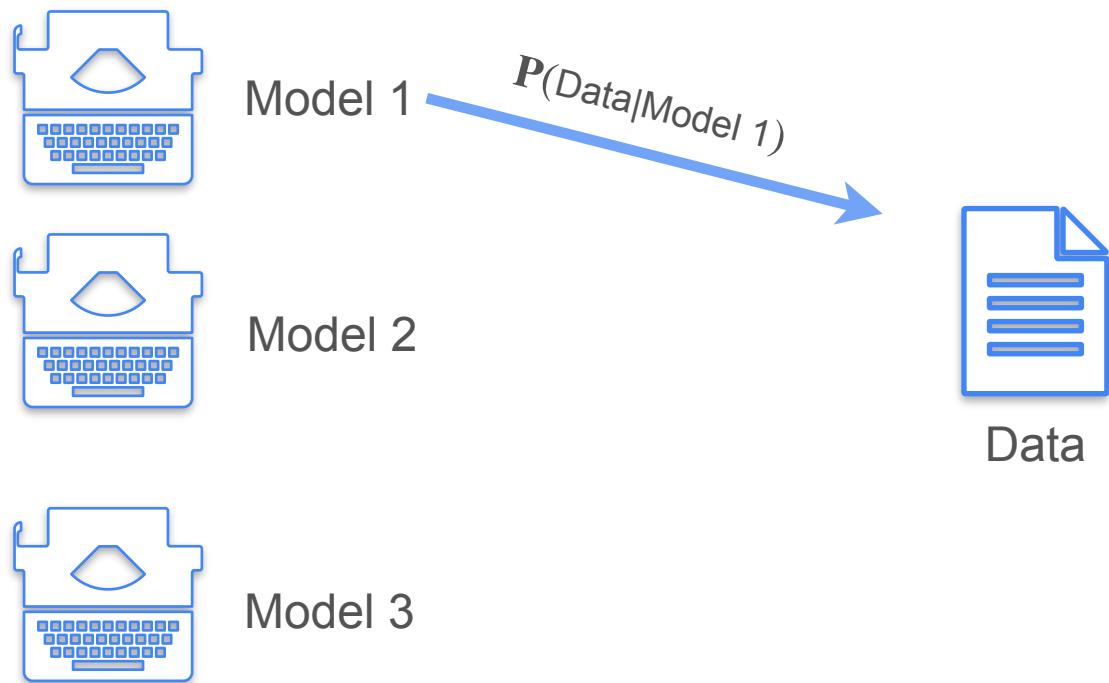


Model 3

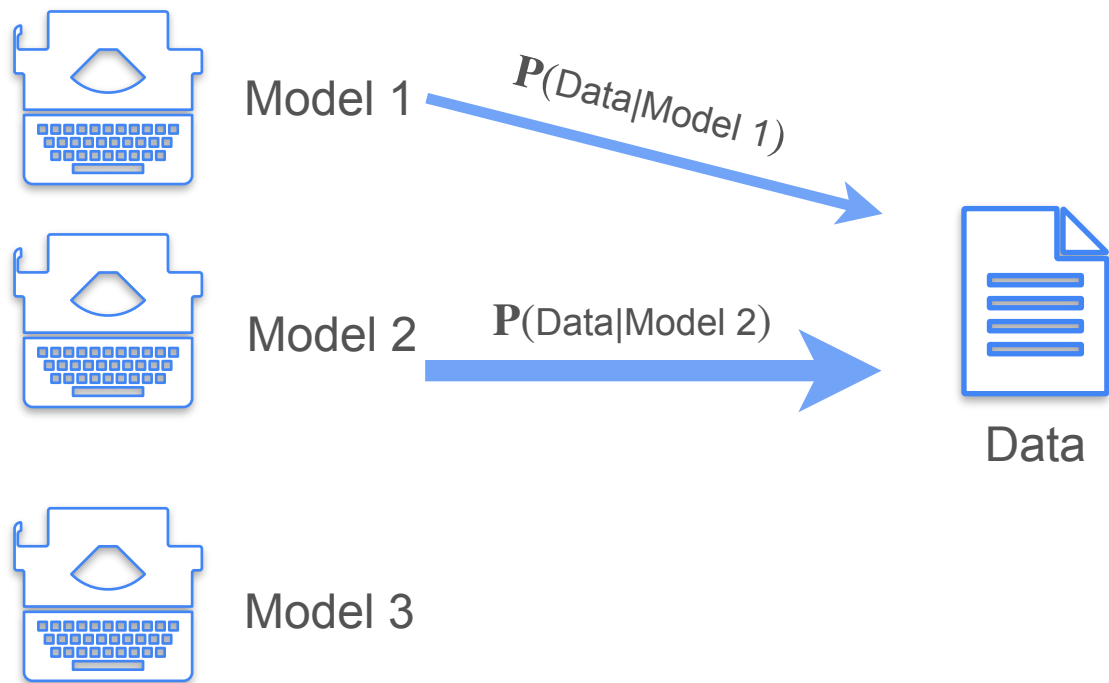


Data

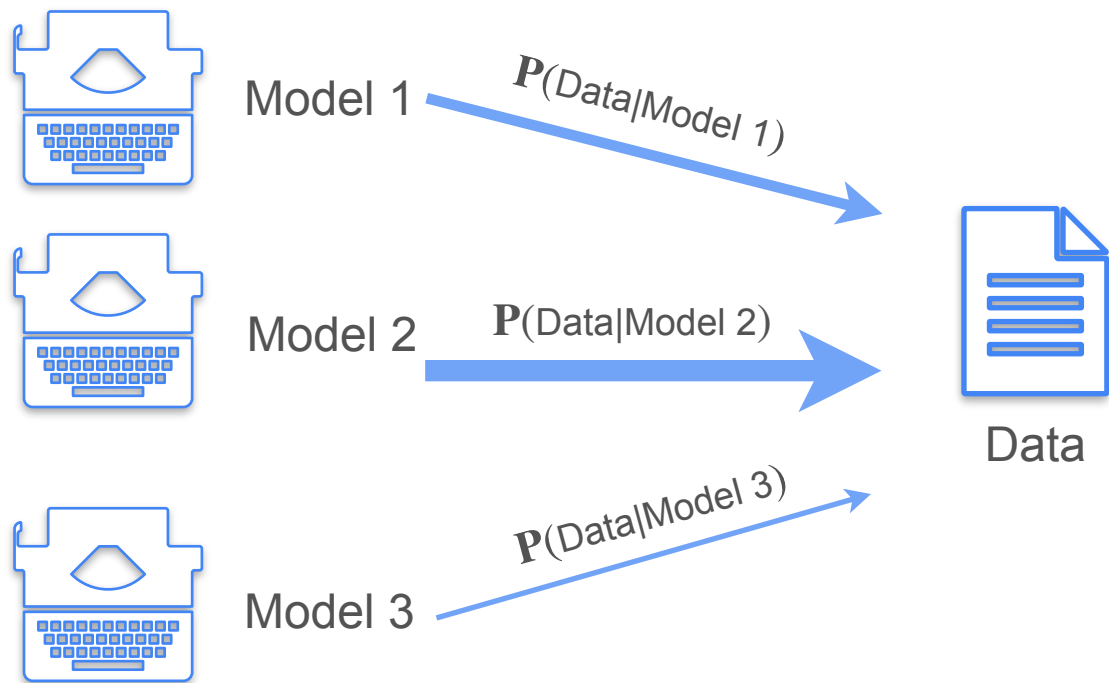
Maximum Likelihood



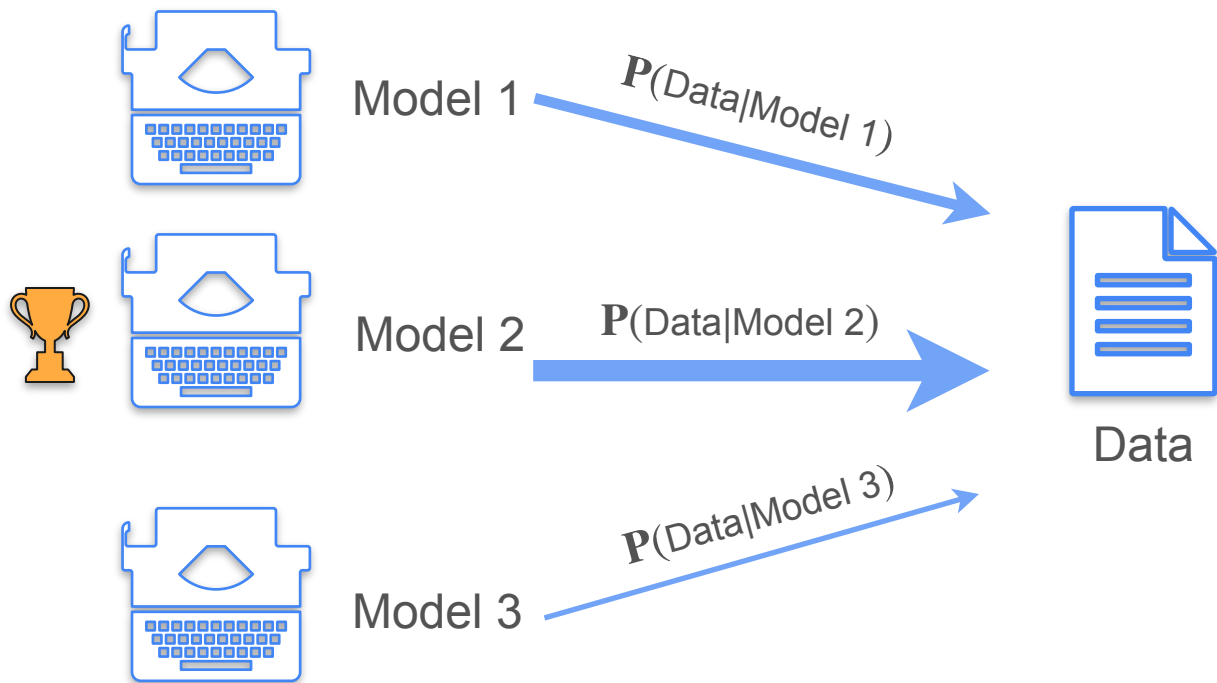
Maximum Likelihood



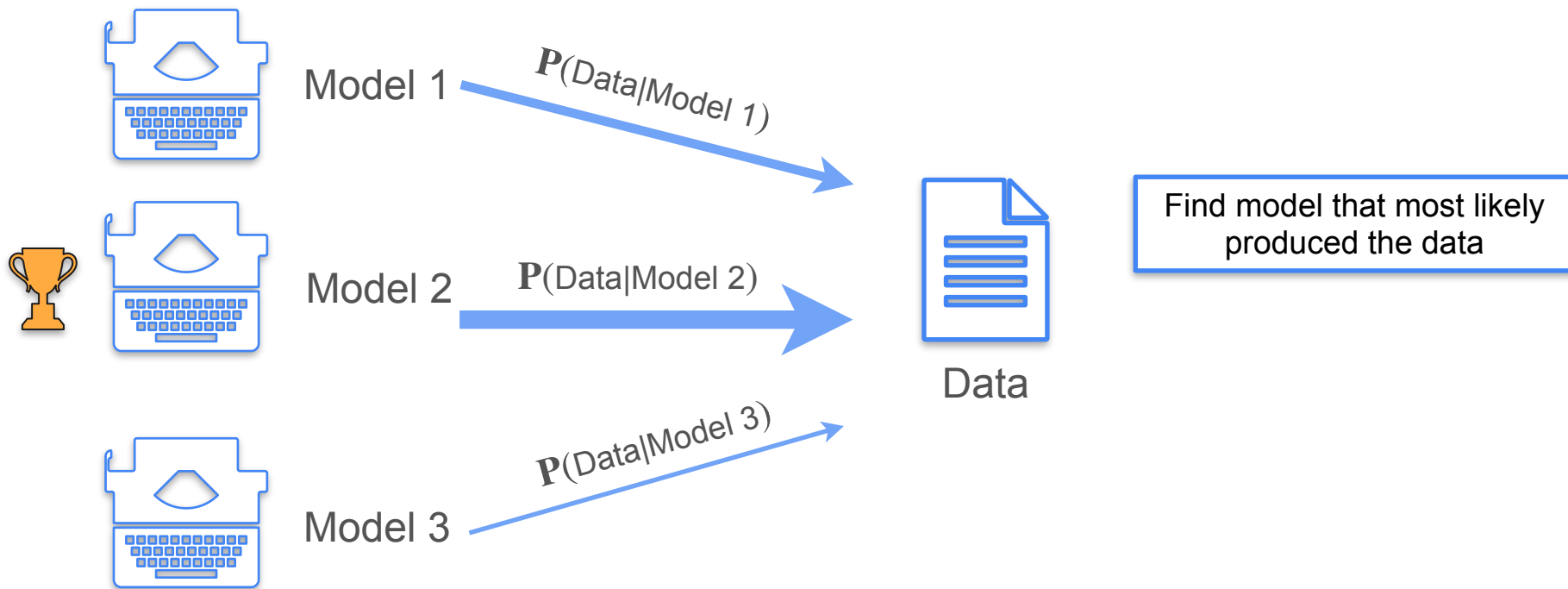
Maximum Likelihood



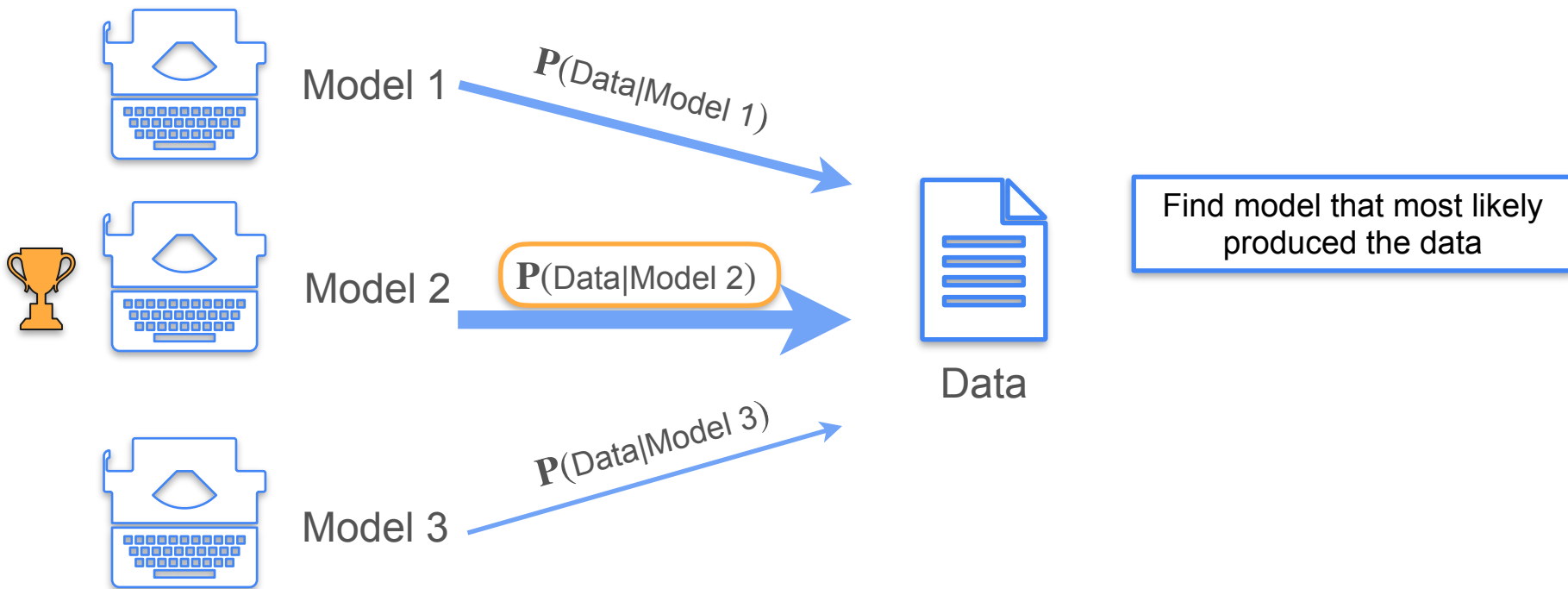
Maximum Likelihood



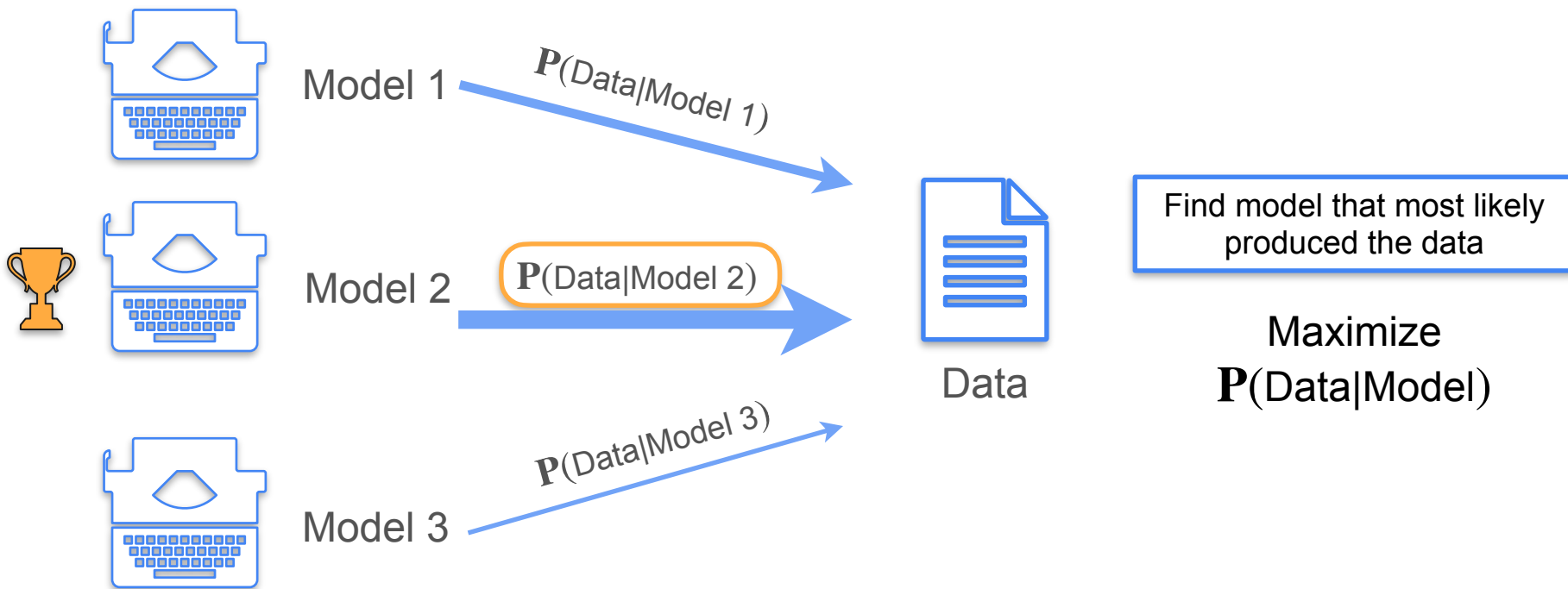
Maximum Likelihood



Maximum Likelihood

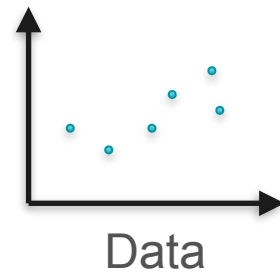


Maximum Likelihood

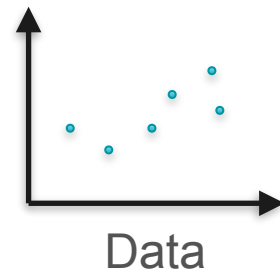
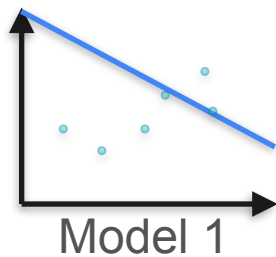


Example: Linear Regression

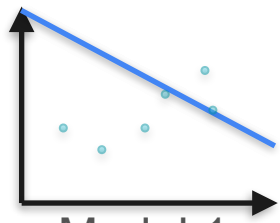
Example: Linear Regression



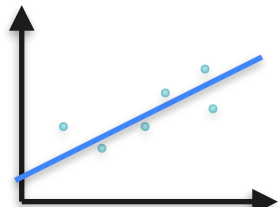
Example: Linear Regression



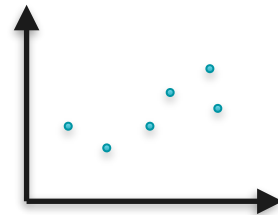
Example: Linear Regression



Model 1

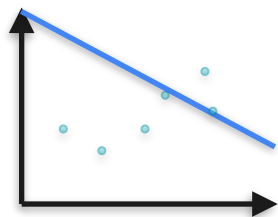


Model 2

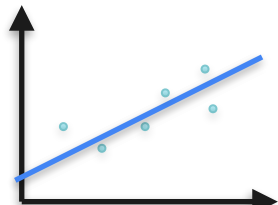


Data

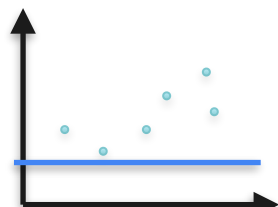
Example: Linear Regression



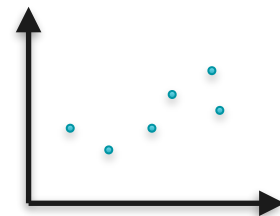
Model 1



Model 2

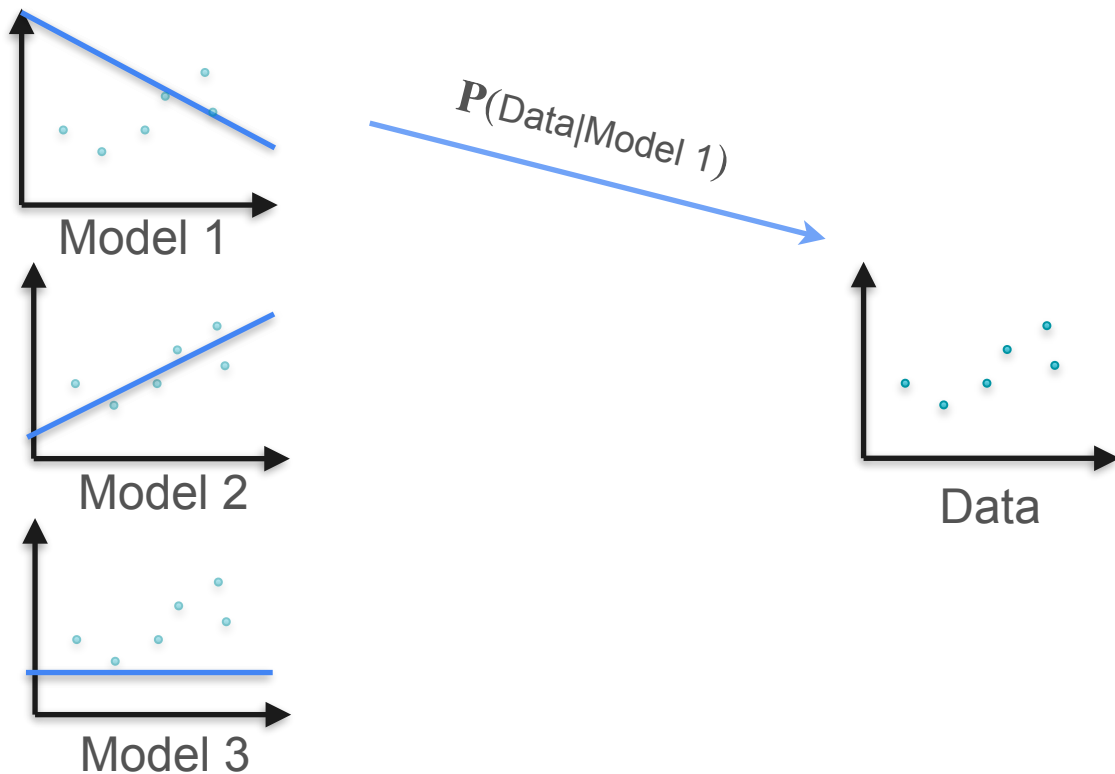


Model 3

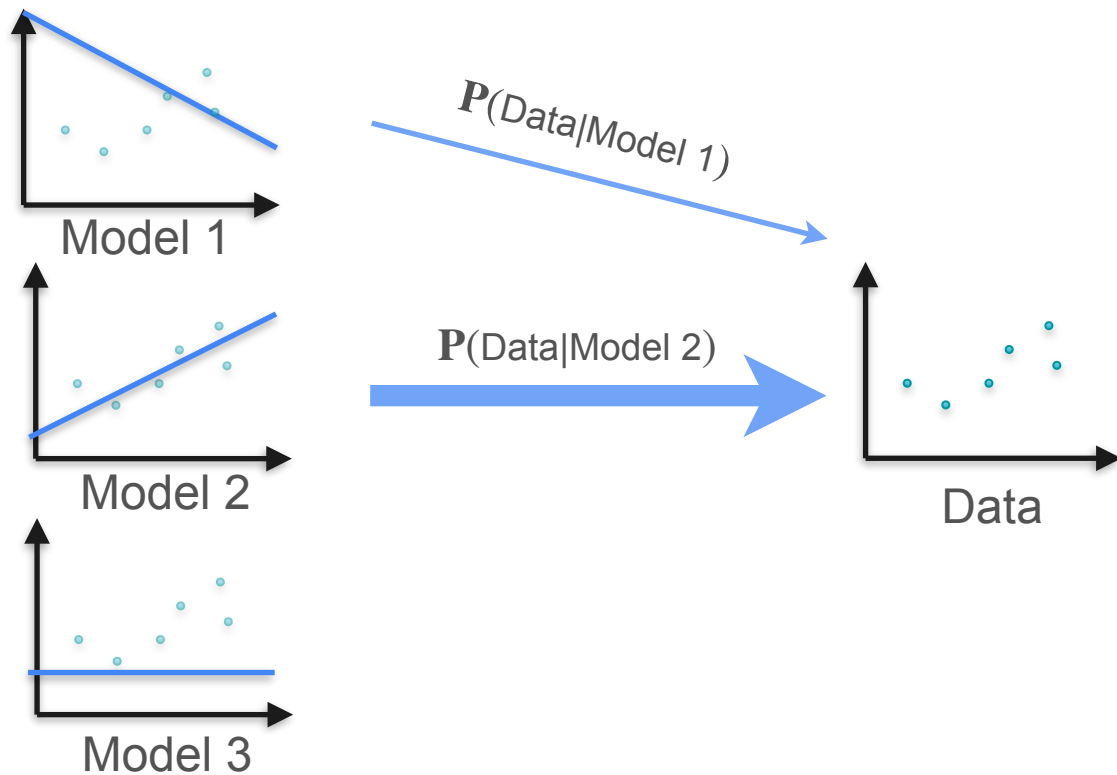


Data

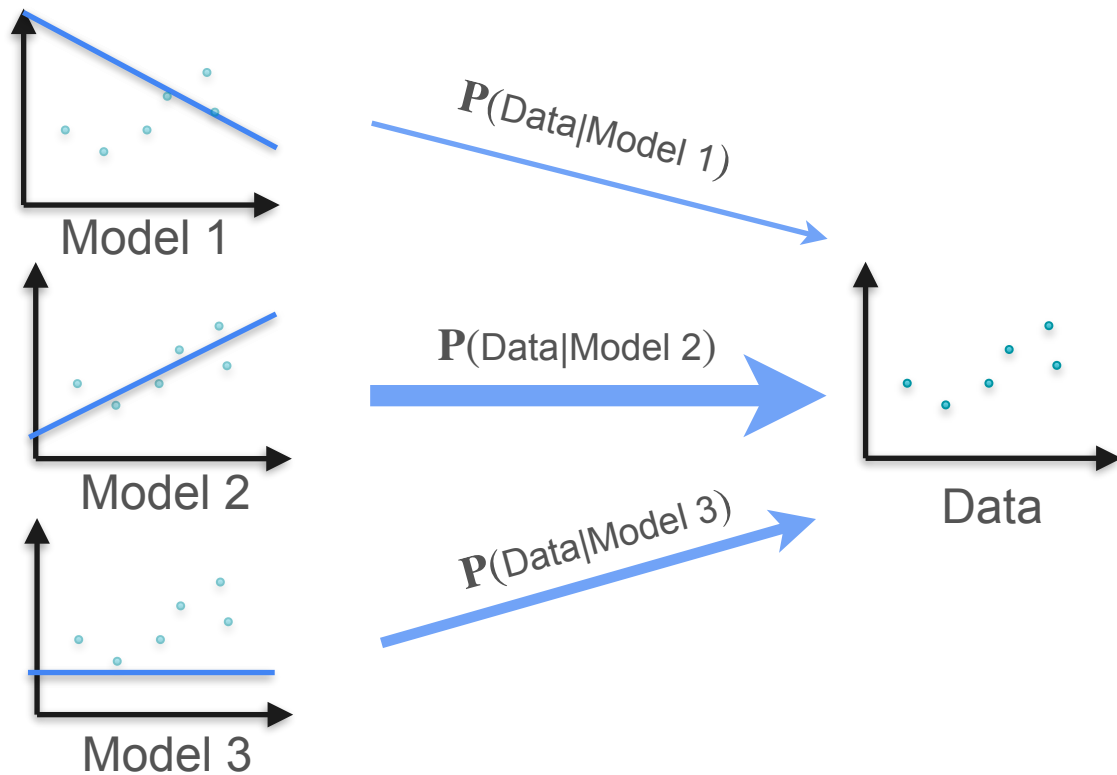
Example: Linear Regression



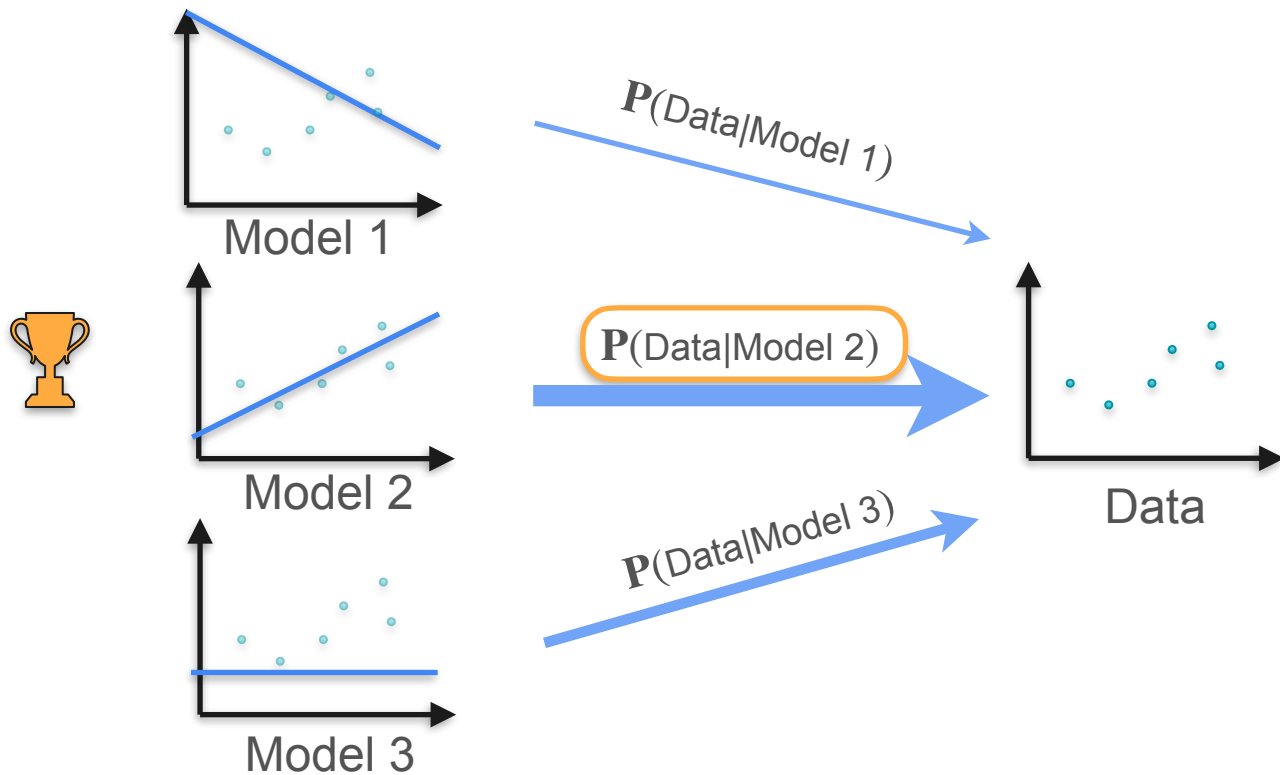
Example: Linear Regression



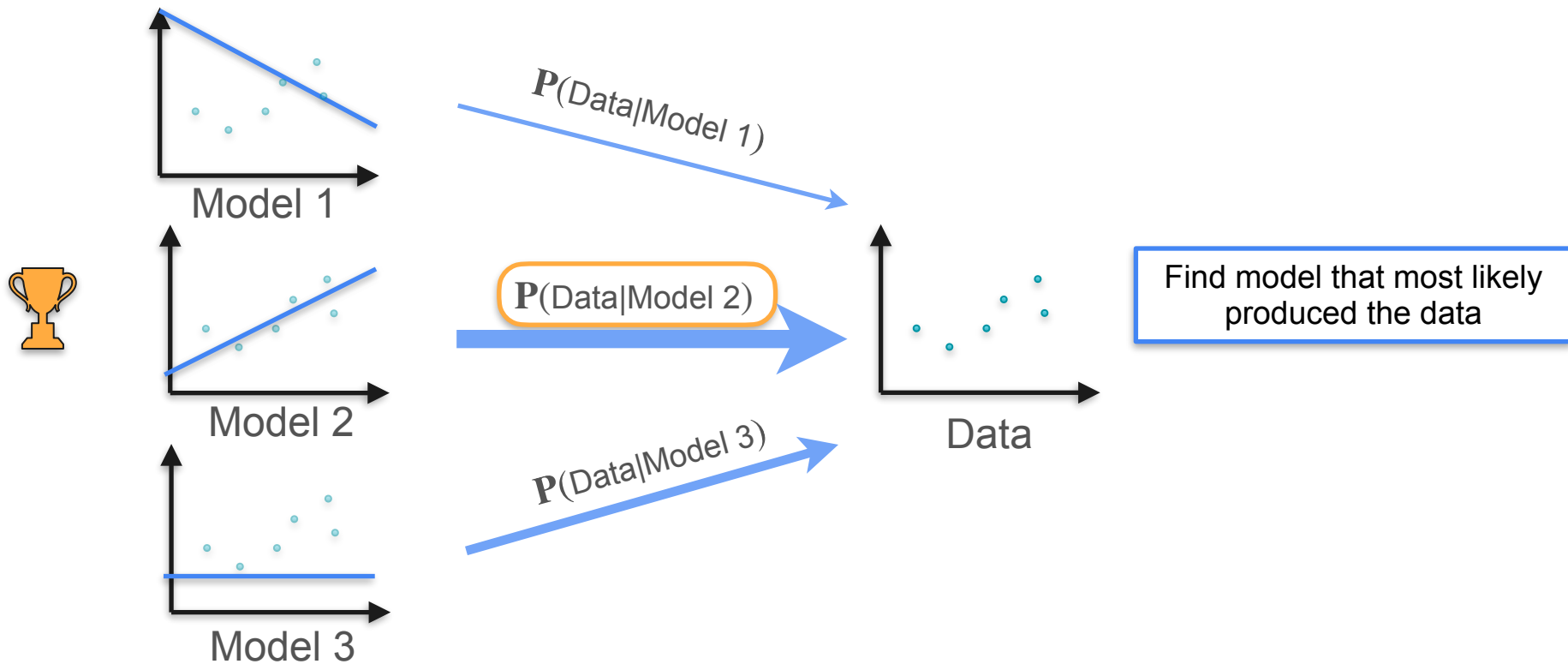
Example: Linear Regression



Example: Linear Regression



Example: Linear Regression





DeepLearning.AI

Point Estimation

MLE: Bernoulli Example

Maximum Likelihood: Bernoulli Example

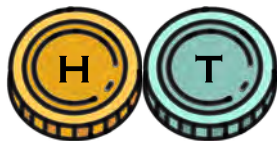
Maximum Likelihood: Bernoulli Example



Maximum Likelihood: Bernoulli Example



Coin 1



$$P(H) = 0.7$$

$$P(T) = 0.3$$

Coin 2



$$P(H) = 0.5$$

$$P(T) = 0.5$$

Coin 3



$$P(H) = 0.3$$

$$P(T) = 0.7$$

Quiz

Which of the coins is more likely?

A. Coin 1 ($P(H) = 0.7$)

B. Coin 2 ($P(H) = 0.5$)

C. Coin 3 ($P(H) = 0.3$)

Maximum Likelihood: Bernoulli Example



Maximum Likelihood: Bernoulli Example



Coin 1 0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.3 0.3 = 0.0051

Maximum Likelihood: Bernoulli Example



Coin 1 0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.3 0.3 = 0.0051

Coin 2 0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 = 0.0010

Maximum Likelihood: Bernoulli Example



Coin 1	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.3	0.3 = 0.0051
Coin 2	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5 = 0.0010
Coin 3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.30	0.7	0.7 = 0.00003

Maximum Likelihood: Bernoulli Example



Coin 1	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.3	$0.3 = 0.0051$
---------------	-----	-----	-----	-----	-----	-----	-----	-----	-----	----------------

Coin 2	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	$0.5 = 0.0010$
---------------	-----	-----	-----	-----	-----	-----	-----	-----	-----	----------------

Coin 3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.30	0.7	$0.7 = 0.00003$
---------------	-----	-----	-----	-----	-----	-----	-----	------	-----	-----------------

Maximum Likelihood: Bernoulli Example



Maximum Likelihood: Bernoulli Example



8H 2T



Maximum Likelihood: Bernoulli Example



Coin 1

$$P(H) = 0.7$$

8H 2T



Maximum Likelihood: Bernoulli Example



Coin 1

$$P(H) = 0.7$$



Coin 2

$$P(H) = 0.5$$

8H 2T



Maximum Likelihood: Bernoulli Example



Coin 1

$$P(H) = 0.7$$



Coin 2

$$P(H) = 0.5$$



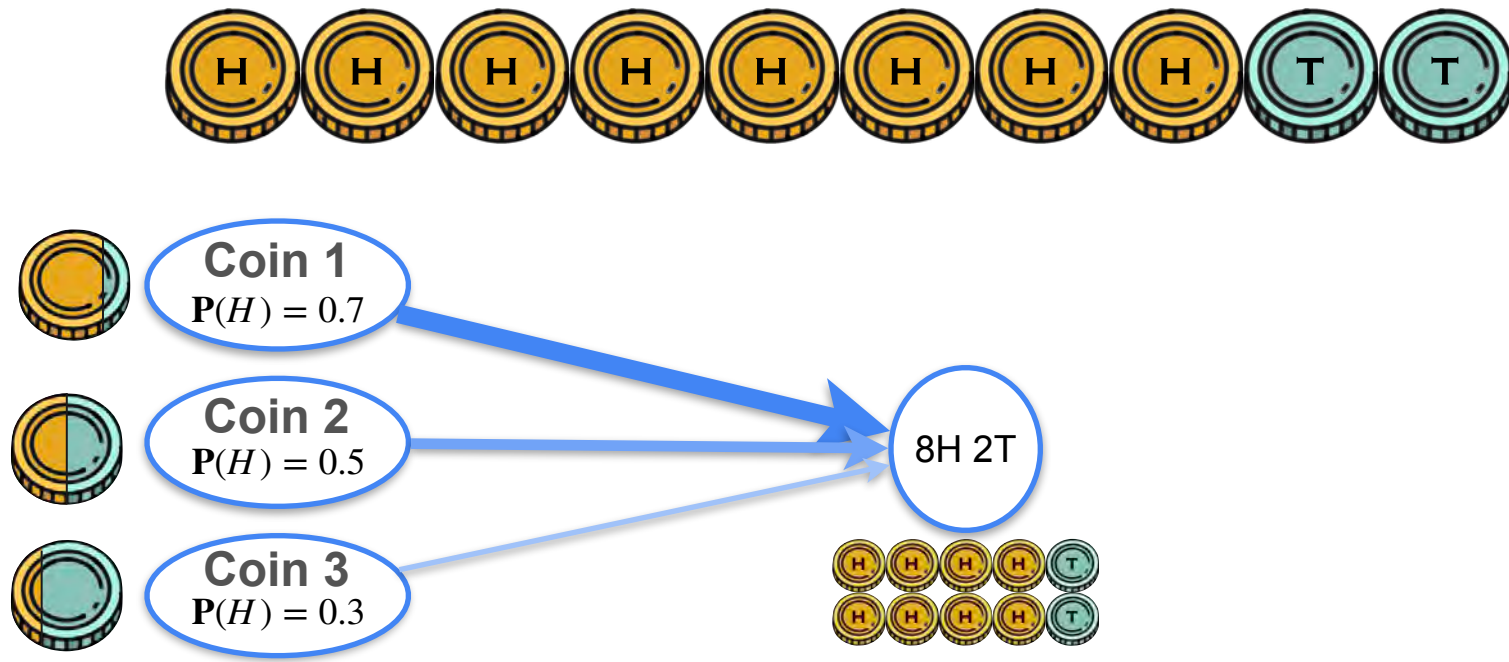
Coin 3

$$P(H) = 0.3$$

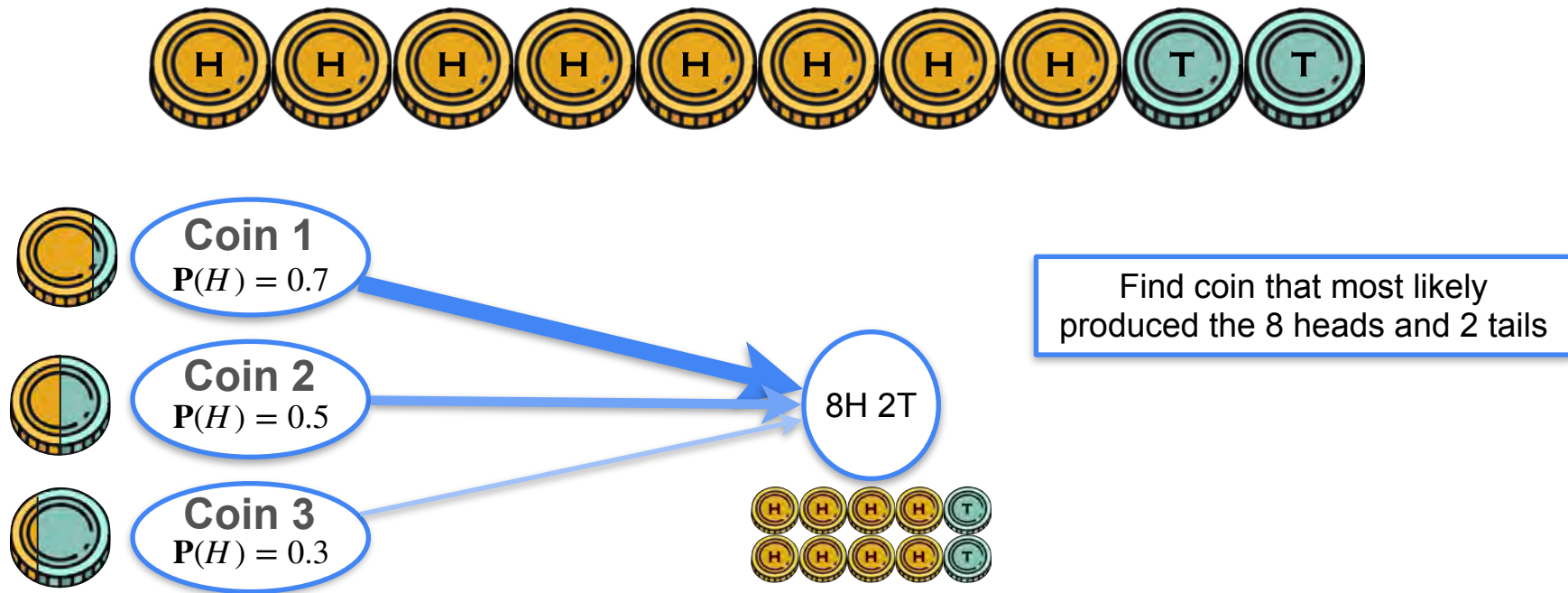
8H 2T



Maximum Likelihood: Bernoulli Example



Maximum Likelihood: Bernoulli Example



Maximum Likelihood: Bernoulli Example



Coin 1

$$P(H) = 0.7$$



Coin 2

$$P(H) = 0.5$$



Coin 3

$$P(H) = 0.3$$

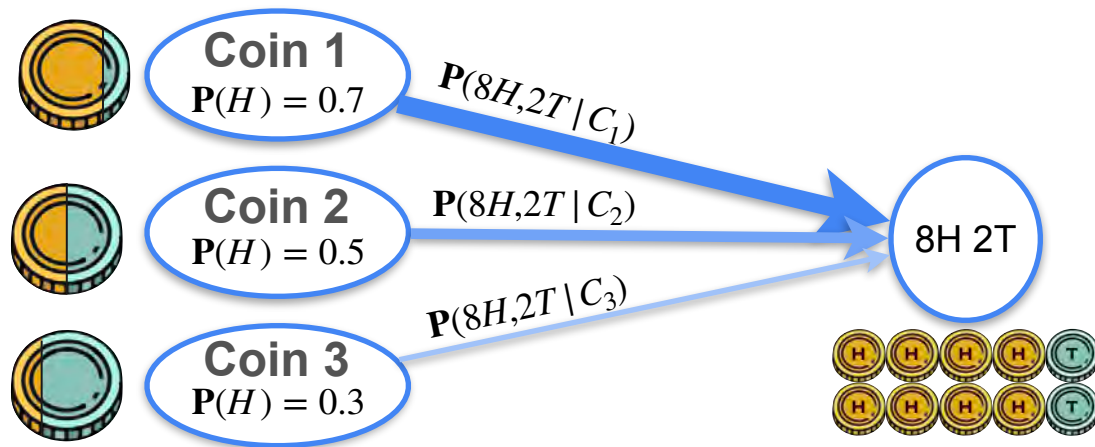
8H 2T



Find coin that most likely
produced the 8 heads and 2 tails

Maximize
 $P(8H, 2T \mid \text{Coin})$

Maximum Likelihood: Bernoulli Example



Find coin that most likely produced the 8 heads and 2 tails

Maximize
 $P(8H, 2T | \text{Coin})$

Maximum Likelihood: Bernoulli Example



Coin 1

$$P(H) = 0.7$$

$$P(8H, 2T | C_1) = 0.0051$$



Coin 2

$$P(H) = 0.5$$

$$P(8H, 2T | C_2) = 0.0010$$



Coin 3

$$P(H) = 0.3$$

$$P(8H, 2T | C_3) = 0.00003$$

8H 2T



Find coin that most likely produced the 8 heads and 2 tails

Maximize
 $P(8H, 2T | \text{Coin})$

Maximum Likelihood: Bernoulli Example



Coin 1

$$P(H) = 0.7$$

$$P(8H, 2T | C_1) = 0.0051$$



Coin 2

$$P(H) = 0.5$$

$$P(8H, 2T | C_2) = 0.0010$$



Coin 3

$$P(H) = 0.3$$

$$P(8H, 2T | C_3) = 0.00003$$

8H 2T



Find coin that most likely produced the 8 heads and 2 tails

Maximize
 $P(8H, 2T | \text{Coin})$

Maximum Likelihood: Bernoulli Example

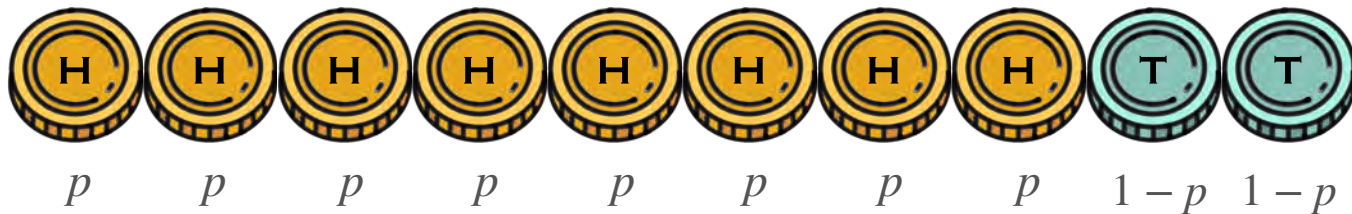


Maximum Likelihood: Bernoulli Example



Can you do any better?

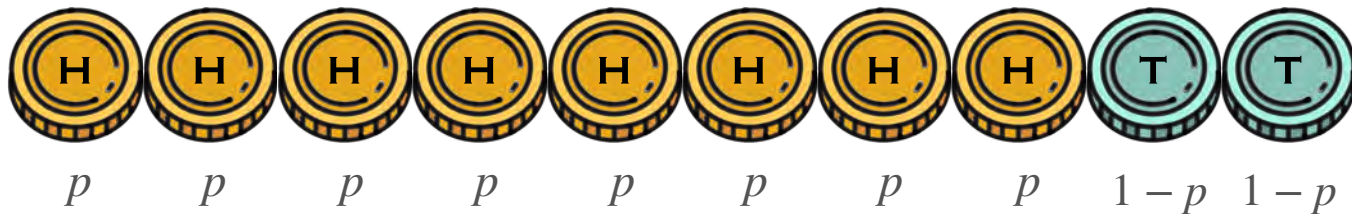
Maximum Likelihood: Bernoulli Example



Can you do any better?

$$p = \mathbf{P}(H)$$

Maximum Likelihood: Bernoulli Example

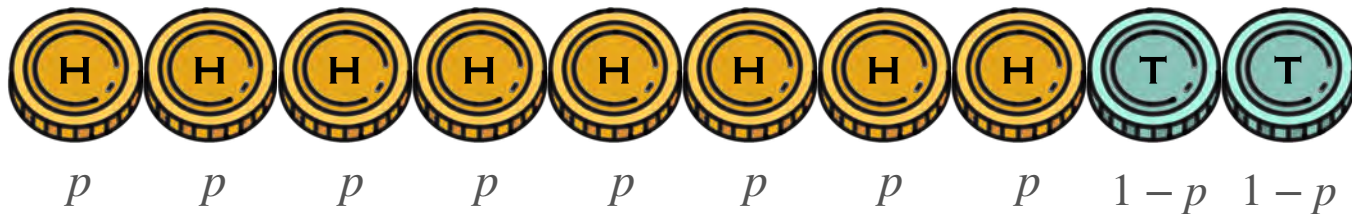


Can you do any better?

$$p = \mathbf{P}(H)$$

$$p^8(1 - p)^2$$

Maximum Likelihood: Bernoulli Example



Can you do any better?

$$p = \mathbf{P}(H)$$

$$p^8(1 - p)^2$$

You want p that maximizes the chances of seeing 8H

Maximum Likelihood: Bernoulli Example



$$p = \mathbf{P}(H)$$

$$p^8(1 - p)^2$$

You want p that maximizes the chances of seeing 8H

Maximum Likelihood: Bernoulli Example



$$p = \mathbf{P}(H) \quad \text{Likelihood} \quad L(p; 8H) = p^8(1-p)^2$$

You want p that maximizes the chances of seeing 8H

Maximum Likelihood: Bernoulli Example



$p = \mathbf{P}(H)$ Likelihood $L(p; 8H) = p^8(1 - p)^2$

You want p that maximizes the chances of seeing 8H

Function of
 p

Maximum Likelihood: Bernoulli Example



$p = \mathbf{P}(H)$ Likelihood $L(p; 8H) = p^8(1 - p)^2$

Function of
 p

You want p that maximizes the chances of seeing 8H

$$\log \left((p^8(1 - p)^2) \right)$$

Maximum Likelihood: Bernoulli Example



$p = \mathbf{P}(H)$ Likelihood $L(p; 8H) = p^8(1 - p)^2$

Function of
 p

You want p that maximizes the chances of seeing 8H

$$\log \left((p^8(1 - p)^2) \right) = 8 \log(p) + 2 \log(1 - p)$$

Maximum Likelihood: Bernoulli Example



$p = \mathbf{P}(H)$ Likelihood $L(p; 8H) = p^8(1 - p)^2$

Function of
 p

You want p that maximizes the chances of seeing 8H

Log-likelihood $\ell(p; 8H) = \log(p^8(1 - p)^2) = 8 \log(p) + 2 \log(1 - p)$

Maximum Likelihood: Bernoulli Example



$p = \mathbf{P}(H)$ Likelihood $L(p; 8H) = p^8(1 - p)^2$

Function of
 p

You want p that maximizes the chances of seeing 8H

Log-likelihood $\ell(p; 8H) = \log((p^8(1 - p)^2)) = 8 \log(p) + 2 \log(1 - p)$

$$\frac{d}{dp} (8 \log(p) + 2 \log(1 - p))$$

Maximum Likelihood: Bernoulli Example



$p = \mathbf{P}(H)$ Likelihood $L(p; 8H) = p^8(1 - p)^2$

Function of
 p

You want p that maximizes the chances of seeing 8H

Log-likelihood $\ell(p; 8H) = \log((p^8(1 - p)^2)) = 8 \log(p) + 2 \log(1 - p)$

$$\frac{d}{dp} (8 \log(p) + 2 \log(1 - p)) = \frac{8}{p} + \frac{2}{1 - p}(-1)$$

Maximum Likelihood: Bernoulli Example



$p = \mathbf{P}(H)$ Likelihood $L(p; 8H) = p^8(1-p)^2$

Function of
 p

You want p that maximizes the chances of seeing 8H

Log-likelihood $\ell(p; 8H) = \log((p^8(1-p)^2)) = 8 \log(p) + 2 \log(1-p)$

$$\frac{d}{dp} (8 \log(p) + 2 \log(1-p)) = \frac{8}{p} + \frac{2}{1-p}(-1) = 0 \Rightarrow \hat{p} = \frac{8}{10}$$

The General Case

Maximum Likelihood: Bernoulli Example

n coins

k heads

Maximum Likelihood: Bernoulli Example

n coins
 k heads



Maximum Likelihood: Bernoulli Example

n coins
 k heads



$$\mathbf{X} = (X_1, \dots, X_n)$$

$$X_i \stackrel{i.i.d}{\sim} \text{Bernoulli}(p)$$

Maximum Likelihood: Bernoulli Example

n coins
 k heads



$$\mathbf{X} = (X_1, \dots, X_n)$$

$$X_i \stackrel{i.i.d}{\sim} \text{Bernoulli}(p)$$

Likelihood

$$L(p; \mathbf{x}) = P_p(\mathbf{X} = \mathbf{x}) = \prod_{i=1}^n p_{X_i}(x_i)$$

Maximum Likelihood: Bernoulli Example

n coins
 k heads



$$\mathbf{X} = (X_1, \dots, X_n)$$

$$X_i \stackrel{i.i.d}{\sim} \text{Bernoulli}(p)$$

Likelihood

$$L(p; \mathbf{x}) = P_p(\mathbf{X} = \mathbf{x}) = \prod_{i=1}^n p_{X_i}(x_i) = \prod_{i=1}^n p^{x_i}(1-p)^{1-x_i}$$

Maximum Likelihood: Bernoulli Example

n coins
 k heads



$$\mathbf{X} = (X_1, \dots, X_n)$$

$$X_i \stackrel{i.i.d}{\sim} \text{Bernoulli}(p)$$

Likelihood

$$L(p; \mathbf{x}) = P_p(\mathbf{X} = \mathbf{x}) = \prod_{i=1}^n p_{X_i}(x_i) = \prod_{i=1}^n \underbrace{p^{x_i}(1-p)^{1-x_i}}$$

$$\text{If } x_i = 1, p^{[x_i]}(1-p)^{[1-x_i]} = p$$

$$\text{If } x_i = 0, p^{[x_i]}(1-p)^{[1-x_i]} = (1-p)$$

Maximum Likelihood: Bernoulli Example

n coins
 k heads



$$\mathbf{X} = (X_1, \dots, X_n)$$

$$X_i \stackrel{i.i.d}{\sim} \text{Bernoulli}(p)$$

Likelihood

$$L(p; \mathbf{x}) = P_p(\mathbf{X} = \mathbf{x}) = \prod_{i=1}^n p_{X_i}(x_i) = \prod_{i=1}^n \underbrace{p^{x_i}(1-p)^{1-x_i}}$$

$$\text{If } x_i = 1, p^{[x_i]}(1-p)^{[1-x_i]} = p$$

$$\text{If } x_i = 0, p^{[x_i]}(1-p)^{[1-x_i]} = (1-p)$$

$$\sum_{i=1}^n x_i = \# \text{ heads}$$

$$n - \sum_{i=1}^n x_i = \# \text{ tails}$$

Maximum Likelihood: Bernoulli Example



$$\mathbf{X} = (X_1, \dots, X_n)$$

$$X_i \stackrel{i.i.d}{\sim} \text{Bernoulli}(p)$$

Likelihood

$$L(p; \mathbf{x}) = P_p(\mathbf{X} = \mathbf{x}) = \prod_{i=1}^n p_{X_i}(x_i) = \prod_{i=1}^n p^{x_i}(1-p)^{1-x_i}$$

Maximum Likelihood: Bernoulli Example



$$\mathbf{X} = (X_1, \dots, X_n)$$

$$X_i \stackrel{i.i.d}{\sim} \text{Bernoulli}(p)$$

Likelihood

$$L(p; \mathbf{x}) = P_p(\mathbf{X} = \mathbf{x}) = \prod_{i=1}^n p_{X_i}(x_i) = \prod_{i=1}^n p^{x_i}(1-p)^{1-x_i} = p^{\left(\sum_{i=1}^n x_i\right)}(1-p)^{\left(n - \sum_{i=1}^n x_i\right)}$$

Maximum Likelihood: Bernoulli Example



$$\mathbf{X} = (X_1, \dots, X_n)$$

$$X_i \stackrel{i.i.d}{\sim} \text{Bernoulli}(p)$$

Likelihood

$$L(p; \mathbf{x}) = P_p(\mathbf{X} = \mathbf{x}) = \prod_{i=1}^n p_{X_i}(x_i) = \prod_{i=1}^n p^{x_i}(1-p)^{1-x_i} = p^{\left(\sum_{i=1}^n x_i\right)} (1-p)^{\left(n - \sum_{i=1}^n x_i\right)}$$

Log-likelihood

$$\ell(p; \mathbf{x}) = \log \left((p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i}) \right)$$

Maximum Likelihood: Bernoulli Example



$$\mathbf{X} = (X_1, \dots, X_n)$$

$$X_i \stackrel{i.i.d}{\sim} \text{Bernoulli}(p)$$

Likelihood

$$L(p; \mathbf{x}) = P_p(\mathbf{X} = \mathbf{x}) = \prod_{i=1}^n p_{X_i}(x_i) = \prod_{i=1}^n p^{x_i}(1-p)^{1-x_i} = p^{\left(\sum_{i=1}^n x_i\right)} (1-p)^{\left(n - \sum_{i=1}^n x_i\right)}$$

Log-likelihood

$$\ell(p; \mathbf{x}) = \log \left((p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i}) \right) = \left(\sum_{i=1}^n x_i \right) \log(p) + \left(n - \sum_{i=1}^n x_i \right) \log(1-p)$$

Maximum Likelihood: Bernoulli Example



$$\mathbf{X} = (X_1, \dots, X_n)$$

$$X_i \stackrel{i.i.d}{\sim} \text{Bernoulli}(p)$$

$$\ell(p; \mathbf{x}) = \log \left((p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i}) \right) = \left(\sum_{i=1}^n x_i \right) \log(p) + \left(n - \sum_{i=1}^n x_i \right) \log(1-p)$$

Maximum Likelihood: Bernoulli Example



$$\mathbf{X} = (X_1, \dots, X_n)$$

$$X_i \stackrel{i.i.d}{\sim} \text{Bernoulli}(p)$$

$$\ell(p; \mathbf{x}) = \log \left((p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i}) \right) = \left(\sum_{i=1}^n x_i \right) \log(p) + \left(n - \sum_{i=1}^n x_i \right) \log(1-p)$$

Find the maximum!

$$\begin{aligned} \frac{d}{dp} \ell(p; \mathbf{x}) &= \frac{d}{dp} \left(\left(\sum_{i=1}^n x_i \right) \log(p) + \left(n - \sum_{i=1}^n x_i \right) \log(1-p) \right) \\ &= \frac{\sum_{i=1}^n x_i}{p} + \frac{n - \sum_{i=1}^n x_i}{1-p} (-1) = 0 \end{aligned}$$

Maximum Likelihood: Bernoulli Example



$$\mathbf{X} = (X_1, \dots, X_n)$$

$$X_i \stackrel{i.i.d}{\sim} \text{Bernoulli}(p)$$

$$\ell(p; \mathbf{x}) = \log \left((p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i}) \right) = \left(\sum_{i=1}^n x_i \right) \log(p) + \left(n - \sum_{i=1}^n x_i \right) \log(1-p)$$

Find the maximum!

$$\begin{aligned} \frac{d}{dp} \ell(p; \mathbf{x}) &= \frac{d}{dp} \left(\left(\sum_{i=1}^n x_i \right) \log(p) + \left(n - \sum_{i=1}^n x_i \right) \log(1-p) \right) \\ &= \frac{\sum_{i=1}^n x_i}{p} + \frac{n - \sum_{i=1}^n x_i}{1-p} (-1) = 0 \end{aligned} \quad \Rightarrow \quad \hat{p} = \frac{\sum_{i=1}^n x_i}{n} = \bar{x}$$

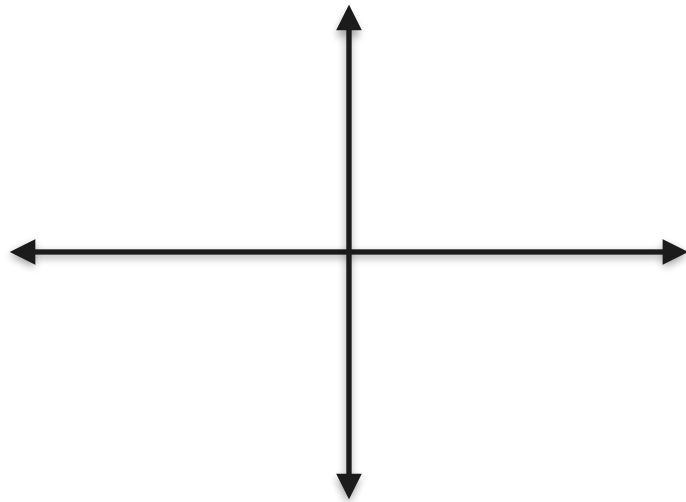


DeepLearning.AI

Point Estimation

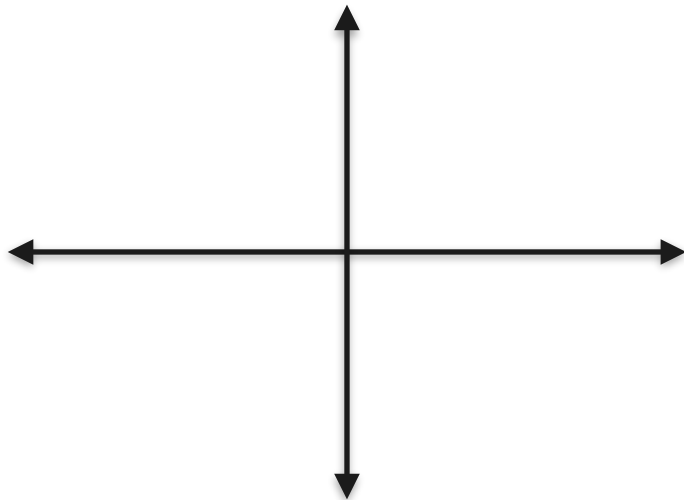
MLE: Gaussian Example

Maximum Likelihood: Gaussian Example



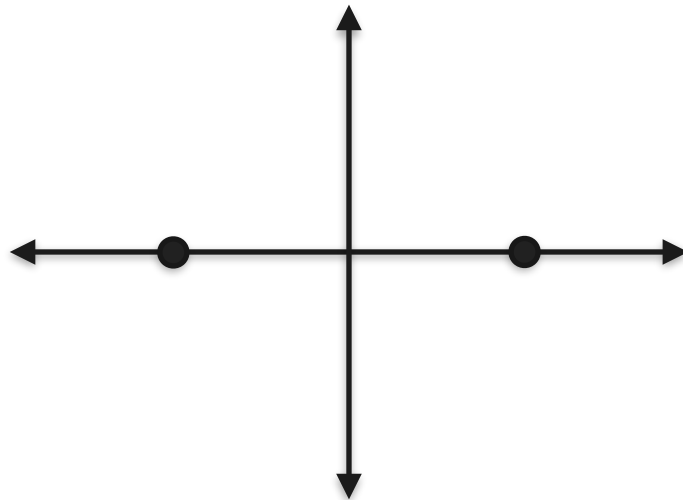
Maximum Likelihood: Gaussian Example

Observations



Maximum Likelihood: Gaussian Example

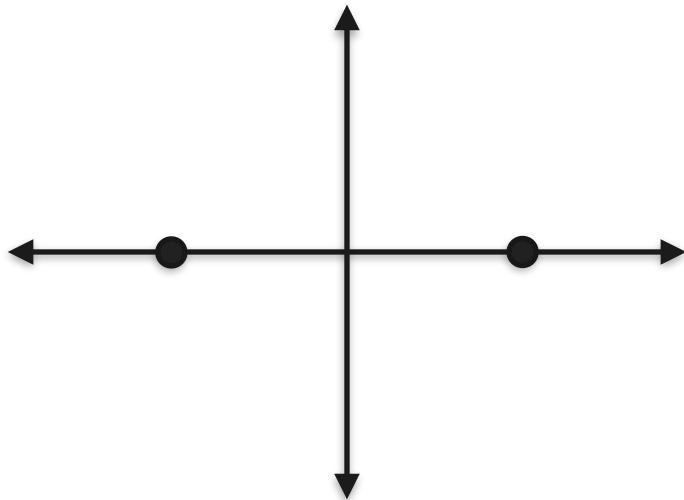
Observations



Maximum Likelihood: Gaussian Example

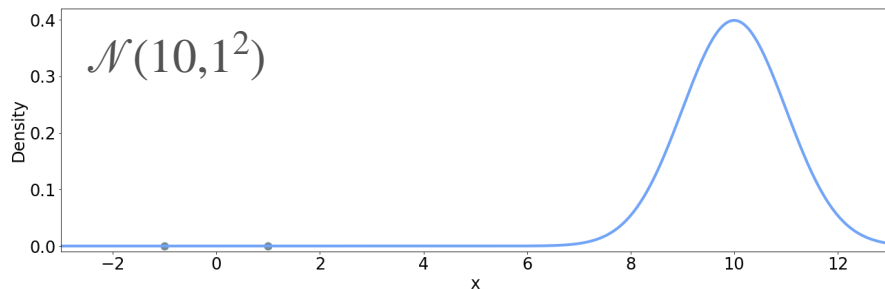
Candidates

Observations

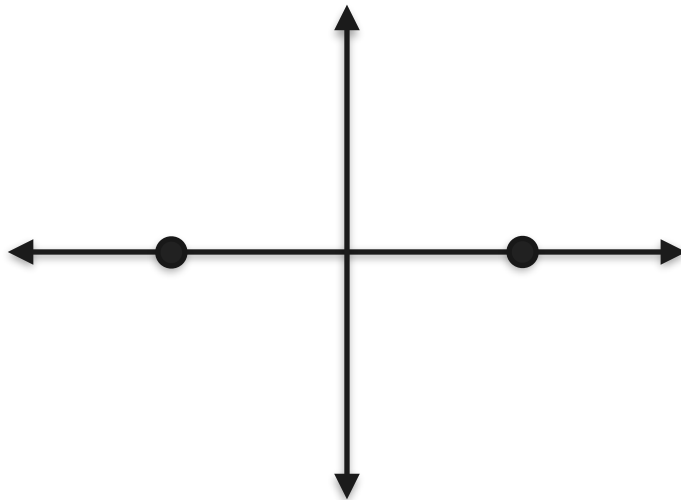


Maximum Likelihood: Gaussian Example

Candidates

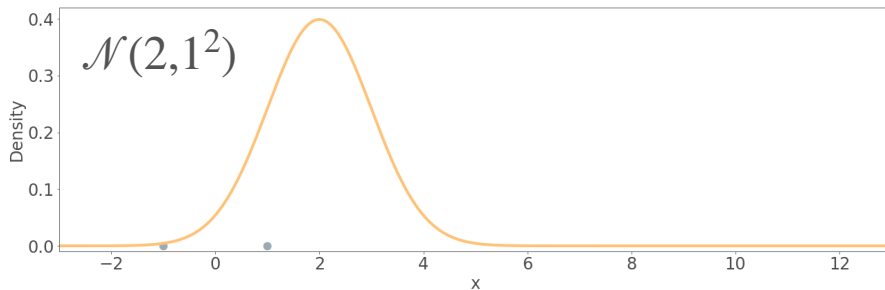
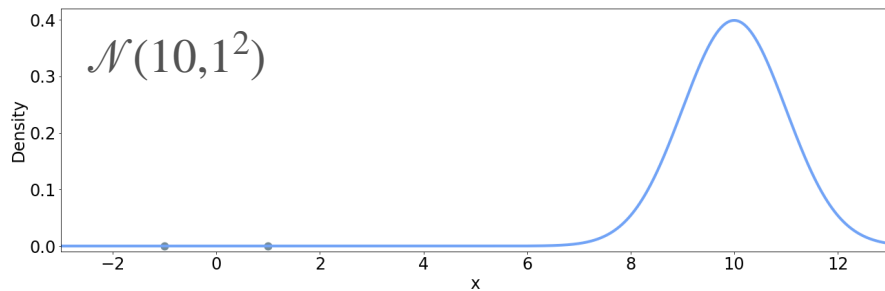


Observations

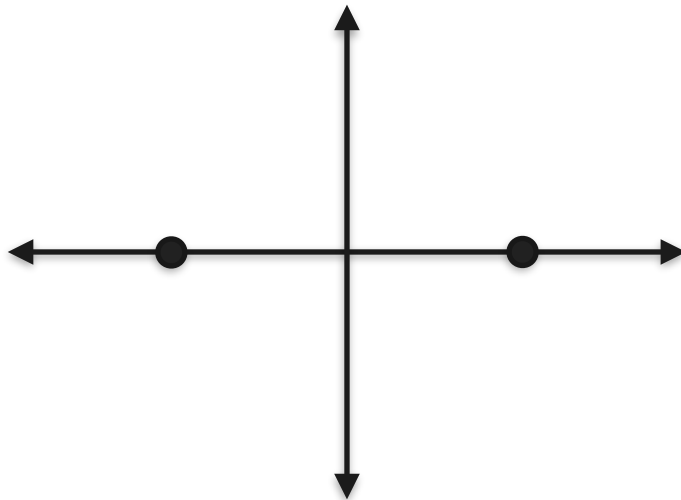


Maximum Likelihood: Gaussian Example

Candidates

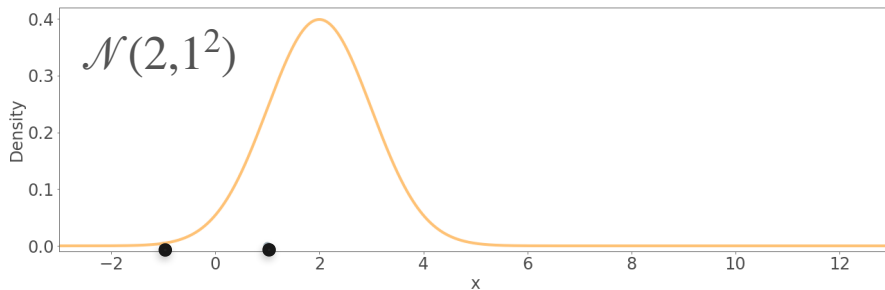
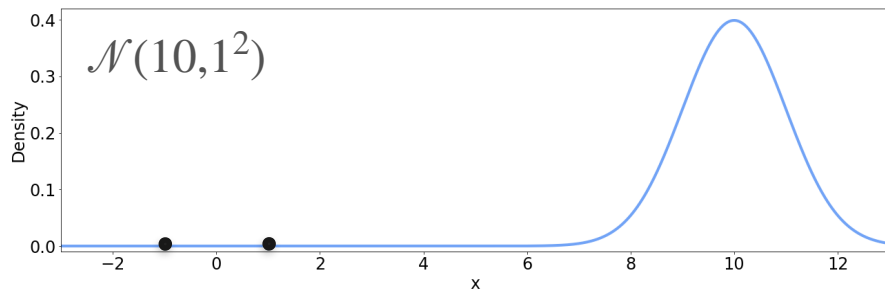


Observations

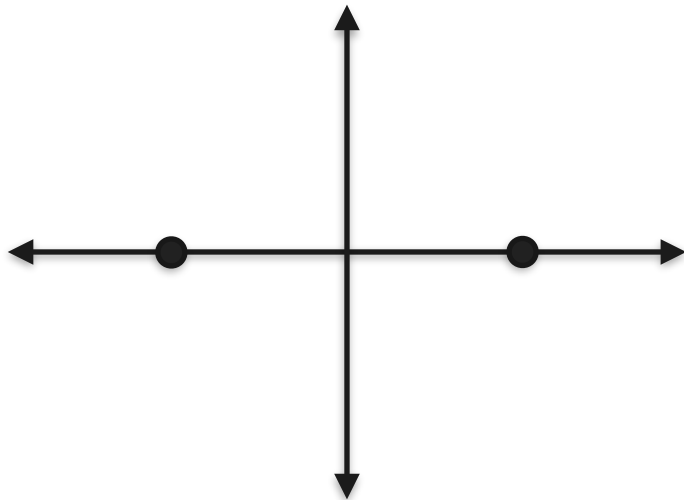


Maximum Likelihood: Gaussian Example

Candidates

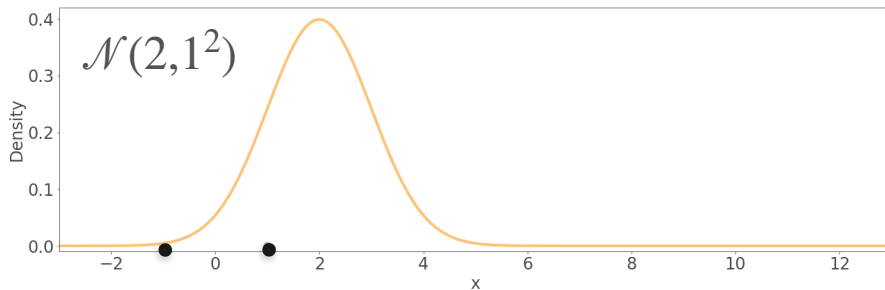
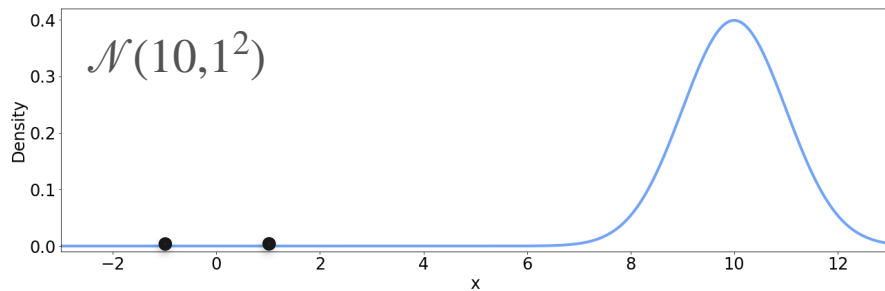


Observations

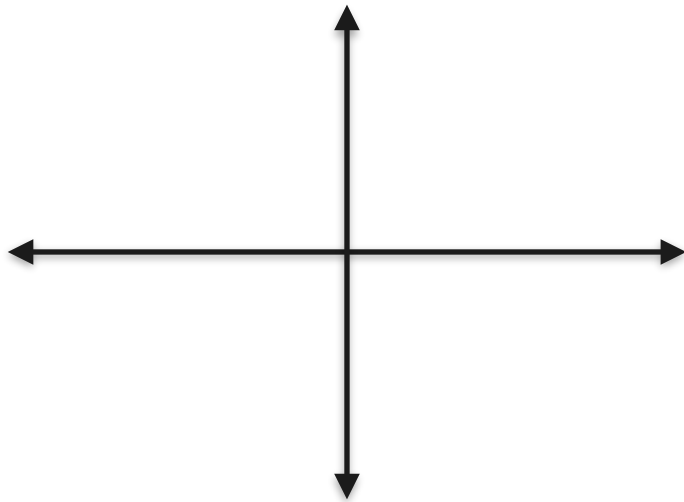


Maximum Likelihood: Gaussian Example

Candidates

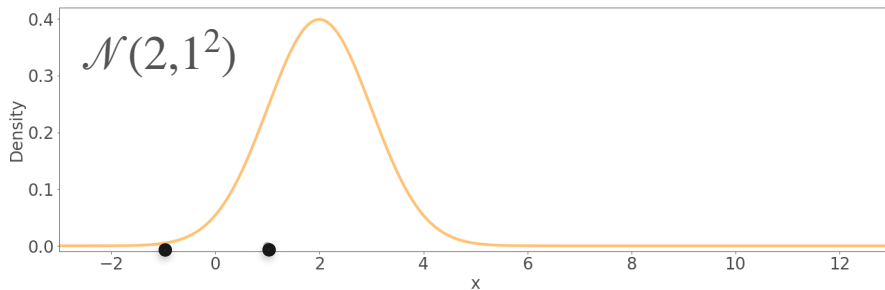
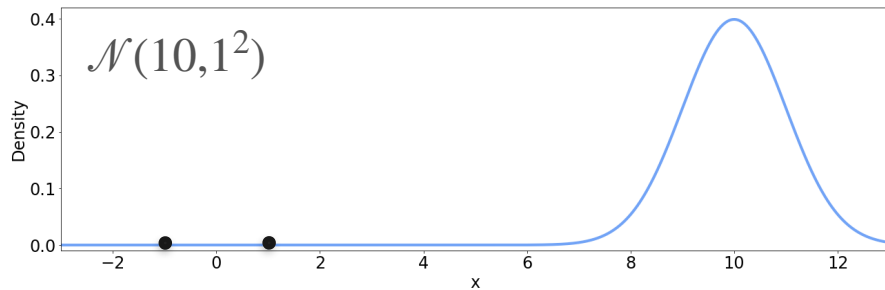


Observations

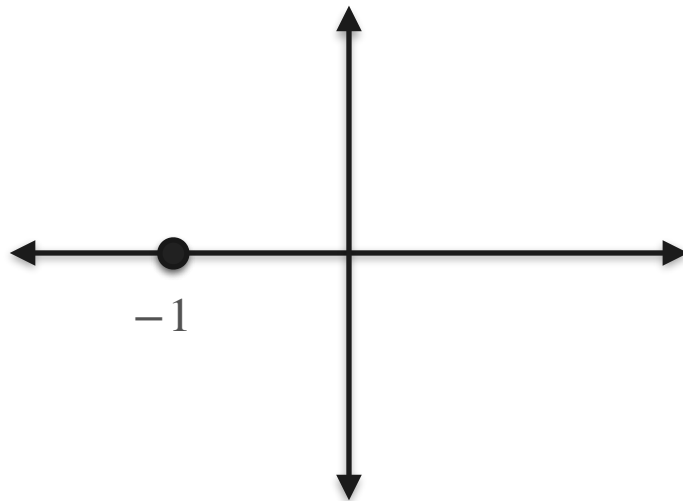


Maximum Likelihood: Gaussian Example

Candidates

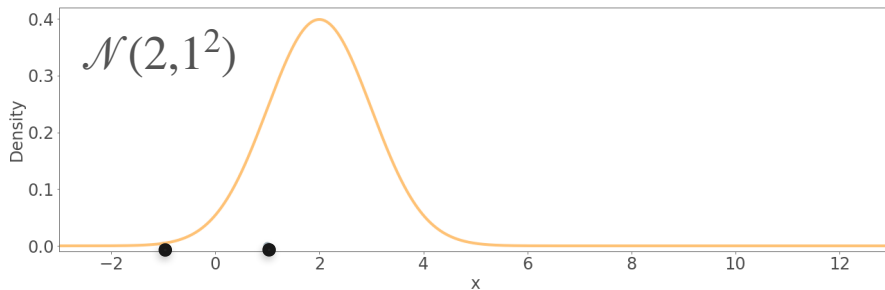
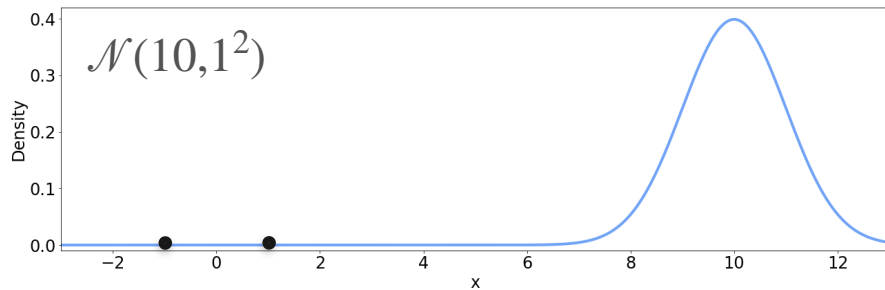


Observations

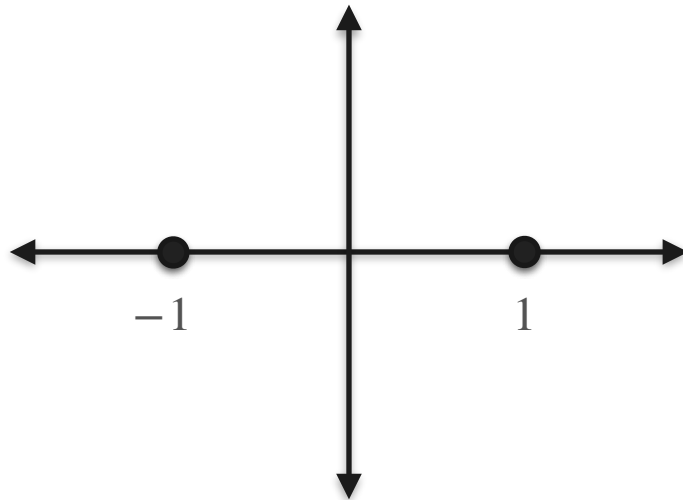


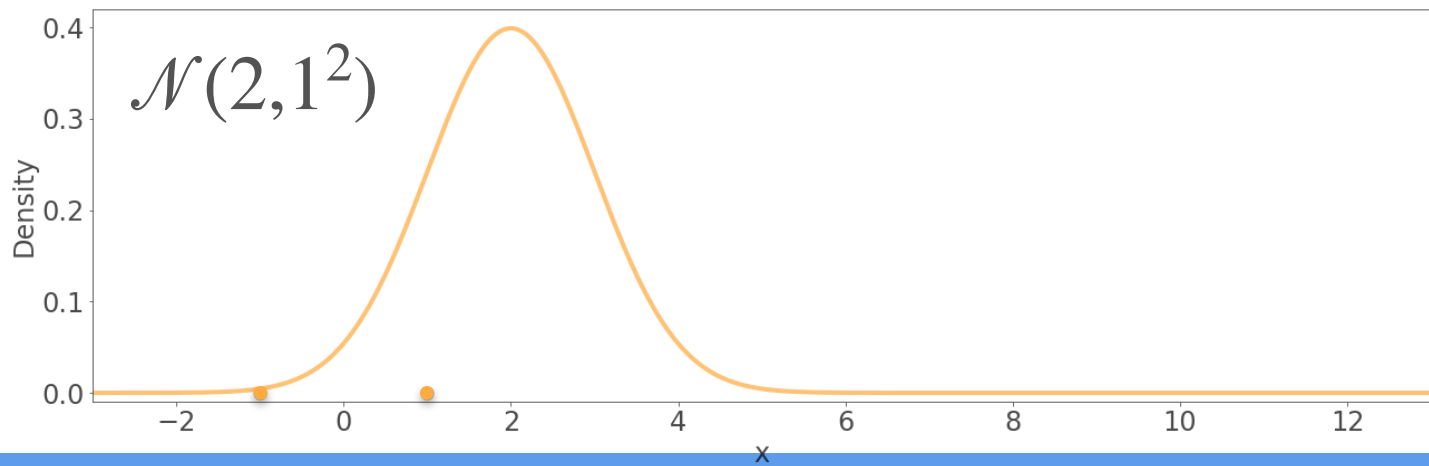
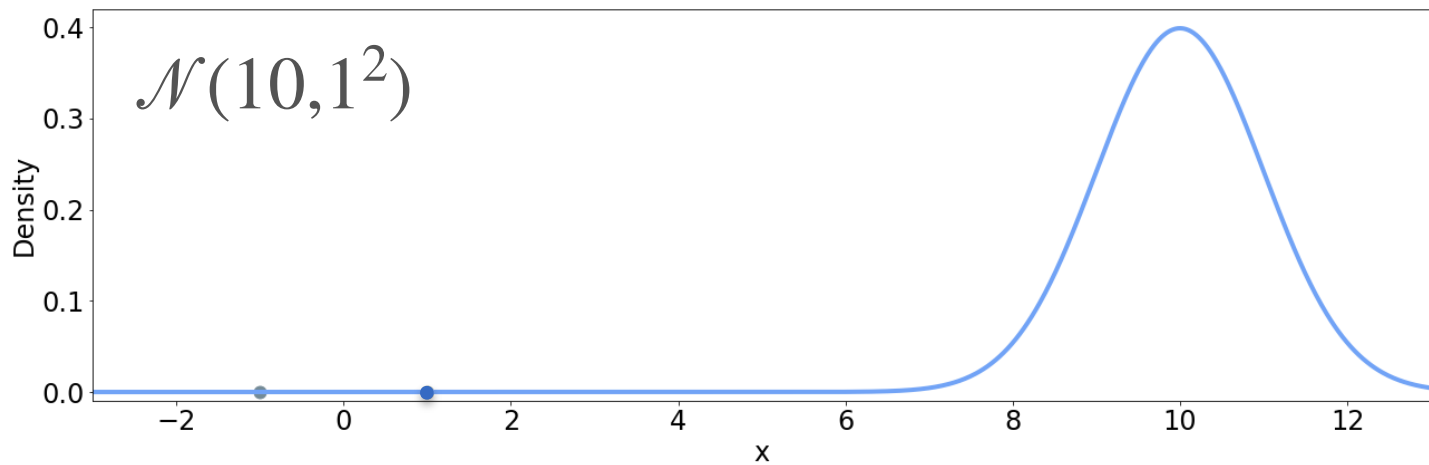
Maximum Likelihood: Gaussian Example

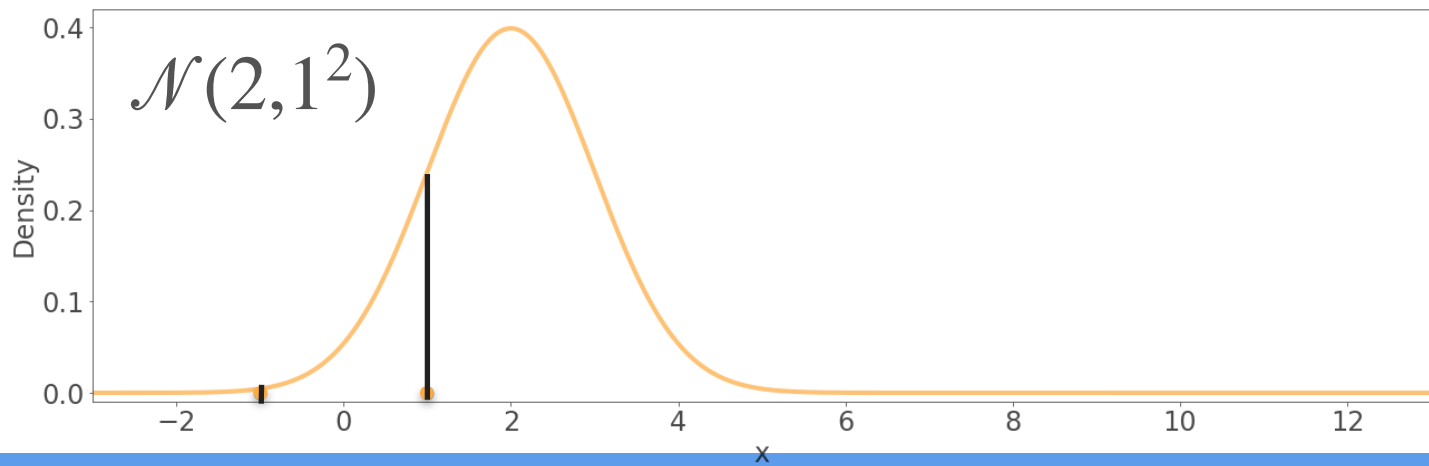
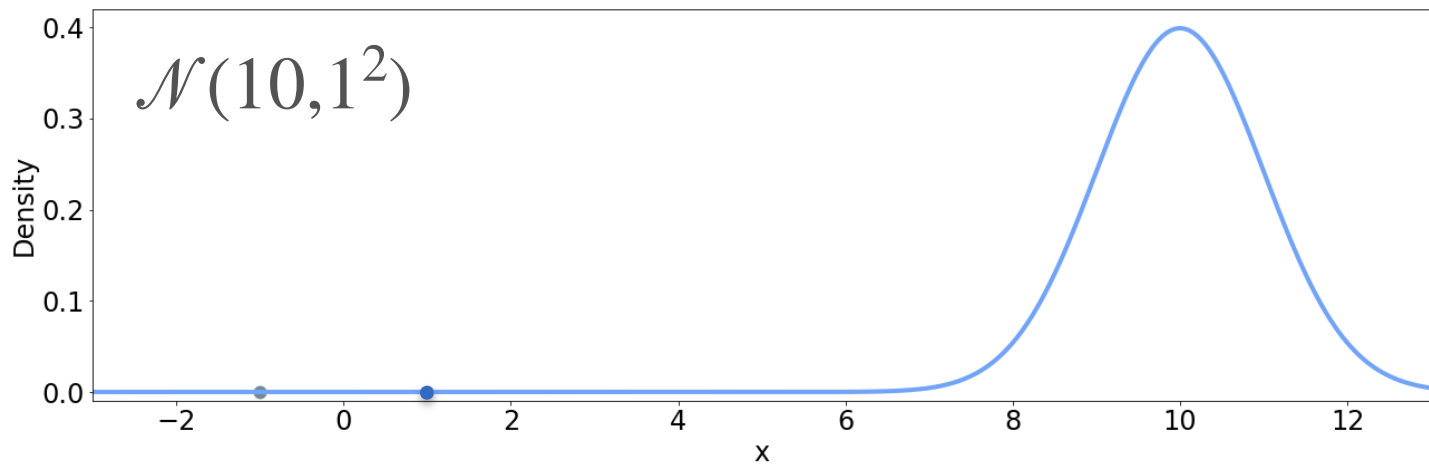
Candidates

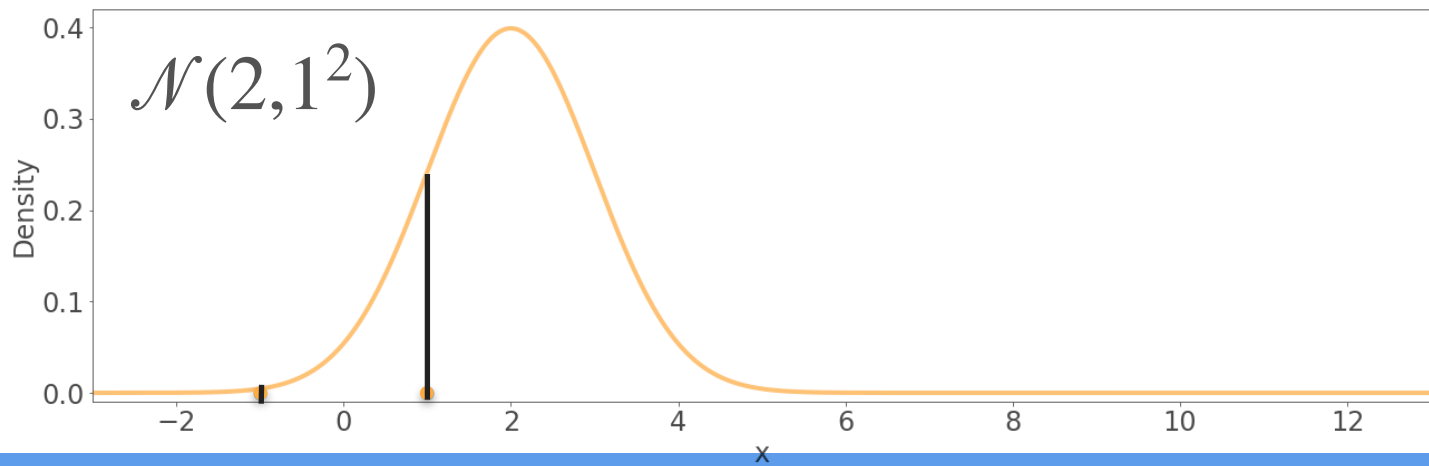
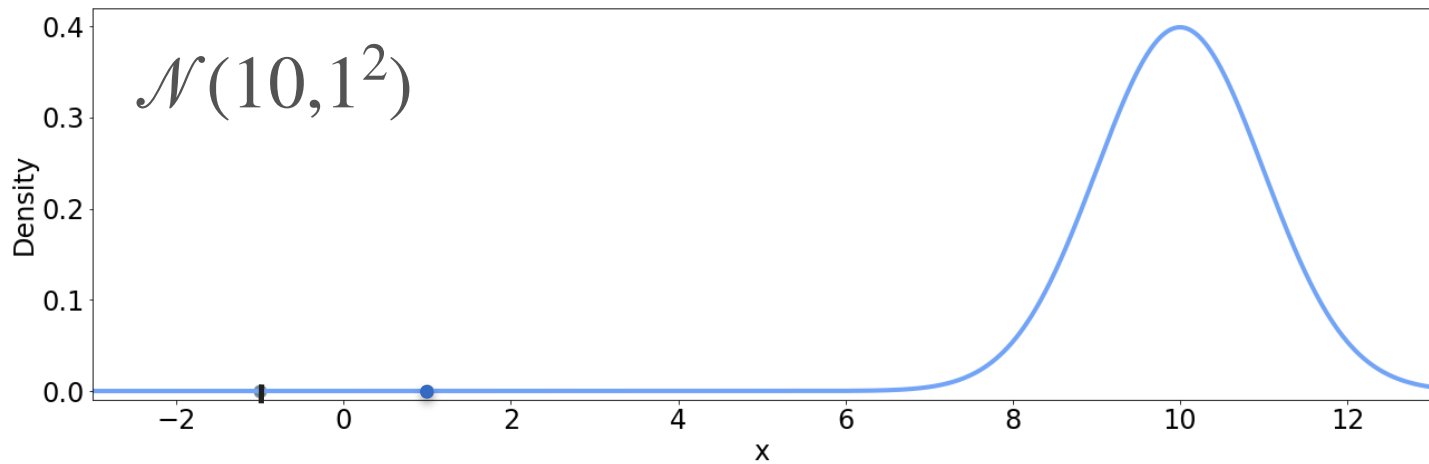


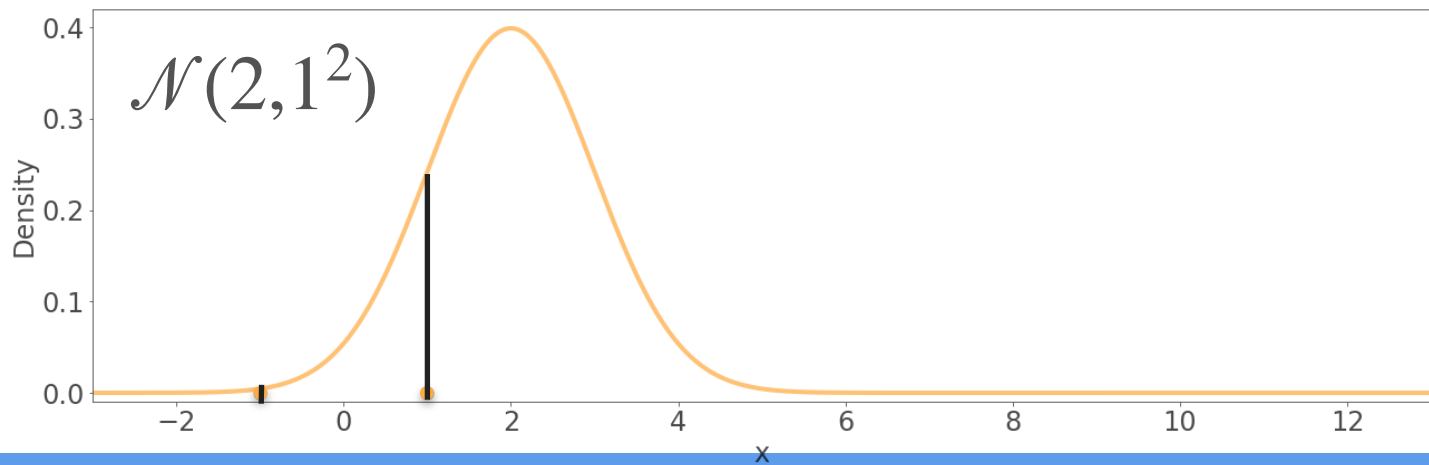
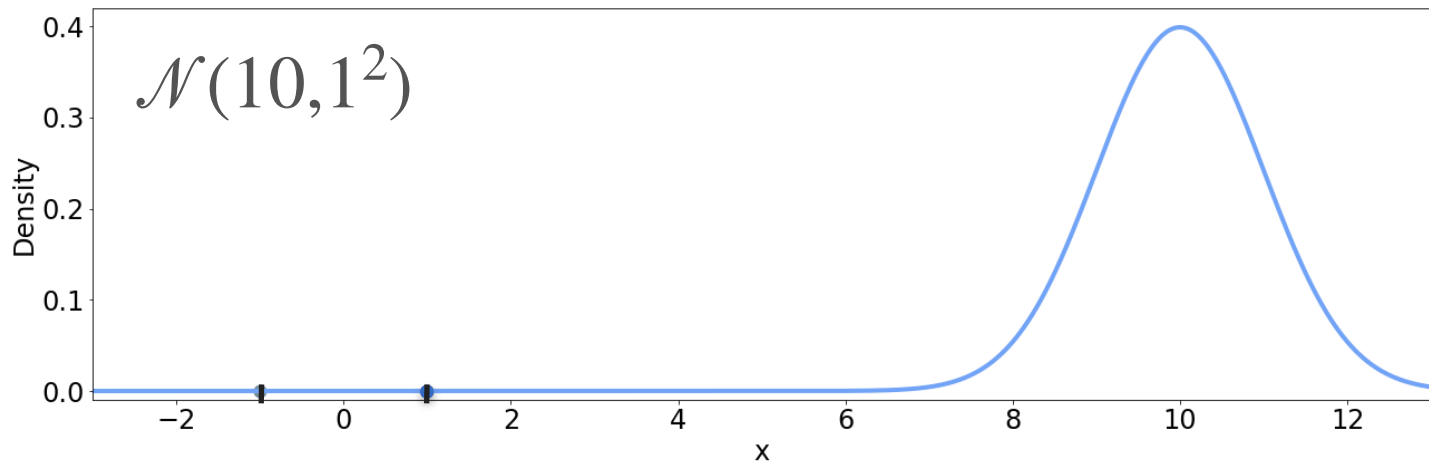
Observations

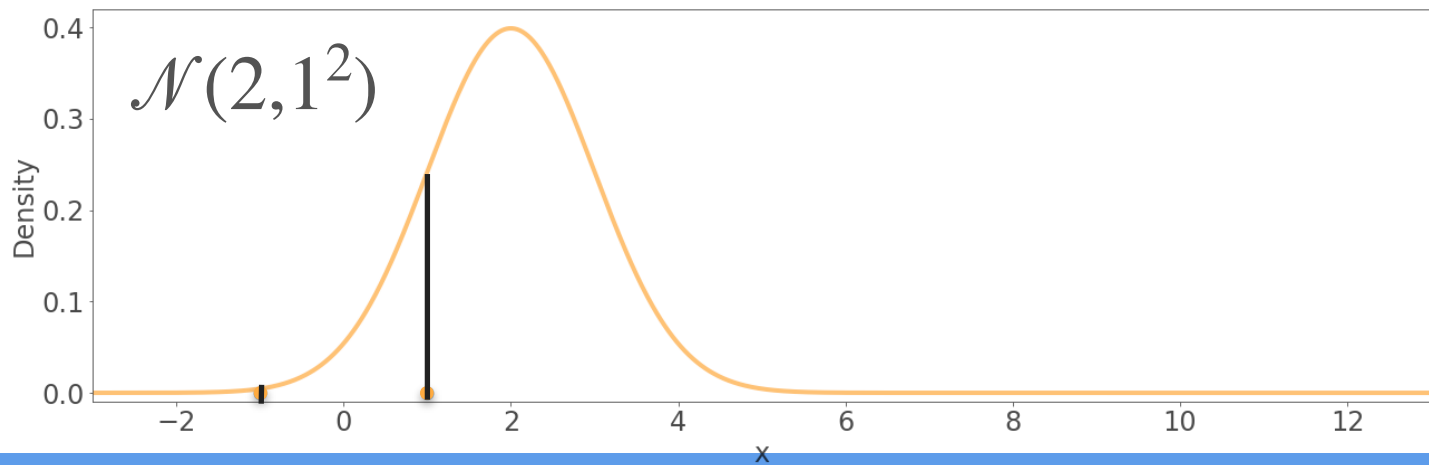
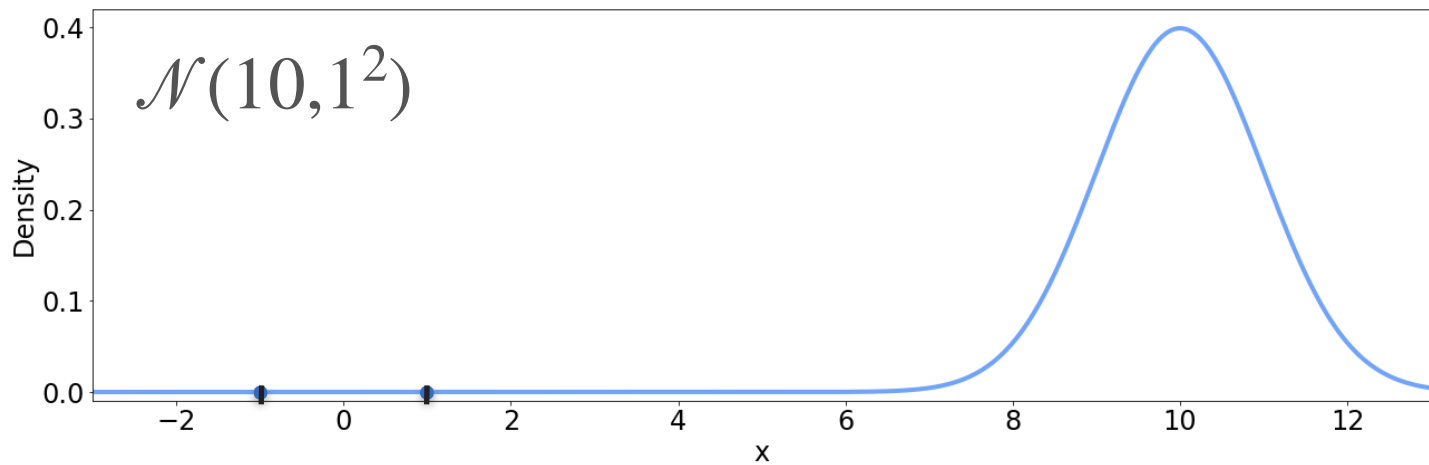


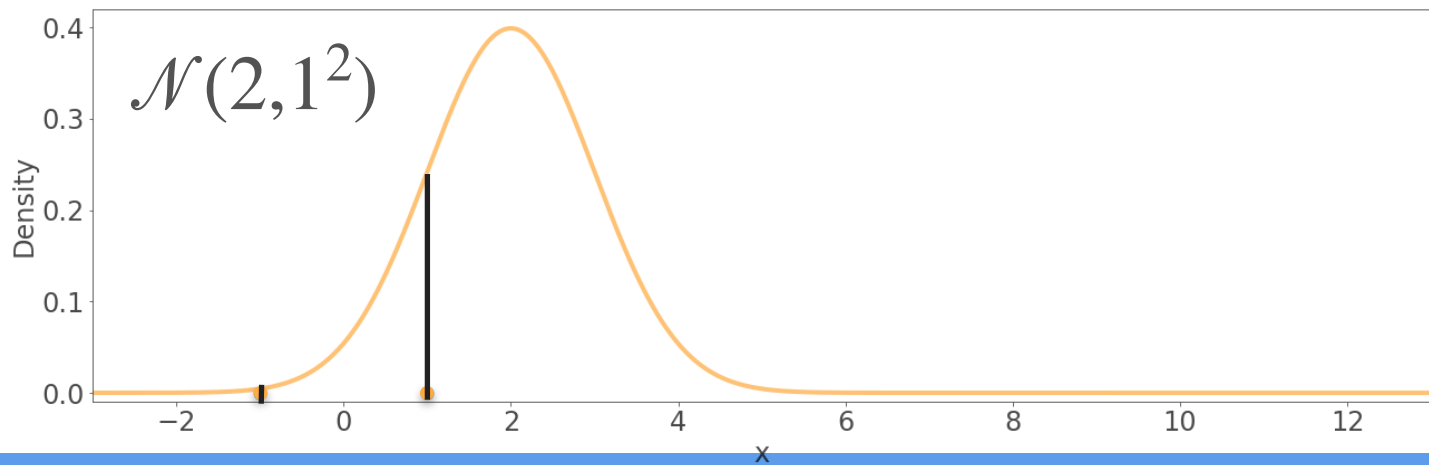
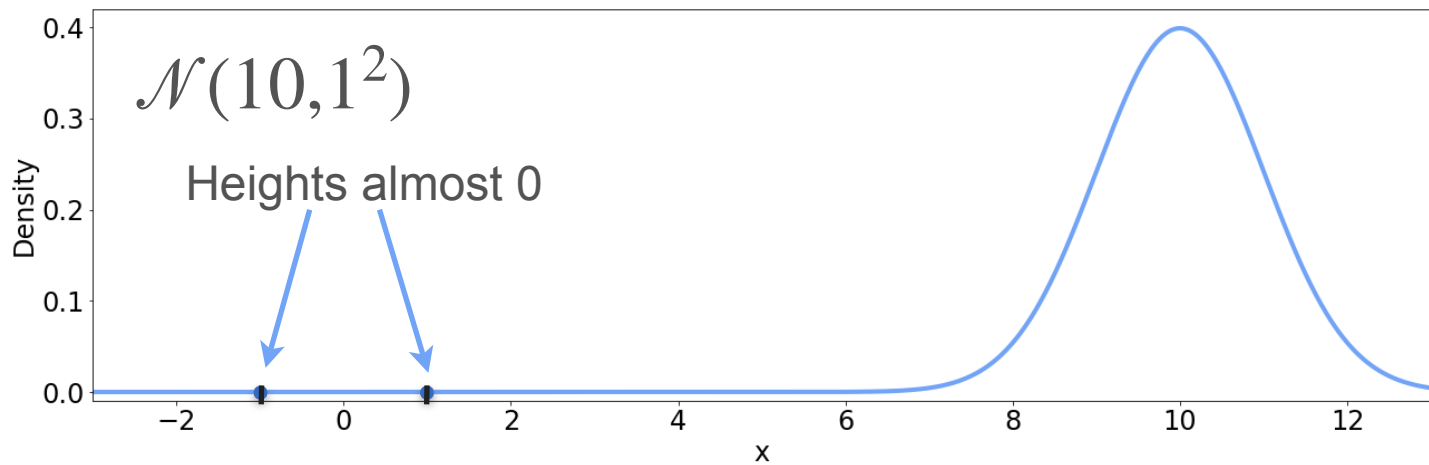


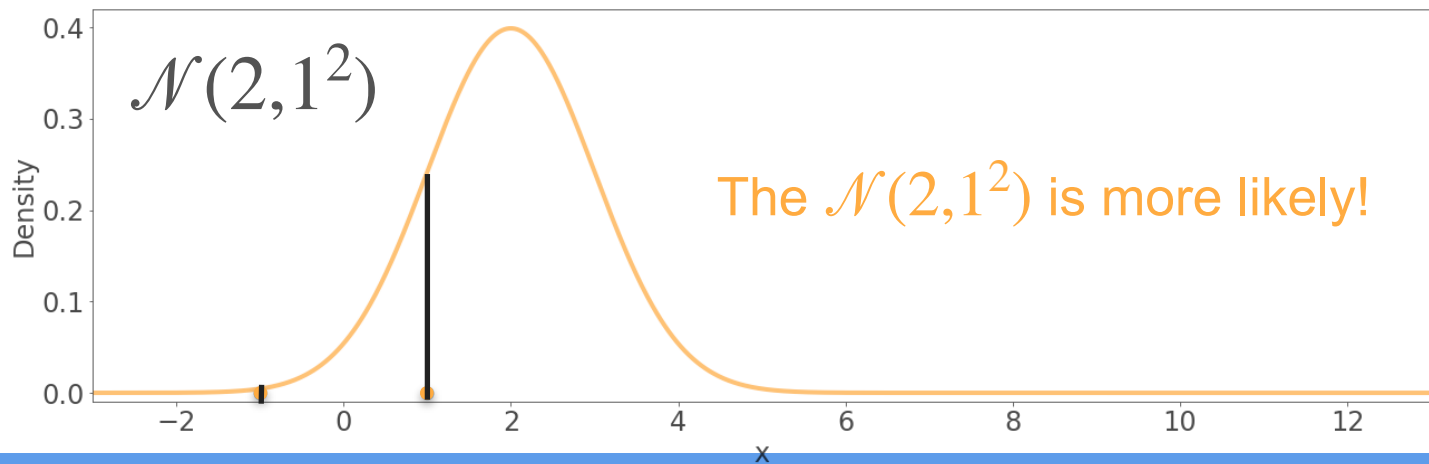
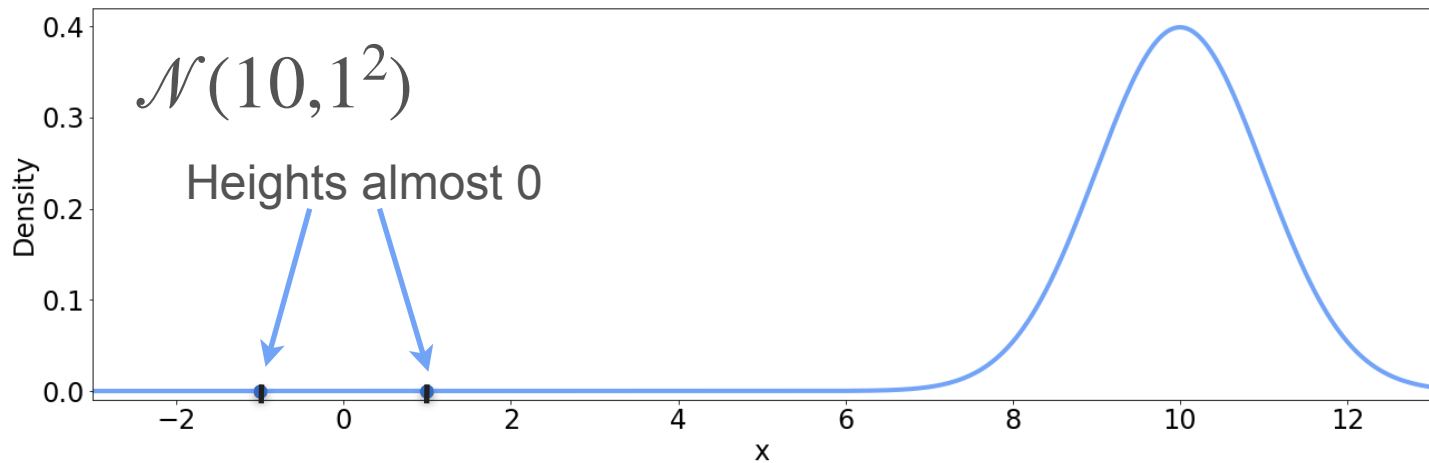


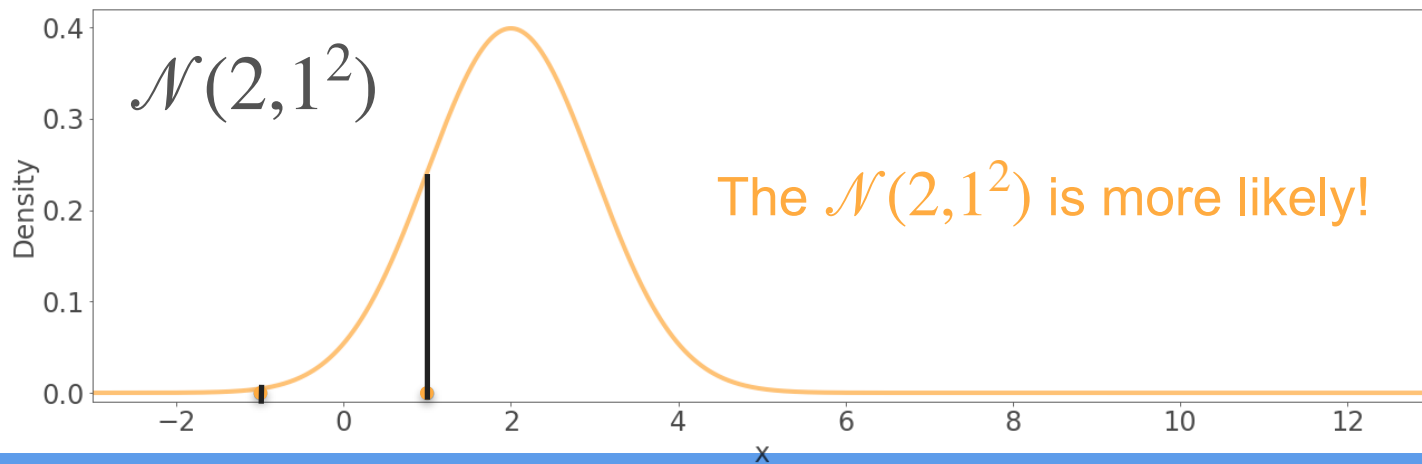
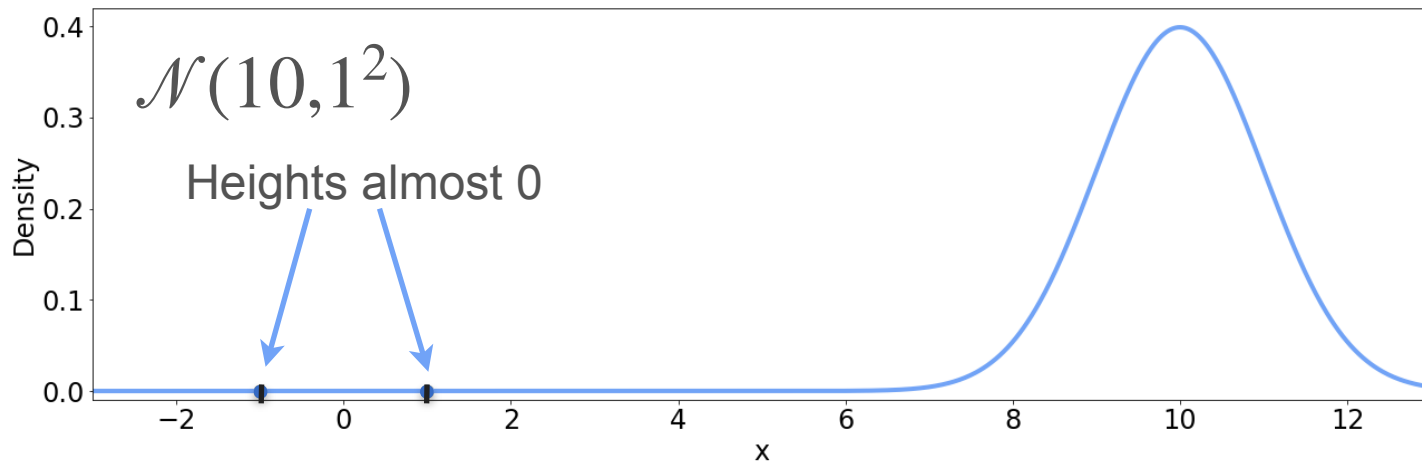






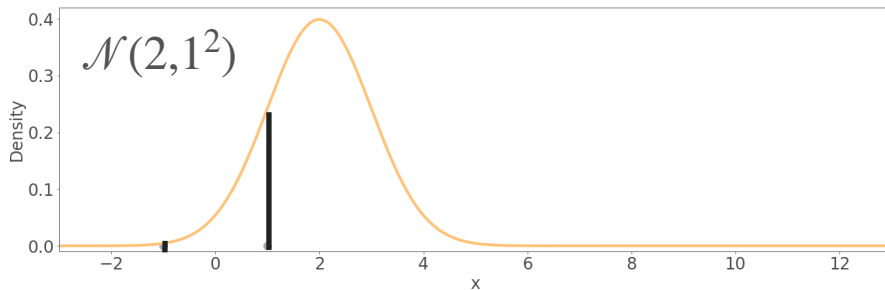
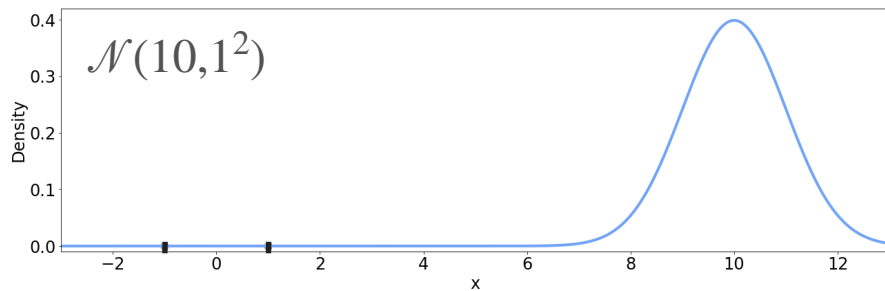




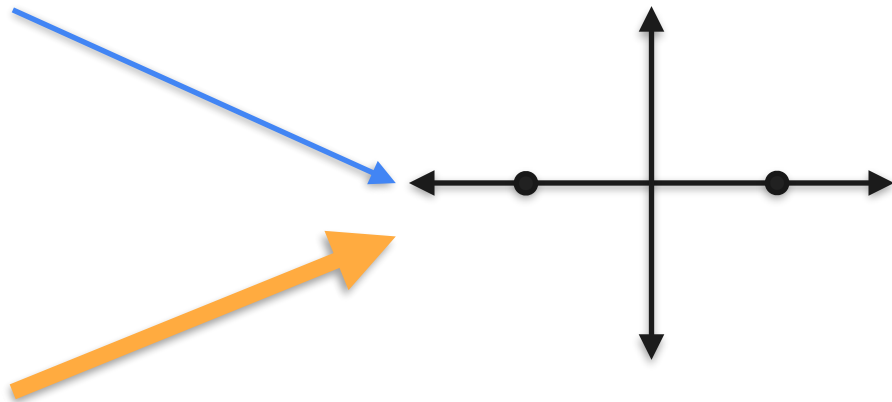


Maximum Likelihood: Gaussian Example

Candidates

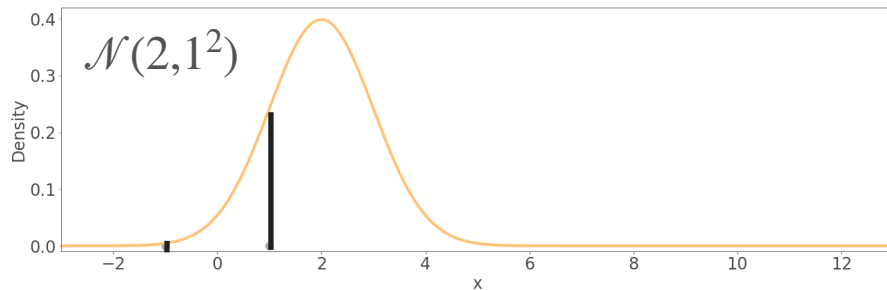
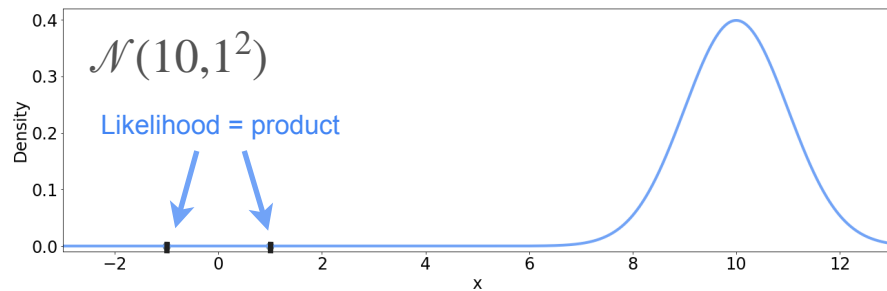


Observations

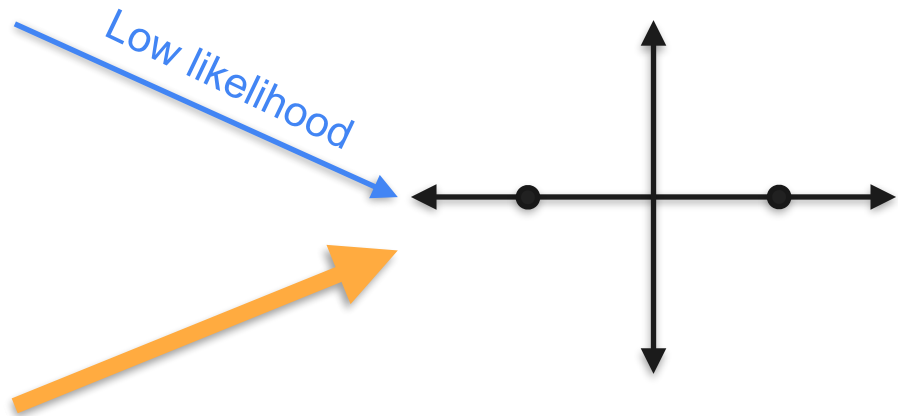


Maximum Likelihood: Gaussian Example

Candidates

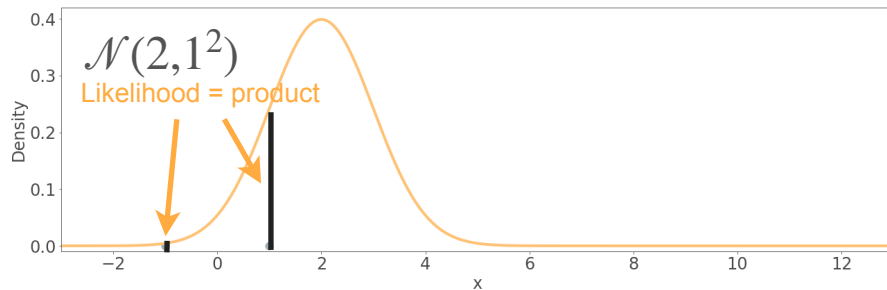
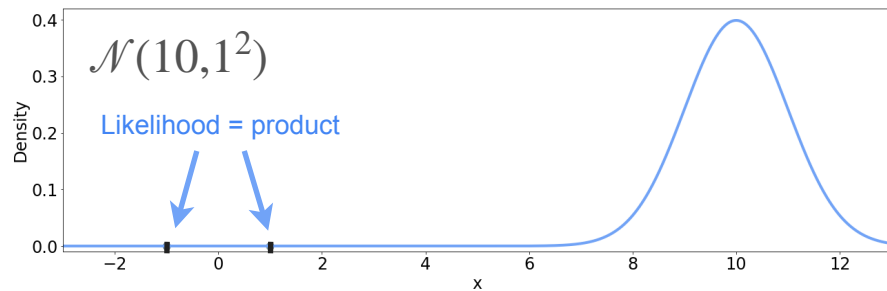


Observations

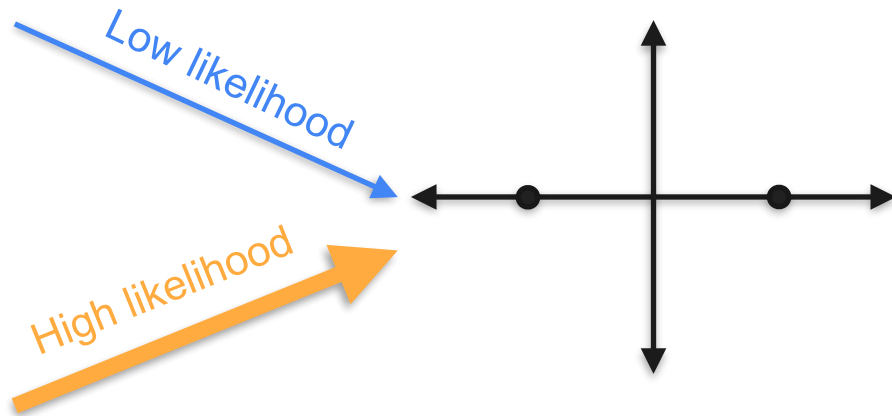


Maximum Likelihood: Gaussian Example

Candidates



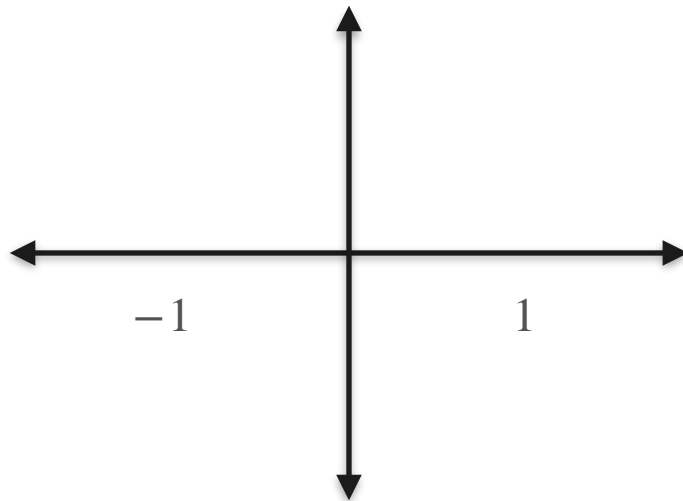
Observations



Gaussians With Three Different Means

Candidates

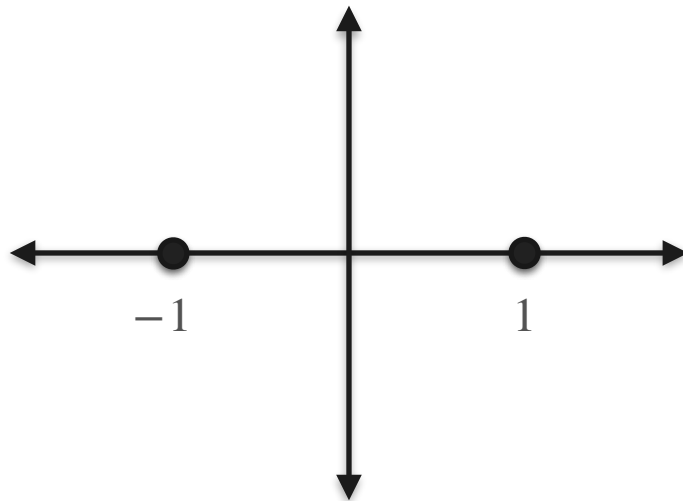
Observations



Gaussians With Three Different Means

Candidates

Observations

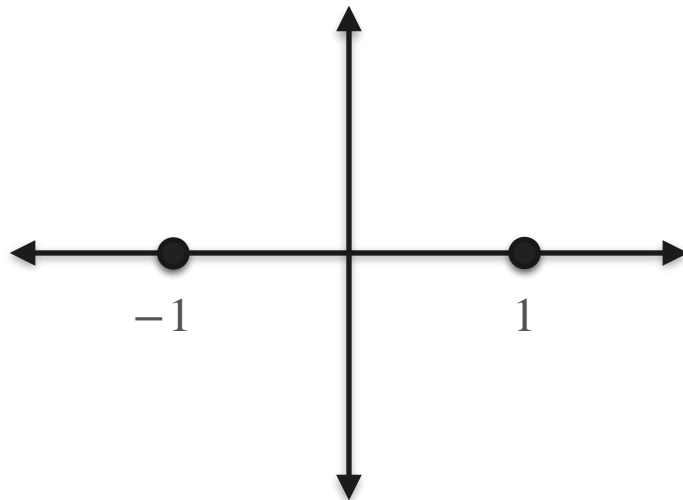


Gaussians With Three Different Means

Candidates

$$\mathcal{N}(-1, 1^2)$$

Observations



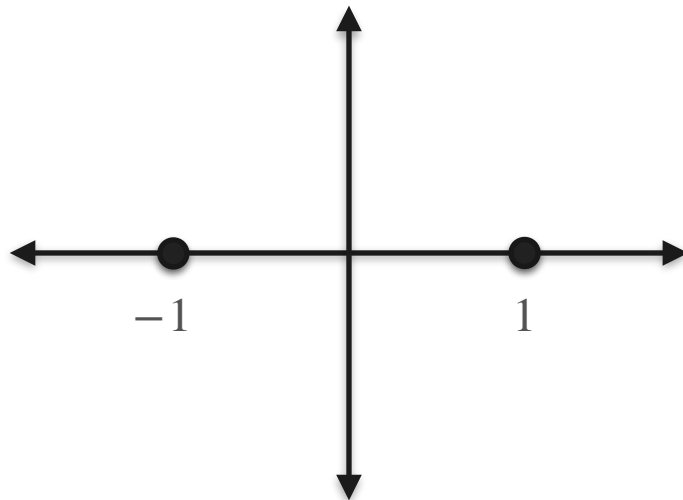
Gaussians With Three Different Means

Candidates

$$\mathcal{N}(-1, 1^2)$$

$$\mathcal{N}(0, 1^2)$$

Observations



Gaussians With Three Different Means

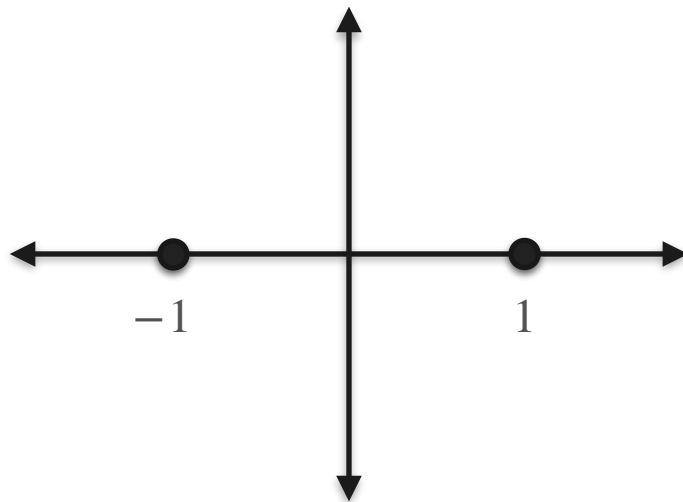
Candidates

Observations

$$\mathcal{N}(-1, 1^2)$$

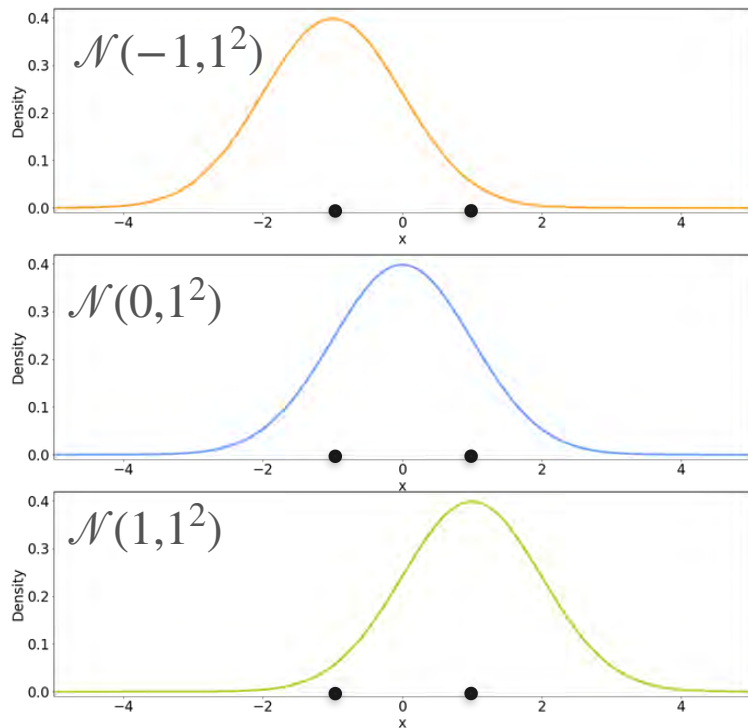
$$\mathcal{N}(0, 1^2)$$

$$\mathcal{N}(1, 1^2)$$

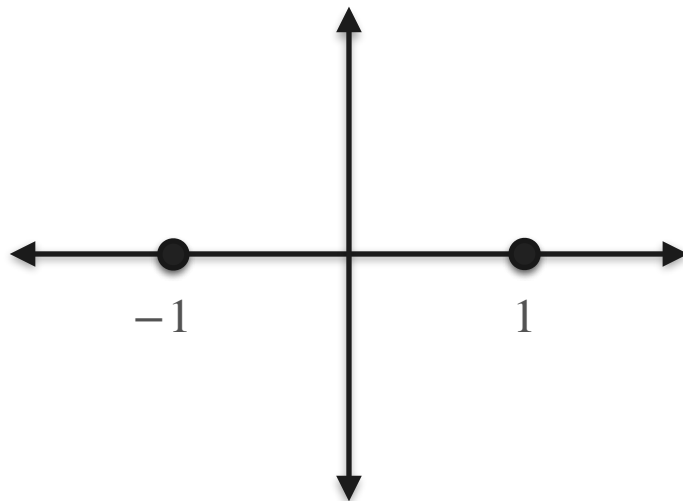


Gaussians With Three Different Means

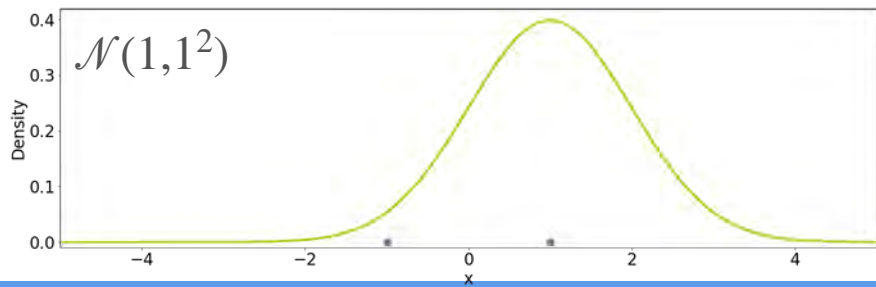
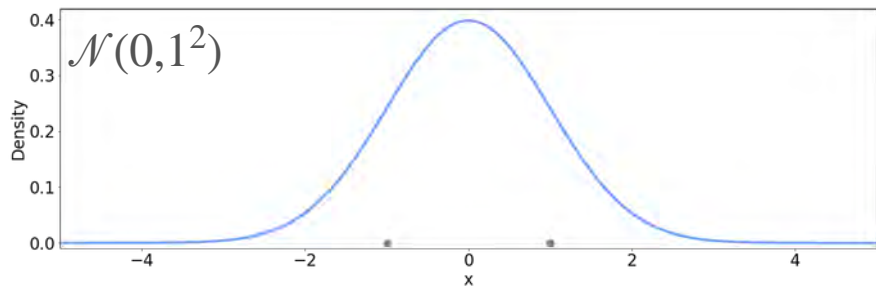
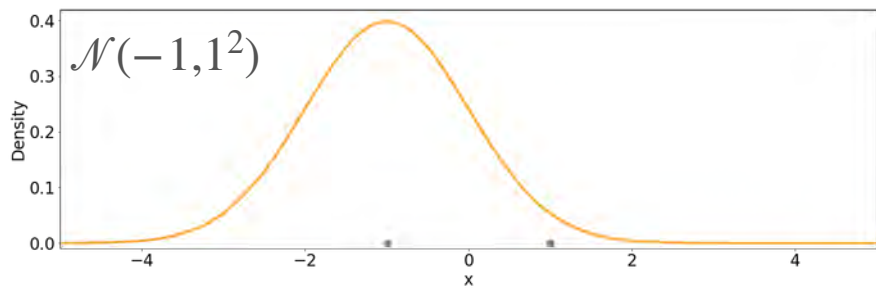
Candidates



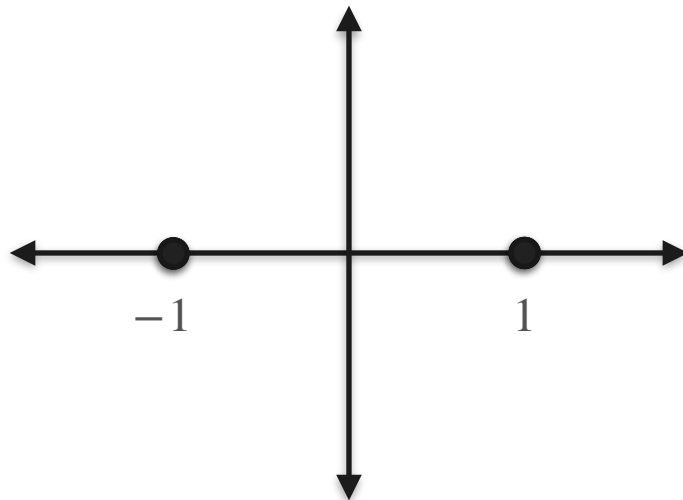
Observations



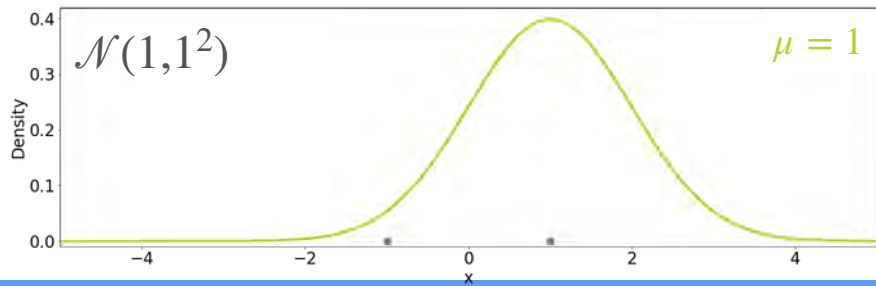
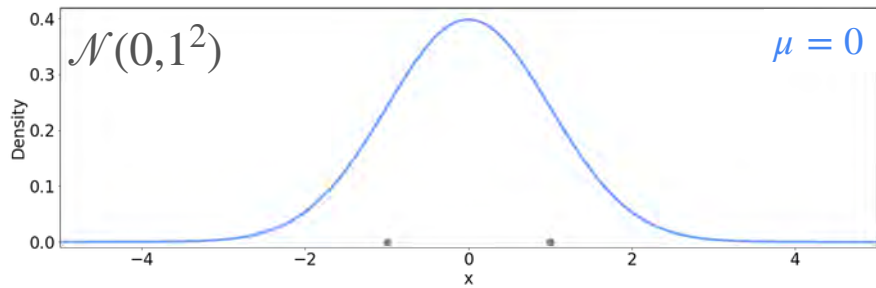
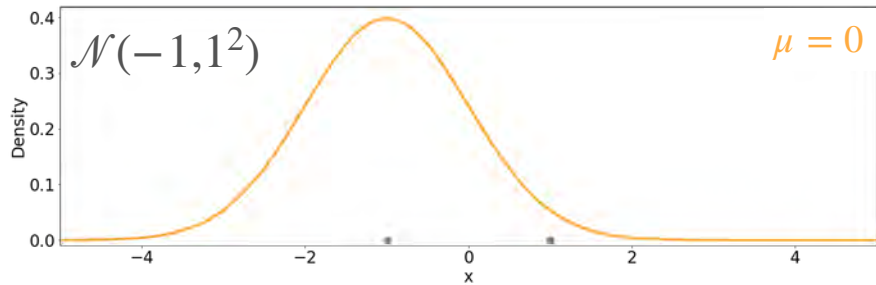
Candidates



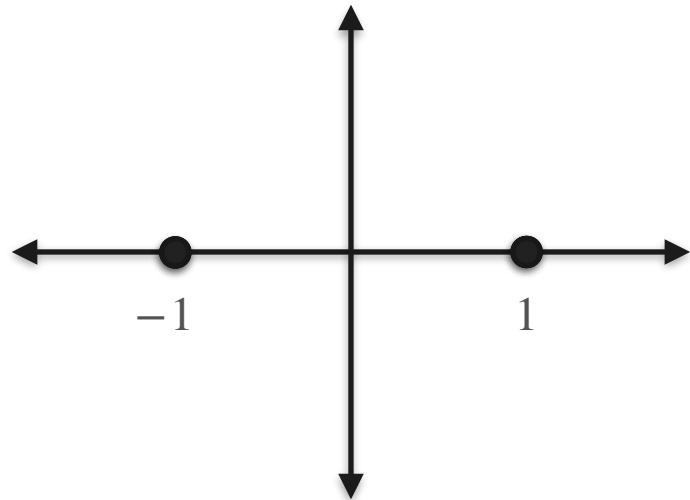
Observations



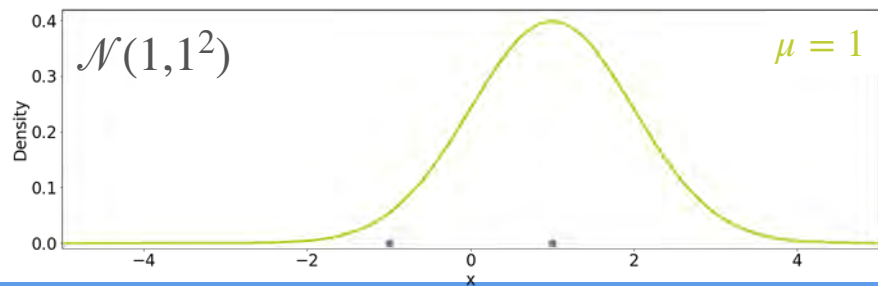
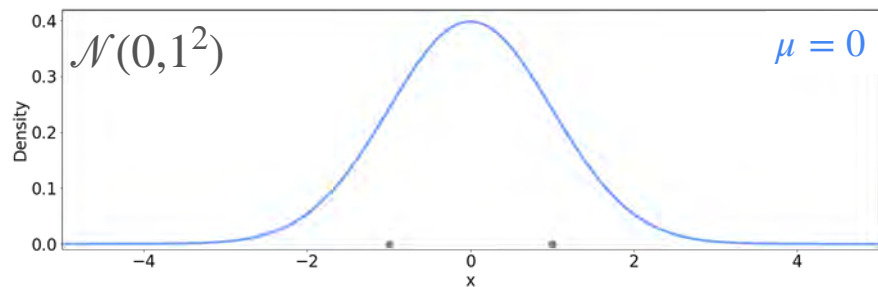
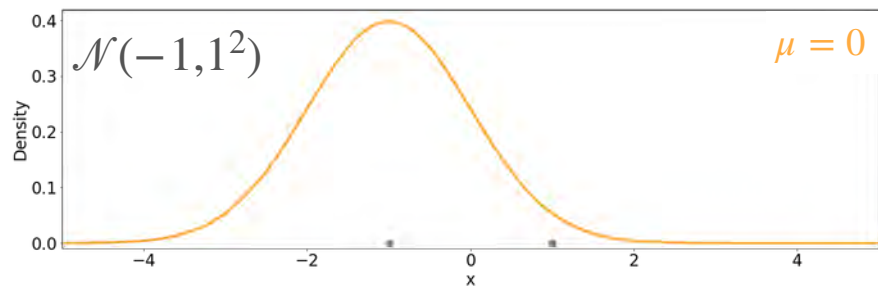
Candidates



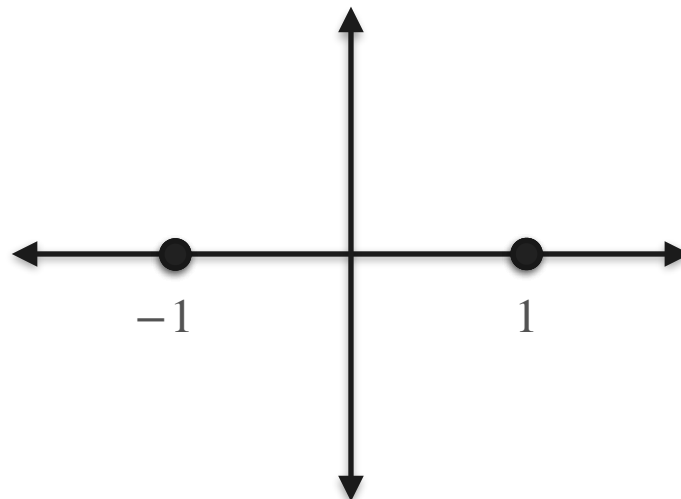
Observations



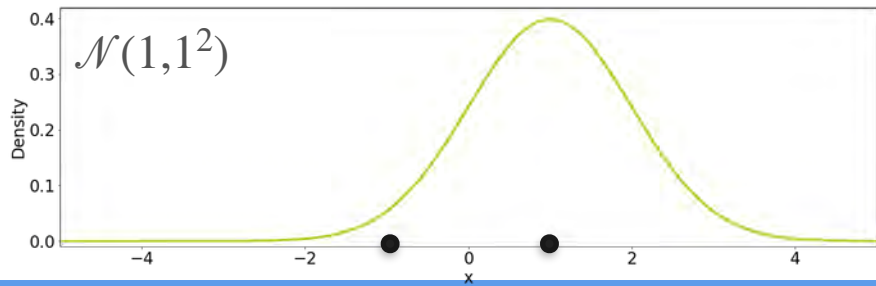
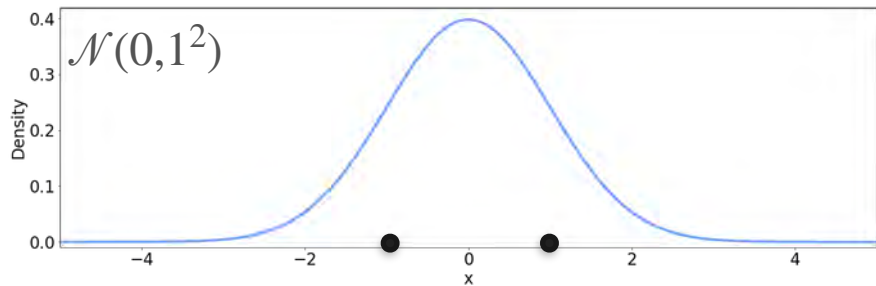
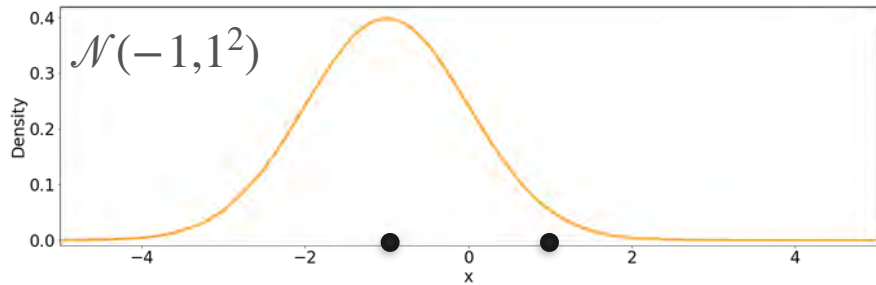
Candidates



Observations

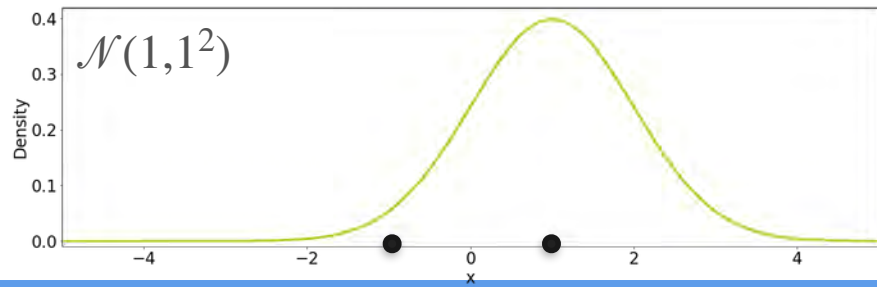
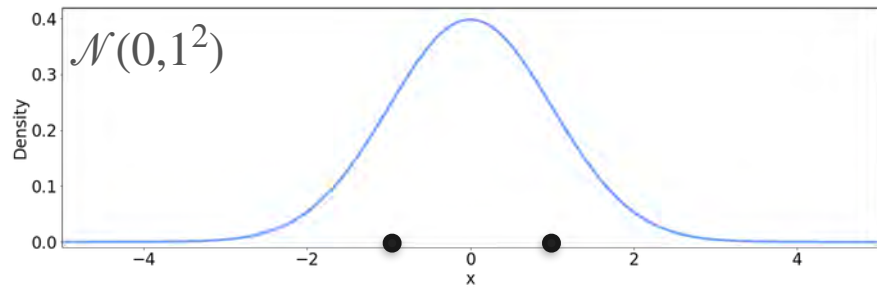
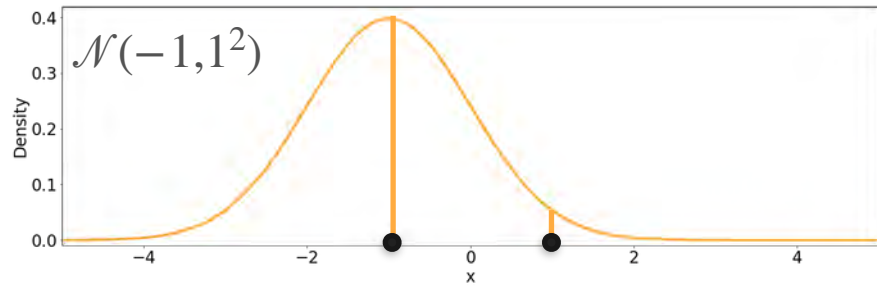


Candidates



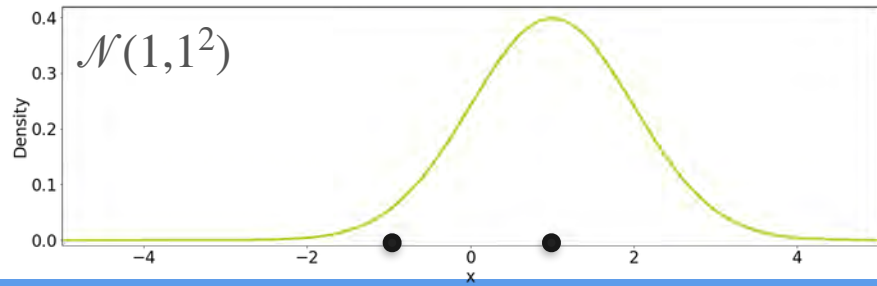
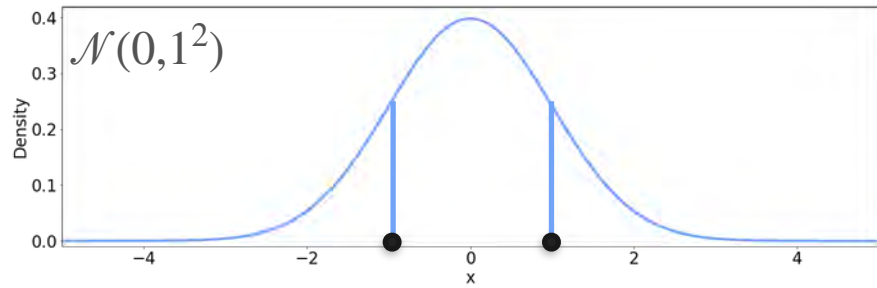
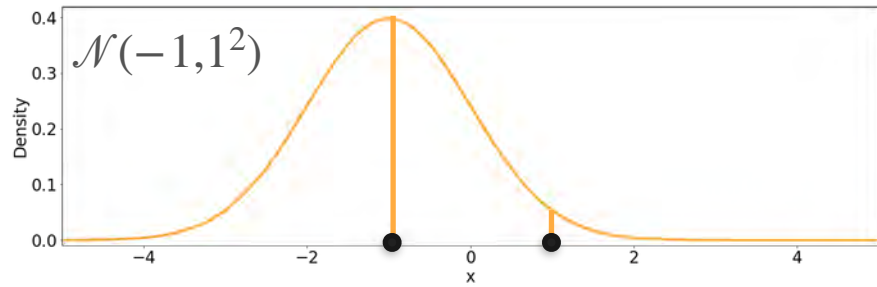
Observations

Candidates



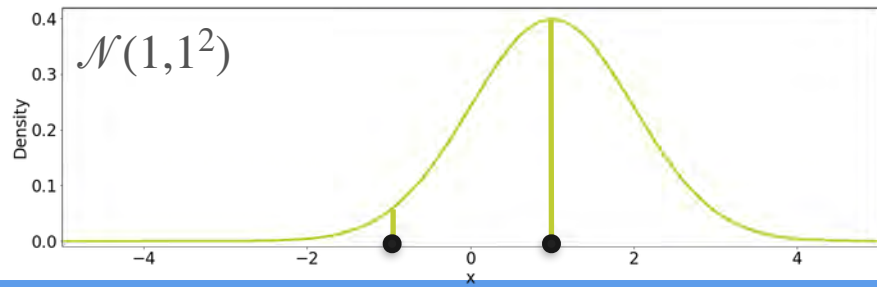
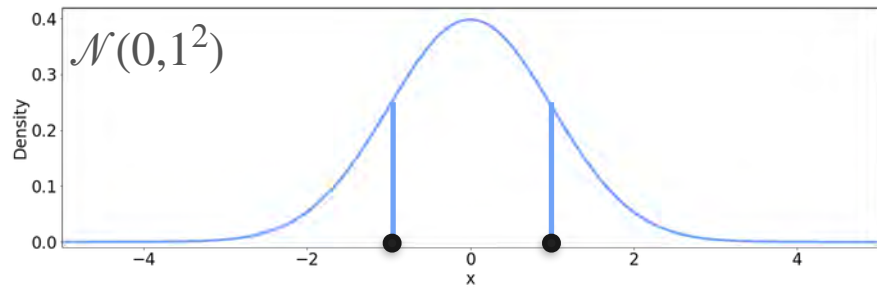
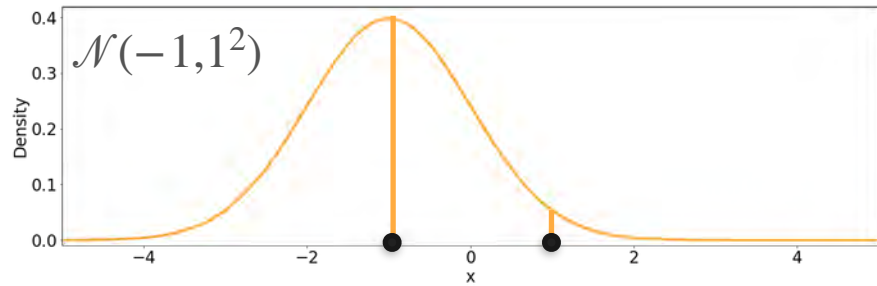
Observations

Candidates



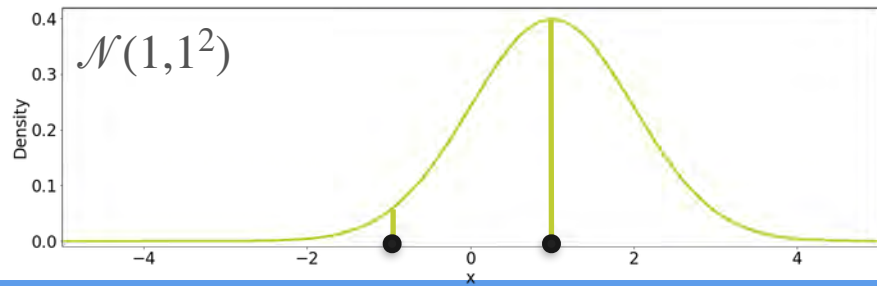
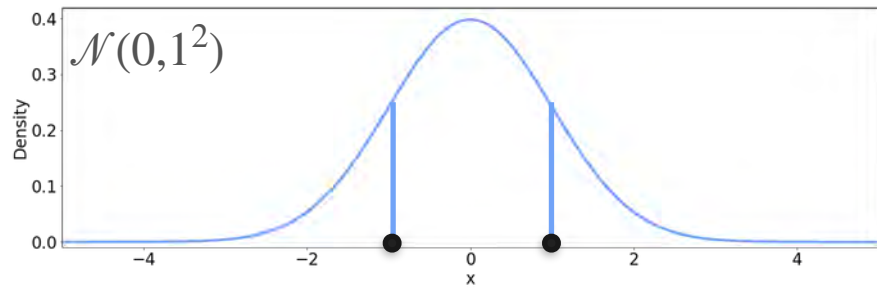
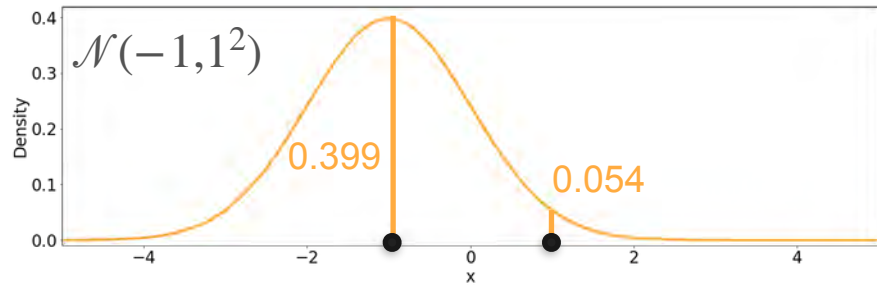
Observations

Candidates



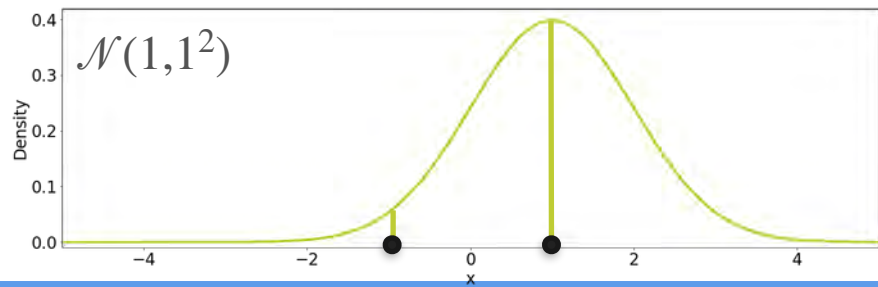
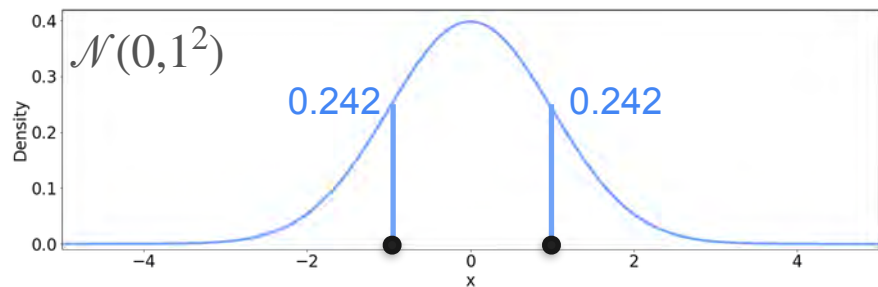
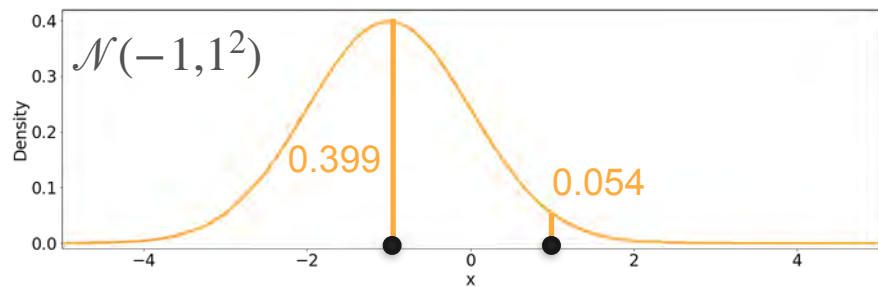
Observations

Candidates



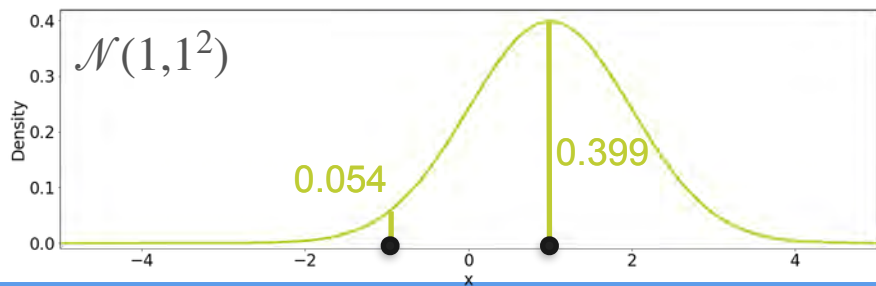
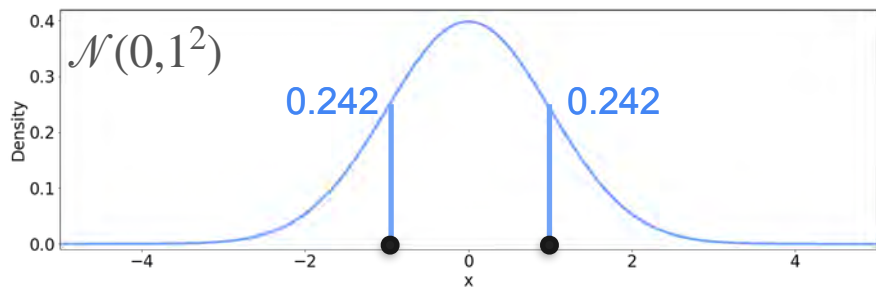
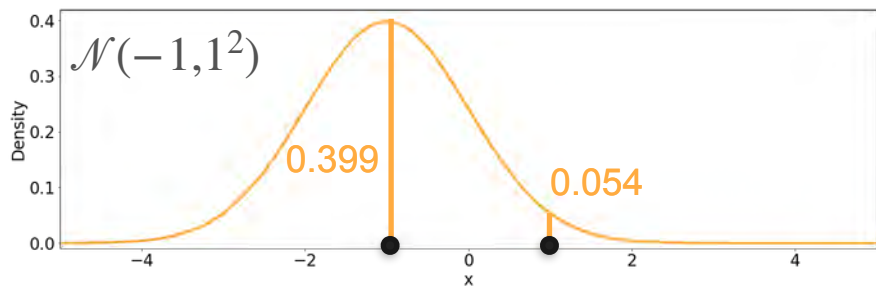
Observations

Candidates



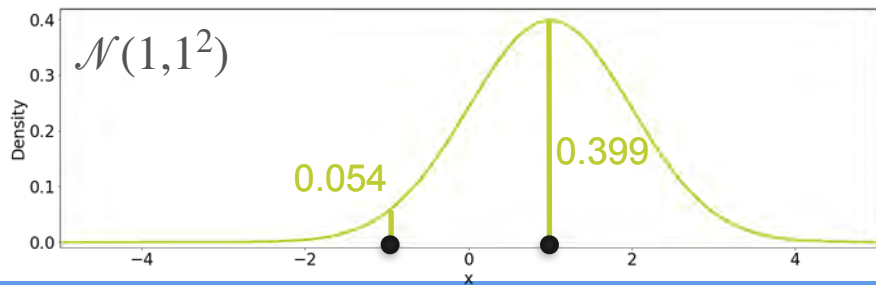
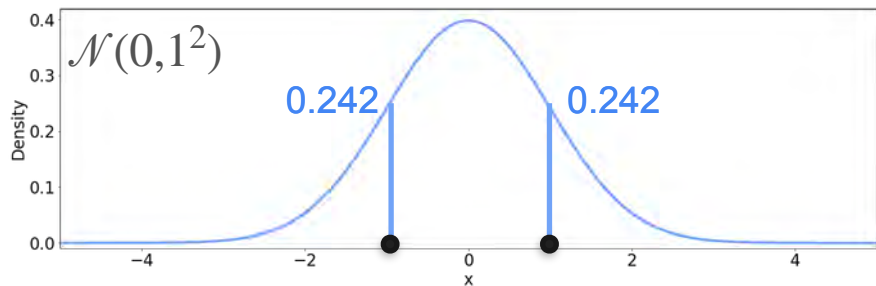
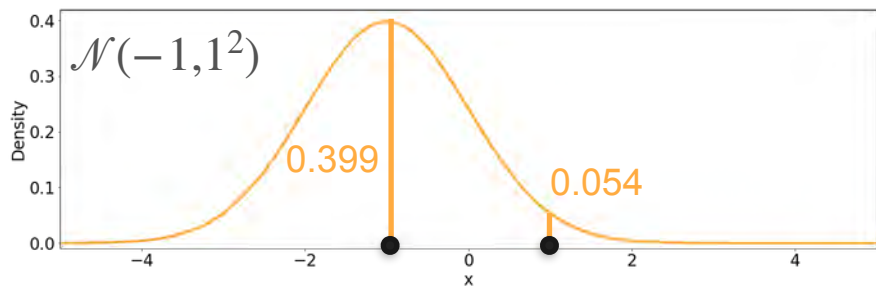
Observations

Candidates



Observations

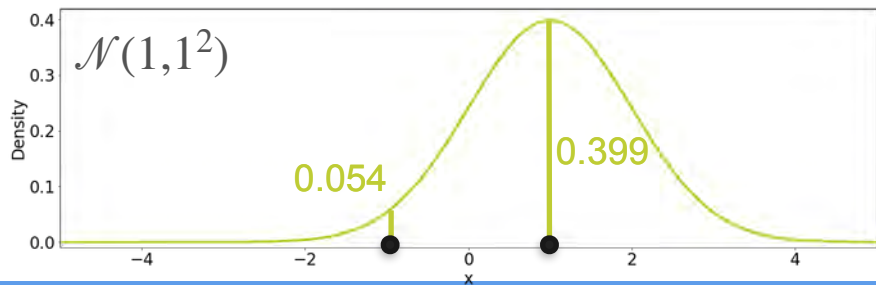
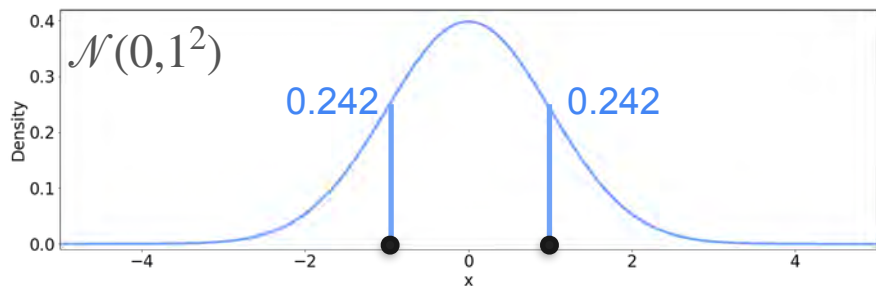
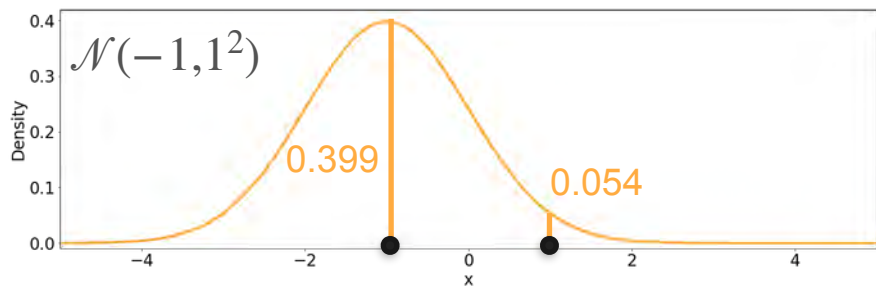
Candidates



Observations

$$0.399 \cdot 0.054 = 0.022$$

Candidates

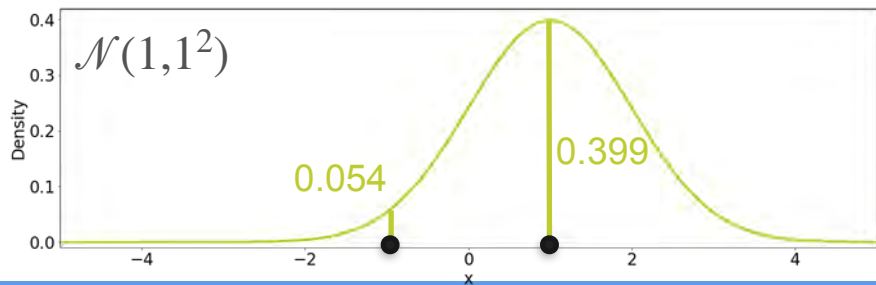
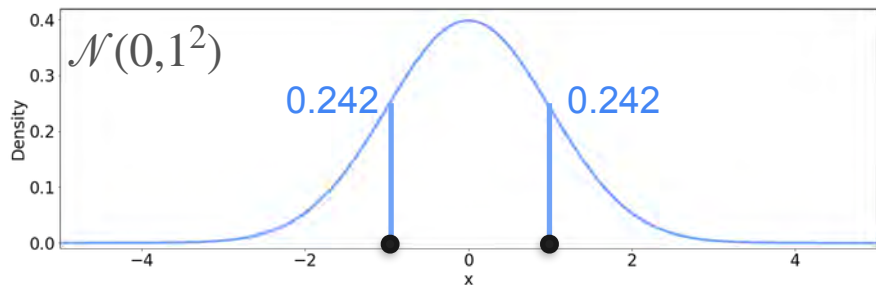
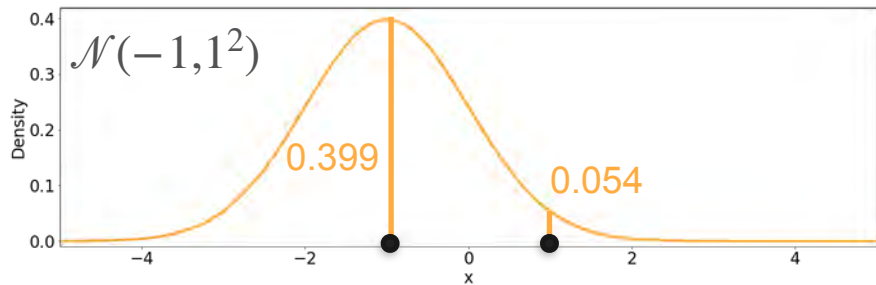


Observations

$$0.399 \cdot 0.054 = 0.022$$

$$0.242 \cdot 0.242 = 0.059$$

Candidates



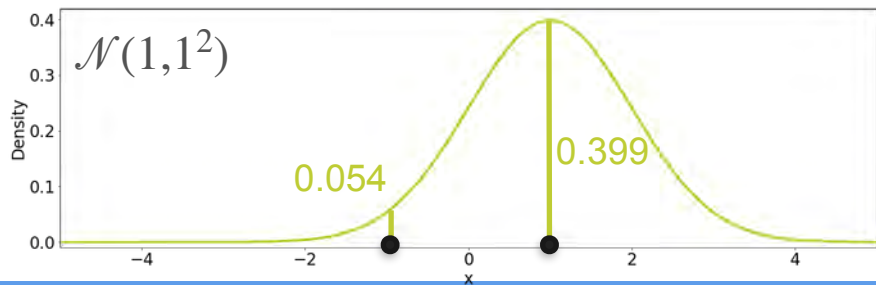
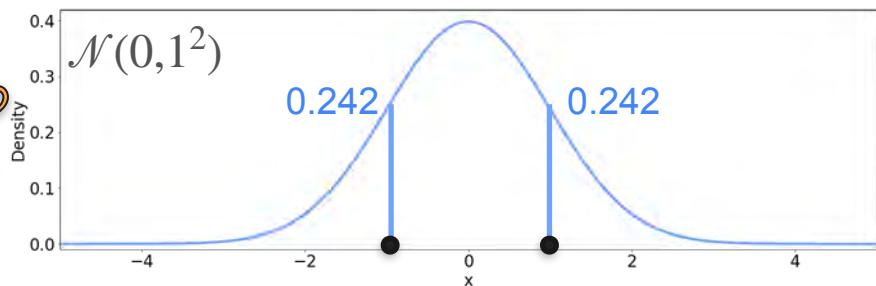
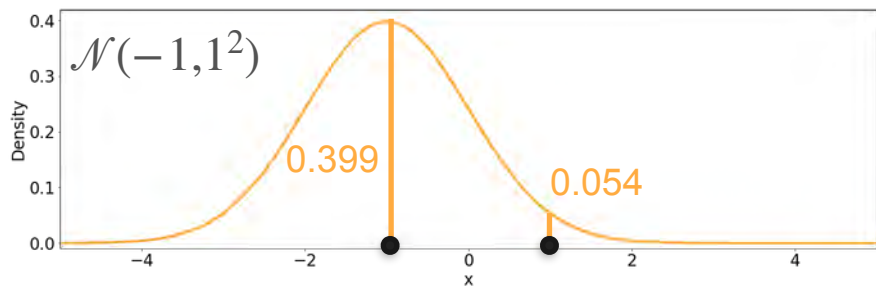
Observations

$$0.399 \cdot 0.054 = 0.022$$

$$0.242 \cdot 0.242 = 0.059$$

$$0.054 \cdot 0.399 = 0.022$$

Candidates



Observations

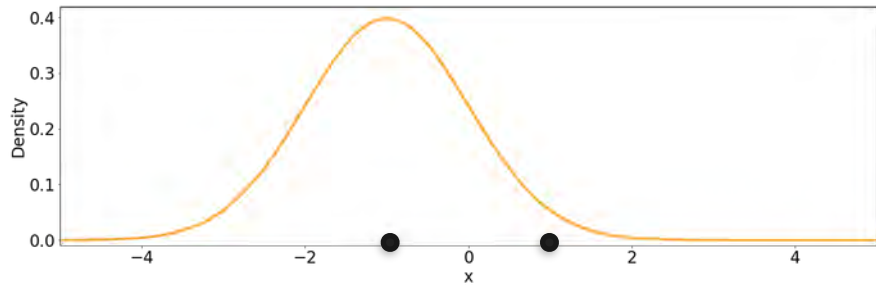
$$0.399 \cdot 0.054 = 0.022$$

The $\mathcal{N}(0, 1)$ is more likely!

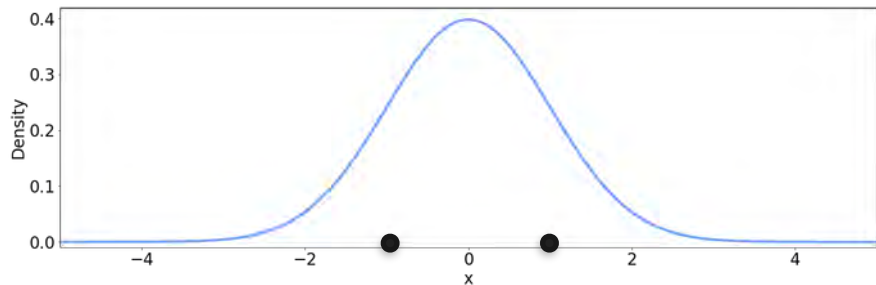
$$0.242 \cdot 0.242 = 0.059$$

$$0.054 \cdot 0.399 = 0.022$$

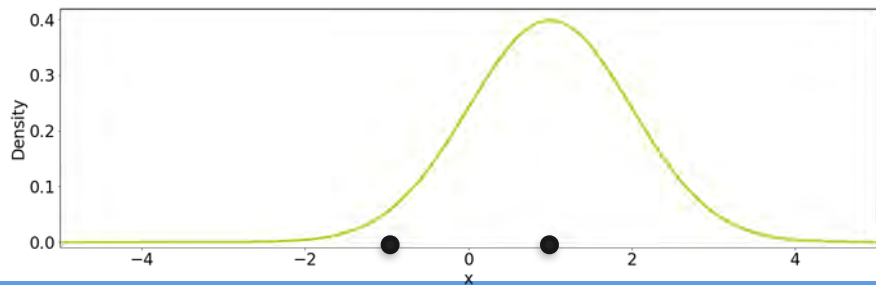
Candidates



Likelihood = 0.022

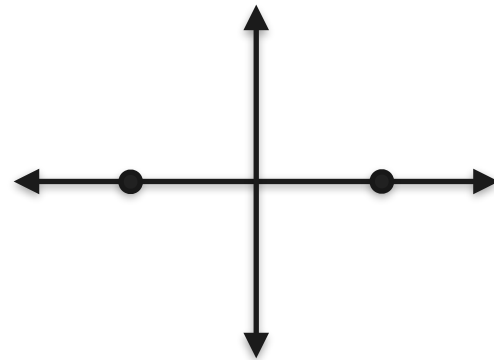


Likelihood = 0.059

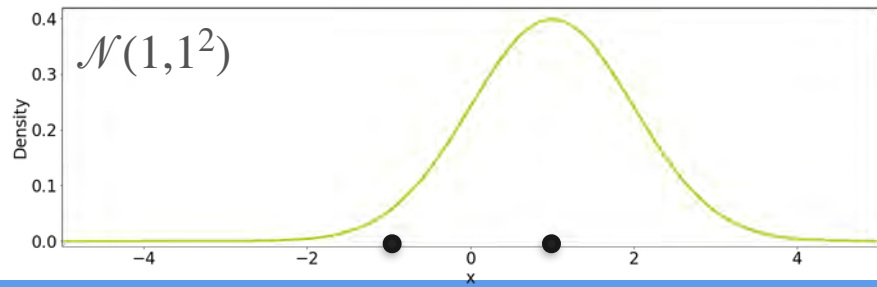
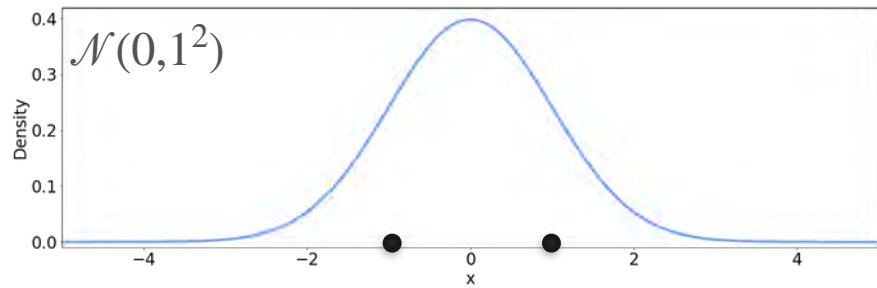
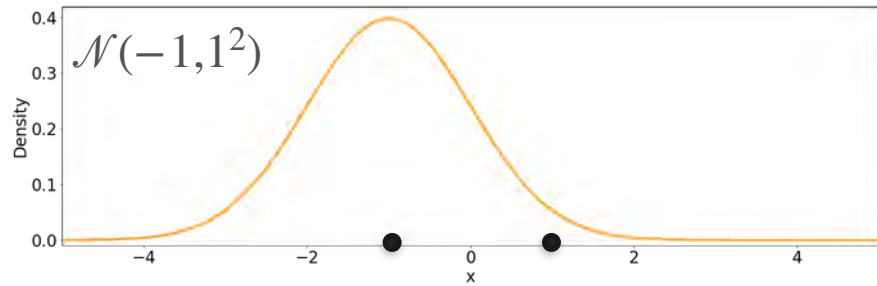


Likelihood = 0.022

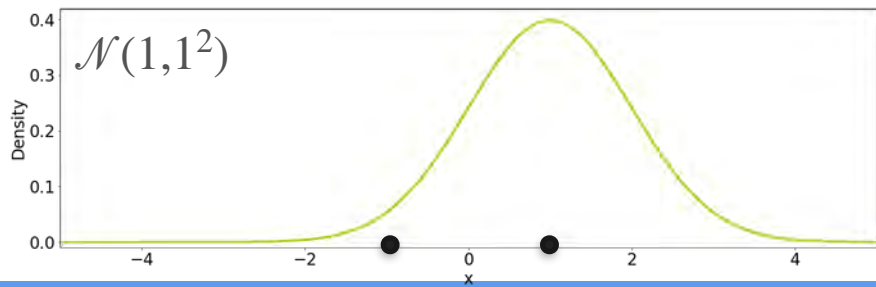
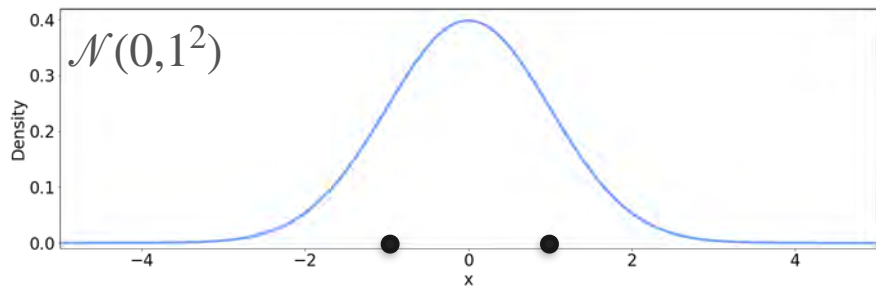
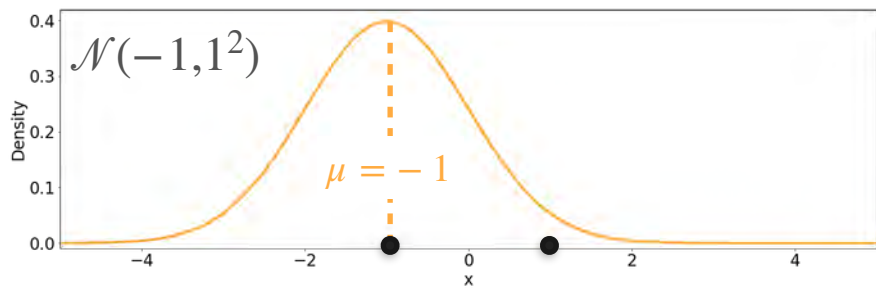
Observations



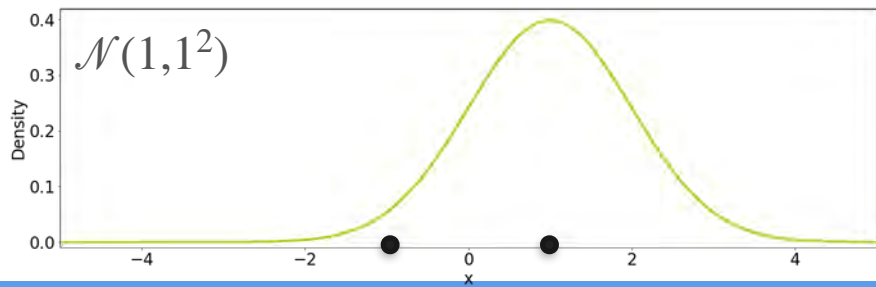
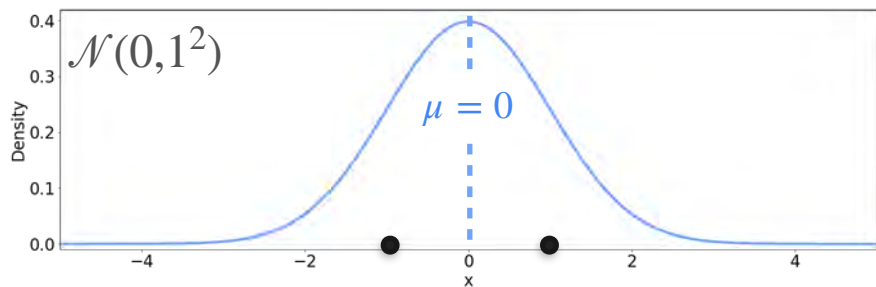
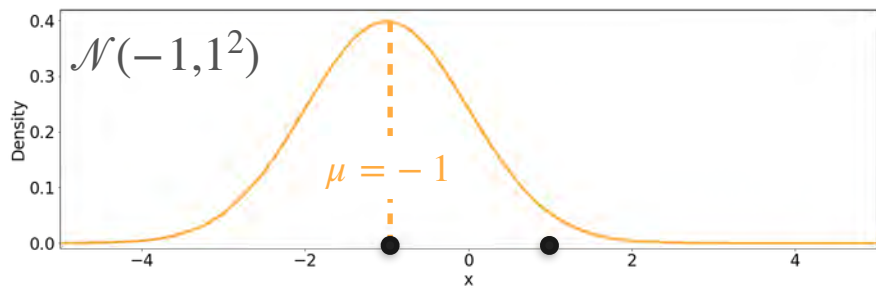
Candidates



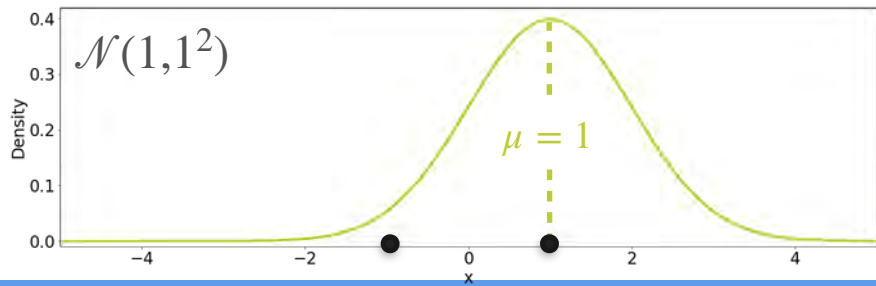
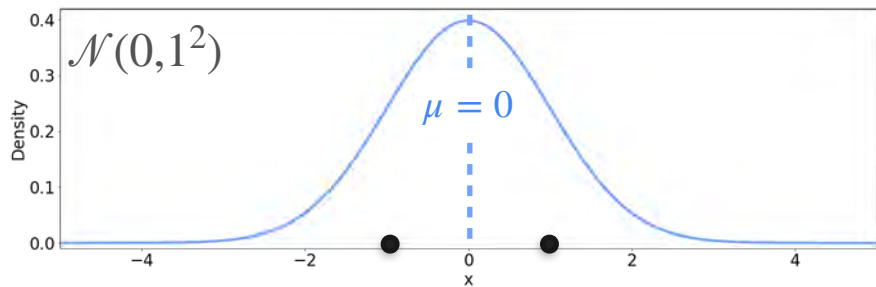
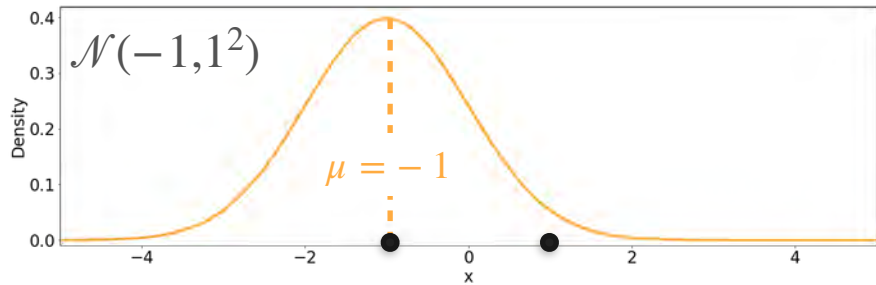
Candidates



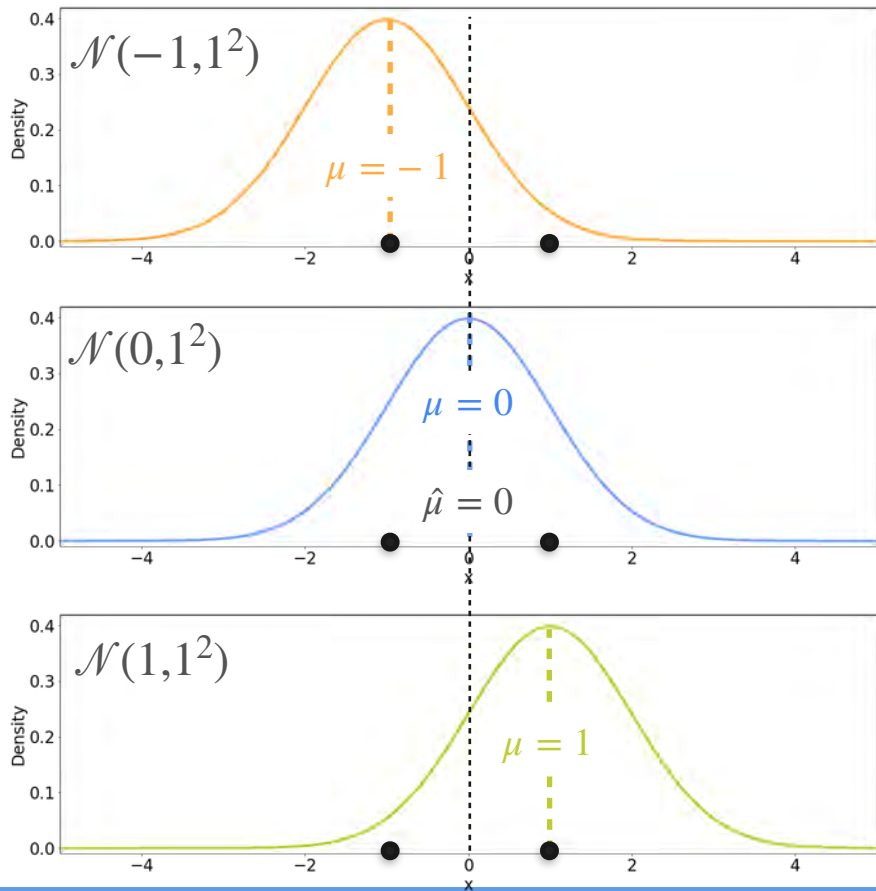
Candidates



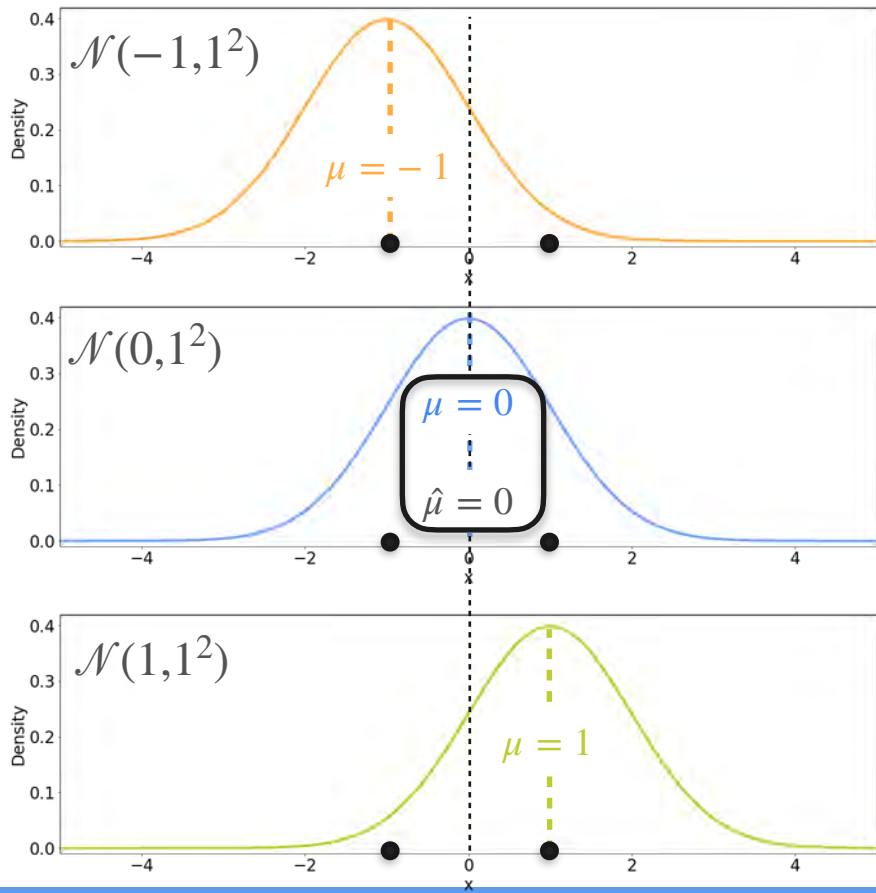
Candidates



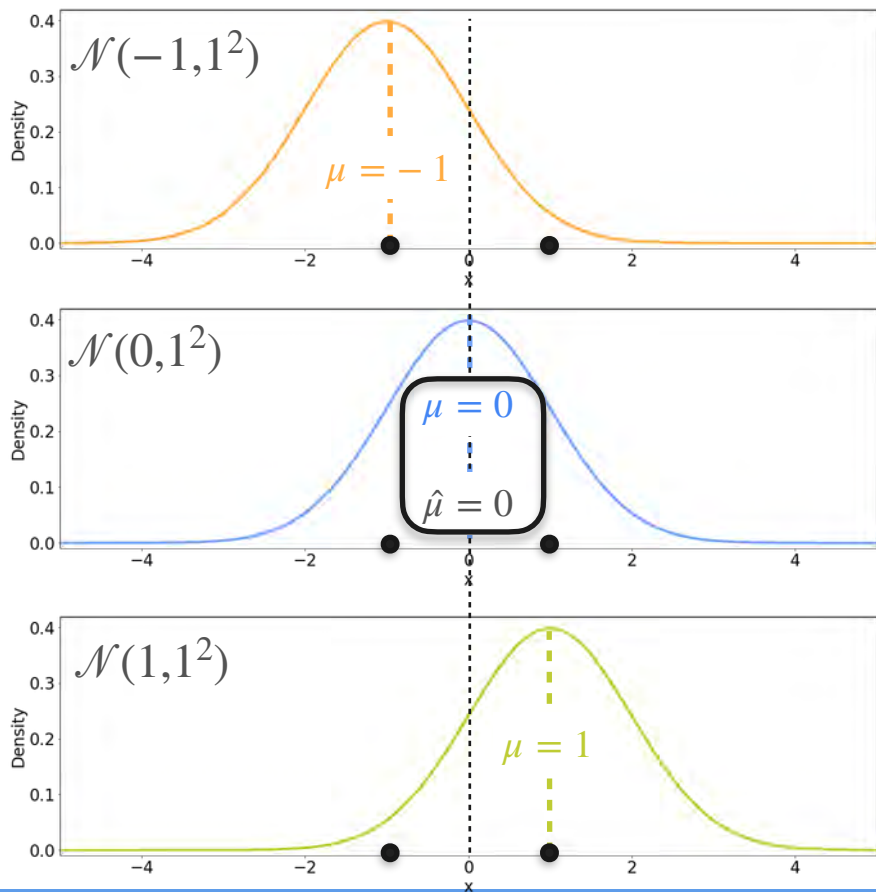
Candidates



Candidates



Candidates

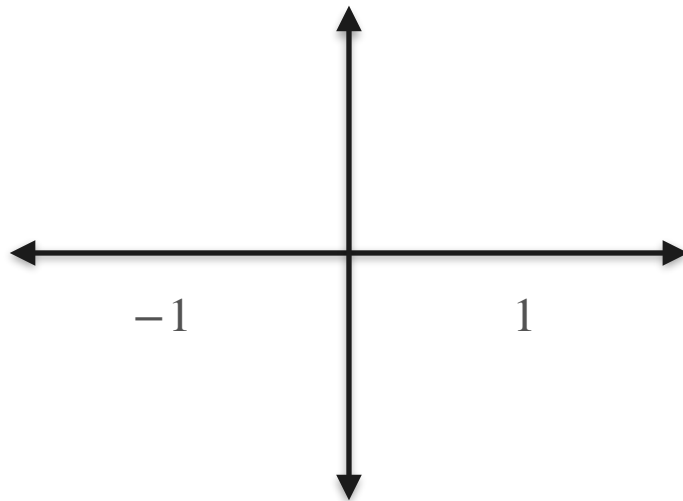


The best distribution is the one where the **mean** of the distribution is the **mean** of the sample

Gaussians With Three Different Variance

Candidates

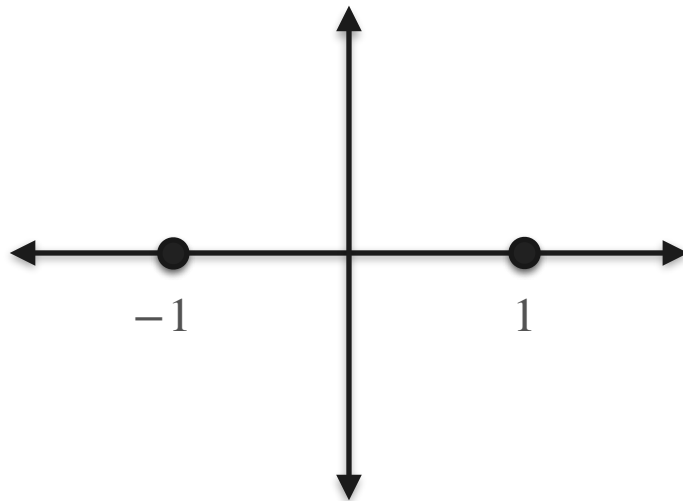
Observations



Gaussians With Three Different Variance

Candidates

Observations

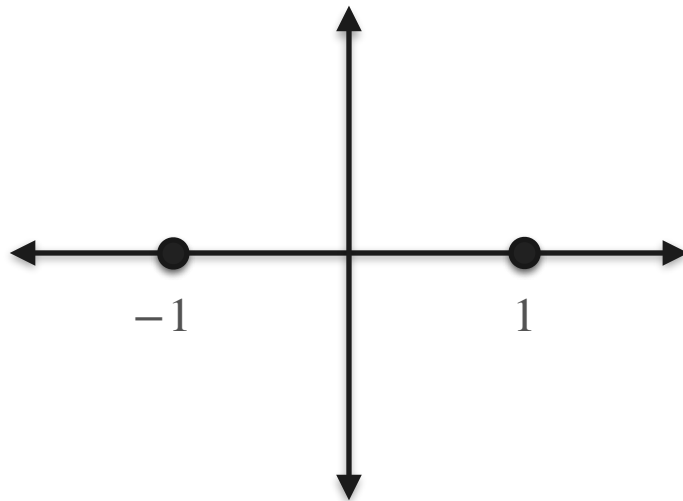


Gaussians With Three Different Variance

Candidates

$$\mathcal{N}(0, 0.5^2)$$

Observations



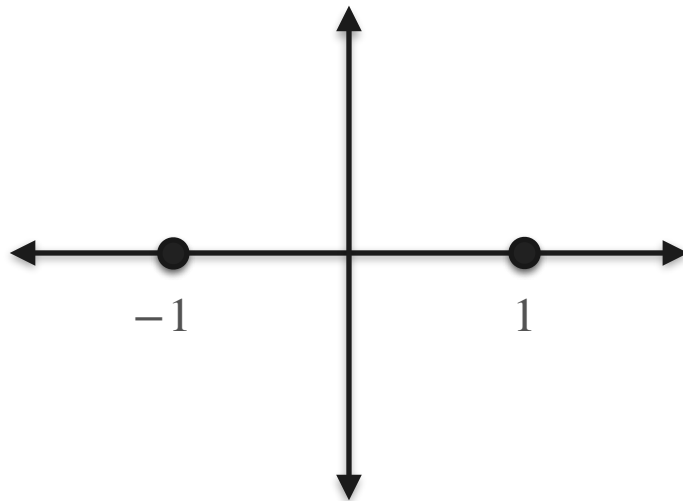
Gaussians With Three Different Variance

Candidates

Observations

$$\mathcal{N}(0, 0.5^2)$$

$$\mathcal{N}(0, 1^2)$$



Gaussians With Three Different Variance

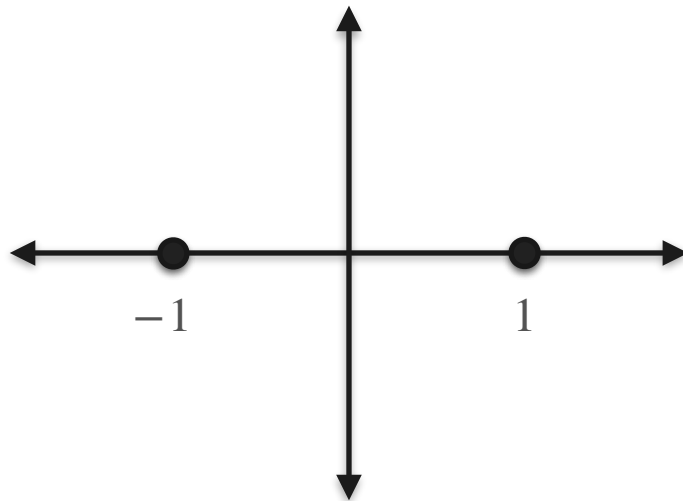
Candidates

Observations

$$\mathcal{N}(0, 0.5^2)$$

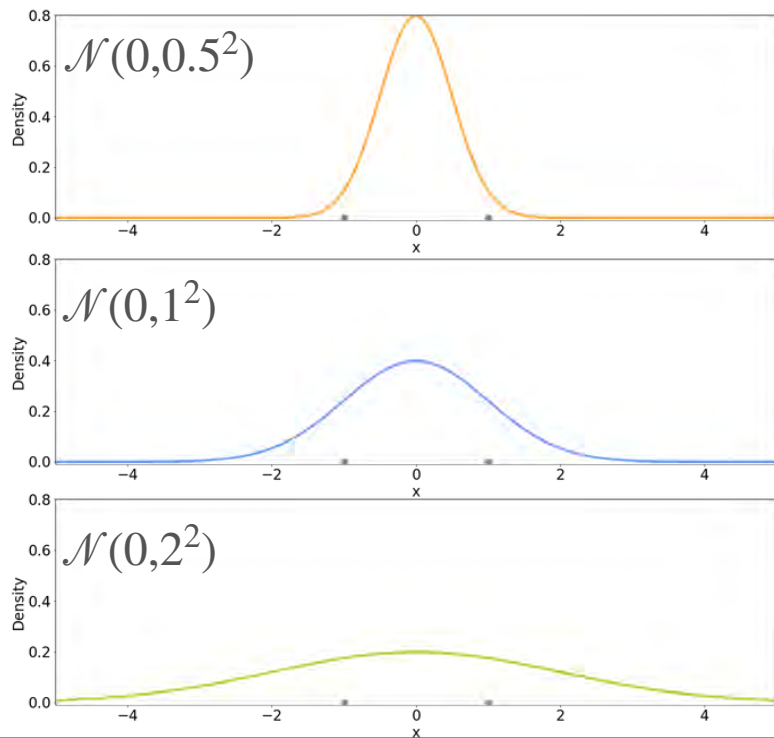
$$\mathcal{N}(0, 1^2)$$

$$\mathcal{N}(0, 2^2)$$

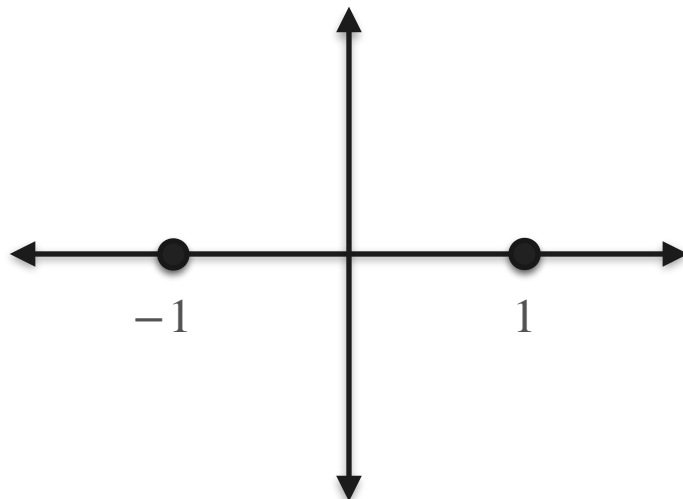


Gaussians With Three Different Variance

Candidates

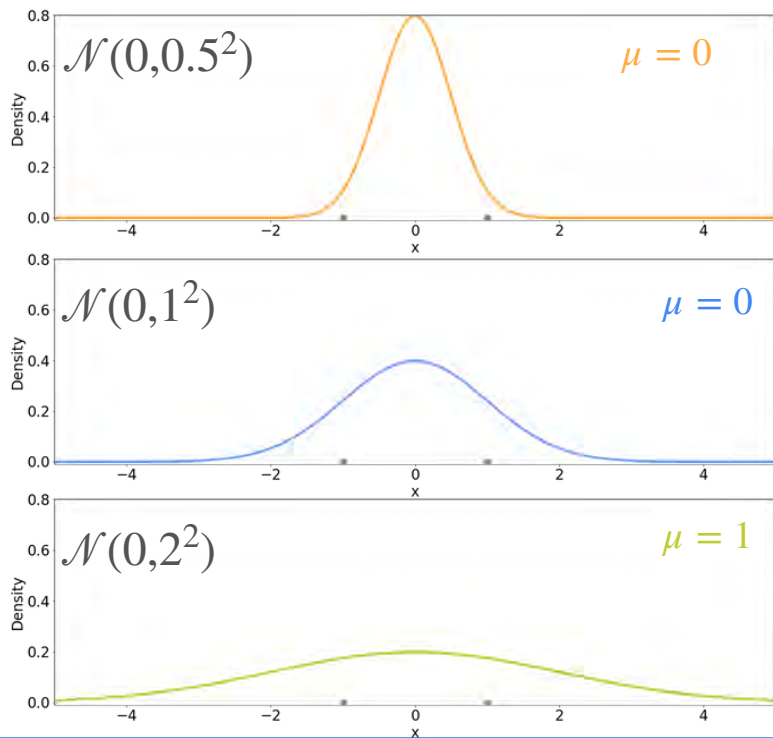


Observations

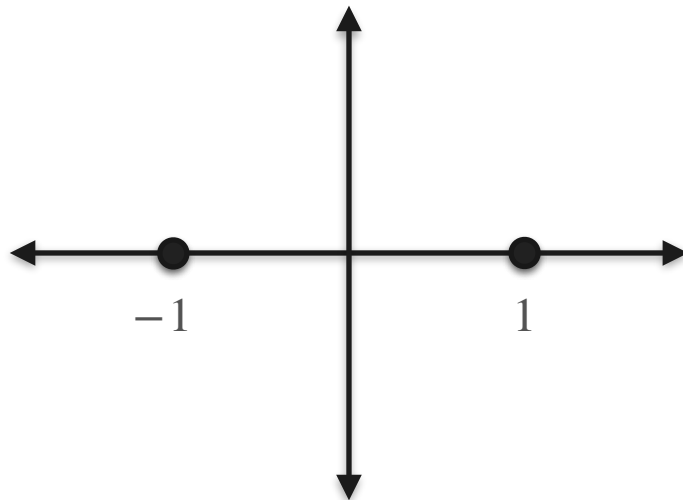


Gaussians With Three Different Variance

Candidates

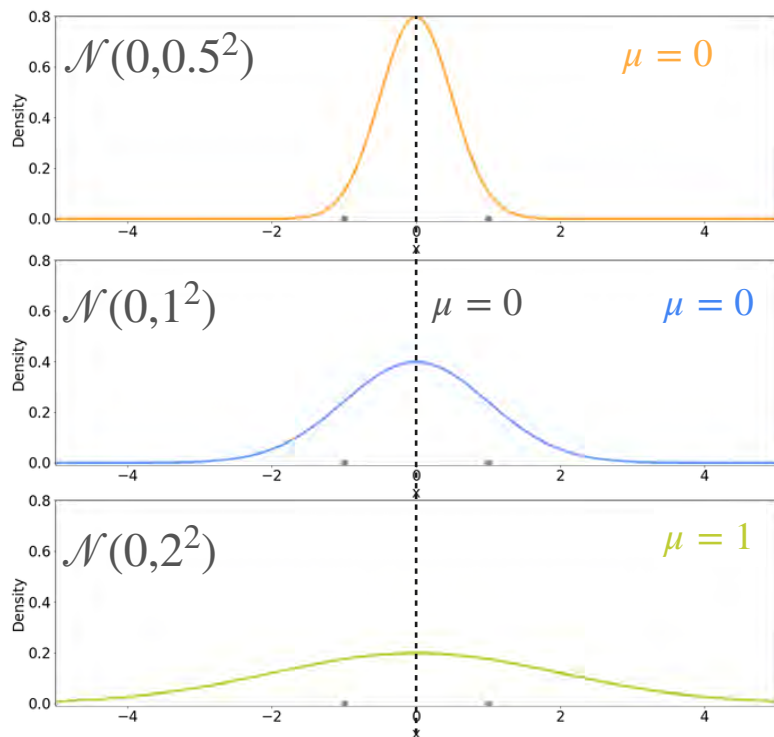


Observations

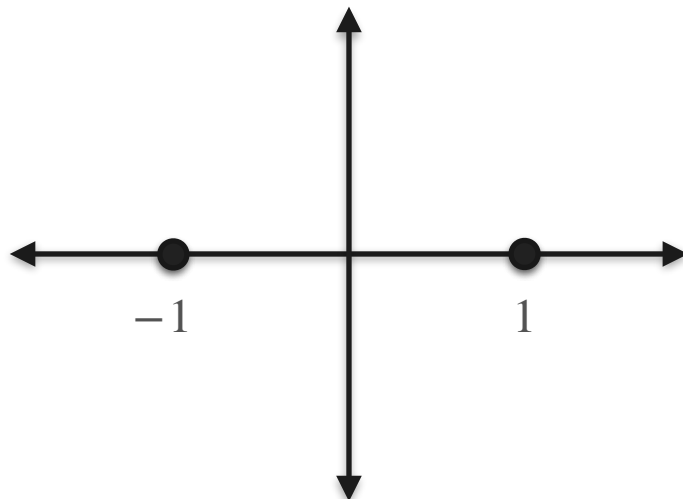


Gaussians With Three Different Variance

Candidates

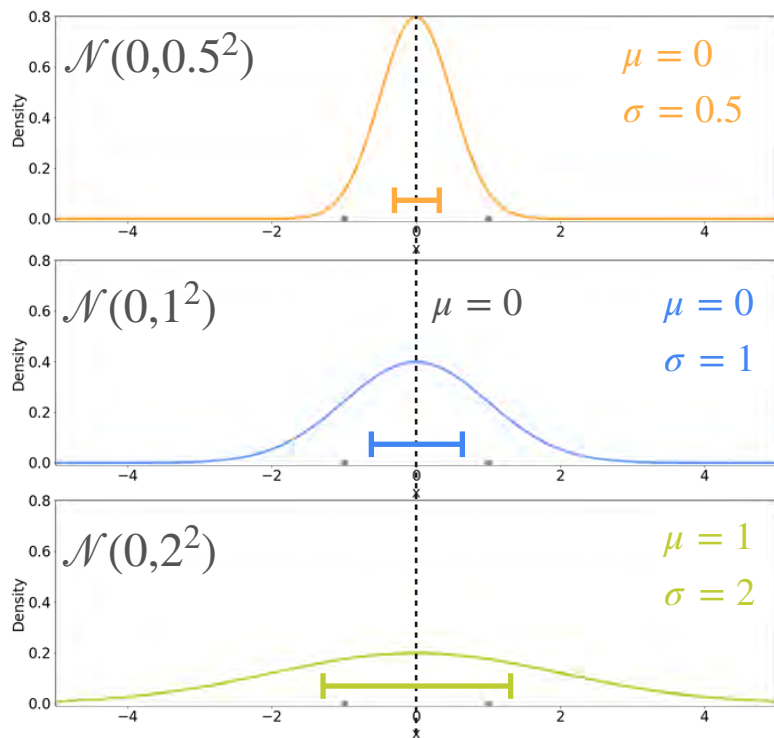


Observations

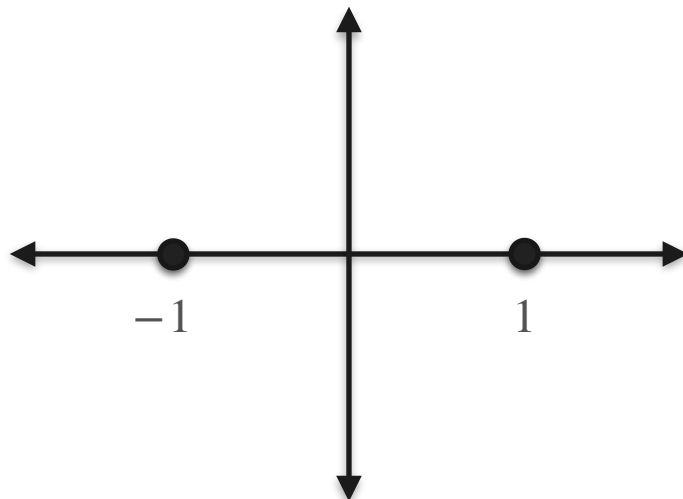


Gaussians With Three Different Variance

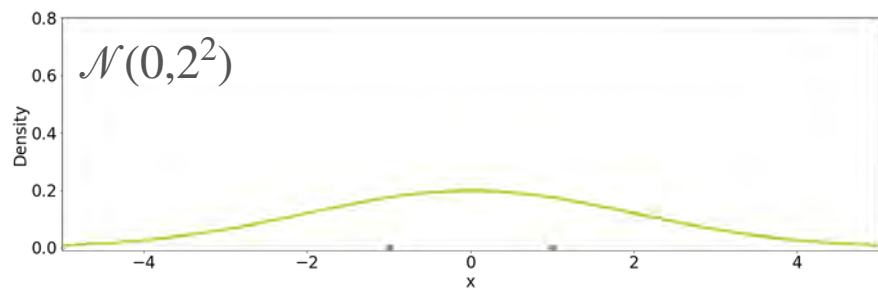
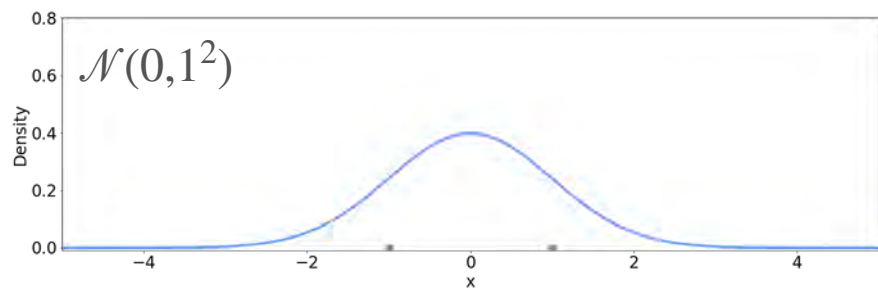
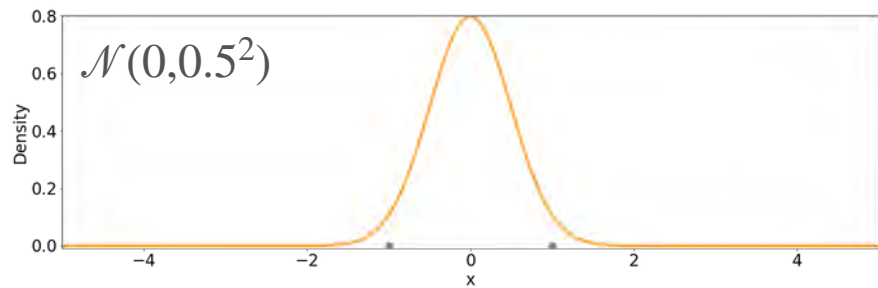
Candidates



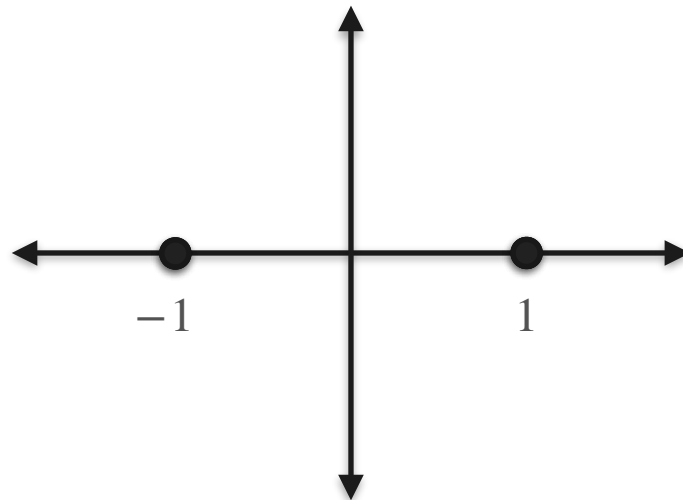
Observations



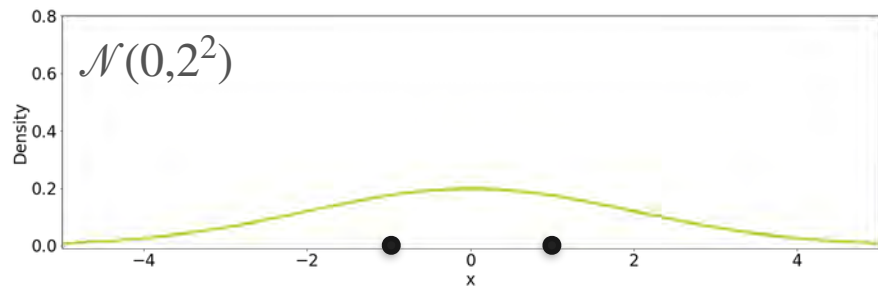
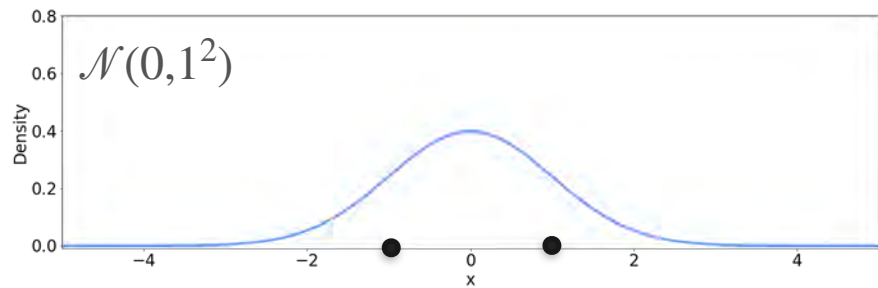
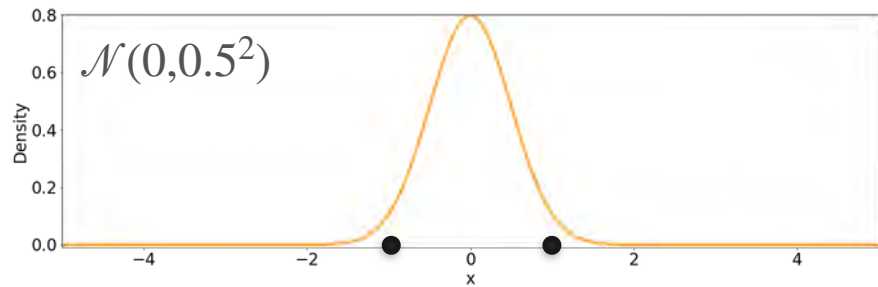
Candidates



Observations

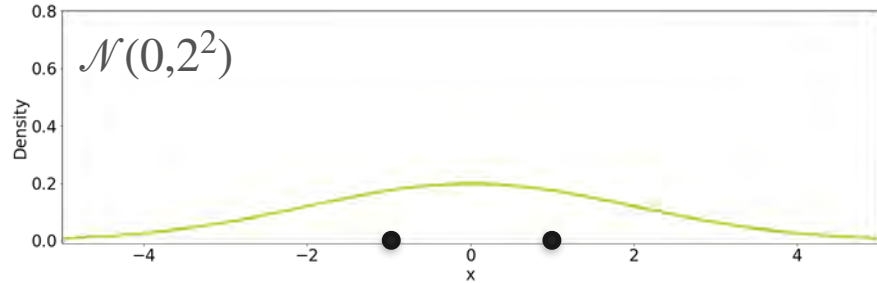
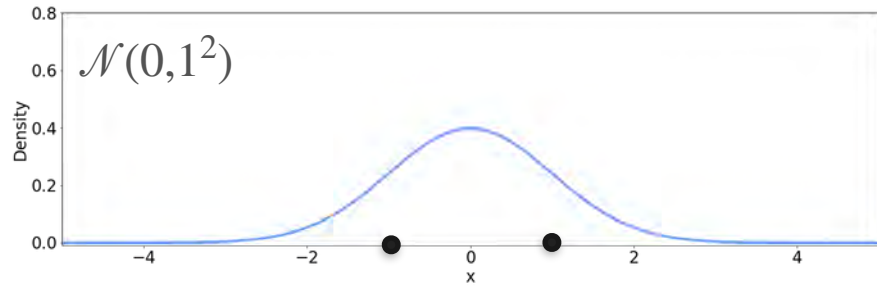
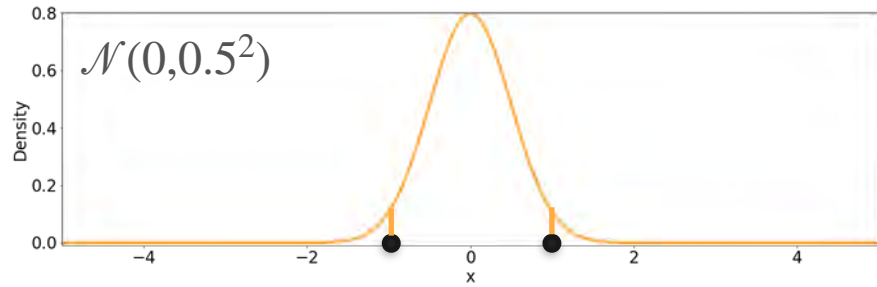


Candidates



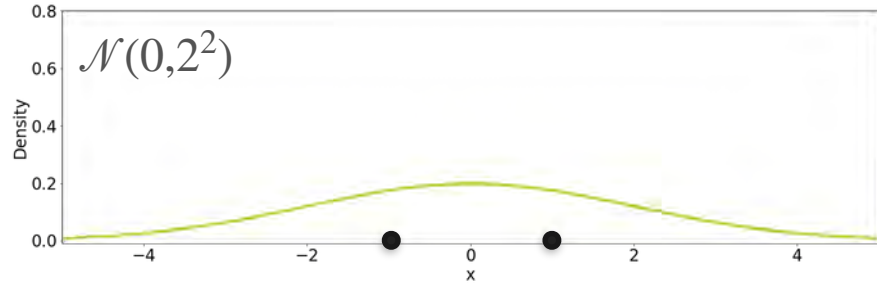
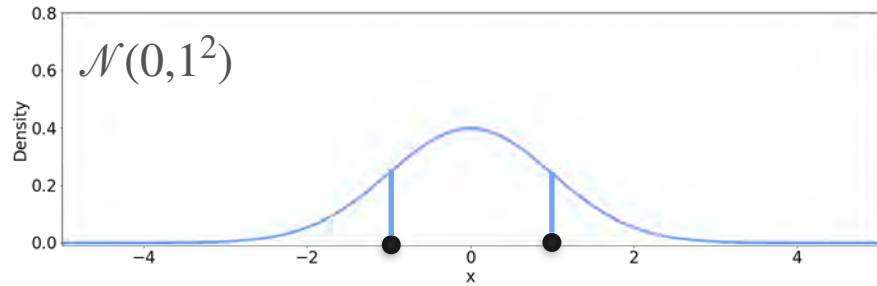
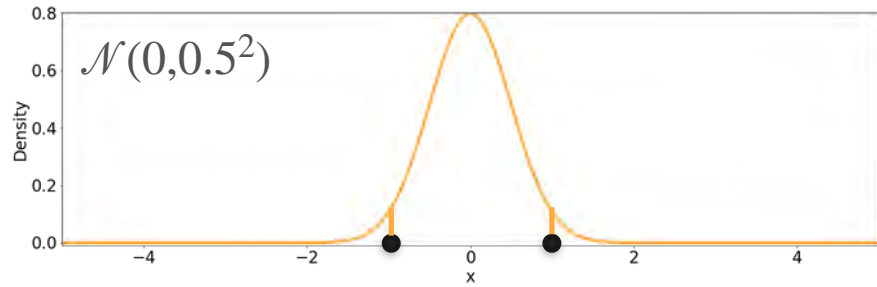
Observations

Candidates



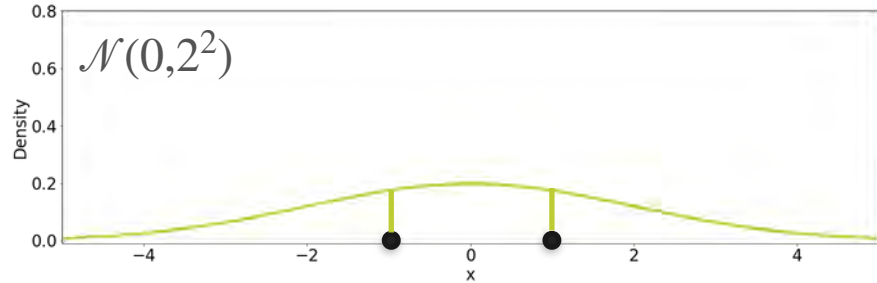
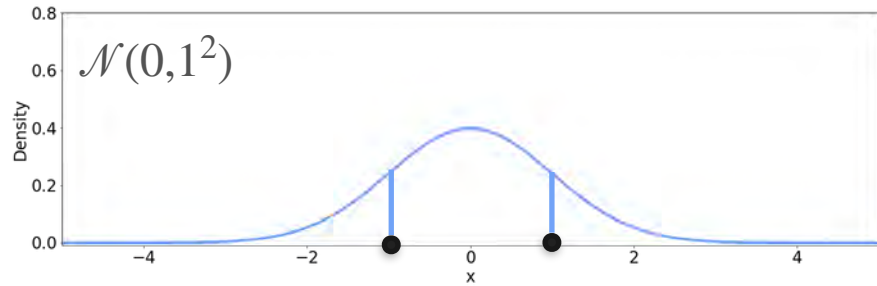
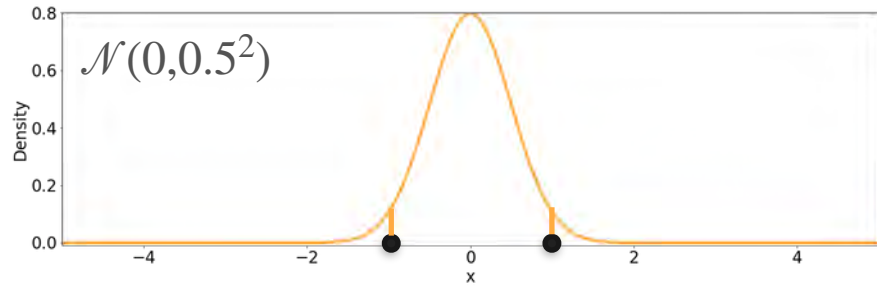
Observations

Candidates



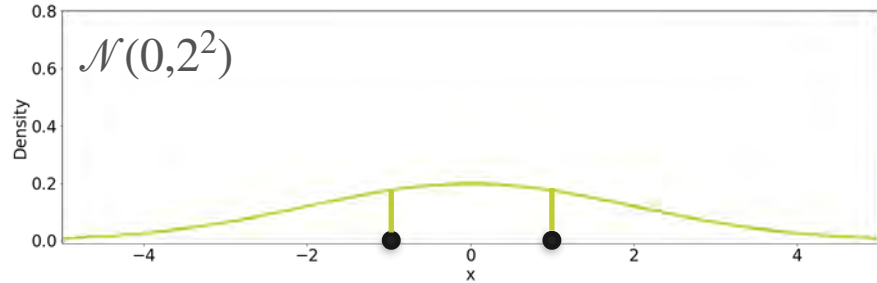
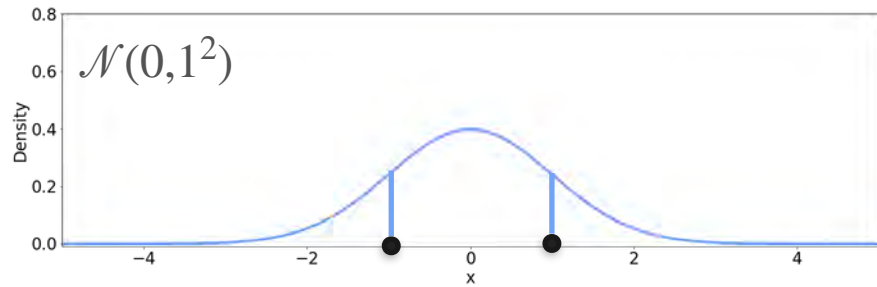
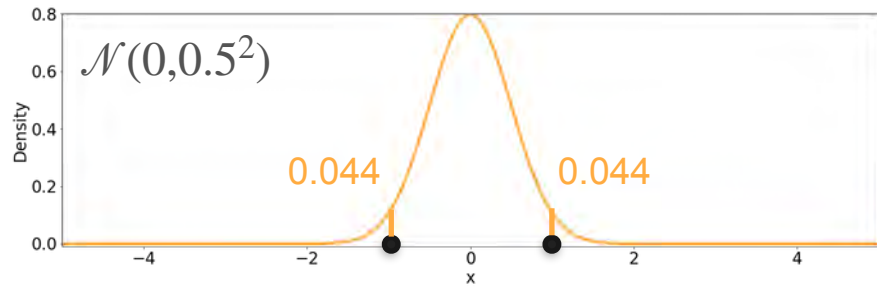
Observations

Candidates



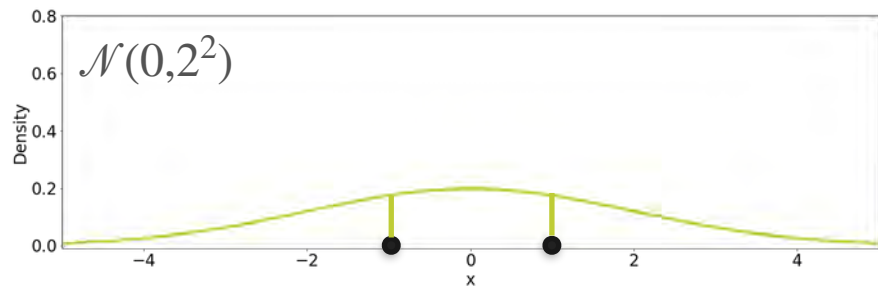
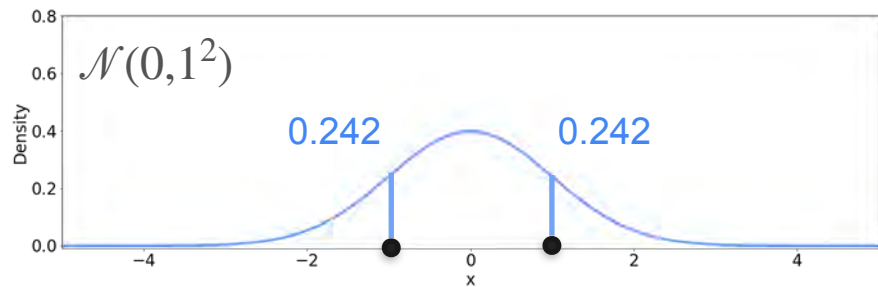
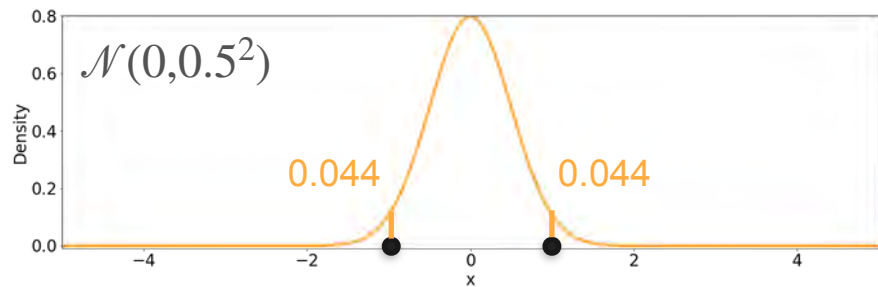
Observations

Candidates



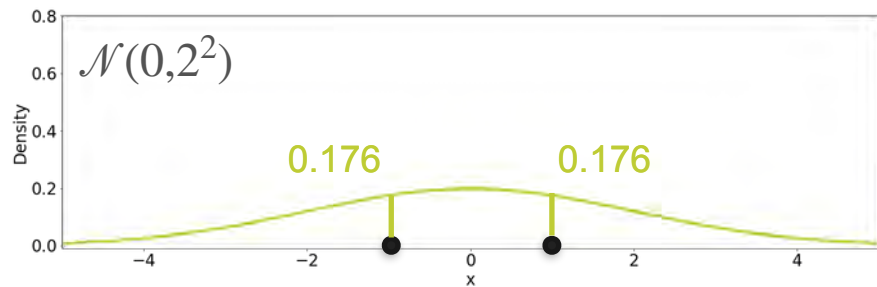
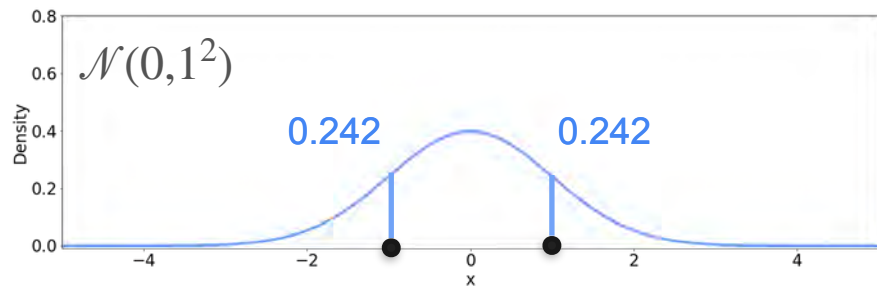
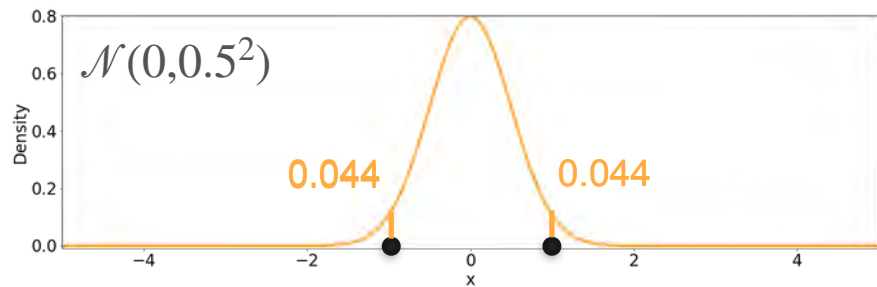
Observations

Candidates



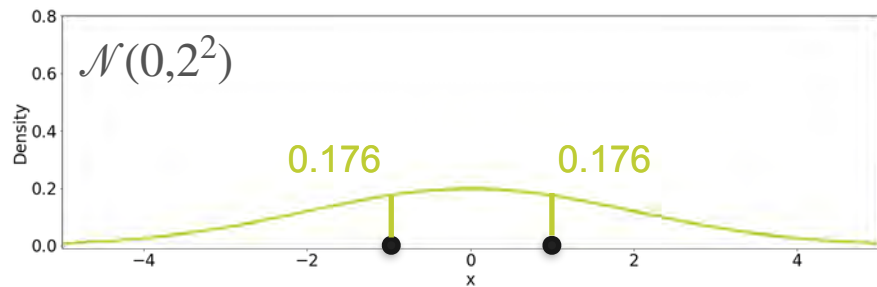
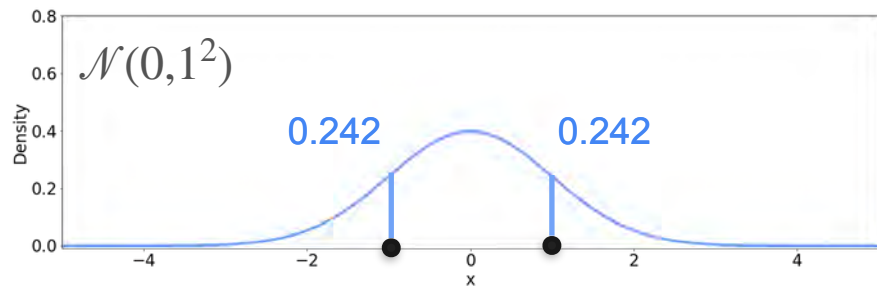
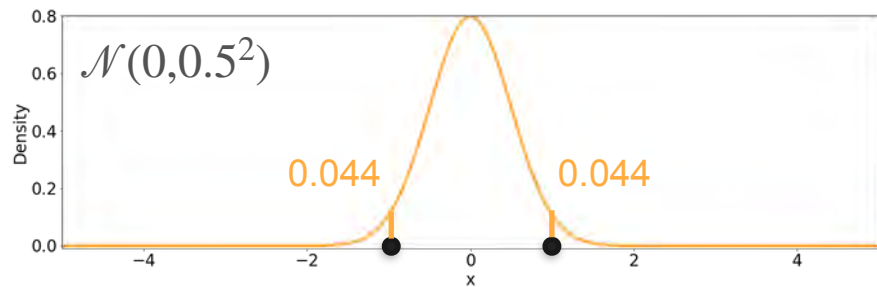
Observations

Candidates



Observations

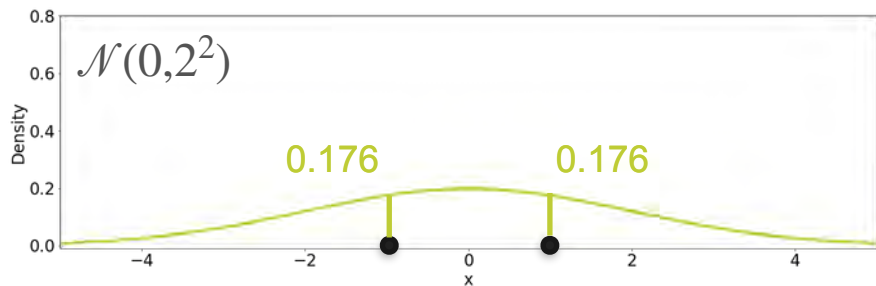
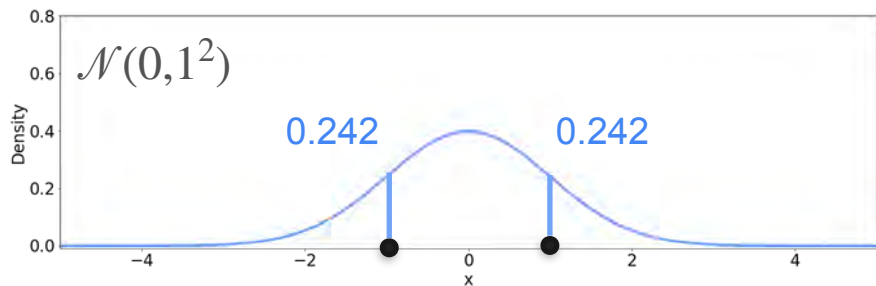
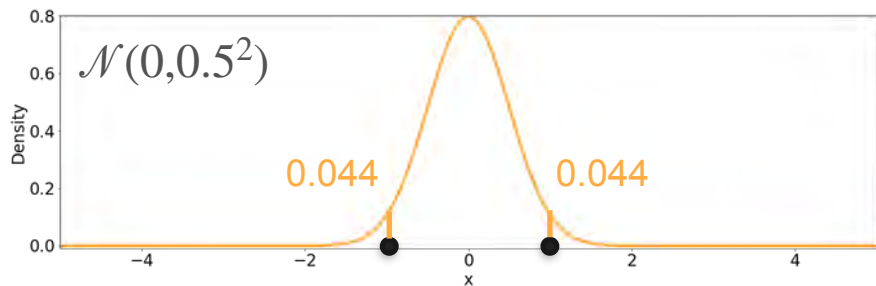
Candidates



Observations

$$0.044 \cdot 0.044 = 0.002$$

Candidates

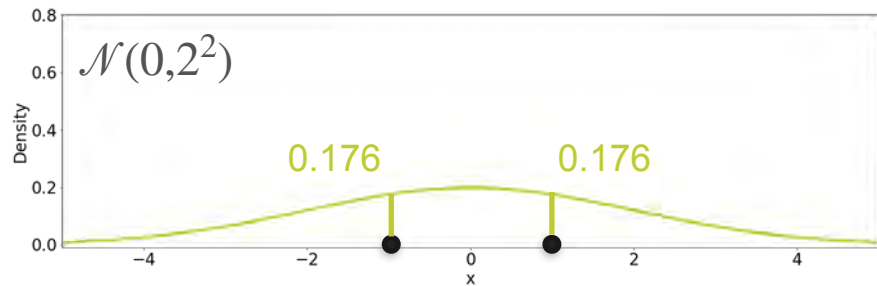
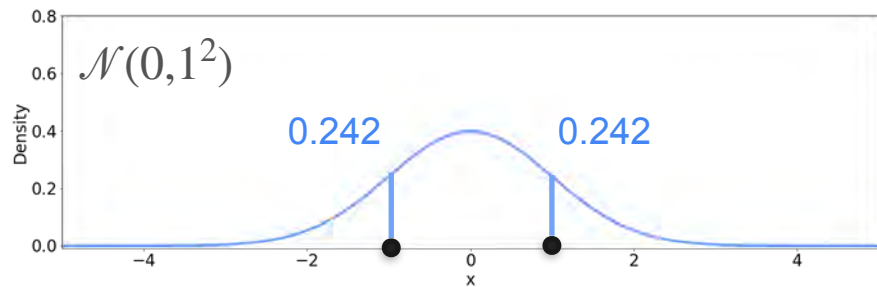
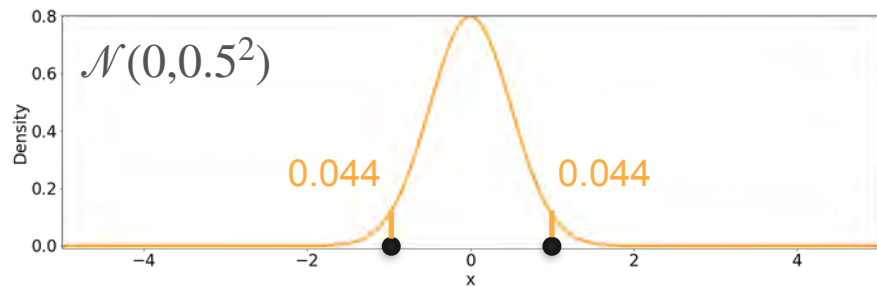


Observations

$$0.044 \cdot 0.044 = 0.002$$

$$0.242 \cdot 0.242 = 0.059$$

Candidates



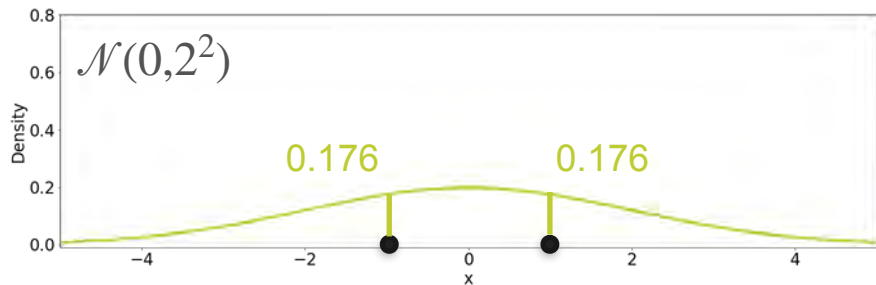
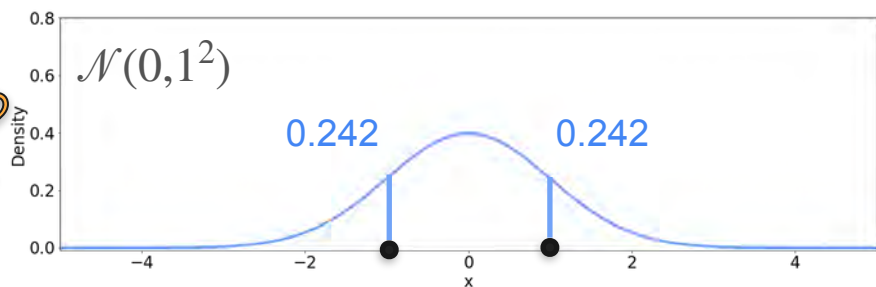
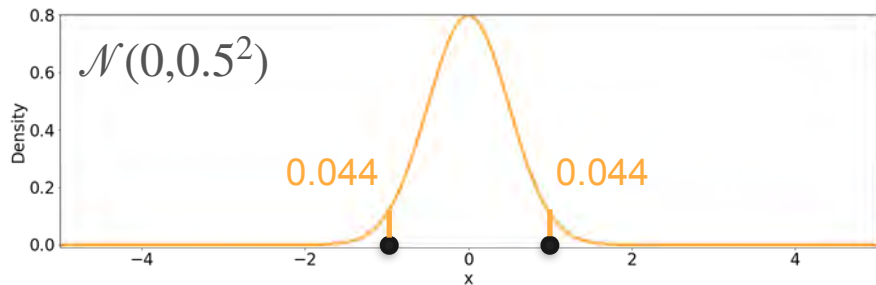
Observations

$$0.044 \cdot 0.044 = 0.002$$

$$0.242 \cdot 0.242 = 0.059$$

$$0.176 \cdot 0.176 = 0.031$$

Candidates



Observations

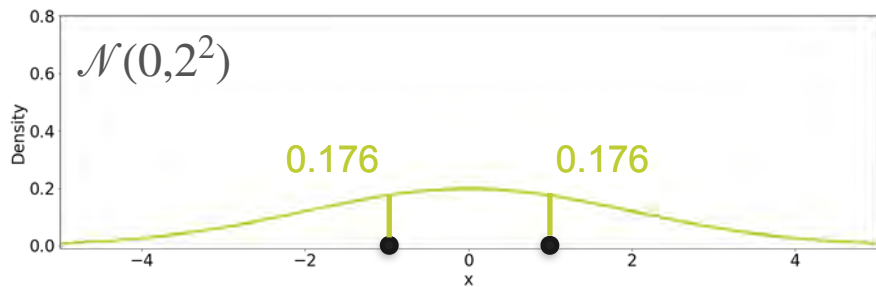
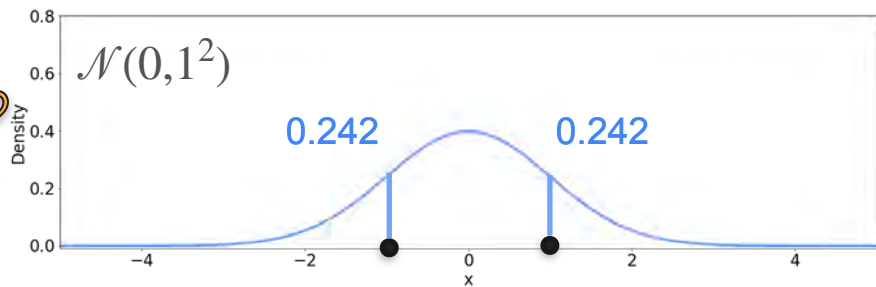
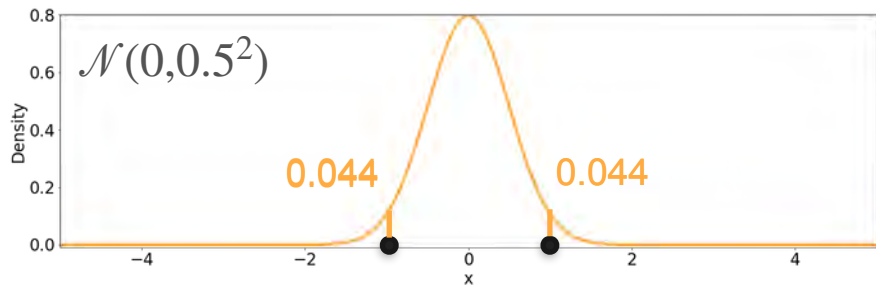
$$0.044 \cdot 0.044 = 0.002$$

The $\mathcal{N}(0, 1^2)$ is more likely!

$$0.242 \cdot 0.242 = 0.059$$

$$0.176 \cdot 0.176 = 0.031$$

Candidates



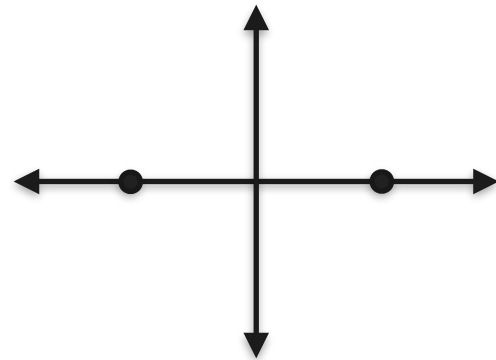
Observations

Likelihood = 0.002

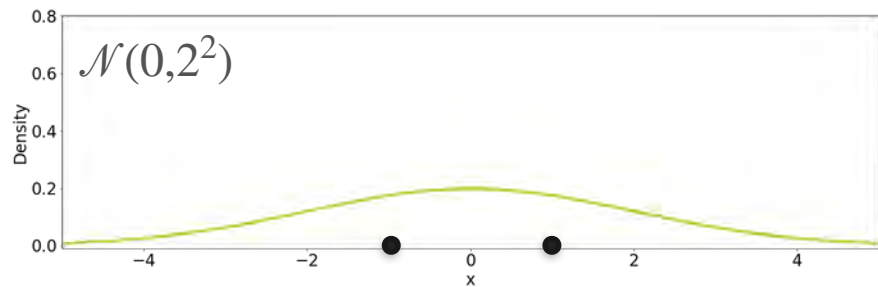
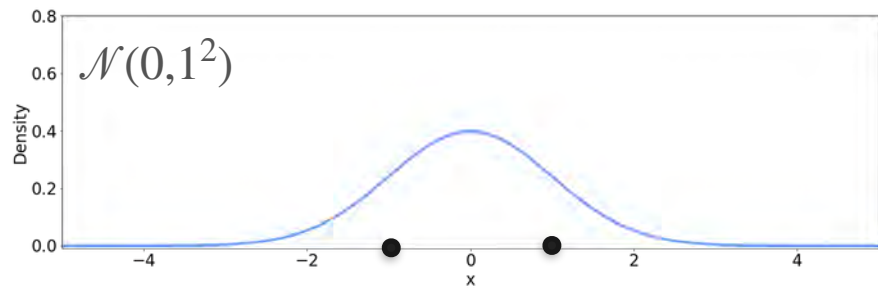
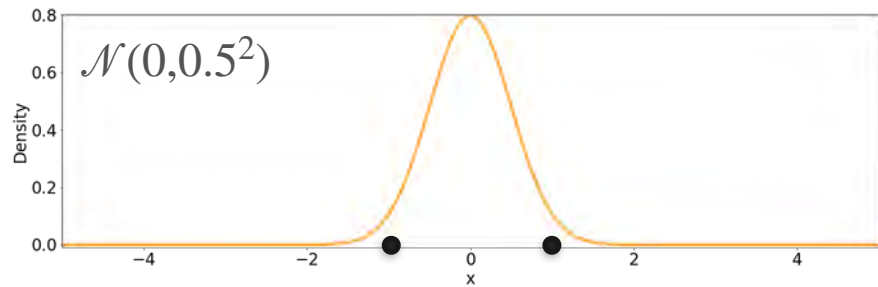
Likelihood = 0.059

Likelihood = 0.031

Observations

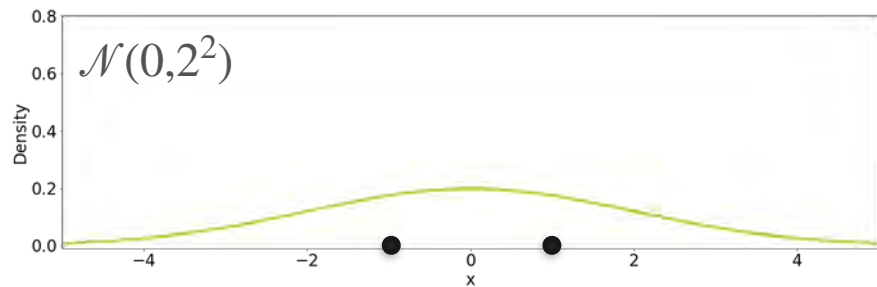
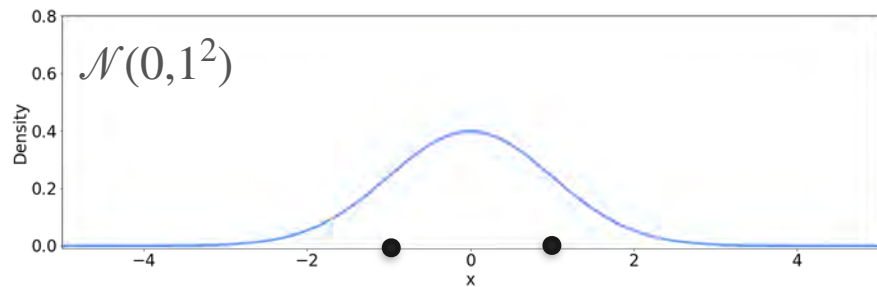
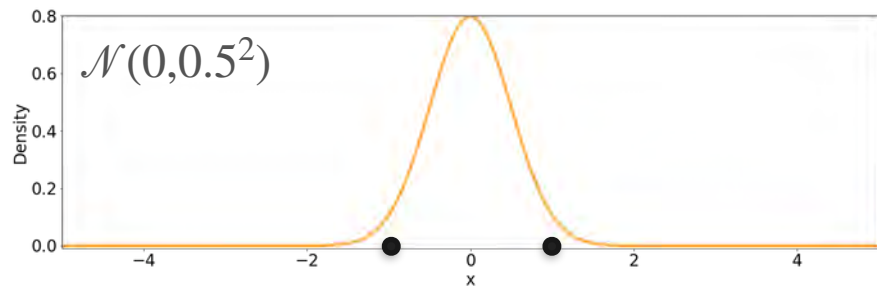


Candidates



Observations

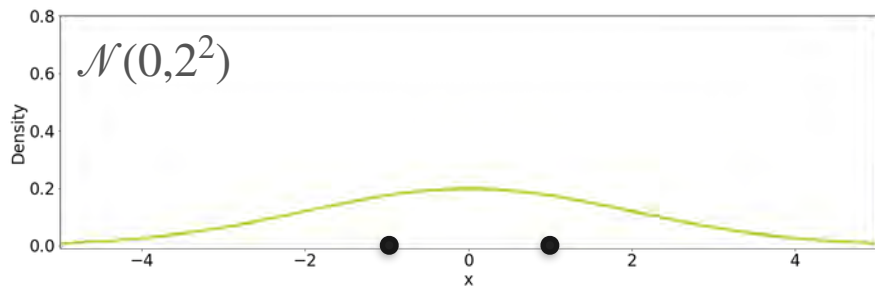
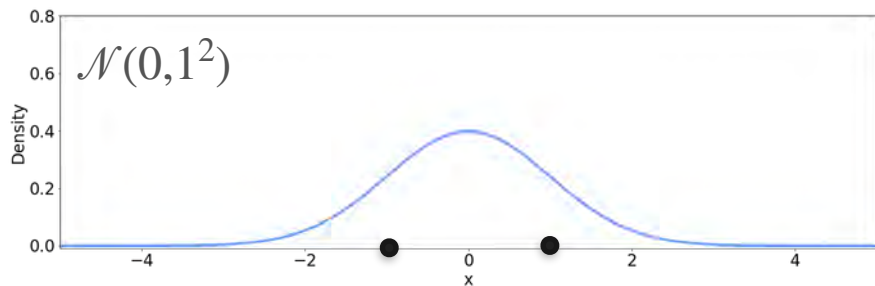
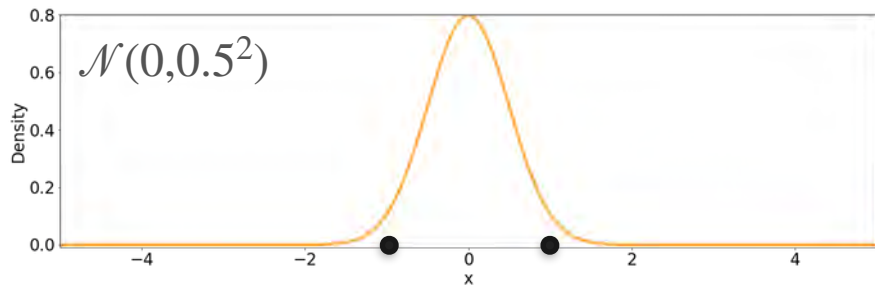
Candidates



Observations

Variance of the observations

Candidates

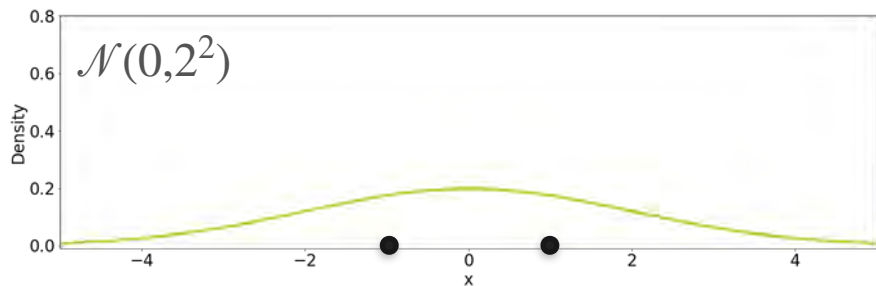
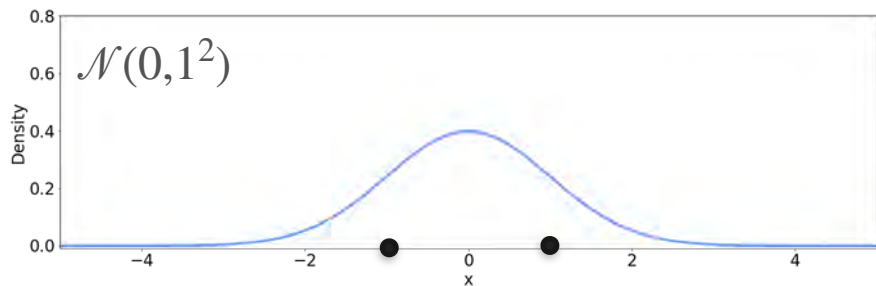
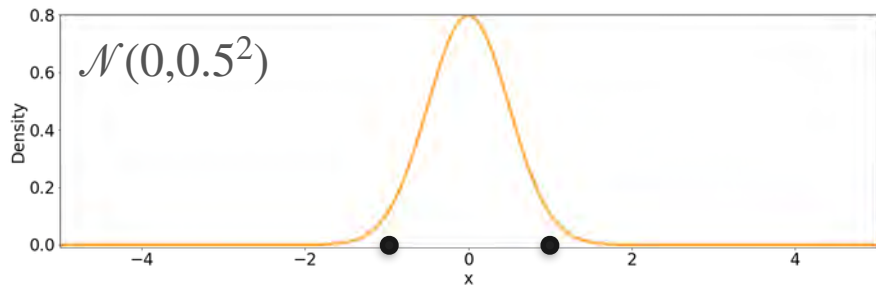


Observations

Variance of the observations

$$\hat{\sigma}^2 = \frac{1}{2} ((0 - 1)^2 + (0 + 1)^2)$$

Candidates

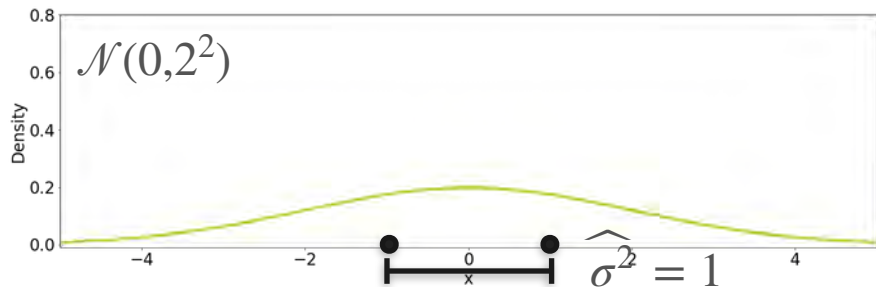
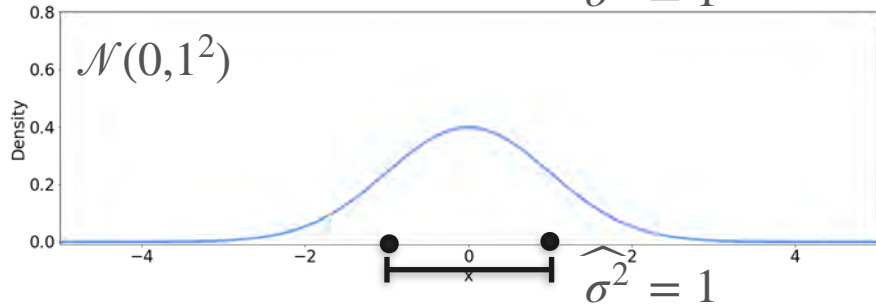
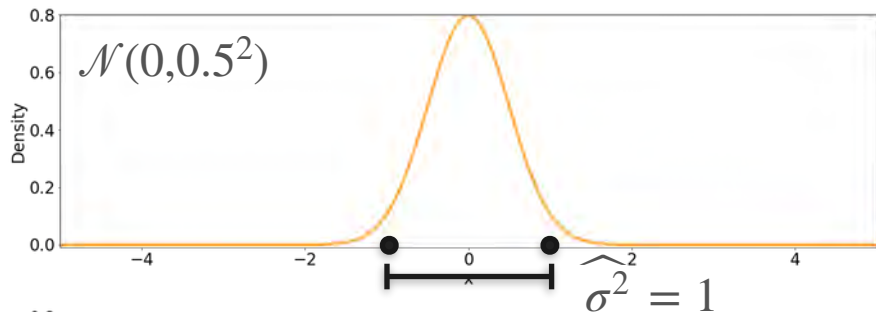


Observations

Variance of the observations

$$\hat{\sigma}^2 = \frac{1}{2} ((0 - 1)^2 + (0 + 1)^2) = 1$$

Candidates

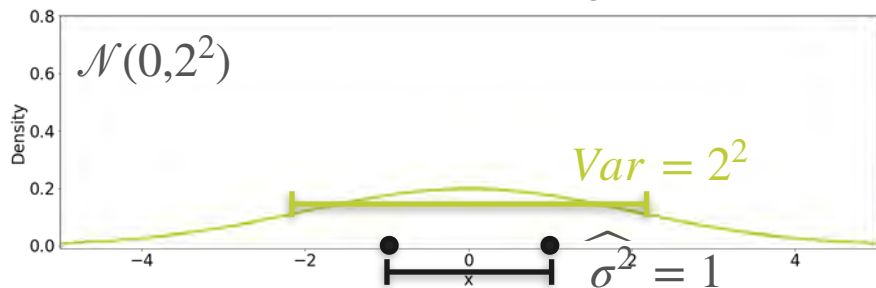
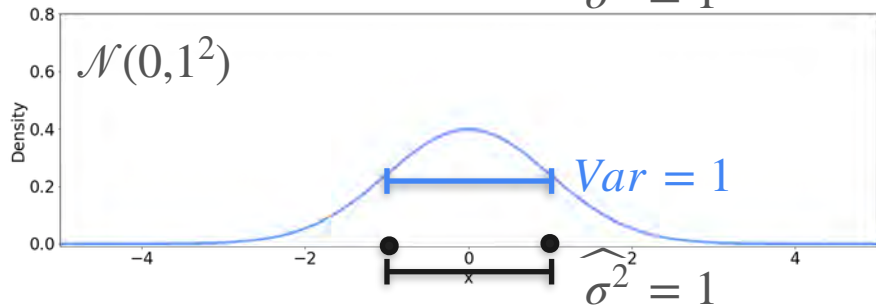
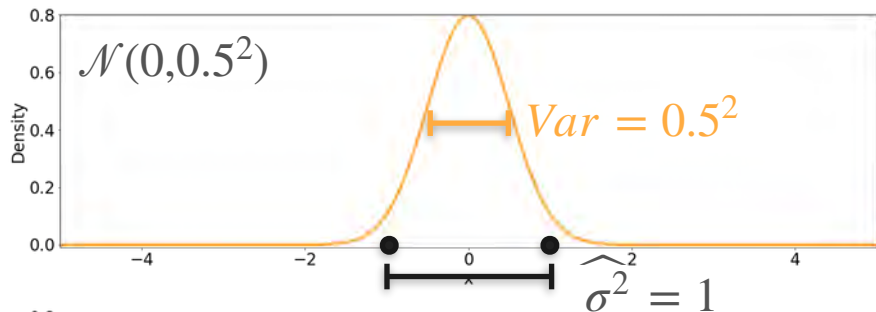


Observations

Variance of the observations

$$\hat{\sigma}^2 = \frac{1}{2} ((0 - 1)^2 + (0 + 1)^2) = 1$$

Candidates

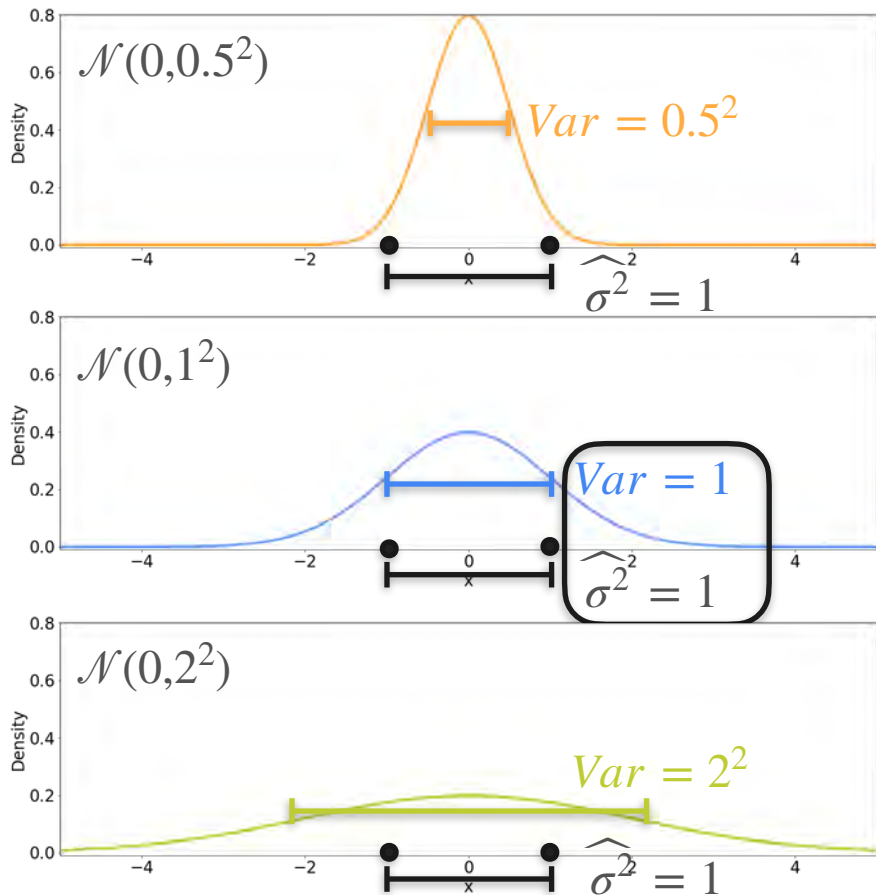


Observations

Variance of the observations

$$\hat{\sigma}^2 = \frac{1}{2} ((0 - 1)^2 + (0 + 1)^2) = 1$$

Candidates

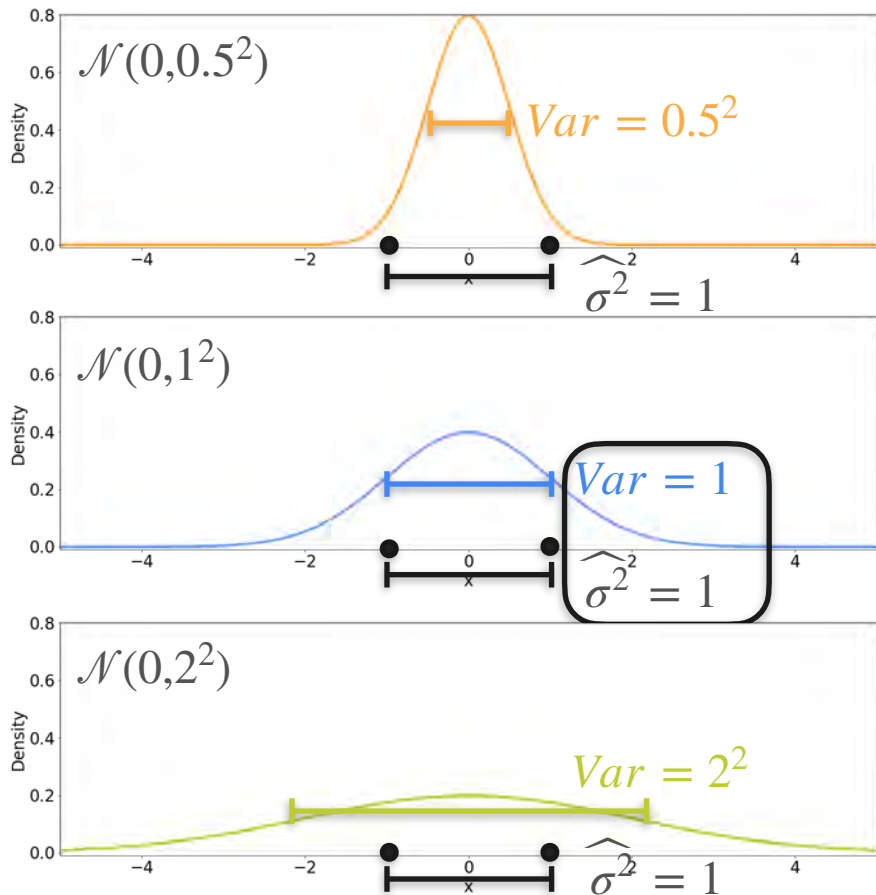


Observations

Variance of the observations

$$\hat{\sigma}^2 = \frac{1}{2} ((0 - 1)^2 + (0 + 1)^2) = 1$$

Candidates



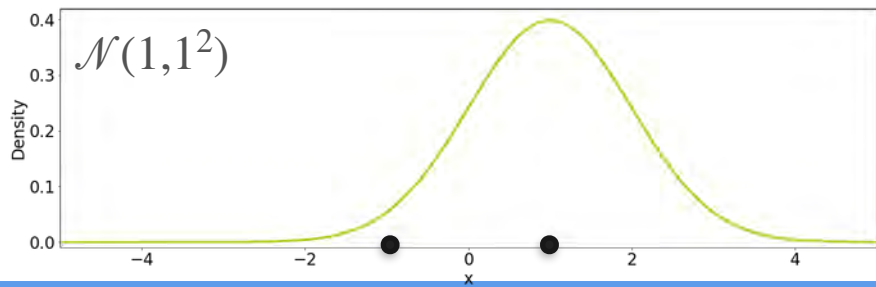
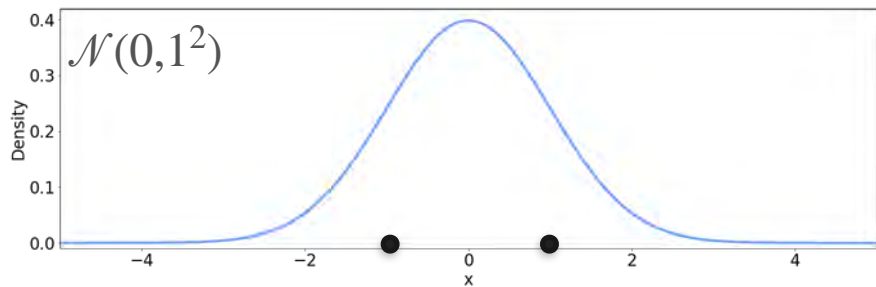
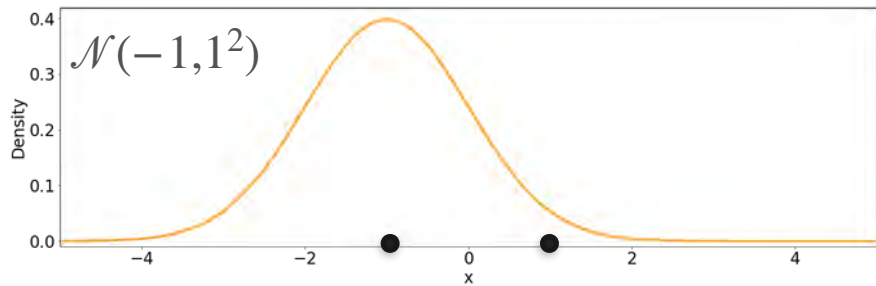
Observations

Variance of the observations

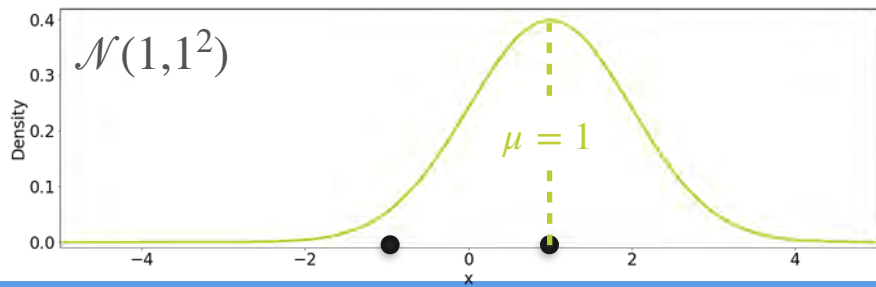
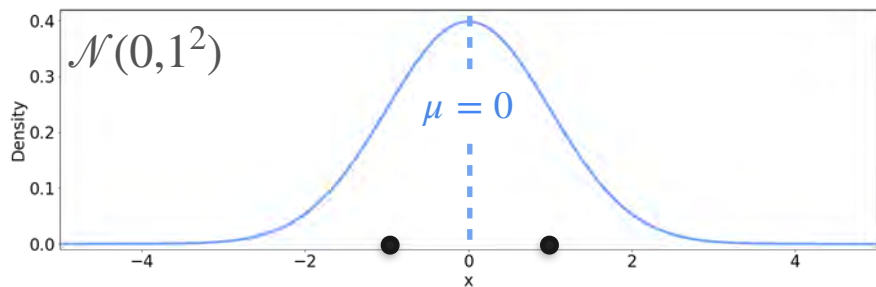
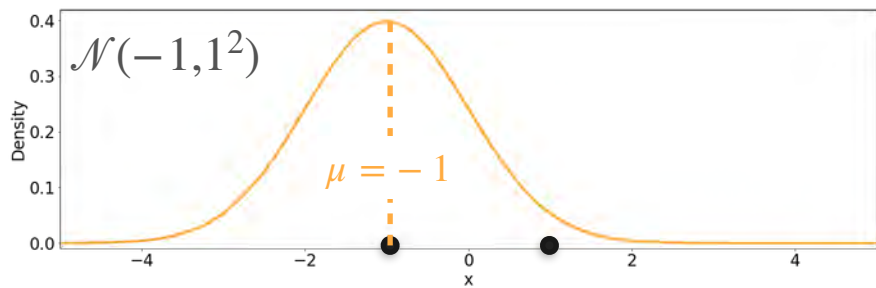
$$\hat{\sigma}^2 = \frac{1}{2} ((0 - 1)^2 + (0 + 1)^2) = 1$$

The best distribution is the one where the **variance** of the distribution is the **variance** of the sample

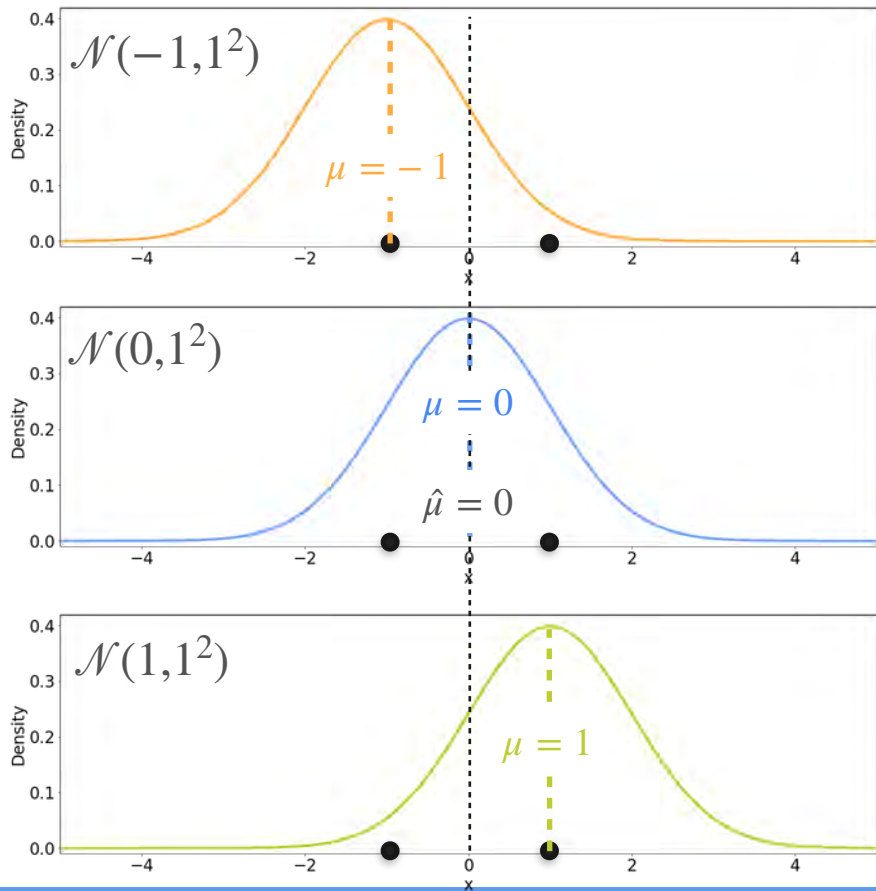
Candidates



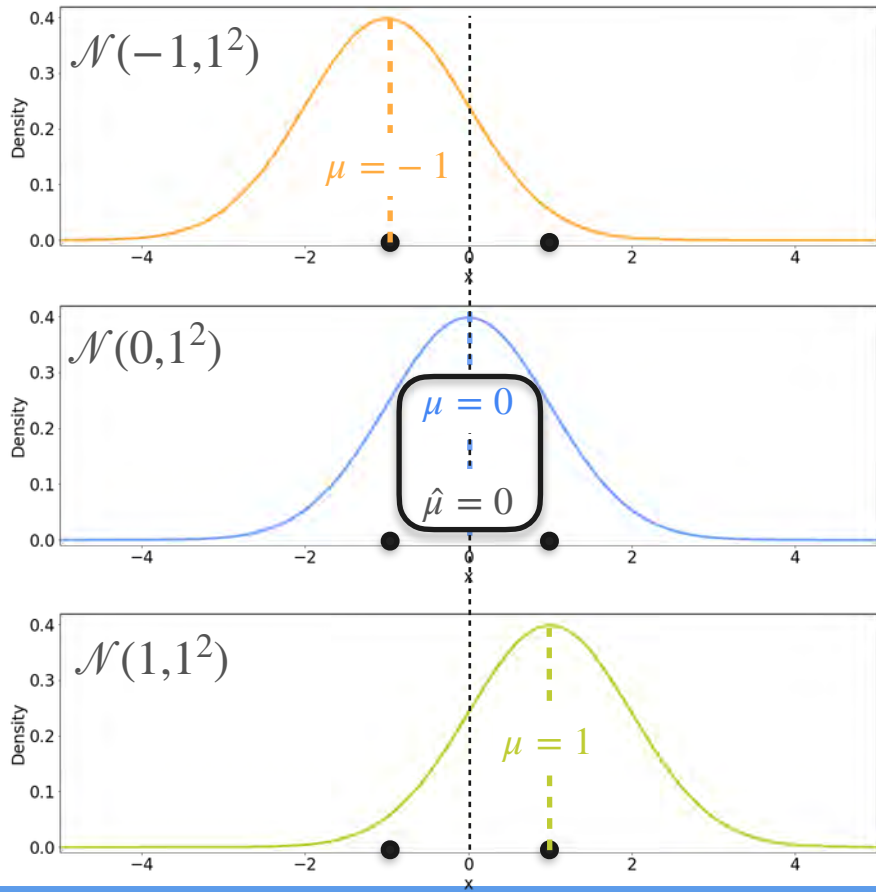
Candidates



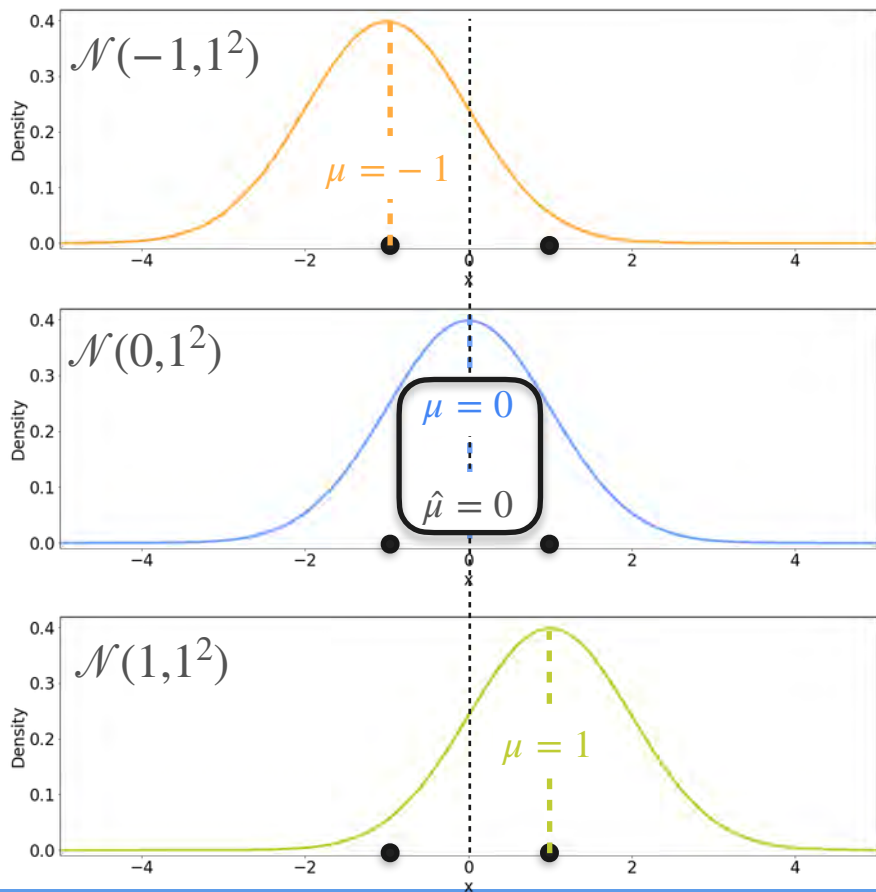
Candidates



Candidates

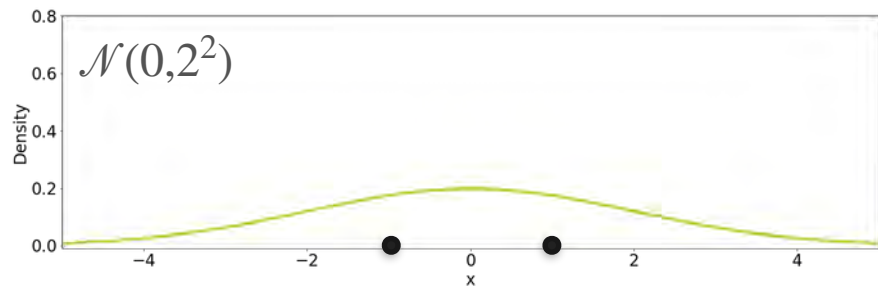
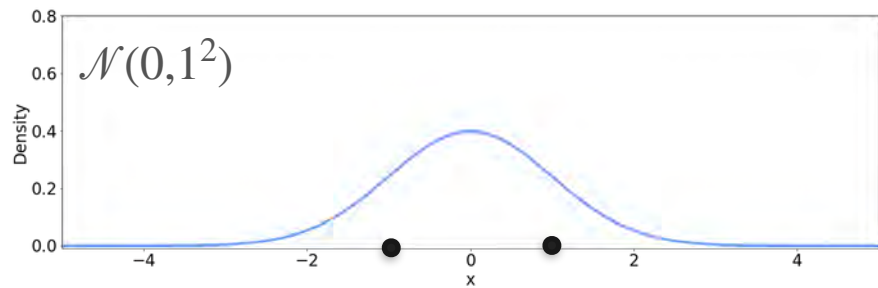
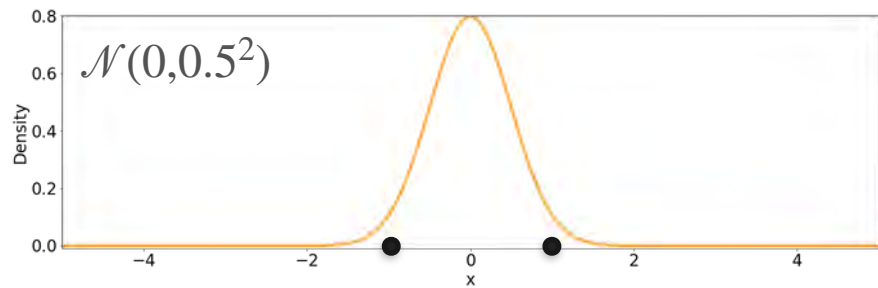


Candidates



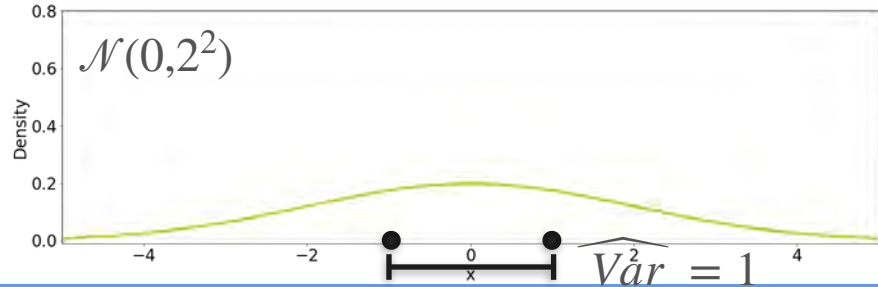
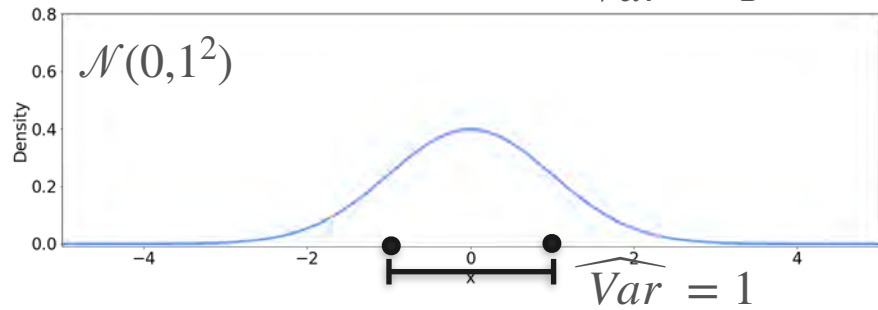
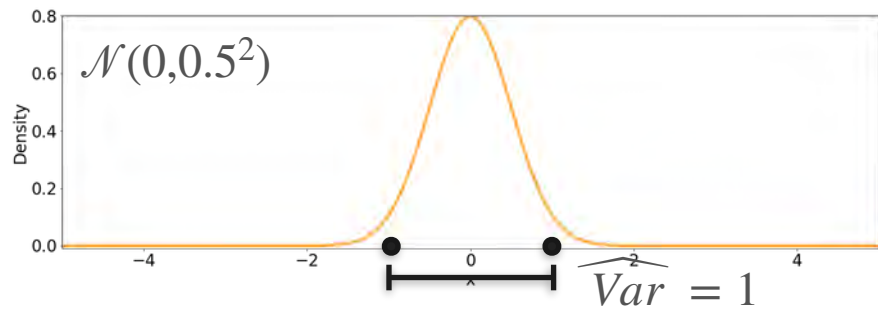
The best distribution is the one where the **mean** of the distribution is the **mean** of the sample

Candidates



Observations

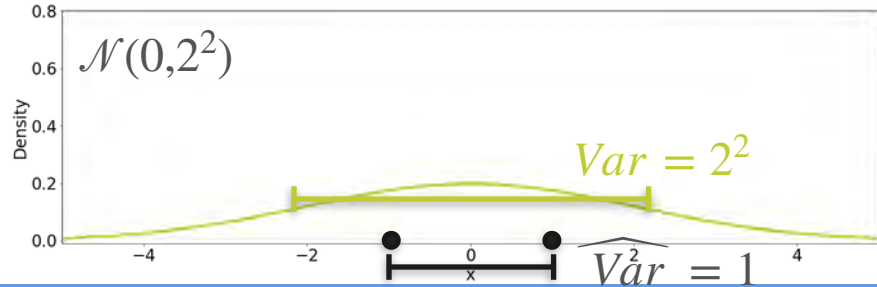
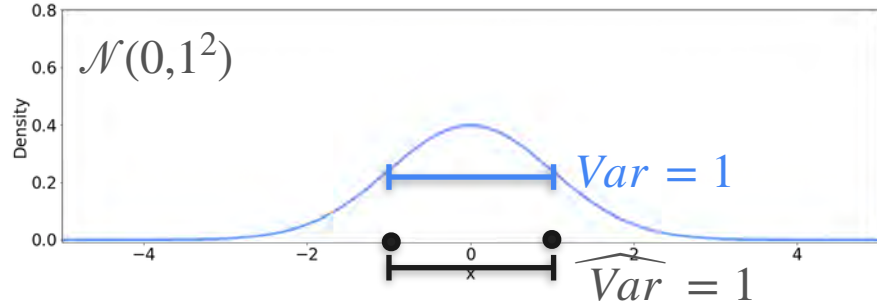
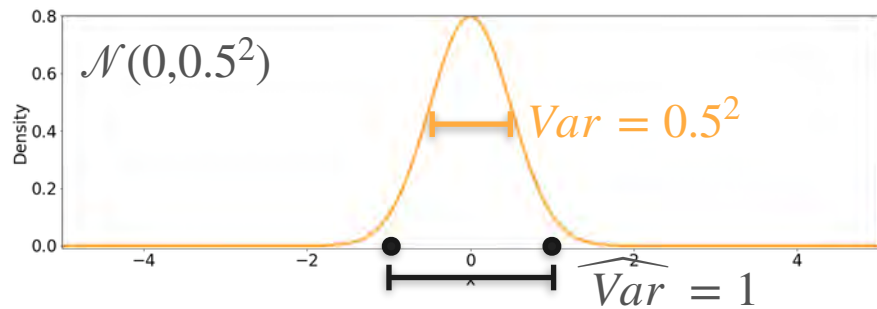
Candidates



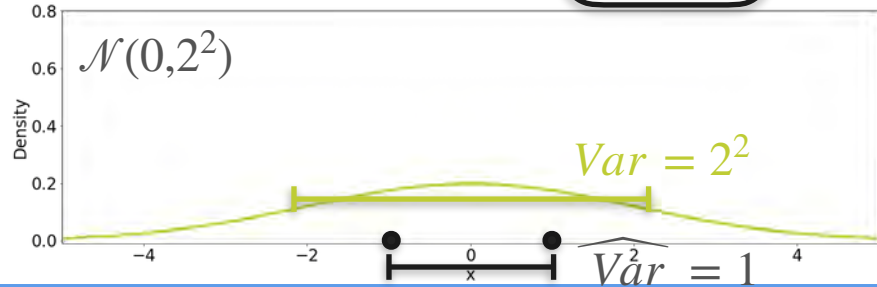
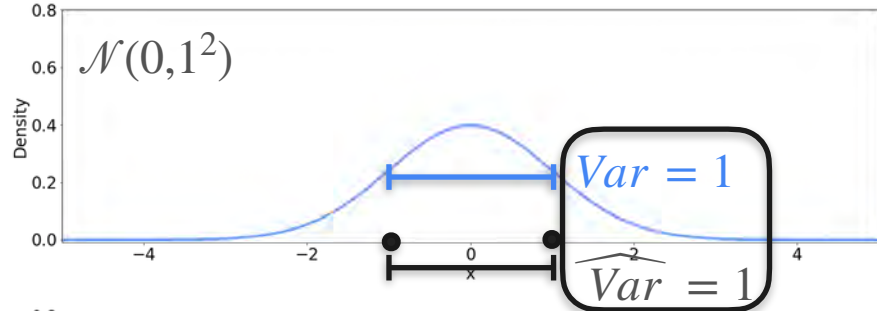
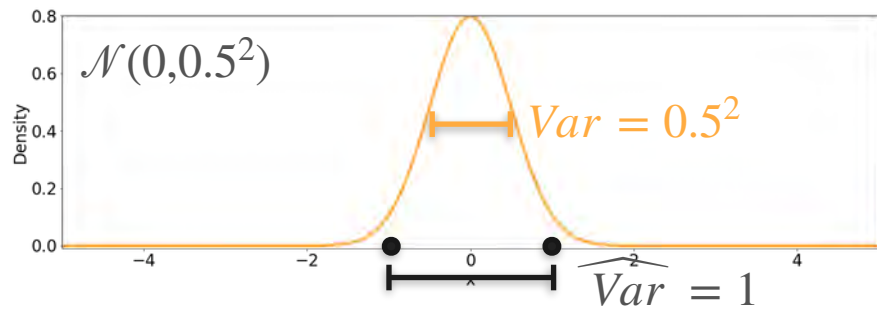
Observations

Candidates

Observations

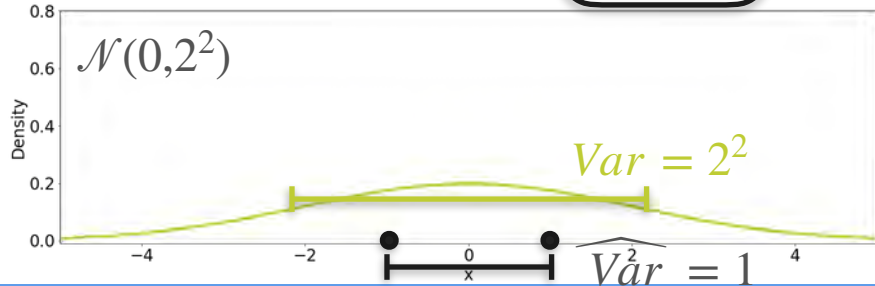
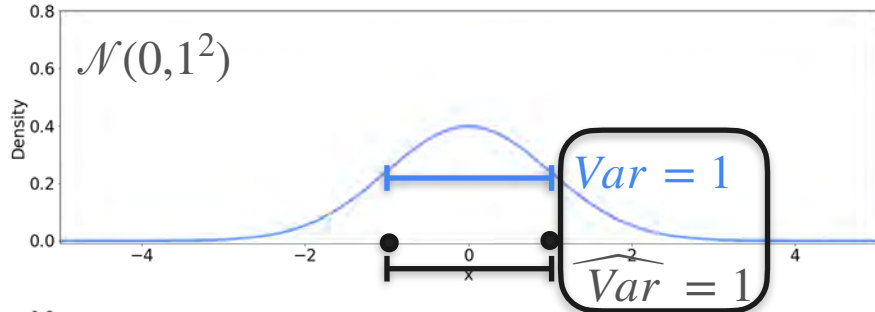
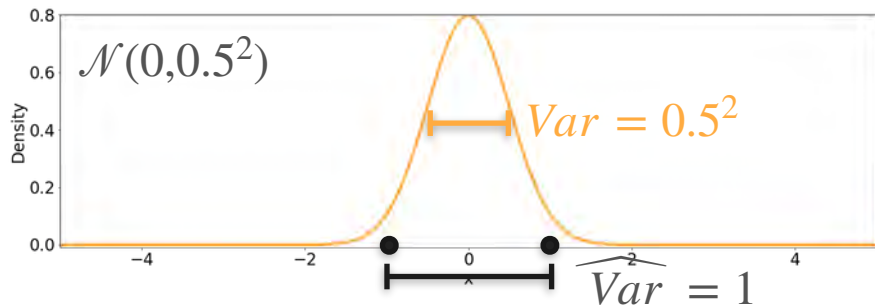


Candidates



Observations

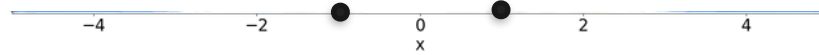
Candidates



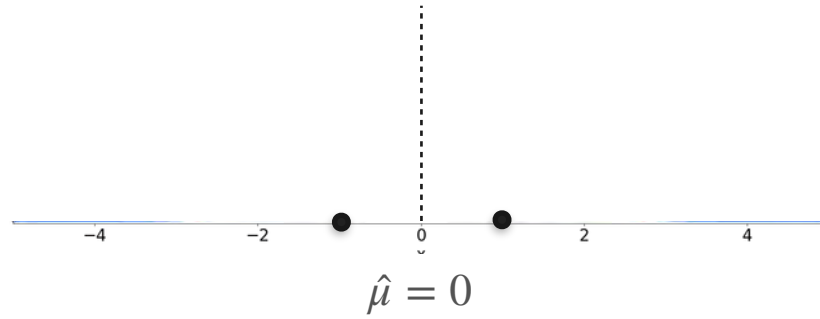
Observations

The best distribution is the one where the **variance** of the distribution is the **variance** of the sample

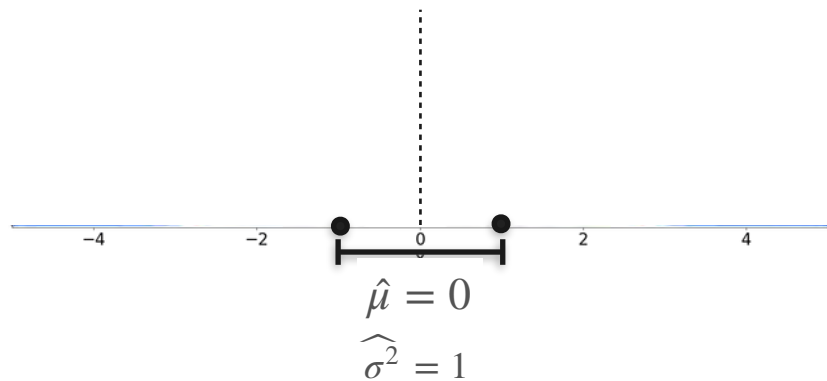
Best Gaussian Fit



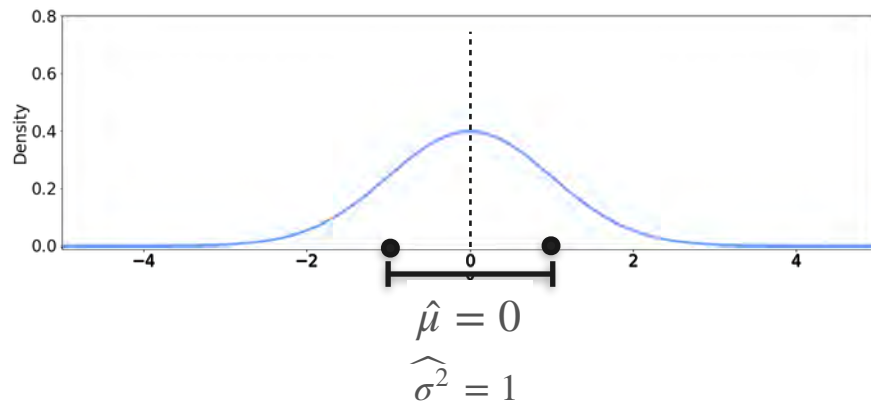
Best Gaussian Fit



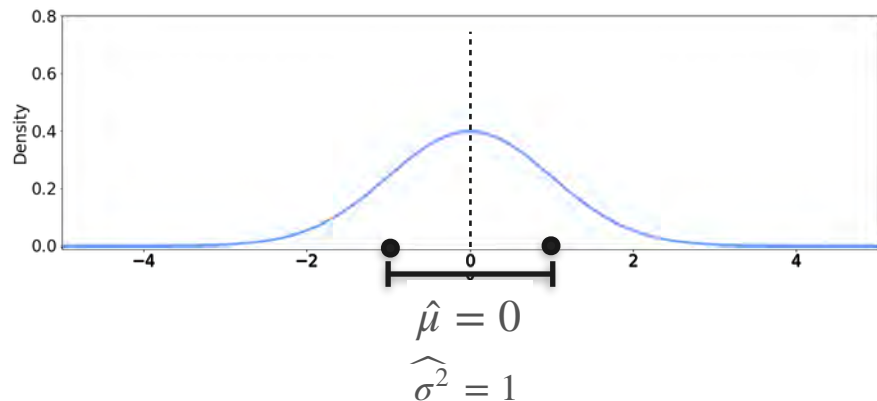
Best Gaussian Fit



Best Gaussian Fit

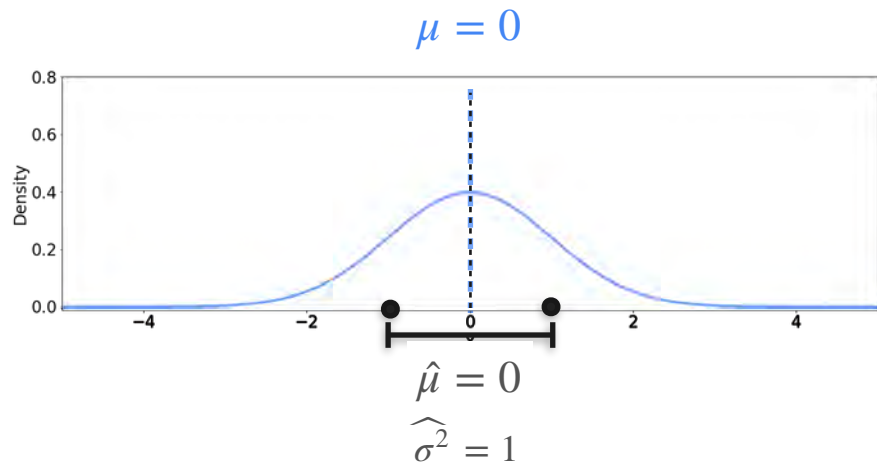


Best Gaussian Fit



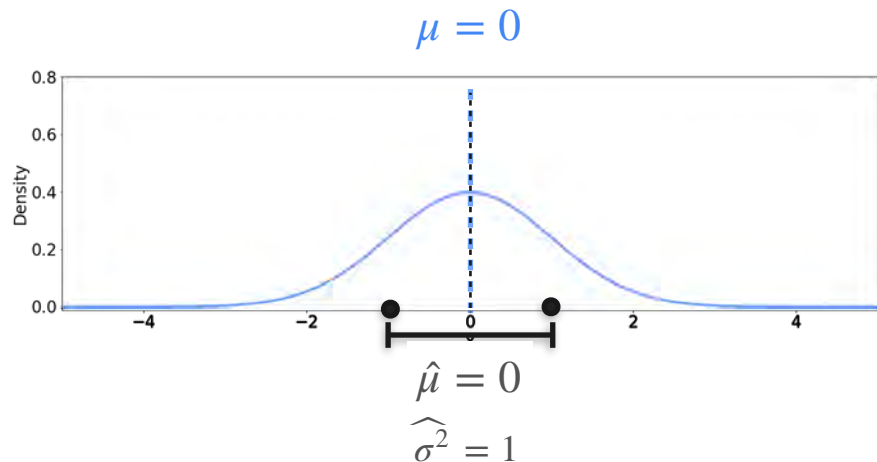
The best distribution is the one where
the **mean** of the distribution is the
mean of the sample

Best Gaussian Fit



The best distribution is the one where
the **mean** of the distribution is the
mean of the sample

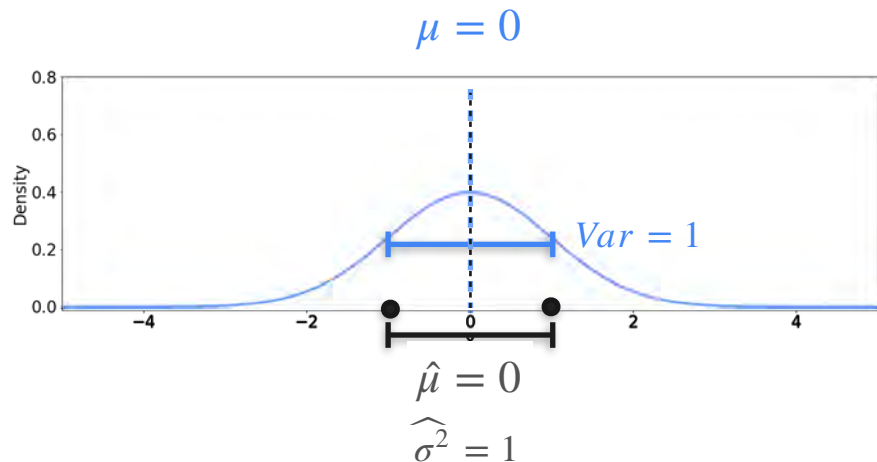
Best Gaussian Fit



The best distribution is the one where the **mean** of the distribution is the **mean** of the sample

The best distribution is the one where the **variance** of the distribution is the **variance** of the sample

Best Gaussian Fit



The best distribution is the one where the **mean** of the distribution is the **mean** of the sample

The best distribution is the one where the **variance** of the distribution is the **variance** of the sample

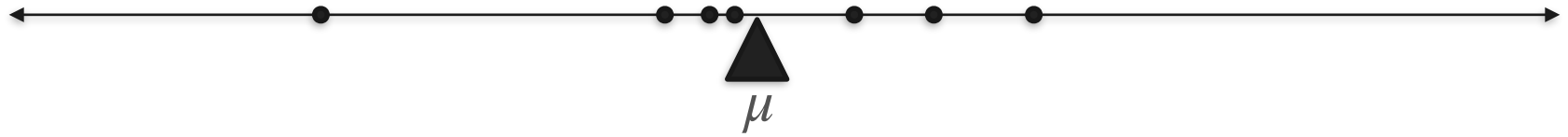
Best Gaussian Fit



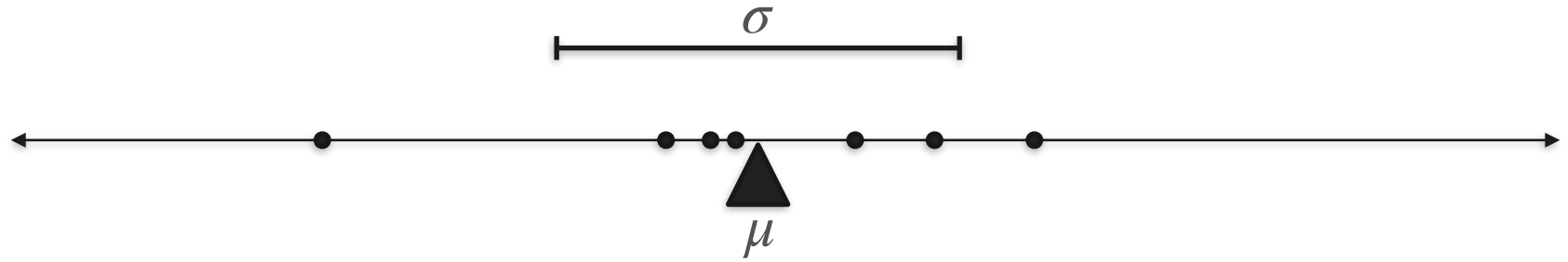
Best Gaussian Fit



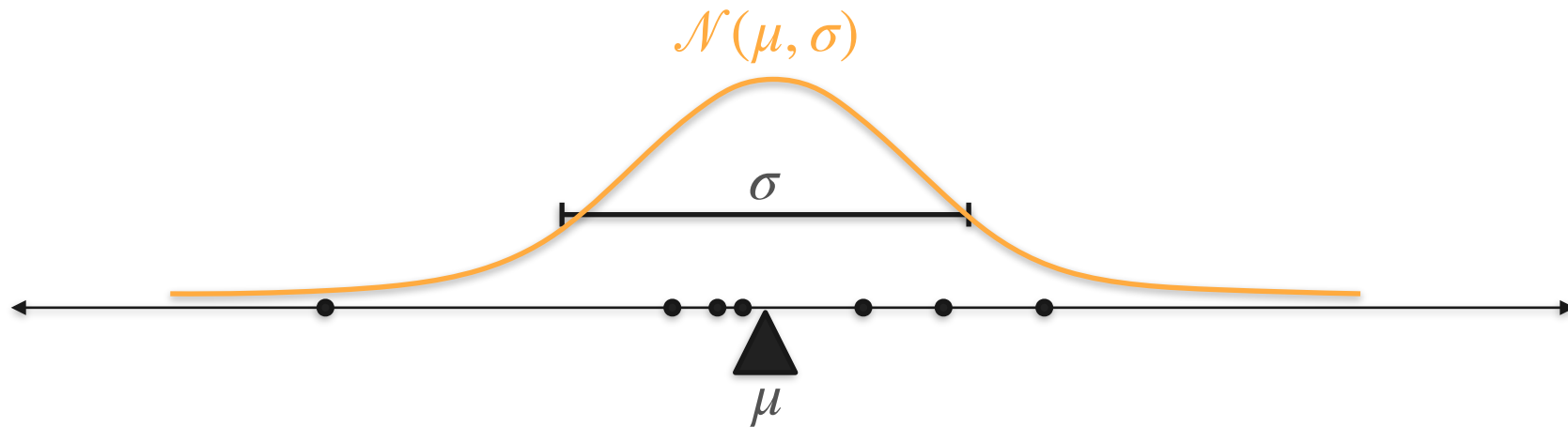
Best Gaussian Fit



Best Gaussian Fit



Best Gaussian Fit



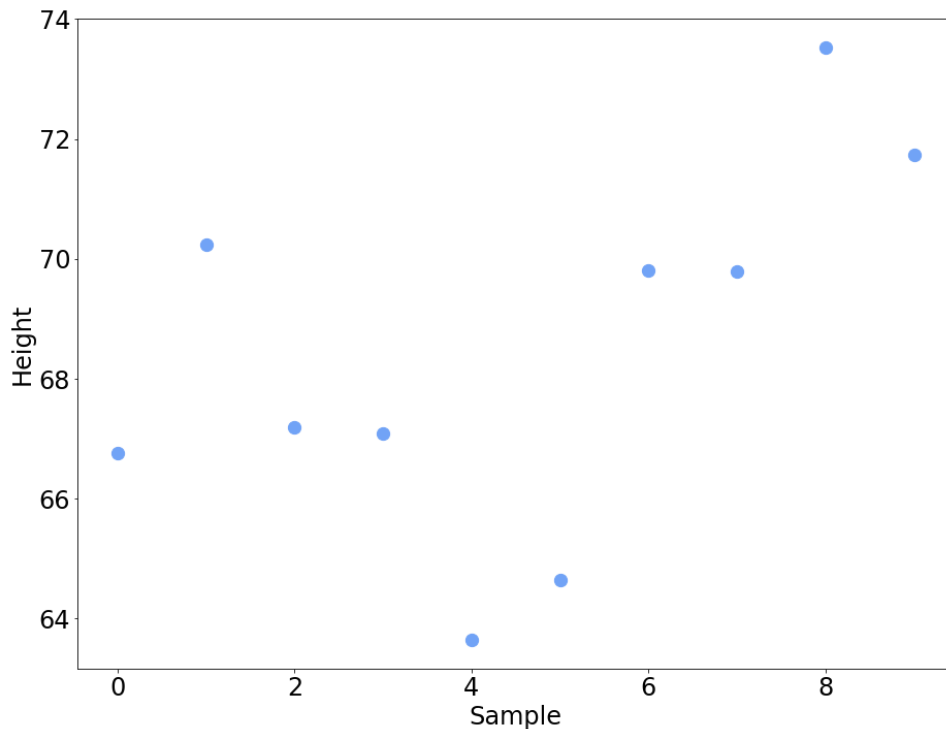
Maximum Likelihood: Gaussian Example

Maximum Likelihood: Gaussian Example

X = "Height of an 18 year old"

You measure 10 people
(i.e. sample size = 10)

$\mathbf{X} = (X_1, X_2, \dots, X_{10})$



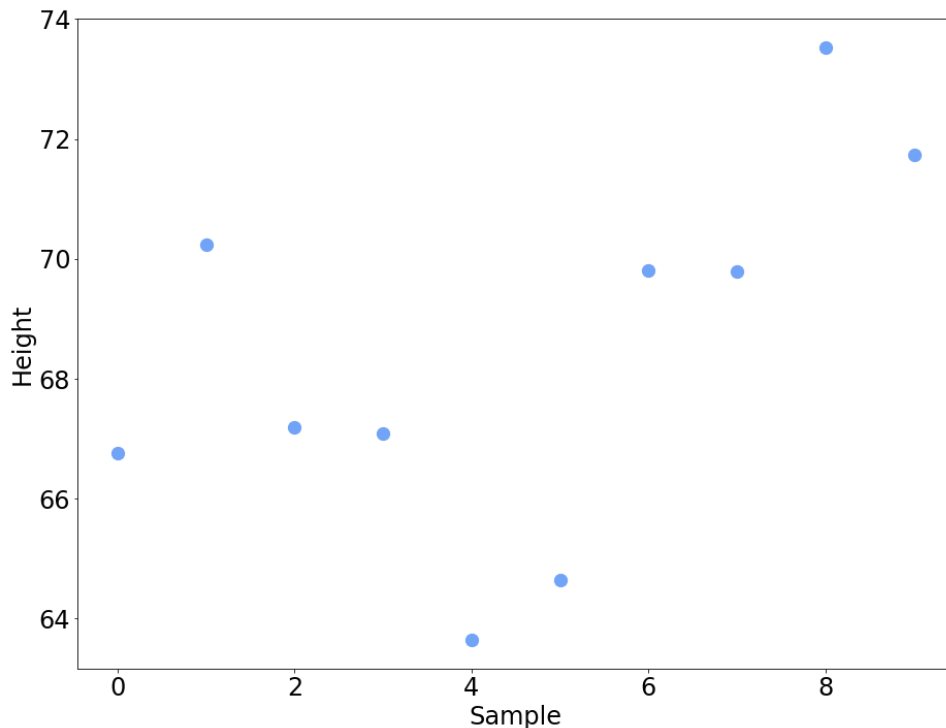
Maximum Likelihood: Gaussian Example

X = "Height of an 18 year old"

You measure 10 people
(i.e. sample size = 10)

$\mathbf{X} = (X_1, X_2, \dots, X_{10})$

$X_1, X_2, \dots, X_{10} \stackrel{i.i.d}{\sim} \mathcal{N}(\mu, \sigma)$



Maximum Likelihood: Gaussian Example

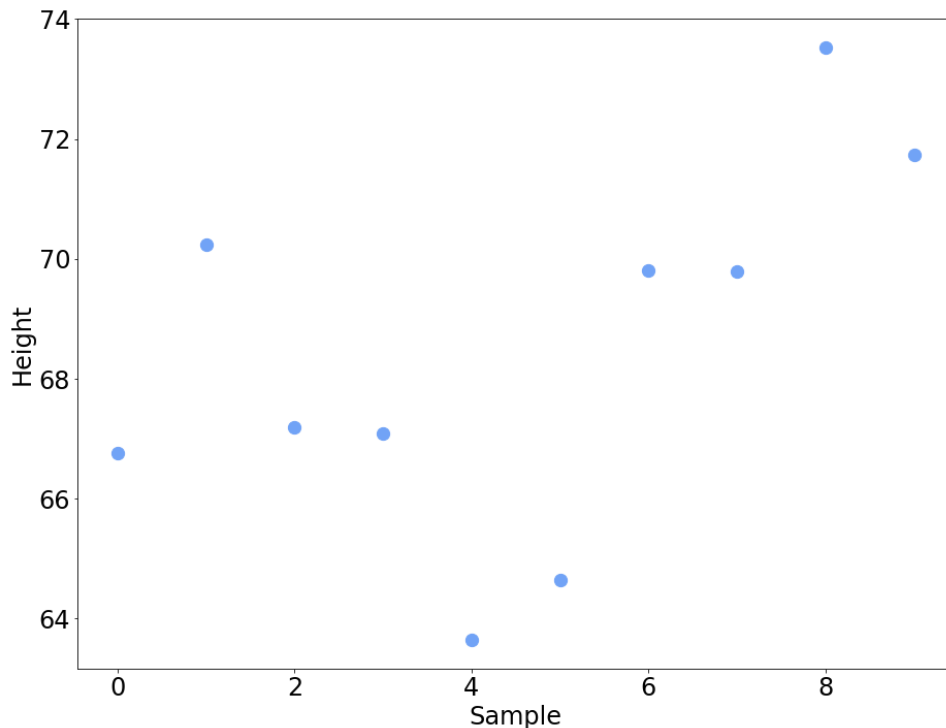
X = "Height of an 18 year old"

You measure 10 people
(i.e. sample size = 10)

$\mathbf{X} = (X_1, X_2, \dots, X_{10})$

$X_1, X_2, \dots, X_{10} \stackrel{i.i.d}{\sim} \mathcal{N}(\mu, \sigma)$

$\mu?$



Maximum Likelihood: Gaussian Example

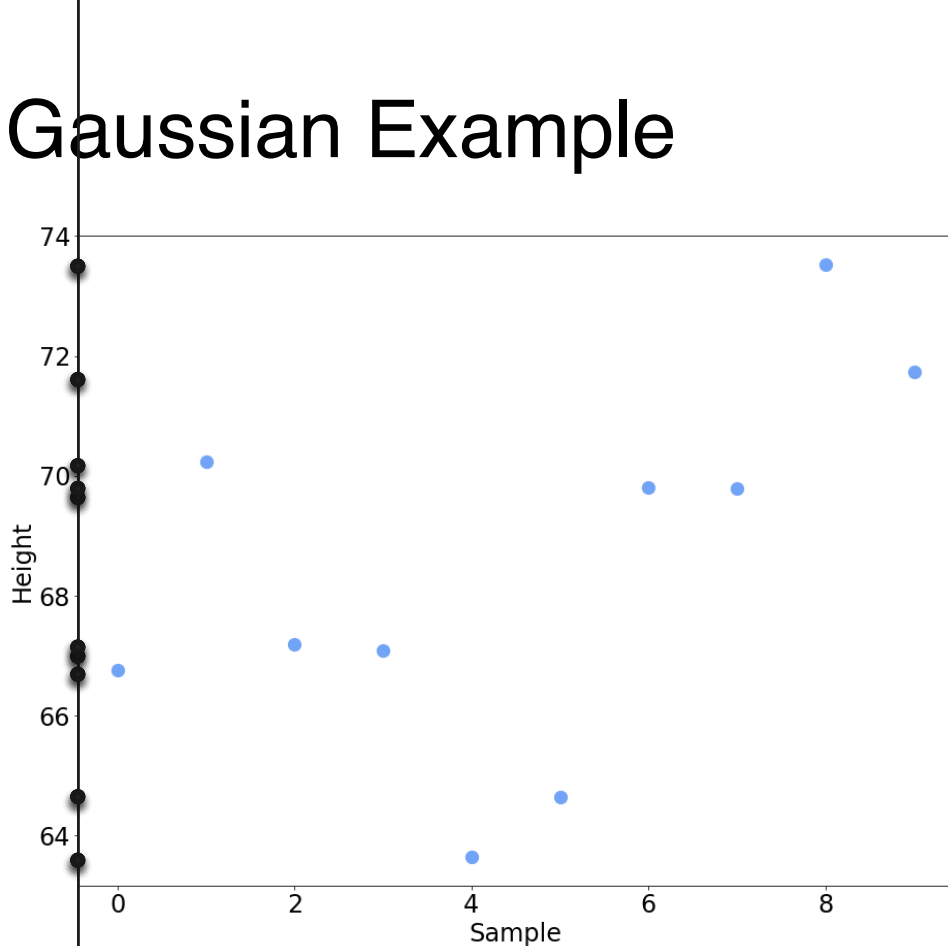
X = "Height of an 18 year old"

You measure 10 people
(i.e. sample size = 10)

$\mathbf{X} = (X_1, X_2, \dots, X_{10})$

$X_1, X_2, \dots, X_{10} \stackrel{i.i.d}{\sim} \mathcal{N}(\mu, \sigma)$

$\mu?$





$$\mu = 64$$



$$\mu = 68$$



$$\mu = 66$$



$$\mu = 70$$



$$\mu = 64$$

$$\sigma = 3.11$$



$$\mu = 68$$

$$\sigma = 3.11$$



$$\mu = 66$$

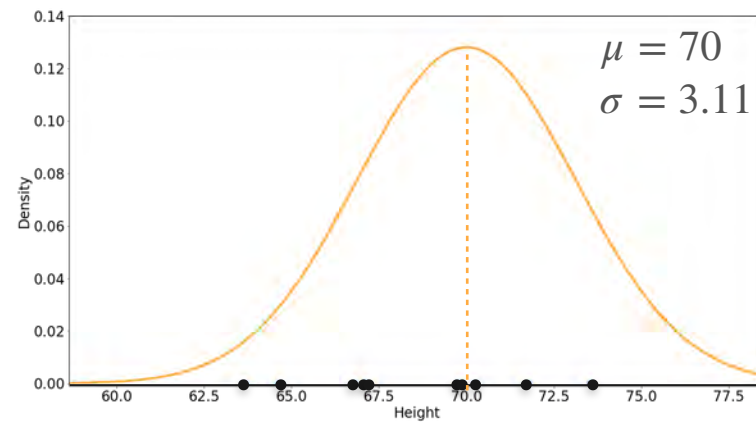
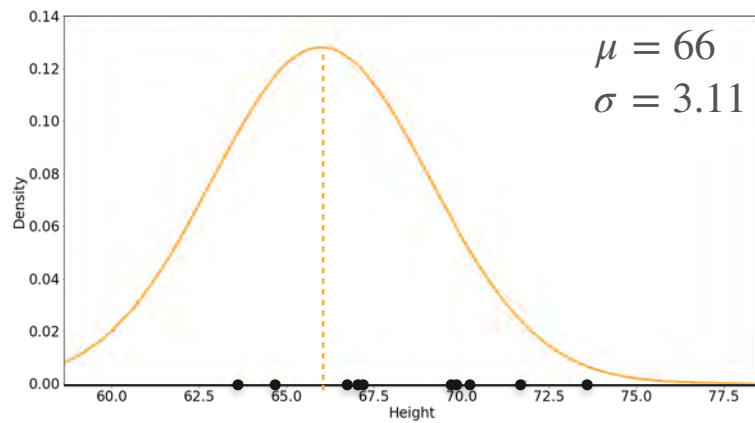
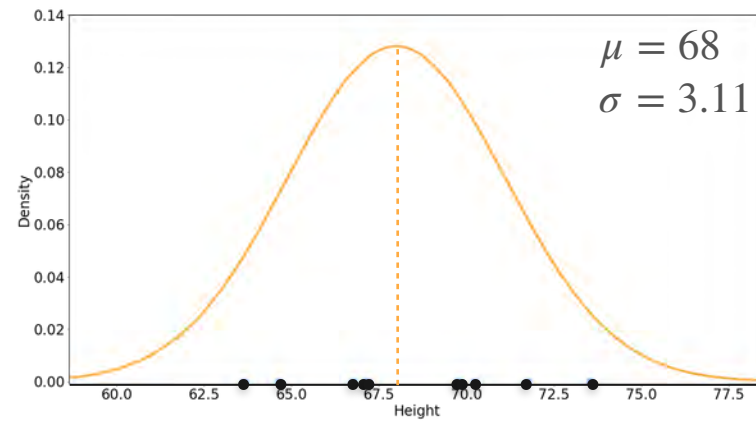
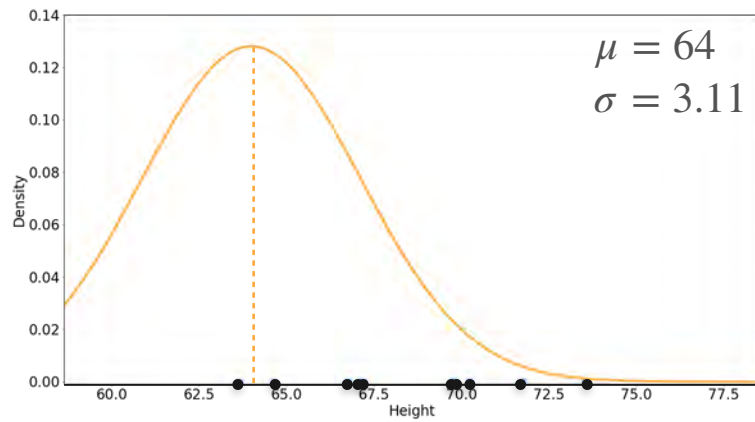
$$\sigma = 3.11$$

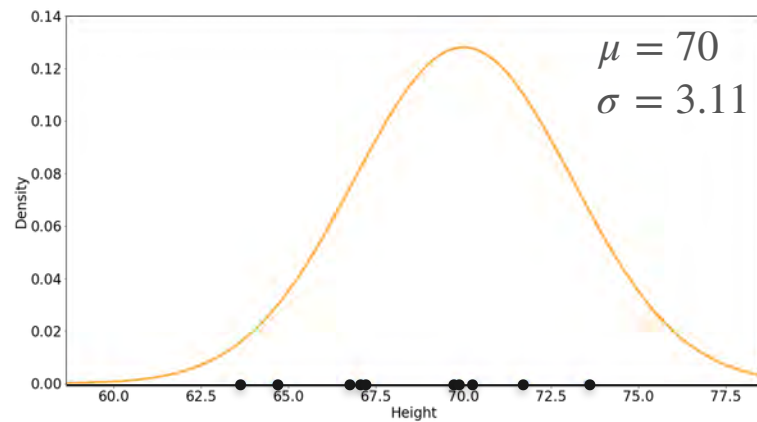
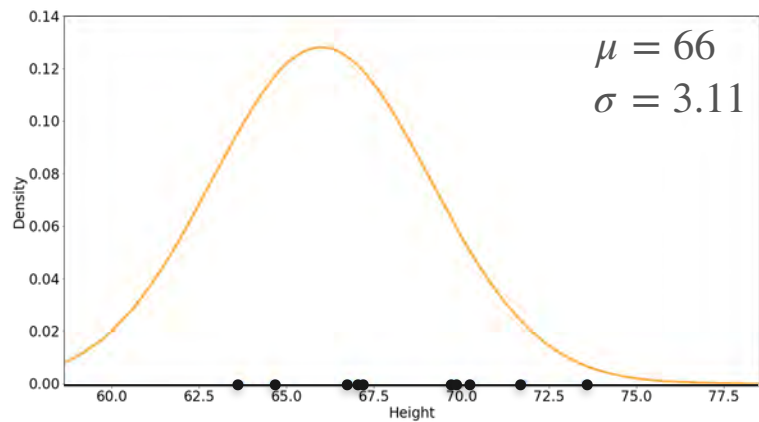
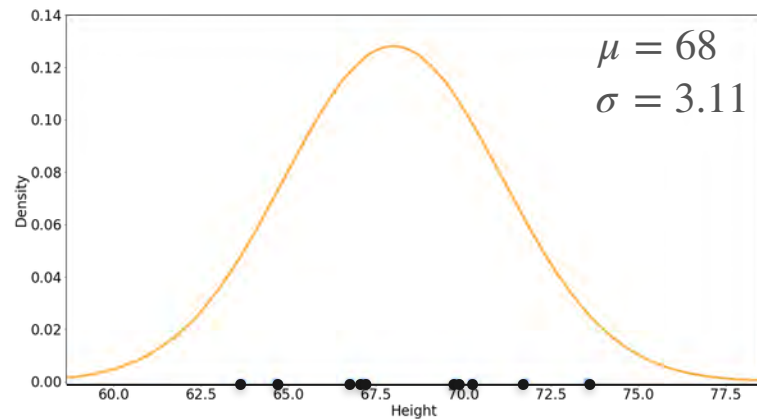
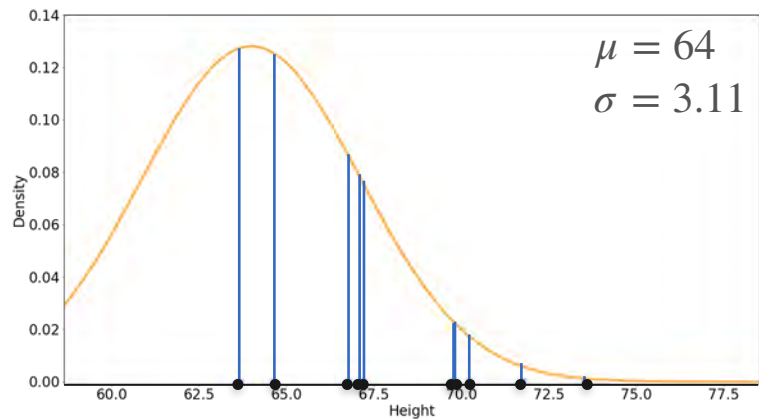


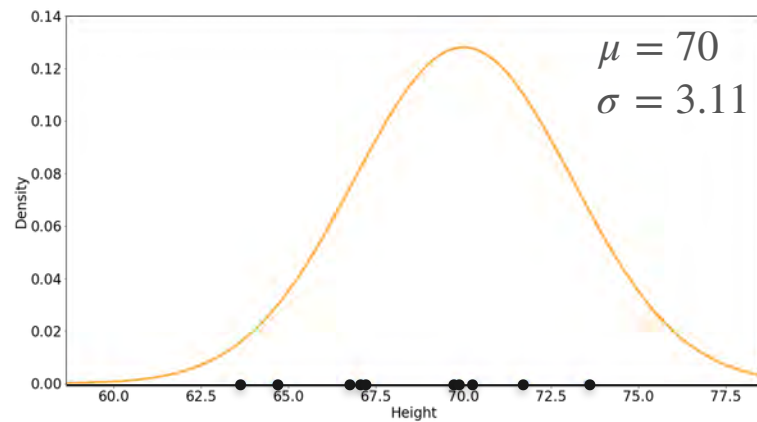
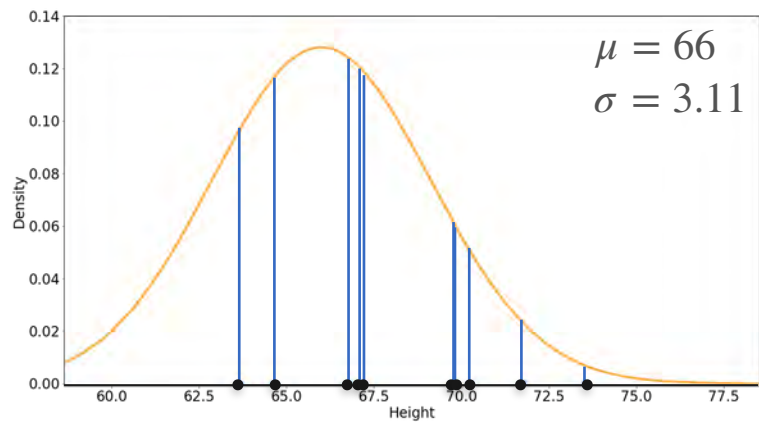
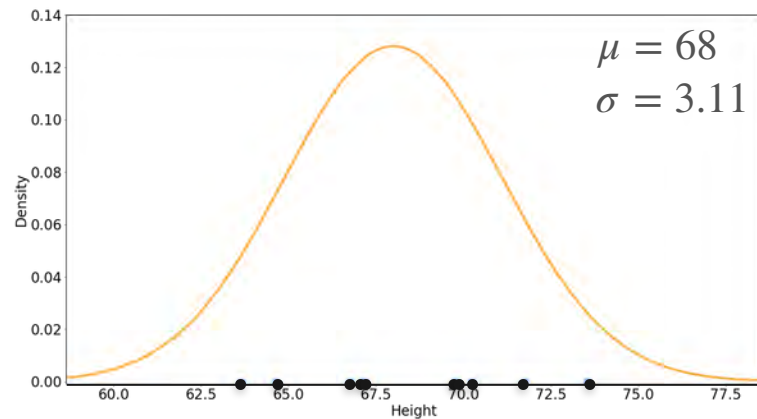
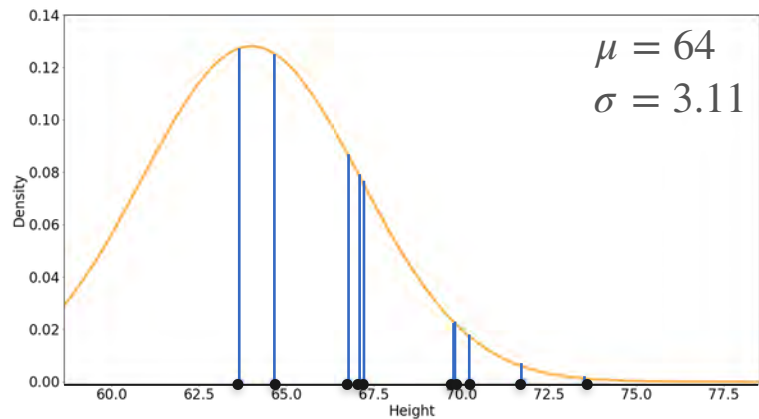
$$\mu = 70$$

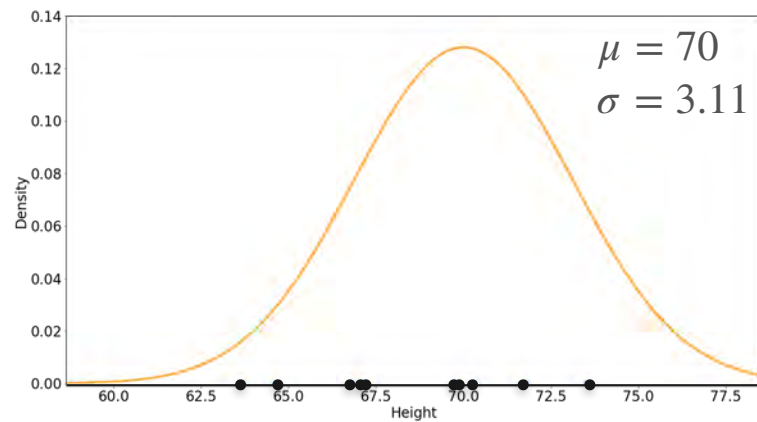
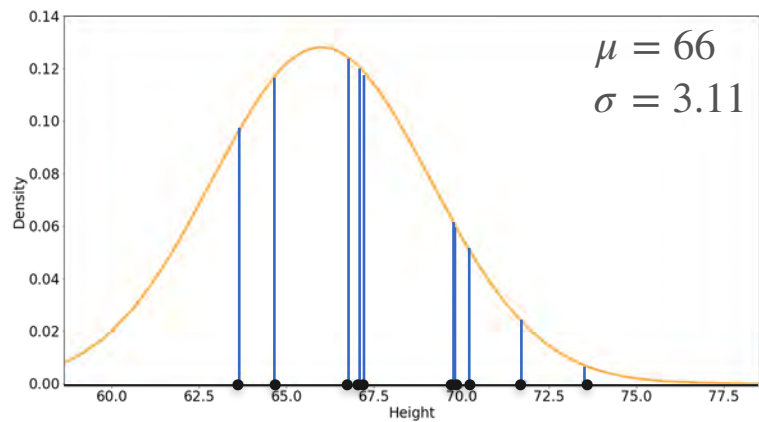
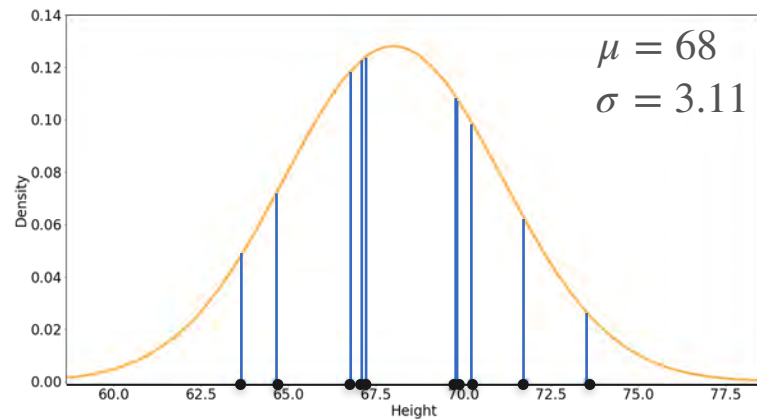
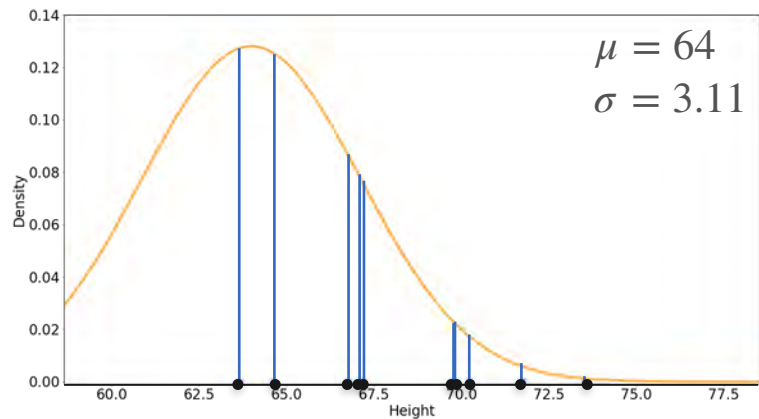
$$\sigma = 3.11$$

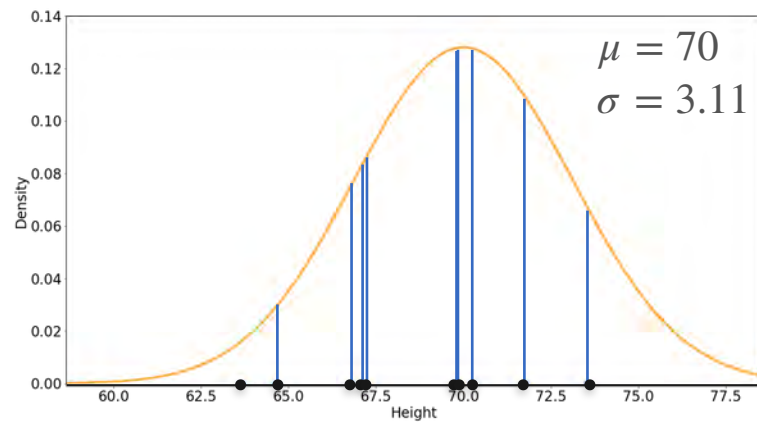
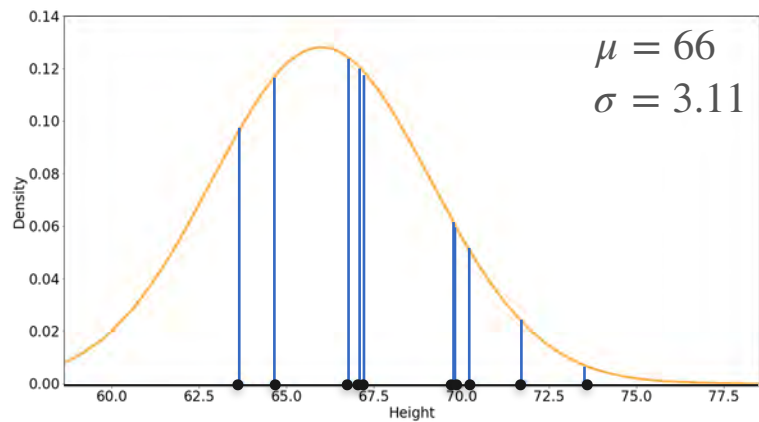
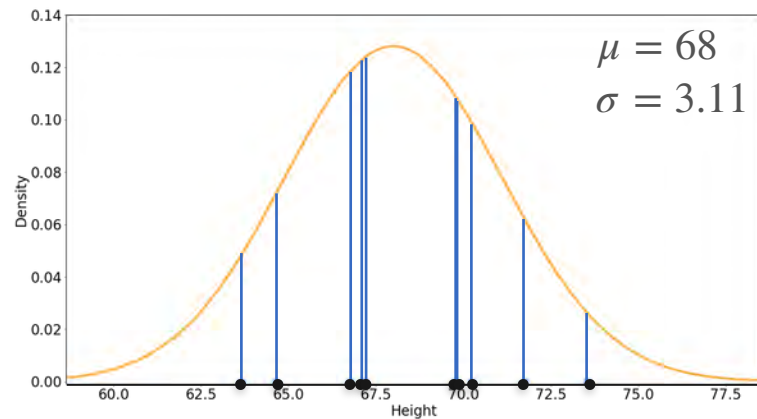
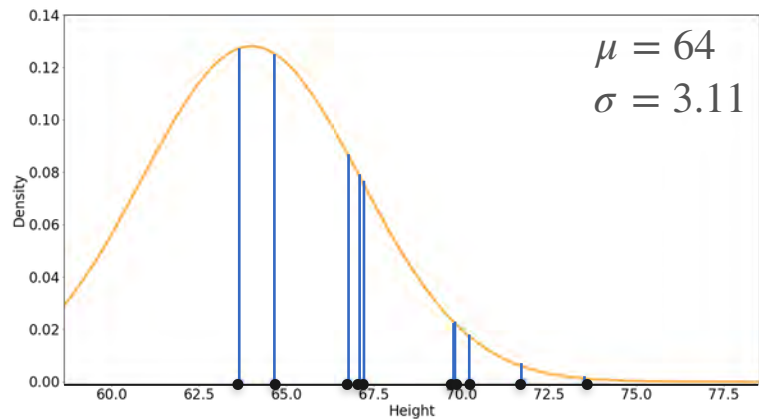


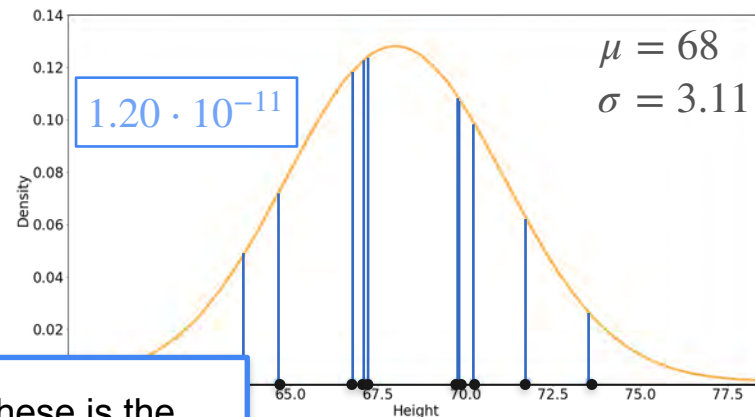
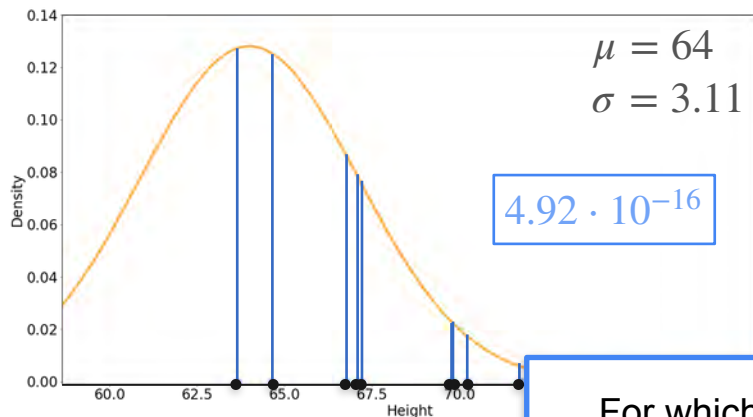




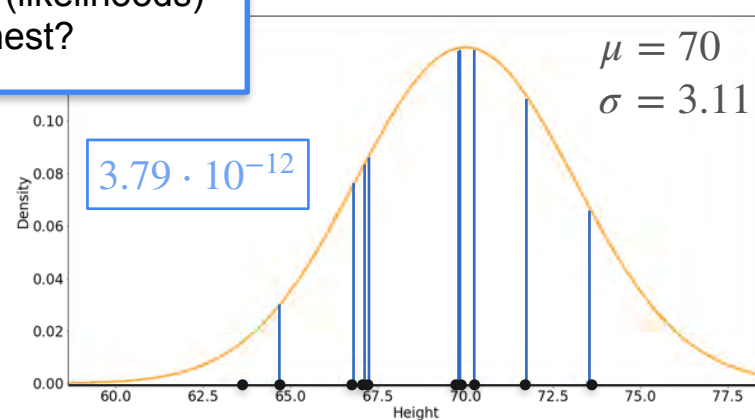
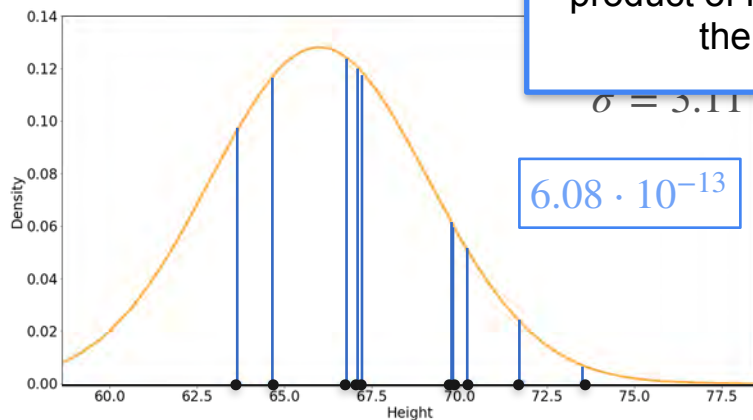


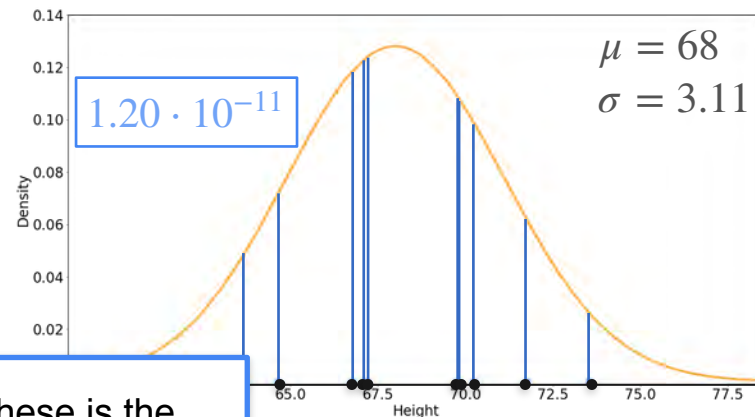
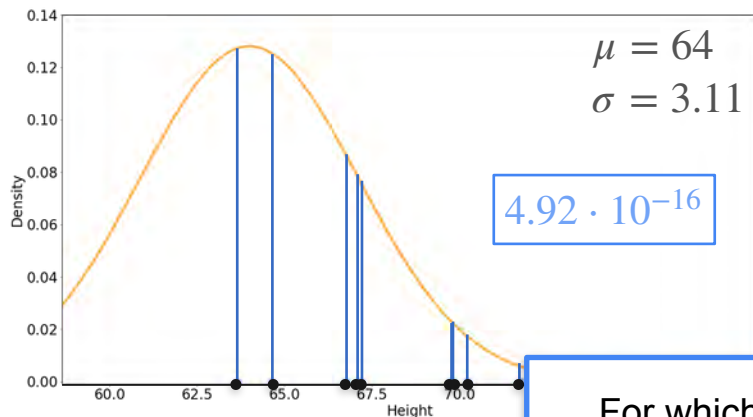




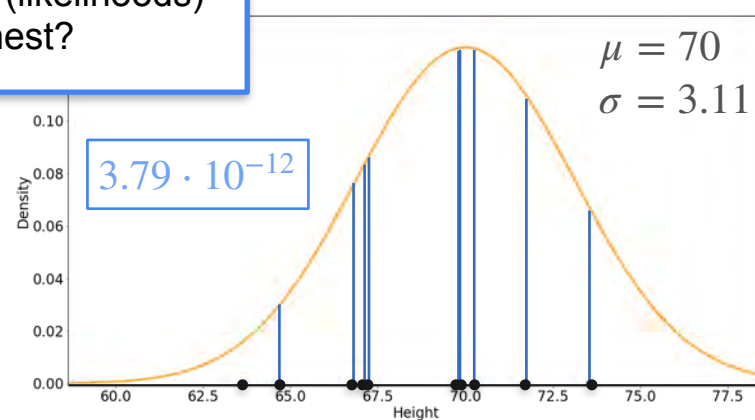
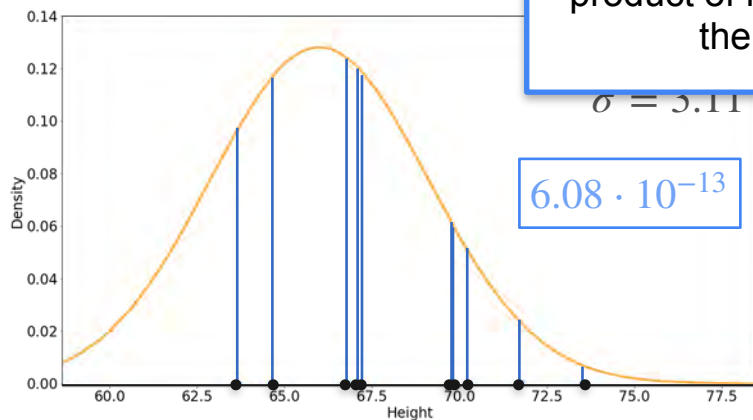


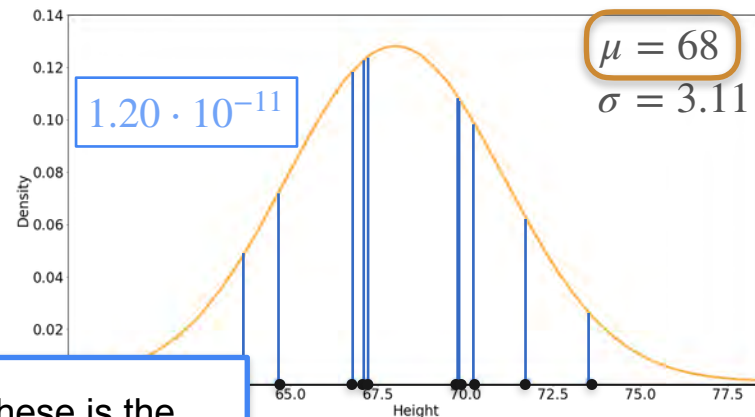
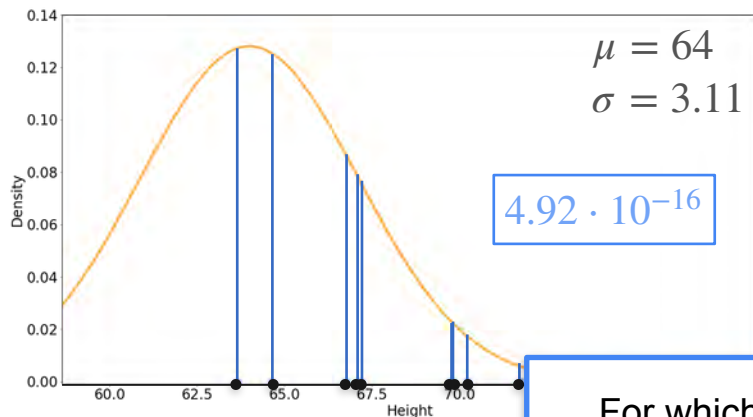
For which of these is the product of lines (likelihoods) the highest?



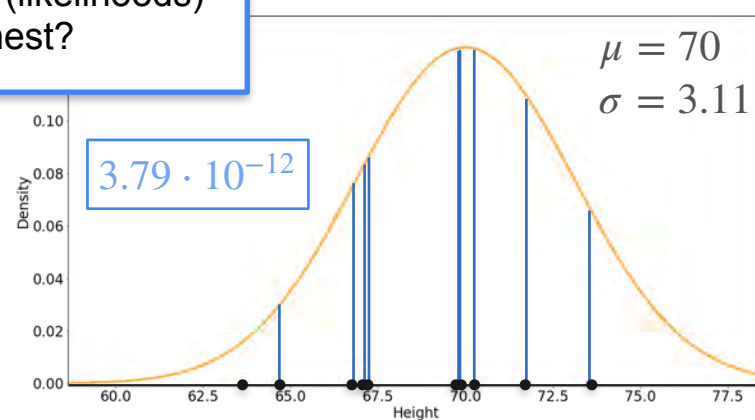
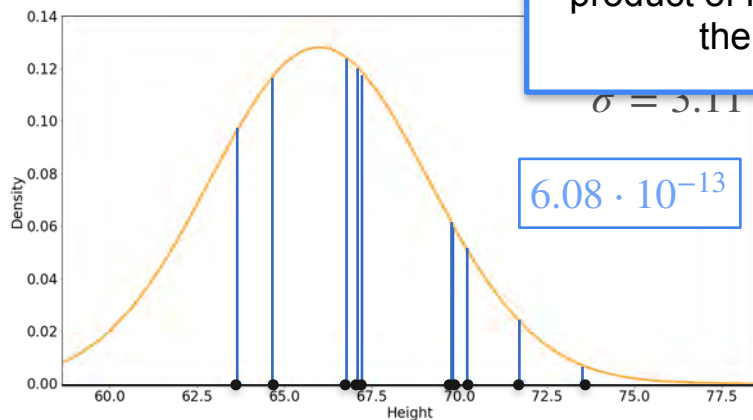


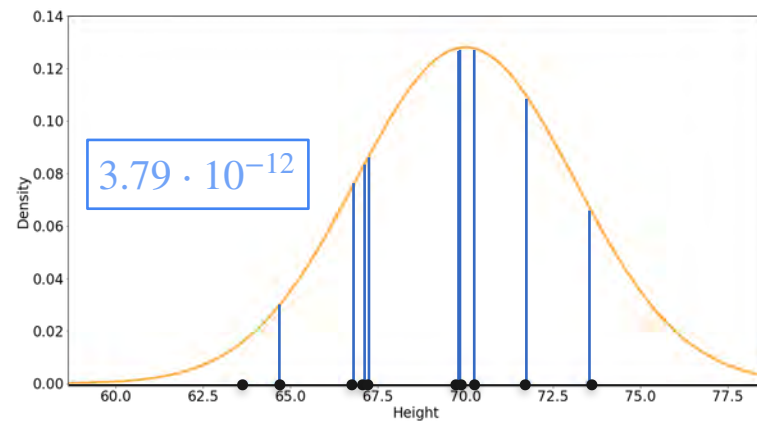
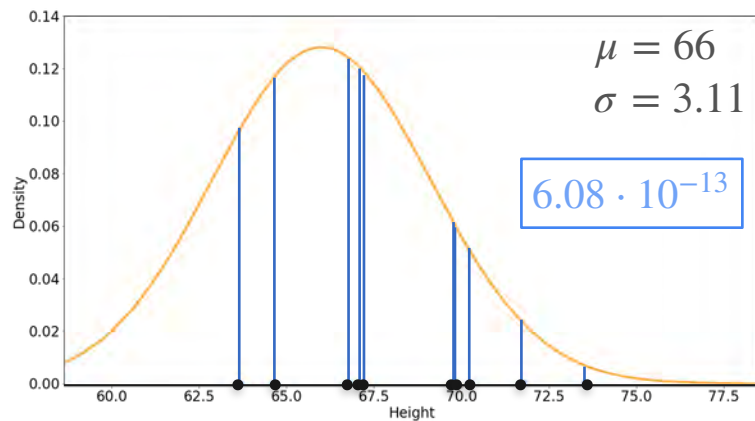
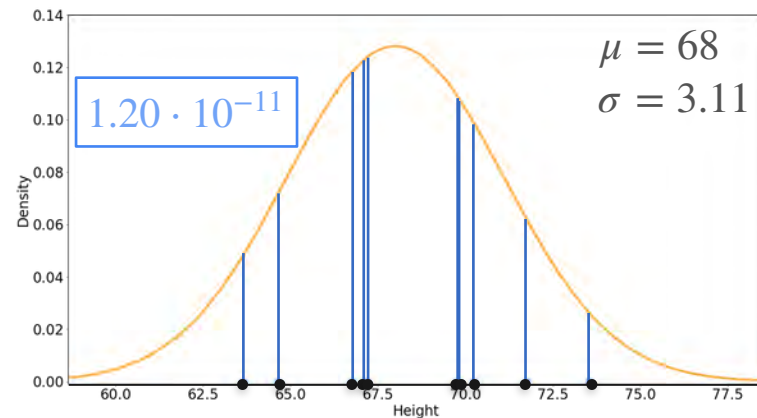
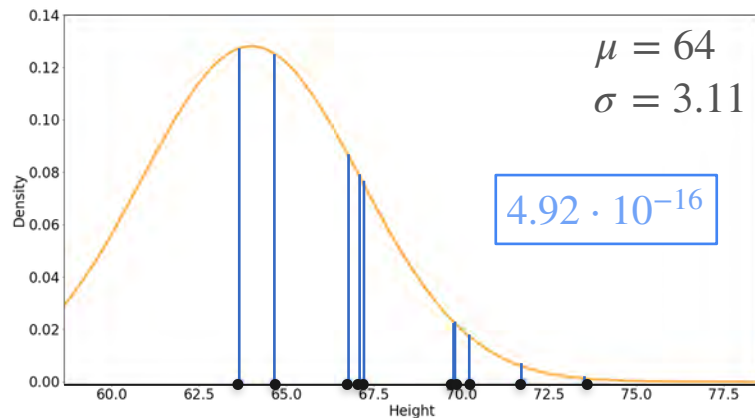
For which of these is the
product of lines (likelihoods)
the highest?

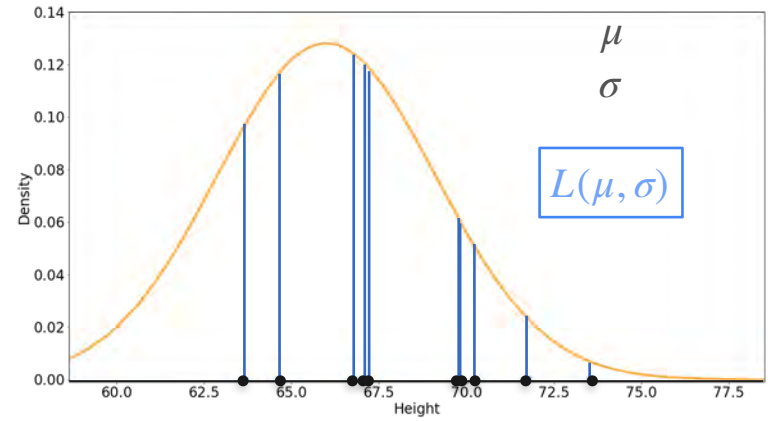




For which of these is the product of lines (likelihoods) the highest?

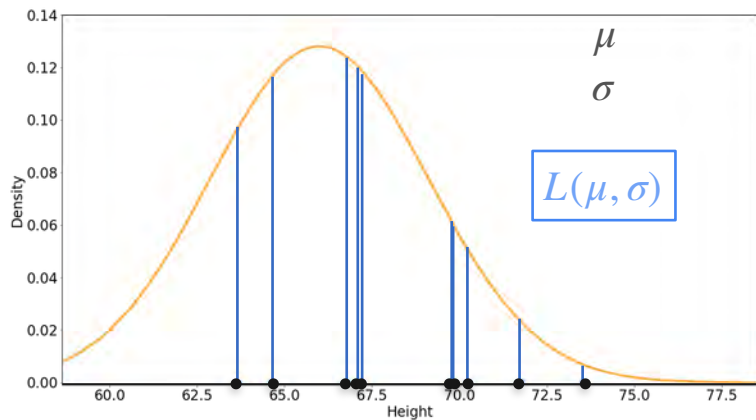






$$\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right]$$

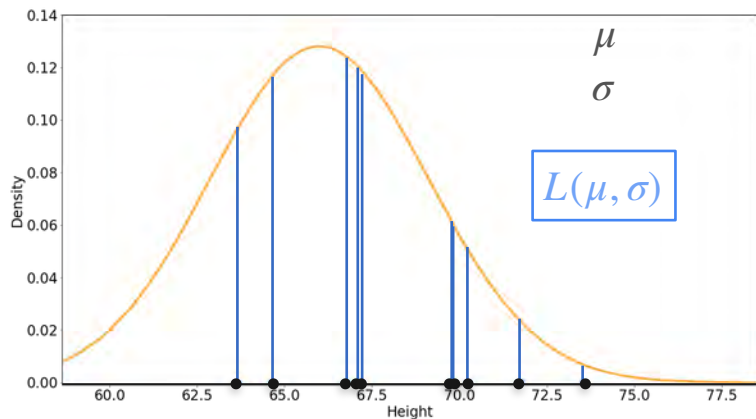
Likelihood $L(\mu; \underline{x}) = \prod_{i=1}^{10} f_{\mu}(x_i)$



$$\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right]$$

Likelihood $L(\mu; \underline{x}) = \prod_{i=1}^{10} f_{\mu}(x_i)$

$$= \prod_{i=1}^{10} \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2} \frac{(x_i - \mu)^2}{\sigma^2}\right)$$

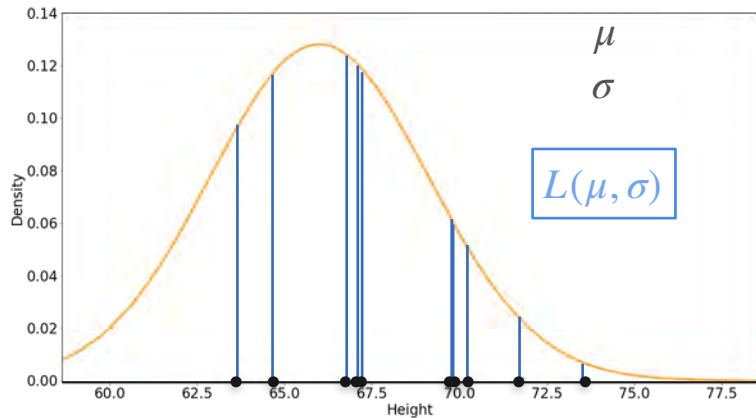


$$\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right]$$

Likelihood $L(\mu; \underline{x}) = \prod_{i=1}^{10} f_{\mu}(x_i)$

Observations
 x_i are fixed!

$$= \prod_{i=1}^{10} \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2} \frac{(x_i - \mu)^2}{\sigma^2}\right)$$



$$\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right]$$

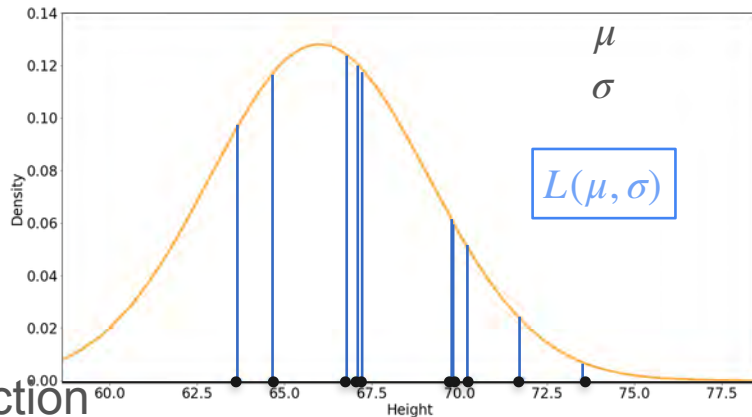
Likelihood $L(\mu; \underline{x}) = \prod_{i=1}^{10} f_{\mu}(x_i)$

$$= \prod_{i=1}^{10} \frac{1}{\sqrt{2\pi} \sigma} \exp \left(-\frac{1}{2} \frac{(x_i - \mu)^2}{\sigma^2} \right)$$

Observations

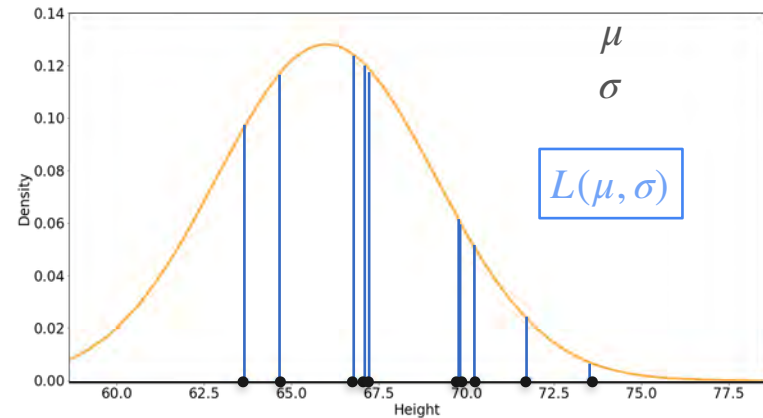
x_i are fixed!

It's a function
of μ



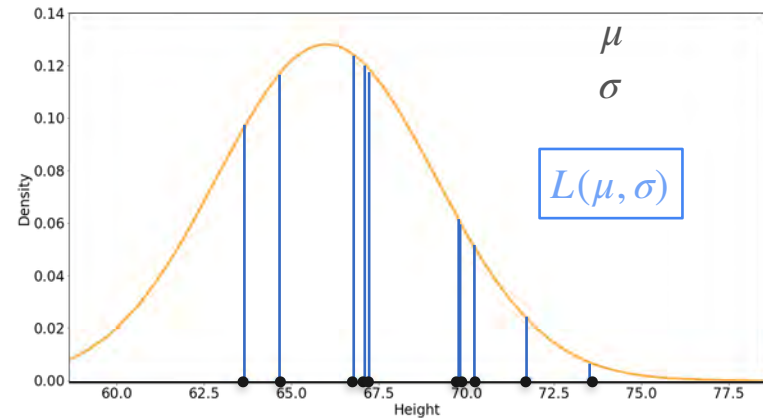
$$\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right]$$

$$\begin{aligned}
 \text{Likelihood } L(\mu; \underline{x}) &= \prod_{i=1}^{10} f_{\mu}(x_i) \\
 &= \prod_{i=1}^{10} \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2} \frac{(x_i - \mu)^2}{\sigma^2}\right) \\
 &\quad \left(\frac{1}{\sqrt{2\pi} \sigma}\right)^{10}
 \end{aligned}$$



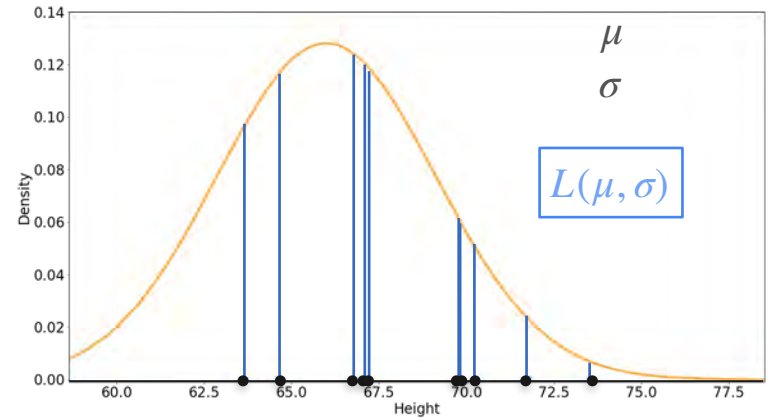
$$\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right]$$

$$\begin{aligned}
 \text{Likelihood } L(\mu; \underline{x}) &= \prod_{i=1}^{10} f_{\mu}(x_i) \\
 &= \prod_{i=1}^{10} \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2} \frac{(x_i - \mu)^2}{\sigma^2}\right) \\
 &= \left(\frac{1}{\sqrt{2\pi} \sigma}\right)^{10} \exp\left(-\frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2}\right)
 \end{aligned}$$



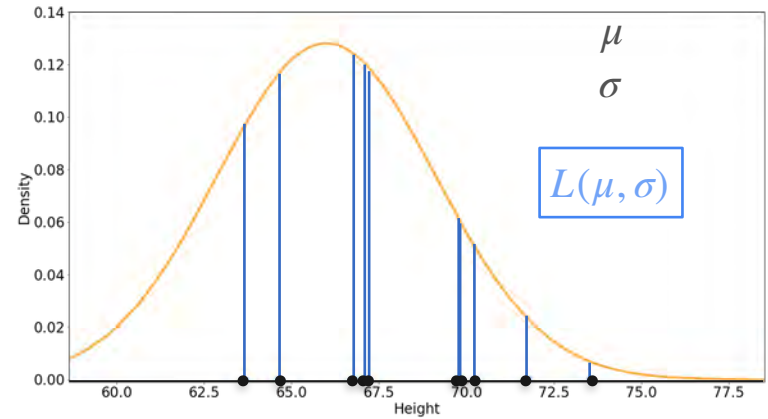
$$\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right]$$

$$\begin{aligned}
 \text{Likelihood } L(\mu; \underline{x}) &= \prod_{i=1}^{10} f_{\mu}(x_i) \\
 &= \prod_{i=1}^{10} \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2} \frac{(x_i - \mu)^2}{\sigma^2}\right) \\
 &= \left(\frac{1}{\sqrt{2\pi} \sigma}\right)^{10} \exp\left(-\frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2}\right)
 \end{aligned}$$



$$\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right]$$

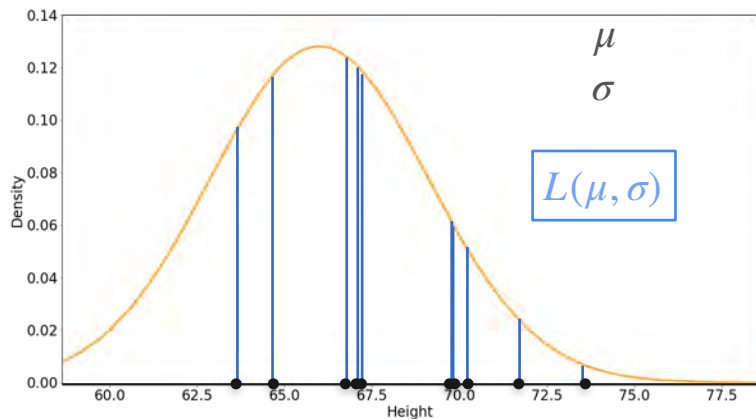
$$\begin{aligned}
 \text{Likelihood } L(\mu; \underline{x}) &= \prod_{i=1}^{10} f_{\mu}(x_i) \\
 &= \prod_{i=1}^{10} \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2} \frac{(x_i - \mu)^2}{\sigma^2}\right) \\
 &= \left(\frac{1}{\sqrt{2\pi} \sigma}\right)^{10} \exp\left(-\frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2}\right)
 \end{aligned}$$



$$\ell(\mu; \mathbf{x}) = \log(L(\mu; \mathbf{x}))$$

$$\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right]$$

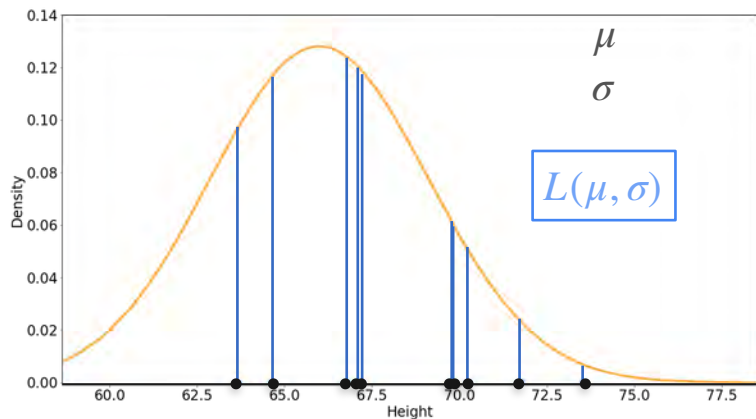
$$\begin{aligned}
 \text{Likelihood } L(\mu; \underline{x}) &= \prod_{i=1}^{10} f_{\mu}(x_i) \\
 &= \prod_{i=1}^{10} \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2} \frac{(x_i - \mu)^2}{\sigma^2}\right) \\
 &= \left(\frac{1}{\sqrt{2\pi} \sigma}\right)^{10} \exp\left(-\frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2}\right)
 \end{aligned}$$



$$\ell(\mu; \mathbf{x}) = \log(L(\mu; \mathbf{x})) = \log\left(\left(\frac{1}{\sqrt{2\pi} \sigma}\right)^{10}\right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2}$$

$$\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log\left(\left(\frac{1}{\sqrt{2\pi} \sigma}\right)^{10}\right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right]$$

$$\begin{aligned}
 \text{Likelihood } L(\mu; \underline{x}) &= \prod_{i=1}^{10} f_{\mu}(x_i) \\
 &= \prod_{i=1}^{10} \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2} \frac{(x_i - \mu)^2}{\sigma^2}\right) \\
 &= \left(\frac{1}{\sqrt{2\pi} \sigma}\right)^{10} \exp\left(-\frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2}\right)
 \end{aligned}$$

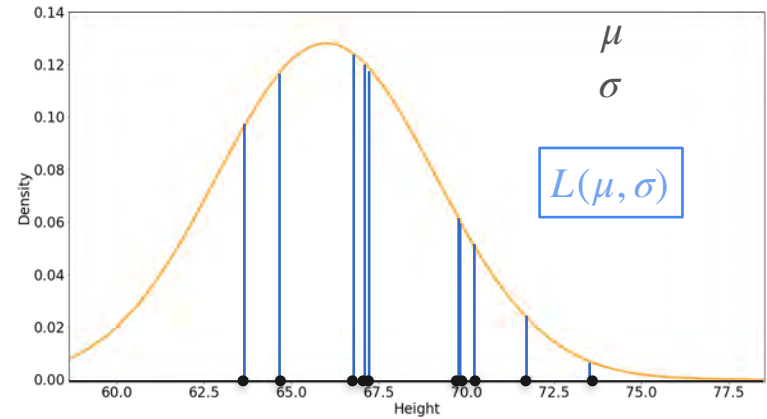


$$\begin{aligned}
 \text{Log-Likelihood } \ell(\mu; \mathbf{x}) &= \log(L(\mu; \mathbf{x})) = \log\left(\left(\frac{1}{\sqrt{2\pi} \sigma}\right)^{10}\right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \\
 \frac{d}{d\mu} \ell(\mu; \mathbf{x}) &= \frac{d}{d\mu} \left[\log\left(\left(\frac{1}{\sqrt{2\pi} \sigma}\right)^{10}\right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right]
 \end{aligned}$$

Likelihood $L(\mu; \underline{x}) = \prod_{i=1}^{10} f_{\mu}(x_i)$

$$= \prod_{i=1}^{10} \frac{1}{\sqrt{2\pi} \sigma} \exp \left(-\frac{1}{2} \frac{(x_i - \mu)^2}{\sigma^2} \right)$$

$$\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \exp \left(-\frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right)$$



Log-Likelihood $\ell(\mu; \mathbf{x}) = \log (L(\mu; \mathbf{x})) = \log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2}$

Derivative $\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right]$

Log-Likelihood $\ell(\mu; \mathbf{x}) = \log(L(\mu; \mathbf{x})) = \log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2}$

Derivative $\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right]$

Log-Likelihood $\ell(\mu; \mathbf{x}) = \log (L(\mu; \mathbf{x})) = \log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2}$

Derivative $\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right] = 0$

Log-Likelihood $\ell(\mu; \mathbf{x}) = \log(L(\mu; \mathbf{x})) = \log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2}$

Derivative $\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right] = 0$

= 0

Log-Likelihood $\ell(\mu; \mathbf{x}) = \log(L(\mu; \mathbf{x})) = \log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2}$

Derivative $\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right] = 0$

$$= 0 - \frac{1}{2} \frac{1}{\sigma^2} \sum_{i=1}^{10} 2(x_i - \mu)(-1)$$

Log-Likelihood $\ell(\mu; \mathbf{x}) = \log(L(\mu; \mathbf{x})) = \log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2}$

Derivative $\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right] = 0$

$$= 0 - \frac{1}{2} \frac{1}{\sigma^2} \sum_{i=1}^{10} 2(x_i - \mu)(-1)$$

This needs to be zero

Log-Likelihood $\ell(\mu; \mathbf{x}) = \log(L(\mu; \mathbf{x})) = \log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right)$

Derivative $\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right) \right] = 0$

=

0

$$-\frac{1}{2} \frac{1}{\sigma^2} \sum_{i=1}^{10} 2(x_i - \mu)(-1)$$

This needs to be zero

Log-Likelihood $\ell(\mu; \mathbf{x}) = \log(L(\mu; \mathbf{x})) = \log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2}$

Derivative $\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right] = 0$

=

0

$$\boxed{-\cancel{\frac{1}{2}} \cancel{\frac{1}{\sigma^2}} \sum_{i=1}^{10} 2(x_i - \mu)(\cancel{-1})}$$

This needs to be zero

Log-Likelihood $\ell(\mu; \mathbf{x}) = \log(L(\mu; \mathbf{x})) = \log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2}$

Derivative $\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right] = 0$

= 0

$$\boxed{-\cancel{\frac{1}{2}} \cancel{\frac{1}{\sigma^2}} \sum_{i=1}^{10} 2(x_i - \mu)(\cancel{-1})}$$

This needs to be zero

$$\sum_{i=1}^{10} (x_i - \mu) = 0$$

Log-Likelihood $\ell(\mu; \mathbf{x}) = \log(L(\mu; \mathbf{x})) = \log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2}$

Derivative $\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} \right) - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right] = 0$

= 0

$$-\frac{1}{2\sigma^2} \sum_{i=1}^{10} 2(x_i - \mu)(-1)$$

This needs to be zero

$$\sum_{i=1}^{10} (x_i - \mu) = 0$$

$$\left(\sum_{i=1}^{10} x_i \right) - 10\mu = 0$$

Log-Likelihood $\ell(\mu; \mathbf{x}) = \log(L(\mu; \mathbf{x})) = \log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right)$

Derivative $\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^{10} - \frac{1}{2} \frac{\sum_{i=1}^{10} (x_i - \mu)^2}{\sigma^2} \right) \right] = 0$

= 0

$$\boxed{-\frac{1}{2\sigma^2} \sum_{i=1}^{10} 2(x_i - \mu)(-1)}$$

This needs to be zero

$$\sum_{i=1}^{10} (x_i - \mu) = 0$$

$$\left(\sum_{i=1}^{10} x_i \right) - 10\mu = 0 \quad \Rightarrow \quad \mu = \frac{1}{10} \sum_{i=1}^{10} x_i$$

Log-Likelihood $\ell(\mu; \mathbf{x}) = \log(L(\mu; \mathbf{x})) = \log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^n \right) - \frac{1}{2} \frac{\sum_{i=1}^n (x_i - \mu)^2}{\sigma^2}$

Derivative $\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^n \right) - \frac{1}{2} \frac{\sum_{i=1}^n (x_i - \mu)^2}{\sigma^2} \right] = 0$

$= 0$

$$-\frac{1}{2} \frac{1}{\sigma^2} \sum_{i=1}^n 2(x_i - \mu)(-1)$$

This needs to be zero

$$\sum_{i=1}^n (x_i - \mu) = 0$$

$$\left(\sum_{i=1}^n x_i \right) - n \mu = 0 \Rightarrow \mu = \frac{1}{n} \sum_{i=1}^n x_i$$

Log-Likelihood $\ell(\mu; \mathbf{x}) = \log(L(\mu; \mathbf{x})) = \log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^n \right) - \frac{1}{2} \frac{\sum_{i=1}^n (x_i - \mu)^2}{\sigma^2}$

Derivative $\frac{d}{d\mu} \ell(\mu; \mathbf{x}) = \frac{d}{d\mu} \left[\log \left(\left(\frac{1}{\sqrt{2\pi} \sigma} \right)^n \right) - \frac{1}{2} \frac{\sum_{i=1}^n (x_i - \mu)^2}{\sigma^2} \right] = 0$

$= 0$

$$-\frac{1}{2\sigma^2} \sum_{i=1}^n 2(x_i - \mu)(-1)$$

This needs to be zero

$$\sum_{i=1}^n (x_i - \mu) = 0$$

$$\left(\sum_{i=1}^n x_i \right) - n \mu = 0 \quad \Rightarrow \quad \mu = \frac{1}{n} \sum_{i=1}^n x_i$$

The best distribution is the one where the **mean** of the distribution is the **mean** of the sample

Maximum Likelihood: Gaussian Example

Maximum Likelihood: Gaussian Example

Heights	66.75	70.24	67.19	67.09	63.65	64.64	69.81	69.79	73.52	71.74
---------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------

Maximum Likelihood: Gaussian Example

Heights	66.75	70.24	67.19	67.09	63.65	64.64	69.81	69.79	73.52	71.74
---------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------

$$\hat{\mu} = \frac{66.75 + 70.24 + 67.19 + 67.09 + 63.65 + 64.64 + 69.81 + 69.79 + 73.52 + 71.74}{10}$$

Maximum Likelihood: Gaussian Example

Heights	66.75	70.24	67.19	67.09	63.65	64.64	69.81	69.79	73.52	71.74
---------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------

$$\hat{\mu} = \frac{66.75 + 70.24 + 67.19 + 67.09 + 63.65 + 64.64 + 69.81 + 69.79 + 73.52 + 71.74}{10}$$
$$= 68.442$$

Maximum Likelihood: Gaussian Example

Heights	66.75	70.24	67.19	67.09	63.65	64.64	69.81	69.79	73.52	71.74
---------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------

Maximum Likelihood: Gaussian Example

What about the variance?

Heights	66.75	70.24	67.19	67.09	63.65	64.64	69.81	69.79	73.52	71.74
---------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------

Maximum Likelihood: Gaussian Example

What about the variance?

Heights	66.75	70.24	67.19	67.09	63.65	64.64	69.81	69.79	73.52	71.74
---------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------

If you repeat the process, assuming unknown variance, you'll get that

Maximum Likelihood: Gaussian Example

What about the variance?

Heights	66.75	70.24	67.19	67.09	63.65	64.64	69.81	69.79	73.52	71.74
---------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------

If you repeat the process, assuming unknown variance, you'll get that

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{10} (x_i - \bar{x})^2}{10}$$

Maximum Likelihood: Gaussian Example

What about the variance?

Heights	66.75	70.24	67.19	67.09	63.65	64.64	69.81	69.79	73.52	71.74
---------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------

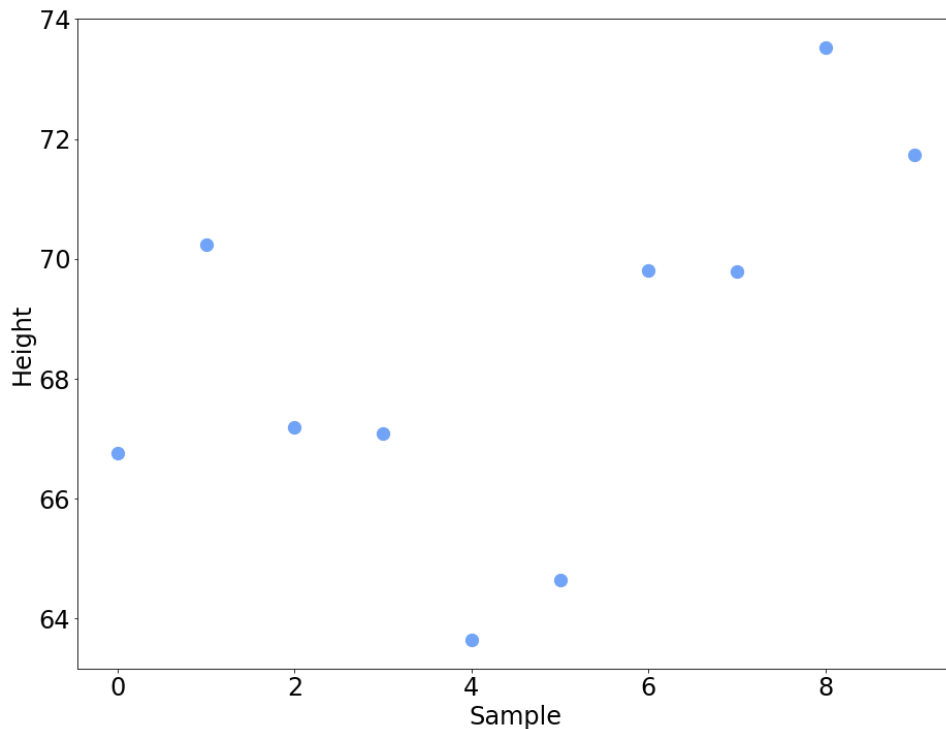
If you repeat the process, assuming unknown variance, you'll get that

$$\begin{aligned}\hat{\sigma}^2 &= \frac{\sum_{i=1}^{10} (x_i - \bar{x})^2}{10} = \frac{1}{10} \left((66.75 - 68.442)^2 + (70.24 - 68.442)^2 + (67.19 - 68.442)^2 + \right. \\ &\quad (67.09 - 68.442)^2 + (63.65 - 68.442)^2 + (64.64 - 68.442)^2 + \\ &\quad (69.81 - 68.442)^2 + (69.79 - 68.442)^2 + (73.52 - 68.442)^2 + \\ &\quad \left. (71.74 - 68.442)^2 \right) \\ &= 8.72\end{aligned}$$

Maximum Likelihood: Gaussian Example

$X = \text{"Height of an 18 year old"}$

$$\mathcal{N}(\mu, \sigma)$$

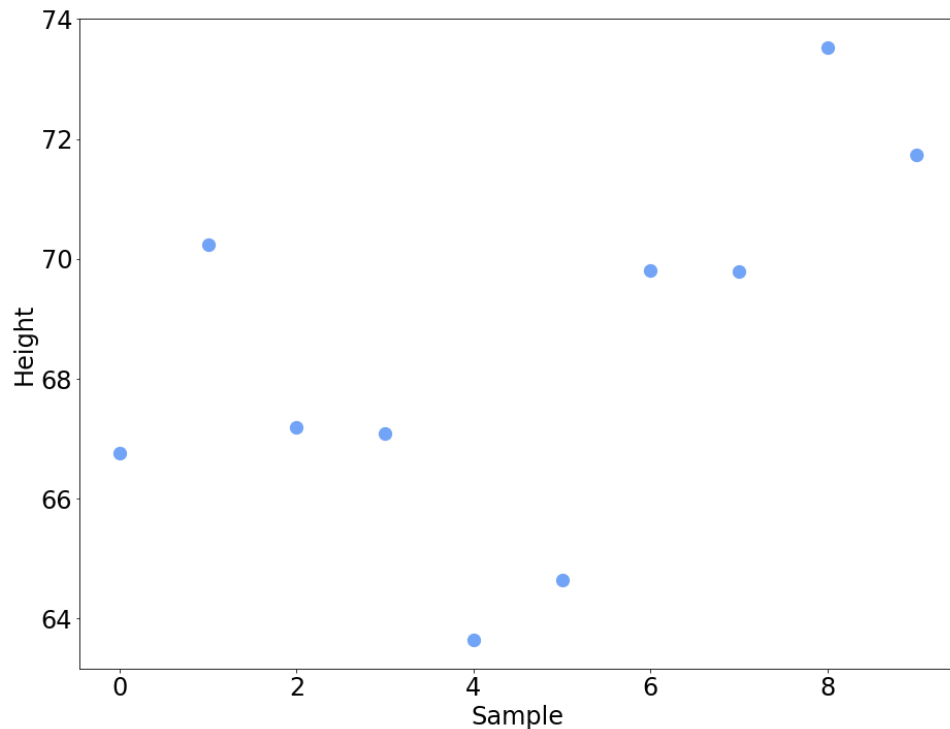


Maximum Likelihood: Gaussian Example

$X = \text{"Height of an 18 year old"}$

$$\mathcal{N}(\mu, \sigma)$$

$$\hat{\mu} = 68.442$$



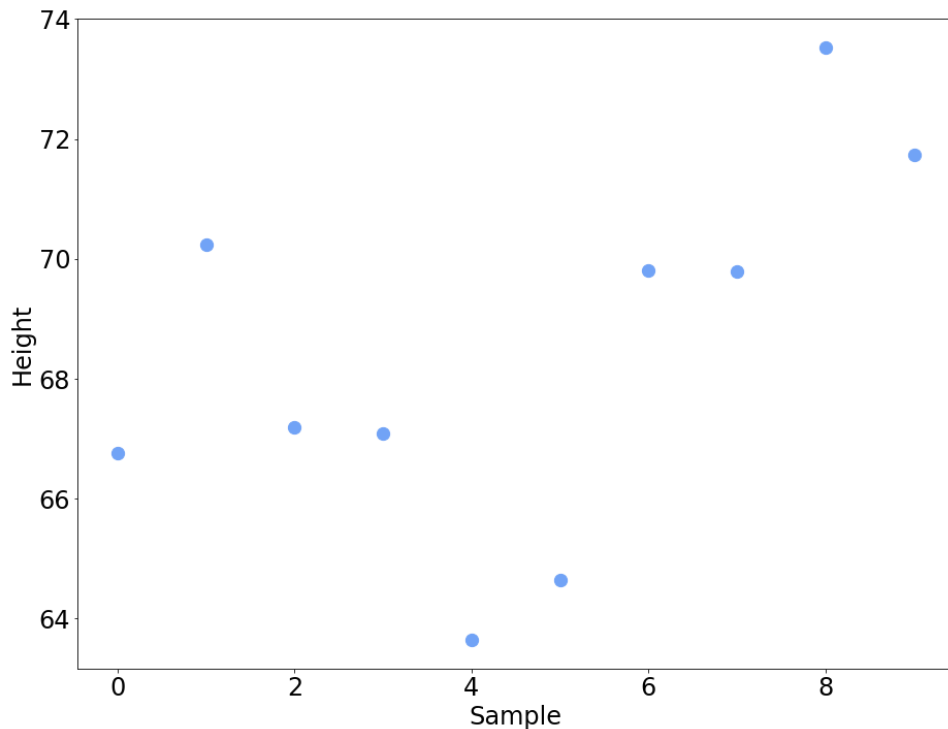
Maximum Likelihood: Gaussian Example

$X = \text{"Height of an 18 year old"}$

$$\mathcal{N}(\mu, \sigma)$$

$$\hat{\mu} = 68.442$$

$$\hat{\sigma} = 8.72$$





DeepLearning.AI

Point Estimation

MLE: Linear Regression

Maximum Likelihood

Maximum Likelihood



Data

Maximum Likelihood



Model 1



Data

Maximum Likelihood



Model 1



Model 2



Data

Maximum Likelihood



Model 1



Model 2

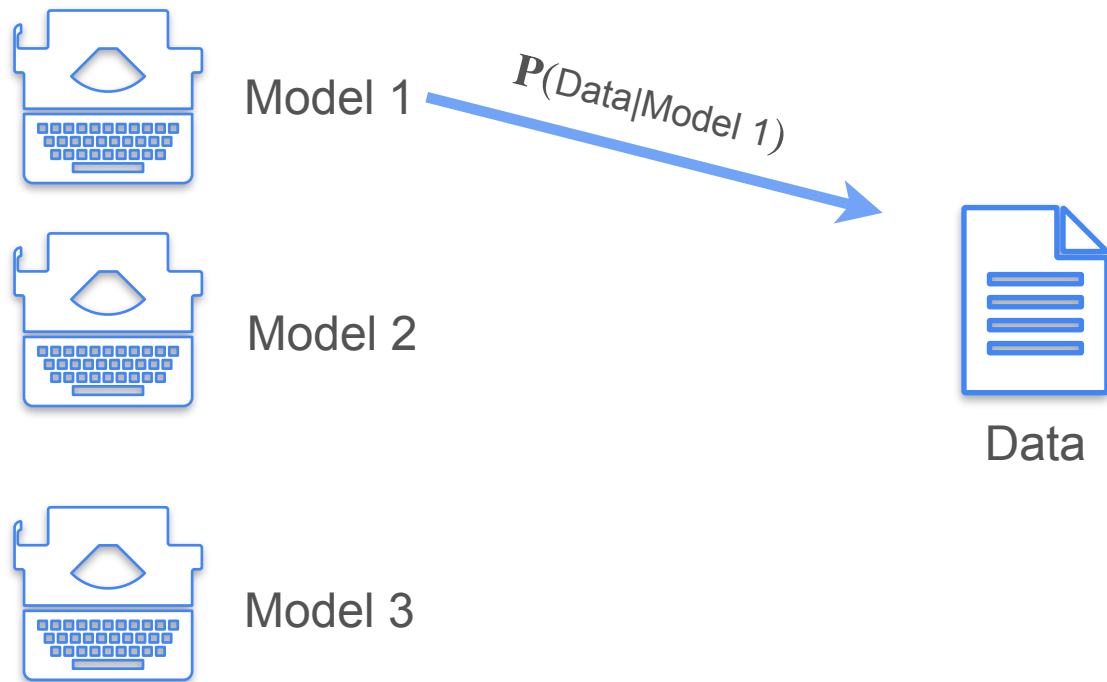


Model 3

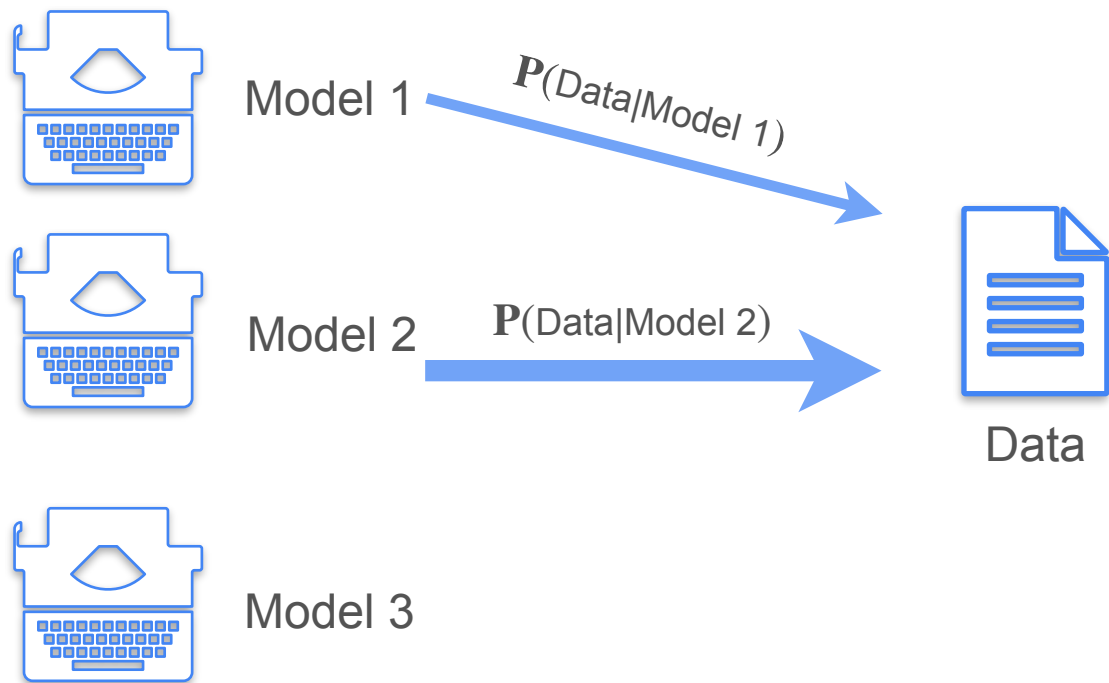


Data

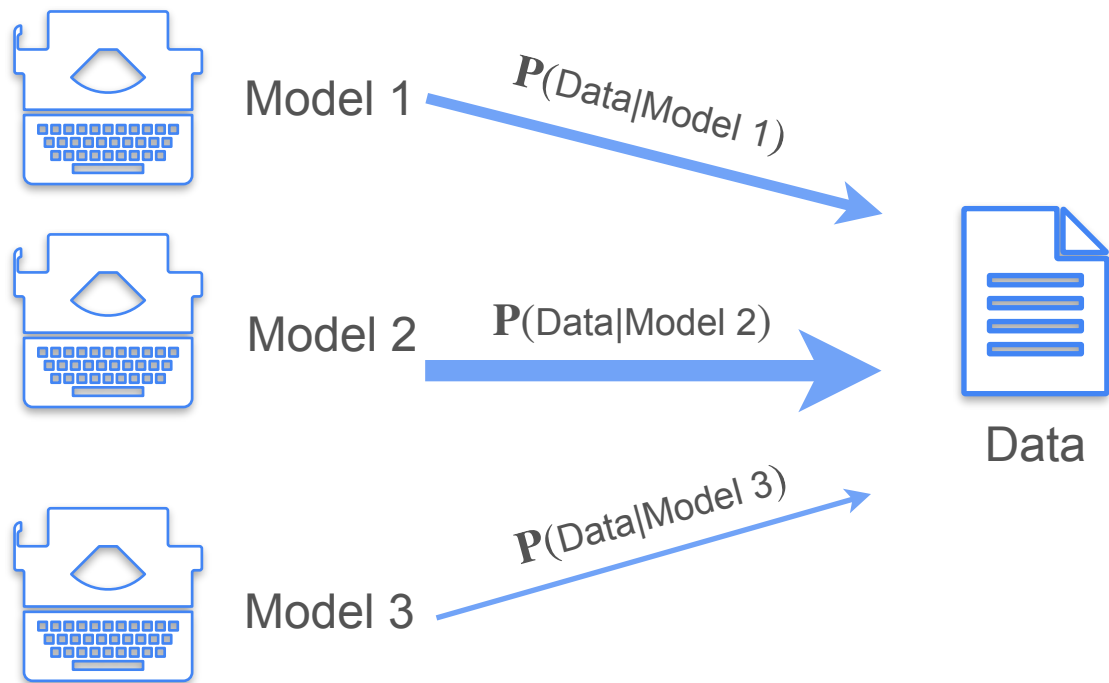
Maximum Likelihood



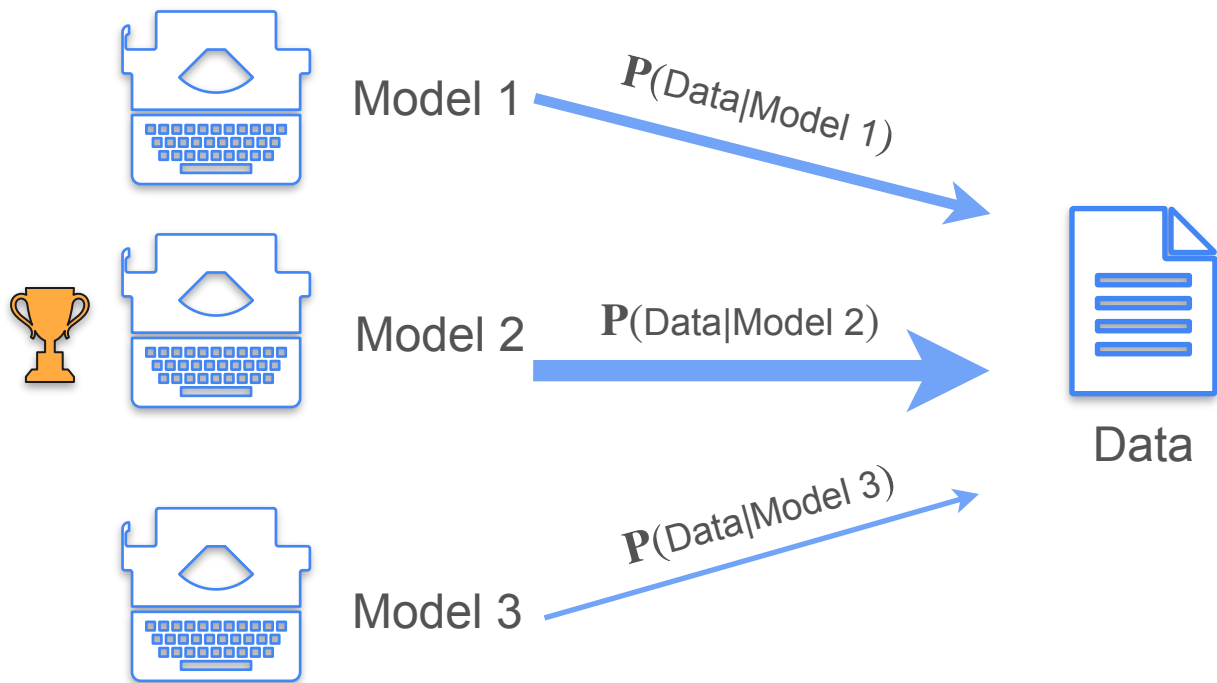
Maximum Likelihood



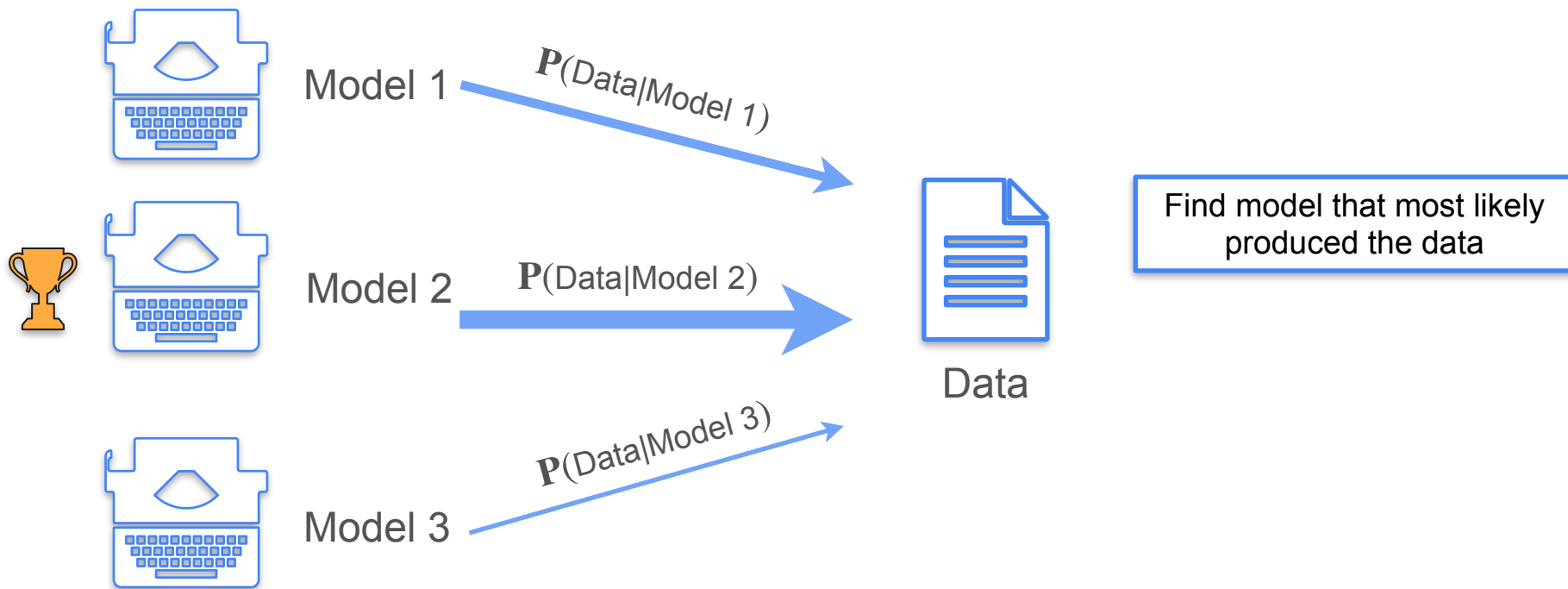
Maximum Likelihood



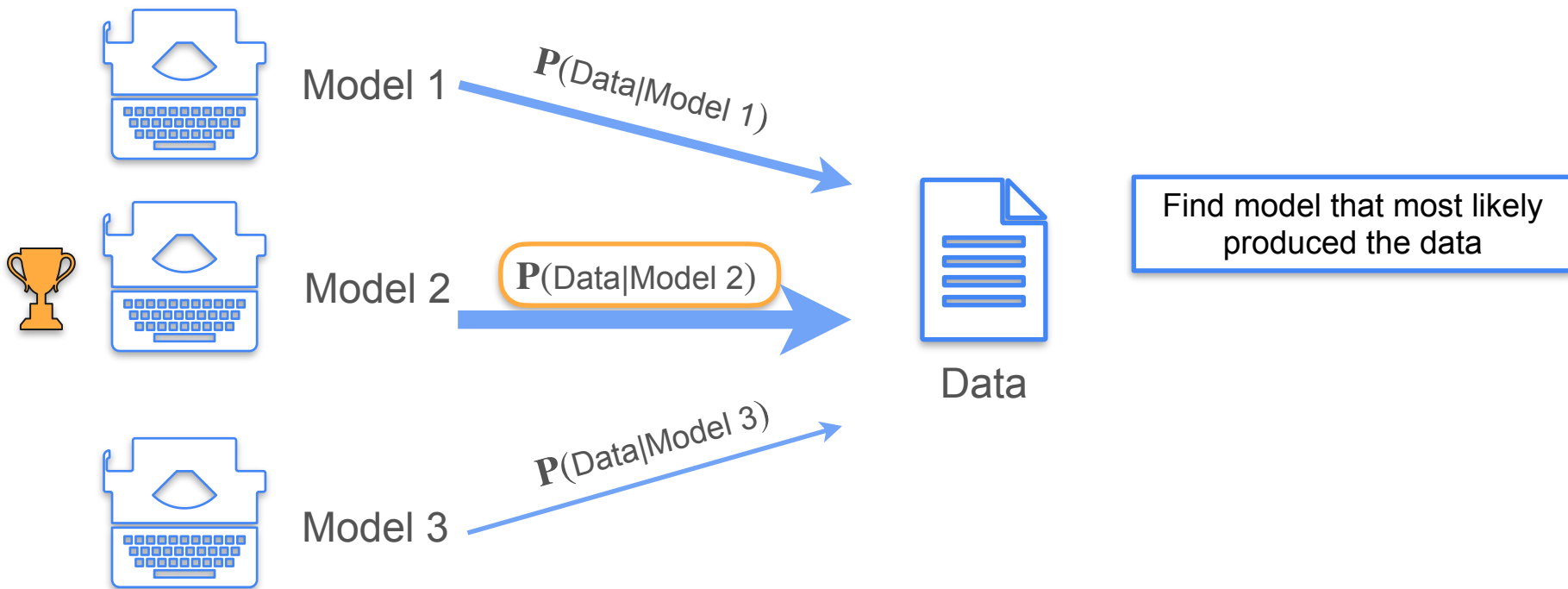
Maximum Likelihood



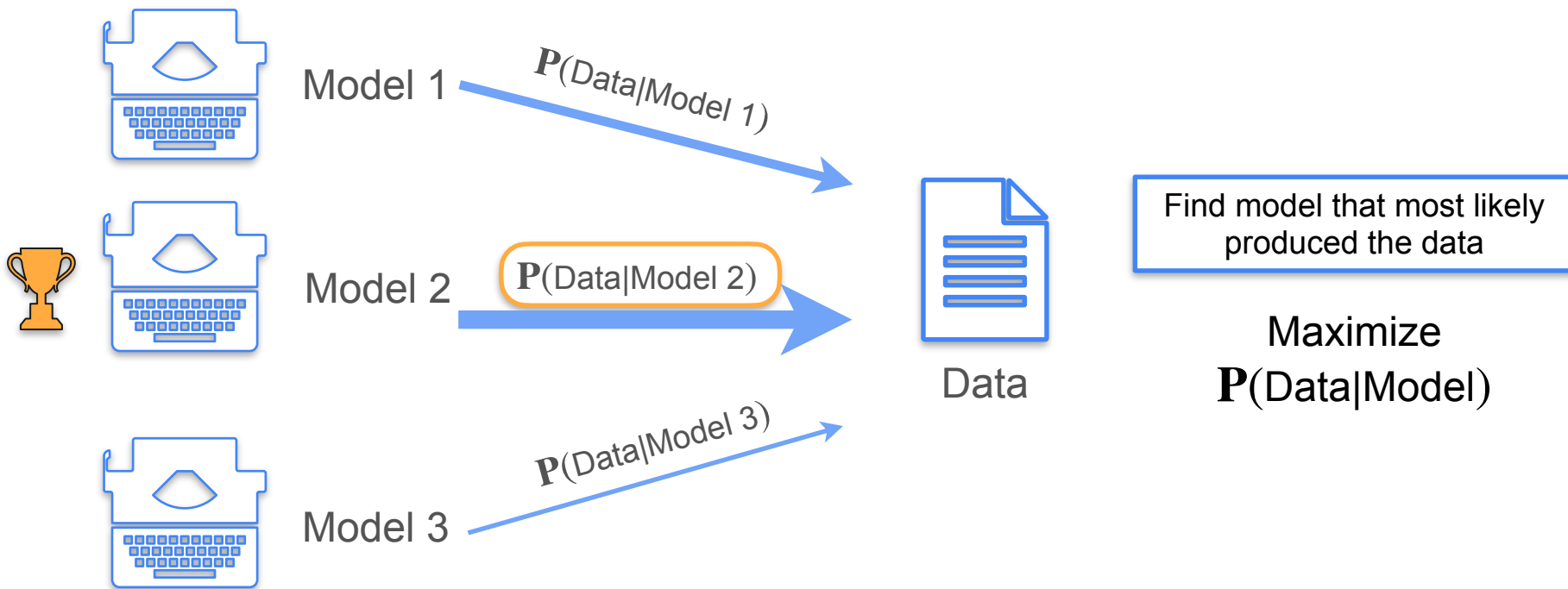
Maximum Likelihood



Maximum Likelihood

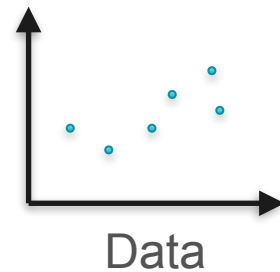


Maximum Likelihood

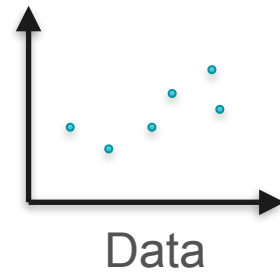
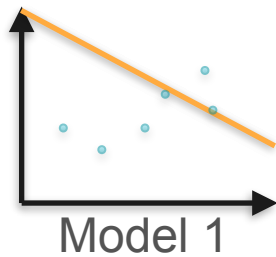


Example: Linear Regression

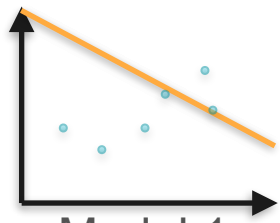
Example: Linear Regression



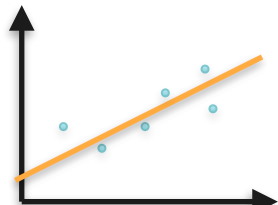
Example: Linear Regression



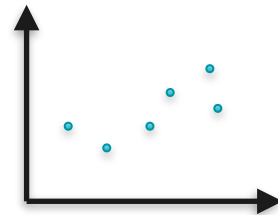
Example: Linear Regression



Model 1

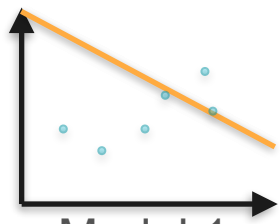


Model 2

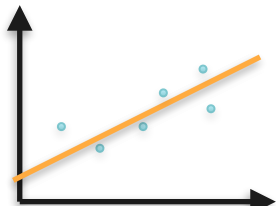


Data

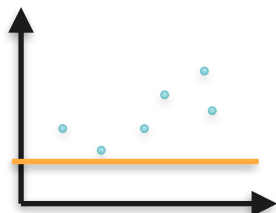
Example: Linear Regression



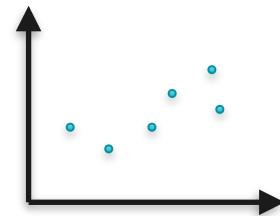
Model 1



Model 2

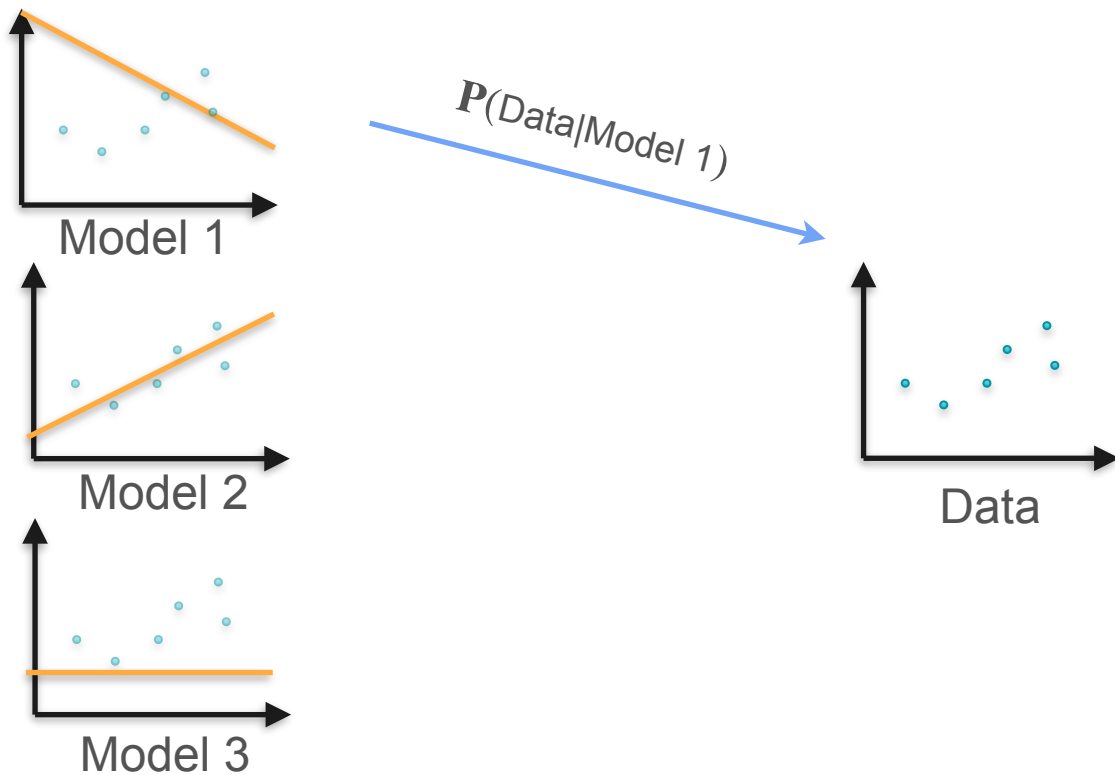


Model 3

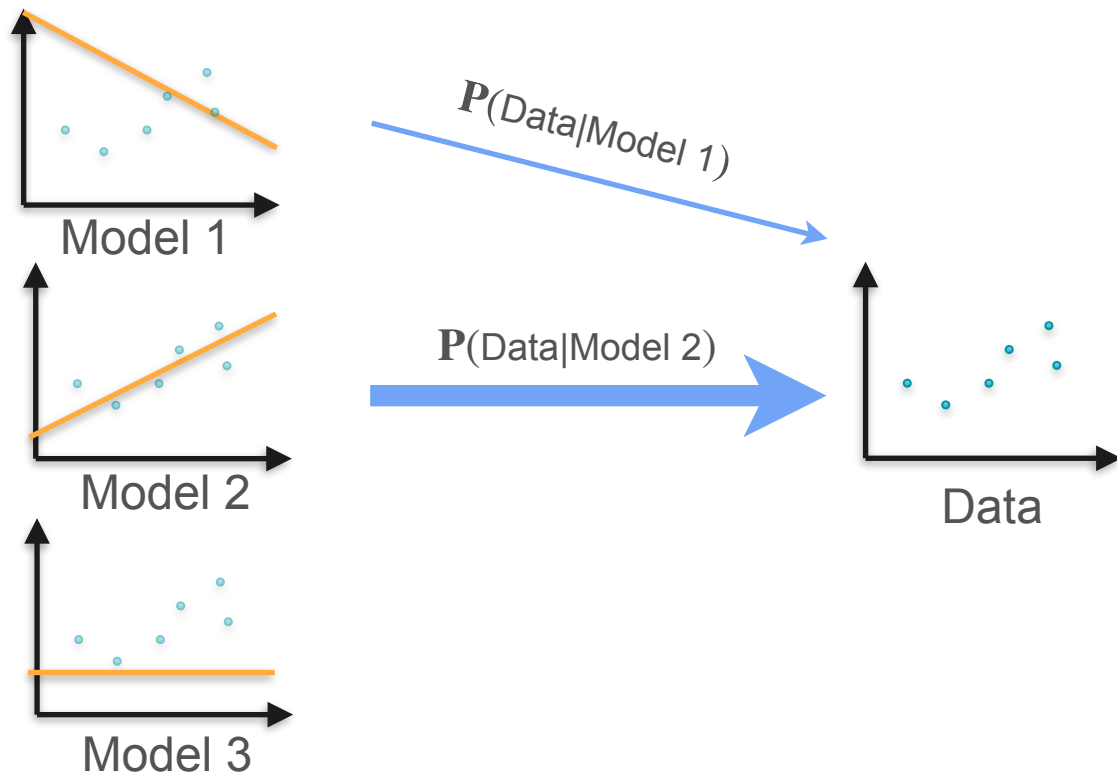


Data

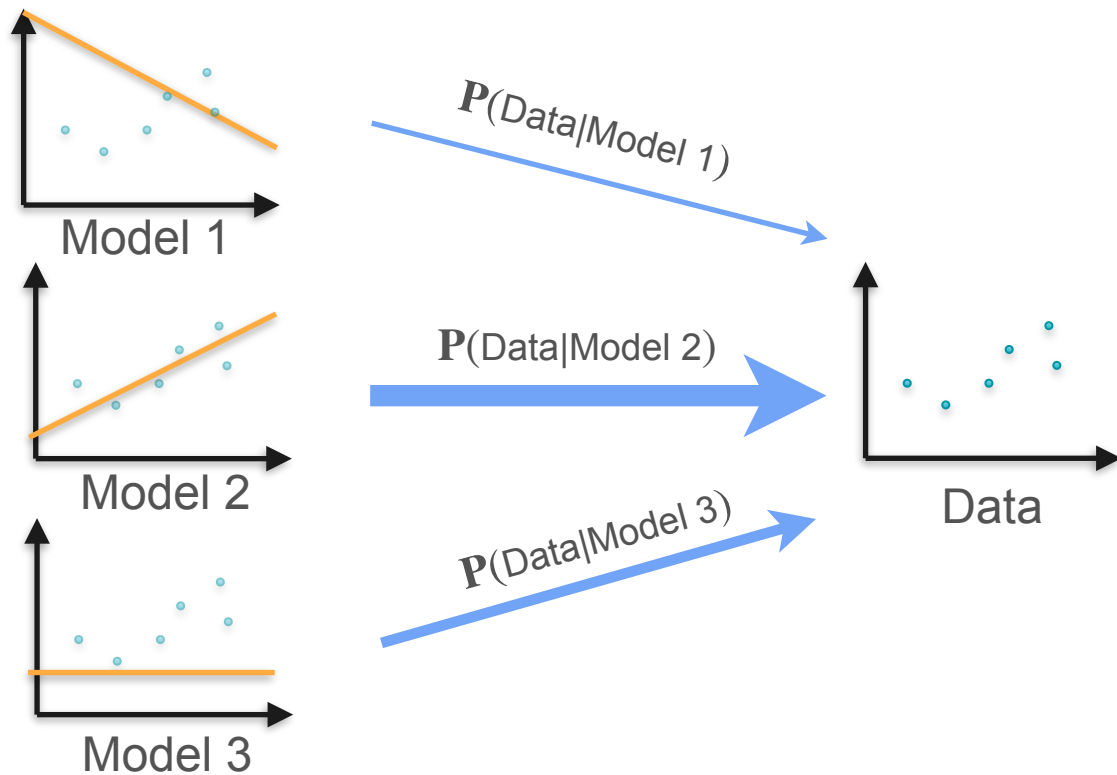
Example: Linear Regression



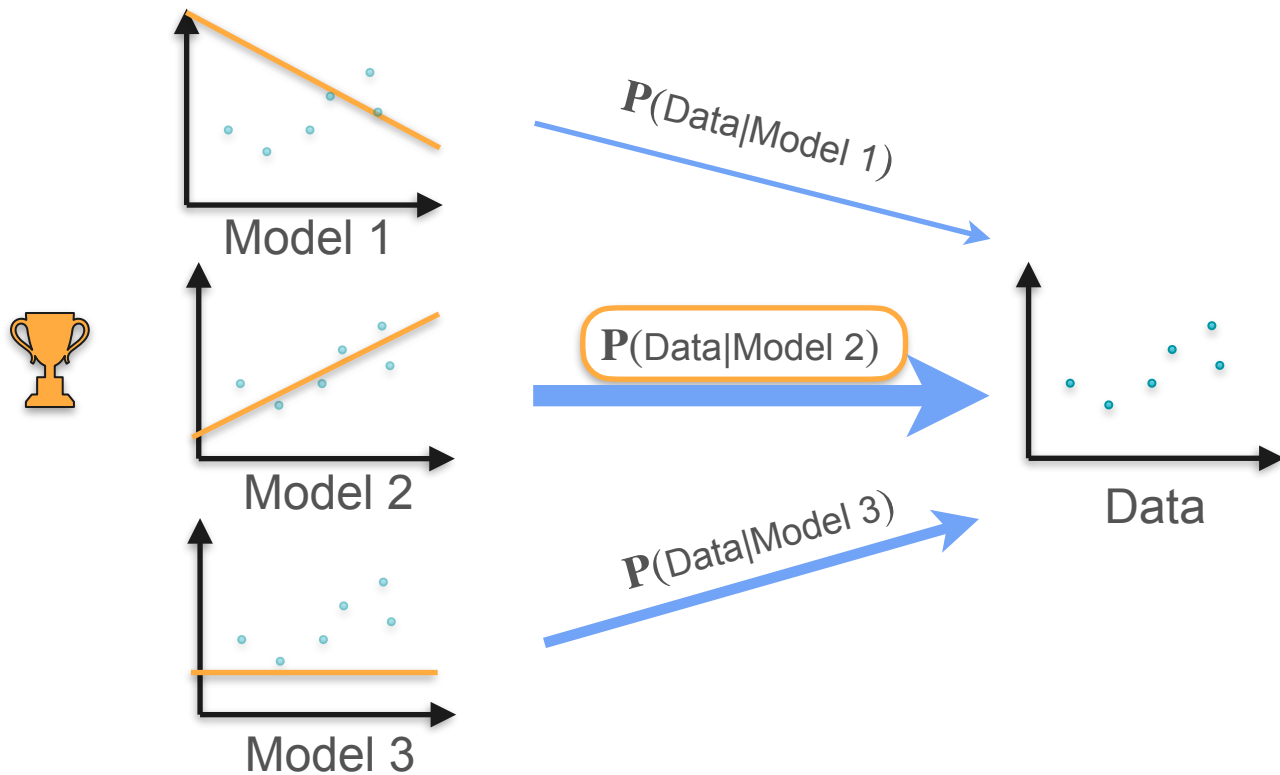
Example: Linear Regression



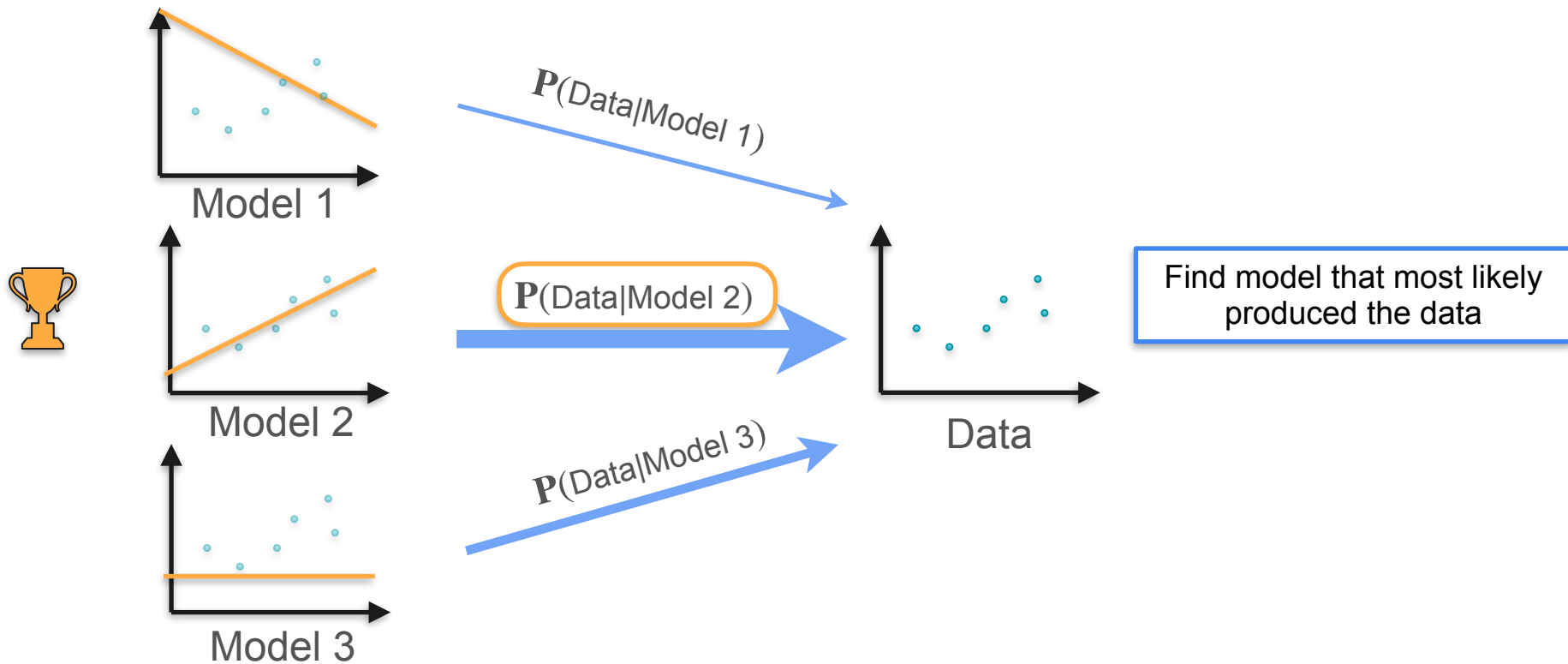
Example: Linear Regression



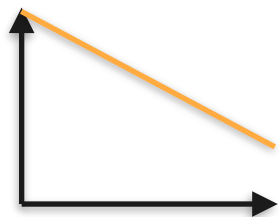
Example: Linear Regression



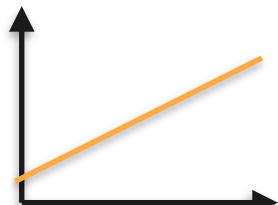
Example: Linear Regression



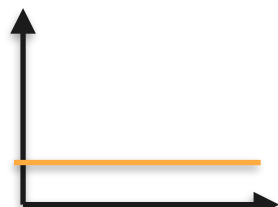
How Exactly Does a Line Produce Points?



Model 1

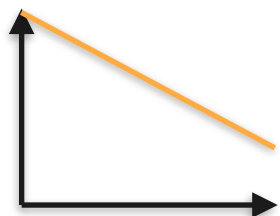


Model 2

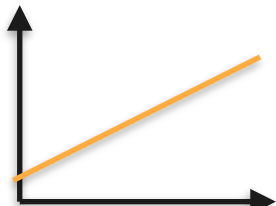


Model 3

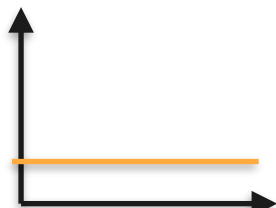
How Exactly Does a Line Produce Points?



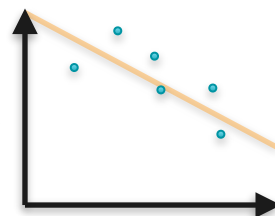
Model 1



Model 2

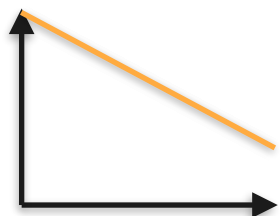


Model 3

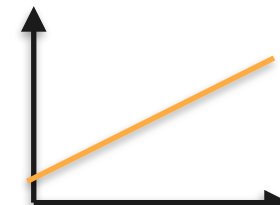


Data 1

How Exactly Does a Line Produce Points?



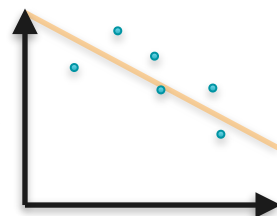
Model 1



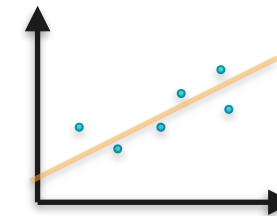
Model 2



Model 3

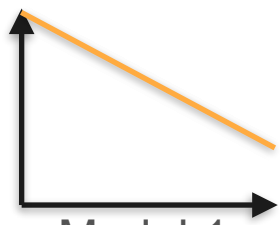


Data 1

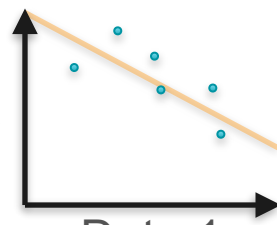


Data 2

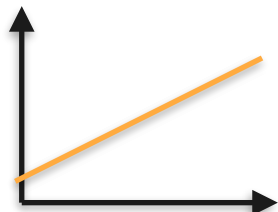
How Exactly Does a Line Produce Points?



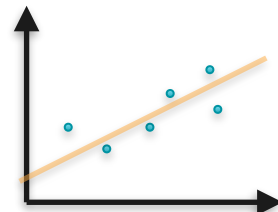
Model 1



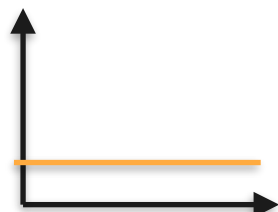
Data 1



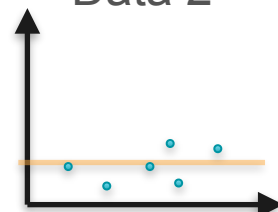
Model 2



Data 2

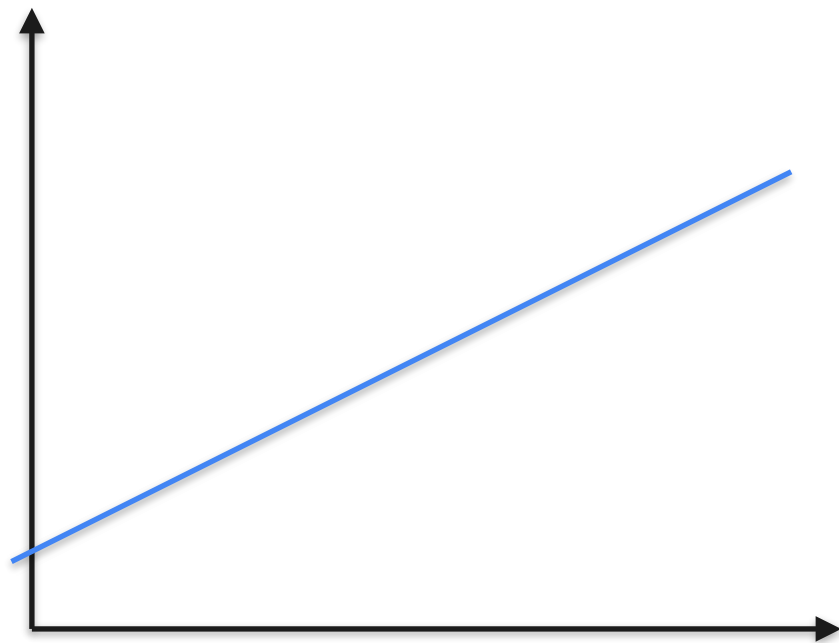
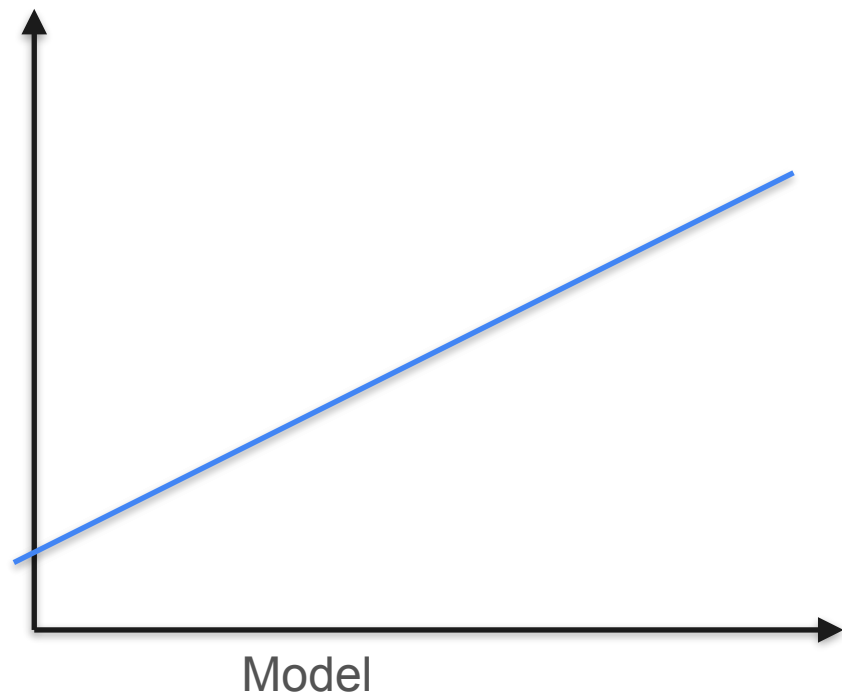


Model 3

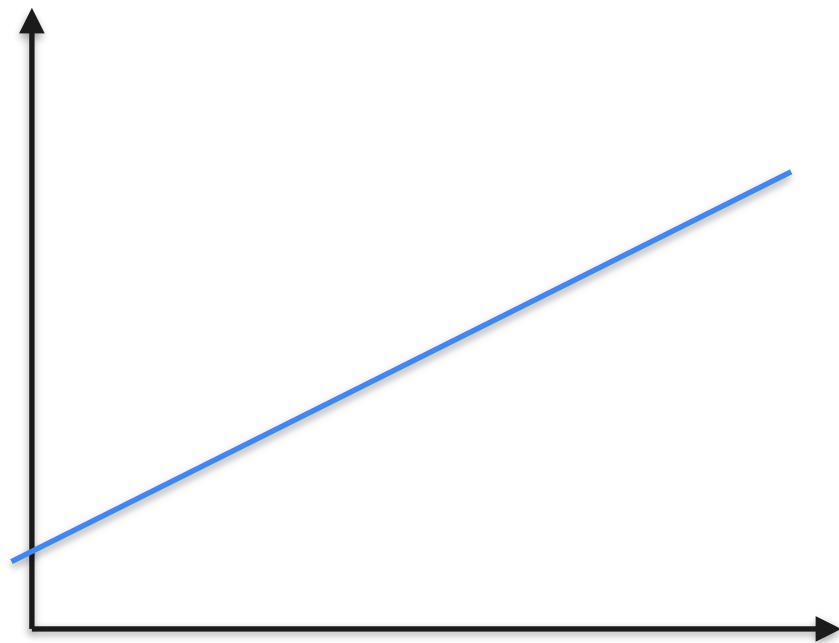
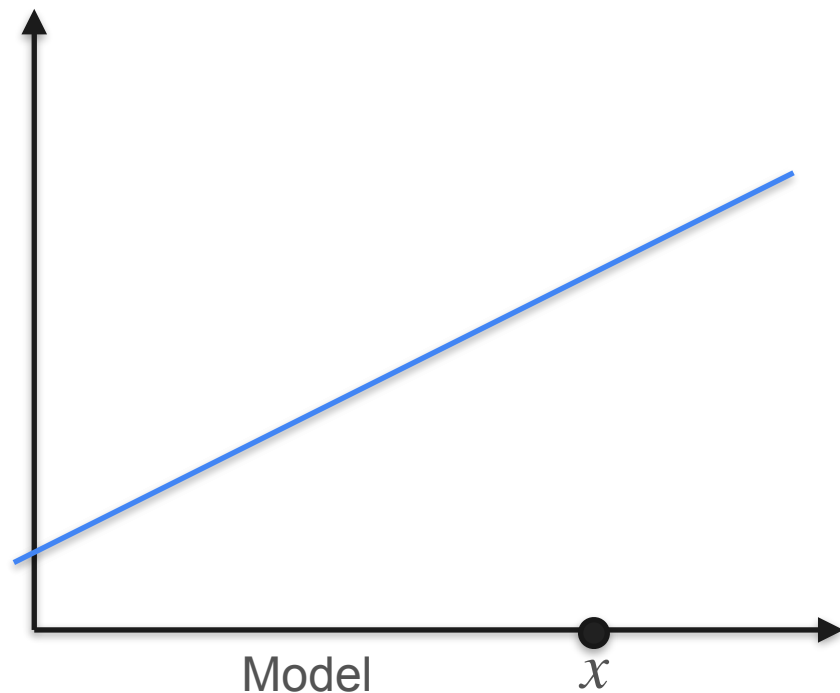


Data 3

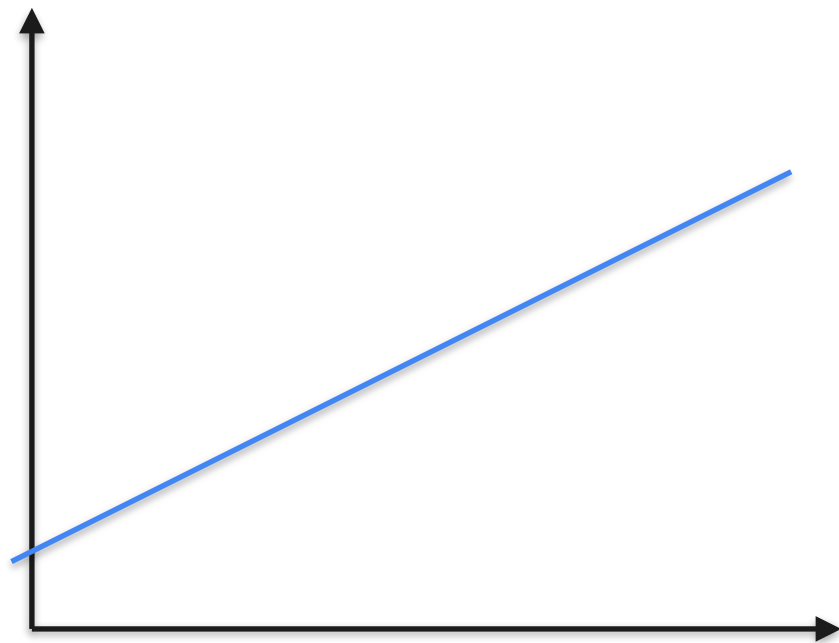
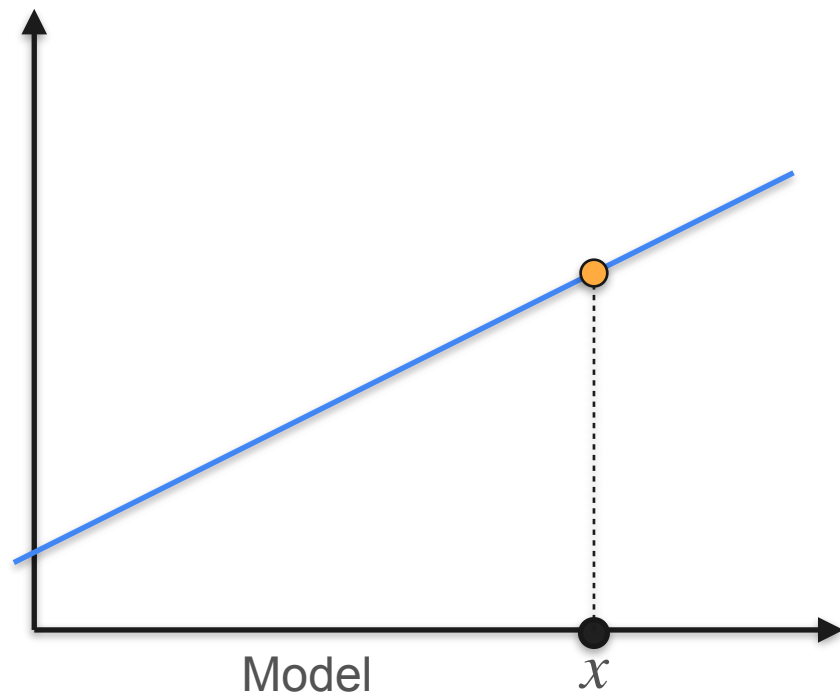
How Exactly Does a Line Produce Points?



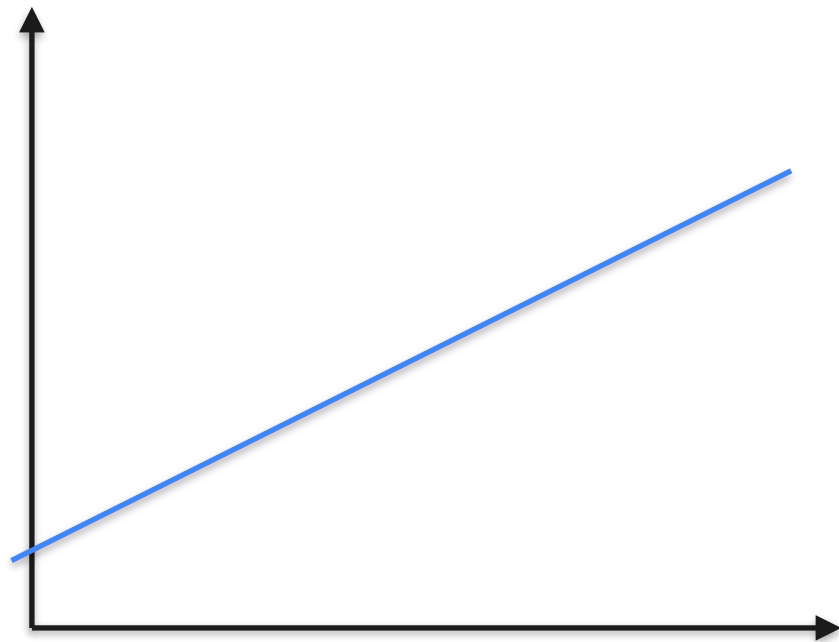
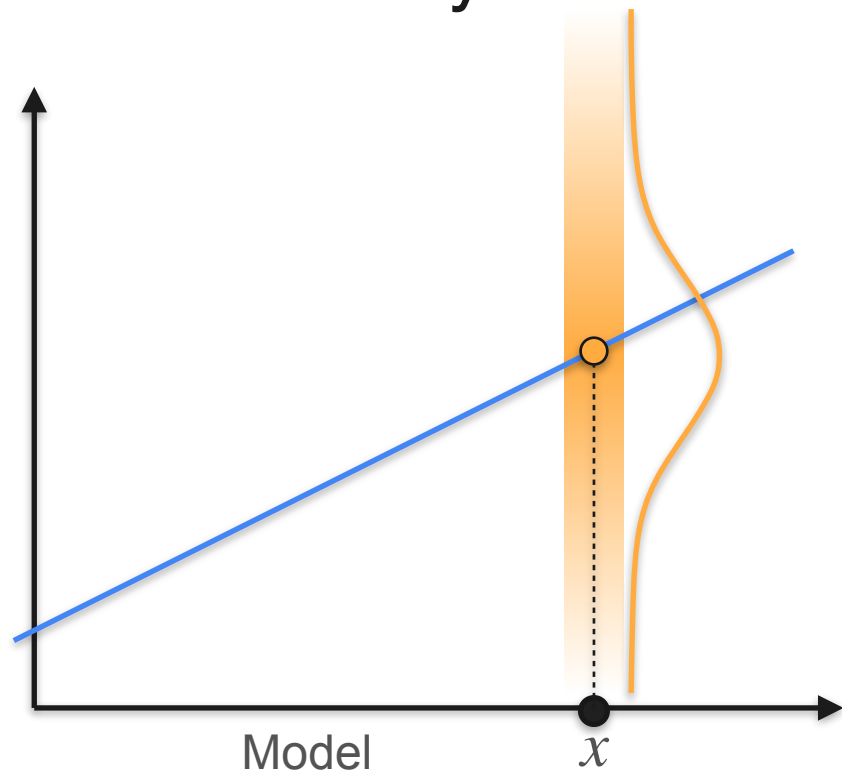
How Exactly Does a Line Produce Points?



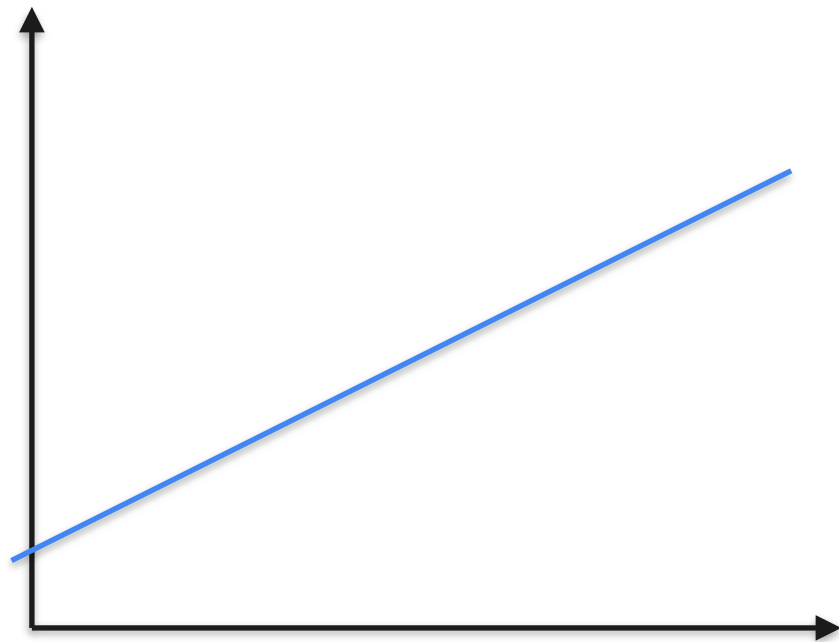
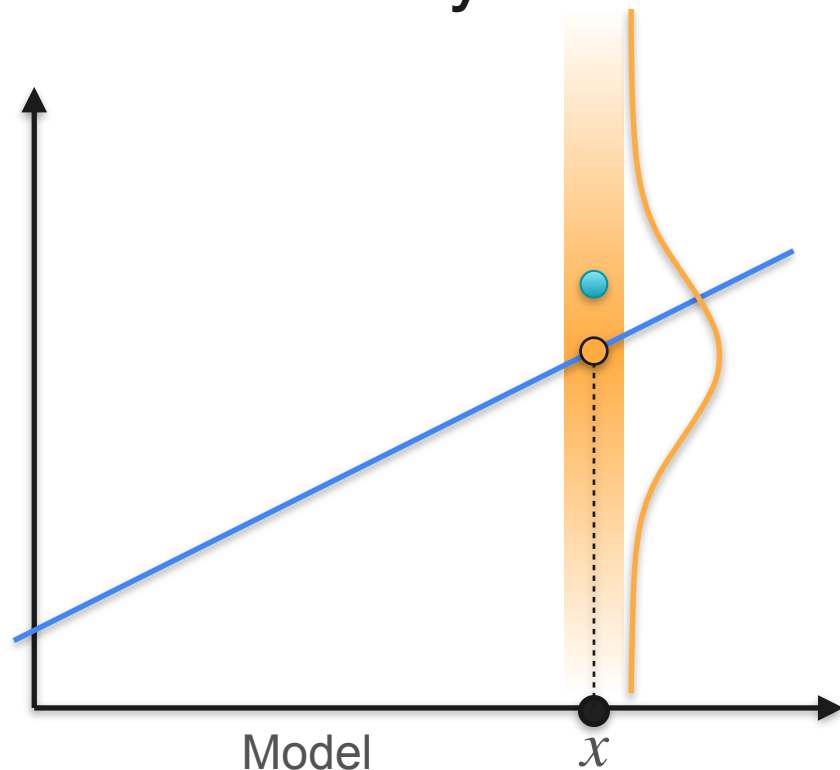
How Exactly Does a Line Produce Points?



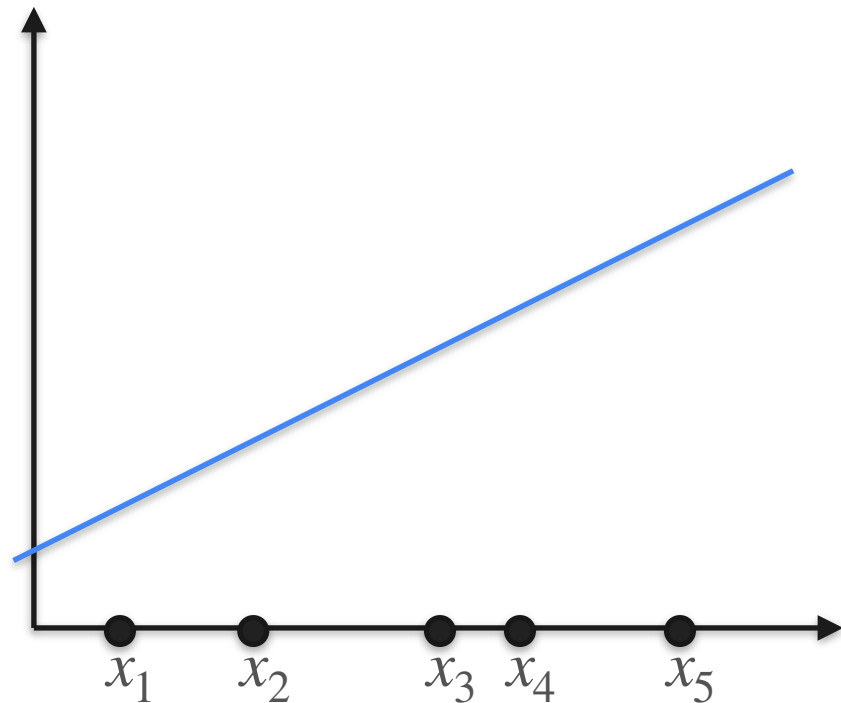
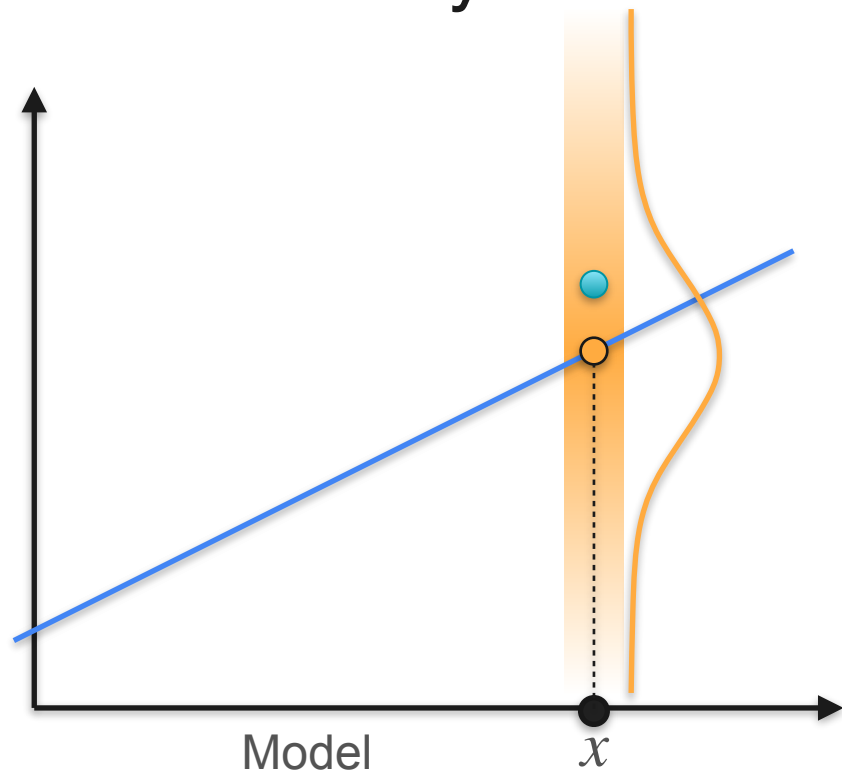
How Exactly Does a Line Produce Points?



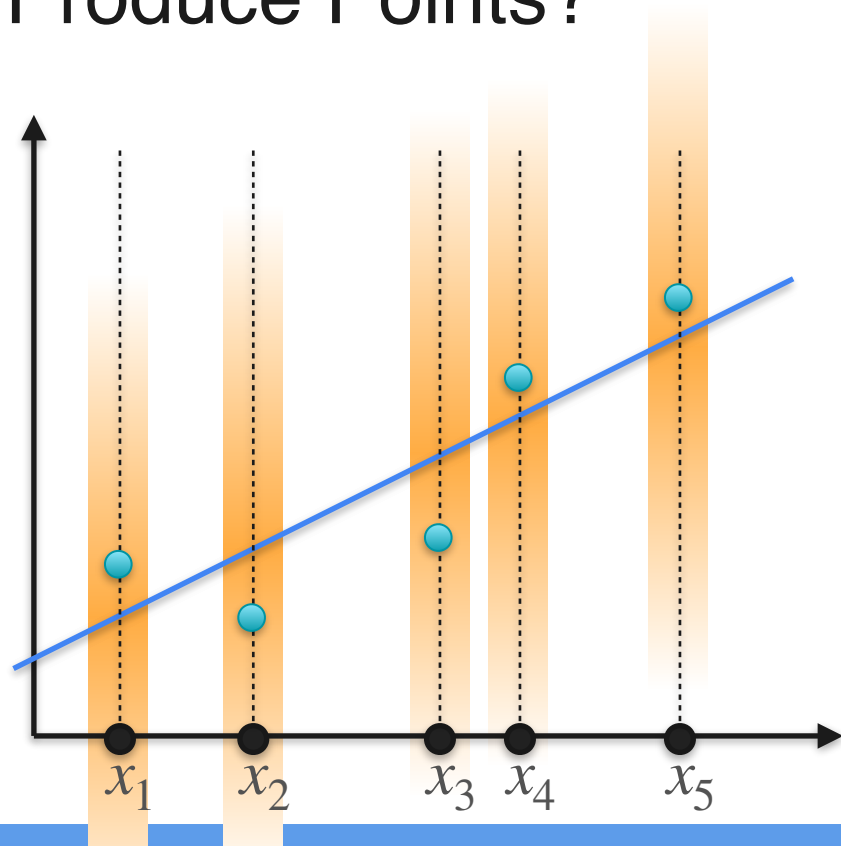
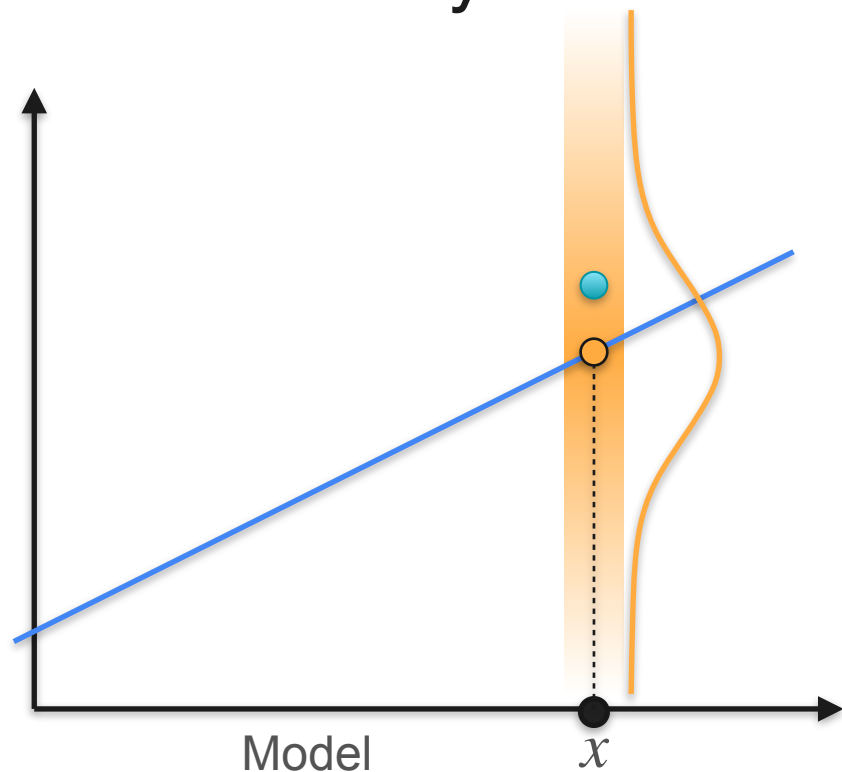
How Exactly Does a Line Produce Points?



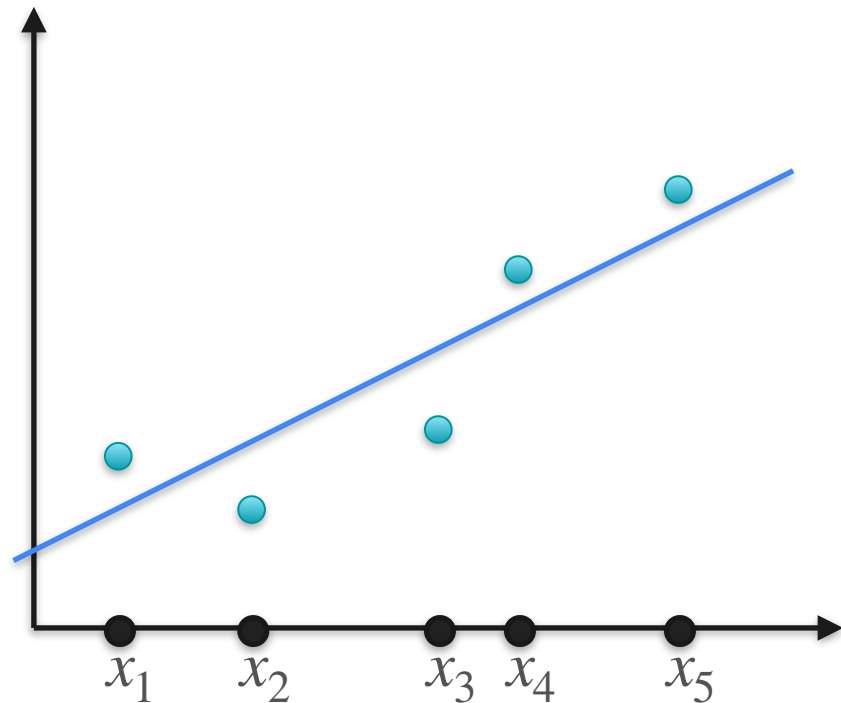
How Exactly Does a Line Produce Points?



How Exactly Does a Line Produce Points?

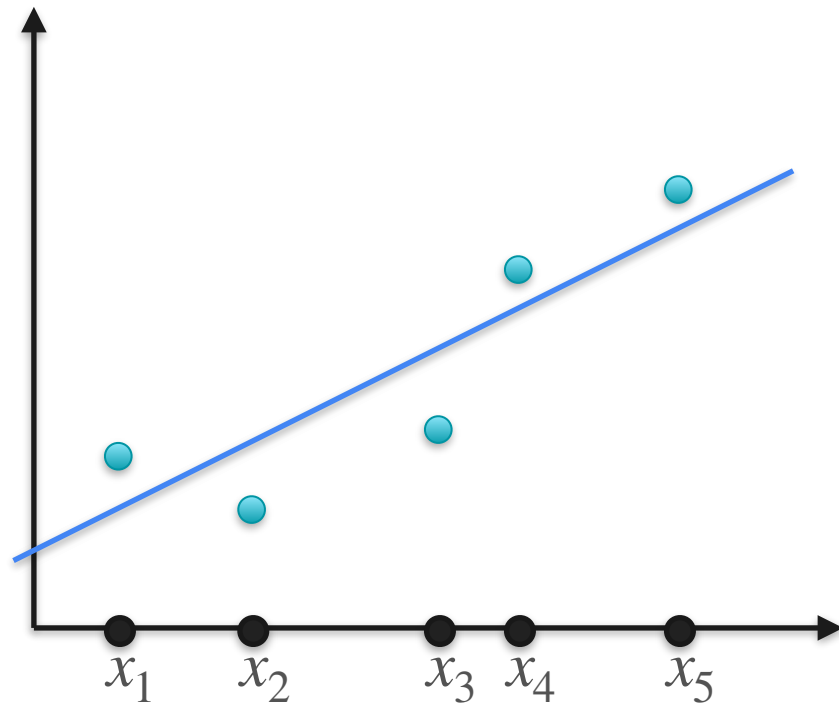


Linear Regression



Linear Regression

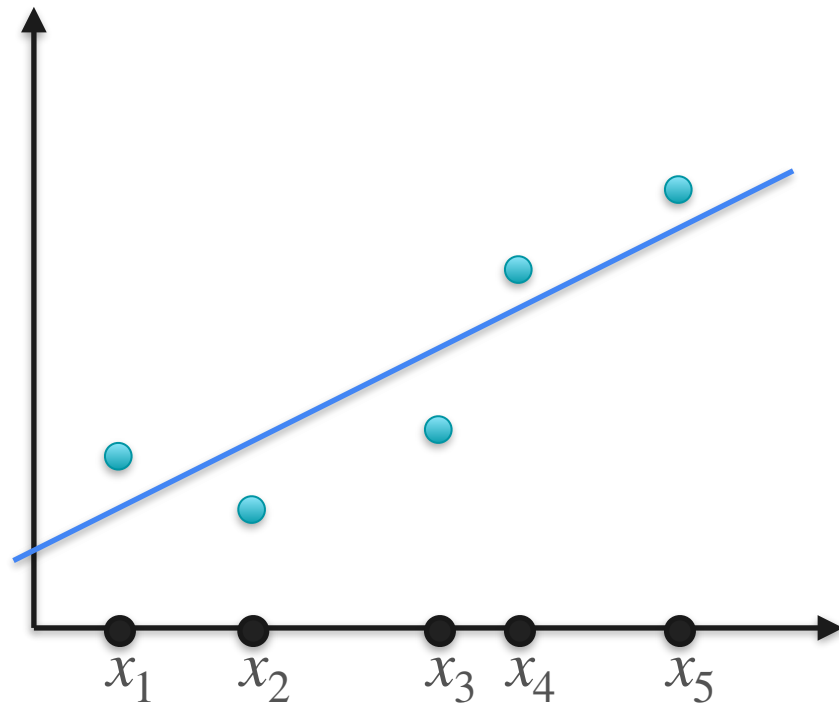
Line that best
produced the points



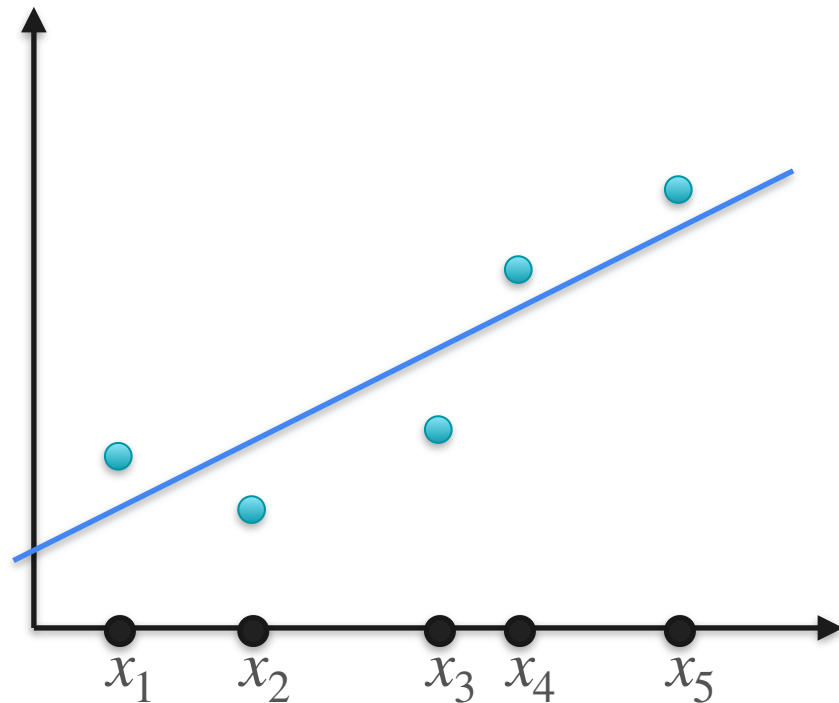
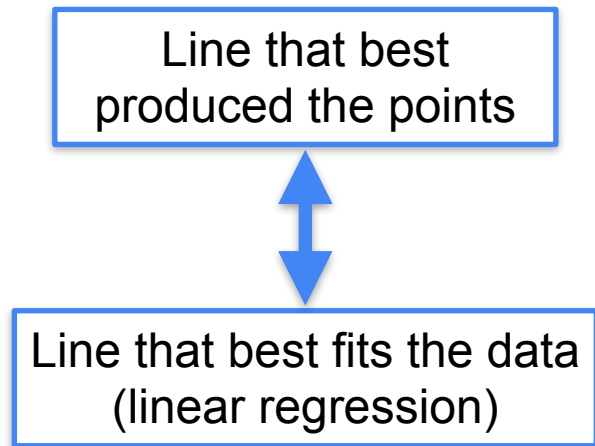
Linear Regression

Line that best
produced the points

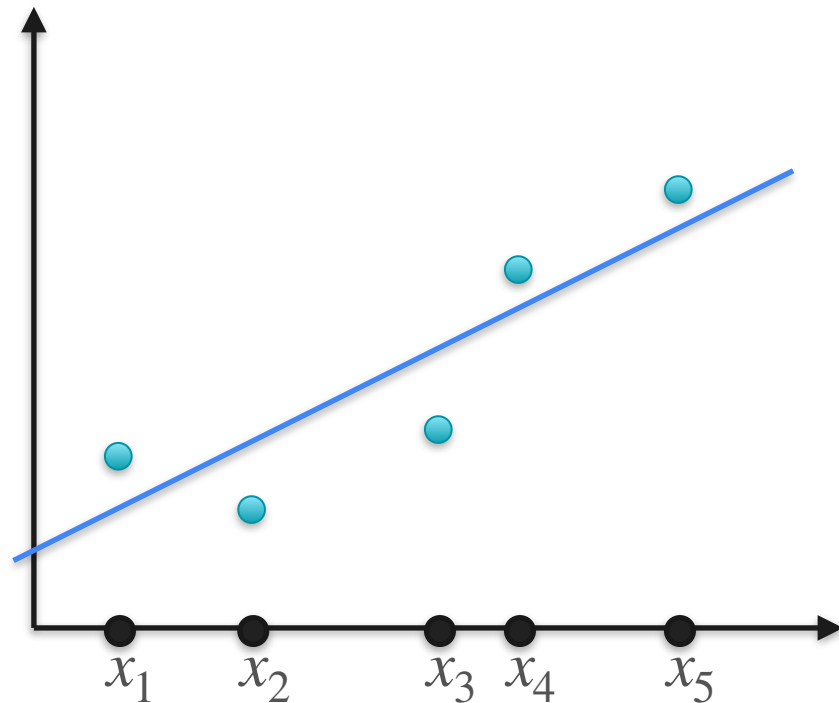
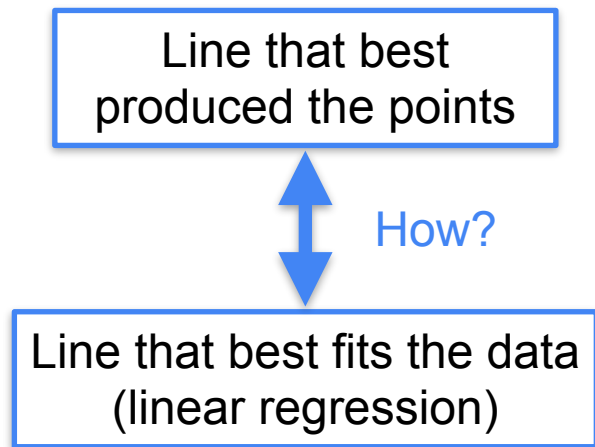
Line that best fits the data
(linear regression)



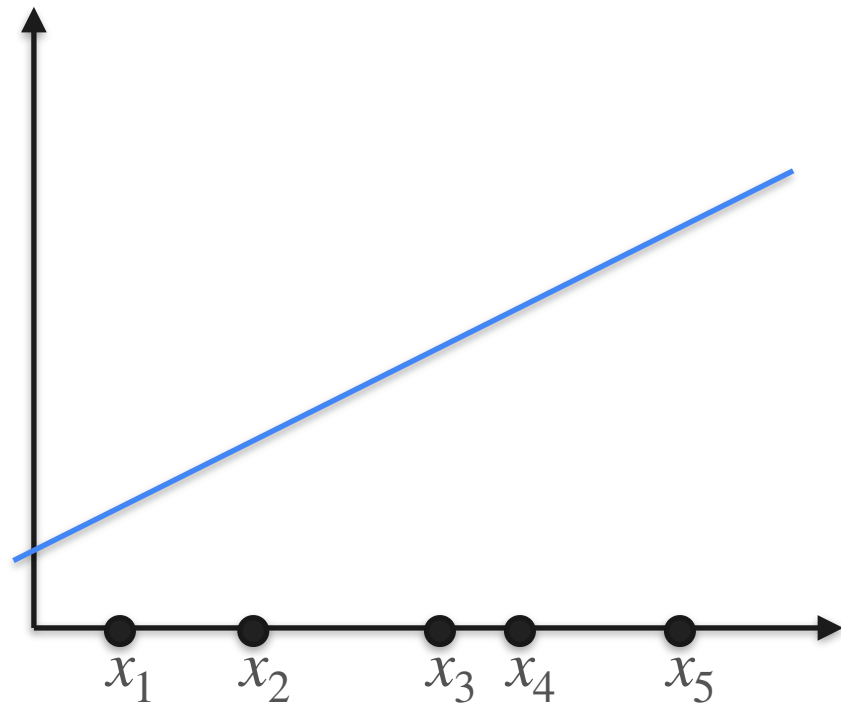
Linear Regression



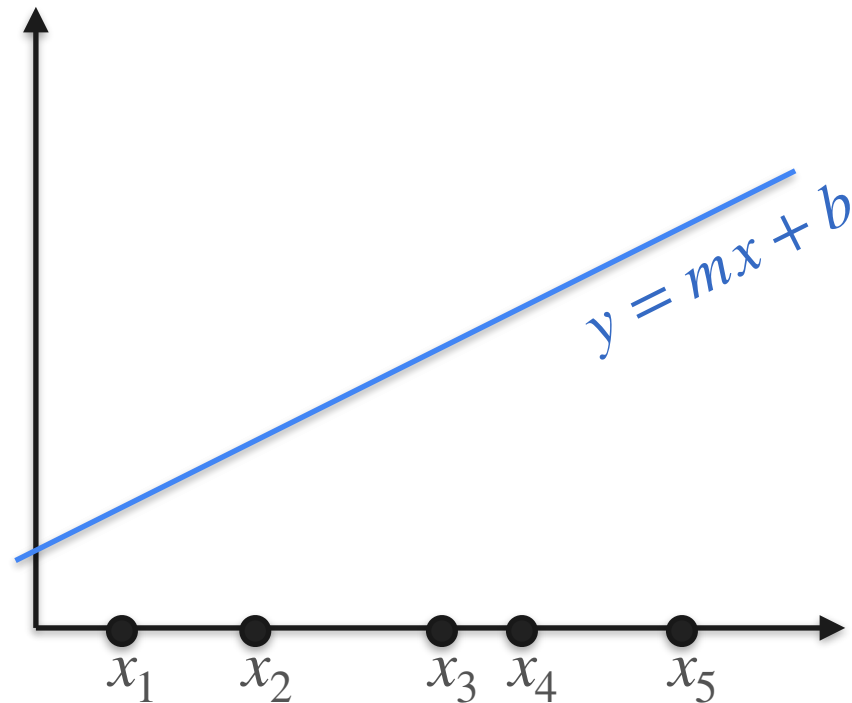
Linear Regression



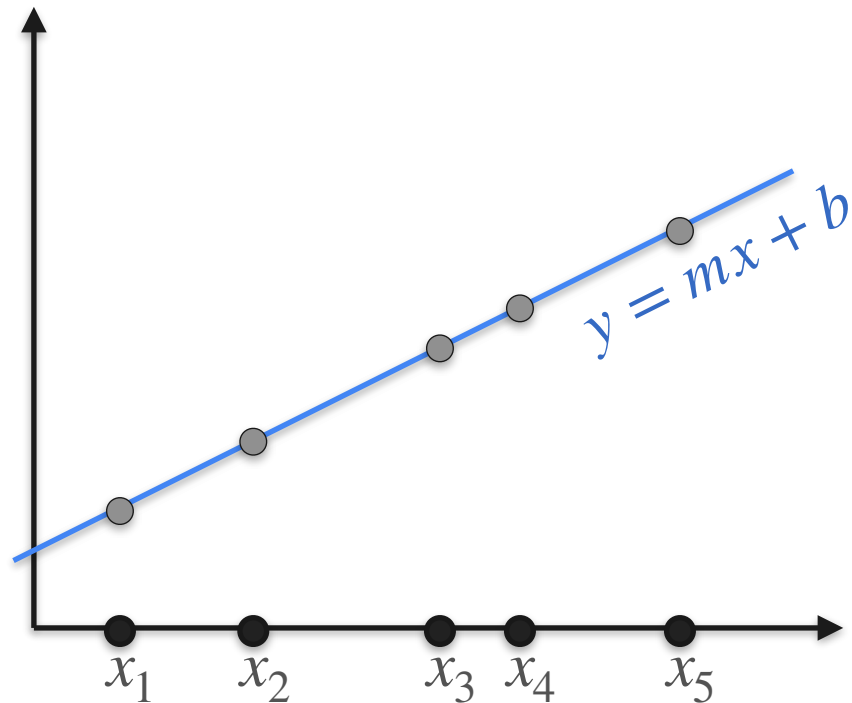
Linear Regression and Likelihood



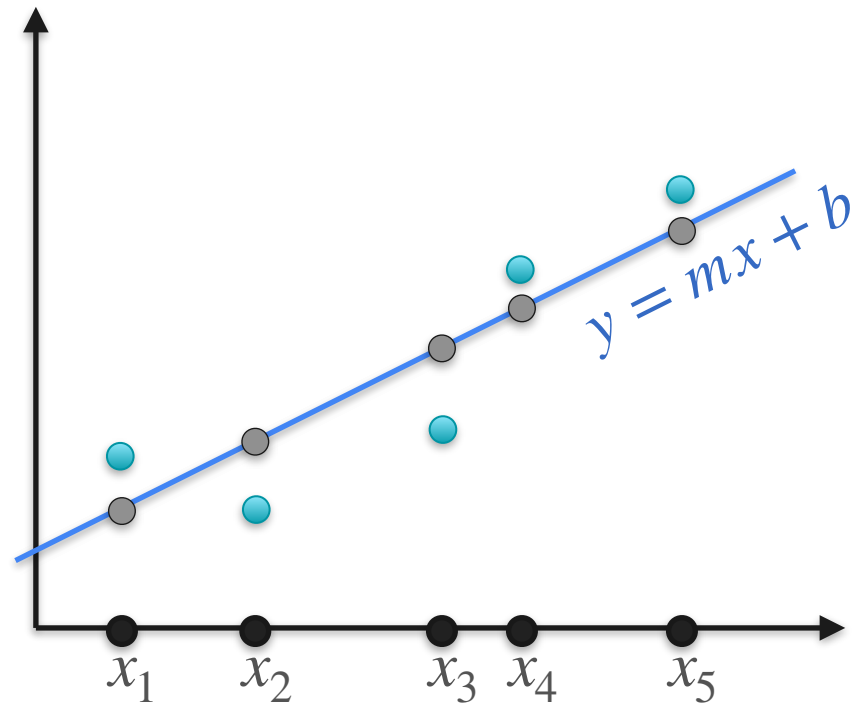
Linear Regression and Likelihood



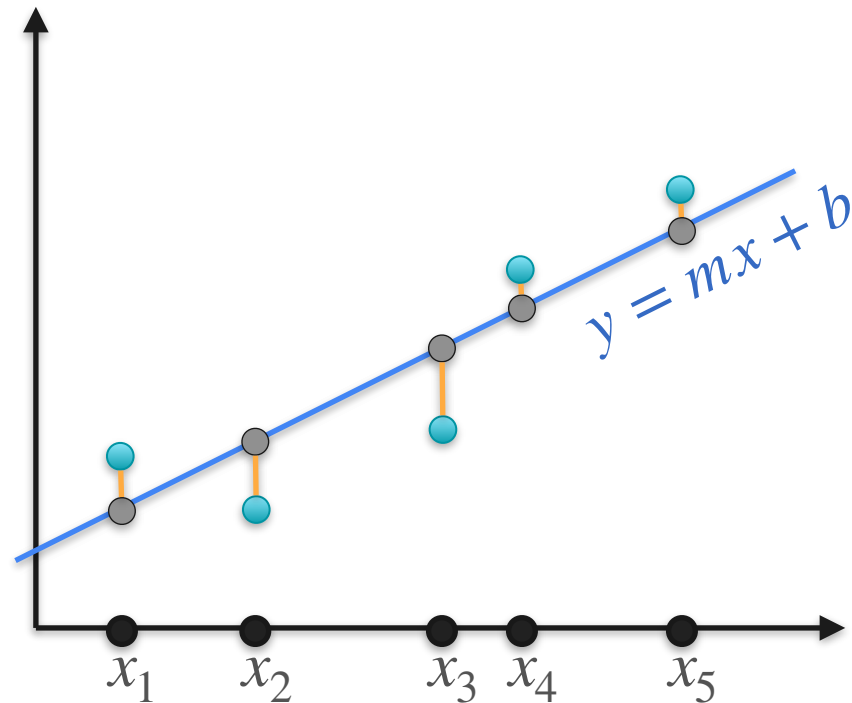
Linear Regression and Likelihood



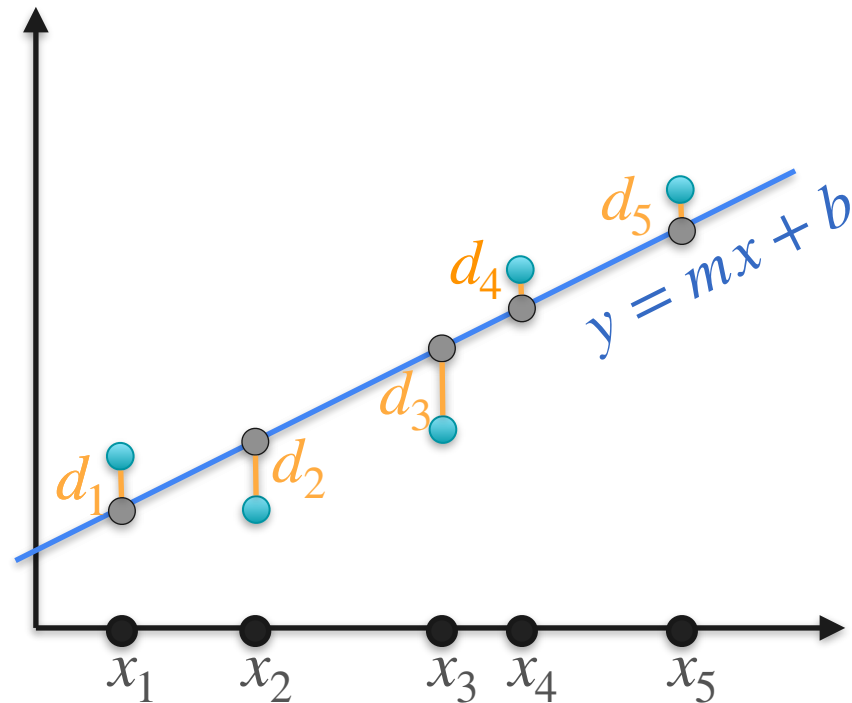
Linear Regression and Likelihood



Linear Regression and Likelihood

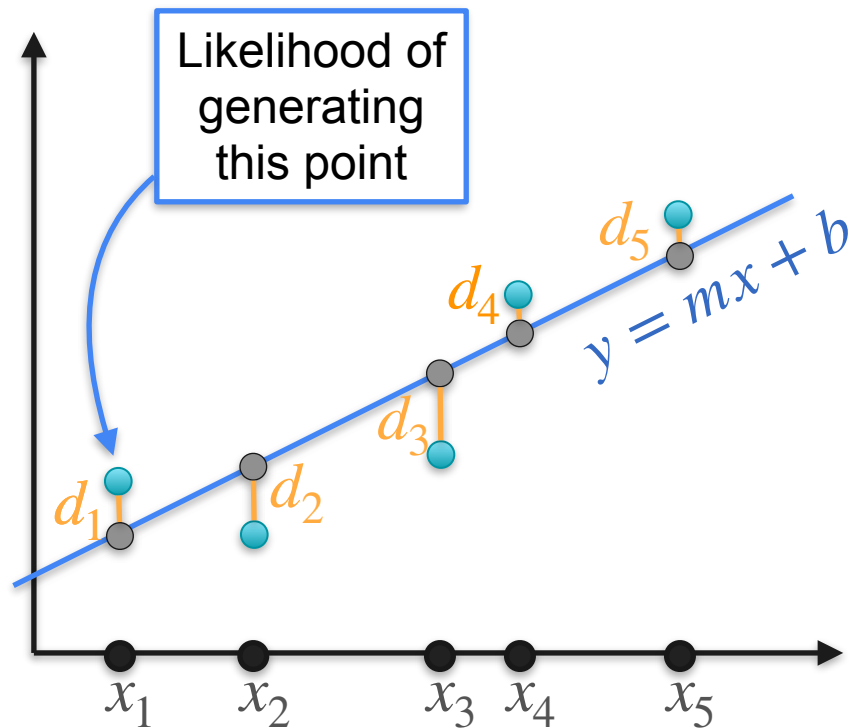


Linear Regression and Likelihood



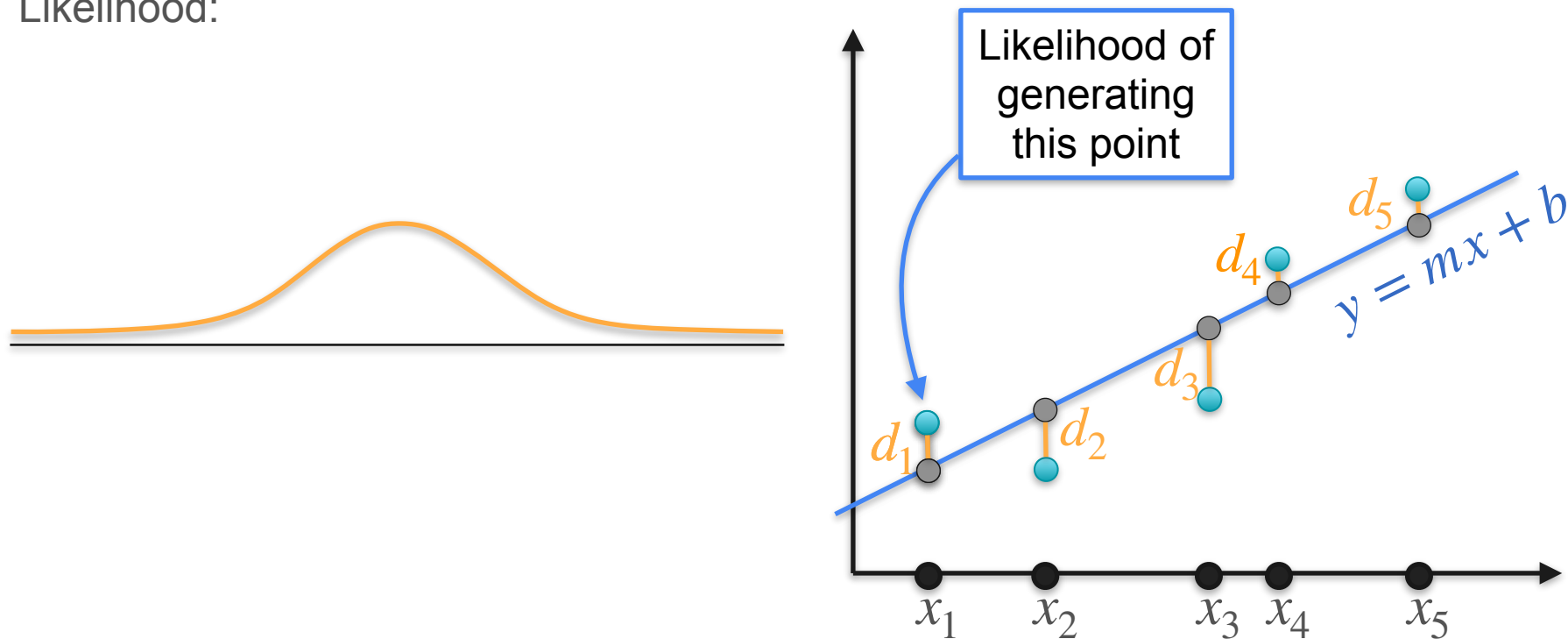
Linear Regression and Likelihood

Likelihood:



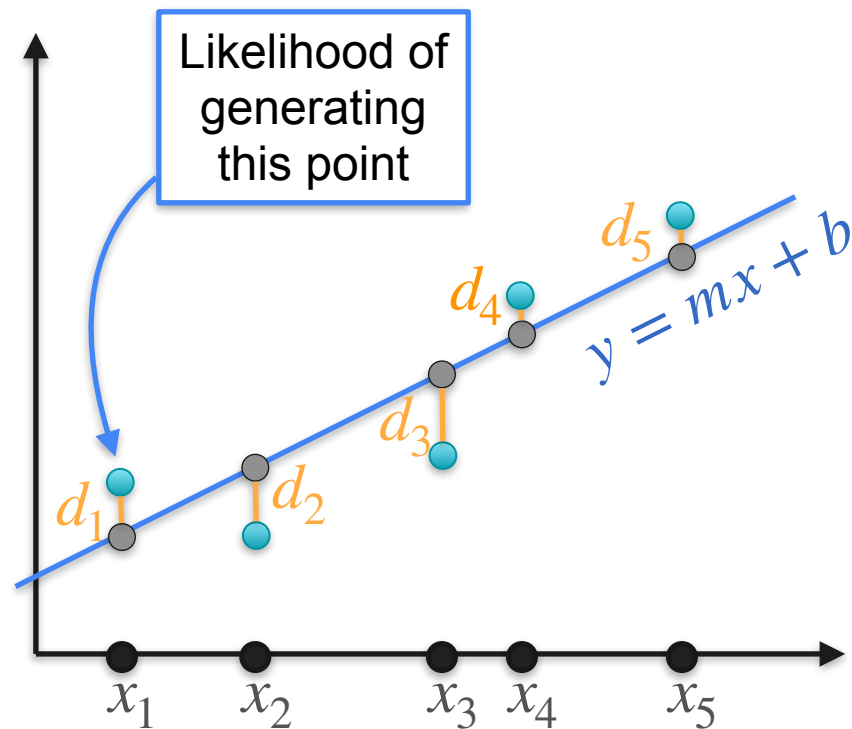
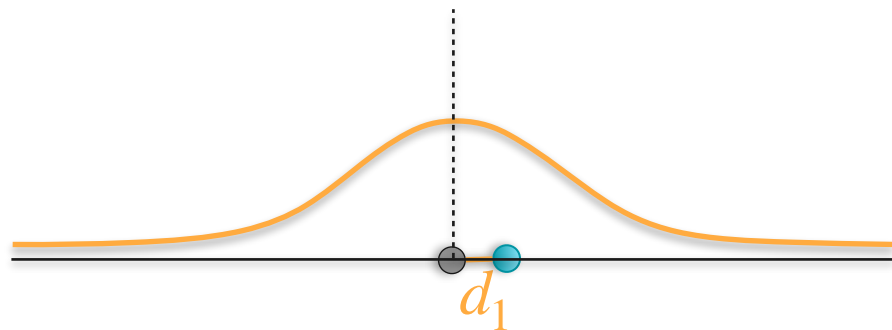
Linear Regression and Likelihood

Likelihood:



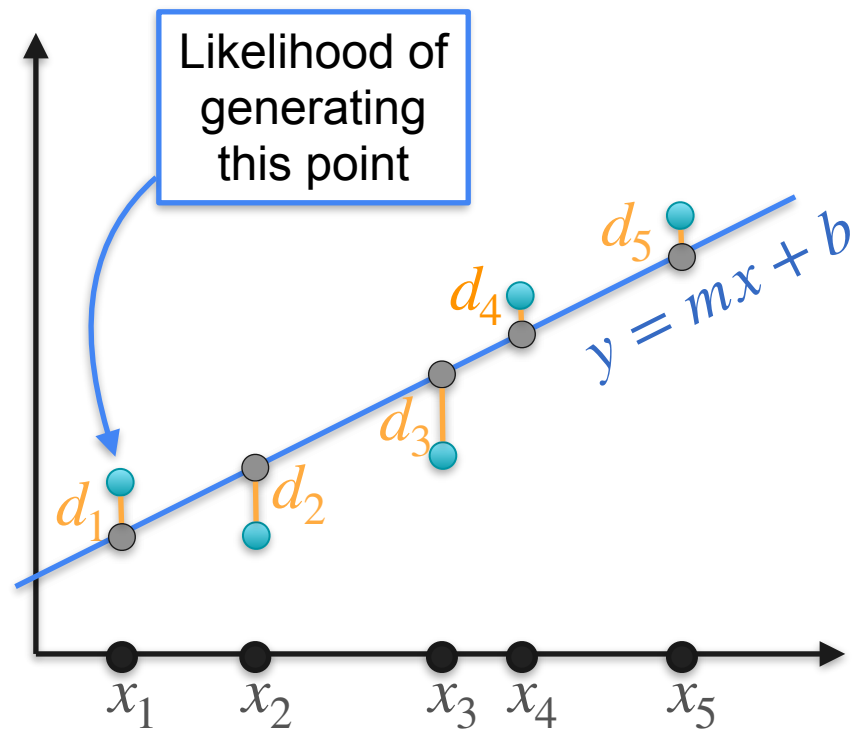
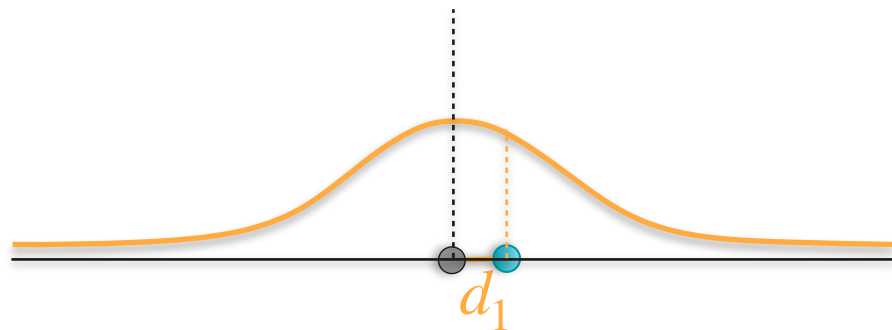
Linear Regression and Likelihood

Likelihood:



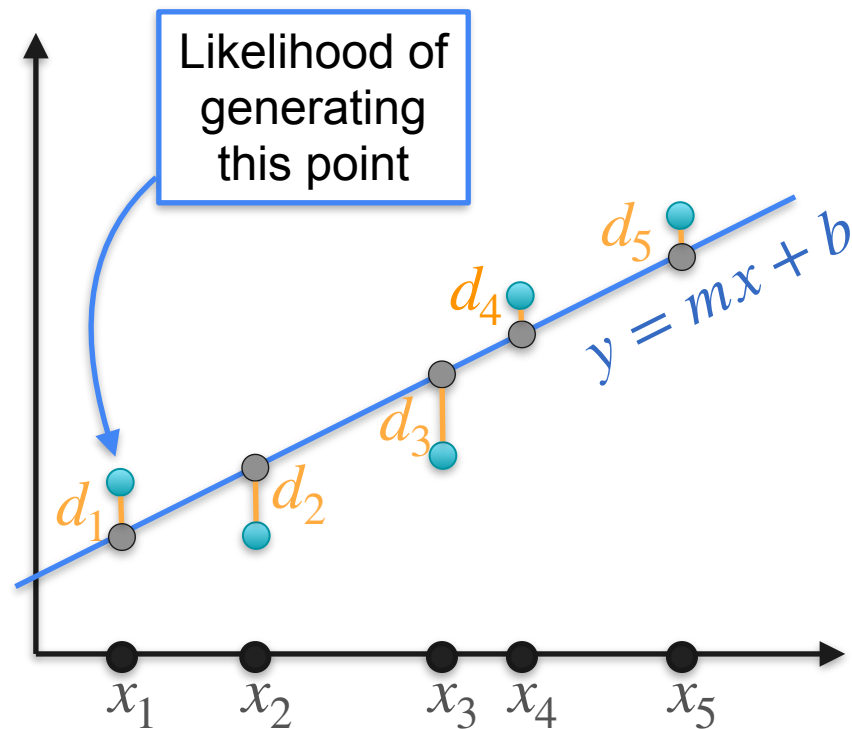
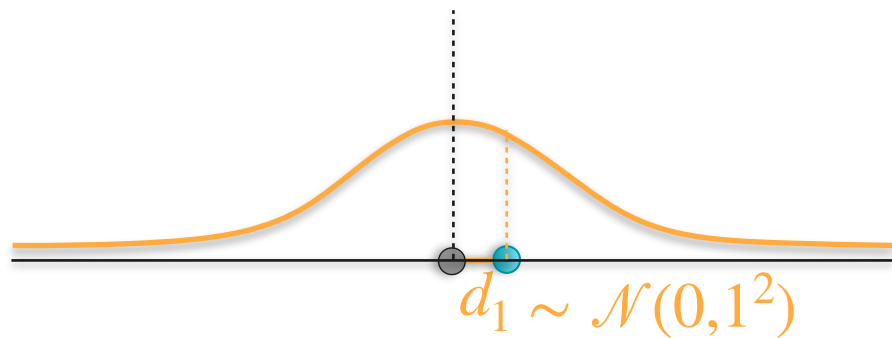
Linear Regression and Likelihood

Likelihood:



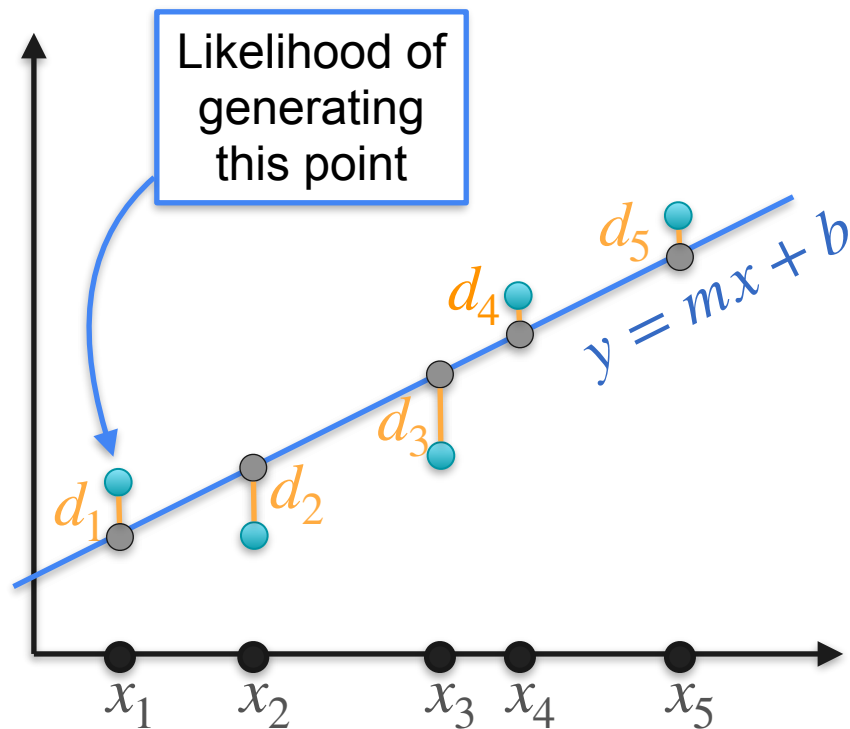
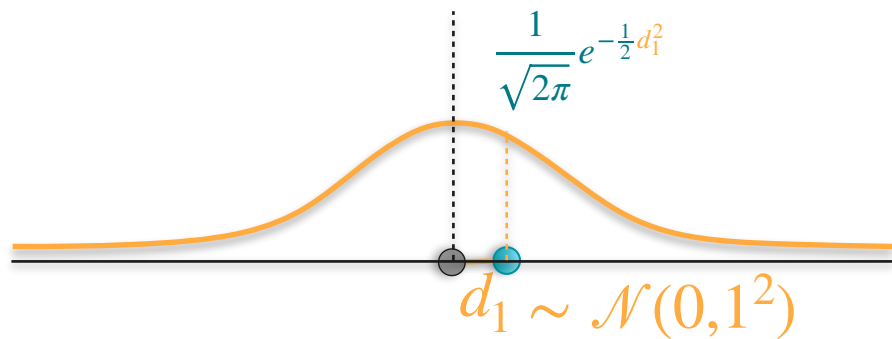
Linear Regression and Likelihood

Likelihood:



Linear Regression and Likelihood

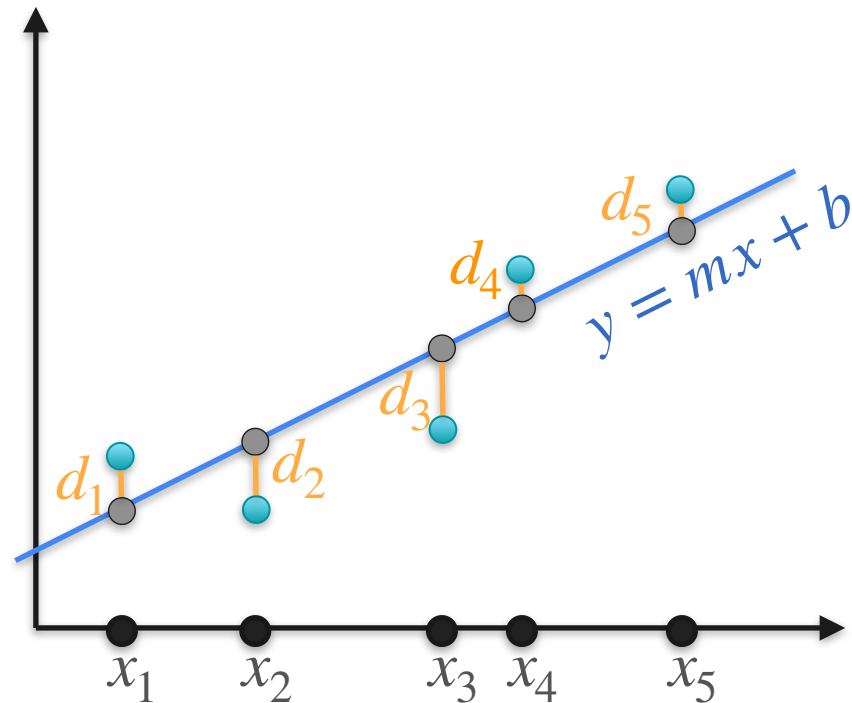
Likelihood:



Linear Regression and Likelihood

Likelihood:

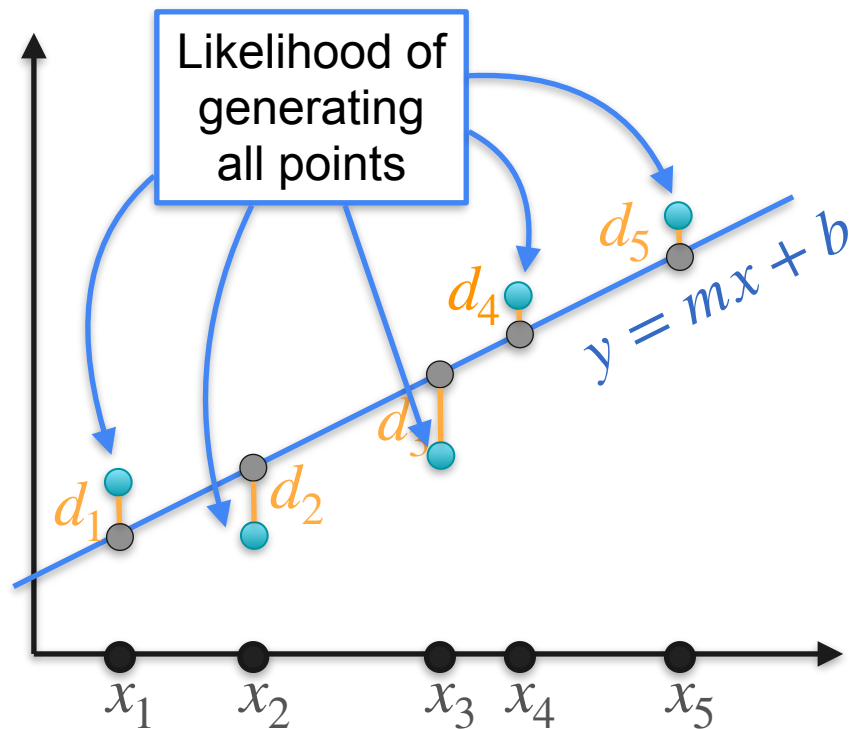
$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_i^2}$$



Linear Regression and Likelihood

Likelihood:

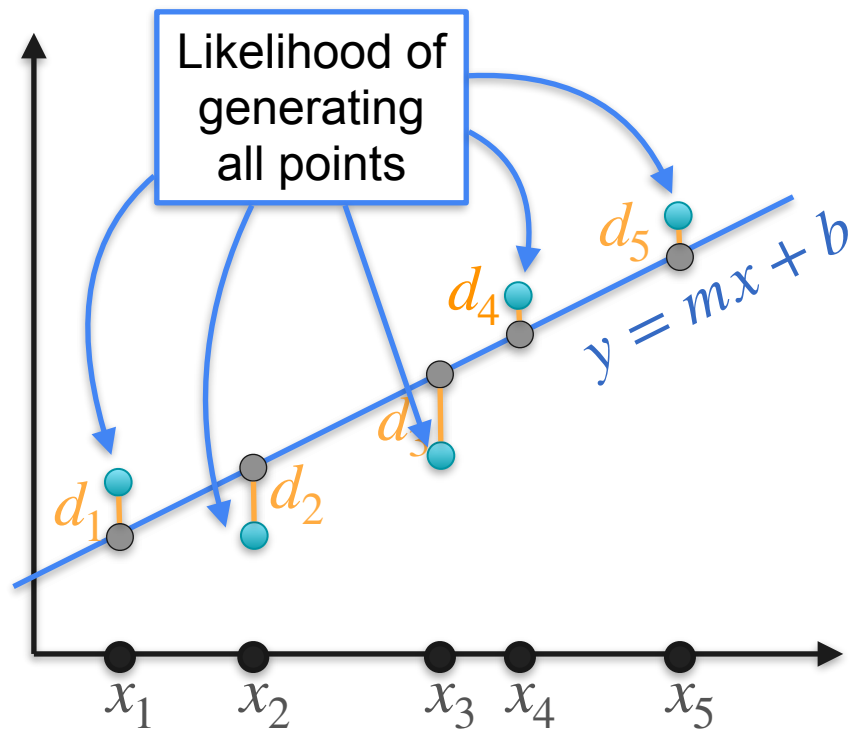
$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_i^2}$$



Linear Regression and Likelihood

Likelihood:

$$\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}d_5^2}$$

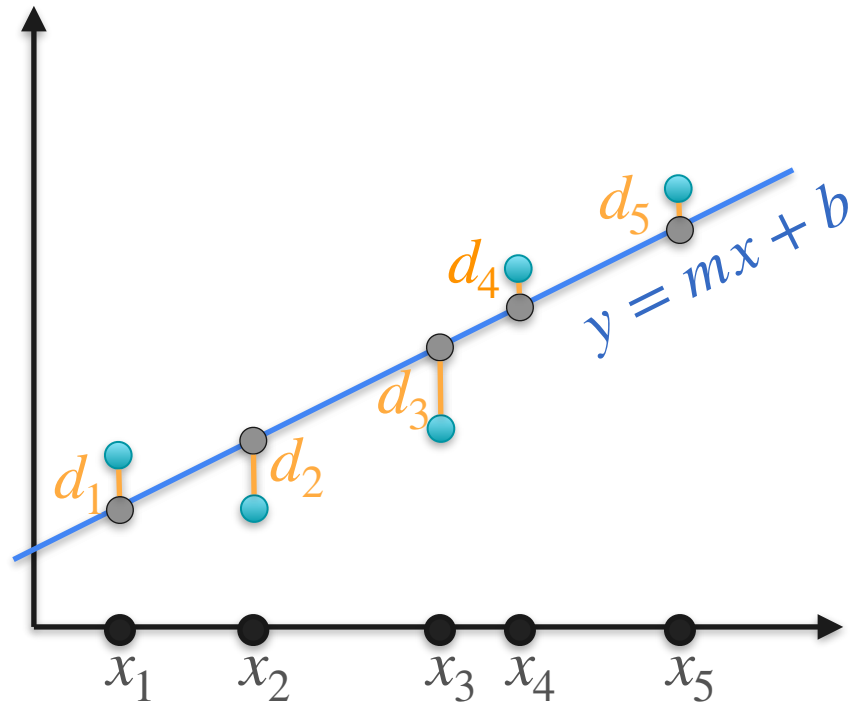


Linear Regression and Likelihood

Likelihood:

$$\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}d_5^2}$$

Maximize

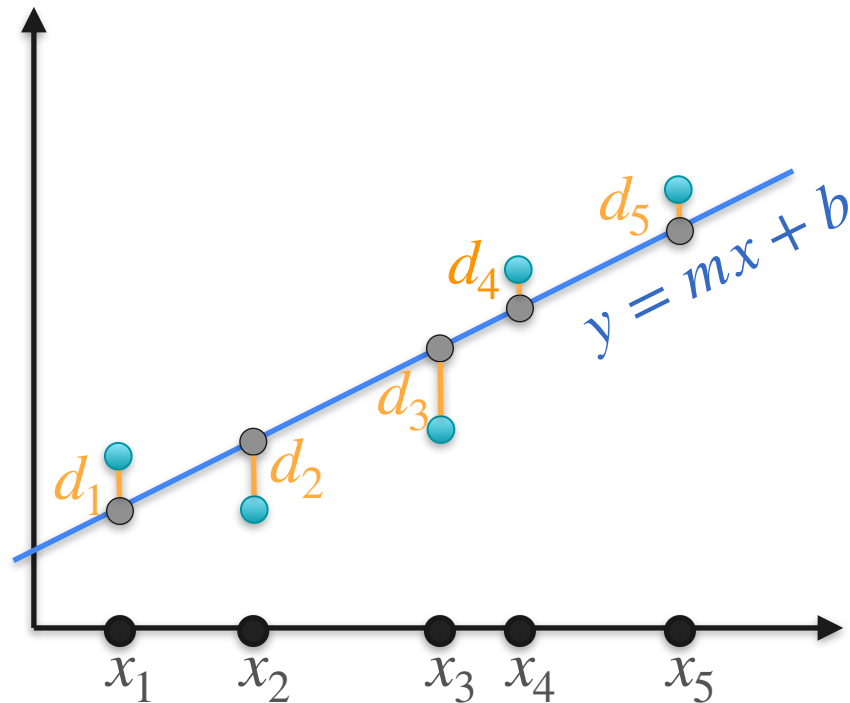


Linear Regression and Likelihood

Likelihood:

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2}$$

Maximize



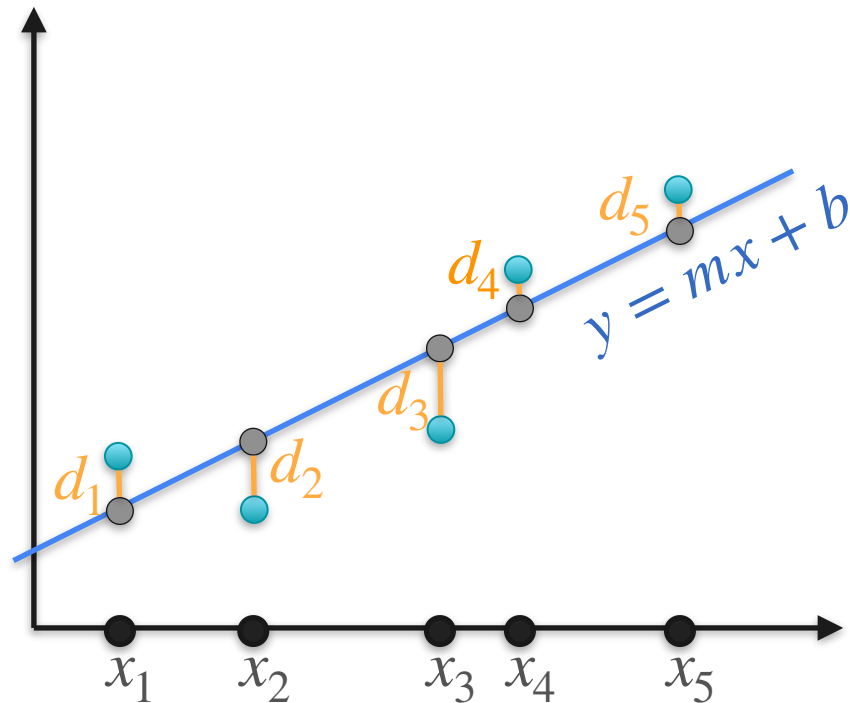
Linear Regression and Likelihood

Likelihood:

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2}$$

Maximize

$$e^{-\frac{1}{2}d_1^2} \cdot e^{-\frac{1}{2}d_2^2} \cdot e^{-\frac{1}{2}d_3^2} \cdot e^{-\frac{1}{2}d_4^2} \cdot e^{-\frac{1}{2}d_5^2}$$



Linear Regression and Likelihood

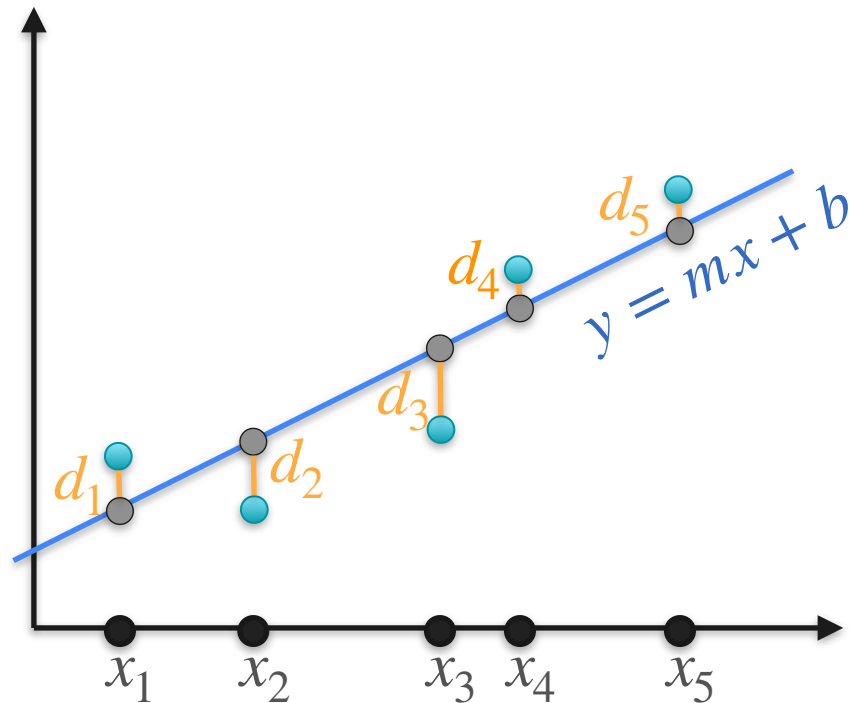
Likelihood:

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2}$$

Maximize

$$e^{-\frac{1}{2}d_1^2} \cdot e^{-\frac{1}{2}d_2^2} \cdot e^{-\frac{1}{2}d_3^2} \cdot e^{-\frac{1}{2}d_4^2} \cdot e^{-\frac{1}{2}d_5^2}$$

$$e^{-\frac{1}{2}(d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2)}$$



Linear Regression and Likelihood

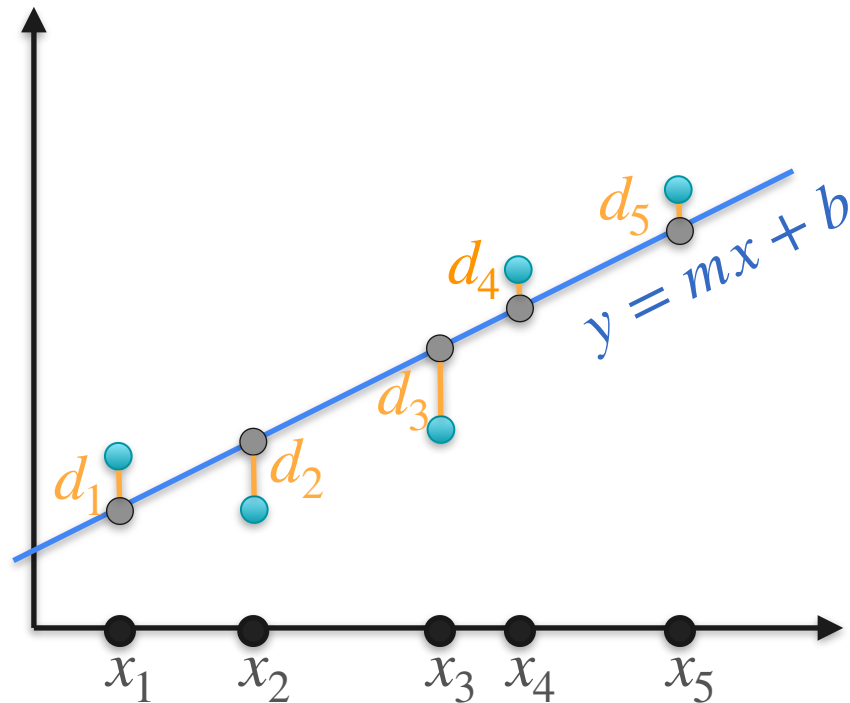
Likelihood:

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2}$$

Maximize

$$e^{-\frac{1}{2}d_1^2} \cdot e^{-\frac{1}{2}d_2^2} \cdot e^{-\frac{1}{2}d_3^2} \cdot e^{-\frac{1}{2}d_4^2} \cdot e^{-\frac{1}{2}d_5^2}$$

$$-\frac{1}{2}(d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2)$$



Linear Regression and Likelihood

Likelihood:

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2}$$

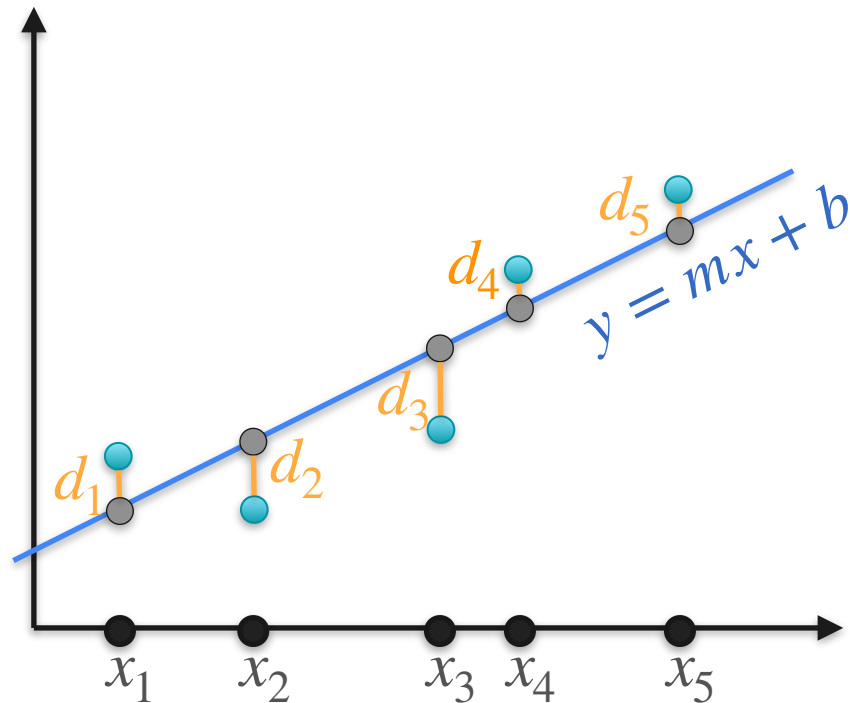
Maximize

$$e^{-\frac{1}{2}d_1^2} \cdot e^{-\frac{1}{2}d_2^2} \cdot e^{-\frac{1}{2}d_3^2} \cdot e^{-\frac{1}{2}d_4^2} \cdot e^{-\frac{1}{2}d_5^2}$$

$$\frac{1}{\sqrt{2\pi}} (d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2)$$

Minimize

$$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$$



Linear Regression and Likelihood

Likelihood:

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2}$$

Maximize

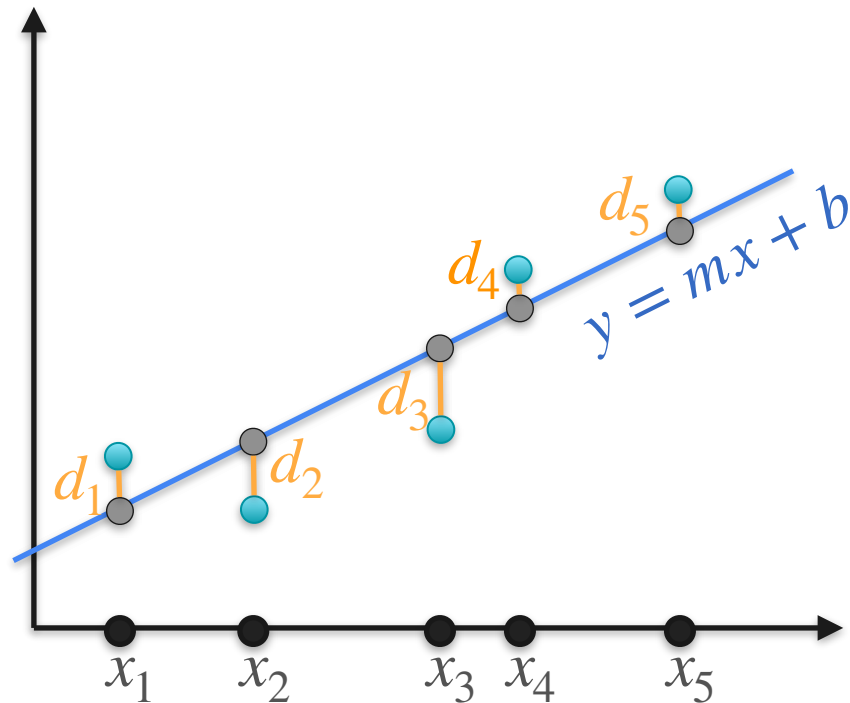
$$e^{-\frac{1}{2}d_1^2} \cdot e^{-\frac{1}{2}d_2^2} \cdot e^{-\frac{1}{2}d_3^2} \cdot e^{-\frac{1}{2}d_4^2} \cdot e^{-\frac{1}{2}d_5^2}$$

$$\frac{1}{2} (d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2)$$

Minimize

$$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$$

Least squares error!



Linear Regression and Likelihood

Likelihood:

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2}$$

Maximize

$$e^{-\frac{1}{2}d_1^2} \cdot e^{-\frac{1}{2}d_2^2} \cdot e^{-\frac{1}{2}d_3^2} \cdot e^{-\frac{1}{2}d_4^2} \cdot e^{-\frac{1}{2}d_5^2}$$

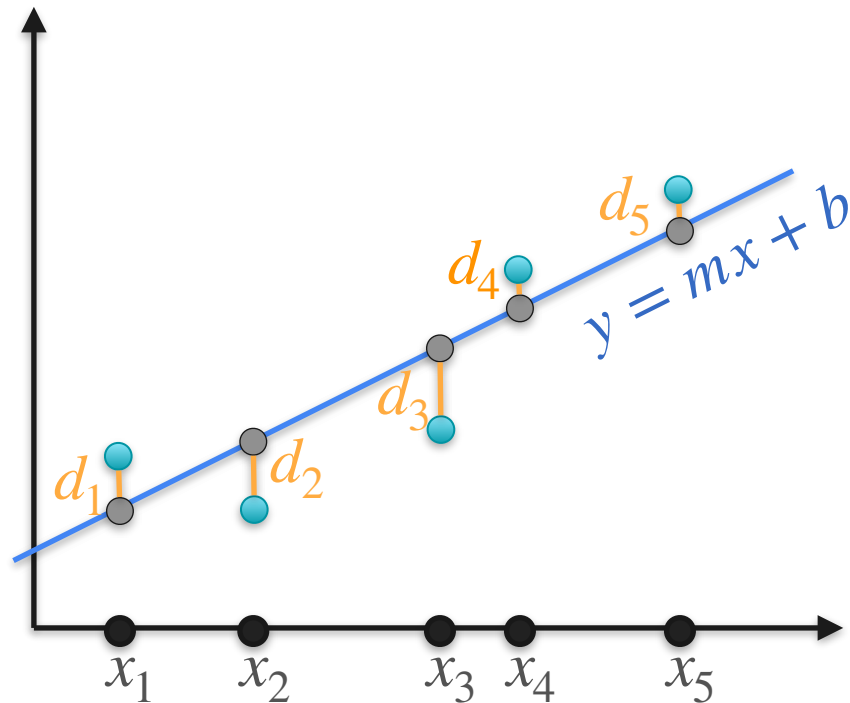
$$\frac{1}{2} (d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2)$$

Minimize

$$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$$

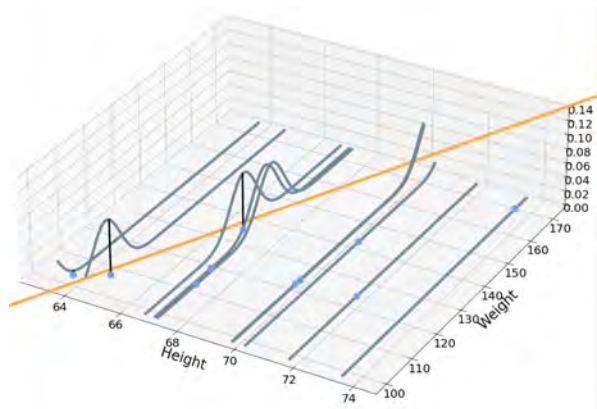
Least squares error!

Linear regression!

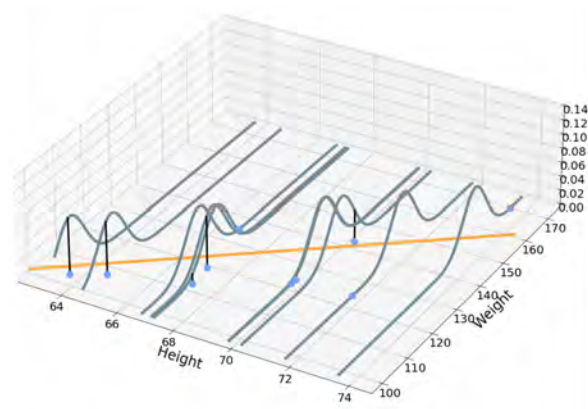
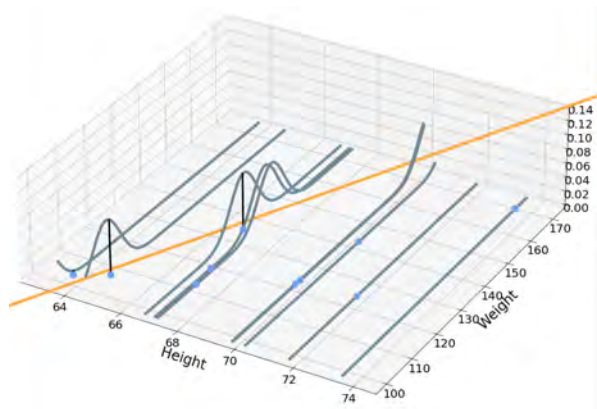


Picking the Right Model

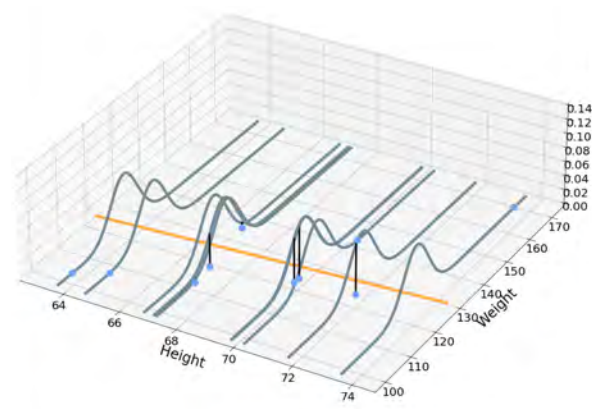
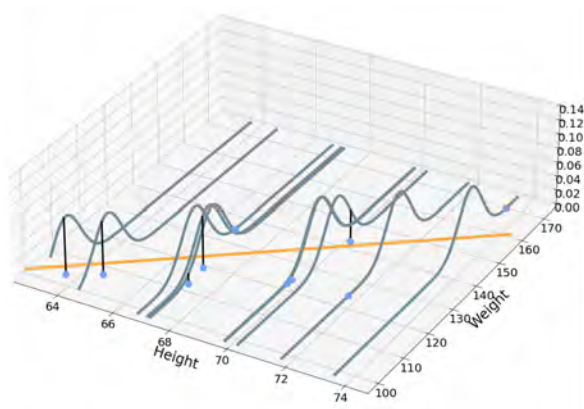
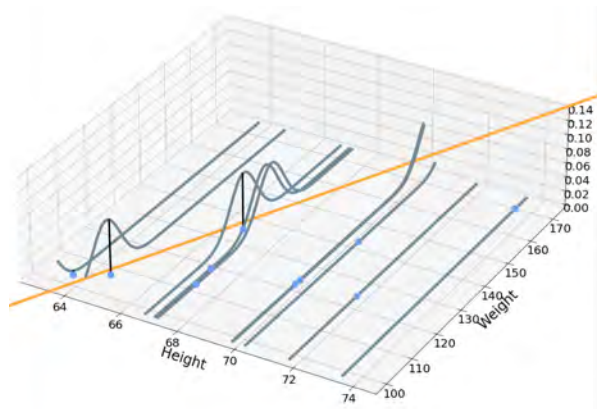
Picking the Right Model



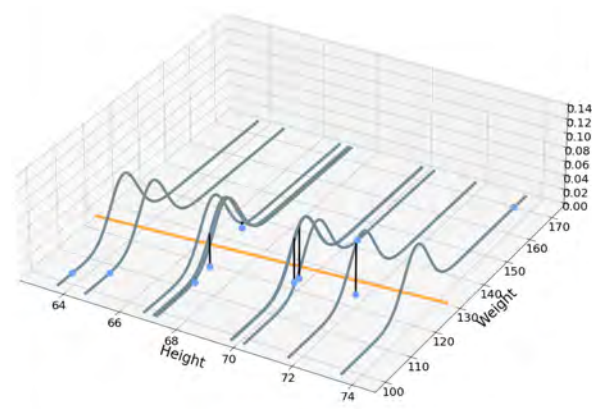
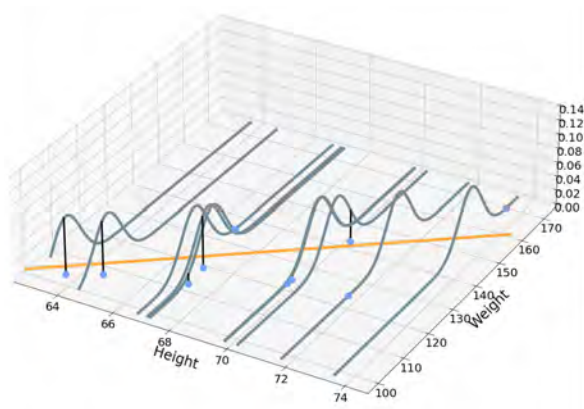
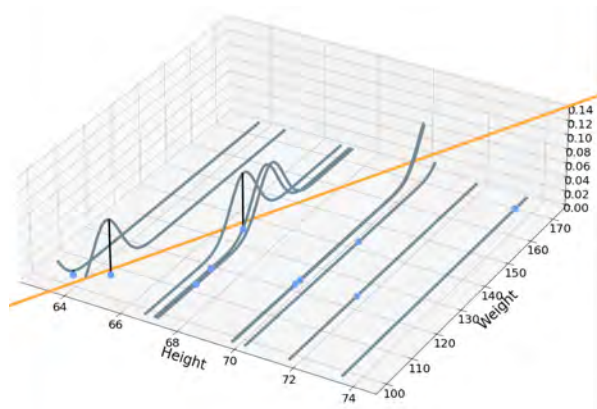
Picking the Right Model



Picking the Right Model



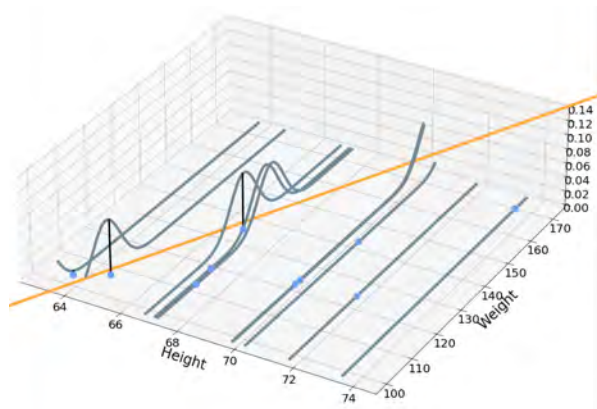
Picking the Right Model



Model 1:

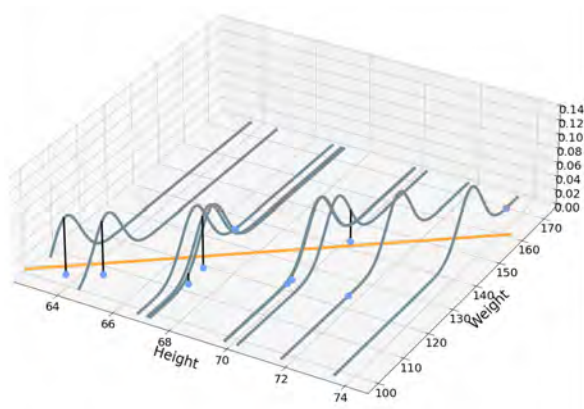
Likelihood = $4.91 \cdot 10^{-260}$

Picking the Right Model



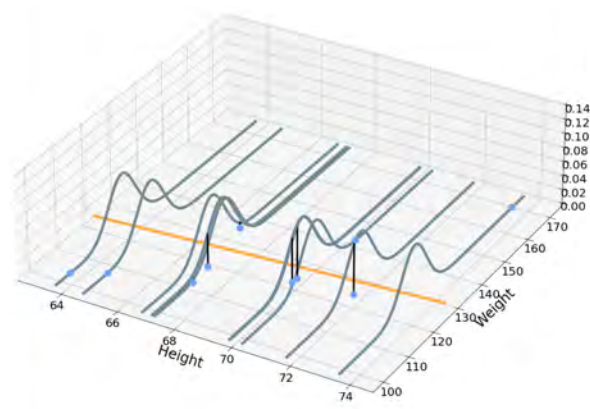
Model 1:

Likelihood = $4.91 \cdot 10^{-260}$

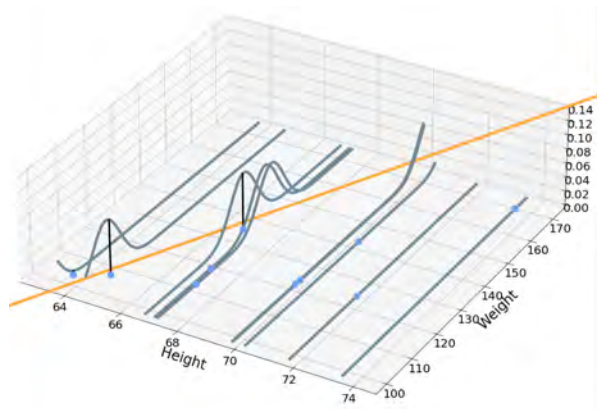


Model 2:

Likelihood = $8.16 \cdot 10^{-28}$

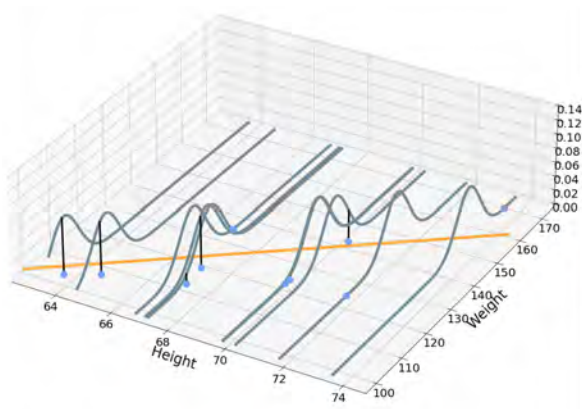


Picking the Right Model



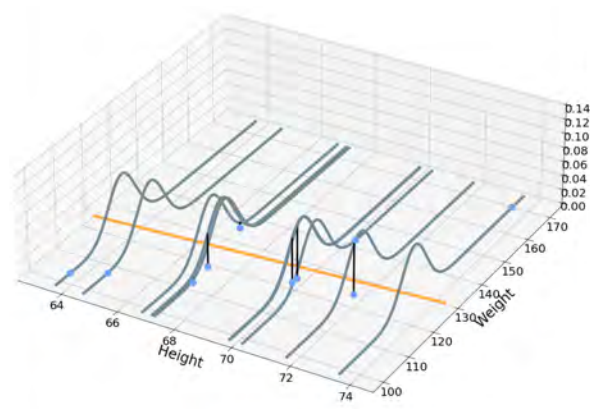
Model 1:

Likelihood = $4.91 \cdot 10^{-260}$



Model 2:

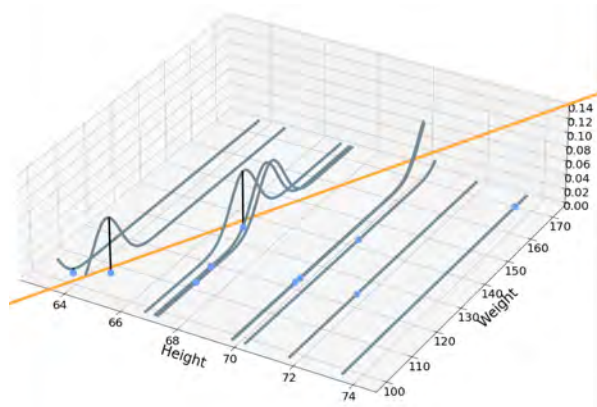
Likelihood = $8.16 \cdot 10^{-28}$



Model 3:

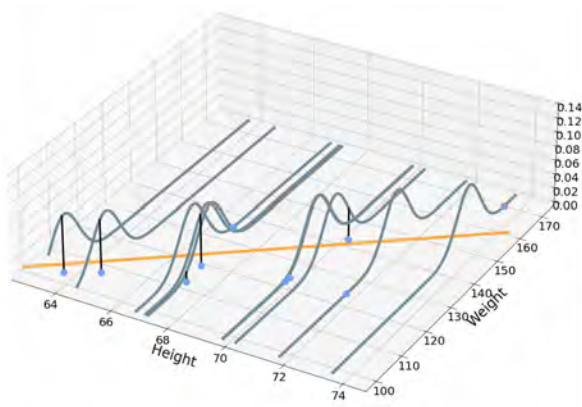
Likelihood = $3.48 \cdot 10^{-49}$

Picking the Right Model



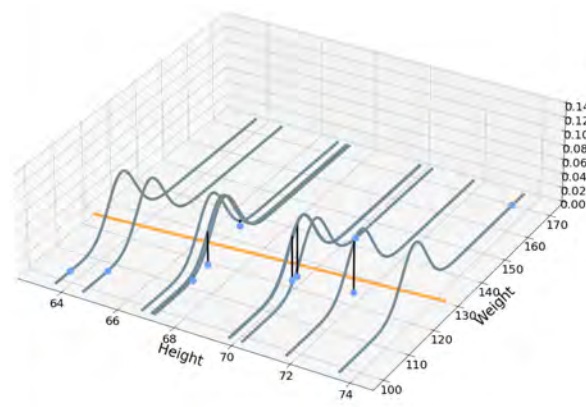
Model 1:

Likelihood = $4.91 \cdot 10^{-260}$



Model 2:

Likelihood = $8.16 \cdot 10^{-28}$



Model 3:

Likelihood = $3.48 \cdot 10^{-49}$



DeepLearning.AI

Point Estimation

Frequentist vs Bayesian Statistics

Frequentist Vs. Bayesian Statistics

Frequentist Vs. Bayesian Statistics

Frequentists

Bayesians

Frequentist Vs. Bayesian Statistics

Frequentists

- Probabilities represent long term frequency of events

Bayesians

Frequentist Vs. Bayesian Statistics

Frequentists

- Probabilities represent long term frequency of events

Bayesians

- Probabilities represent the degree of belief (or certainty)

Frequentist Vs. Bayesian Statistics

Frequentists

- Probabilities represent long term frequency of events
- Concept of Likelihood

Bayesians

- Probabilities represent the degree of belief (or certainty)

Frequentist Vs. Bayesian Statistics

Frequentists

- Probabilities represent long term frequency of events
- Concept of Likelihood

Bayesians

- Probabilities represent the degree of belief (or certainty)
- Concept of Prior

Frequentist Vs. Bayesian Statistics

Frequentists

- Probabilities represent long term frequency of events
- Concept of Likelihood

Bayesians

- Probabilities represent the degree of belief (or certainty)
- Concept of Prior
- Goal: update prior belief based on observations

Frequentist Vs. Bayesian Statistics

Frequentists

- Probabilities represent long term frequency of events
- Concept of Likelihood
- Goal: Find the model that most likely generated the observed data

Bayesians

- Probabilities represent the degree of belief (or certainty)
- Concept of Prior
- Goal: update prior belief based on observations

Frequentist Approach

Frequentist Approach



Frequentist Approach



p = probability of heads

Frequentist Approach



p = probability of heads

$$p = 0.8$$

Frequentist Approach



p = probability of heads

$$p = 0.8$$

Maximum likelihood

Bayesian Approach

p = probability of heads



Bayesian Approach

p = probability of heads



p could be anything between 0 and 1

Bayesian Approach

p = probability of heads



p could be anything between 0 and 1

p could still be anything between 0 and 1, but it may be closer to 1

Bayesian Approach

p = probability of heads



p could be anything between 0 and 1

p could still be anything between 0 and 1, but it may be closer to 1

p could still be anything between 0 and 1, but it's probably around 0.5

Bayesian Approach

p = probability of heads



p could be anything between 0 and 1

p could still be anything between 0 and 1, but it may be closer to 1

p could still be anything between 0 and 1, but it's probably around 0.5

p could still be anything between 0 and 1, but it's more likely closer to 0.66

Bayesian Approach

p = probability of heads



p could be anything between 0 and 1

p could still be anything between 0 and 1, but it may be closer to 1

p could still be anything between 0 and 1, but it's probably around 0.5

p could still be anything between 0 and 1, but it's more likely closer to 0.66

...

Bayesian Approach

p = probability of heads



p could be anything between 0 and 1

p could still be anything between 0 and 1, but it may be closer to 1

p could still be anything between 0 and 1, but it's probably around 0.5

p could still be anything between 0 and 1, but it's more likely closer to 0.66

...

p could still be anything between 0 and 1, but it's very probably close to 0.8

Bayesian Approach

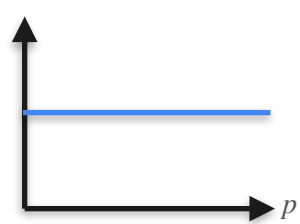


p = probability of heads

Bayesian Approach



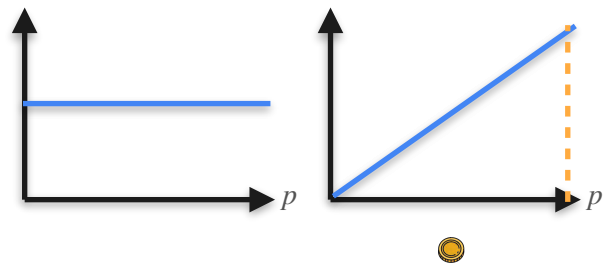
p = probability of heads



Bayesian Approach



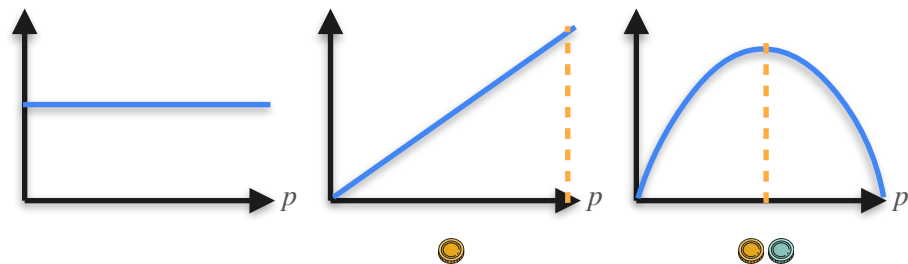
p = probability of heads



Bayesian Approach



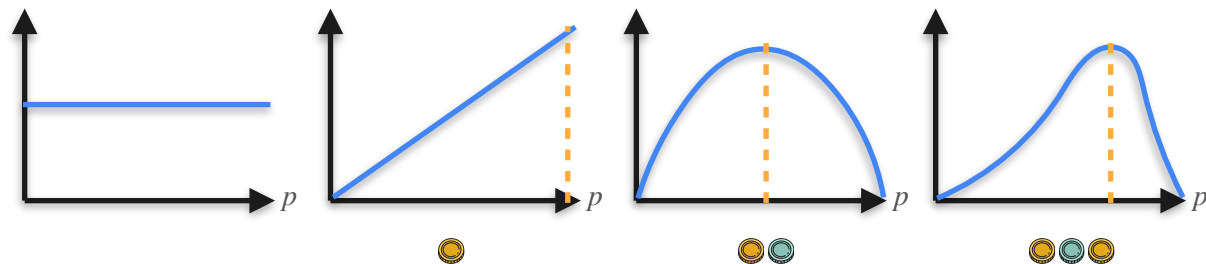
p = probability of heads



Bayesian Approach



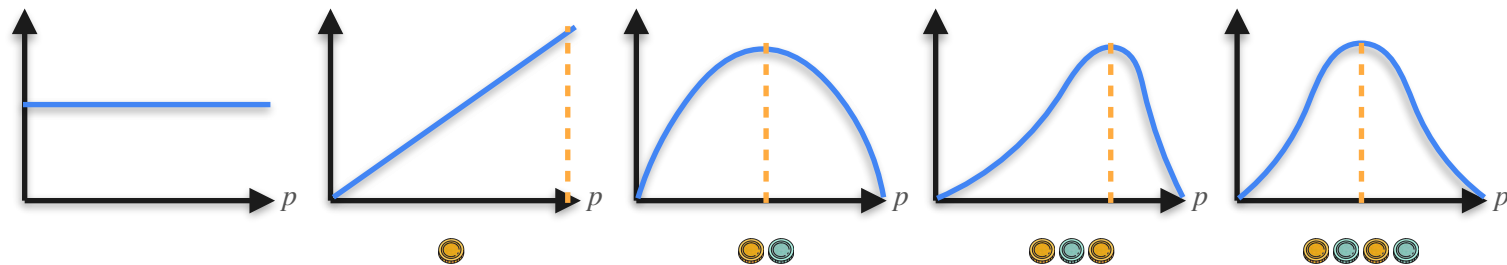
p = probability of heads



Bayesian Approach



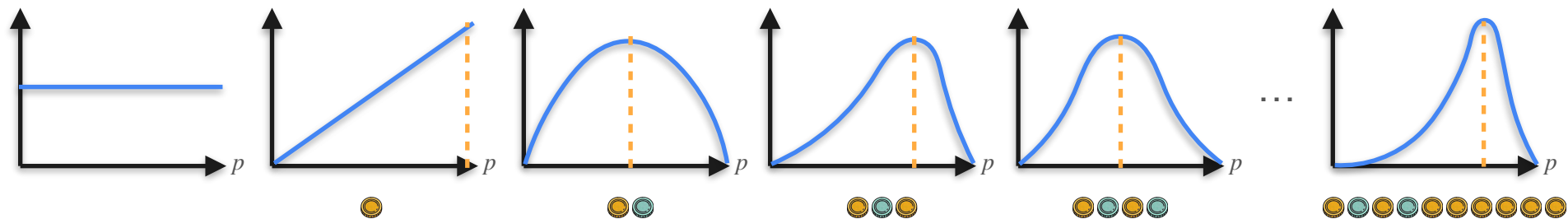
p = probability of heads



Bayesian Approach



p = probability of heads



Bayesian Statistics

Bayesian Statistics

Remember Bayes' theorem? $\mathbf{P}(B | A) = \frac{\mathbf{P}(A | B)\mathbf{P}(B)}{\mathbf{P}(A)}$

Bayesian Statistics

Remember Bayes' theorem? $\mathbf{P(B|A)} = \frac{\mathbf{P(A|B)P(B)}}{\mathbf{P(A)}}$

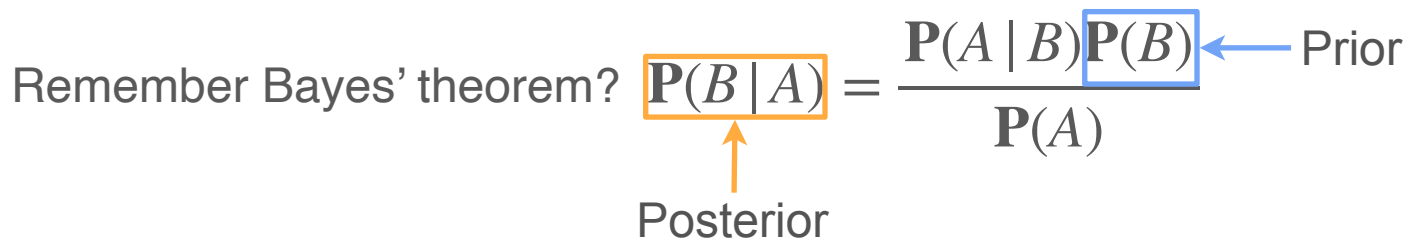
↑
Posterior

Bayesian Statistics

Remember Bayes' theorem? $\mathbf{P(B|A)} = \frac{\mathbf{P(A|B)P(B)}}{\mathbf{P(A)}}$

Posterior

Prior



Bayesian Statistics

Remember Bayes' theorem? $\mathbf{P(B | A)}$ = $\frac{\mathbf{P(A | B)P(B)}}{\mathbf{P(A)}}$

Posterior

Normalizing constant

Prior

The diagram illustrates Bayes' theorem with the equation $P(B | A) = \frac{P(A | B)P(B)}{P(A)}$. The term $P(B | A)$ is enclosed in an orange box, with an orange arrow pointing to it from the label 'Posterior' below. The term $P(A)$ is enclosed in a green box, with a green arrow pointing to it from the label 'Normalizing constant' below. The term $P(B)$ is enclosed in a blue box, with a blue arrow pointing to it from the label 'Prior' to its right. The text 'Remember Bayes' theorem?' is positioned to the left of the equation.

Bayesian Statistics

The parameters
(or model) you
want to estimate

Remember Bayes' theorem? $\mathbf{P(B | A)}$ = $\frac{\mathbf{P(A | B)P(B)}}{\mathbf{P(A)}}$

Posterior

Normalizing constant

Prior

Bayesian Statistics

The parameters (or model) you want to estimate

Your sample vector

Remember Bayes' theorem?

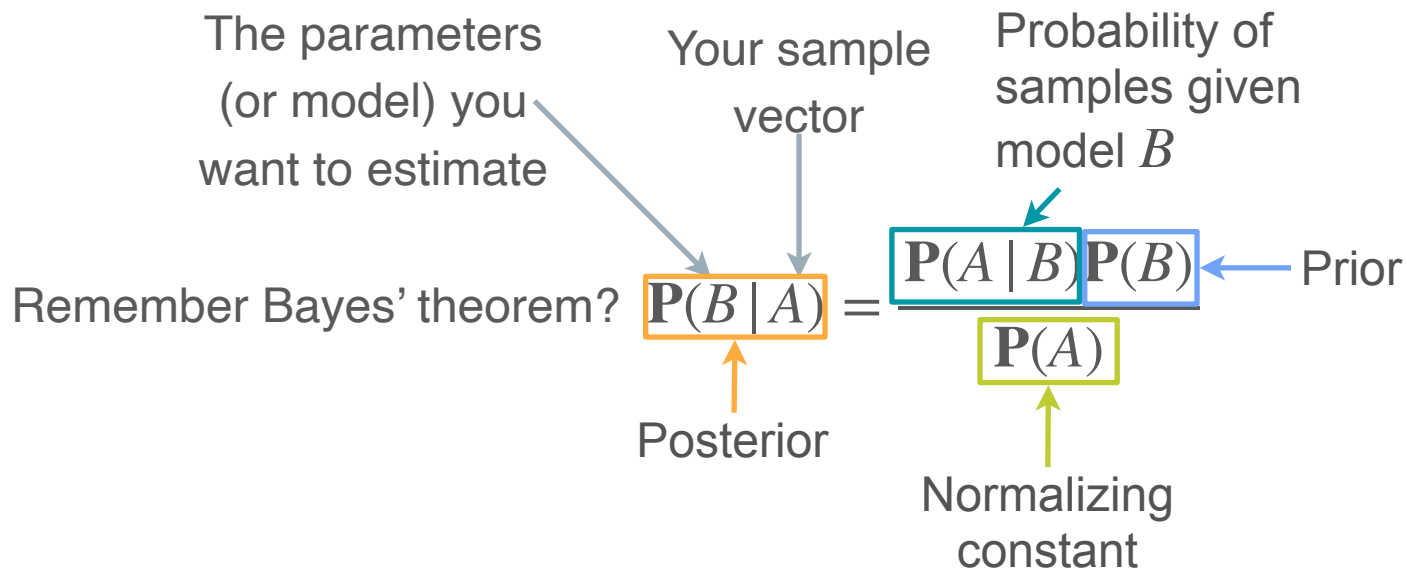
$$\mathbf{P}(B | A) = \frac{\mathbf{P}(A | B) \mathbf{P}(B)}{\mathbf{P}(A)}$$

Posterior

Normalizing constant

Prior

Bayesian Statistics



Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$



Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$



$$\Theta = P(H)$$

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$



Θ is a random variable $\rightarrow \Theta = P(H)$

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$



Θ is a random variable



$$\Theta = P(H)$$

$$X_i = 1 \text{ if } H, 0 \text{ if } T$$

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$



Θ is a random
variable



$$\Theta = P(H)$$

$$X_i = 1 \text{ if } H, 0 \text{ if } T$$

Observations:

$$\sum_{i=1}^{10} X_i = 8$$

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$



Θ is a random variable $\rightarrow \Theta = P(H)$ $X_i = 1$ if H , 0 if T

Observations: $\sum_{i=1}^{10} X_i = 8$

Suppose that Θ takes on the particular value θ

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$



Θ is a random variable $\rightarrow \Theta = P(H)$ $X_i = 1$ if H , 0 if T
Observations: $\sum_{i=1}^{10} X_i = 8$

Suppose that Θ takes on the particular value θ

$P(A|B)$

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$



Θ is a random variable

$$\rightarrow \Theta = P(H) \quad X_i = 1 \text{ if } H, 0 \text{ if } T$$

Observations: $\sum_{i=1}^{10} X_i = 8 \leftarrow A$

Suppose that Θ takes on the particular value θ

$$P(A|B)$$

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$



Θ is a random variable

$$\rightarrow \Theta = P(H) \quad X_i = 1 \text{ if } H, 0 \text{ if } T$$

Observations: $\sum_{i=1}^{10} X_i = 8 \leftarrow A$

Suppose that Θ takes on the particular value $\theta \leftarrow B$

$$P(A|B)$$

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$



Θ is a random variable

$\rightarrow \Theta = P(H) \quad X_i = 1 \text{ if } H, 0 \text{ if } T$

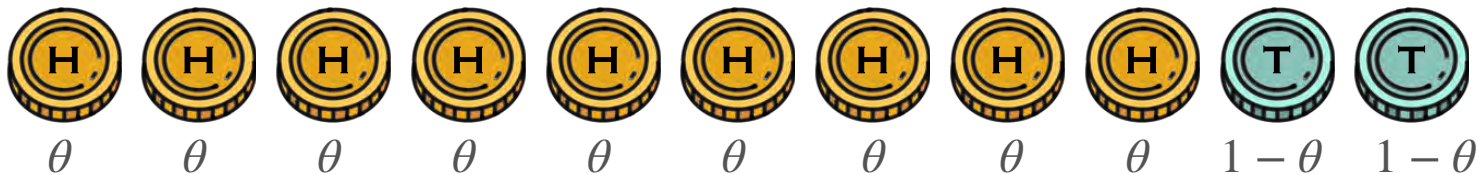
Observations: $\sum_{i=1}^{10} X_i = 8 \leftarrow A$

Suppose that Θ takes on the particular value $\theta \leftarrow B$

$$P(A|B) \rightarrow P\left(\sum_{i=1}^{10} X_i = 8 \mid \Theta = \theta\right) =$$

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$



Θ is a random variable

$\rightarrow \Theta = P(H) \quad X_i = 1 \text{ if } H, 0 \text{ if } T$

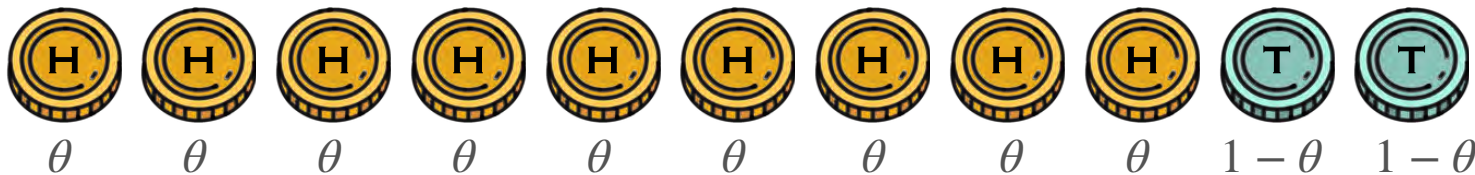
Observations: $\sum_{i=1}^{10} X_i = 8 \leftarrow A$

Suppose that Θ takes on the particular value $\theta \leftarrow B$

$$P(A|B) \rightarrow P\left(\sum_{i=1}^{10} X_i = 8 \mid \Theta = \theta\right) = \theta^8 (1 - \theta)^2$$

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$



Θ is a random variable

$\rightarrow \Theta = P(H) \quad X_i = 1 \text{ if } H, 0 \text{ if } T$

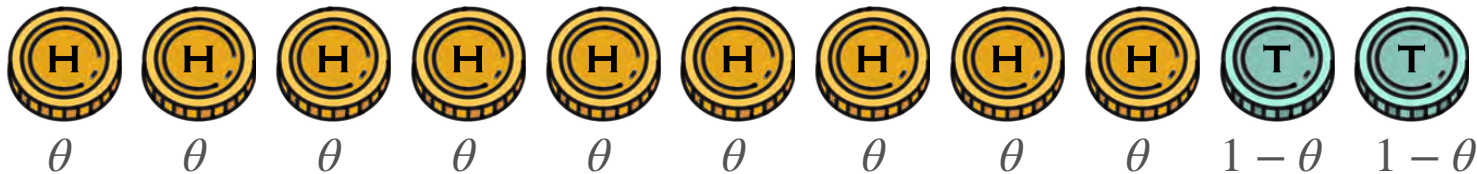
Observations: $\sum_{i=1}^{10} X_i = 8 \leftarrow A$

Suppose that Θ takes on the particular value $\theta \leftarrow B$

$$P(A|B) \rightarrow P\left(\sum_{i=1}^{10} X_i = 8 \mid \Theta = \theta\right) = \theta^8 (1 - \theta)^2 \quad X_i \mid \Theta = \theta \sim \text{Bernoulli}(\theta)$$

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$

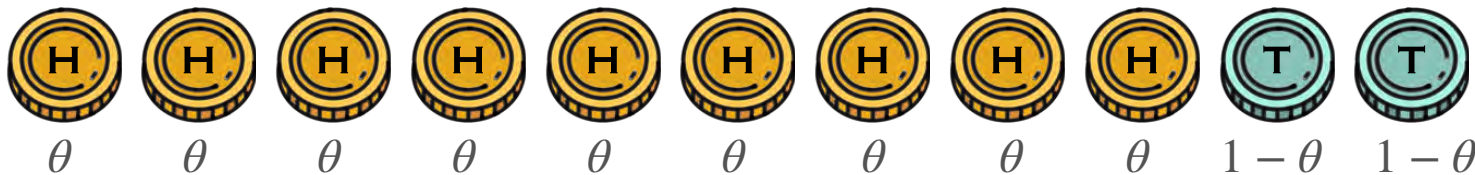


$X_i | \Theta = \theta \sim \text{Bernoulli}(\theta)$

How do you choose the prior?

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(\textcolor{brown}{B} | \textcolor{teal}{A}) = \frac{P(\textcolor{teal}{A} | \textcolor{brown}{B})P(\textcolor{brown}{B})}{P(\textcolor{teal}{A})}$



$X_i | \Theta = \theta \sim \textit{Bernoulli}(\theta)$

How do you choose the prior?

If before tossing the coins you have no idea whether the coin is fair or not, then your prior belief about Θ should not favor any value.

You can model this with a uniform distribution

$$\Theta \sim \textit{Uniform}(0,1) \qquad f_{\Theta}(\theta) = 1, \quad 0 \leq \theta \leq 1$$

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $P(\mathbf{B}|\mathbf{A}) = \frac{P(\mathbf{A}|\mathbf{B})P(\mathbf{B})}{P(\mathbf{A})}$



$X_i | \Theta = \theta \sim \text{Bernoulli}(\theta)$

How do you choose the prior?

If before tossing the coins you have no idea whether the coin is fair or not, then your prior belief about Θ should not favor any value.

You can model this with a uniform distribution

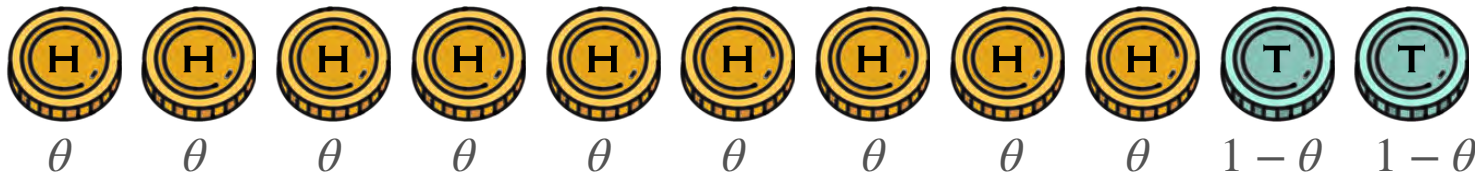
$\Theta \sim \text{Uniform}(0,1)$

$$f_{\Theta}(\theta) = 1, \quad 0 \leq \theta \leq 1$$

← Prior ($P(\mathbf{B})$)

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $\mathbf{P(B|A)} = \frac{\mathbf{P(A|B)P(B)}}{\mathbf{P(A)}}$

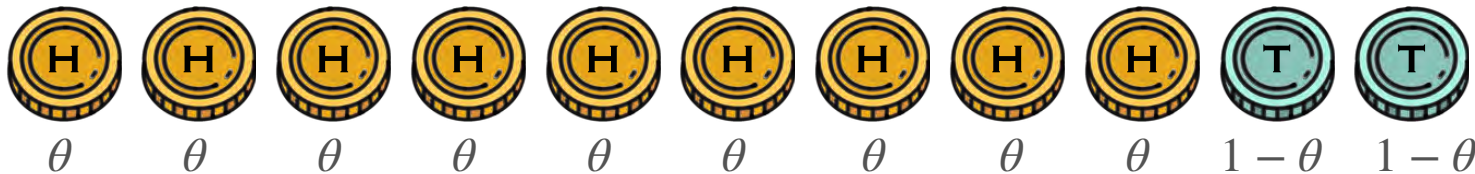


$X_i | \Theta = \theta \sim \text{Bernoulli}(\theta) \leftarrow (\mathbf{P(A | B)})$

$\Theta \sim \text{Uniform}(0,1) \leftarrow \text{Prior } (\mathbf{P(B)})$

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $\mathbf{P(B|A)} = \frac{\mathbf{P(A|B)P(B)}}{\mathbf{P(A)}}$



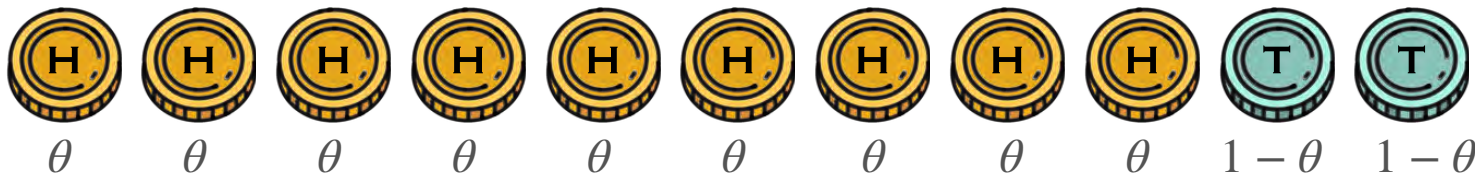
$X_i | \Theta = \theta \sim \text{Bernoulli}(\theta) \leftarrow (\mathbf{P(A | B)})$

$\Theta \sim \text{Uniform}(0,1) \leftarrow \text{Prior } (\mathbf{P(B)})$

How do you get the posterior? Bayes!

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $\mathbf{P(B|A)} = \frac{\mathbf{P(A|B)P(B)}}{\mathbf{P(A)}}$



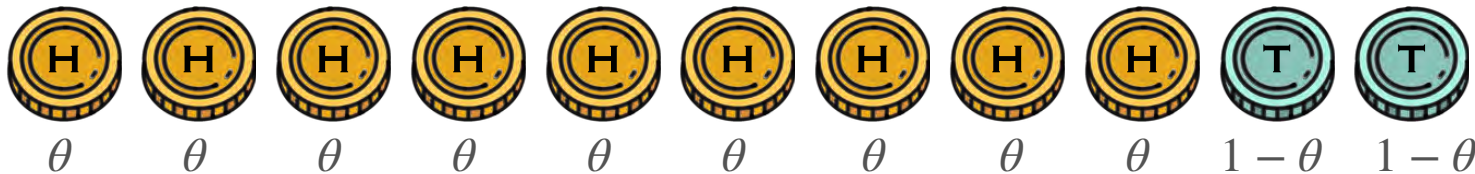
$X_i | \Theta = \theta \sim \text{Bernoulli}(\theta) \leftarrow (\mathbf{P(A | B)})$ $\Theta \sim \text{Uniform}(0,1) \leftarrow \text{Prior } (\mathbf{P(B)})$

How do you get the posterior? Bayes!

$$f_{\Theta|\mathbf{X}=\mathbf{x}}(\theta) = \frac{p_{\mathbf{X}|\Theta=\theta}(\mathbf{x})f_{\Theta}(\theta)}{p_{\mathbf{X}}(\mathbf{x})}$$

Bayesian Statistics: Bernoulli Example

Remember Bayes' theorem? $\mathbf{P(B|A)} = \frac{\mathbf{P(A|B)P(B)}}{\mathbf{P(A)}}$



$X_i | \Theta = \theta \sim \text{Bernoulli}(\theta) \leftarrow (\mathbf{P(A | B)})$ $\Theta \sim \text{Uniform}(0,1) \leftarrow \text{Prior } (\mathbf{P(B)})$

How do you get the posterior? Bayes!

$$\boxed{f_{\Theta|\mathbf{X}=\mathbf{x}}(\theta)} = \frac{p_{\mathbf{X}|\Theta=\theta}(\mathbf{x})f_{\Theta}(\theta)}{\boxed{p_{\mathbf{X}}(\mathbf{x})} \leftarrow \text{Normalizing constant}}$$

↑
Posterior ($\mathbf{P(B | A)}$)

Bayesian Statistics: MAP Estimator

$$f_{\Theta|\mathbf{X}=\mathbf{x}}(\theta) \propto p_{\mathbf{X}|\Theta=\theta}(\mathbf{x})f_{\Theta}(\theta)$$

Bayesian Statistics: MAP Estimator

$$f_{\Theta|\mathbf{X}=\mathbf{x}}(\theta) \propto p_{\mathbf{X}|\Theta=\theta}(\mathbf{x}) f_{\Theta}(\theta)$$

$\mathbf{P}(\text{data} \mid \text{model})$ $\mathbf{P}(\text{model})$

Bayesian Statistics: MAP Estimator

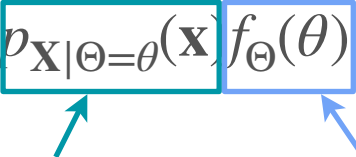
$$f_{\Theta|\mathbf{X}=\mathbf{x}}(\theta) \propto p_{\mathbf{X}|\Theta=\theta}(\mathbf{x}) f_{\Theta}(\theta)$$

Does this look familiar?

P(data | model)

P(model)

Bayesian Statistics: MAP Estimator

$$f_{\Theta|\mathbf{X}=\mathbf{x}}(\theta) \propto p_{\mathbf{X}|\Theta=\theta}(\mathbf{x}) f_{\Theta}(\theta)$$


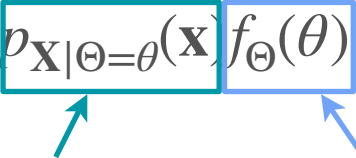
Does this look familiar?

P(data | model)

P(model)

Point estimators?

Bayesian Statistics: MAP Estimator

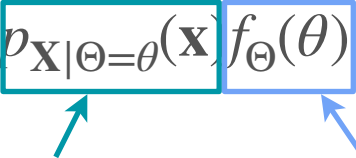
$$f_{\Theta|\mathbf{X}=\mathbf{x}}(\theta) \propto p_{\mathbf{X}|\Theta=\theta}(\mathbf{x}) f_{\Theta}(\theta)$$


Does this look familiar? **P(data | model)** **P(model)**

Point estimators?

$$\hat{\theta} = \arg \max_{\theta} f_{\Theta|\mathbf{X}=\mathbf{x}}(\theta) = \arg \max_{\theta} p_{\mathbf{X}|\Theta=\theta}(\mathbf{x}) f_{\Theta}(\theta)$$

Bayesian Statistics: MAP Estimator

$$f_{\Theta|\mathbf{X}=\mathbf{x}}(\theta) \propto p_{\mathbf{X}|\Theta=\theta}(\mathbf{x}) f_{\Theta}(\theta)$$


Does this look familiar? **P(data | model)** **P(model)**

Point estimators?

$$\hat{\theta} = \arg \max_{\theta} f_{\Theta|\mathbf{X}=\mathbf{x}}(\theta) = \arg \max_{\theta} p_{\mathbf{X}|\Theta=\theta}(\mathbf{x}) f_{\Theta}(\theta)$$

This is called the Maximum a Posteriori (MAP) estimator

Bayesian Statistics: MAP Estimator

$$f_{\Theta|\mathbf{X}=\mathbf{x}}(\theta) \propto p_{\mathbf{X}|\Theta=\theta}(\mathbf{x}) f_{\Theta}(\theta)$$


Does this look familiar? **P(data | model)** **P(model)**

Point estimators?

$$\hat{\theta} = \arg \max_{\theta} f_{\Theta|\mathbf{X}=\mathbf{x}}(\theta) = \arg \max_{\theta} p_{\mathbf{X}|\Theta=\theta}(\mathbf{x}) f_{\Theta}(\theta)$$

This is called the Maximum a Posteriori (MAP) estimator

Notice that if the prior is non informative, then MAP = MLE

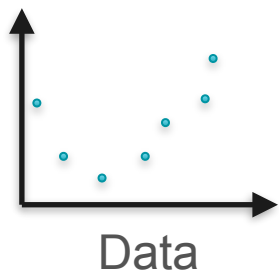


DeepLearning.AI

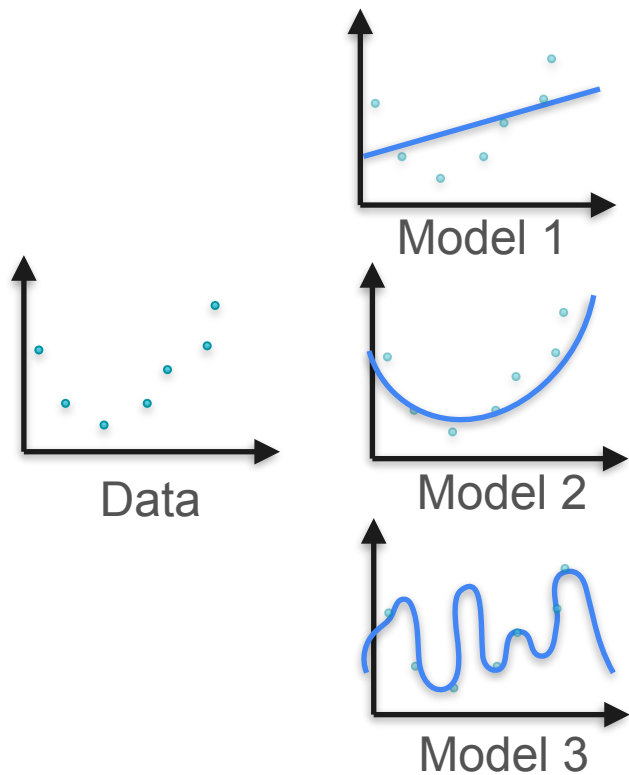
Point Estimation

Regularization

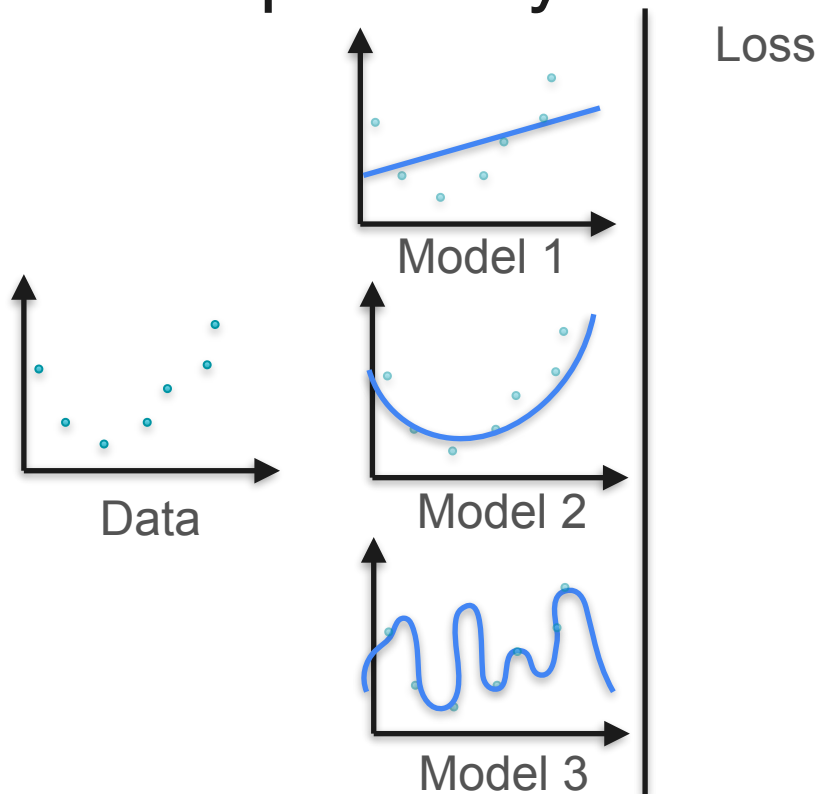
Example: Polynomial Regression



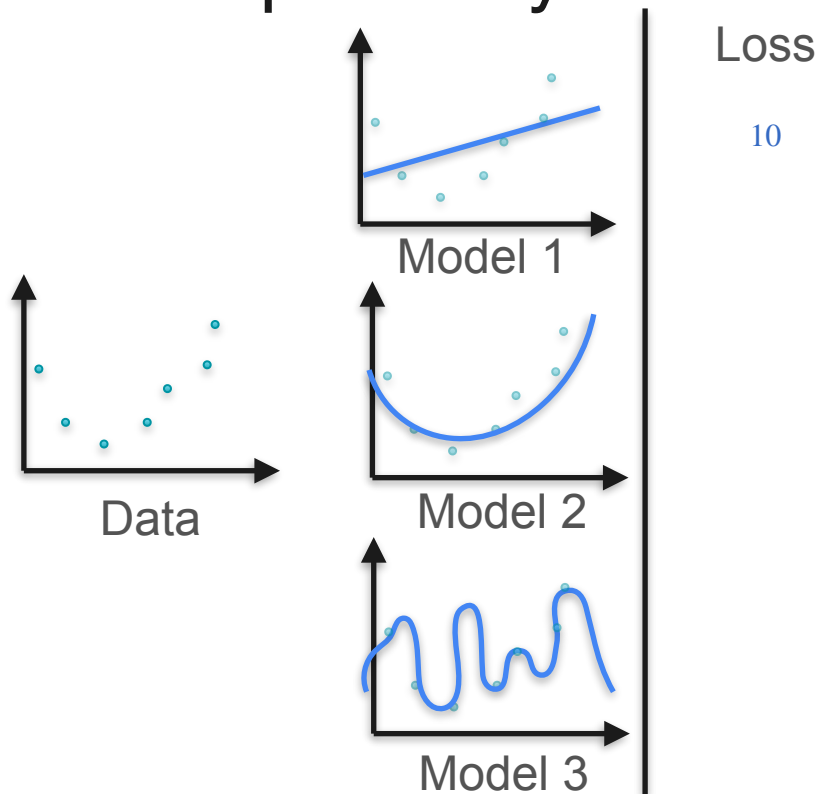
Example: Polynomial Regression



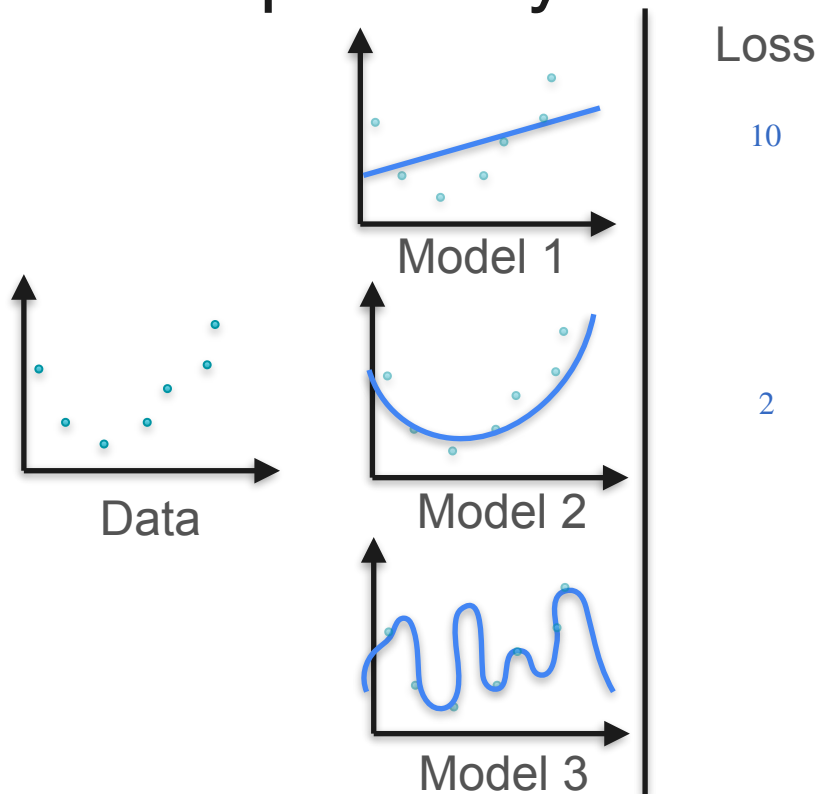
Example: Polynomial Regression



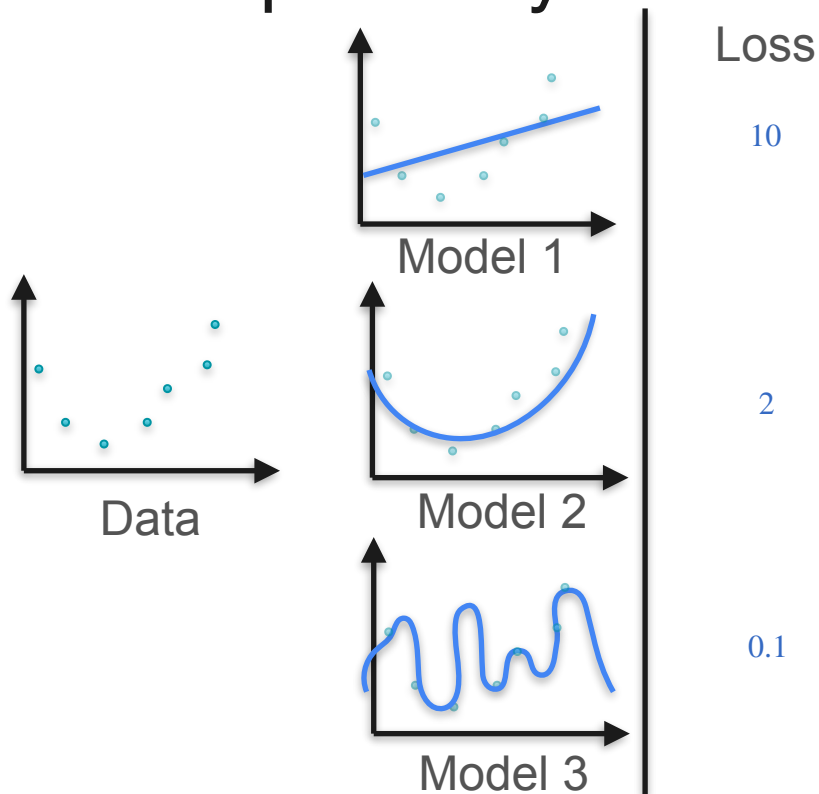
Example: Polynomial Regression



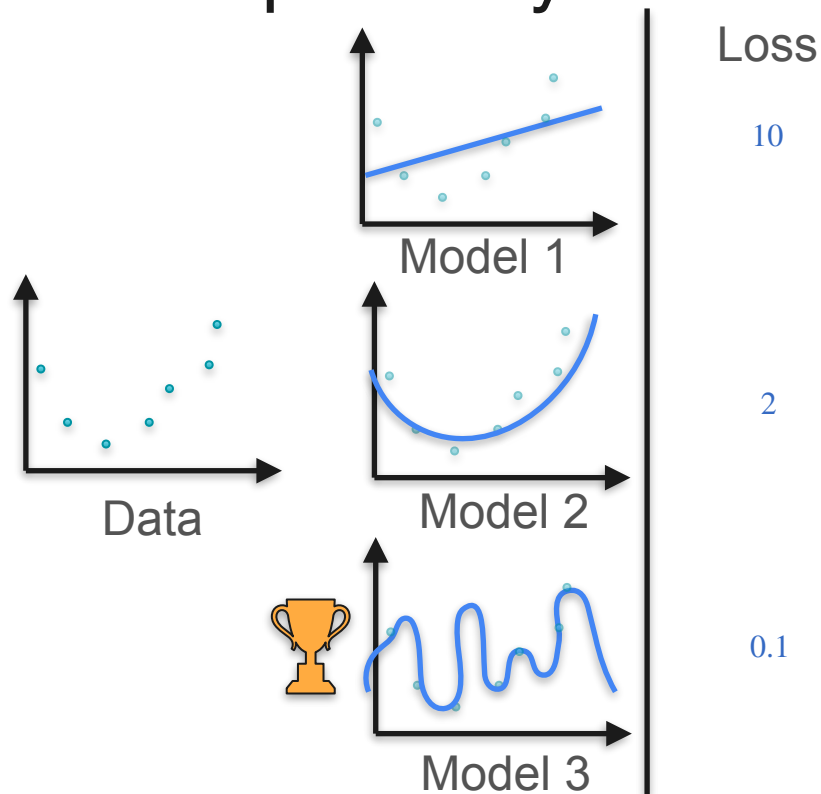
Example: Polynomial Regression



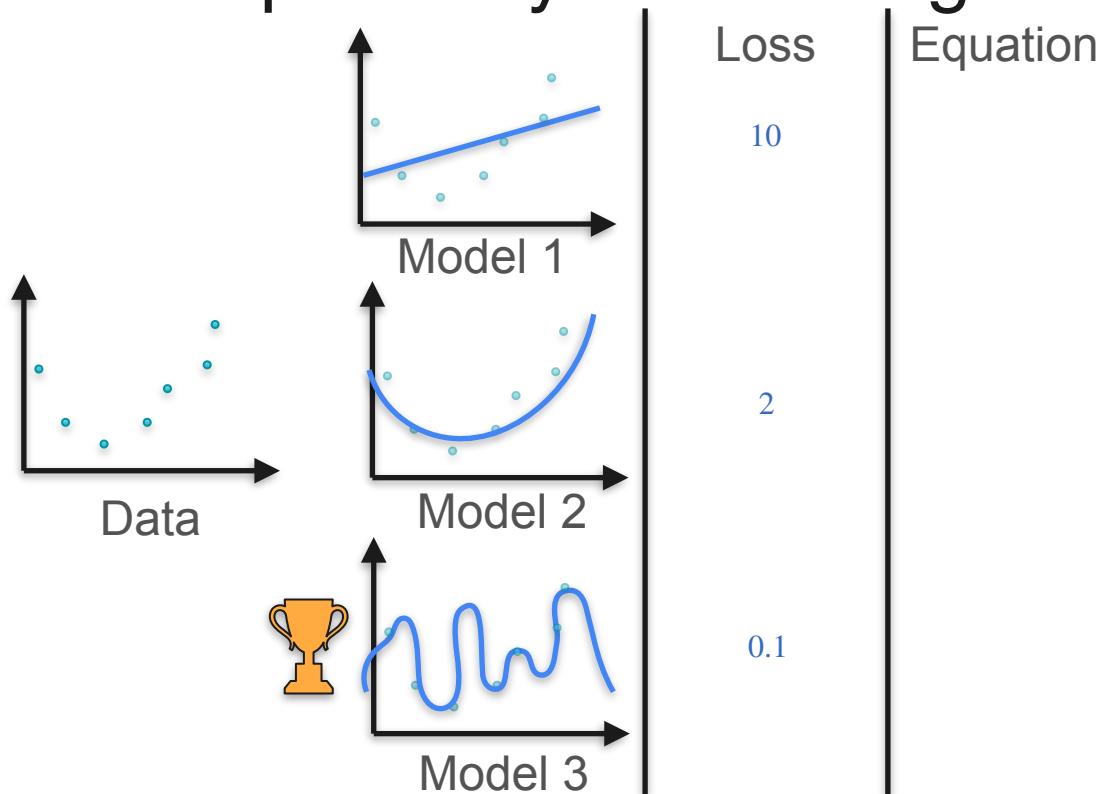
Example: Polynomial Regression



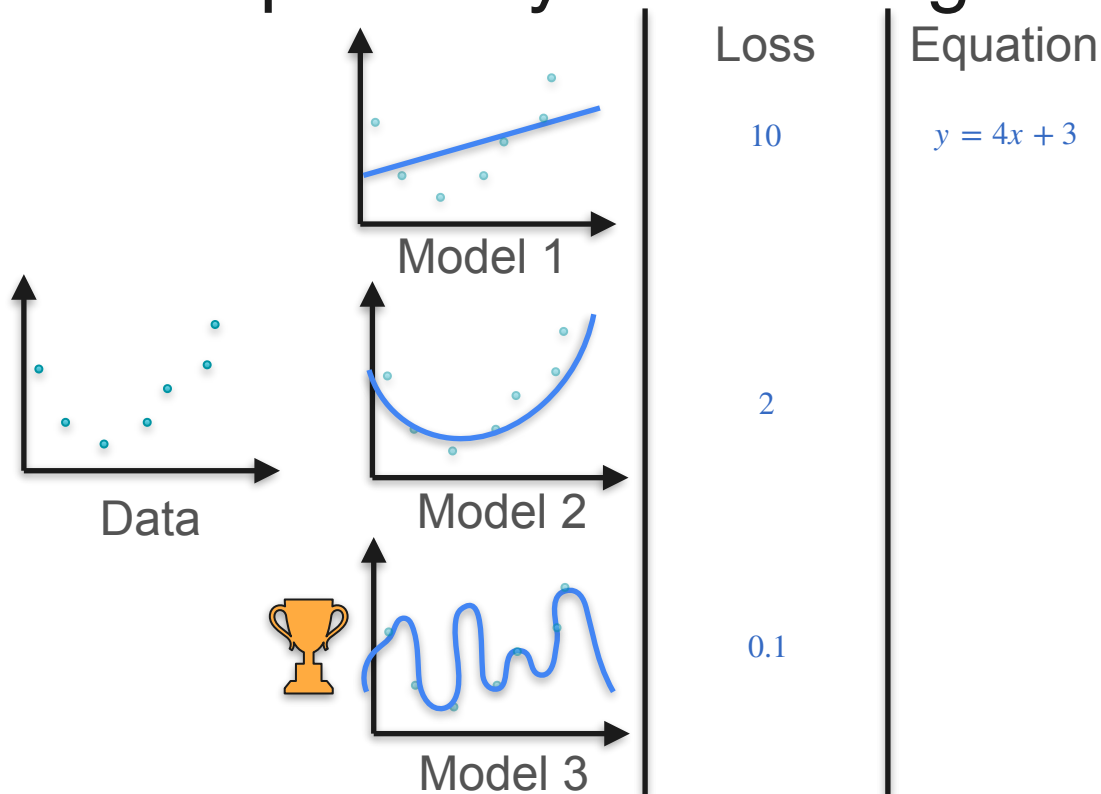
Example: Polynomial Regression



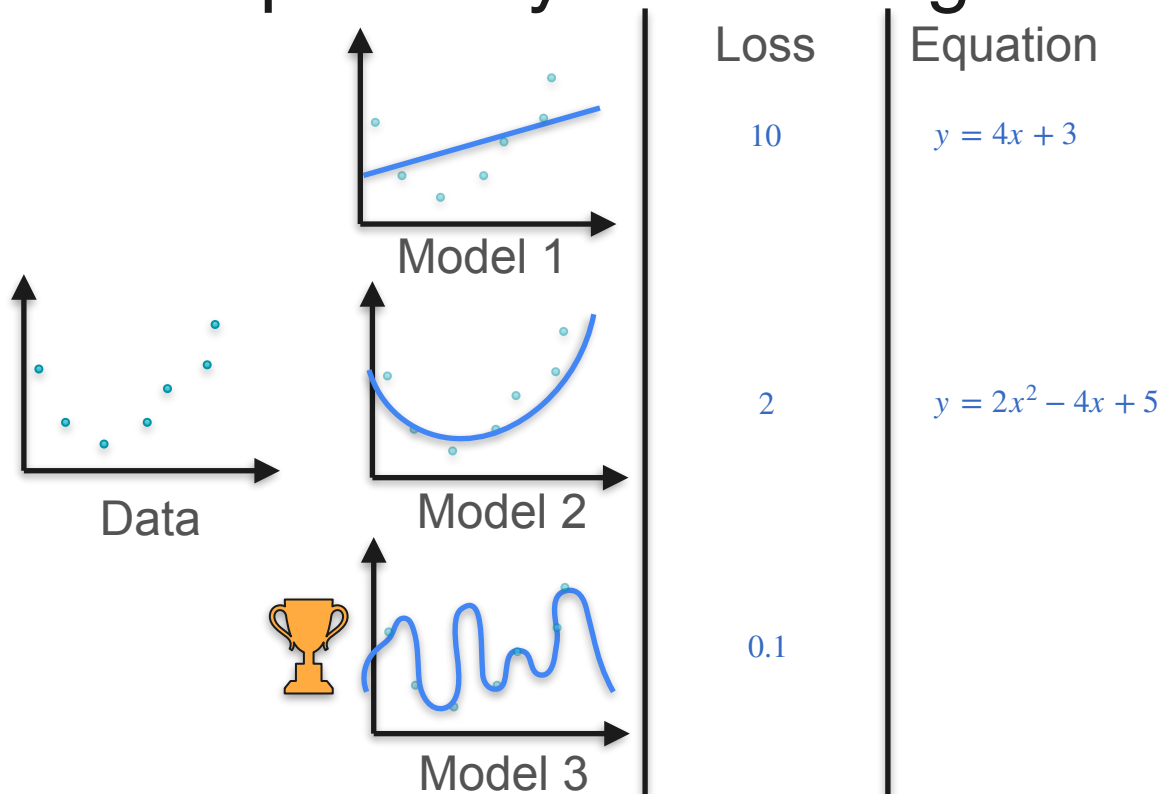
Example: Polynomial Regression



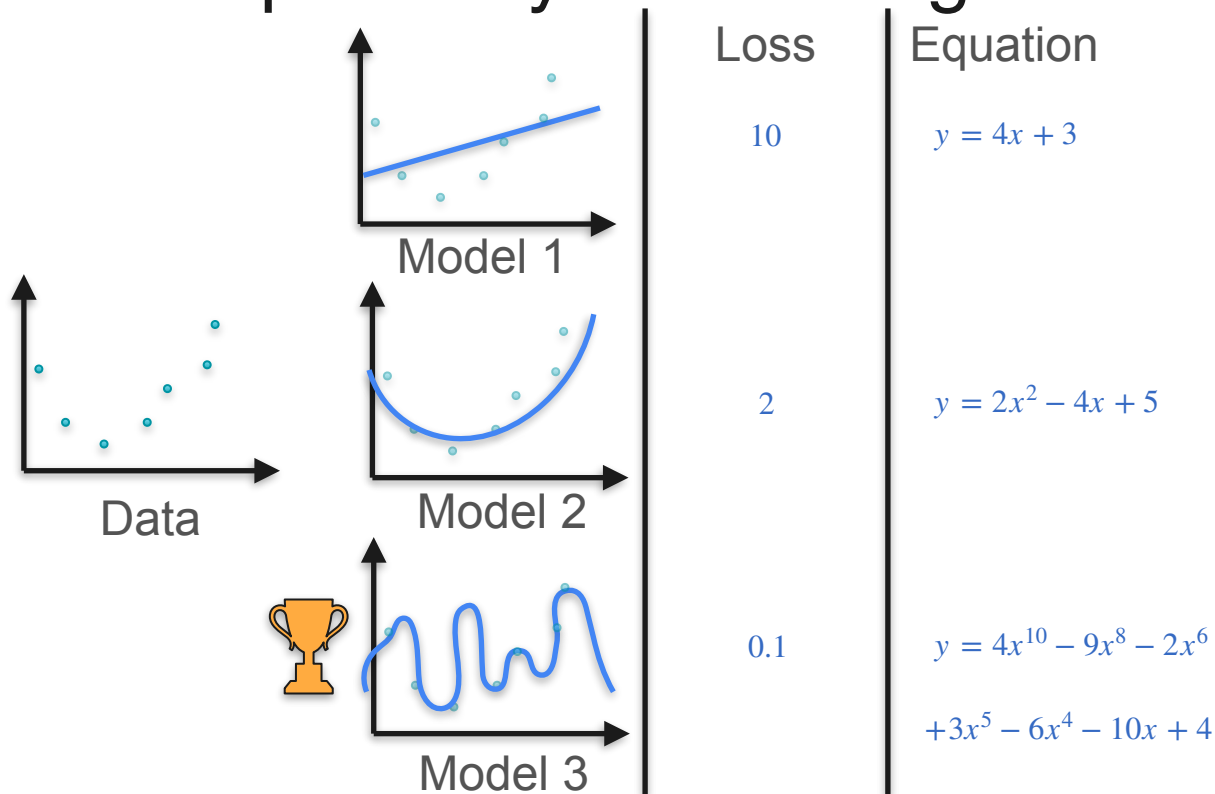
Example: Polynomial Regression



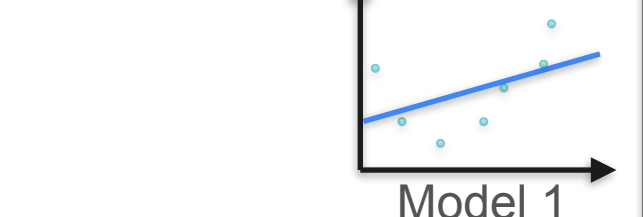
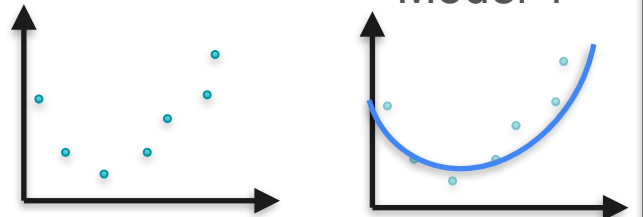
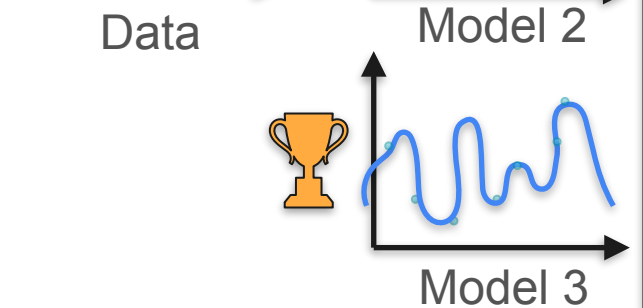
Example: Polynomial Regression



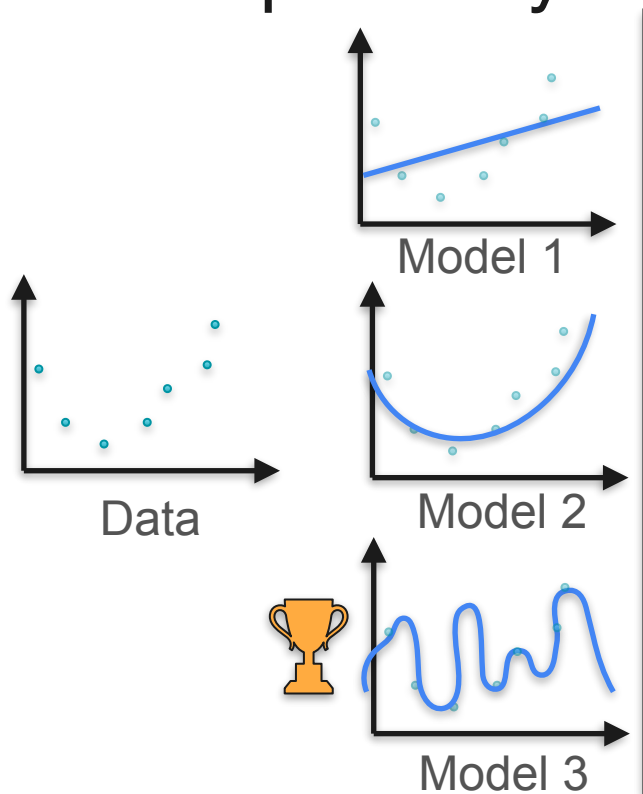
Example: Polynomial Regression



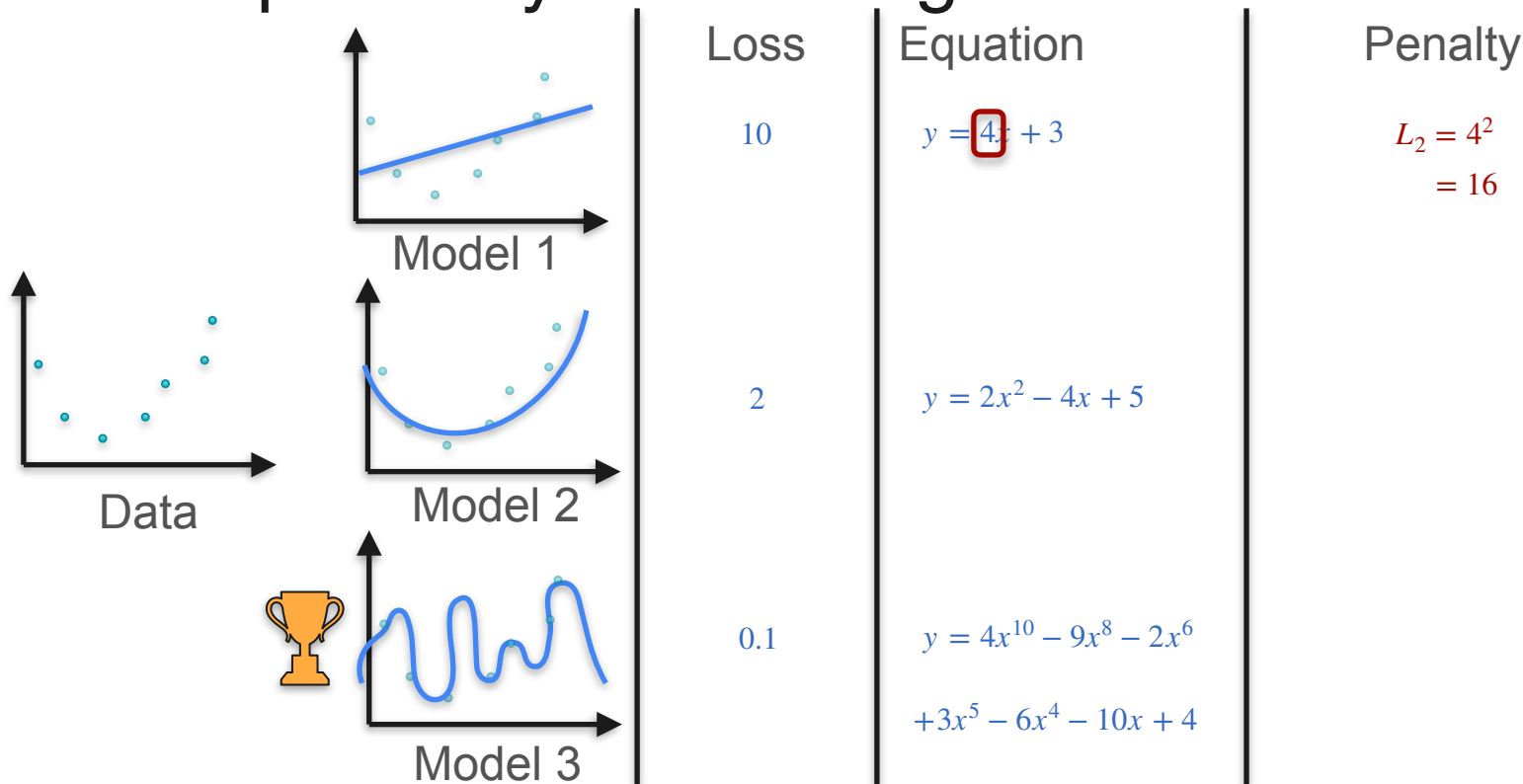
Example: Polynomial Regression

	Loss	Equation	Penalty
 A scatter plot of data points and a linear regression line (Model 1) that does not fit the data well.	10	$y = 4x + 3$	
 A scatter plot of data points and a quadratic regression curve (Model 2) that fits the data better than Model 1.	2	$y = 2x^2 - 4x + 5$	
 A scatter plot of data points and a high-degree polynomial regression curve (Model 3) that fits the data perfectly, indicated by a trophy icon.	0.1	$y = 4x^{10} - 9x^8 - 2x^6 + 3x^5 - 6x^4 - 10x + 4$	

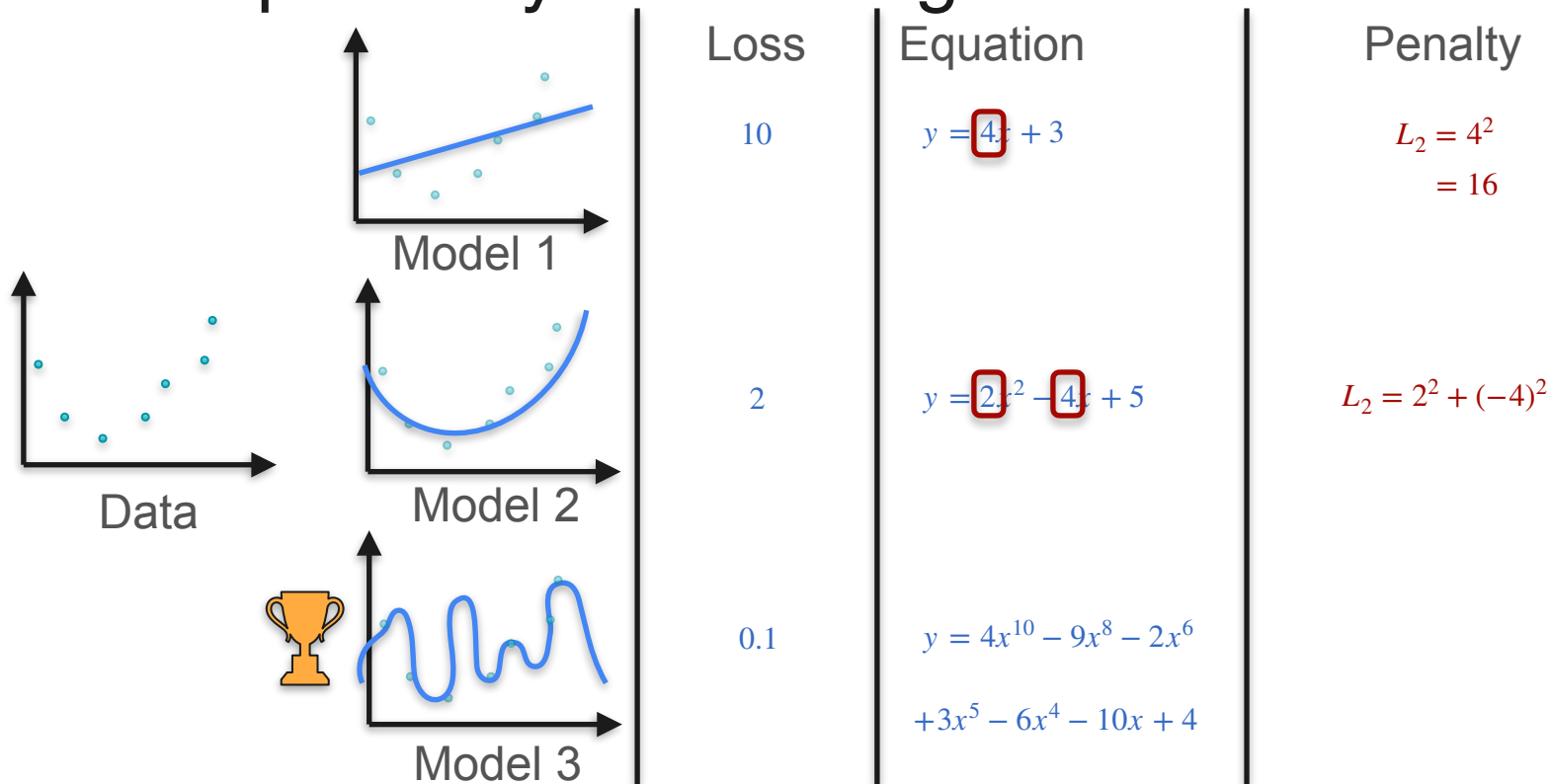
Example: Polynomial Regression

		Loss	Equation	Penalty
	Model 1	10	$y = 4x + 3$	$L_2 = 4^2$
	Model 2	2	$y = 2x^2 - 4x + 5$	
	Model 3	0.1	$y = 4x^{10} - 9x^8 - 2x^6 + 3x^5 - 6x^4 - 10x + 4$	

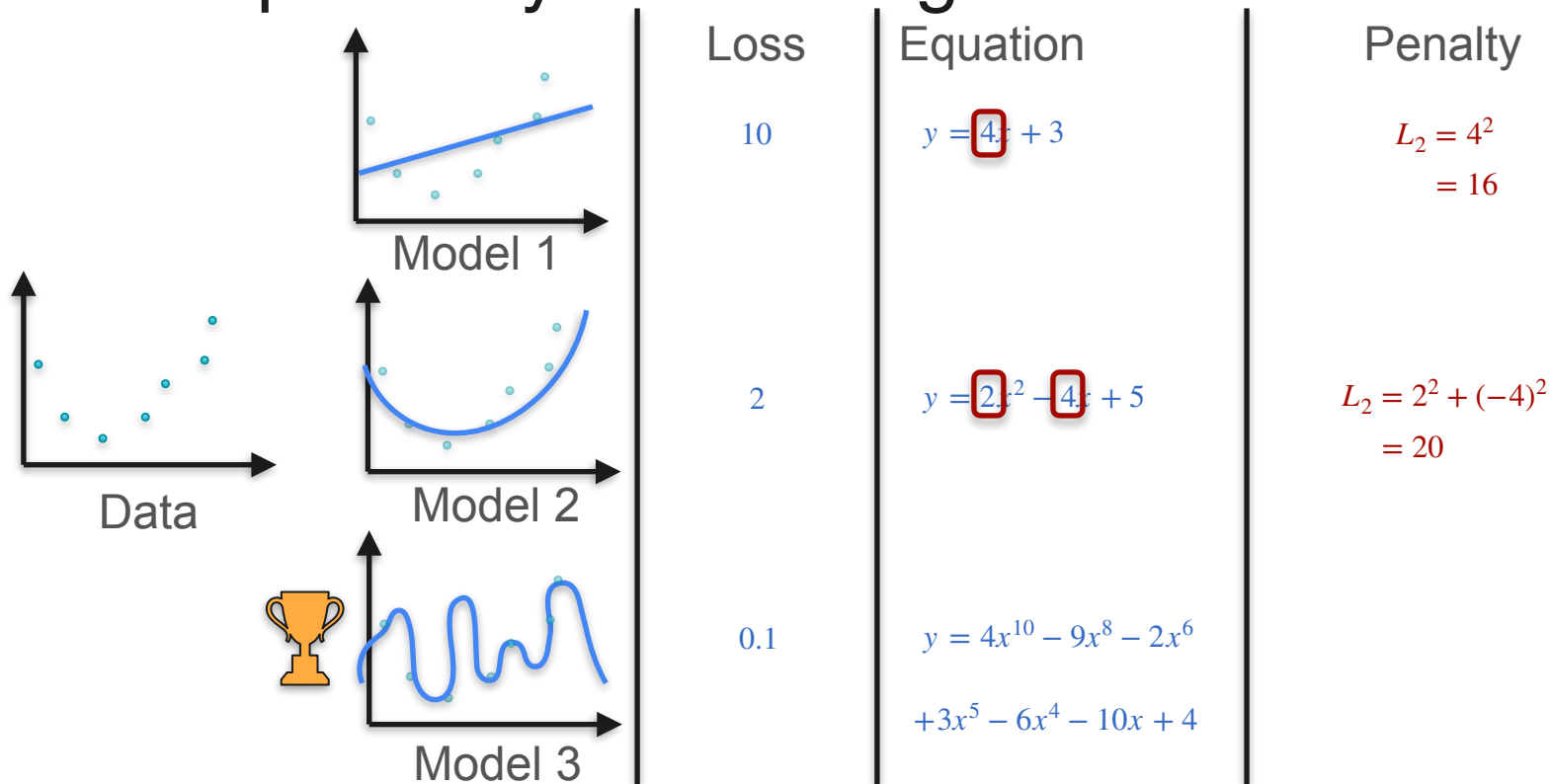
Example: Polynomial Regression



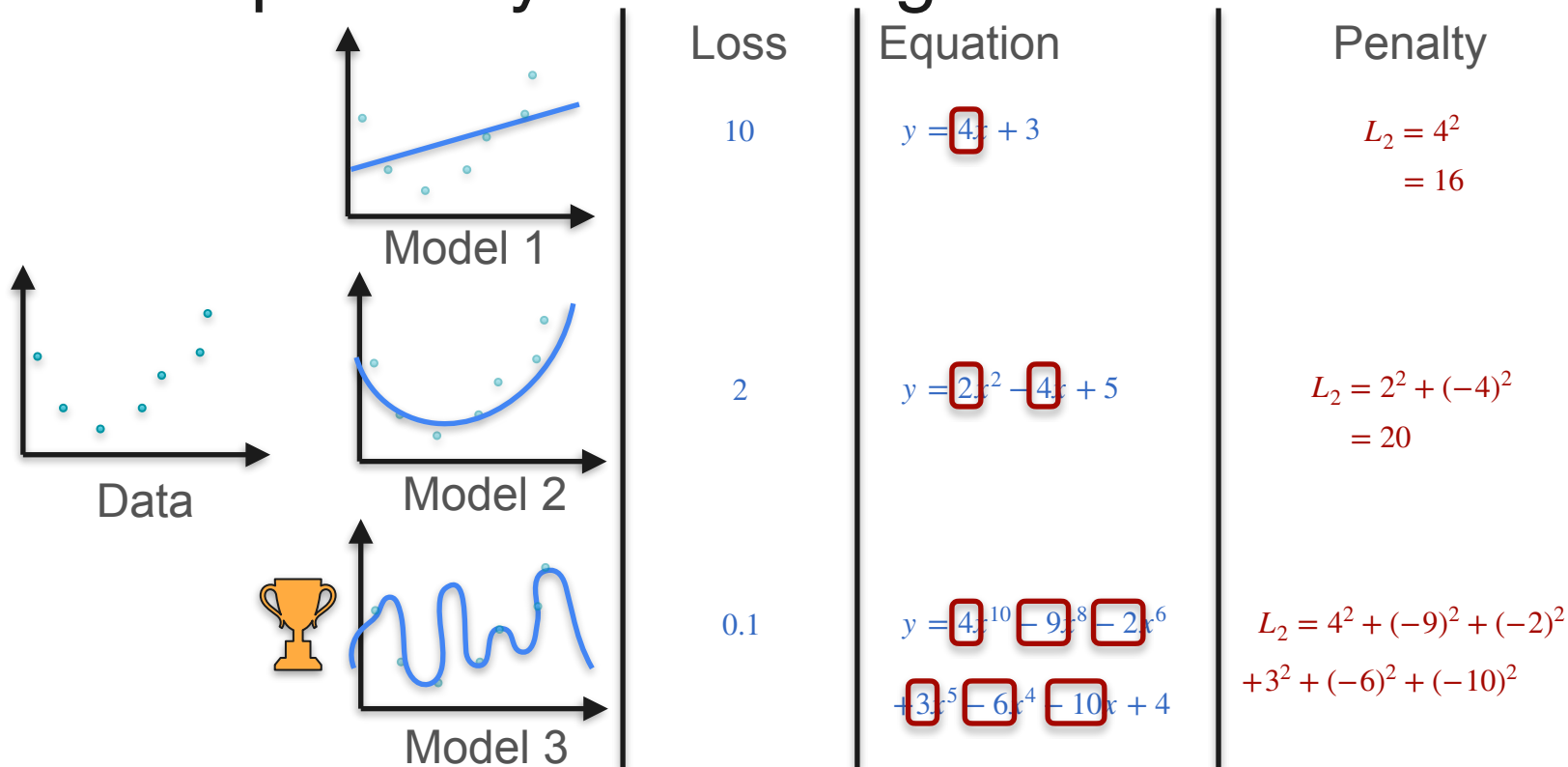
Example: Polynomial Regression



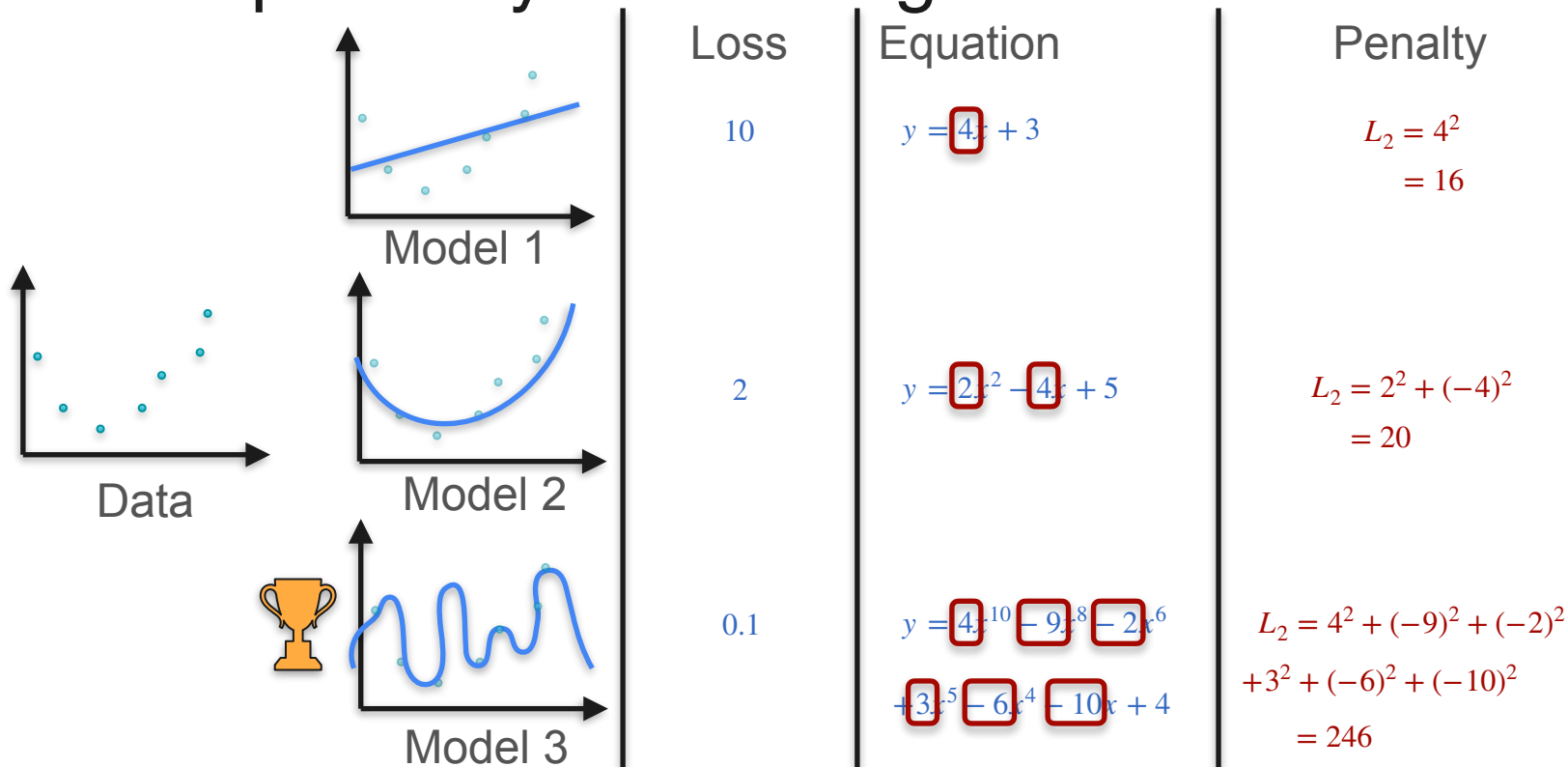
Example: Polynomial Regression



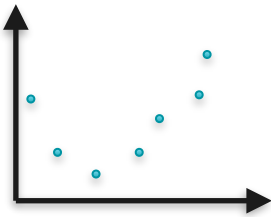
Example: Polynomial Regression



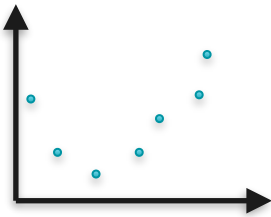
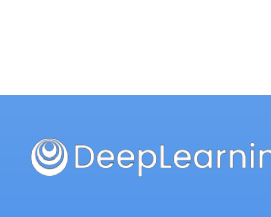
Example: Polynomial Regression



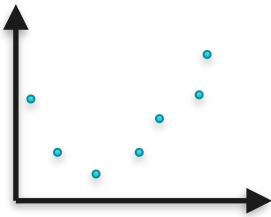
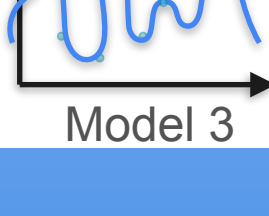
Example: Polynomial Regression

	Loss	Equation	Penalty	New loss
 Data	10	$y = 4x + 3$	$L_2 = 4^2$ $= 16$	
 Model 1	2	$y = 2x^2 - 4x + 5$	$L_2 = 2^2 + (-4)^2$ $= 20$	
 Model 2	0.1	$y = 4x^{10} - 9x^8 - 2x^6$ $+ 3x^5 - 6x^4 - 10x + 4$	$L_2 = 4^2 + (-9)^2 + (-2)^2$ $+ 3^2 + (-6)^2 + (-10)^2$ $= 246$	
 Model 3				

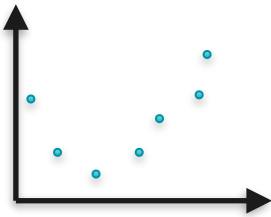
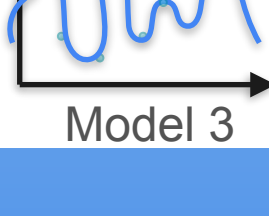
Example: Polynomial Regression

Data	Model 1	Loss	Equation	Penalty	New loss
		10	$y = 4x + 3$	$L_2 = 4^2 = 16$	26
		2	$y = 2x^2 - 4x + 5$	$L_2 = 2^2 + (-4)^2 = 20$	
		0.1	$y = 4x^{10} - 9x^8 - 2x^6 + 3x^5 - 6x^4 - 10x + 4$	$L_2 = 4^2 + (-9)^2 + (-2)^2 + 3^2 + (-6)^2 + (-10)^2 = 246$	

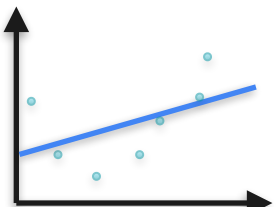
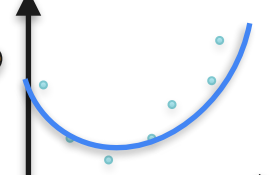
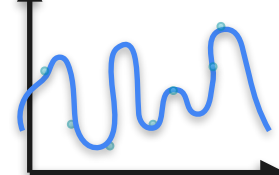
Example: Polynomial Regression

Data	Model 1	Loss	Equation	Penalty	New loss
		10	$y = 4x + 3$	$L_2 = 4^2 = 16$	26
		2	$y = 2x^2 - 4x + 5$	$L_2 = 2^2 + (-4)^2 = 20$	22
		0.1	$y = 4x^{10} - 9x^8 - 2x^6 + 3x^5 - 6x^4 - 10x + 4$	$L_2 = 4^2 + (-9)^2 + (-2)^2 + 3^2 + (-6)^2 + (-10)^2 = 246$	

Example: Polynomial Regression

Data	Model 1	Loss	Equation	Penalty	New loss
		10	$y = 4x + 3$	$L_2 = 4^2 = 16$	26
		2	$y = 2x^2 - 4x + 5$	$L_2 = 2^2 + (-4)^2 = 20$	22
		0.1	$y = 4x^{10} - 9x^8 - 2x^6 + 3x^5 - 6x^4 - 10x + 4$	$L_2 = 4^2 + (-9)^2 + (-2)^2 + 3^2 + (-6)^2 + (-10)^2 = 246$	246.1

Example: Polynomial Regression

 Data	 Model 1	Loss 10	Equation $y = 4x + 3$	Penalty $L_2 = 4^2 = 16$	New loss 26
	 Model 2	2	$y = 2x^2 - 4x + 5$	$L_2 = 2^2 + (-4)^2 = 20$	22
	 Model 3	0.1	$y = 4x^{10} - 9x^8 - 2x^6 + 3x^5 - 6x^4 - 10x + 4$	$L_2 = 4^2 + (-9)^2 + (-2)^2 + 3^2 + (-6)^2 + (-10)^2 = 246$	246.1

Regularization Term

Regularization Term

Model: $y = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

Regularization Term

Model: $y = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

Log-loss: $\ell\ell$

Regularization Term

Model: $y = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

Log-loss: $\ell\ell$

L2 Regularization Error: $a_n^2 + a_{n-1}^2 + \dots + a_1^2$

Regularization Term

Model: $y = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

Log-loss: $\ell\ell$

L2 Regularization Error: $a_n^2 + a_{n-1}^2 + \dots + a_1^2$

Regularization parameter: λ

Regularization Term

Model: $y = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

Log-loss: $\ell\ell$

L2 Regularization Error: $a_n^2 + a_{n-1}^2 + \dots + a_1^2$

Regularization parameter: λ

Regularization Term

Model: $y = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

Log-loss: $\ell\ell$

L2 Regularization Error: $a_n^2 + a_{n-1}^2 + \dots + a_1^2$

Regularization parameter: λ

Regularized error:

Regularization Term

Model: $y = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

Log-loss: $\ell\ell$

L2 Regularization Error: $a_n^2 + a_{n-1}^2 + \dots + a_1^2$

Regularization parameter: λ

Regularized error: $\ell\ell + \lambda (a_n^2 + a_{n-1}^2 + \dots + a_1^2)$



DeepLearning.AI

Point Estimation

Back to Bayesics

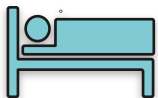
There's Popcorn on the Floor. What Happened?



Movies



Board Games



Nap



There's Popcorn on the Floor. What Happened?

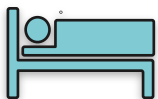


Movies

High



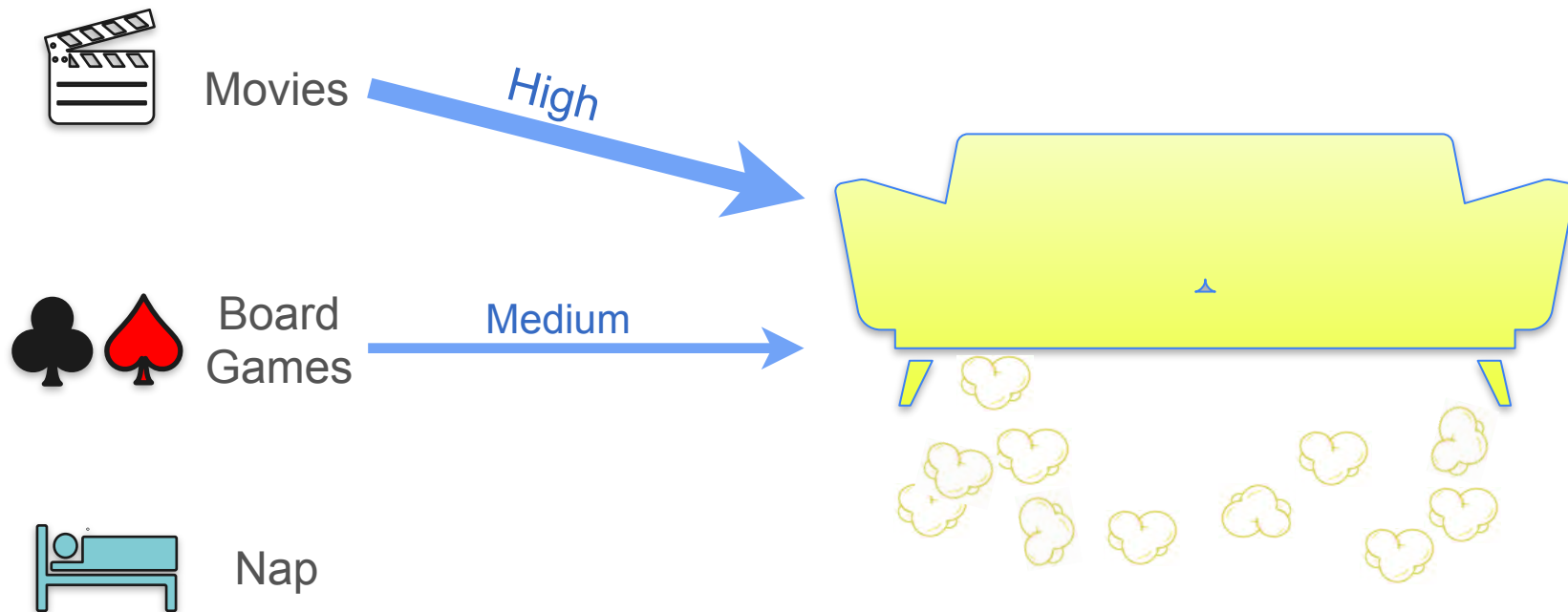
Board
Games



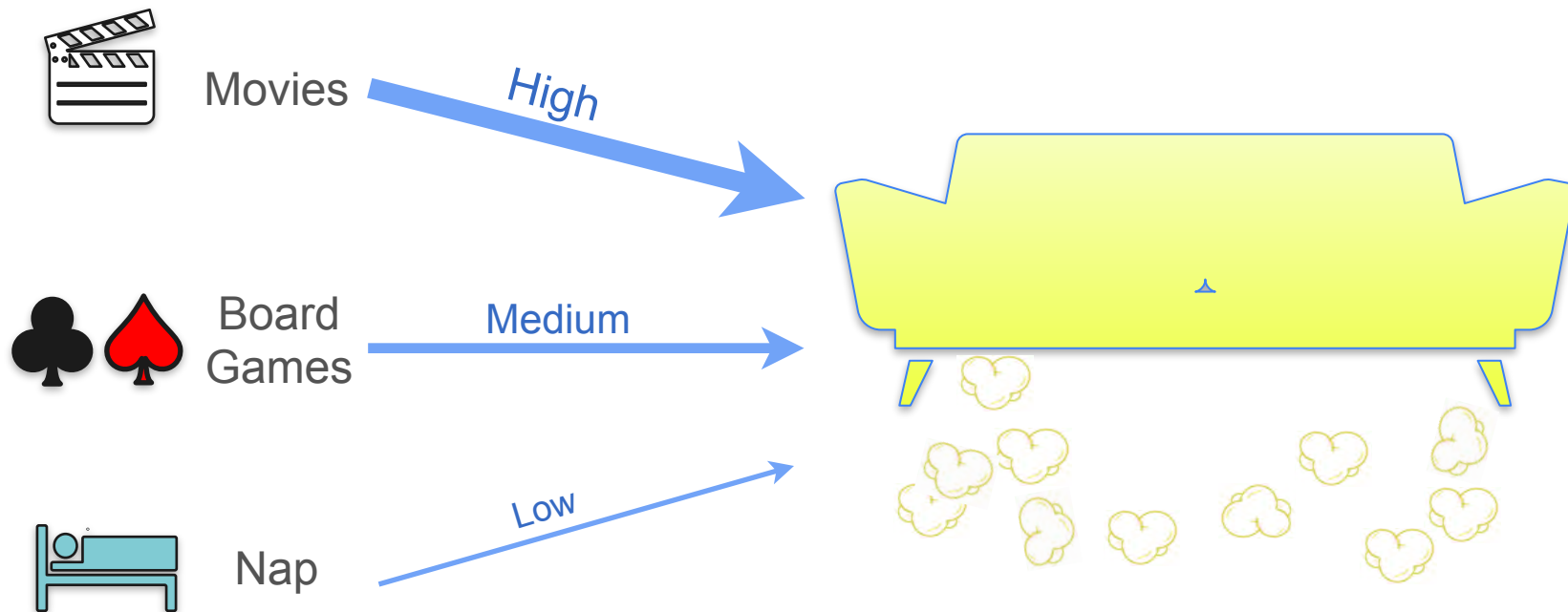
Nap



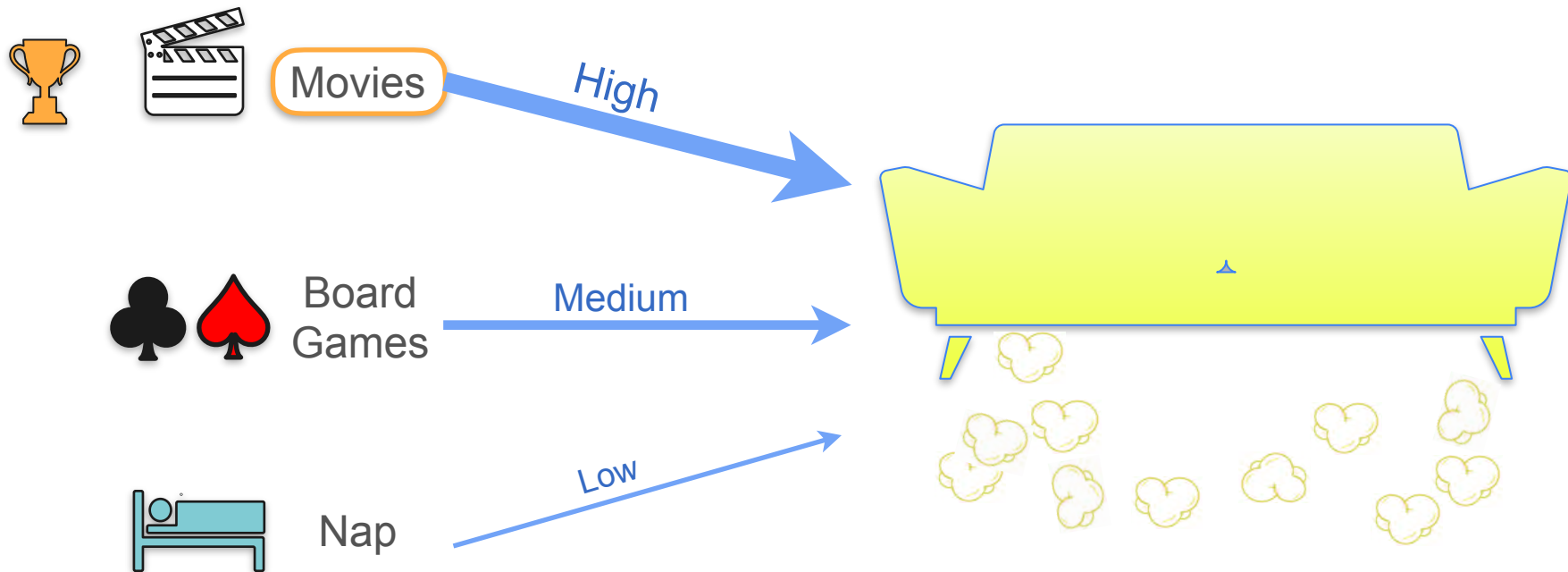
There's Popcorn on the Floor. What Happened?



There's Popcorn on the Floor. What Happened?



There's Popcorn on the Floor. What Happened?



There's Popcorn on the Floor. What Happened?



Movies



Popcorn
throwing
contest

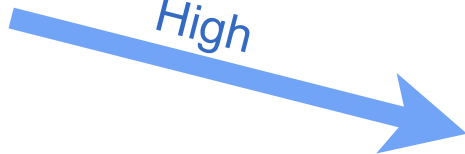


There's Popcorn on the Floor. What Happened?



Movies

High



Popcorn
throwing
contest



There's Popcorn on the Floor. What Happened?



Movies

High



Popcorn
throwing
contest

Very high



There's Popcorn on the Floor. What Happened?



Movies

High

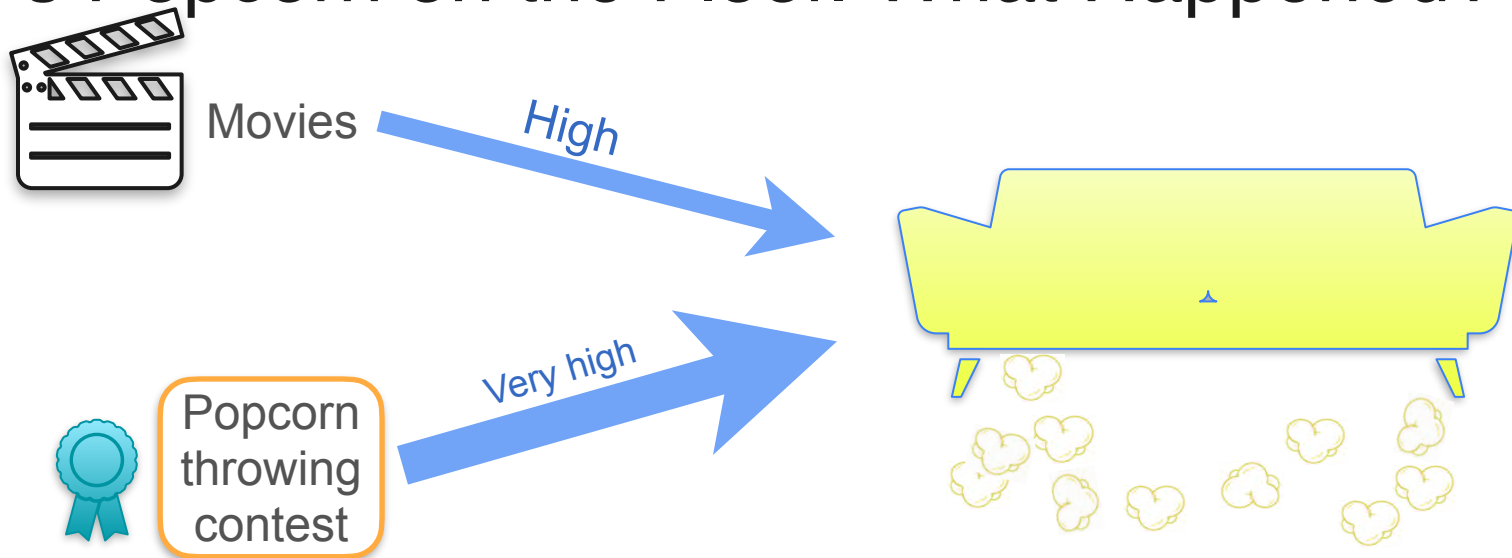


Popcorn
throwing
contest

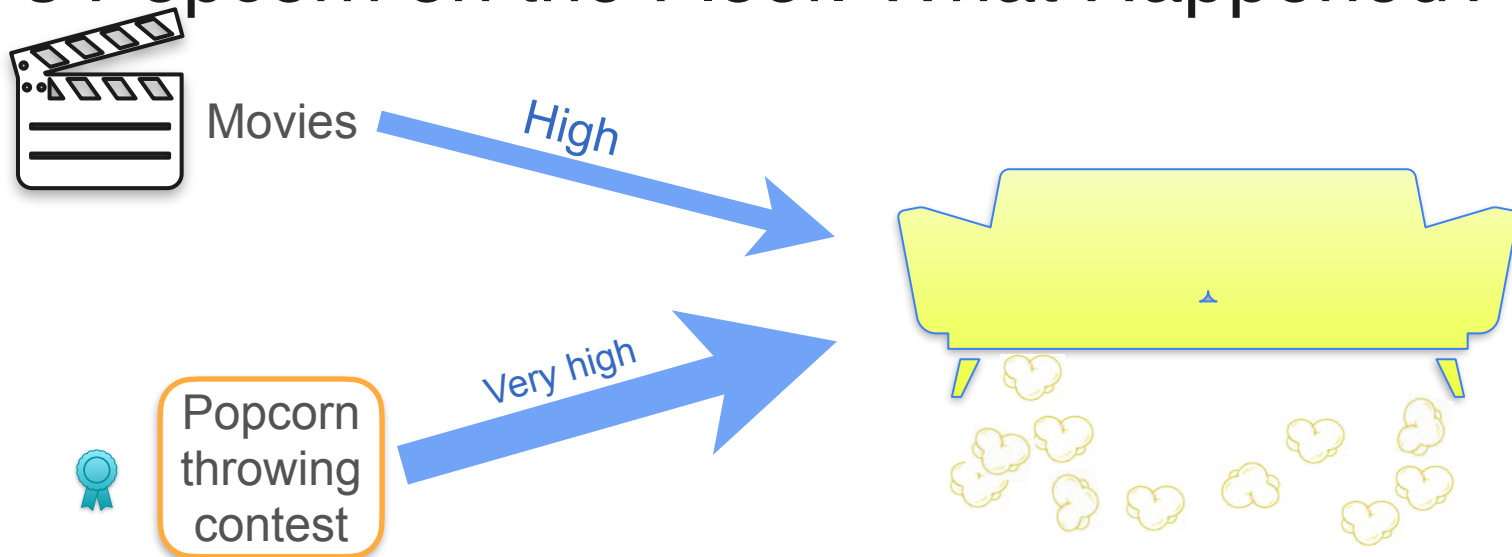
Very high



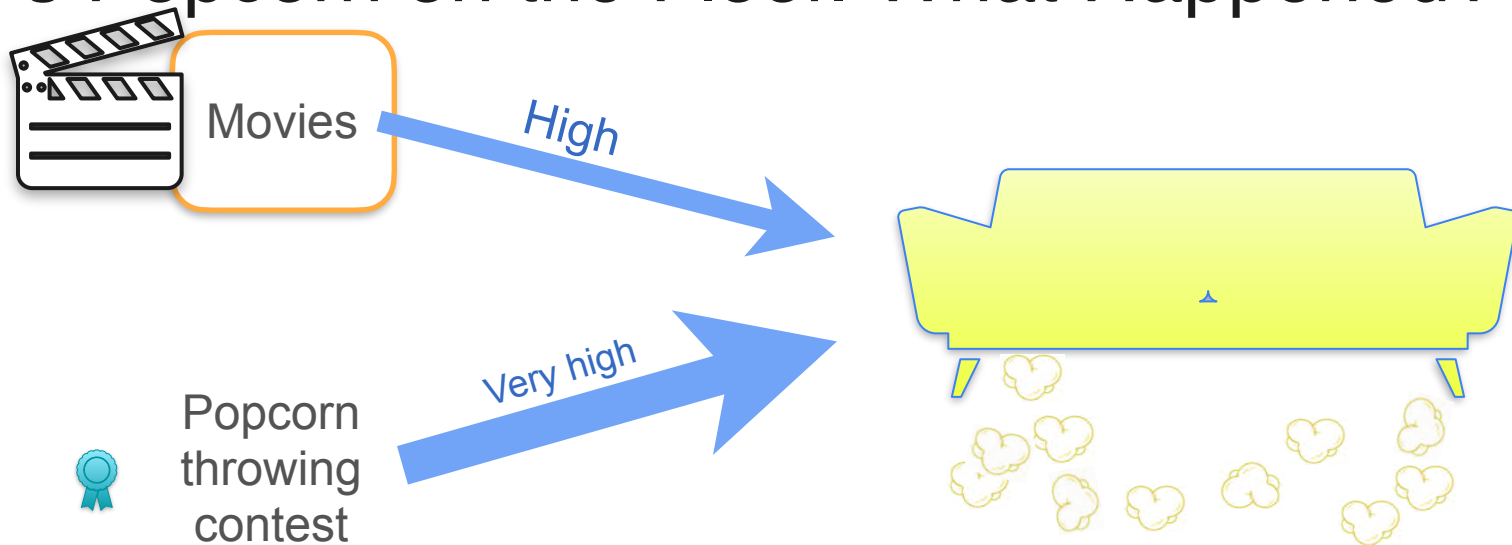
There's Popcorn on the Floor. What Happened?



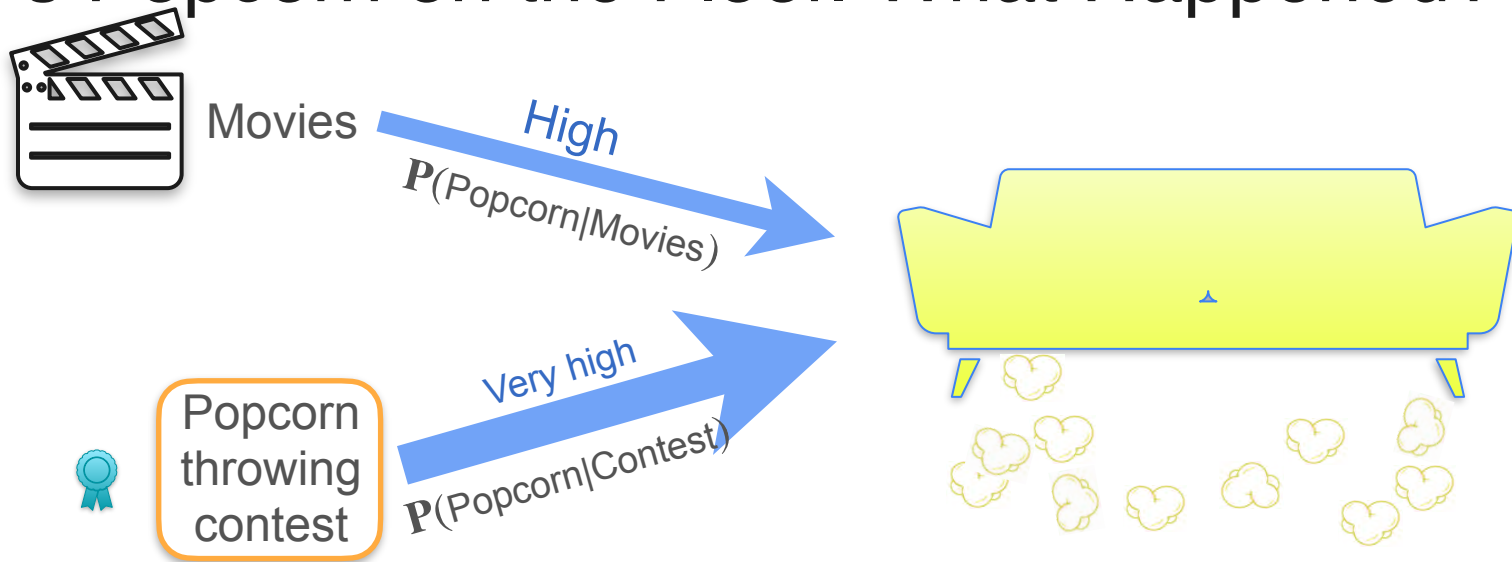
There's Popcorn on the Floor. What Happened?



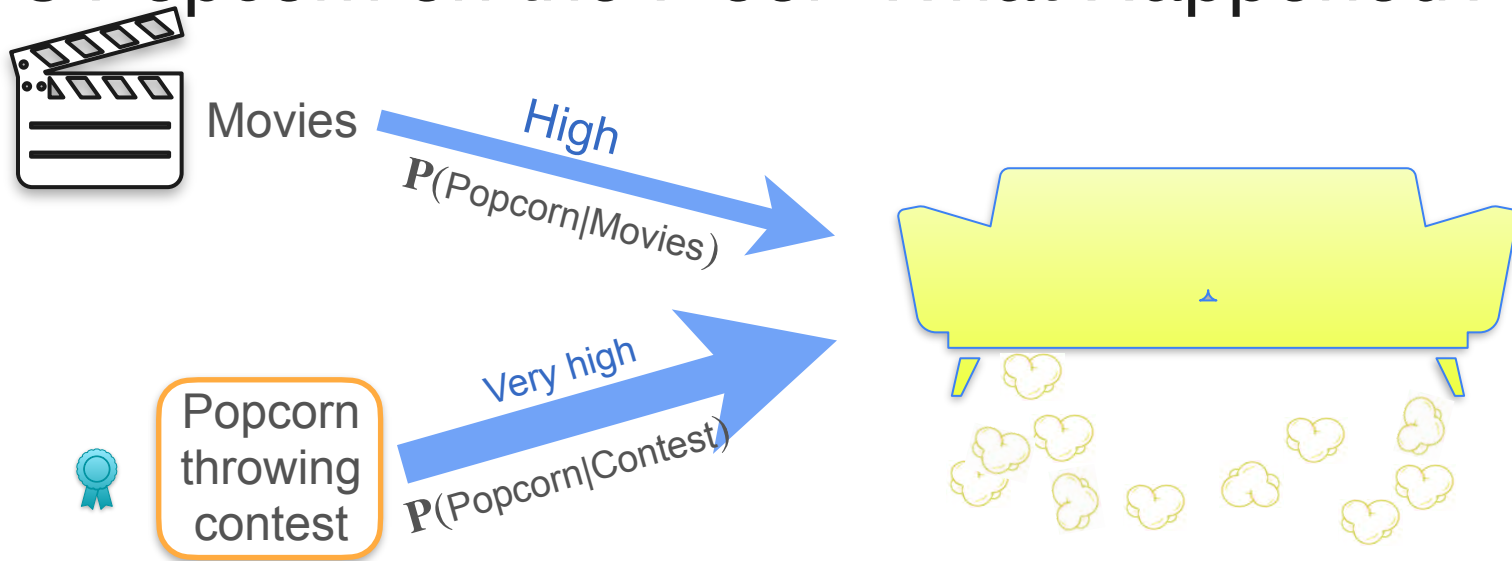
There's Popcorn on the Floor. What Happened?



There's Popcorn on the Floor. What Happened?

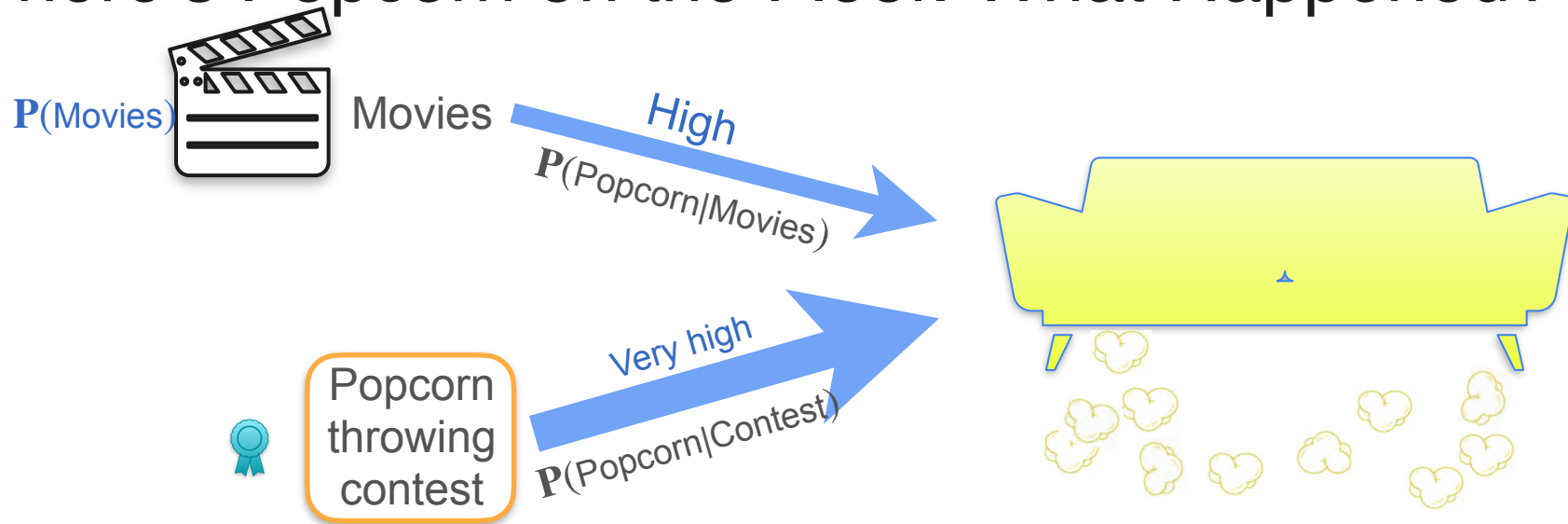


There's Popcorn on the Floor. What Happened?



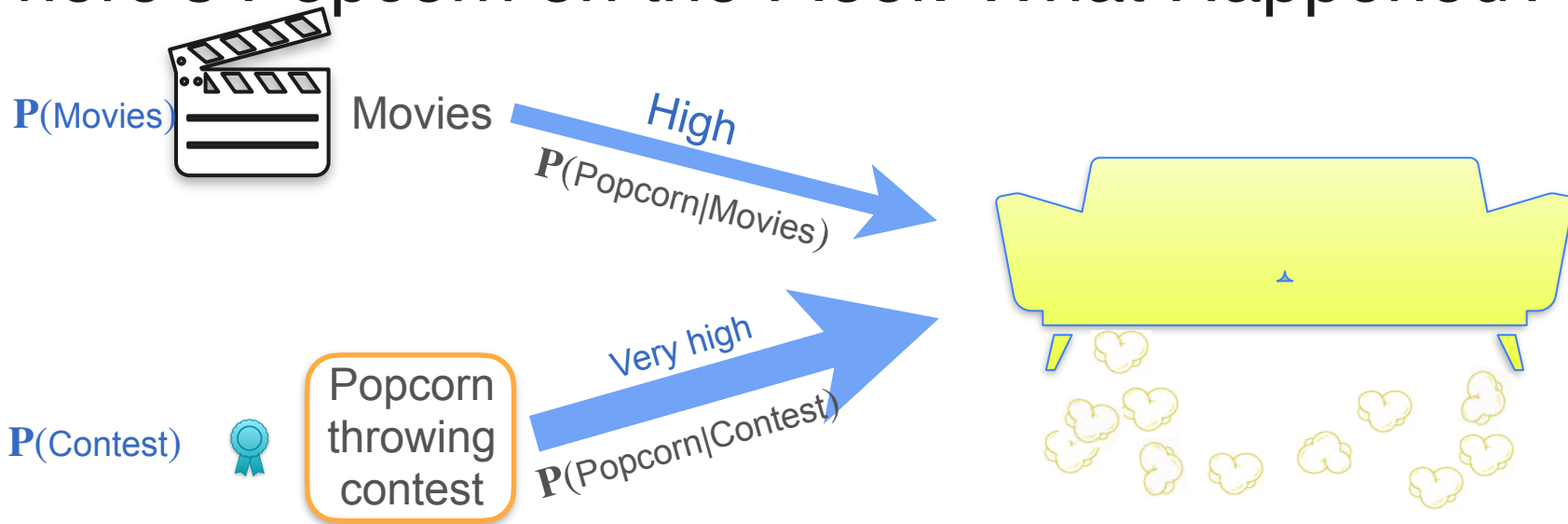
$$P(\text{Popcorn}|\text{Movies}) < P(\text{Popcorn}|\text{Contest})$$

There's Popcorn on the Floor. What Happened?

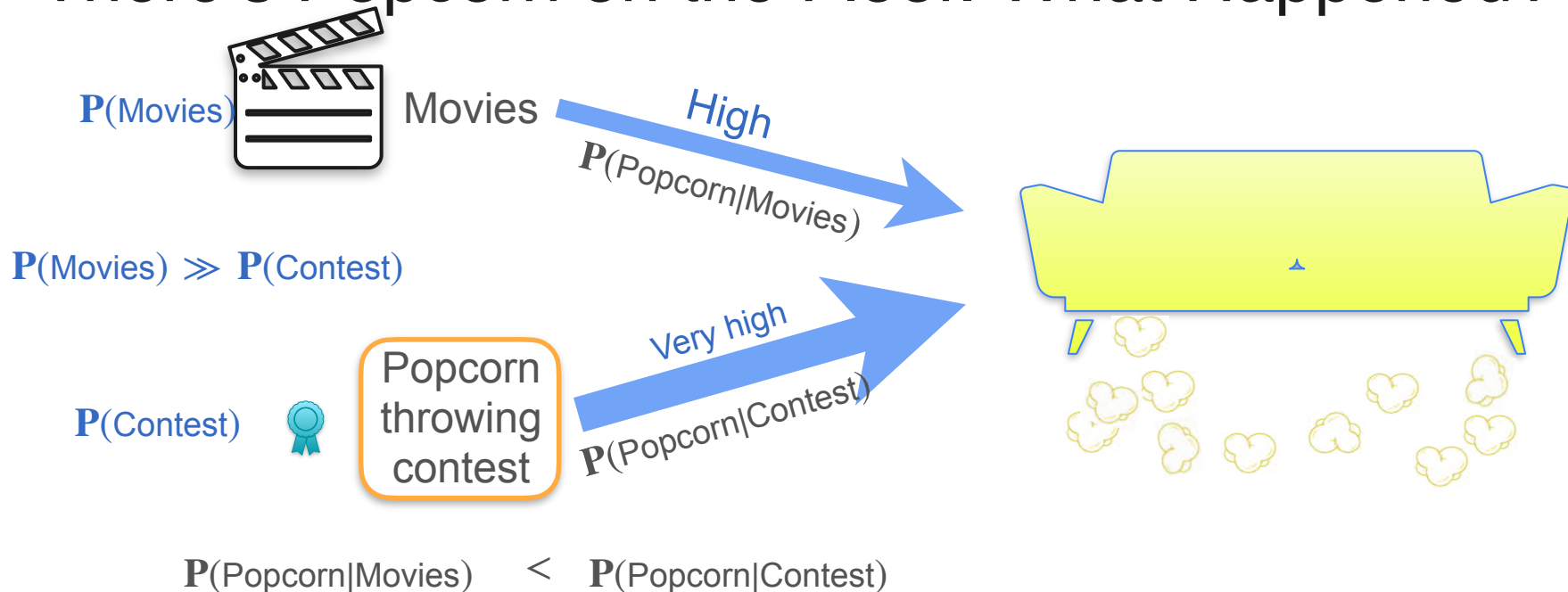


$$P(\text{Popcorn}|\text{Movies}) < P(\text{Popcorn}|\text{Contest})$$

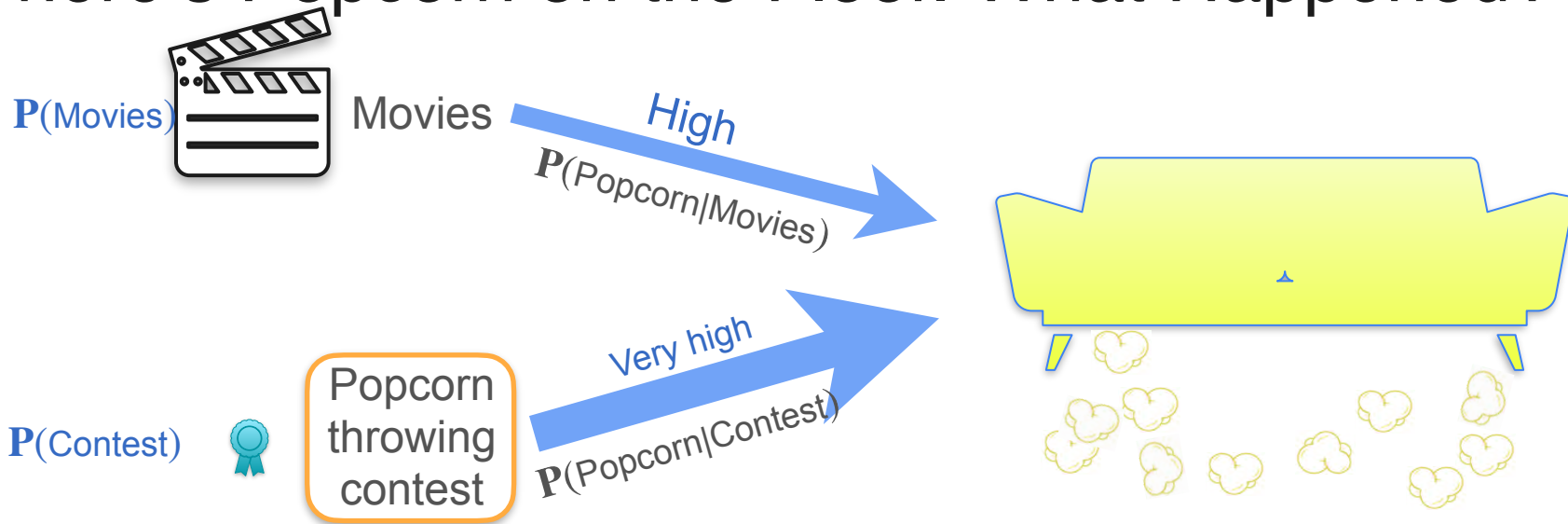
There's Popcorn on the Floor. What Happened?



There's Popcorn on the Floor. What Happened?

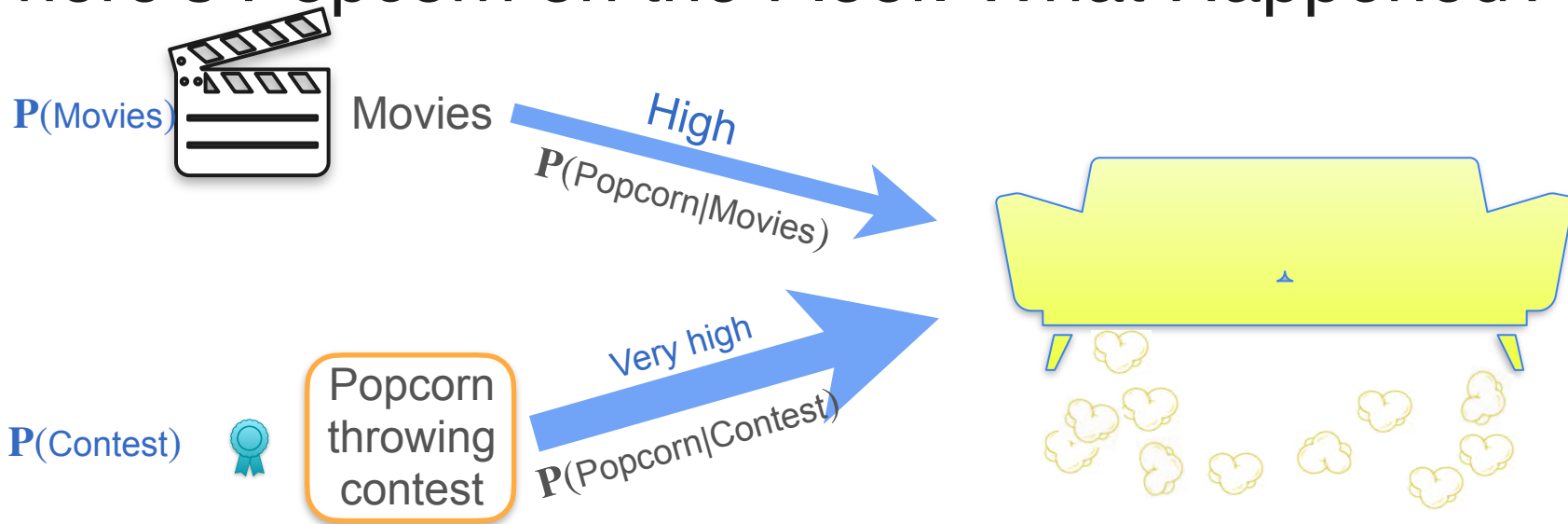


There's Popcorn on the Floor. What Happened?



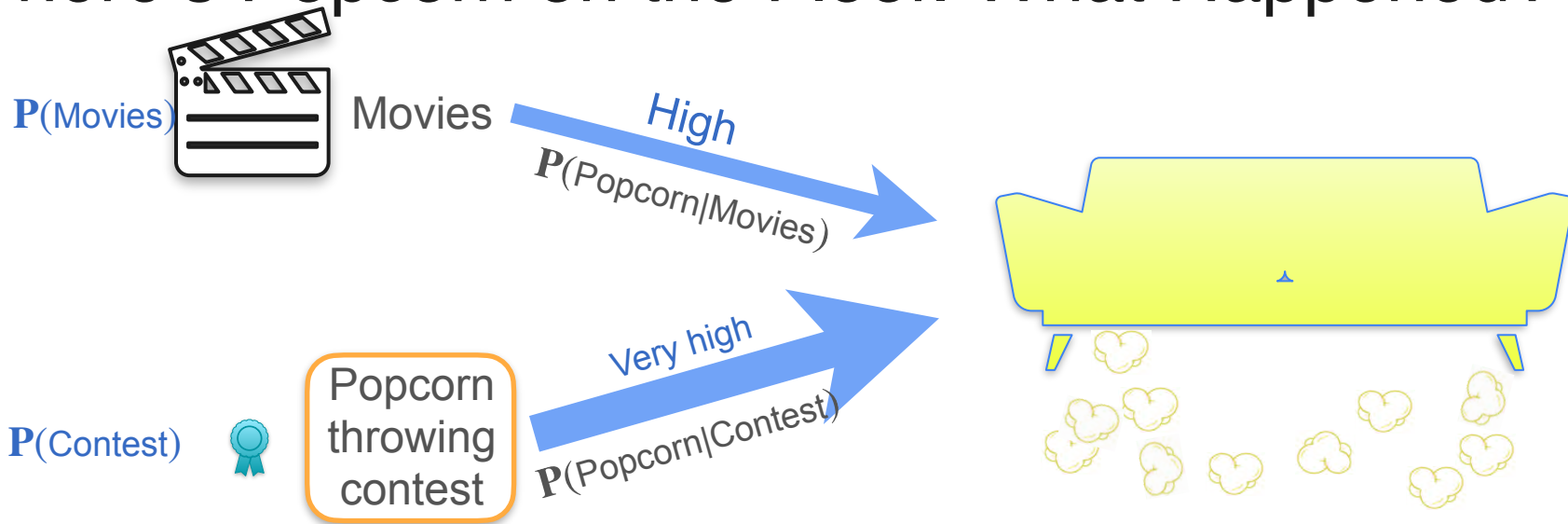
$$P(\text{Popcorn}|\text{Movies}) P(\text{Movies}) < P(\text{Popcorn}|\text{Contest}) P(\text{Contest})$$

There's Popcorn on the Floor. What Happened?



$$P(\text{Popcorn}|\text{Movies}) P(\text{Movies}) > P(\text{Popcorn}|\text{Contest}) P(\text{Contest})$$

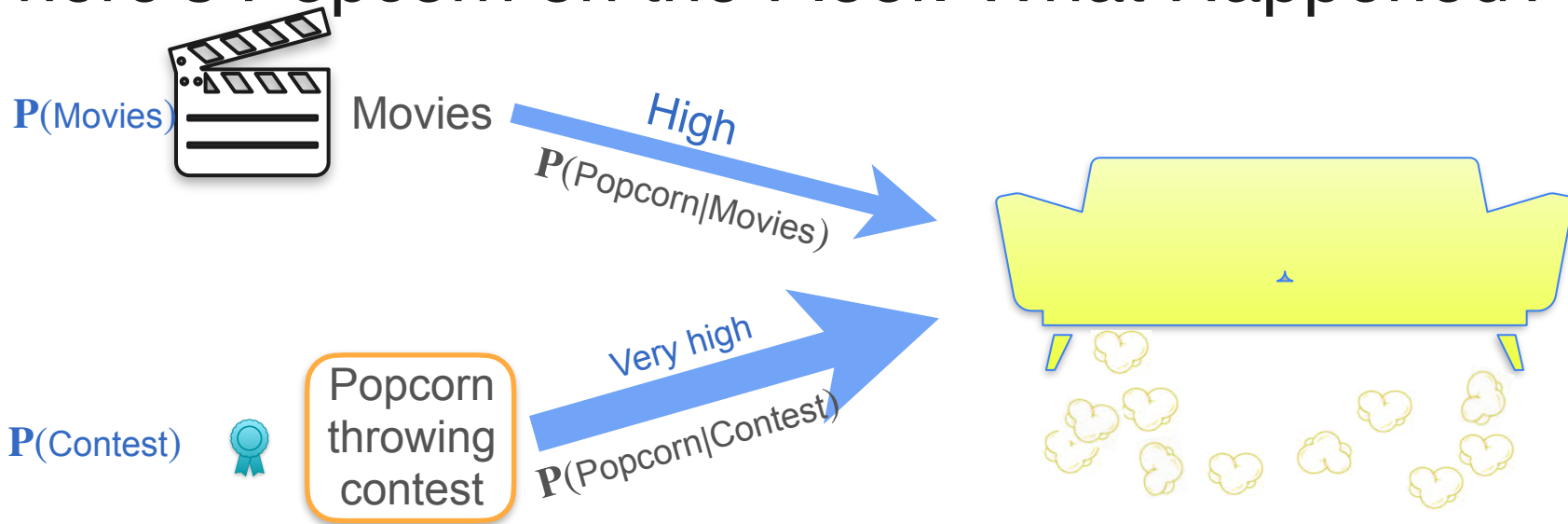
There's Popcorn on the Floor. What Happened?



$$P(\text{Popcorn}|\text{Movies}) P(\text{Movies}) > P(\text{Popcorn}|\text{Contest}) P(\text{Contest})$$

$$P(A | B)P(B) = P(A \cap B)$$

There's Popcorn on the Floor. What Happened?

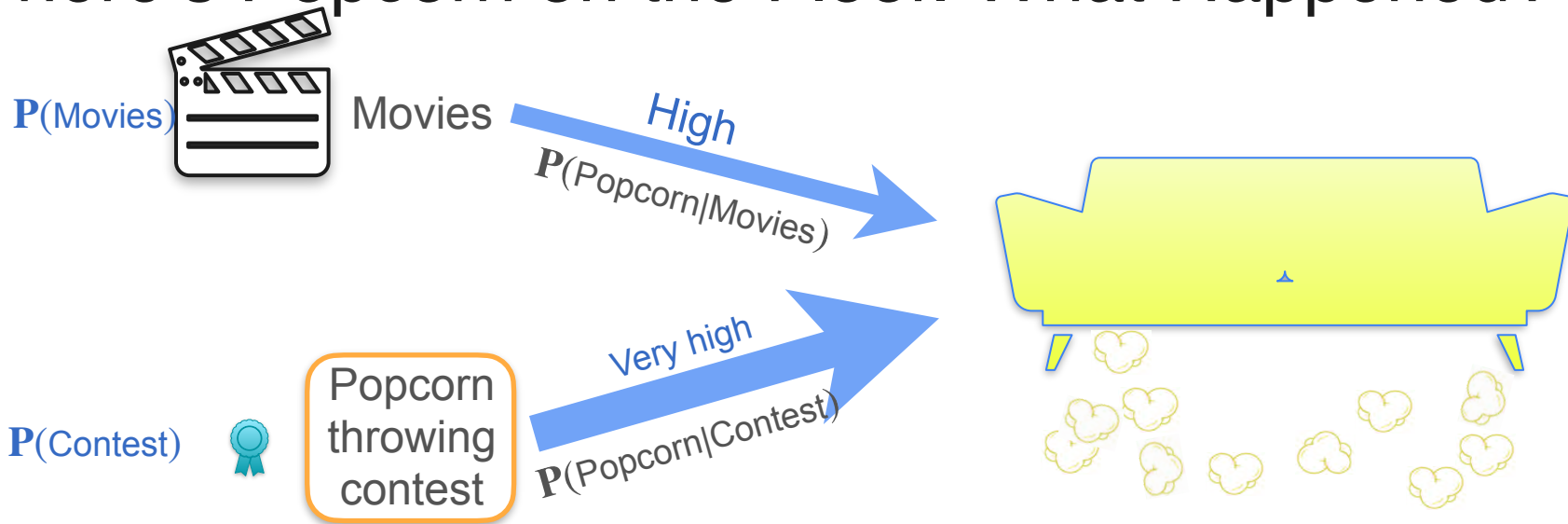


$$P(\text{Popcorn}|\text{Movies}) P(\text{Movies}) > P(\text{Popcorn}|\text{Contest}) P(\text{Contest})$$

$$P(\text{Popcorn} \cap \text{Movie})$$

$$P(A | B)P(B) = P(A \cap B)$$

There's Popcorn on the Floor. What Happened?



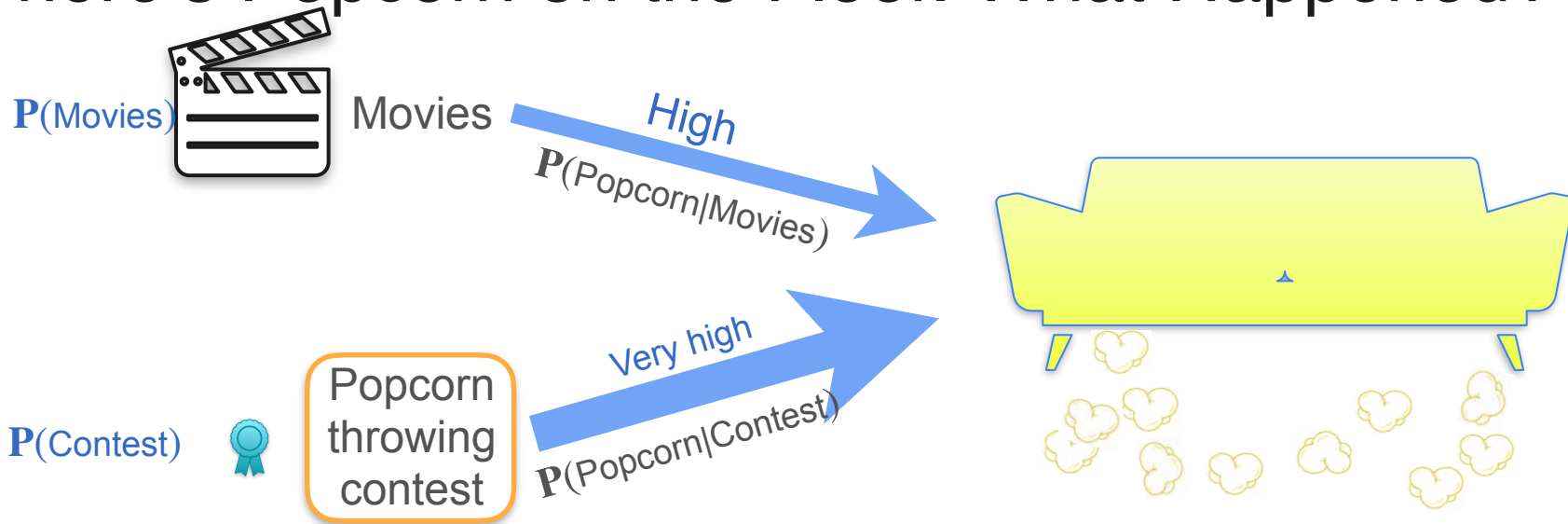
$$P(\text{Popcorn}|\text{Movies}) P(\text{Movies}) > P(\text{Popcorn}|\text{Contest}) P(\text{Contest})$$

$$P(\text{Popcorn} \cap \text{Movie})$$

$$P(\text{Popcorn} \cap \text{Contest})$$

$$P(A|B)P(B) = P(A \cap B)$$

There's Popcorn on the Floor. What Happened?



$$P(\text{Popcorn}|\text{Movies}) P(\text{Movies}) > P(\text{Popcorn}|\text{Contest}) P(\text{Contest})$$

$$P(\text{Popcorn} \cap \text{Movie}) > P(\text{Popcorn} \cap \text{Contest})$$

$$P(A|B)P(B) = P(A \cap B)$$

What Does This Have To Do With Regularization?



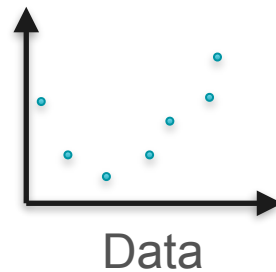
DeepLearning.AI

Point Estimation

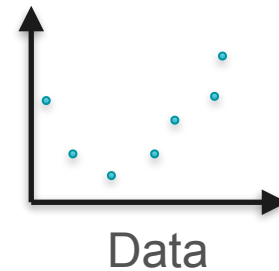
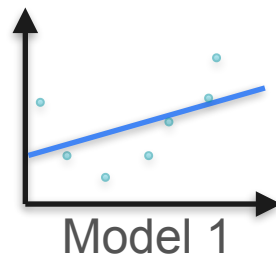
Bayes Theorem and Regularization

Example: Polynomial Regression

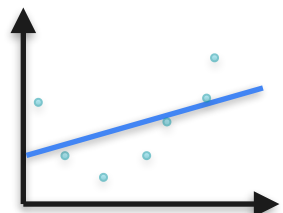
Example: Polynomial Regression



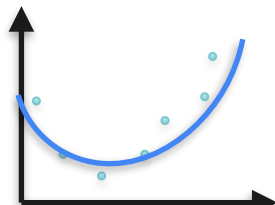
Example: Polynomial Regression



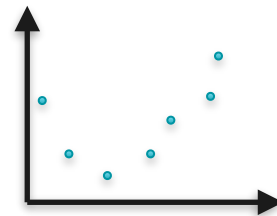
Example: Polynomial Regression



Model 1

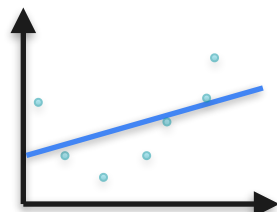


Model 2

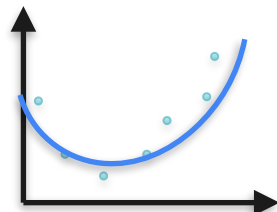


Data

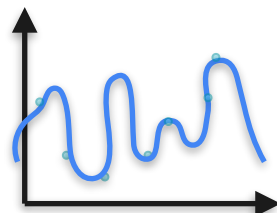
Example: Polynomial Regression



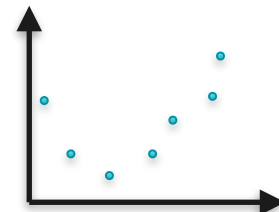
Model 1



Model 2

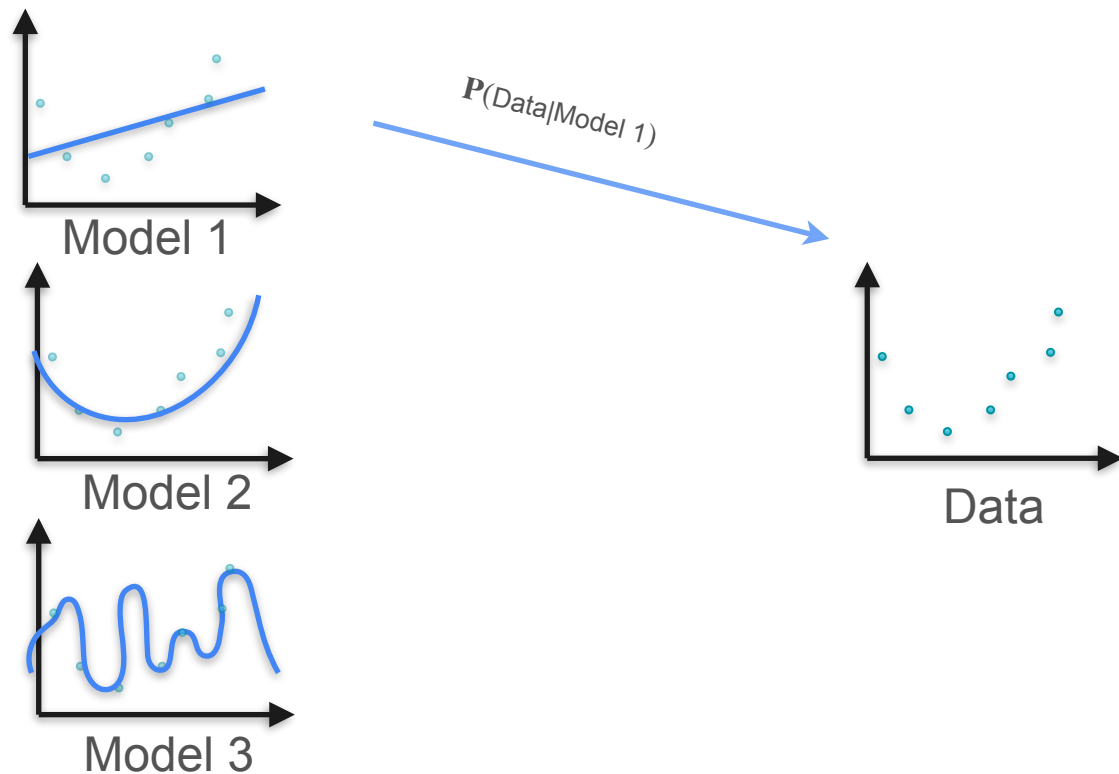


Model 3

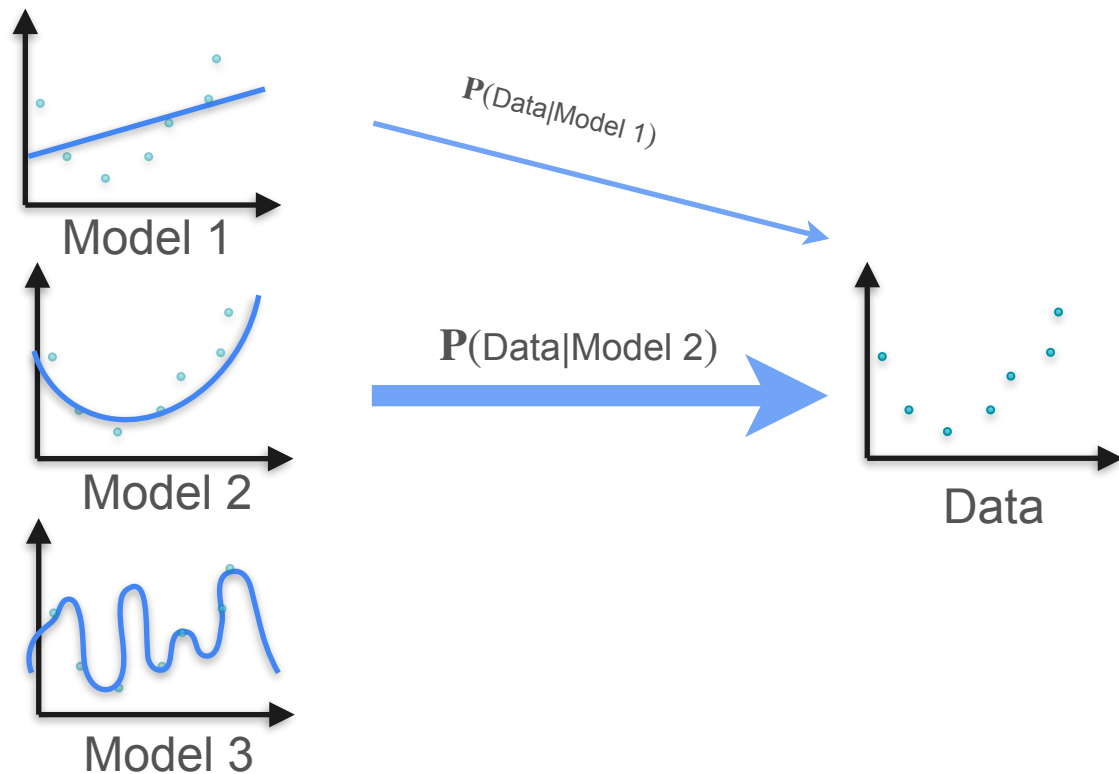


Data

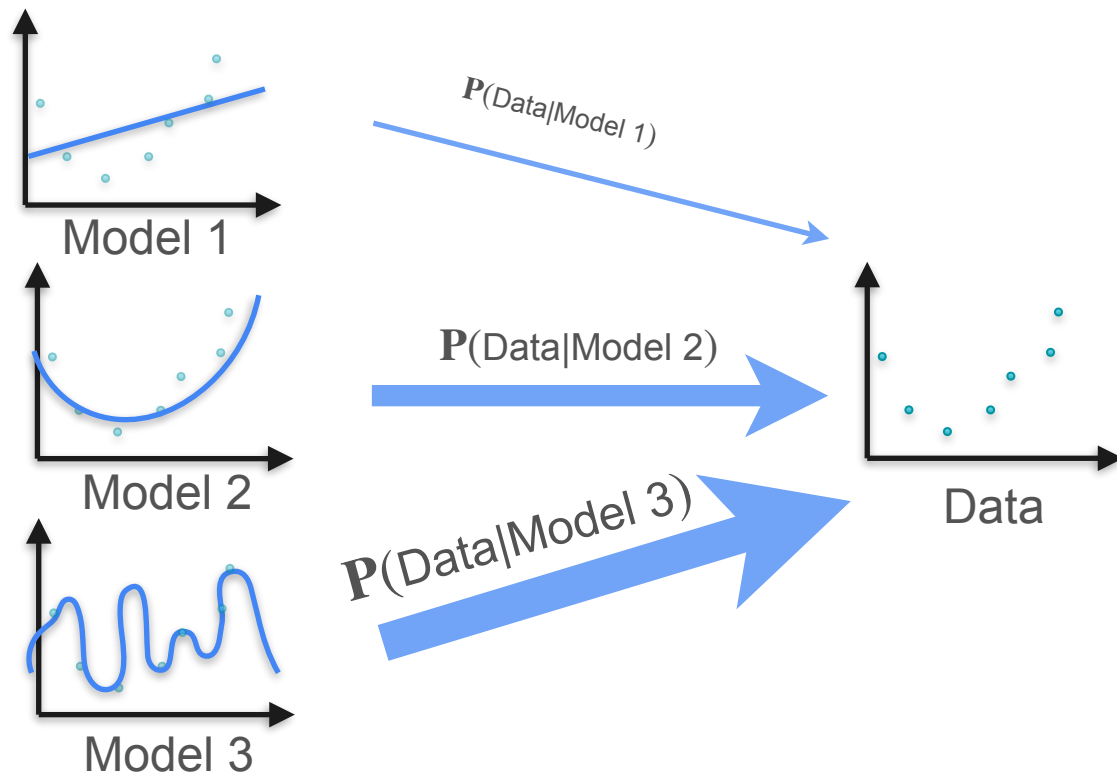
Example: Polynomial Regression



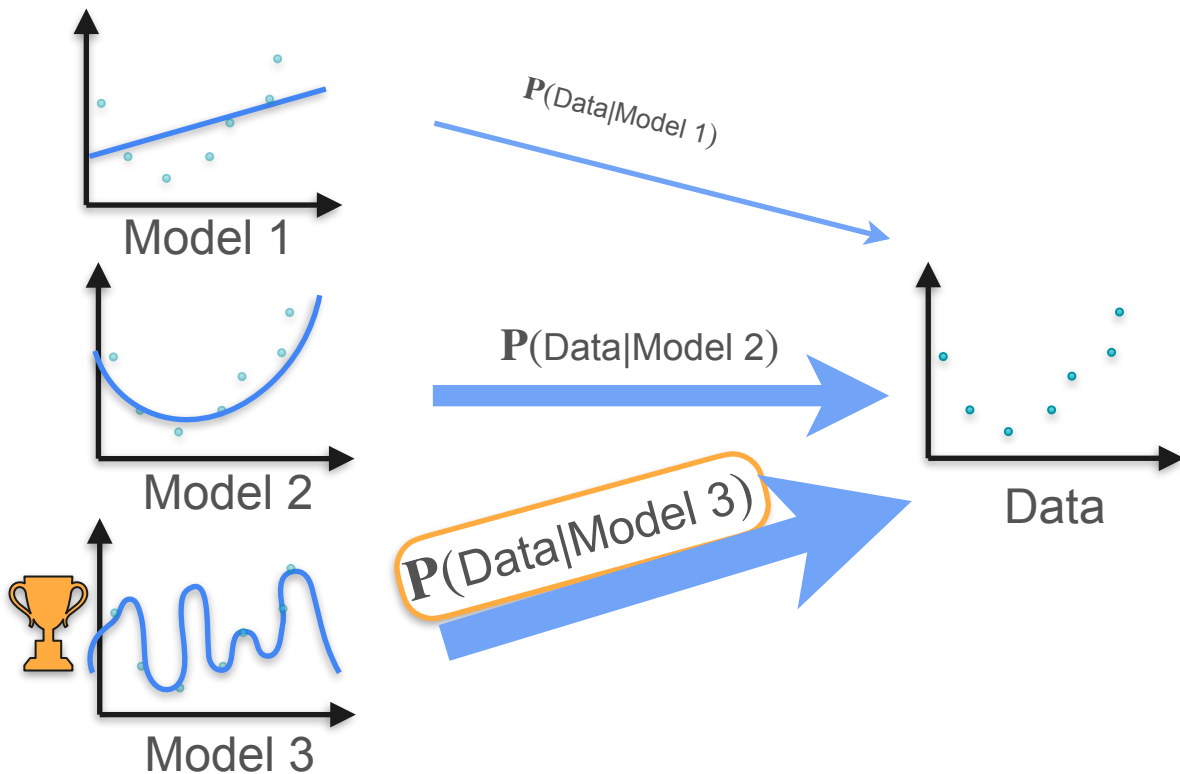
Example: Polynomial Regression



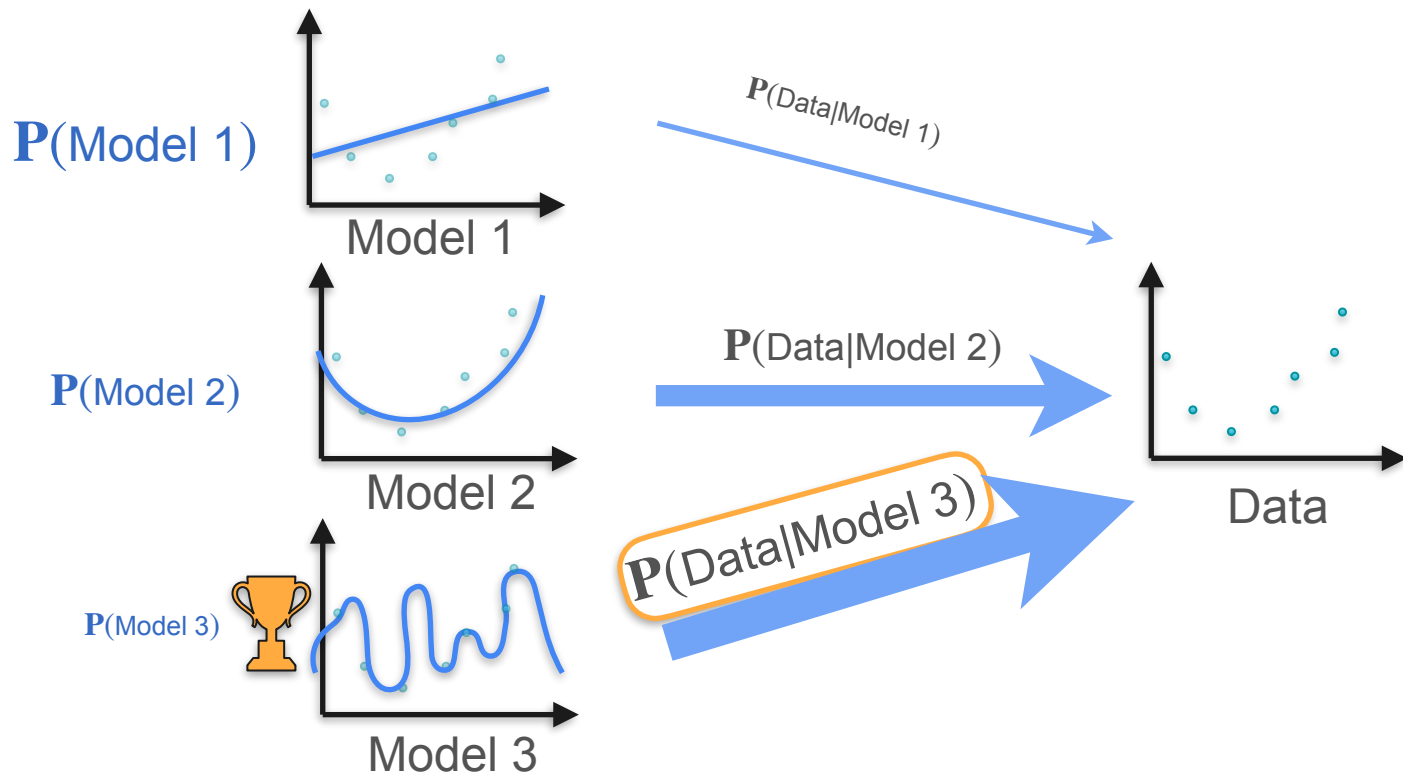
Example: Polynomial Regression



Example: Polynomial Regression

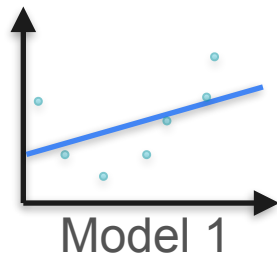


Example: Polynomial Regression

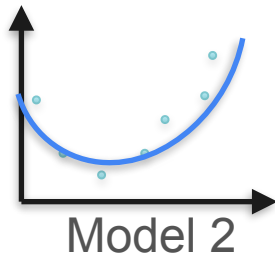


Example: Polynomial Regression

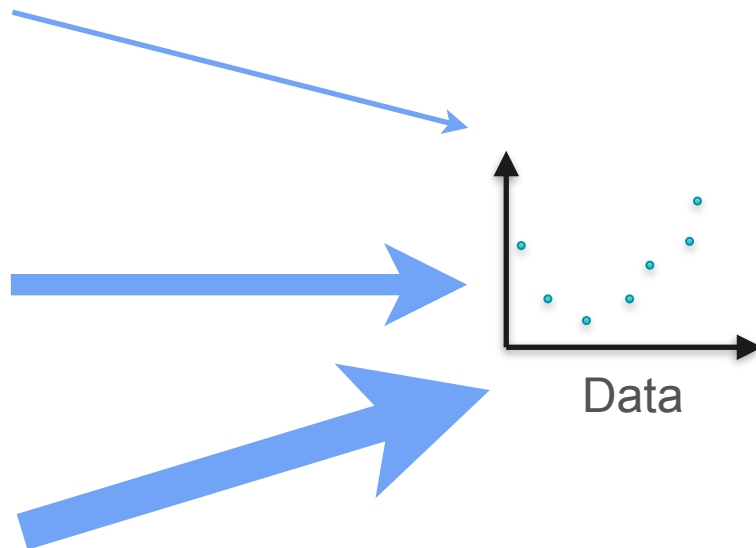
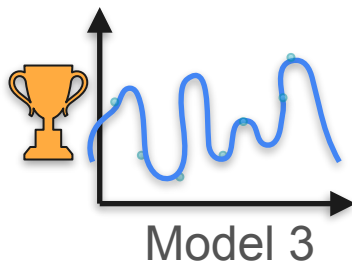
$P(\text{Model 1})$ $P(\text{Data}|\text{Model 1})$



$P(\text{Model 2})$ $P(\text{Data}|\text{Model 2})$

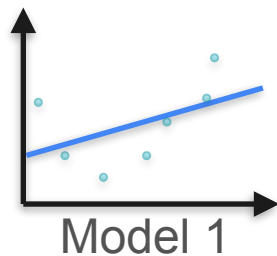


$P(\text{Model 3})$ $P(\text{Data}|\text{Model 3})$

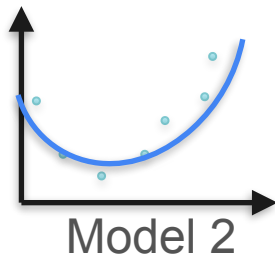


Example: Polynomial Regression

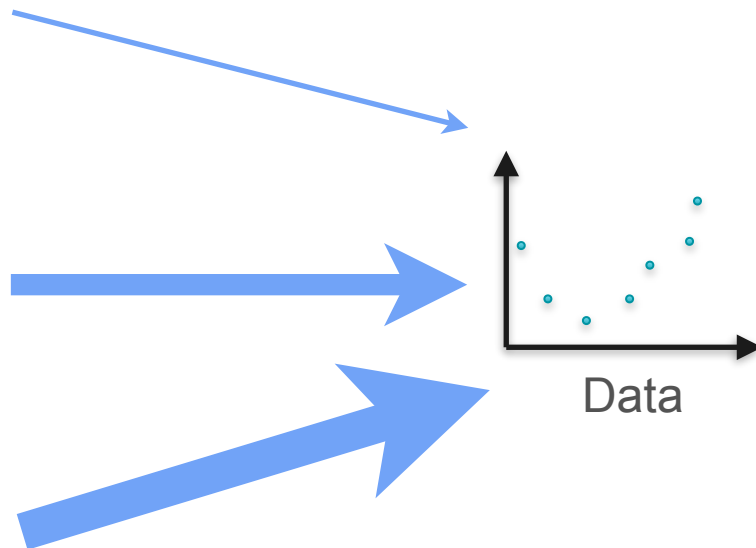
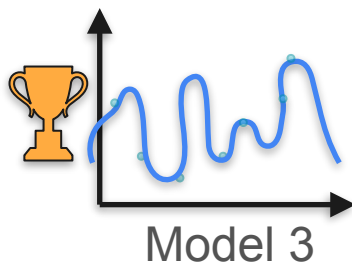
$P(\text{Model 1})$ $P(\text{Data}|\text{Model 1})$



$P(\text{Model 2})$ $P(\text{Data}|\text{Model 2})$

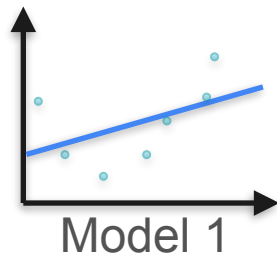


$P(\text{Model 3})$ $P(\text{Data}|\text{Model 3})$

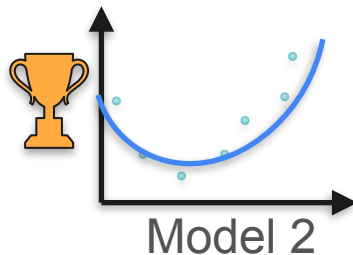


Example: Polynomial Regression

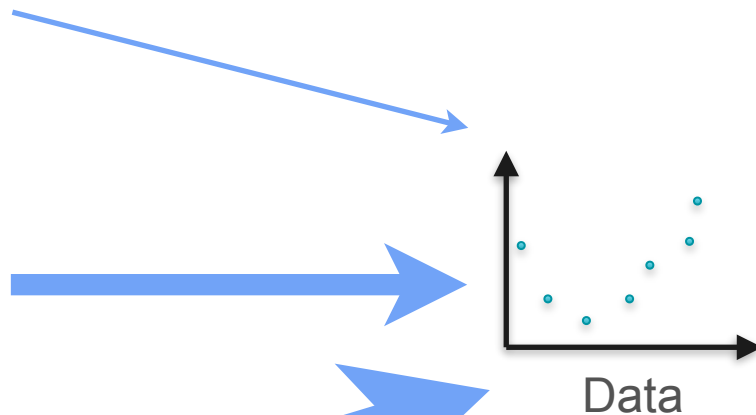
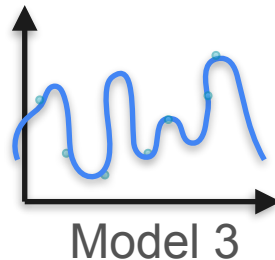
$P(\text{Model 1})$ $P(\text{Data}|\text{Model 1})$



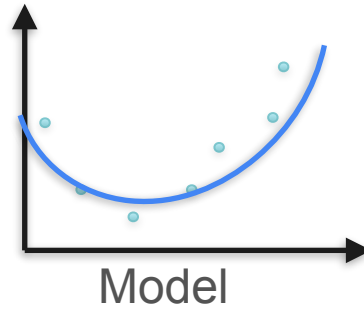
$P(\text{Model 2})$ $P(\text{Data}|\text{Model 2})$



$P(\text{Model 3})$ $P(\text{Data}|\text{Model 3})$

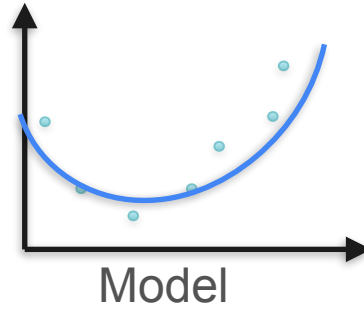


Maximum likelihood



Polynomial regression

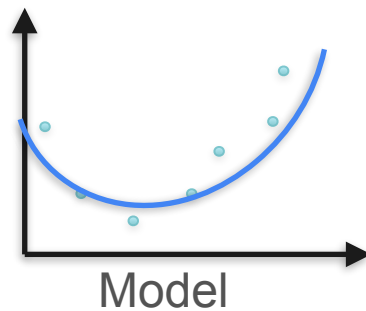
Maximum likelihood



Polynomial regression

Log-loss

Maximum likelihood

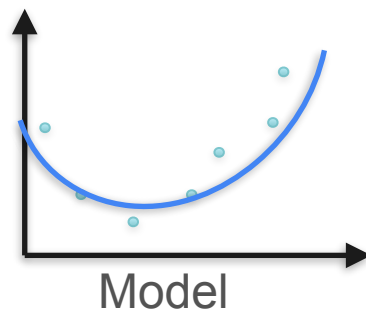


Polynomial regression

$\mathbf{P}(\text{Data}|\text{Model})$

Log-loss

**Maximum likelihood
with Bayes**

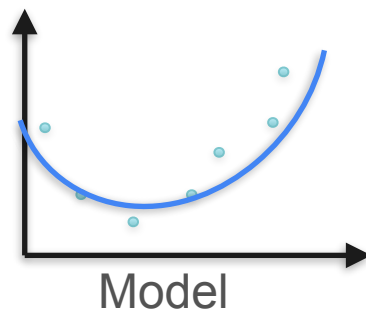


Polynomial regression

$\mathbf{P}(\text{Data}|\text{Model})$

Log-loss

**Maximum likelihood
with Bayes**



Polynomial regression

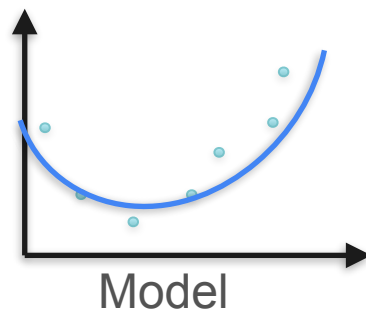
$P(\text{Data}|\text{Model})$

Log-loss

•

$P(\text{Model})$

**Maximum likelihood
with Bayes**



**Polynomial regression
with regularization**

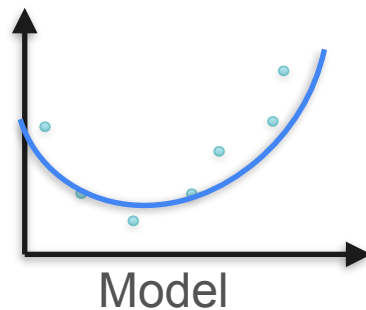
$P(\text{Data}|\text{Model})$

Log-loss

•

$P(\text{Model})$

**Maximum likelihood
with Bayes**



$P(\text{Data}|\text{Model})$

•

$P(\text{Model})$

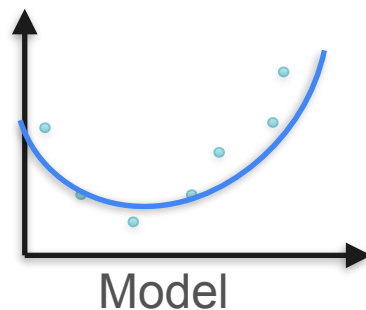
**Polynomial regression
with regularization**

Log-loss

+

Regularization term

Maximum likelihood
with Bayes



Polynomial regression
with regularization

$P(\text{Data}|\text{Model})$

•

$P(\text{Model})$

Take logarithms!

Log-loss

+

Regularization term

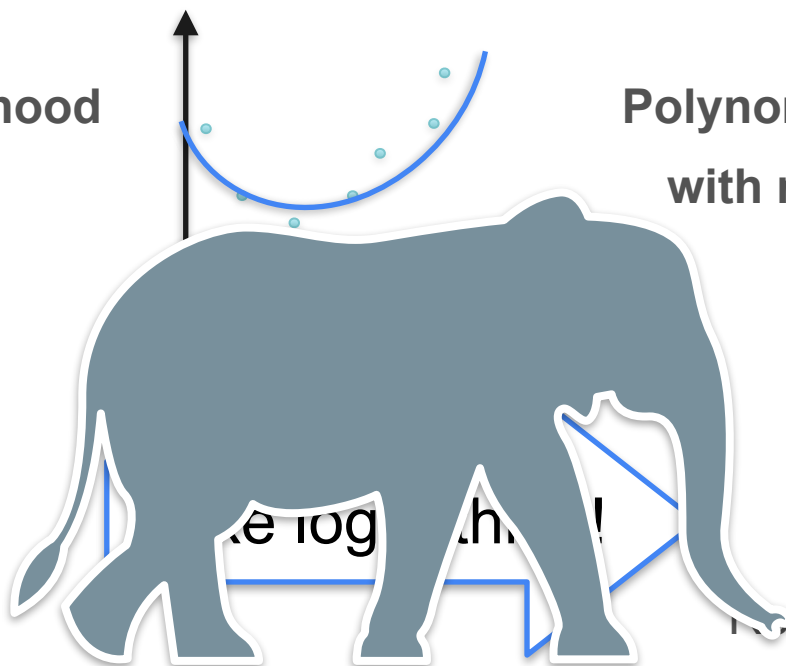
Maximum likelihood
with Bayes

Polynomial regression
with regularization

$P(\text{Data}|\text{Model})$

•

$P(\text{Model})$



Log-loss

+

Regularization term

Maximum likelihood
with Bayes

Polynomial regression
with regularization

$P(\text{Data}|\text{Model})$

$P(\text{Model})$

?

$\propto e^{\log \text{likelihood}}$

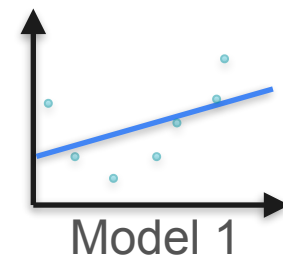
Log-loss

+

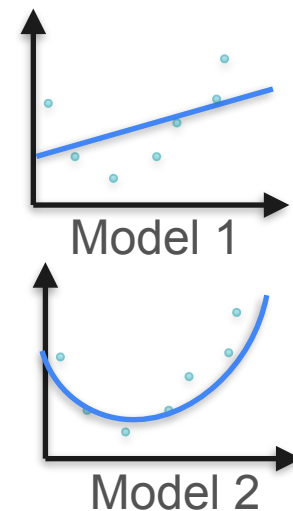
regularization term

What Is the Probability of a Model?

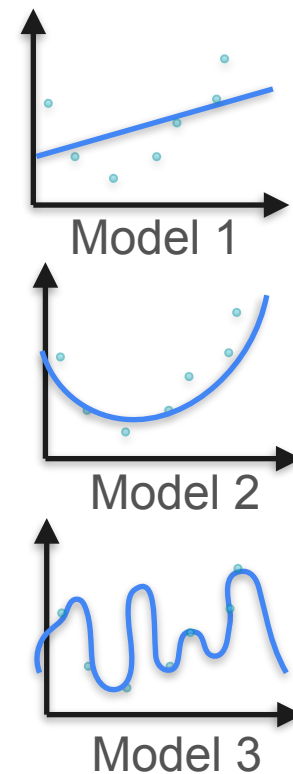
What Is the Probability of a Model?



What Is the Probability of a Model?

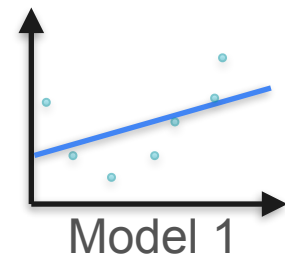


What Is the Probability of a Model?

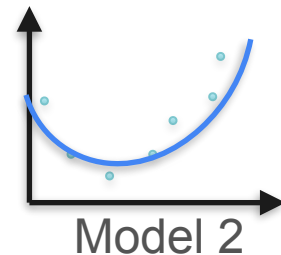


What Is the Probability of a Model?

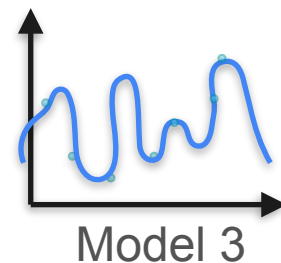
$P(\text{Model 1})$



$P(\text{Model 2})$



$P(\text{Model 3})$



What Is the Probability of a Model?

$P(\text{Model 1})$

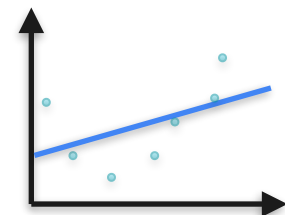
$$a_1x + b$$

$P(\text{Model 2})$

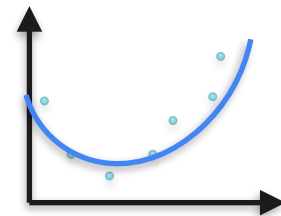
$$a_1x^2 + a_2x + b$$

$P(\text{Model 3})$

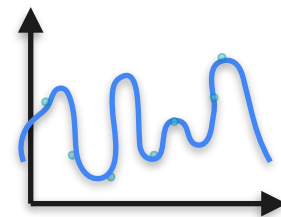
$$a_1x^{10} + a_2x^9 + \dots + a_{10}x + b$$



Model 1



Model 2

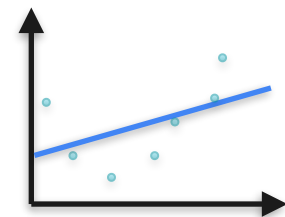


Model 3

What Is the Probability of a Model?

$P(\text{Model 1})$

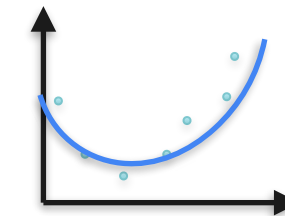
$$a_1x + b$$



Model 1

$P(\text{Model 2})$

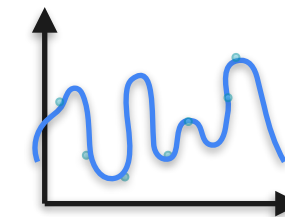
$$a_1x^2 + a_2x + b$$



Model 2

$P(\text{Model 3})$

$$a_1x^{10} + a_2x^9 + \dots + a_{10}x + b$$



Model 3

What Is the Probability of a Model?

$\mathbf{P}(\text{Model 1})$

$$a_1x + b$$

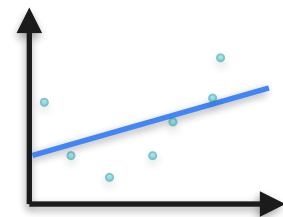
$\mathbf{P}(\text{Model 2})$

$$a_1x^2 + a_2x + b$$

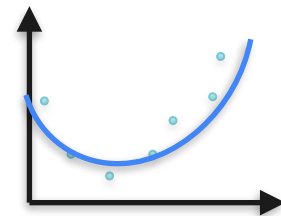
$\mathbf{P}(\text{Model 3})$

$$a_1x^{10} + a_2x^9 + \cdots + a_{10}x + b$$

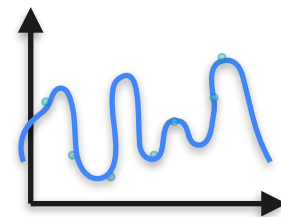
$$a_i \sim \mathcal{N}(0,1)$$



Model 1



Model 2

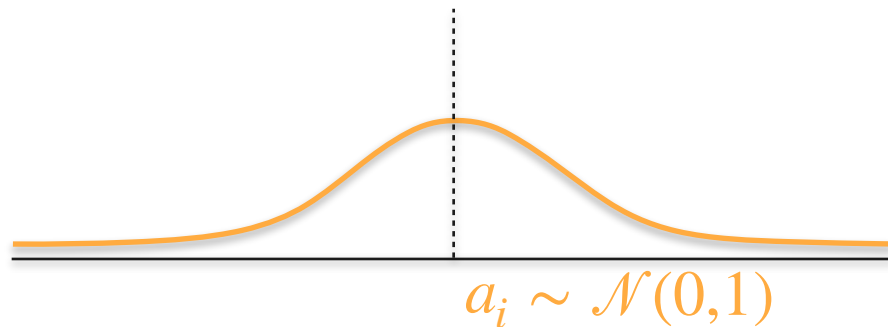


Model 3

What Is the Probability of a Model?

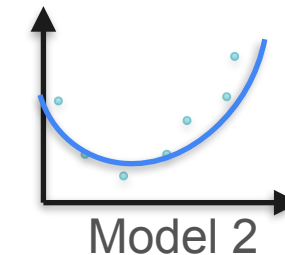
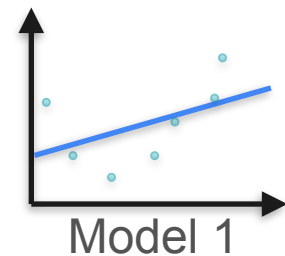
$P(\text{Model 1})$

$$a_1x + b$$



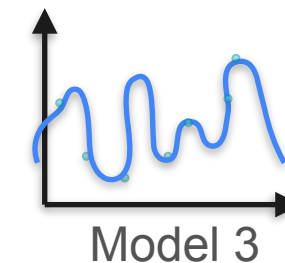
$P(\text{Model 2})$

$$a_1x^2 + a_2x + b$$



$P(\text{Model 3})$

$$a_1x^{10} + a_2x^9 + \dots + a_{10}x + b$$



What Is the Probability of a Model?

P(Model 1)

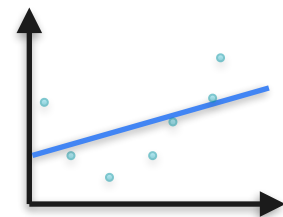
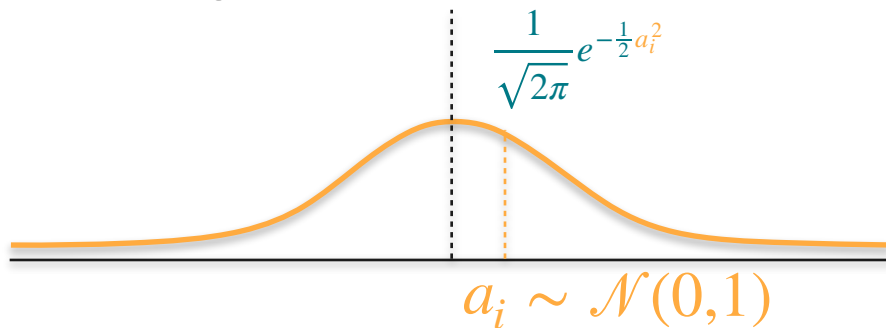
$$\boxed{a_1}x + b$$

P(Model 2)

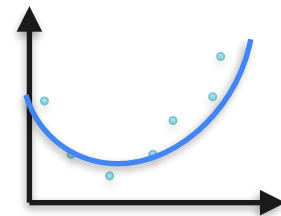
$$a_1x^2 + a_2x + b$$

P(Model 3)

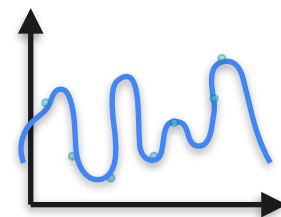
$$a_1x^{10} + a_2x^9 + \cdots + a_{10}x + b$$



Model 1



Model 2

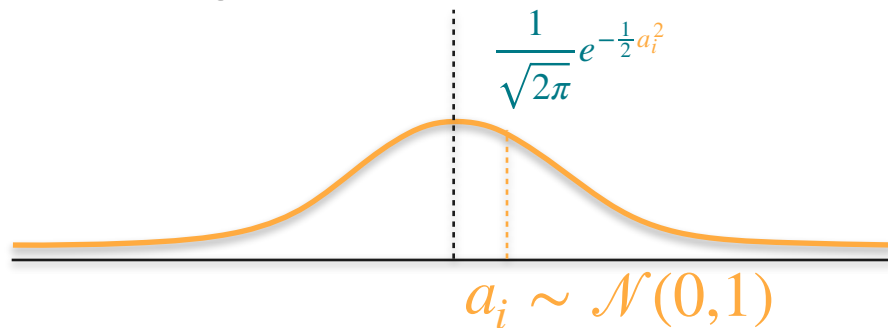


Model 3

What Is the Probability of a Model?

$$\mathbf{P}(\text{Model 1}) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2}$$

$$a_1x + b$$

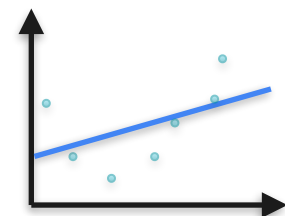


$\mathbf{P}(\text{Model 2})$

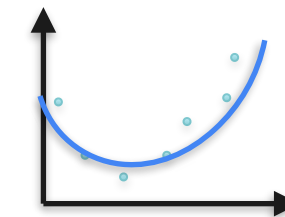
$$a_1x^2 + a_2x + b$$

$\mathbf{P}(\text{Model 3})$

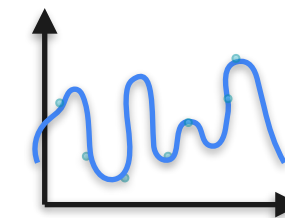
$$a_1x^{10} + a_2x^9 + \cdots + a_{10}x + b$$



Model 1



Model 2

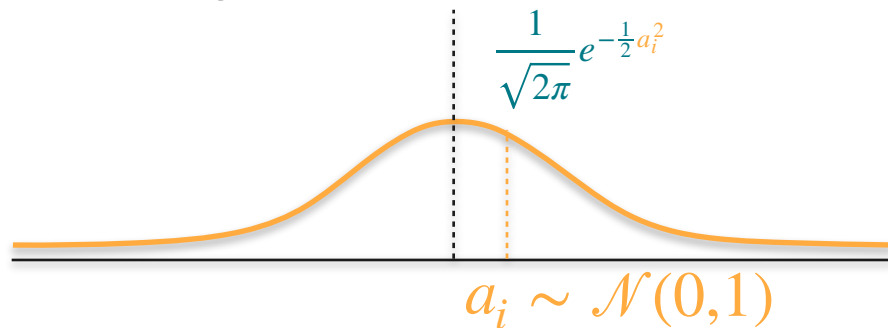


Model 3

What Is the Probability of a Model?

$$\mathbf{P}(\text{Model 1}) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2}$$

$$a_1x + b$$

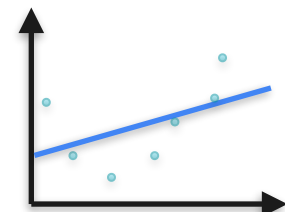



$\mathbf{P}(\text{Model 2})$

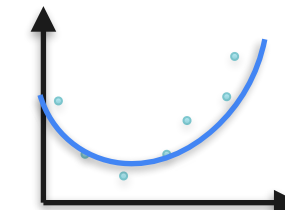
$$a_1x^2 + a_2x + b$$

$\mathbf{P}(\text{Model 3})$

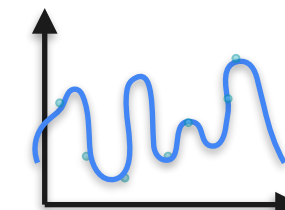
$$a_1x^{10} + a_2x^9 + \cdots + a_{10}x + b$$



Model 1



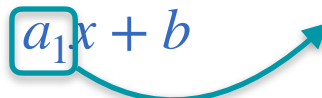
Model 2

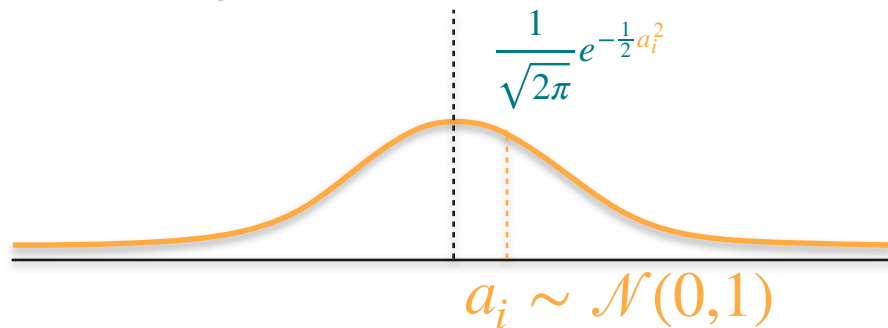


Model 3

What Is the Probability of a Model?

$$\mathbf{P}(\text{Model 1}) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2}$$

$$a_1x + b$$


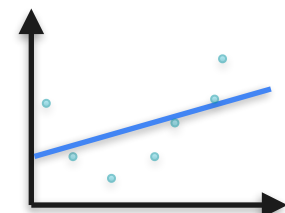


$\mathbf{P}(\text{Model 2})$

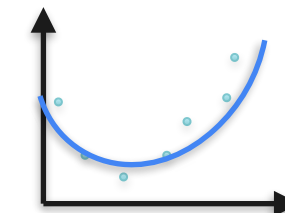
$$a_1x^2 + a_2x + b$$

$\mathbf{P}(\text{Model 3})$

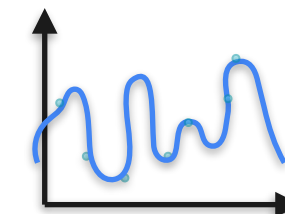
$$a_1x^{10} + a_2x^9 + \cdots + a_{10}x + b$$



Model 1



Model 2

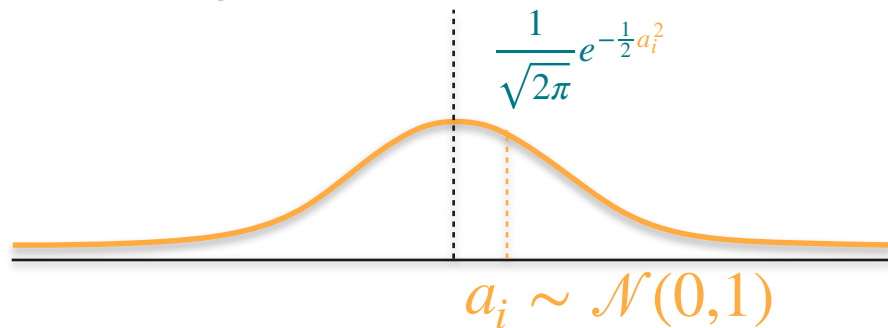


Model 3

What Is the Probability of a Model?

$$P(\text{Model 1}) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2}$$

$$a_1x + b$$

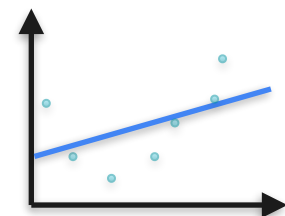


$$P(\text{Model 2})$$

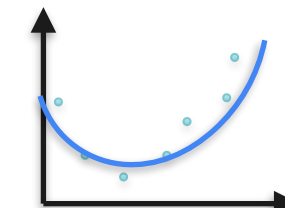
$$a_1x^2 + a_2x + b = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2}$$

$$P(\text{Model 3})$$

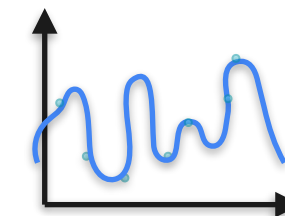
$$a_1x^{10} + a_2x^9 + \dots + a_{10}x + b$$



Model 1



Model 2

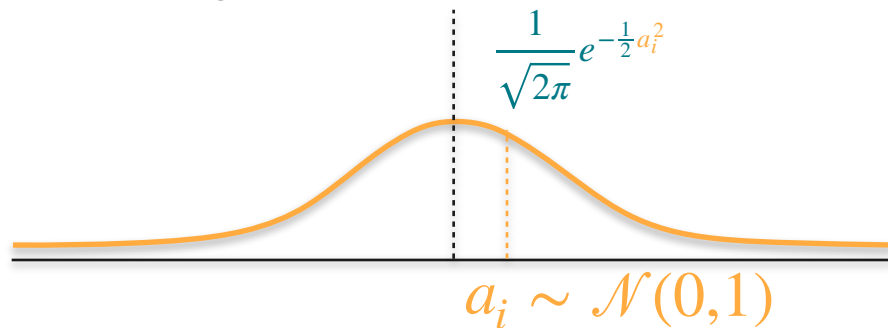


Model 3

What Is the Probability of a Model?

$$P(\text{Model 1}) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2}$$

$$a_1 x + b$$



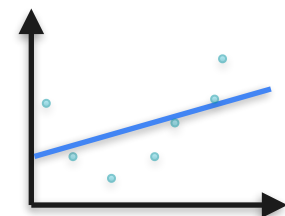
$P(\text{Model 2})$

$$a_1 x^2 + a_2 x + b = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2}$$

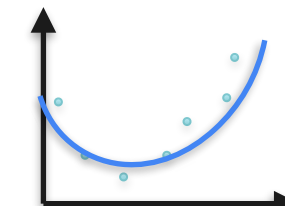
$$a_1 x^2 + a_2 x + b$$

$P(\text{Model 3})$

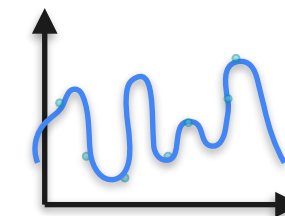
$$a_1 x^{10} + a_2 x^9 + \dots + a_{10} x + b$$



Model 1



Model 2

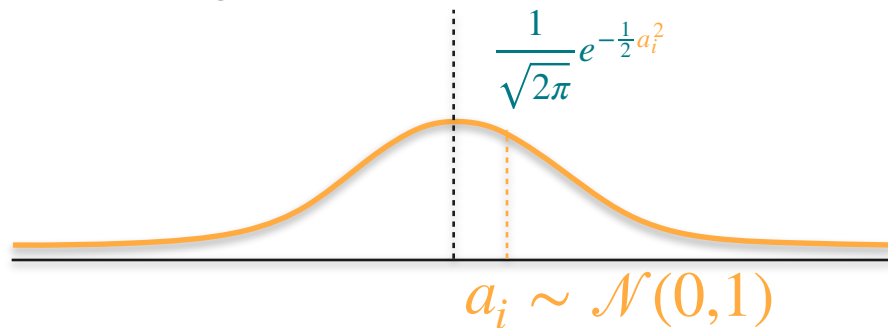


Model 3

What Is the Probability of a Model?

$$P(\text{Model 1}) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2}$$

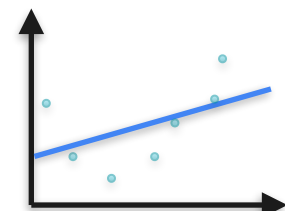
$$a_1x + b$$



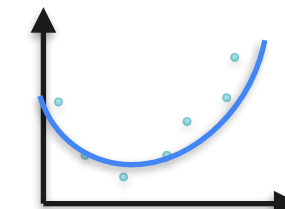
$P(\text{Model 2})$

$$a_1x^2 + a_2x + b = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2}$$

$$a_1x^2 + a_2x + b$$



Model 1

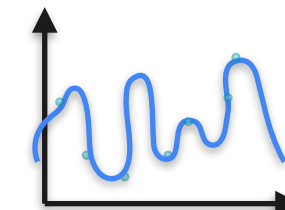


Model 2

$P(\text{Model 3})$

$$a_1x^{10} + a_2x^9 + \dots + a_{10}x + b = \prod_{i=1}^{10} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_i^2}$$

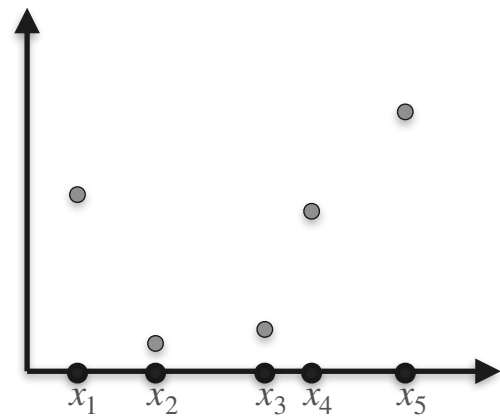
$$a_1x^{10} + a_2x^9 + \dots + a_{10}x + b$$



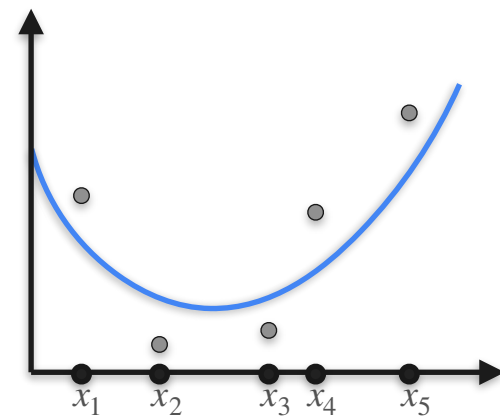
Model 3

Bayes and Regularization

Bayes and Regularization

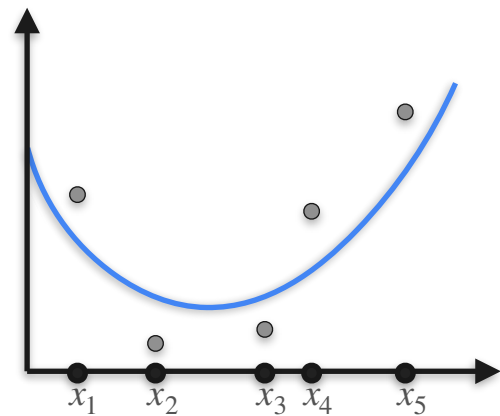


Bayes and Regularization



Bayes and Regularization

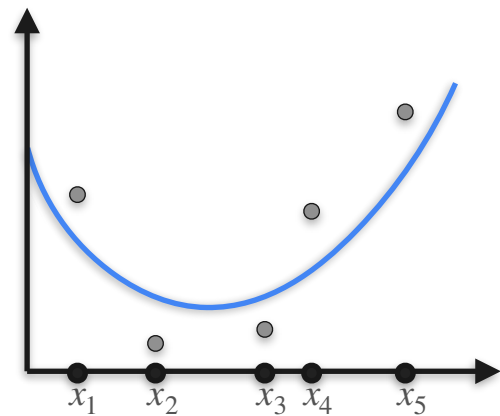
$\mathbf{P}(\text{Data}|\text{Model})$



Bayes and Regularization

$P(\text{Data}|\text{Model})$

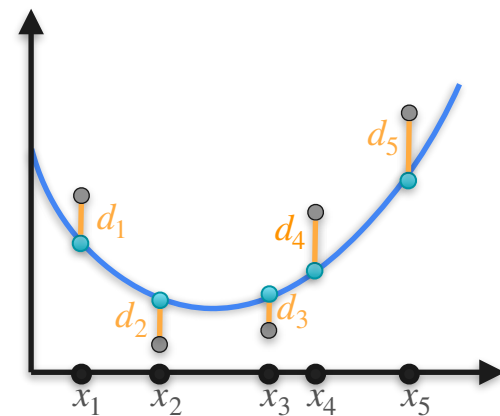
$P(\text{Model})$



Bayes and Regularization

$P(\text{Data}|\text{Model})$

$P(\text{Model})$

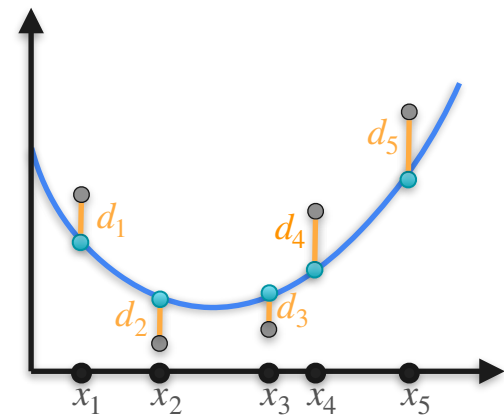


Bayes and Regularization

$P(\text{Data}|\text{Model})$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2}$$

$P(\text{Model})$

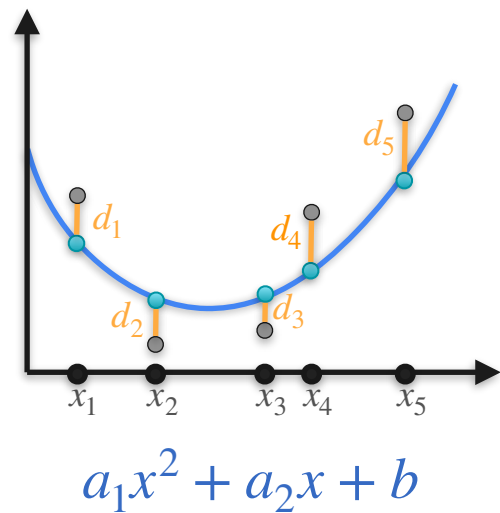


Bayes and Regularization

$P(\text{Data}|\text{Model})$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2}$$

$P(\text{Model})$

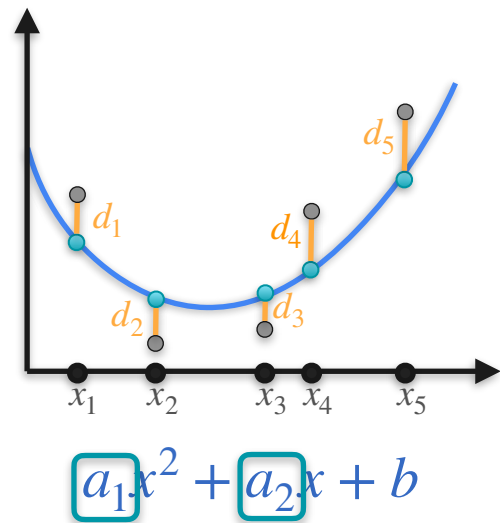


Bayes and Regularization

$P(\text{Data}|\text{Model})$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2}$$

$P(\text{Model})$

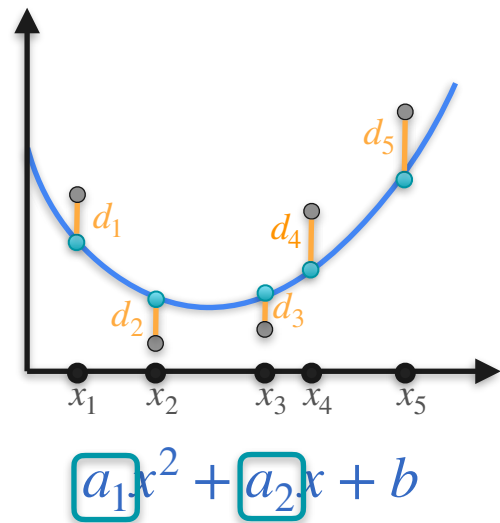


Bayes and Regularization

$P(\text{Data}|\text{Model})$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2}$$

$P(\text{Model})$

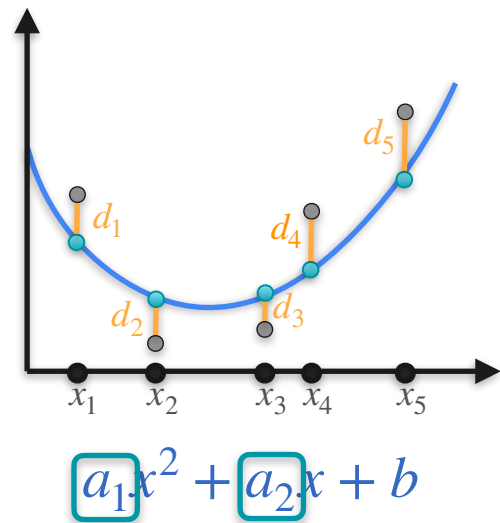


Bayes and Regularization

$P(\text{Data}|\text{Model})$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2}$$

$P(\text{Model})$

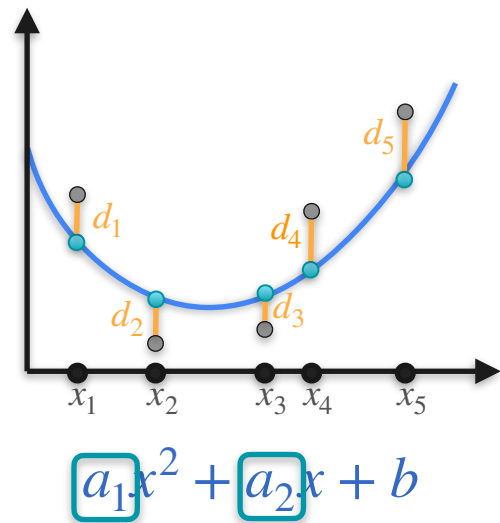


Bayes and Regularization

$P(\text{Data}|\text{Model})$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2}$$

$P(\text{Model})$



Bayes and Regularization

$P(\text{Data}|\text{Model})$

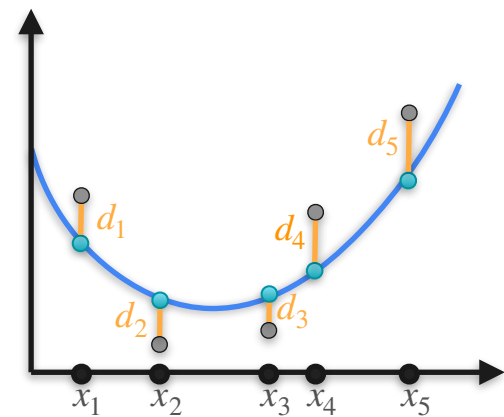
$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2}$$

log

$$-\frac{1}{2}(d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2)$$

$P(\text{Model})$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2}$$



$$a_1x^2 + a_2x + b$$

Bayes and Regularization

$P(\text{Data}|\text{Model})$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2}$$

log

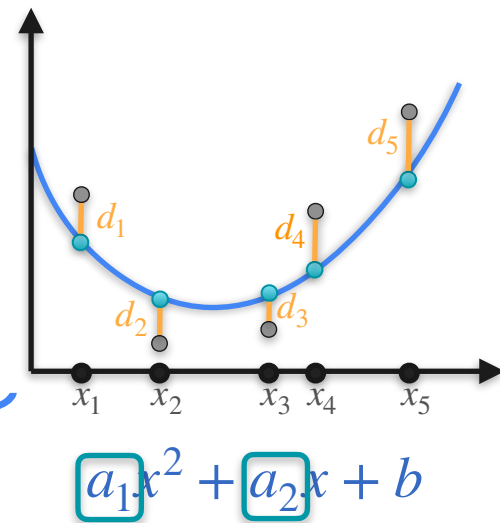
$$-\frac{1}{2}(d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2)$$

$P(\text{Model})$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2}$$

log

$$-\frac{1}{2}(a_1^2 + a_2^2)$$

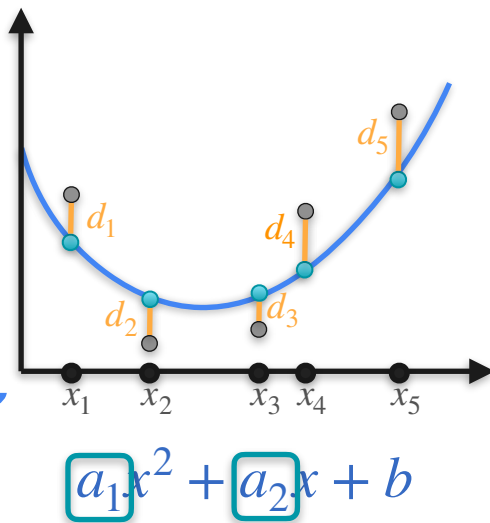


Bayes and Regularization

$P(\text{Data}|\text{Model})$

$P(\text{Model})$

$$\begin{aligned}
 & \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2} \\
 & \quad \downarrow \log \quad \quad \quad \downarrow \log \quad \quad \quad \downarrow \log \\
 & -\frac{1}{2}(d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2) \quad + \quad -\frac{1}{2}(a_1^2 + a_2^2)
 \end{aligned}$$



Bayes and Regularization

$P(\text{Data}|\text{Model})$

$P(\text{Model})$

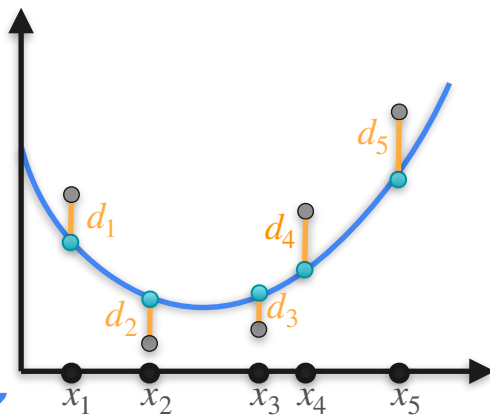
$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2}$$

Maximize

$$-\frac{1}{2}(d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2)$$

+

$$-\frac{1}{2}(a_1^2 + a_2^2)$$



$$a_1x^2 + a_2x + b$$

Bayes and Regularization

$P(\text{Data}|\text{Model})$

$P(\text{Model})$

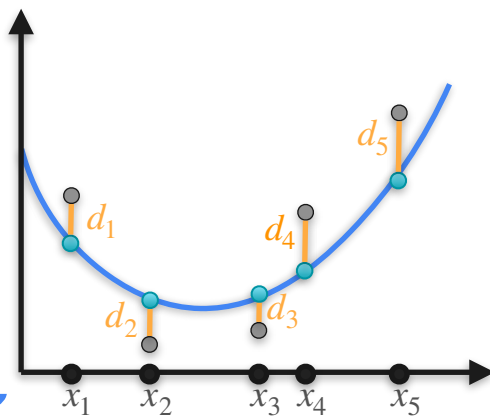
$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2}$$

Maximize

$$-\frac{1}{2}(d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2)$$

+

$$-\frac{1}{2}(a_1^2 + a_2^2)$$



$$a_1x^2 + a_2x + b$$

Minimize

Bayes and Regularization

$P(\text{Data}|\text{Model})$

$P(\text{Model})$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2}$$

Maximize

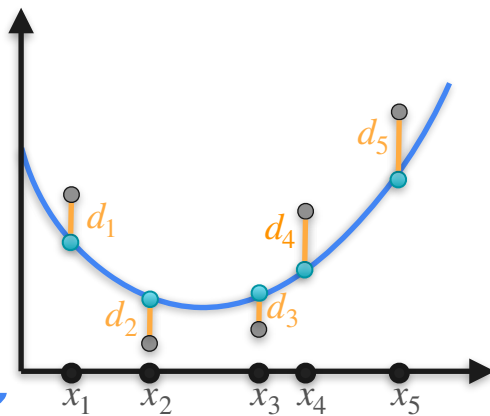
$$-\frac{1}{2}(d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2)$$

+

$$-\frac{1}{2}(a_1^2 + a_2^2)$$

Minimize

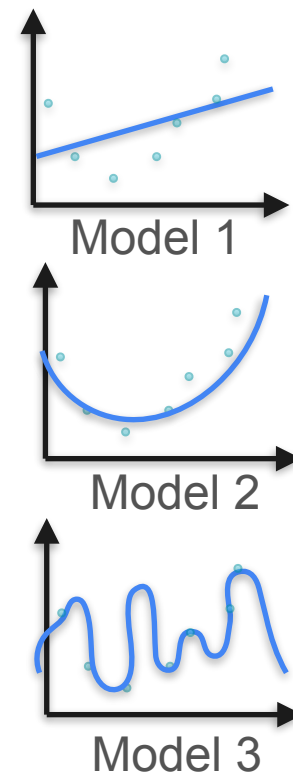
$$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + a_1^2 + a_2^2$$



$$a_1x^2 + a_2x + b$$

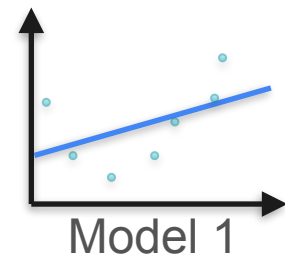
Regularization

Regularization

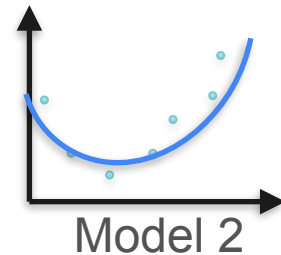


Regularization

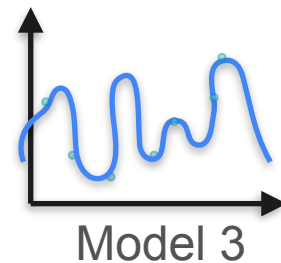
$P(\text{Model 1})$



$P(\text{Model 2})$



$P(\text{Model 3})$



Regularization

P(Model 1)

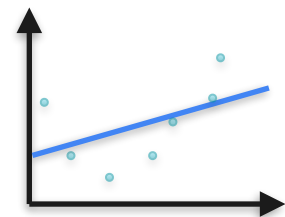
Minimize x_1^2

P(Model 2)

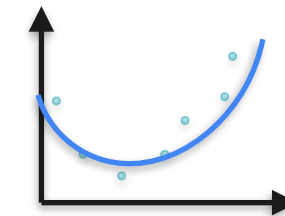
Minimize $x_1^2 + x_2^2$

P(Model 3)

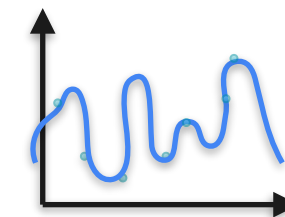
Minimize $x_1^2 + \dots + x_{10}^2$



Model 1



Model 2



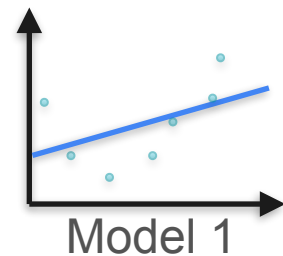
Model 3

Regularization

P(Model 1)

Minimize x_1^2

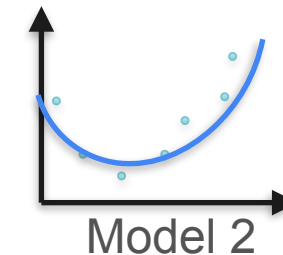
P(Data|Model 1)



P(Model 2)

Minimize $x_1^2 + x_2^2$

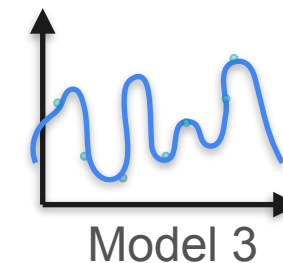
P(Data|Model 2)



P(Model 3)

Minimize $x_1^2 + \dots + x_{10}^2$

P(Data|Model 3)



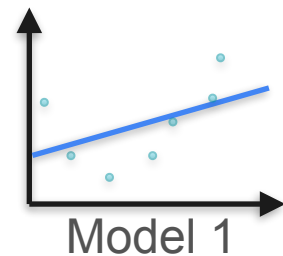
Regularization

P(Model 1)

Minimize x_1^2

P(Data|Model 1)

$$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$$

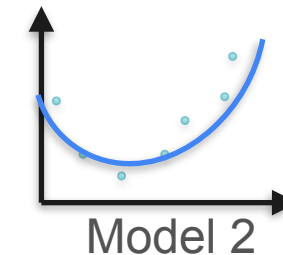


P(Model 2)

Minimize $x_1^2 + x_2^2$

P(Data|Model 2)

$$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$$

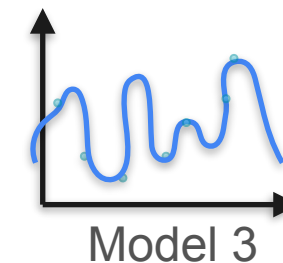


P(Model 3)

Minimize $x_1^2 + \dots + x_{10}^2$

P(Data|Model 3)

$$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$$



Regularization

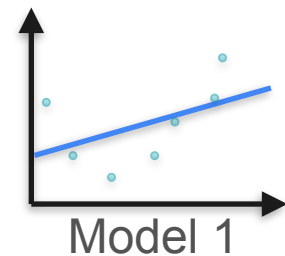
New Loss

P(Model 1)

Minimize x_1^2

P(Data|Model 1)

$$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$$

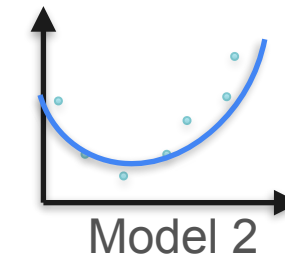


P(Model 2)

Minimize $x_1^2 + x_2^2$

P(Data|Model 2)

$$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$$

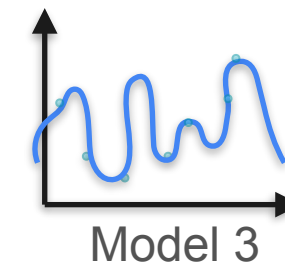


P(Model 3)

Minimize $x_1^2 + \dots + x_{10}^2$

P(Data|Model 3)

$$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$$



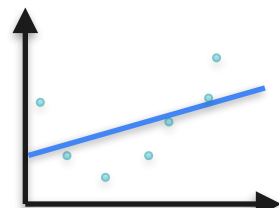
Regularization

New Loss

$P(\text{Model 1})$

Minimize x_1^2 + $d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$

$P(\text{Data}|\text{Model 1})$

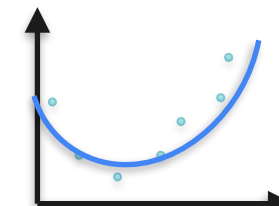


Model 1

$P(\text{Model 2})$

Minimize $x_1^2 + x_2^2$ + $d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$

$P(\text{Data}|\text{Model 2})$

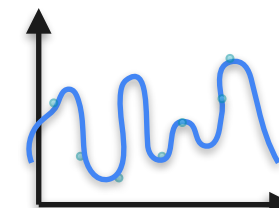


Model 2

$P(\text{Model 3})$

Minimize $x_1^2 + \dots + x_{10}^2$ + $d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$

$P(\text{Data}|\text{Model 3})$



Model 3



DeepLearning.AI

Point Estimation

Conclusion